

XXV International Symposium on Solid State Dosimetry

Local Venue: Centro de Desenvolvimento da Tecnologia Nuclear (CDTN)
September 22-26, 2025 — Belo Horizonte, Brazil

Book of Proceedings: Volume 4

Editors

Telma C. F. Fonseca

Bruno M. Mendes

Lucas F. Araújo

Ester M. R. Andrade



BOOK OF PROCEEDINGS

Volume 4

XXV International Symposium on Solid State Dosimetry

Edited by

Telma C. F. Fonseca

Bruno M. Mendes

Lucas F. Araújo

Ester M. R. Andrade





Contents

1	Preface	6
2	Summary and Perspectives	7
3	Organization, Committees, and Reviewers	8
3.1	Chair	8
3.2	Co-Chair	8
3.3	Organizing Committee	8
3.4	Scientific Committee	8
3.5	Publicity committee	9
3.6	Social Committee	9
3.7	Sponsorship Committee	9
3.8	Website Committee	9
3.9	Conference Volunteers	9
3.10	Reviewers	10
4	Event Organization, Supporting Institutions, and Sponsors	12
4.1	Organization	12
4.2	Supporting Institutions	12
4.3	Sponsors	12
5	Schedule	13
6	Pre-Symposium Courses	32
6.1	Development of PET Radiopharmaceuticals	32
6.2	Introduction to radiation detectors	33
6.3	Importancia de la física en una imagen por resonancia magnética en el diagnóstico clínico. . .	33
6.4	Solid State Dosimetry: Principles and Applications & OSL Advanced Materials.	34
7	Symposium Speakers	35
7.1	Dr. Ademar Benévolo Lugão: Use of ionizing radiation for the production of nanoparticles and advanced curatives.	35
7.2	Dr. Bernardo Dantas: Disseminación de técnicas de monitoreo in vivo ocupacional en una Red de Clinicas de Medicina Nuclear	35
7.3	Dr. Carmen Bueno: Radiation processing with electron beams: On the feasibility of using electronic dosimeters for real-time dosimetry	36
7.4	Dr. Geórgia Santos Joana: WiN Brasil - Diversity & Inclusion Management	36
7.5	Dr. Helio Yoriyaz: Proton Beam Therapy: Main Characteristics and Challenges	37
7.6	Dr. Luciana Carvalheira: Neutrons, Knowledge, and New Generations: How Brazil's Argonauta Reactor is Shaping the Future of Radiochemistry and Its Impact on Global Dosimetry Challenges.	38



7.7	Dr. Maria Neves: Range radiation exposure indicator	38
7.8	Dr Richard Hugtenburg: New technologies as a route to particle therapy: Laser-hybrid Acceleration for Radiobiology Applications - The LhARA project.	38
7.9	Dr. Rolf Behrens: From Dosemeter Development to Routine use - Standards and Uncertainties	39
7.10	Dr. Silvia Hidalgo Tobón: State of the art of magnetic resonance imaging: from fundamental physics to clinical applications.	39
7.11	Dr. Susana Lalic: Plant Stem Cells as an Ethical Alternative for Biodosimetry	40
7.12	Dr. Tarcísio Campos and Dr. Alberto Campos: Challenges in radioisotope decontamination and production through neutron spectral selection	40
7.13	Dr. Thiago Lima: Current and future trends in Nuclear medicine Dosimetry	41
7.14	Dr. Yvone Mascarenhas: Personal Dosimetry - The Importance of Regulation, Technology, and Culture. Evaluation of Dosimetry Services in Latin America and the Caribbean	41
7.15	Dr. Wilson Calvo: Current Status of Strategic Projects at CNEN	42
7.16	Dr. Bruno Melo Mendes	43
7.17	Dr. Telma Cristina Ferreira Fonseca	43
8	Round Tables	44
8.1	AI's Challenges and Opportunities: Connecting Research, Industry, and Society	44
8.2	Radon in Environment: Tracer Applications and Public Health Risks	45
8.3	Leadership in Scientific and Technological Research	46
9	Mentorship Helen Khoury WIN-Brasil	47
10	Honors Awards	48
10.1	Prof. Dr. Tarcisio Passos Ribeiro de Campos	48
10.2	Prof. Dr. Teógenes Augusto Da Silva	49
10.3	Prof. Dr. Helen Jamil Khoury - In Memoriam	50
11	Helen Khoury Award for Best Works	51
11.1	Best Work in Communication	51
11.1.1	Honorable Mentions in Communication	51
11.2	Best Work in Panel	52
11.2.1	Honorable Mentions in Panel	52
12	Symposium Papers	54
G05-014:	Proposal for the use of thorium as fuel in TRIGA® research reactors	54
G05-015:	Quality assurance in ⁹⁰ Y radioembolization: an intercomparison protocol for activity optimization	65
G05-016:	Radiological Characterization and Dosimetric Assessment of Liquid Radioactive Waste from BN-350 Reactor	72
G05-018:	Reference Measurements of Thermal Neutron Flux Using a Cadmium-Encapsulated ⁶ LiI (Eu) Detector at the Neutron Metrology Laboratory	81
G05-019:	Structural and compositional changes in zeolitic materials exposed to uranium mine effluents: implications for environmental radioprotection	92
G05-020:	Survey of radiation exposure from portable X-ray fluorescence	102
G05-021:	Transfer Factors of Natural Radionuclides in Organic Foods from Rio de Janeiro, Brazil .	107
G05-023:	Analysis of Natural Radioactivity in Beach Sands Along the Rio de Janeiro State Coast by Gamma Spectrometry	114
G06-002:	Estimación del término fuente de un irradiador gamma de ⁶⁰ Co utilizando el código MCNP	125
G06-004:	Alpha particle spectrometry and Monte Carlo evaluation of ²⁴¹ Am sources for use in Elastic Recoil Detection Analysis (ERDA)	138

G06-00: Axial neutron flux mapping in the Argonauta Research Reactor at IEN-CNEN using neutron activation and γ -ray spectrometry techniques	146
G06-006: Comparative study of gamma efficiency curves in HPGe detector with different calibration standards	156
G06-008: Determination of the mass attenuation coefficient of polymers using Monte Carlo N-Particle simulation	164
G06-011: Identification and quantification of K, Na and Mn in textured soy protein sold in the city of Rio de Janeiro using the Argonauta research reactor of IEN/CNEN and the techniques of instrumental neutron activation analysis and γ -ray spectrometry	172
G06-012: Neutron Source Applied for Eccentric Scale Prediction in Oil Pipelines using Deep Learning	182
G06-017: Response Function Validation of HPGe Detector Through GADRAS and MCNP Simulation	193
G07-001: Pulsed X-ray-Induced Nuclear Fragmentation in a Radioresistant Ovarian Cancer Cell Line	203
G07-002: Low-Dose Neonatal Ionizing Radiation Induces Modest, Sex-Specific Changes in Adult Mouse Behavior and Hippocampal Oxidative Stress	213
G07-007: Germination response of 'Micro-Tom' to controlled magnetization by using a Rodin coil .	221
G07-008: Impact of Physical Exercise on Irisin Expression in the Brain of Irradiated Mice: Effects on Behavior and Memory	232
G07-010: Magnetic stimulation as a promoting factor for germination and development in Guazuma ulmifolia L. seeds	239
G07-014: Radiometric analysis of sediments and water from the Lagoons of East Fluminense, Rio de Janeiro	246
G07-015: Radiometric analysis of the bivalve mollusk Perna	256
G07-017: Study of Drinking Water Quality in Rio de Janeiro: Assessment of Gamma Radiation Levels and Physicochemical Parameters	263
G07-018: Growth Inhibition of <i>S. cerevisiae</i> BY4741 Exposed to Gold Nanoparticles and Low-Energy Radiation	271
G08-001: Cathodoluminescence behavior of PVDF films doped with CuSO_4 under high-dose gamma irradiation	282
G08-006: Dosimetric Properties of nanoDot OSL Detectors for Radiotherapy Quality Control: An Initial Characterization and Uncertainty Analysis	292
G08-008: Investigation of the luminescent properties of zirconia-enriched dental resins	303
G08-013: Radiation Detection of Lead-Free BaZrO_3 Perovskite Scintillators	309
G08-014: Feasibility of Retrospective Dosimetry Using Ceramic Material from Spark Plugs	318
G08-019: Unsupervised learning for pattern recognition in K-feldspar using cathodoluminescence emission and EDX composition mapping	325
G08-023: Repuesta Termoluminiscente ante rayos gamma de la HAp pura y dopada con tierras raras (Dy^{3+} y Eu^{3+}) a dos concentraciones 0.1 M y 0.5 M	333

CHAPTER 1

Preface

It is with great pleasure that we welcome all participants to the XXV International Symposium on Solid State Dosimetry (ISSSD) in September of 2025, for the first time, at Centro de Desenvolvimento da Tecnologia Nuclear (CDTN/CNEN) and Universidade Federal de Minas Gerais (UFMG) in Belo Horizonte - Minas Gerais - Brazil. This milestone edition marks over two decades of continuous commitment to the advancement of radiation dosimetry through solid-state materials, techniques, and applications.

Since its inception, the ISSSD has served as a vital platform for scientists, engineers, and professionals to share knowledge, discuss new developments, and foster collaborations in the field of radiation detection and measurement. This 25th edition continues this proud tradition, bringing together leading experts and emerging researchers from across the globe to present their latest findings and insights.

The program of this symposium reflects the diversity and depth of the field, covering topics ranging from fundamental studies of dosimetric materials to applied research in medical physics, environmental monitoring, and radiation protection. Special attention has been given to emerging technologies and interdisciplinary approaches that are shaping the future of dosimetry.

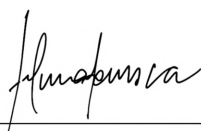
We are especially grateful to all contributors, authors, reviewers, session chairs, and members of the organizing, scientific, and support committees, whose dedication and effort have made this event possible. We also extend our sincere appreciation to our institutional partners and sponsors for their continued support.

We hope that this symposium will not only be an opportunity for scientific enrichment but also a space for meaningful exchanges, new collaborations, and the strengthening of our global community.

Welcome to the XXV ISSSD, and may this be a rewarding and inspiring experience for all.

Organizing Committee

XXV International Symposium on Solid State Dosimetry



Telma C. F. Fonseca
Chair at XXV ISSSD



Bruno M. Mendes
Chair at XXV ISSSD



Juan Azorin Nieto
President of the Sociedad Mexicana de
Irradiación y Dosimetría (SMID)



CHAPTER 2

Summary and Perspectives

The XXV International Symposium on Solid State Dosimetry (ISSSD-2025) brought together researchers, professionals, and students from across the globe, reinforcing the vitality and relevance of solid state dosimetry in advancing radiation science and applications.

The 250 submitted works for the event were distributed across eight thematic areas, reflecting a wide range of research interests. Dosimetry (G2) had the highest representation with 90 works (36% of total), followed by Medical Physics (G4) with 36 contributions (14.4%) and Applications of Thermoluminescence (G1) with 35 works (14%). Other areas included Luminescent Materials (G8, 24 works, 9.6%), Radiation Protection (G5, 21 works, 8.4%), Radiobiology (G7, 19 works, 7.6%), Radiation Sources (G6, 15 works, 6%), and Ionizing and Non-Ionizing Radiation (G3, 10 works, 4%).

Regarding the presentation format, a total of 99 oral communications were delivered, with 54 in person and 45 online, while 151 poster presentations were held, including 96 in person and 55 online. Dosimetry (G2) again showed strong engagement with 23 in-person oral presentations, 10 online oral presentations, 38 posters in person, and 19 online posters. Applications of Thermoluminescence (G1) accounted for 8 oral in person, 7 online, 12 posters in person, and 8 online, while Medical Physics (G4) registered 7 oral in person, 9 online, 11 posters in person, and 9 online. Other areas contributed more modest numbers, such as Luminescent Materials (G8) with 19 works, mainly presented as posters, and Radiobiology (G7) and Radiation Protection (G5) with 19 and 21 contributions, respectively, shared between oral and poster sessions. Radiation Sources (G6) and Ionizing and Non-Ionizing Radiation (G3) had smaller representations but maintained coverage of fundamental aspects of the field. Overall, the data demonstrate diverse scientific participation, with a healthy balance between oral and poster formats, and a substantial proportion of online presentations, highlighting the hybrid and inclusive nature of the event.

Reflecting the truly international character of the symposium, participants represented 18 countries, totaling 248 presentations, ensuring broad participation and accessibility despite geographical distances. An analysis of the data reveals a notable predominance from Brazil, with 175 presentations. Mexico followed with a significant participation of 37 presentations. Other Latin American countries also had a notable presence, including Colombia (7), Chile (4), Peru (4), and Argentina (3). The international representation was broadened by Saudi Arabia and Kazakhstan, both with 3 presentations, and Indonesia with 2. Finally, contributing to the event's diversity, Bolivia, Canada, Ecuador, Spain, Guatemala, Malaysia, the United Kingdom, Switzerland, and Uruguay each participated with a single presentation.

CHAPTER 3

Organization, Committees, and Reviewers

3.1 CHAIR

Telma Cristina Ferreira Fonseca (DEN/UFMG-Brazil)

Silvia S. Hidalgo Tobón (UAM/México)

3.2 CO-CHAIR

Bruno Melo Mendes (CDTN/CNEN-Brazil)

Pedro Ramón González Martínez (México)

3.3 ORGANIZING COMMITTEE

Telma Cristina Ferreira Fonseca (DEN/UFMG-Brazil)

Bruno Melo Mendes (CDTN/CNEN-Brazil)

Lucas Fabrício de Araújo (DEN/UFMG-Brazil)

Ester Maria Rodrigues de Andrade (CDTN/CNEN-Brazil)

Ana Carolina Costa da Silva (WiN-Brazil)

Teógenes Augusto da Silva (CDTN/CNEN-Brazil)

Juan Azorín Nieto (UAMI-México)

3.4 SCIENTIFIC COMMITTEE

Andrea Mantuano¹ (UERJ-Brazil)

Ester Maria Rodrigues de Andrade¹ (CDTN/CNEN-Brazil)

Lucas Paixão Reis (IMA/UFMG-Brazil)

Luiz Cláudio Meira-Belo (CDTN/CNEN-Brazil)

Rita de Cássia de Oliveira Sebastião (Chemical/ICEX/UFMG-Brazil)

Sonia Seger Mercedes (DEN/UFMG-Brazil)

¹President



3.5 PUBLICITY COMMITTEE

Davi Silveira Azevedo¹ (CDTN/CNEN-Brazil)
Bruna Lima Figueiredo¹ (DEN/UFMG-Brazil)
Jaime Torres Juárez (UAMI-México)
Fernando Soares Santos (CDTN/CNEN-Brazil)
Sara Suely Ventura da Silva Lima (CDTN/CNEN-Brazil)
Beatriz Diniz de Oliveira Guedes (CDTN/CNEN-Brazil)
Fernanda Rocha Cavalcante (CDTN/CNEN-Brazil)

3.6 SOCIAL COMMITTEE

Camila Engler¹ (CDTN/CNEN-Brazil)
Elydio Soares¹ (CDTN/CNEN-Brazil)
Laura Takahashi (CDTN/CNEN-Brazil)
Thayná Felizardo (CDTN/CNEN-Brazil)

3.7 SPONSORSHIP COMMITTEE

Bruno Gallotti (DEN/UFMG-Brazil)
Pâmera Paola Silva (Brazil)

3.8 WEBSITE COMMITTEE

Gabriel Góes Braga (COLTEC-Brazil)
Maria Luisa de Paula Ferreira da Silva (UFMG-Brazil)
Norberto Ferreira da Silva (Brazil)

3.9 CONFERENCE VOLUNTEERS

Anna Clara Moreira Da Silva (IMA/UFMG-Brazil)
Asif Amin (Farmácia/UFMG-Brazil)
Bianca De Oliveira Gomes (IB/UNESP-Brazil)
Bruna Lima Figueiredo (DEN/UFMG-Brazil)
Carolina Santos Lamounier (CDTN/CNEN-Brazil)
Danielly Tamires Da Silva Costa (IPEN/CNEN-Brazil)
Daniele Gonçalves Mesquita (UFPEL-Brazil)
Davi Silveira Azevedo (CDTN/CNEN-Brazil)
Fernanda Rocha Cavalcante (CDTN/CNEN-Brazil)
Fernando Soares Santos (CDTN/CNEN-Brazil)
João Victor Ramos Oliveira (ICEX/UFMG-Brazil)
José Eliseu de Souza Junior (DEN/UFMG-Brazil)
Ketrim Regina Souza (INB-Brazil)
Lucas Beloti Silva (ICEX/UFMG-Brazil)
Maria Luisa de Paula Ferreira da Silva (ICEX/UFMG-Brazil)
Mateus de Souza Resende (IMA/UFMG-Brazil)
Sâmella de Andrade Brum (DEN/UFMG-Brazil)
Thamyra Cybelle Vieira Dos Santos (INMETRO-Brazil)

¹ President



3.10 REVIEWERS

Abner Duarte Soares (CNEN-Brazil)
Adriana de Souza Medeiros Batista (UFMG-Brazil)
Alex Cristóvão Holanda de Oliveira (FNE-Brazil)
Ana Letícia Almeida Dantas (IRD/CNEN-Brazil)
Andrea Mantuano (UERJ-Brazil)
Angel Ramírez (IGEF/UNAM-Mexico)
Ângela Marques (UFMG-Brazil)
Bernardo Dantas (IRD/CNEN-Brazil)
Bruno Cordeiro Silva (CEFET/MG-Brazil)
Bruno Gallotti (UFMG-Brazil)
Bruno Mendes (CDTN/CNEN-Brazil)
Camila Engler (CDTN/CNEN-Brazil)
Carlos Velasquez (UFMG-Brazil)
Carmen C. Bueno (IPEN/CNEN-Brazil)
Carolina Viloría Barragán (ITM-Colômbia)
Cássio Miri Oliveira (UNIFESP-Brazil)
Crissia Carem Paiva Fontainha (UFMG-Brazil)
Daniel Calheiro (UFMG-Brazil)
Daniel Campolina (CDTN/CNEN-Brazil)
Danilo Junot (UERJ-Brazil)
Denison Santos (IRD/CNEN-Brazil)
Elydio José Soares (CDTN/CNEN-Brazil)
Ester Maria Rodrigues de Andrade (CDTN/CNEN-Brazil)
Estefania Reyes Soto (UAM-México)
Evelise Gomes Lara (RMB-Brazil)
Evelyn Huerta (UAM-México)
Fernando Leyton (UTA-Chile)
Francisco Torres Hoyos (Universidad de Córdoba-Colômbia)
Helio Yoriyaz (IPEN/CNEN-Brazil)
Ivonne Berenice Lozano Rojas (IPN-México)
Jaime Torres Juárez (UAM-México)
Janaína Mendes (INCA-Brazil)
Jhon Brandon Felix Ferreira (UERJ-Brazil)
Jhonny Antonio Benavente Castillo (IPEN/CNEN-Brazil)
Jony Marques (UFMG-Brazil)
José Marques Lopes (IME-Brazil)
José Wilson Vieira (IFPE and UPE-Brazil)
Juan Azorín (UAM-México)
Kelly Cristina Martins Faêda (PUC/MG-Brazil)
Laura Cardoso Takahashi (CDTN/CNEN-Brazil)
Leanderson Cordeiro (UFPE-Brazil)
Leonardo Silva (UFMG-Brazil)
Lucas Daniel Cunha de Paula (CDTN/CNEN-Brazil)
Lucas Fabrício de Araújo (UFMG-Brazil)
Lucas Paixão (UFMG-Brazil)
Luciana Moreira (CEFET/MG-Brazil)
Luciana Nogueira (UFMG-Brazil)
Luciana Penedo (UERJ-Brazil)



Luciene Mota (UFMG-Brazil)
Luís Alexandre Gonçalves Magalhães (LCR/UERJ)
Luiz Claudio Meira-Belo (CDTN/CNEN-Brazil)
Marco Antonio Polo Labarrios (UAM Iztapalapa)
Marco Aurelio de Sousa Lacerda (CDTN/CNEN-Brazil)
Marcia Regina Pereira Attie (UFS-Brazil)
Maria Da Penha Albuquerque Potiens (IPEN/CNEN-Brazil)
Maria do Socorro Nogueira (CDTN/CNEN-Brazil)
Maria Gabrielle de Lima Rocha (FAFAR/UFMG-Brazil)
Maria José Neves (CDTN/CNEN-Brazil)
Marissa Anabel Rivera Cardona (CNEN-Brazil)
Mauricio Moacir da Silva Borges (UERJ-Brazil)
Modesto Sosa (UG-México)
Natalia Regina Souza Araujo (UFMG-Brazil)
Natasha Briggs e Silva (UERJ-Brazil)
Paulo Márcio de Oliveira (UFMG-Brazil)
Pedro R. González (ININ-México)
Priscila Santana (UFMG-Brazil)
Rita Sebastião (UFMG-Brazil)
Rodrigo Gontijo (UFMG-Brazil)
Samara Cristina Ferreira Machado (UERJ-Brazil)
Silvia Hidalgo (UAM-Mexico)
Sonia Mercedes (UFMG-Brazil)
Talita De Oliveira Santos (UFMG-Brazil)
Tarcisio Passos Ribeiro de Campos (CDTN/CNEN and UFMG-Brazil)
Telma Fonseca (UFMG-Brazil)
Teógenes da Silva (CDTN/CNEN-Brazil)
Tiago Jornada (HC/UFPE-Brazil)

CHAPTER 4

Event Organization, Supporting Institutions, and Sponsors

4.1 ORGANIZATION



4.2 SUPPORTING INSTITUTIONS



4.3 SPONSORS





CHAPTER 5

Schedule

General Schedule

Hour

201100

Monday, 09/22/2025

08:00 Registration

Courses Pre-Symposium		
	Spectrum Auditorium	Lattice Auditorium
	Chair: Marina Bicalho	Chair: Álvaro M. L. Gómez
10:30 COFFE BREAK	08:00 Dr. Marina Bicalho (CDTN/CNEN-Brazil), Dr. Isabel Dregely (Fachhochschule Technikum Wien-Austria) and Dr. Emerson Bernardes (IPEN/CNEN-Brazil) <i>Development of PET Radiopharmaceuticals</i>	08:00 Dr. Marco Antonio Polo Labarrios (UAM-Mexico) and Dr. Álvaro M. L. Gómez (DEN/UFMG-Brazil) <i>Introduction to Radiation Detectors</i>

12:00 LUNCH

Courses Pre-Symposium		
	Spectrum Auditorium	Lattice Auditorium
	Chair: Angel Ramirez	Chair: Carlos E. Velasquez
15:00 COFFE BREAK	13:00 Dr. Silvia S. Hidalgo Tobón (UAM/Mexico) <i>Importancia de la física en una imagen por resonancia magnética en el diagnóstico clínico</i>	13:00 Dr. Ivonne Berenice Lozano Rojas (CICATA-Mexico) <i>Solid State Dosimetry: Principles and Applications & OSL Advanced Materials</i>

Tuesday, 09/23/2025

08:00 Registration

Opening Ceremony - Lumina Auditorium

09:00 Opening Ceremony – **Amenônia Maria Ferreira Pinto** – Centro do Desenvolvimento da Tecnologia Nuclear (CDTN/CNEN), **Bruno Melo Mendes (ISSSD 2025 – Chair)**, **Henrique Resende Martins** – Vice-director of Escola de Engenharia of Universidade Federal de Minas Gerais (UFMG), **Silvia S. Hidalgo Tobón (ISSSD 2025 – Chair)** – Universidad Autónoma Metropolitana (UAM), **Telma Cristina Ferreira Fonseca (ISSSD 2025 – Chair)** – Universidade Federal de Minas Gerais (UFMG), **Wilson Aparecido Parejo Calvo** – Comissão Nacional de Energia Nuclear (CNEN).

10:00 COFFEE BREAK

Invited Speaker - Lumina Auditorium

Chair: Silvia Hidalgo

10:30 *Invited Speaker* – Wilson Aparecido Parejo Calvo (CNEN-Brazil),
Current Status of Strategic Projects at CNEN.

11:20 *Invited Speaker* – Yvone Maria Mascarenhas (SAPRA Landauer-Brazil),
Personal Dosimetry: The Importance of Regulation, Technology, and Culture. Evaluation of Dosimetry Services in Latin America and the Caribbean.

12:30 LUNCH

Roundtable Session (Lattice Auditorium)

13:30 Radon in Environment: Tracer Applications and Public Health Risks
Chair: Talita Santos (CDTN/CNEN-Brazil),
Panelists: Elydio Soares, Laura Takahashi, and Nathan da Silva

Radiobiology	Luminescent Materials/ Applications of Thermoluminescence
Spectrum Auditorium	Lumina Auditorium
Chair: Tarcísio Campos	Chair: Susana Lalic

13:30 Jhon Brandon Felix Ferreira
G07-002 - *Low-Dose Neonatal Ionizing Radiation Induces Modest, Sex-Specific Changes in Adult Mouse Behavior and Hippocampal Oxidative Stress*

13:45 Lucas Siqueira de Lima Neto
G07-013 - *Radiometric Analysis by gamma spectrometry of copaiba oil using a Hyper*

14:00 Iris Bomfim Martins Ferreira
G07-015 - *Radiometric analysis of the bivalve mollusk Perna*

13:30 João Pedro
VanellusRad - *The First Brazilian Real-Time IoT Electronic Dosimeter for Radiation Monitoring*

14:00 Ángel Ramírez Luna
G01-002 - *Afinando la Cronología Relativa de Cerámicas de la Zona de Playa Vicente, Veracruz, México Aplicando el Método de Datación Absoluta por Termoluminiscencia*

Roundtable Session (Lumina Auditorium)

14:30 AI's Challenges and Opportunities: Connecting Research, Industry, and Society
Chair: Telma Cristina Ferreira Fonseca (DEN/UFMG-Brazil),
Panelists: Ana Carolina Costa, Andre Henrique Alves Carneiro and Rita Sebastião

13:30 to 16:30 - Poster Session (Main Hall)

**Note: 14:30 to 16:30 - Referees for evaluation
In-Person**

G1 - Applications of thermoluminescence (dosimetry, dating, industrial, etc.)

- G01-001** - Olania Herrera González, *Involvement of SMARCB1 subunit of the mSWI/SNF and GLI-1 epigenetic complex and their radiolabeling as a strategy for ionizing therapy in lung cancer*
- G01-011** - Samira Batista da Silva, *Enzymatic biocatalysis of alginate spheres and iodinated amino acids for applications in embolization guided by SPECT*
- G01-026** - Roseli Künzel, *A low-cost integrated system for thermoluminescence and optically stimulated luminescence measurements*
- G01-029** - Yuri Avelino Costa, *Analysis of the average geometric parameters of the closed pores of the fine aggregate matrix using X-ray micro-computed tomography and linear regression*
- G01-032** - Danilo O. Junot, *Crab shell biowaste composites as luminescent radiation detectors*
- G01-035** - Luiz Matheus de Souza Aguiar Leite, *Development of a radiochromic film using a polyacetylene compound*
- G01-036** - Thalles Oliveira Campagnani, *Development of a test bench using a CsI/SiPM detector for mass flow measurement in industrial processes*
- G01-038** - Ketrin Regina Souza, *Empowering the Future of Nuclear Energy: A Mentorship Program for the Development of Female Leadership*
- G01-040** - Raissa Alexia Camargo Guassu, *Estimation of absorbed dose in critical organs during interventional procedures: experimental and computational approach*
- G01-041** - Danilo O. Junot, *Influence of precursor purity on the synthesis and TL/OSL response of CaSO₄-based radiation detectors*
- G01-044** - Vinicius Costa Silva, *Radiological characterization of the waters feeding the Camorim Reservoir by alpha-beta – Pedra Branca State Park*
- G01-045** - Margarete Cristina Guimarães, Thessa Cristina Alonso, *Standardization of gamma irradiation procedures for the preservation of bibliographic heritage*

G5 - Radiological Protection

- G05-003** - Adriana de Souza Medeiros Batista, *Assessment of volcanic rock adsorbents for the remediation of acid mine drainage in uranium mining*
- G05-006** - Daniel de Castro Pacheco, *Development of a distributed monitoring system for active radiological protection*
- G05-007** - Cássio Miri Oliveira, *Dose evaluation and suggestion of typical values for head CT headache protocols: a preliminary study*
- G05-015** - Thamyra Cybelle Vieira dos Santos, *Quality assurance in ⁹⁰Y radioembolization: an intercomparison protocol for activity optimization*
- G05-017** - Fábio Lima Guimarães, *Radiometric survey at the Stone House Lookout, Saquarema-RJ*
- G05-019** - Marcela Morais Freitas, *Structural and compositional changes in zeolitic materials exposed to uranium mine effluents: implications for environmental radioprotection*
- G05-022** - Bianca Rossini Marques, *Validation of analytical methods using gamma spectrometry applied to the radiometric characterization of monazite sand*

G8 - Luminescent materials

- G08-004** - Divanizia do Nascimento Souza, *Development of high-sensitivity CaSO₄:Tm,Mn detectors for thermoluminescent and optically stimulated luminescent dosimetry*
- G08-007** - Roseli Künzel, *Effect of silver doping concentrations on the structural and luminescence properties of CaMoO₄ powders*
- G08-008** - Paula Thays Santos Malachias, *Investigation of the luminescent properties of zirconia-enriched dental resins*
- G08-009** - Bruno Araújo, *Optically stimulated luminescence (OSL) characteristics of seven different color types of quartz samples under 405 nm stimulation*
- G08-010** - Rafael dos Santos Viana, *Promotion of the Compton effect of ¹³⁷Cs photons to enhance the effi-*

ciency of radioluminescent batteries

G08-011 - Anderson Bezerra, *Properties of $\text{CaSO}_4\text{:Eu,Mn}$ for Luminescent Dosimetry*

G08-012 - Patrícia de Lara Antonio, *PTTL and PTOSL of TLD-100 exposed to ^{60}Co and lighting with LEDs*

G08-013 - Daniele Gonçalves Mesquita, *Radiation Detection of Lead-Free BaZrO_3 Perovskite Scintillators*

G08-014 - Claudete Roberta Evangelista, *Feasibility of Retrospective Dosimetry Using Ceramic Material from Spark Plugs*

G08-015 - Anderson Manoel Bezerra da Silva, *Structural, Morphological, Luminescent, and Dosimetric Properties of $\text{CaSO}_4\text{:Yb,Mn}$ Phosphors*

G08-016 - André Felipe de Oliveira, *Tailored theranostic nanocomposites: mesoporous silica, gold nanoparticles, and gadolinium complexes for cancer diagnosis and treatment*

G08-020 - Jéssica Pauline Nunes Marinho, *Potential theranostic nanoparticles of HAp-Gd^{3+} with curcumin and folic acid for cancer therapy*

15:30 COFFEE BREAK

Invited Speaker - Lumina Auditorium

Chair: Ivonne B. L. Rojas

16:00 Invited Speaker – Silvia S. Hidalgo Tobón (UAM-Mexico),

State of the art of magnetic resonance imaging: from fundamental physics to clinical applications.

17:00 - 20:00 Welcome & Honors Night - Lumina Auditorium

- For the Ceremony: Bruno M. Mendes, Marco Aurélio de Sousa Lacerda and Telma Cristina Ferreira Fonseca
- Award Ceremony for Honored Researchers: Dr. Tarcísio P. R. de Campos and Dr. Teógenes Augusto da Silva
- Cultural Performance by Aruanda Group
- Welcome Cocktail Gathering

Wednesday, 09/24/2025

Applications of Thermoluminescence
Spectrum Auditorium
Chair: Andrea Mantuano

08:00 Rafael Cogollo

G01-004 - *Dosimetric characteristics and kinetic analysis of pure and cerium-doped alumina (α - Al_2O_3) for low-dose thermoluminescent dosimetry applications*

08:15 Héctor Maya

G01-006 - *Kinetic analysis method for general order TL glow curves*

08:30 Natalia Barbosa Gonzaga Mendes

G01-024 - *Using gamma spectrometry as a tool for assessing residual dose in irradiated topaz*

08:45 Alberto Mizrahy Campos

G01-027 - *Random walk of neutrons in a rotational environment*

09:00 Enrique Alejandro Chávez Camacho

G01-030 - *Analysis of the effects of gamma radiation on amaranth seeds and sprouts: a design of experiments (DoE) approach*

Dosimetry

Lattice Auditorium
Chair: Ana Carolina Costa

08:00 Berenice Guadalupe Olmos Grimaldo

G02-006 - *Medición de radón en la zona del Peñón Blanco, Salinas de Hidalgo, San Luis Potosí, México*

08:15 Renato Monaco Nunes

G02-018 - *A Monte Carlo-based software for photon transport in water using the CUDA platform*

08:30 Iasmin Vieira Nishibayaski

G02-019 - *A new formulation of Fricke gel dosimeter development using salicylic acid*

08:45 Camilla da Silva Sampaio

G02-026 - *Assessing the applicability of TDCR counting for nonpure beta emitters: the case of ^{210}Pb in liquid scintillation cocktails*

09:00 Laura Cardoso Takahashi

G02-040 - *Development of an integrated national database for effective dose calculation from environmental radiation exposure in Brazil*

09:15 Laisa de Almeida Neves

G05-023 - *Analysis of Natural Radioactivity in Beach Sands Along the Rio de Janeiro State Coast by Gamma Spectrometry*

10:00 COFFEE BREAK

Radiation Sources
Spectrum Auditorium
Chair: Denison Santos

10:30 Nayelly Nikole Díaz Gutiérrez

G06-002 - *Estimación del término fuente de un irradiador gamma de ^{60}Co utilizando el código MCNP*

10:45 Ismael Teixeira Vicente

G06-004 - *Alpha particle spectrometry and Monte Carlo evaluation of ^{241}Am sources for use in Elastic Recoil Detection Analysis (ERDA)*

11:00 Estela Maria de Oliveira

G06-006 - *Comparative study of gamma efficiency curves in HPGe detector with different calibration standards*

Medical Physics
Lattice Auditorium
Chair: Lucas Paixão

10:30 Sara Suely Ventura da Silva Lima

G04-004 - *Comparative Monitoring of Lens, Thyroid, and Whole Body in Interventional Cardiology Procedures*

10:45 Prycylla Gomes Creazolla

G02-089 - *Volumetric dose evaluation using 6 MV linear accelerators for blood irradiation*

11:00 Juraci Passos dos Reis Junior

G04-011 - *An AI revolution in radiotherapy in SUS: The example of Mário Kroeff Hospital*

11:15 Natasha Briggs e Silva

G04-020 - *Determination of the mass attenuation coefficient of thermoset and thermoplastic resins at energies used in conventional mammography*

11:15 Roos S. de Freitas Dam
G06-012 - *Neutron Source Applied for Eccentric Scale Prediction in Oil Pipelines using Deep Learning*

11:30 Roger Ferreira da Silva
G04-033 - *Mammography quality assurance based on phantom image with deep learning*

08:00 to 12:30 - Poster Session (Main Hall)

Note: 10:00 to 12:30 - Referees for evaluation In-Person

G2 - Dosimetry (environmental, personal, internal, external, computational, etc.)

G02-003 - Emilia Suárez, *Development and Assessment of Perovskite-based X-Ray Detectors for In Vivo Dosimetry Applications*

G02-017 - Tales Noujaim Silveira, *^{68}Ga Production with the GE PETtrace-8 Cyclotron: a Monte Carlo study*

G02-020 - Leanderson Pereira Cordeiro, *AI-based segmentation performance and implications for organ-level dosimetry in radionuclide therapy*

G02-021 - José Marques Lopes, *An Environmental Radiometric Assessment Based on Airborne Gamma Spectrometry Data in the State of Rio de Janeiro*

G02-023 - Hugo Leonardo Lemos Silva, *Analytical and simulation methods in determining radiation exposure time of healthcare waste*

G02-027 - Alessandra Vaz dos Anjos Santos, *Automated System for Visual Detection of Personal Dosimeters Using Convolutional Neural Networks*

G02-028 - Tiago Macedo da Cunha e Silva, *Bayesian-Optimized Monte Carlo Modeling of CyberKnife Collimators and Source Parameters for Small-Field Dosimetry*

G02-030 - Heloísa Milena Caldeira Rhodes, *Comparison of Methods for Determining the Half-Value Layer (HVL) in Computed Tomography Equipment*

G02-031 - Nicolle Marques Cardoso Ignacio, *Comparison of the Dosimetric Properties of EBT-3 and EBT-4 Radiochromic Films for Application in Hemocomponent Dosimetry*

G02-032 - Fernanda Rocha Cavalcante, *Comparison of X-ray spectral correction software for a CdTe detector*

G02-033 - Guilherme Gazolla Santana, *Computational Modeling of an Anthropomorphic Phantom for Applications in Aerospace Dosimetry*

G02-034 - José Marques Lopes, *Computational Simulation of a NaI(Tl) Detector and Definition of Its Main Parameters for Application in a Neutron Activation Analysis Laboratory*

G02-035 - Lucas Beloti Silva, *Computational Tools for Automatic Conversion of Microscopy Images into Voxelized Cellular Phantoms*

G02-043 - João Vinícius Batista Valença, *Dose estimation in organs of a pediatric patient undergoing brain radiotherapy via Monte Carlo simulation*

G02-045 - Guilherme Cavalcante de Albuquerque Souza, *Dosimetric comparison of methods for obtaining dose profiles in head CT scans using 120 kV X-ray beams*

G02-046 - Frederico Possato Tagliaferro, *Dosimetry Of Electret Microphones Irradiated By X-Rays*

G02-047 - Josemary A. C. Gonçalves, *Energy Dependence of an Epitaxial Diode for Radioprotection Dosimetry*

G02-048 - Enzo Cortez, *Estimating the dose rate constant for Iodine-131 via Monte Carlo simulation: a first step towards pediatric radioiodine therapy shielding evaluation*

G02-050 - João Vinícius Batista Valença, *Evaluating Radiation Dose in Veterinary Professionals: MCNPX-Based Simulation of Equine Radiography Exposure*

G02-051 - Camila Nunes Carvalho Souza, *Evaluating the Performance of Bayesian Optimization in High-Dose-Rate Brachytherapy Planning*

G02-052 - Maria Rita Pascoal de Souza Barros, *Evaluation of Absorbed Dose Uniformity in Irradiated Foods with Different Packaging Geometries Using PMMA Dosimetry*

G02-054 - Gabriel Gomes do Nascimento, *Evaluation of the reproducibility of a new lens dosimeter holder*

G02-061 - João Victor Ramos Oliveira, *Influence of detector geometry on small-field radiotherapy simulations*

G02-066 - Daniel de Castro Pacheco, *Modeling and validation of a Whole Body Counter for in vivo measurements using CAD and MCNP6*

- G02-070** - Greiciane de Jesus Cesário, *Monte Carlo Simulation of Depth Dose in Mammography Using Geant4 and 3D-Printed ABS and PETG Phantoms*
- G02-071** - Kyssylla Monnyelle Araujo Silva, *Multielectrode Array Devices for dosimetry and dose enhancement factor analysis in nanoparticle radiotherapy*
- G02-072** - Mateus de Souza Resende, *^{99m}Tc Radiolabeled Papain Nanoparticle: in silico Dosimetry Evaluation using DM BRA phantom and the experimental biodistribution data*
- G02-074** - Henrique Chaves Gulino, *New Low Cost Process of PIN Diode Fabrication for Radiation Detection*
- G02-077** - Carmen C. Bueno, *Online response of a homemade diode dosimetry system in a mobile 500-700 keV electron beam facility*
- G02-078** - Anderson Manoel Bezerra da Silva, *Optimization of Mn²⁺ and Tb³⁺ Concentrations in CaSO₄:Mn,Tb and Advanced Investigation of its TL/OSL Dosimetric Properties*
- G02-079** - Rômulo Monteiro Lopes, *Parametrization and Validation of a HPGe Detector by Monte Carlo Simulation*
- G02-080** - Hugo Leonardo Lemos Silva, *Assessing Photon and Electron Ranges in Healthcare Waste for Radiation Sterilization*

Note: 10:00 to 12:30 - Referees for evaluation
On-line

G1 - Applications of thermoluminescence (dosimetry, dating, industrial, etc.)

- G01-007** - Lenin Estuardo Cevallos Robalino, *Monte Carlo characterization of neutron personal dosimeter based on TLD-600 and TLD-700 irradiated with ²⁵²Cf neutron source on the ISO-phantom at LPN-CIEMAT*
- G01-010** - Abel Fernando Martínez Cruz, *Thermoluminescence of MgO Chemically Modified with Nd and Li Ions, Obtained by Glycine-based Solution Combustion Synthesis*
- G01-013** - Hassan Salah, *Assessing Patient Exposure in PET/CT scans: A Head and Neck Dose Parameter Study*
- G01-019** - Mohammed, *Evaluation of Radiation Dose Reduction Strategies in Whole-Body PET/CT Imaging*
- G01-021** - Hassan Salah, *Optimizing Radiation Dose Parameters in PSMA PET/CT for Prostate Cancer: Balancing Diagnostic Efficacy and Patient Safety*
- G01-022** - Abdelmoneim Sulieman, *Patient-Specific Dose Optimization in Breast PET/CT: Towards ALARA-Compliant Protocols*
- G01-033** - Edwar Alonzzo Canaza Mamani, *Datación OSL-SAR y caracterización tecnológica de las cerámicas arqueológicas de los grupos de tradición ceramista Tupiguarani: Pirajuí I y Arataca II, Pernambuco, Brasil*
- G01-034** - David Alexander Urbina Leal, *Design and Analysis of FDM Prototypes for Gold-198 Nanoparticles*

G2 - Dosimetry (environmental, personal, internal, external, computational, etc.)

- G02-001** - Alejandro Ferreira Tapia, *Caracterización Física y Computacional de Dispositivos Modeladores de Radiación Electromagnética mediante Simulación Monte Carlo*
- G02-008** - Woody Alem Wanderley Araripe Farias, *A platform for creating computational exposure models: personalized dosimetry with Holmium-166*
- G02-009** - Woody Alem Wanderley Araripe Farias, *All about MARTIN: a BREP phantom with blood and lymphatic vessels applied to Yttrium-90 radioembolization*
- G02-012** - Aline Fabiane Gonçalves de Oliveira, *Geospatial assessment of radionuclide dispersion and associated effective dose modeled with ARTM View: a case study at CDTN's radiopharmaceutical unit*
- G02-015** - Jardel Lemos Thalhoffer, *Radiometric Profile Analysis of Soils from the São José de Itaboraí Paleontological Natural Park*
- G02-016** - Marcelo Franklin da Silva Santos, *Synthesis and Luminescent Characterization of CaSO₄:Tm,Li via Slow Evaporation, Co-precipitation, and Solid-State Diffusion Routes*
- G02-022** - Akemi Yagui, *Organ Dose Estimation in Pediatric Abdomen and Pelvis Radiographs Using Caldose_X Monte Carlo Simulations*
- G02-025** - Ranulfo da Silva Dias, *Assembly and Geant4-based computational simulation of a Helium-3 de-*

tector for ground-level cosmic radiation monitoring

G02-029 - Daniel Vieira Lamas, *Characterization of the Ubatuba Formation: Integration of Gamma Radiometry, Total Organic Carbon, and Well Logs from the Upper Cretaceous to Lower Paleogene Intervals of the Campos Basin*

G02-037 - Paulo Sérgio de Abreu Junior, *Cytogenetic assessment of ionizing radiation exposure in *Allium cepa* using micronucleus frequency: a comparative study between Antarctic and control conditions*

G02-038 - Ana Laura Burin, *Development of a Computational Phantom of the Vertebral Column for Dosimetric Investigations*

G02-039 - Leonardo da Silva Boia, *Development of a Dedicated System for Generating Input Files for the MCNP and Adaptation of a Phantom with Edema for Simulation of Prostate Brachytherapy with ^{125}I*

G02-044 - Wallacy Viana, *Dose Rate Estimation by Extracting and Analyzing Colors from an Image Generated by a CMOS Image Sensor*

G02-059 - Lucas Fabricio de Araujo, *In Silico Investigation of the Impact of Gold Nanoparticles on the Radiosensitivity of 3D HeLa Cells Using the Monte Carlo Method*

G02-060 - Gabriela Rodé de Assis da Silva, *In Situ Gamma-Ray Spectrometric Mapping of Beach Sand Radioactivity in Niterói, Brazil*

G02-067 - Pedro Henrique Avelino de Andrade, *Monte Carlo Modeling of the BPW34 Photodiode for X-ray Dosimetry*

G02-073 - Nadia Rodrigues dos Santos, *Natural Radionuclide Activity and Radiological Risk Assessment in Brazilian Coffee Samples*

G02-086 - Marcelo Luís dos Santos Filho, *Study of $\text{CaSO}_4:\text{Tm},\text{Li}$ Based Luminescent Composites Applied to TL/OSL Dosimetry in Mammography*

G02-087 - Leonardo da Silva Boia, *MCNP simulation of the effects of prostatic edema in LDR brachytherapy with ^{125}I using the adapted ICRP 110 voxel male phantom*

G02-093 - Daniel V. Lamas, *Quantification of Gamma Radiation on Drilling Cuttings Through HPGe Detector: Comparative Study With Well Logs Spectrometry Data*

G3 - Ionizing and non-ionizing radiation

G03-005 - Samara Mendes Matos, *Effect of Neutron Activation in the Biodegradable Chitosan polymer associated with Gold Nanoparticles.*

G03-010 - Angel Garcia-Duran, *Simulador de detector de Germanio de alta pureza con fuentes radiactivas, embebido en FPGA*

G4 - Medical Physics

G04-001 - Nicolás Eugenio Martín, *Adaptation and performance of CdTe-based XFCT detection system using Ag-tacers for Biomedical Applications*

G04-007 - Priscila Santos Amorim, *Implementation of free software for managing imaging protocols*

G04-008 - Angelica Viridiana Roman Martinez, *Magnetic Resonance Imaging Assessment of Corpus Callosum Volume in Pediatric Epilepsy Patients*

G04-009 - Jaime Torres Juárez, *Optimization of the Pulse Sequence for Visualization of Myocardial Fiber Architecture Using Diffusion Tensor Imaging in the Study of Cardiomyopathies*

G04-012 - Gisell Ruiz Boiset, *Assessment of radiologically tissue-equivalent materials for breast imaging phantoms using X-ray spectrometry*

G04-024 - Jesús Emmanuel Mares Ponce, *Evaluation of elemental Iron content in pharmaceutical formulations using magnetic susceptibility technique*

G04-026 - Camila Hitomi Murata, *Evaluation of the affect on performance of a 3T MRI by the proximity of a subway station and an adjacent MRI scanner*

G04-027 - Carolina Salinas Domján, *Experimental characterization of magnetic fields for dosimetric modeling with Monte Carlo methods in MR-Linac radiotherapy*

G04-029 - Ana Carolina Costa da Silva, *Global Usability of Mobile CT Scanners in Rapid Stroke Diagnosis: A Literature Review*

G04-030 - Raissa Alexia Camargo Guassu, *Impact of Body Mass Index on Radiation Doses in Interventional Radiology*

G5 - Radiological Protection

- G05-002** - Wallacy Viana, *Analysis of the characteristics of the Ingeo™ 3D850 polymer in the context of its use in visual inspections in the TRIGA IPR-R1 nuclear reactor*
- G05-004** - Hericka O. Kenup-Hernandes, *Computational Modeling for Radiological Risk Assessment: Simulation of Plume Dispersion in Radioactive Waste Fires Using HotSpot*
- G05-008** - Akemi Yagui, *Evaluation of radiation exposure in pediatric interventional cardiology procedures*
- G05-009** - Hericka O. Kenup-Hernandes, *Gadolinium Adsorption by Functionalized Mesoporous Silicas (MCM- 41/SBA-15): A Proposal for Volume and Dose Reduction in Radioactive Waste*
- G05-018** - Watila Lins Silva, *Reference Measurements of Thermal Neutron Flux Using a Cadmium-Encapsulated ^6LiI (Eu) Detector at the Neutron Metrology Laboratory*
- G05-020** - Fábio Sabará Dias, *Survey of radiation exposure from portable X-ray fluorescence*

G6 - Radiation Sources

- G06-005** - Gláucia de Oliveira Barreto, *Axial neutron flux mapping in the Argonauta Research Reactor at IEN-CNEN using neutron activation and γ -ray spectrometry techniques*
- G06-008** - João Pedro Carvalho Andrade, *Determination of the mass attenuation coefficient of polymers using Monte Carlo N-Particle simulation*
- G06-011** - Gabriel Tura Echeverria, *Identification and quantification of K, Na and Mn in textured soy protein sold in the city of Rio de Janeiro using the Argonauta research reactor of IEN/CNEN and the techniques of instrumental neutron activation analysis and γ -ray spectrometry*

G7 - Radiobiology

- G07-004** - Ivana Mara Gomes Andersen Cavalcanti, *Determination of mechanical and physical properties of pig ear cartilage through modeling and simulation using DFT*
- G07-014** - Yasmym Sarmiento Pereira, *Radiometric analysis of sediments and water from the Lagoons of East Fluminense, Rio de Janeiro.*
- G07-016** - Gabriela Rodé de Assis da Silva, *Radiometric characterization of Erythrina verna Vell.: A study of natural radioactivity in plant tissues*

G8 - Luminescent materials

- G08-003** - Marcelo Luis dos Santos Filho, *Development of $\text{CaSO}_4\text{:Li}$ Composites by Different Synthesis Routes for Applications in TL and OSL Dosimetry*
- G08-005** - Brunno Florencia de Oliveira, *Dosimetric Potential of Cowrie Shell Biomass under High Radiation Doses using TL and OSL Techniques*
- G08-018** - Caroline Pereira dos Santos, *Repuesta Termoluminiscente ante rayos gamma de la HAp pura y dopada con tierras raras (Dy^{3+} y Eu^{3+}) a dos concentraciones 0.1 M y 0.5 M*
- G08-019** - André Luiz Tavares e Silva, *Unsupervised learning for pattern recognition in K-feldspar using cathodoluminescence emission and EDX composition mapping*
- G08-021** - Joel Arturo Rivera-Garcia, *Estudio de las propiedades luminiscentes del aluminosilicato de estroncio dopado con Ce ($\text{SrAl}_2\text{Si}_2\text{O}_8\text{:Ce}$) sintetizado por el método de reacción en estado sólido*

12:30 LUNCH

Applications of Thermoluminescence / Luminescent Materials Virtual Room 1 Chair: Daniel Calheiro/Rodrigo Gadelha	Luminescent Materials / Dosimetry Virtual Room 2 Chair: Marco A. Labarrios/Álvaro M. L. Gómez	Medical Physics / Ionizing and non-Ionizing / Radiobiology Virtual Room 3 Chair: Jaime T. Juárez/Bruno Gallotti
13:30 Katherine G. Delgado G01-005 - <i>Effective dose received by Dentistry students in intraoral dental radiology using TLD dosimetry</i> 13:45 Nayely D. M. Ramirez G01-008 - <i>Synthesis of Y₂O₃ nanophosphors and evaluation of their thermoluminescent response for use in radiation dosimetry</i> 14:00 Abel Fernando Martínez Cruz G01-009 - <i>Thermally Stimulated Luminescence of Chemically Modified Magnesium Oxide with Gd³⁺ and Li⁺, Obtained by a Glycine-based Combustion Process</i> 14:15 Radmila Sabitova G01-016 - <i>Characterization of additional irradiation positions in the IVG.1M research reactor</i> 14:30 E. Cruz-Zaragoza G01-020 - <i>Optically -and thermally stimulated luminescence detection of irradiated oregano (Lippia graveolens) polymineral</i> 14:45 Umme Muslima G01-023 - <i>Thermoluminescence Characterization of Natural Flake Graphite Under 6 MeV Electron Beam Exposure from Medical LINAC for Dosimetric Applications</i> 15:00 André Luiz Tavares e Silva G01-025 - <i>A Data Science Approach on the Study of Al₂O₃ Thermoluminescent Glow Curves Under Varying Heating Rates</i> 15:15 Gabriela Pontes Cardoso G08-001 - <i>Cathodoluminescence behavior of PVDF films doped with CuSO₄ under high-dose gamma irradiation</i>	13:30 Jessica Paola C. Fraga G02-005 - <i>La termoluminiscencia del Grafito para su posible aplicación dosimétrica</i> 13:45 José Edgardo Arellano Hernández G02-004 - <i>Estudio de la respuesta termoluminiscente de NA2O-Y2O3-P2O5-SIO2 irradiados con Electrones</i> 14:00 Marcelo F. da Silva Santos G02-014 - <i>New Approaches in the Synthesis of CaSO₄:Tm,Li Phosphors with Sugars as Chelating Agents for Dosimetry Applications</i> 14:15 Vinícius R. A. A. Martins G02-055 - <i>Experimental Evaluation of the Mechanical and Dosimetric Properties of the Red Perspex 4034 Dosimeter After 15 Years of Aging</i> 14:30 Rayline Leite da Silva G02-084 - <i>Standardized Protocol for the Analysis of Cutting Samples and Lithological Materials by Gamma Spectrometry</i> 14:45 Nadia Rodrigues dos Santos G02-049 - <i>Estimation of the soil-to-plant transfer factor of natural radionuclides in organic crop from Pedra Branca State Park</i> 15:00 Leonardo da Silva Boia G02-036 - <i>Construction of the Glioblastoma scenario in the right optic nerve region using DIP techniques for simulation in the MCNP code with conversion to lattice structure by SCAN2MCNP</i> 15:15 Leonardo Santiago Melgaço Silva G02-041 - <i>Development of an optical sensor prototype for dose readings in Fricke gel dosimeters</i>	13:30 Erika Patricia Azorín Vega G04-005 - <i>Dosimetric and Biological Characterization of a 3D Cell Culture Model for the Evaluation of Radiopharmaceutical Induced Bystander Effect</i> 13:45 Leticia González Zamora G04-010 - <i>Study of Prefractal Antennas for Multiband Radiofrequency Applications</i> 14:00 Maria L. Miranda Maciel G04-014 - <i>Calibration of radiochromic films EBT2, EBT3, EBT4 and RTQA2 and assessment of uncertainties for radiotherapy doses</i> 14:15 Elsa Bifano Pimenta G04-022 - <i>Evaluating Lung CT Protocols Using the Detectability Index: An Anthropomorphic Phantom-Based Evaluation of Photon-Counting and Conventional CT</i> 14:30 Estefania Reyes Soto G04-002 - <i>Automated Classification and Manual Segmentation of Pediatric Brain Tumors Using Deep Learning on MRI Data</i> 14:45 Giulia Rita de Souza Faés G04-035 - <i>Radiotherapy Planning with VMAT in Halcyon 2.0: Impacts of Isocenter Configuration on Dosimetric Parameters and Treatment Efficiency</i> 15:00 Natalia Matos Orozco G04-003 - <i>Diseño y cálculo de blindajes para garantizar la protección radiológica en una radiofarmacia productora de radiofármacos</i> 15:15 Kuanysh Samarkhanov G05-016 - <i>Radiological Characterization and Dosimetric Assessment of Liquid Radioactive</i>

15:30 Júlia Carina Orfão Costa
G08-002 - *Characterizations of SrMoO₄:Dy³⁺ powders synthesized by hydrothermal method*

15:45 Daniele da Silva Machado
G08-006 - *Dosimetric Properties of nanoDot OSL Detectors for Radiotherapy Quality Control: An Initial Characterization and Uncertainty Analysis*

16:00 Carina Oliva Torres Cortés
G08-023 - *Repuesta Termoluminiscente ante rayos gamma de la HAp pura y dopada con tierras raras (Dy³⁺ y Eu³⁺) a dos concentraciones 0.1 M y 0.5 M*

16:15 Caroline Pereira dos Santos
G08-017 - *Uncovering the intrinsic nature of the ternary metallic alloys CuNiPd and Cu₂PPd₂: A DFT study*

16:30 Rodrigo Martínez Baltezar
G08-022 - *Point defects study on Aluminum nitride for applications in luminescence dosimetry*

16:45 Pedro R. González
G08-024 - *Thermoluminescent properties of MgO:Sm,Tb+PTFE dosimeters*

17:00 Sandrith M. Pérez Ramos
G01-003 - *Análisis cinético de la curva de brillo termoluminiscente del Óxido de Berilio (BeO) irradiado con Rayos X en un rango de baja dosis*

15:30 Carolina Salinas Domján
G02-090 - *Assessing Water-Equivalence in Gram-Negative Bacteria-Hydrogel Radiation Dosimeters: A Comparative Analysis with Gram-Positive Bacterial Systems*

15:45 Rayline Leite da Silva
G07-017 - *Study of Drinking Water Quality in Rio de Janeiro: Assessment of Gamma Radiation Levels and Physicochemical Parameters*

16:00 Teodoro Córdova Fraga
G04-019 - *COVID largo y fibrosis pulmonar: clasificación basada en tomografía en población mexicana*

16:15 Teodoro Córdova Fraga
G04-013 - *Automated detection and characterization of renal calculi using U-Net and Hounsfield unit analysis in computed tomography scans*

16:30 Priscila Santos Rodrigues
G04-036 - *Three-dimensional dose mapping of gold-198 nanoparticles: A comparative study of 3D-printed matrices using Fricke xylenol gel and Monte Carlo simulations*

16:45 Arthur Reis Martins
G06-015 - *Production of Radioisotopes in Nuclear Microreactors for Medical, Industrial, and Scientific Applications*

17:00 Jardel Lemos Thalhoffer
G02-010 - *Assessment of the level of natural radioactivity in the region of the future Municipal Environmental Protection Area (APA) Ilha Verde, Muriqui, Mangaratiba - RJ*

Waste from BN-350 Reactor

15:30 Stanislav Svetachev
G06-007 - *Determination of detection limits of elements in sediment and rock standards for INAA using the IGR reactor*

15:45 Jonathan O. dos Santos
G06-017 - *Response Function Validation of HPGe Detector Through GADRAS and MCNP Simulation*

16:00 Roberto Méndez Villafañe
G06-003 - *Neutron measurements around the research reactor RA-6 at Centro Atómico Bariloche with a Bonner Sphere Spectrometer*

16:15 Roberto Méndez Villafañe
G06-001 - *Development of a detailed model with MCNP of the Neutron Standards Laboratory at CIEMAT and validation with a Bonner Sphere Spectrometer System*

16:30 Fatima Castañeda Ortiz
G07-001 - *Pulsed X-ray-Induced Nuclear Fragmentation in a Radioresistant Ovarian Cancer Cell Line*

16:45 Daniele G. Mesquita
G07-006 - *Gamma Radiation Modulates Genomic Maintenance Pathways in T-Lymphocytes: An Analysis of Clinical Doses*

17:00 Emilio Romero Muñoz
G07-007 - *Germination response of 'Micro-Tom' to controlled magnetization by using a Rodin coil*

17:15 Raúl Ken Delgado Velázquez
G07-010 - *Magnetic stimulation as a promoting factor for germination and development in Guazuma ulmifolia L. seeds*

13:30 to 16:00 TECHNICAL VISIT

Chair: Camila Engler

Places to visit at CDTN:

- Gamma Irradiation Laboratory - [More information, access here](#)
- Dosimeter Calibration Laboratory - [More information, access here](#)
- TRIGA IPR-R1 Reactor - [More information, access here](#)

Thursday, 09/25/2025

Invited Speaker - Lumina Auditorium

Chair: Bernardo Dantas

08:00 *Invited Speaker* – Luciana Carvalheira (IEN/CNEN-Brazil),
Neutrons, Knowledge, and New Generations: How Brazil's Argonauta Reactor is Shaping the Future of Radiochemistry and Its Impact on Global Dosimetry Challenges.

09:00 *Invited Speaker* – Bernardo Maranhão Dantas (IRD/CNEN-Brazil),
Diseminación de técnicas de monitoreo in vivo ocupacional en una Red de Clínicas de Medicina Nuclear.

09:25 *Invited Speaker* – Maria Jose Neves (CDTN/CNEN-Brazil),
Range Radiation Exposure Indicator.

10:00 COFFEE BREAK

Invited Speaker - Lumina Auditorium

Chair: Telma Cristina Ferreira Fonseca

10:30 *Invited Speaker* – Ademar Lugão (IPEN/CNEN-Brazil),
Use of ionizing radiation for the production of nanoparticles and advanced curatives.

11:30 *Invited Speaker* – Richard Hugtenburg (Swansea University-United Kingdom),
New Technologies as a Route to Particle Therapy: Laser-Hybrid Acceleration for Radiobiology Applications – The LhARA Project.

12:30 LUNCH

Invited Speaker - Lumina Auditorium

Chair: Richard Hugtenburg

13:30 *Invited Speaker* – Hélio Yoriyaz (IPEN/CNEN-Brazil),
Proton Beam Therapy: Main Characteristics and Challenges.

Dosimetry
Spectrum Auditorium
Chair: Peterson Squair

Medical Physics/Dosimetry
Lattice Auditorium
Chair: Lucas Paixão

13:30 Divanizia do Nascimento Souza
G02-042 - *Dose Distribution in an Anthropomorphic Cervix Phantom Using Two Computer Modeling Approaches*

13:45 Francisco Harley Dantas Hauradou Xavier
G02-057 - *Generator of Health Optimized Simulation Templates – GHOST a 3D Slicer plugin to voxelized medical images for MCNP*

14:00 Francisco H. D. Hauradou Xavier
G02-094 - *Optimization of dose rate map acquisition time for internal dosimetry in nuclear medicine using 3D U-Net convolutional neural networks*

14:15 José Elizeu de Souza Junior
G02-062 - *Influence of Photon Emission Spectra on Air Kerma Rate Constants for Radionuclides: A Monte Carlo EGSnrc Study*

13:30 Ana Beatriz Caldeira de Andrade
G04-025 - *Evaluation of spatial resolution in PET equipment with a Jaszczak-analog phantom built by additive manufacturing*

13:45 Crislane de Jesus Cesário
G04-028 - *Experimental Evaluation of 3D-Printed Materials (ABS and PETG) for Mammographic Phantom Applications Using CaSO₄:Dy Dosimetry*

14:00 Nathalia Luzia Aidar Alves
G04-032 - *Machine Learning in PSQA: Performance of a Random Forest Model on Radiotherapy Metrics*

14:15 André Luiz Espindola Fidelis
G04-037 - *Validation of a 3D-Printed Anthropomorphic Eye Phantom for Quality Assurance in External Beam Radiotherapy Using TLDs and Radiochromic Film*

14:30 Leonardo Gualberto Silva Macedo
G02-095 - *Mass Analysis of Computational Mouse Models Developed at CDTN: A Comparative Study of Male and Female Versions*

14:45 Beatriz Diniz de Oliveira Guedes
G02-082 - *Spectrophotometric Analysis of a 3D-Printed Photopolymerizable Resin Exposed to High-Dose Gamma Radiation*

15:00 Hirys Sales
G02-053 - *Evaluation of Polymeric Materials as Muscle Tissue Equivalents for Physical Phantoms Used in the Calibration of In Vivo Monitoring Systems*

13:30 - 17:00 - Poster Session (Main Hall)

Note: 14:30 to 17:00 - Referees for evaluation In-Person

G2 - Dosimetry (environmental, personal, internal, external, computational, etc.)

- G02-081** - Felipe Mamed de Souza, *SMS Notification System for Potential Radiological Risk Scenarios*
G02-085 - Alcilene Cristina da Silva, *Study and Characterization of the EPR Response of New Materials for High Dose Dosimetry*
G02-088 - Ana Letícia Almeida Dantas, *The IRD In Vivo Monitoring Laboratory - LABMIV: Instrumental Capabilities applied in routine and emergency situations*
G02-091 - Maryanna Regina de Souza Roberto, *OSL Tandem Dosimeter for ICRU 95 Operational Quantities*
G02-092 - Thalys Matheus Gama de Oliveira, *Photons Response of an Albedo Monitor*

G3 - Ionizing and non-ionizing radiation

- G03-001** - Paula Regina Rodriguez Coelho, *"Analysis of spatial resolution in two different phantoms as a quality parameter in MRI pediatric pathology"*
G03-002 - José Marques Lopes, *Adjustment Factor for Gamma Spectra Due to the Influence of Sample Density on the Shielding Background Radiation*
G03-003 - Davyd Soares Dunda, *Application of Steam Methodology in Science Teaching: Prototyping a Cloud Chamber*
G03-004 - José Marques Lopes, *Comparison of environmental sampling results using portable and benchtop scintillation detectors*
G03-006 - Adriana de Souza Medeiros Batista, *Effect of repolymerization and gamma irradiation on the morphology of styrene-divinylbenzene composites with magnetite*
G03-007 - Paulo Cezar Rocha Silveira, *Gamma Irradiation Effects on the Mechanical and Ballistic Performance of 3D-Printed Polylactic Acid*
G03-008 - Luciana Carvalheira, *Gold nanoparticles with potential for use in multimodal combat of head and neck cancer*

G4 - Medical Physics

- G04-006** - Otávio Gabriel Moreira Silva, *SNR and CTDIvol study as a function of the Noise Index in Computed Tomography exams*
G04-016 - Rute Cristina de Oliveira Soares, *Chest Computed Tomography in Oncology: Clinical Contributions and Dosimetric Assessment*

G04-017 - Gustavo Freire Pereira da Silva, *Comparative Validation of the DEEDZ Methodology and a Photon-Interaction-Based Model Implemented in TEMPy for Estimating Effective Atomic Number and Electron Density*

G04-018 - Rute Cristina de Oliveira Soares, *Contributions of Computed Tomography to Breast Cancer Staging and Follow-Up: Justification of Practice and Dosimetric Analysis*

G04-021 - Iasmin Vieira Nishibayaski, *Dosimetric evaluation of EBT3 and EBT4 in superficial HDR brachytherapy for keloids*

G04-023 - Camila Engler, *Evaluation of Breast Positioning Quality and Its Association with Technical Parameters in Screening Mammography*

G04-031 - Douglas Philipe Martinho Dos Santos, *Investigation of Diagnostic Reference Levels (DRL) in medical radiography: dose-area product (DAP)*

G04-034 - Kellen Adriana Curci Daros, *Radiation dose from pediatric coronary artery calcium scoring CT scan to evaluate chronic kidney disease-mineral and bone disorder: Observational study*

G04-038 - Arícia Ravane Pereira da Cruz, *Validation of an alanine-based postal dosimetry system for quality control in radiotherapy*

G6 - Radiation Sources

G06-013 - Kathleen Martins, *Neutronic Analysis of an Accelerator-Driven Subcritical Reactor Utilizing Thorium Dioxide and Reprocessed Nuclear Fuels*

G7 - Radiobiology

G07-003 - Christiana da Silva Leite, *Comparative Study of Organ Mass in Healthy and Tumor-Bearing Mice for Preclinical Dosimetry Purposes*

G07-005 - Camila Ramos Silva, *Development and validation of a body-shielding device for radiotherapy of breast cancer in mice using a panoramic gamma source*

G07-008 - Luciana Penedo de Melo Teixeira, *Impact of Physical Exercise on Irisin Expression in the Brain of Irradiated Mice: Effects on Behavior and Memory*

G07-009 - Olanía Herrera González, *Involvement of SMARCB1 subunit of the mSWI/SNF and GLI-1 epigenetic complex and their radiolabeling as a strategy for ionizing therapy in lung cancer*

G07-011 - Camila de Almeida Salvego, *Nitric Oxide-Donor Nanoparticles Potentiate Radiotherapy of Triple-Negative Breast Cancer*

G07-018 - Bruno Gallotti, *Growth Inhibition of S. cerevisiae BY4741 Exposed to Gold Nanoparticles and Low-Energy Radiation*

G07-019 - Maurício Moacir da Silva Borges, *The Role of Caffeine in Reducing TNF- α and Improving Cognitive in Irradiated Mice in the Brain*

G07-020 - Luis Martín García Ortiz, *Cellular response in human leukocytes (in vitro), due to exposure to UVB radiation using apoptotic biomarkers*

Invited Speaker - Lumina Auditorium

Chair: Teógenes da Silva

14:30 *Invited Speaker* – Georgia Joana (CNEN-Brazil) and Ketrim Souza (INB-Brazil), *WiN Brasil - Diversity & Inclusion Management*.

15:30 COFFEE BREAK

16:00 - Mentorship Helen Khoury WiN Brasil (Spectrum Auditorium)

Empowering the Future of Nuclear Energy: A Mentorship Program for the Development of Female Leadership
Moderator: Telma Cristina Ferreira Fonseca (DEN/UFMG-Brazil),
Panelists: Ketrim Souza and Winners (WiN Brasil)

Roundtable Session (Lumina Auditorium)

16:00 Leadership in Scientific and Technological Research

Moderator: Luciana Carvalheira (IEN/CNEN-Brazil),

Panelists: Adriana Marques and Anna Letícia Barbosa de Sousa

**Radiation Protection
Lumina Auditorium
Chair: Teógenes da Silva**

**Radiation Sources / Radiation Protection
Lattice Auditorium
Chair: Jhonny A. Benavente Castillo**

17:00 Mirella Maturo da Cruz

G05-021 - *Transfer Factors of Natural Radionuclides in Organic Foods from Rio de Janeiro, Brazil*

17:15 Elydio J. D. Soares

G05-010 - *Investigation of radon exhalation from mortars made from niobium tailings and the mitigating role of coating materials*

17:30 Raphael Francisco Gomes dos Santos

G05-005 - *Determination of Radiation Protection Procedures for an Offshore Plant Inspection in Brazil Using Radiotracer Techniques*

17:00 Gabriel do Nascimento Freitas

G06-014 - *Nuclear simulation workflow to correct gamma ray well logs for radioactive contaminated drilling fluids*

17:15 José Alípio dos Santos Filho

G06-016 - *Proposal for the construction of a neutron generator for application in research and industry*

17:30 Welen Nunes de Lima

G05-014 - *Proposal for the use of thorium as fuel in TRIGA[®] research reactors*

Friday, 09/26/2025

Invited Speaker - Lumina Auditorium

Chair: Richard Hugtenburg

08:00 *Invited Speaker* – Rolf Behrens (PTB-Germany),
From dosemeter development to routine use – Standards and Uncertainties.

08:40 *Invited Speaker* – Susana Lalic (DFI/UFS-Brazil),
Plant Stem Cells as an Ethical Alternative for Biodosimetry.

09:20 *Invited Speaker* – Thiago Viana Miranda Lima (University of Lucerne-Switzerland),
Current and Future Trends in Nuclear Medicine Dosimetry.

10:00 COFFEE BREAK

Radiological Sources
Spectrum Auditorium
Chair: Carmen C. Bueno

Dosimetry
Lumina Auditorium
Chair: Peterson Squair

10:30 Matheus R. do Nascimento
G03-009 - *Speknife: open-source software for X-ray spectra correction in solid state dosimetry*

10:30 Jhonny A. Benavente Castillo
G02-007 - *Simulación del blindaje del búnker de un acelerador ciclotrón utilizando el código MCNP*

10:45 Eduardo B. de Paula
G05-011 - *Metrological Enhancement of Radiopharmaceuticals Half-Lives Determination via γ -ray Spectrometry and Algorithmic Data Processing*

10:45 Natasha Briggs e Silva
G02-011 - *Preliminary Evaluation of a Positioning Phantom for Dosimetric Audit in Ir-192 HDR Brachytherapy: Results from an IAEA Collaborative Project*

11:00 Igor Andrade Machado
G02-083 - *SpecUnPy code: updates and what is coming*

Invited Speaker - Lumina Auditorium

Chair: Bruno M. Mendes

11:30 *Invited Speaker* – Tarcísio P. R. de Campos (DEN/UFGM and CDTN/CNEN-Brazil) and Alberto Mizrahy Campos (Augsburg-Germany),
Challenges in Radioisotope Decontamination and Production through Neutron Spectral Selection.

12:30 LUNCH

Dosimetry
Spectrum Auditorium
Chair: Maria da Penha A. Potiens

Dosimetry
Lattice Auditorium
Chair: Bruno M. Mendes

Applications of Thermoluminescence
Lumina Auditorium
Chair: Teógenes da Silva

13:30 Isabela N. S. Ferreira
G02-064 - *Mapping of natural radionuclides U, Th and K in the city of Belo Horizonte/Minas Gerais*

13:45 Claudio A. Federico
G02-065 - *Measurements of Ionizing Radiation Dose Onboard*

13:30 Celso G. de Paulo
G02-069 - *Monte Carlo simulation of a HPGe detector*

13:45 Ana G. Leão Ferro
G02-075 - *Occupational exposure assessment in ROLL procedures with Tc-99m: A Monte Carlo simulation approach*

13:30 Tarcísio P. R. Campos
G01-046 - *The reduction of the radiological impact of nuclear fuel waste*

14:00 Júlia B. Severo
G01-039 - *Environmental Gamma Radiation and Radon Exposure in Areas Surrounding Urban*

*an Aircraft from Southeastern
Brazil to the Antarctic Continent*

14:00 Paulo J. Iack Silva
G02-068 - *Monte Carlo N-Particle
(MCNP) Simulations for
Scintillation Emission in
Quantum Dot-Doped Systems*

14:00 Rafael S. Oliveira
G02-076 - *On the adaptation of a
commercial Bonner Sphere
Spectrometer to be used with
thermoluminescent detectors*

*Quarries in Belo Horizonte,
Brazil*

Invited Speaker - Lumina Auditorium

Chair: Telma Cristina Ferreira Fonseca

14:30 *Invited Speaker* – Carmen Bueno (IPEN/CNEN-Brazil),
Radiation processing with electron beams: On the feasibility of using electronic dosimeters for real-time dosimetry.

15:30 COFFEE BREAK

16:00 CLOSING SESSION

- Honorary Award in Memory of Helen Jamil Khoury: Carmen Bueno
- Helen Khoury Award for Best Works: Andrea Mantuano & Ester Andrade
- Final Session: Summary and Perspectives: Telma Cristina Ferreira Fonseca & Bruno M. Mendes

CHAPTER 6

Pre-Symposium Courses

6.1 DEVELOPMENT OF PET RADIOPHARMACEUTICALS

Pre-symposium course speakers

- Dr. Marina Bicalho (CDTN/CNEN-Brazil)
- Special Guest: Dr. Isabel Dregely (Fachhochschule Technikum Wien-Austria)
- Special Guest: Dr Emerson Bernardes (IPEN/CNEN-Brazil)

Dr. Marina Bicalho Silveira is an Industrial Pharmacist (UFMG, 2007) with an M.Sc. (2010) and Ph.D. (2016) in Radiation Science & Technology (CNEN). Specialized in Radiopharmacy (SBRAFH, 2023), she works as a Senior Technologist at CDTN/CNEN, leading Quality Assurance for PET radiopharmaceutical production. Her expertise covers GMP, regulatory compliance, and R&D of novel radiopharmaceuticals, including preclinical studies. She also serves as a Professor in CDTN's Graduate Program, bridging industry and academia.

Dr. Isabel Dregely is Head of the Competence Center for Artificial Intelligence and Data Analytics at Fachhochschule Technikum Wien. She previously served as Senior Lecturer at King's College London and was a UKRI Future Leaders Fellow. Her research background includes advanced MRI techniques, particularly in prostate and cardiovascular imaging, with postdoctoral work at UCLA and experience at the Technical University of Munich. She holds a PhD in Physics from the University of New Hampshire, where she focused on hyperpolarized xenon MRI and RF coil development. Dr. Dregely combines expertise in biomedical imaging with AI and data analytics applications.

Dr. Emerson Soares Bernardes is a pharmacist and researcher with over two decades of experience in immunology, molecular biology, and nuclear medicine. He served as Manager of the Radiopharmacy Center at the Nuclear and Energy Research Institute (IPEN), where he led pioneering projects in the development of radiopharmaceuticals. He currently holds a position as a Visiting Researcher at Erasmus Medical Center in the Netherlands, contributing to innovation initiatives and patent development. His career is distinguished by the integration of scientific research, strategic management, and entrepreneurship.

Description: This course covers the basics of PET radiopharmaceuticals-production, clinical applications, and their impact on nuclear medicine-while showcasing CDTN'S cutting-edge technologies and their role in enhancing diagnostics and patient care in Brazil.

Dr. Marina Bicalho - Theoretical Aspects and State of the Art in PET Radiopharmaceuticals.

Dr. Emerson Bernardes - Advances in new PET Radiopharmaceuticals

Dr. Isabel Dregely - Artificial Intelligence Applied to PET Imaging



6.2 INTRODUCTION TO RADIATION DETECTORS

Pre-symposium course speakers

- Dr. Marco Antonio Polo Labarrios (UAM-Mexico)
- Dr. Álvaro M. L. Gómez (DEN/UFGM-Brazil)

Dr. Marco Antonio Polo Labarrios holds a Ph.D. and Master's degree in Energy Engineering from UNAM and a Bachelor's in Energy Engineering from UAM Iztapalapa. He teaches part-time at Universidad Iberoamericana and UNAM, focusing on Transport Phenomena and Renewable Energy. His research involves the simulation of energy processes based on mass, momentum, and energy transport. He has published 21 scientific articles, participated in multiple national and international conferences, and supervised four undergraduate theses. Dr. Polo Labarrios was part of the National Commission on Nuclear Safety and has been a Level 1 member of the National System of Researchers since 2021.

Dr. Álvaro M. L. Gómez holds a Ph.D. and M.Sc. in Nuclear Science and Technology from the Federal University of Minas Gerais – UFGM (2017; 2021). Holds a Bachelor's degree in Physics (diploma validated by UFGM) and two postgraduate specializations: Bioengineering – Universidad Distrital Francisco José de Caldas (2008; 2014) (Colombia), and University Teaching – Universidad Cooperativa de Colombia (2013). Has research experience in ionizing radiation dosimetry, development of tissue-equivalent materials for computed tomography phantoms, material characterization, and teaching experience at the Department of Nuclear Engineering (DEN) at UFGM, lecturing on radiation protection, radiation detection and nuclear instrumentation, radioisotope applications, and introduction to nuclear energy.

Description: Throughout the course, we will explore different types of radiation detectors, their operating principles, and criteria for their proper selection and use. The goal is to provide students with a solid foundation that enables them to identify the most suitable equipment for each situation and use it efficiently, maximizing both measurement accuracy and radiological protection.

6.3 IMPORTANCIA DE LA FÍSICA EN UNA IMAGEN POR RESONANCIA MAGNÉTICA EN EL DIAGNÓSTICO CLÍNICO.

Pre-symposium course speakers

- Dr. Silvia S Hidalgo Tobón (UAM-Mexico)
- Msc. Jaime Torres Juárez (UAM-Mexico)

Dr. Silvia Hidalgo earned her Bachelor's degree in Physics in 1998 from the University of the Americas-Puebla, Mexico. In 2000, she earned her Master of Science in Medical Physics from the Institute of Physics at the National Autonomous University of Mexico (the first Master of Science in Medical Physics from the UNAM program). In 2005, she earned her PhD in Physics, specializing in Magnetic Resonance Imaging from the Sir Peter Mansfield Magnetic Resonance Center at the University of Nottingham, England. She completed a postdoctoral fellowship in Magnetic Resonance Imaging at the Robarts Research Institute, University of Western Ontario, Canada. She also completed a postdoctoral fellowship at the Autonomous Metropolitan University, Iztapalapa. Her scientific interests are related to advanced functional magnetic resonance imaging, neural networks applied to neuroscience, and magnetic resonance safety in daily clinical practice. She has taught courses and supervised theses at the undergraduate, master's, and doctoral levels in magnetic resonance imaging at the UAM, UAEM, and UNAM. She has given lectures at various national and international forums on nuclear magnetic resonance imaging. She is a member of the International Society of Magnetic Resonance in Medicine, RSNA, the Mexican Physical Society, IEEE, and the Neuroscience Society, among others.



Descripción: Se presentarán las bases de la física de una imagen por resonancia magnética y su relación con el radio diagnóstico. La seguridad en resonancia magnética que evita complicaciones y accidentes que con la continua capacitación disminuyen notablemente. Así como las técnicas nuevas que están incluyendo inteligencia artificial para un mejor diagnóstico por imagen.

6.4 SOLID STATE DOSIMETRY: PRINCIPLES AND APPLICATIONS & OSL ADVANCED MATERIALS.

Pre-symposium course speakers

- Dr. Ivonne Berenice Lozano Rojas (IPN-Mexico)

Dr. Ivonne, a distinguished Mexican scientist holding a Ph.D. from Centro de Investigación en Ciencia Aplicada y Tecnología Avanzada, Unidad Legaria, conducts pioneering research in TL/OSL dosimetry, radiation detectors, and food irradiation detection, significantly advancing medical physics and safety.

Description: This minicourse focuses on quartz as a natural radiation dosimeter, exploring its defects and their role in optically stimulated luminescence (OSL). Advanced OSL materials, including synthetic and natural quartz, are analyzed for radiation dose assessment in dating and dosimetry applications.

CHAPTER 7

Symposium Speakers

7.1 DR. ADEMAR BENÉVOLO LUGÃO: USE OF IONIZING RADIATION FOR THE PRODUCTION OF NANOPARTICLES AND ADVANCED CURATIVES.

Dr. Ademar Lugão holds Bachelor's in Chemical Engineering from the Polytechnic School of the University of São Paulo – USP (1980), earned a Master's degree in Nuclear Technology from IPEN-USP (1985), and a Ph.D. in Nuclear Materials Technology, also from IPEN-USP (2004). With more than 30 years of experience as a research leader in polymer irradiation, he is currently the Manager of the Center for Chemistry and the Environment at the Nuclear and Energy Research Institute – IPEN. His main research focuses on the synthesis, processing, and characterization of polymers; structural modification of polymers through energetic processes, including crosslinking, grafting, branching, and chain scission; development of hydrogels for health and environmental applications; clean technologies for producing polyolefins with controlled rheology; starch foams for sustainable packaging; advanced low-cost dressings tailored to the needs of the Brazilian Public Health System (SUS); albumin and papain nanoparticles for encapsulating radiotherapeutics and chemotherapeutics; and radioactive gold nanoparticles (Au-198) for applications in radiodiagnosis and radiotherapy (theranostics). He currently serves as the Coordinator of the Nanomaterials and Nanocomposites Coordination Hub of SISNANO (National System of Nanotechnology Laboratories) of the Ministry of Science, Technology, and Innovation (MCTIC).

Description: Use of ionizing radiation for the production of nanoparticles and advanced curatives.

7.2 DR. BERNARDO DANTAS: DISEMINACIÓN DE TÉCNICAS DE MONITOREO IN VIVO OCUPACIONAL EN UNA RED DE CLINICAS DE MEDICINA NUCLEAR

Dr. Bernardo Maranhão Dantas holds a PhD in Sciences (Biology – Nuclear Biosciences) from the State University of Rio de Janeiro (1998). He is a Senior Technologist at the Brazilian National Nuclear Energy Commission (CNEN) and a faculty member of the Graduate Program at the Institute of Radiation Protection and Dosimetry, focusing on Radiation Biophysics. He has published 87 peer-reviewed articles and has supervised numerous students across undergraduate, master's, and doctoral levels. He has led five research and development projects and currently coordinates the CNPq-certified research group "Occupational Internal Dosimetry."



Description: Professionals who handle radiopharmaceuticals for diagnostic and therapeutic purposes in nuclear medicine are subject to the intake of radionuclides through inhalation and ingestion. The radiological protection aspects of this practice are regulated in Brazil by the National Nuclear Energy Commission (CNEN). The International Atomic Energy Agency (IAEA) recommends the implementation of occupational monitoring programs when there is a risk of annual effective doses exceeding 1 mSv. The control of radionuclide intake can be carried out using internal dosimetry techniques. Currently, in Brazil, there are not enough laboratories capable of providing internal monitoring services to meet the entire demand if this requirement were to be enforced by CNEN. This paper presents an overview of the current situation in Brazil and economically viable technical alternatives, aiming to make the implementation of routine internal monitoring programs for workers occupationally exposed to significant risks of radionuclide exposure in nuclear medicine feasible.

7.3 DR. CARMEN BUENO: RADIATION PROCESSING WITH ELECTRON BEAMS: ON THE FEASIBILITY OF USING ELECTRONIC DOSIMETERS FOR REAL-TIME DOSIMETRY

Dr. Carmen holds Bachelor's and Licentiate degrees in Physics from PUC/SP, as well as Master's and Doctorate degrees in Nuclear Instrumentation from the same university. She began her academic career at PUC/SP, where she was appointed as a Full Professor of Nuclear Physics. In 1995, she started working as a researcher at the Nuclear and Energy Research Institute (IPEN-CNEN/SP) and later completed her postdoctoral work at the Instrumentation and Experimental Particle Physics Laboratory (LIP) at the University of Coimbra. She also holds a second Doctorate in nuclear technology from IPEN-USP. She has published nearly 150 full-length articles in international journals and conference proceedings and has supervised numerous master's, doctoral, and postdoctoral students. Currently, she is a senior researcher at IPEN with over 30 years of experience in nuclear instrumentation for radiation measurement and dosimetry. Most of her research focuses on developing semiconductor-based dosimetry systems for medical and radiation processing fields. She currently serves on the Steering Committee of the National Institute of Science and Technology (INCT) project, "Nuclear Instrumentation and Applications in Industry and Health" (INAIS). Additionally, she is a member of the Advisory Boards of the Brazilian Society of Solid-State Dosimetry (SSD) and the Brazilian Association of Nuclear Energy (ABEN).

Description: In this lecture, we will briefly introduce the radiation processing of products, a relatively new industry with wide applications and significant commercial success, primarily involving gamma rays and electron beams (EB). The main features of the irradiation facilities, the doses delivered, and the regulatory requirements for the processed products will be discussed to highlight the role of dosimetry in this field. Several passive solid-state dosimeters, which are well-established in radiation processing dosimetry, will be introduced to emphasize the need for online dosimeters, whether as routine tools or for monitoring the operating conditions of EB facilities. An overview of efforts to develop online dosimeters, with a focus on electronic devices, will be provided. Subsequently, the lecture will explore the operational characteristics and the physics behind the dose and dose rate response of diodes, explaining how our group has addressed issues related to their use as real-time dosimeters in fixed and mobile EB facilities. The latest data obtained with these diodes, both as routine dosimeters and as electron beam monitors, will be presented to assess the feasibility of their application. The current status of online high-dose dosimeters and the challenges of meeting the metrological standards required in radiation processing dosimetry will also be discussed.

7.4 DR. GEÓRGIA SANTOS JOANA: WIN BRASIL - DIVERSITY & INCLUSION MANAGEMENT

Bachelor's degree in Physics from the Federal University of Minas Gerais - UFMG (2003), master's degree in Radiation, Mineral, and Materials Science and Technology from the Nuclear Technology Development Center of the National Nuclear Energy Commission - CDTN/CNEN (2005), and a Ph.D. in Nuclear Science and



Technology from the Postgraduate Program of the Department of Nuclear Engineering at UFMG (2018). She is currently a technologist at the National Nuclear Energy Commission. She previously worked in the field of Medical Physics at the Superintendence of Sanitary Surveillance of the Minas Gerais State Health Secretariat, where she coordinated the State Quality Control Program in Mammography - PECQMamo. She has experience in Medical Physics and Health Surveillance, with an emphasis on Radiological Protection, Regulation, Safety, and Quality in health services. Her research focuses on Radiological Protection, Quality Control, Radiation Dosimetry and Metrology, and Risk Analysis and Management in the field of medical applications of radiation.

Description: WiN Brasil - Diversity & Inclusion Management

7.5 DR. HELIO YORIYAZ: PROTON BEAM THERAPY: MAIN CHARACTERISTICS AND CHALLENGES

Bachelor in Physics (University of São Paulo, 1979–1982), Master's (1984–1986) and PhD(1998–2000) in Nuclear Technology from the University of São Paulo. Former RHAE fellow at the Paul Scherrer Institute (Switzerland, 1993–1994) and IAEA Fellow at Oak Ridge Associated Universities (USA, 1996–1997). Researcher at the Nuclear and Energy Research Institute (IPEN), working in the field of Nuclear Engineering with an emphasis on Medical Physics, particularly in Monte Carlo simulation, dosimetry and nuclear medicine. Currently a faculty member of the Graduate Program in Nuclear Technology at USP and the Professional Master's Program in Radiation Technology in Health at IPEN. Bachelor's in Physics (University of São Paulo, 1979–1982), Master's (1984–1986) and PhD(1998–2000) in Nuclear Technology from the University of São Paulo. Former RHAE fellow at the Paul Scherrer Institute (Switzerland, 1993–1994) and IAEA Fellow at Oak Ridge Associated Universities (USA, 1996–1997). Researcher at the Nuclear and Energy Research Institute (IPEN), working in the field of Nuclear Engineering with an emphasis on Medical Physics, particularly in Monte Carlo simulation, dosimetry and nuclear medicine. Currently a faculty member of the Graduate Program in Nuclear Technology at USP and the Professional Master's Program in Radiation Technology in Health at IPEN.

Description: Proton therapy is an increasingly relevant form of radiotherapy because it enables a reduction in the total dose compared to photons, preserving critical structures while simultaneously delivering higher and better-conformed doses to the tumor volume. This effect is achieved through the superposition of several weighted Bragg curves with different incident energies, resulting in a broadened Bragg peak, known as the Spread-Out Bragg Peak (SOBP). Although already considered technologically mature, proton therapy still presents important challenges that need to be investigated. Among these, the precise calculation of the relative biological effectiveness (RBE) stands out, defined as the ratio between the doses required for two different types of radiation to produce the same biological effect. This aspect is directly linked to the physical interaction processes between the proton and matter. When a proton with kinetic energy passes through a medium, it interacts with electrons and nuclei of the target, causing its deceleration. These interactions are known as electronic stopping and nuclear stopping, respectively. The stopping power describes the force the medium exerts on the ion, corresponding to the average energy loss per unit path length. In the case of proton-matter interaction, electronic stopping is dominant. This energy loss dynamic is fundamental to understanding the dose deposition profile. As the proton slows down, the energy transferred to biological tissue per unit path length increases, reaching a maximum at a specific depth. This region, known as the Bragg peak, coincides with the maximum stopping power and represents the area of greatest interest for clinical applications. Therefore, its precise positioning is critical during treatment planning. To study these processes in greater detail, advanced approaches have been employed, such as real-time Time-Dependent Density Functional Theory (TDDFT). This is a non-perturbative method based on modern quantum simulations, which has proven particularly useful for investigating electronic stopping in complex systems like water and DNA, bringing computational modeling closer to the real conditions found in biological tissues.

7.6 DR. LUCIANA CARVALHEIRA: NEUTRONS, KNOWLEDGE, AND NEW GENERATIONS: HOW BRAZIL'S ARGONAUTA REACTOR IS SHAPING THE FUTURE OF RADIOCHEMISTRY AND ITS IMPACT ON GLOBAL DOSIMETRY CHALLENGES.

Dr. Luciana Carvalho is a Black Brazilian scientist and singer from Rio de Janeiro, leading pioneering projects at the Institute of Nuclear Engineering and serving as a professor and vice-coordinator in postgraduate programs for Nuclear Sciences. She co-founded startups like Magtech Brasil and Radíon, driving innovation in radiochemistry and nanomaterials. Her research spans neutron activation, medical polymers, and environmental monitoring, alongside mentoring future scientists through WiN Brasil's Helen Khoury Program. As a member of MunaN, she advocates for the inclusion of young Black women in nuclear sciences. Honored by ALERJ for empowering women and recognized as a leader by ENAP's LideraGOV 4.0, she embodies audacity, joy, and a relentless pursuit of scientific freedom and innovation.

Description: For over 60 years, Brazil's Argonauta research reactor has addressed hands-on education in physics and engineering of reactors, and radiochemistry. As a hub for RD&I, Argonauta drives partnerships with institutions like IME, UFRJ and startups in radiotracer technologies. Modernization efforts, including fuel improvements and new radioisotopes production, exemplify its commitment to advancing radiation science nuclear applications worldwide.

7.7 DR. MARIA NEVES: RANGE RADIATION EXPOSURE INDICATOR

Dr. Maria José Neves is a Senior Researcher at the Brazilian National Nuclear Energy Commission (CNEN), based at the Centre for the Development of Nuclear Technology (CDTN), where she leads the Radiobiology Laboratory. She holds a Bachelor's degree in Medical Biological Sciences from the Faculty of Medicine of Ribeirão Preto, University of São Paulo (USP-RP), as well as a Master's and Ph.D. in Physiology from the same institution, with a focus on metabolism. She completed her postdoctoral training in Biochemistry at the Katholieke Universiteit Leuven (KUL), Belgium. Her main research interests include cancer cell metabolism and the biological effects of ionizing radiation.

Description: A small plastic or glass tube (with a final volume of 1.5mL or smaller) containing a solution, composed of sodium selenite and glycose, can be fixed to the surface of a material and will indicate whether it was itself exposed to moderate doses of range radiation (5 – 100 kGy) in a range irradiator contained in the source of ^{60}Co . This type of irradiator is widely used in the irradiation of foods such as dried grains, fruits, tubers, etc., sterilization of surgical material, among other activities. Given its low cost and ease of preparation, it can be frozen and kept in stock at the time of use, which is why we consider an alternative option to confirm exposure to high-range radiation.

7.8 DR RICHARD HUGTENBURG: NEW TECHNOLOGIES AS A ROUTE TO PARTICLE THERAPY: LASER-HYBRID ACCELERATION FOR RADIOBIOLOGY APPLICATIONS - THE LHARA PROJECT.

Dr Richard Hugtenburg is a medical physics expert and research academic at Swansea University, UK, specialising in radiotherapy physics. Dr Hugtenburg began his career in Christchurch, New Zealand, working as a medical physicist while studying for a PhD on the use of computational methods in radiation oncology. He moved to the UK in 1997 and continued to practice in radiotherapy physics; first at the Queen Elizabeth Medical Centre in Birmingham, then at Singleton Hospital in Swansea. His research examines dosimetry methods associated with emergent radiotherapy technologies, utilising Monte Carlo modelling for a diverse range of radiation physics problems, including the analysis of energy deposition in biological systems, the optimisation of dosimeters and detectors, nanoparticulates, shielding and therapy systems.

Description: Particle therapy is a form of external beam radiotherapy that utilises ions to achieve precise (sub-mm) physical distributions of physical dose with the benefits of high LET to combat radioresistant tumours. Particle therapies are also expected to offer better normal-tissue sparing and reduce long-term side-effects in patients, where the current state-of-the-art in radiotherapy has contributed to significant improvements in the survivability of many cancers in recent years. The LhARA project is a consortium of UK universities that considers three new technologies as a means to reduce the cost, physical footprint and flexibility of particle therapies, where currently treatments, such as carbon ion therapy, are restricted to a few laboratories world-wide. The laser-hybrid method uses high-powered lasers to generate ions with kinetic energies of 10s of MeV and at high dose-rates suitable to examine high dose-rate sparing effects such as FLASH radiotherapy.

7.9 DR. ROLF BEHRENS: FROM DOSEMETER DEVELOPMENT TO ROUTINE USE - STANDARDS AND UNCERTAINTIES

Dr. Rolf Behrens holds a diploma at Technical University of Braunschweig (Germany) and a doctorate (Dr. rer. nat.) in physics from the University of Jena (Germany). He is a senior scientist at the Physikalisch-Technische Bundesanstalt (PTB), Germany's national metrology institute, where he leads the Working Group on Dosimetry for Brachytherapy and Beta Radiation Protection. Dr. Behrens is responsible for quality assurance in the Ionizing Radiation division, serves as Radiation Protection Officer for brachytherapy and beta radiation, and acts as an auditor for the German Accreditation Service (DAkkS). He heads and participates in several national and international standardization committees, including the IEC and ISO working groups on dosimetry and uncertainties, and chairs the German Committee on Radiation Protection Dosimetry. His research focuses on dosimetry of beta and photon radiation for radiation protection and brachytherapy, Monte Carlo simulations, spectrometry, Bayesian data evaluation, measurement uncertainty, and dosimetry for the lens of the eye.

Description: Radiation protection uses protection quantities (not directly measurable) and operational quantities (measured to estimate protection quantities). Standards ensure the quality of dose measurements, established by IEC and ISO, and adopted by regional organizations (CEN, CENELEC, GCC, SARSO, ARSO). Type tests for photon dosimeters use ISO 4037 reference fields; beta and neutron fields follow ISO 6980 and ISO 8529. Standards like IEC 62387 and IEC 61526 address other influencing factors such as angle, dose rate, temperature, and electromagnetic immunity. Uncertainties are calculated by combining the effects of these influences, as described in IEC TS 62461 (updated 2025). Performance is regularly checked using comparison measurements following ISO 14146 (updated 2024, includes EURADOS). [Click here to view the presentation slides.](#)

7.10 DR. SILVIA HIDALGO TOBÓN: STATE OF THE ART OF MAGNETIC RESONANCE IMAGING: FROM FUNDAMENTAL PHYSICS TO CLINICAL APPLICATIONS.

Dr. Silvia Hidalgo earned her Bachelor's degree in Physics in 1998 from the University of the Americas-Puebla, Mexico. In 2000, she earned her Master of Science in Medical Physics from the Institute of Physics at the National Autonomous University of Mexico (the first Master of Science in Medical Physics from the UNAM program). In 2005, she earned her PhD in Physics, specializing in Magnetic Resonance Imaging from the Sir Peter Mansfield Magnetic Resonance Center at the University of Nottingham, England. She completed a postdoctoral fellowship in Magnetic Resonance Imaging at the Robarts Research Institute, University of Western Ontario, Canada. She also completed a postdoctoral fellowship at the Autonomous Metropolitan University, Iztapalapa. Her scientific interests are related to advanced functional magnetic resonance imaging, neural networks applied to neuroscience, and magnetic resonance safety in daily clinical practice. She has taught courses and supervised theses at the undergraduate, master's, and doctoral levels in magnetic resonance imaging at the UAM, UAEM, and UNAM. She has given lectures at various national and international forums on nuclear magnetic resonance imaging. She is a member of the International Society of Magnetic Resonance in Medicine, RSNA, the Mexican Physical Society, IEEE, and the Neuroscience Society, among others.

Description: State of the art of magnetic resonance imaging: from fundamental physics to clinical applications.

7.11 DR. SUSANA LALIC: PLANT STEM CELLS AS AN ETHICAL ALTERNATIVE FOR BIODOSIMETRY

Dr. Susana de Souza Lalic is Full Professor of Physics at the Federal University of Sergipe. She holds a B.Sc. (1997) and Ph.D. in Physics (2002) from the University of São Paulo and a postdoctoral fellowship at the University of Pisa (2011). She is a faculty member of the Ph.D. Program in Industrial Engineering (Nuclear Engineering and Industrial Safety) at the University of Pisa and on the Board of Directors of the Master Course in CBRNe Protection at Tor Vergata University of Rome. She is Counselor of the Brazilian Physical Society (SBF), Chair of its Medical Physics Committee, and Vice-chair of the International Solid State Dosimetry Organization (ISSDO). She previously served as Secretary-General of SBF and Coordinator of the Graduate Program in Physics at UFS. Her research focuses on medical physics, condensed matter physics, and applied nuclear physics, with emphasis on radiation detectors, material characterization, radiobiology, and radiation metrology.

Description: This study investigates an ethical approach to biodosimetry, utilizing plants as a sustainable alternative to traditional animal models. Plants, known for their sensitivity to environmental stressors such as radiation, provide a valuable platform for identifying genotoxic agents while reducing reliance on animal testing. The research highlights the utility of the onion (*Allium cepa*) as an *in vivo* model for assessing radiation-induced genetic and chromosomal damage. While *Allium cepa* is a widely recognized standard in environmental contamination studies, it remains underexplored as a tool for radiation detection. A comprehensive evaluation of radiation effects and their temporal progression is made possible by analyzing key variables, including linear energy transfer, dose, and dose rate. The study also examines the potential of photobiomodulation to enhance biological recovery in tissues damaged by radiation, offering a novel intervention strategy. To ensure robustness, findings from the *Allium cepa* model are cross-validated against other *in vivo*, *in vitro*, and *in silico* systems, establishing its efficacy as an ethical and effective alternative for radiation biodosimetry.

7.12 DR. TARCÍSIO CAMPOS AND DR. ALBERTO CAMPOS: CHALLENGES IN RADIOISOTOPE DECONTAMINATION AND PRODUCTION THROUGH NEUTRON SPECTRAL SELECTION

Dr. Tarcisio P. R. de Campos is retired full professor from the Federal University of Minas Gerais (UFMG), currently collaborating with both UFMG and the Nuclear Technology Development Center (CDTN). He holds a degree in Engineering from UFMG, an MSc in Nuclear Engineering from UFRJ, a PhD from the University of Illinois at Urbana-Champaign, and a postdoctoral fellowship at MIT as a Fulbright Fellow. His expertise spans nuclear engineering, radiobiology, radiochemistry, radiotherapy, and the development of radioisotopes, simulators, and medical software. He has published extensively, holds 14 patents, and has supervised over 75 graduate theses. He has coordinated over 28 research projects and actively collaborates with national and international research institutions.

Dr. Alberto M. Campos is Postdoctoral researcher in Augsburg, Germany (2024–present), with a PhD in Mathematics from the Institute for Pure and Applied Mathematics (IMPA, 2024). He completed a research stay at Instituto Superior Técnico (IST), Lisbon (2022), holds a Master's and Bachelor's degree in Mathematics from UFMG, and a technical degree in Electrotechnics from CEFET-MG. He has served as a teaching assistant in various probability and analysis courses at IMPA and UFMG, and has presented talks at the Erdős Institute (Budapest) and the Department of Nuclear Engineering at UFMG. His research interests lie broadly in Probability Theory, with a focus on random covering problems, percolation, and random walks.



Description: We present investigations into the production and decontamination of radioisotopes using a selective Spectral Resonant Selector (RSS), focusing on neutron radiative capture reactions in precursor nuclides and transmutation reactions of long-lived radioactive waste. The physical and mathematical aspects of biased Brownian motion are addressed, through which a tailored neutron spectrum can be generated to target specific nuclides. Reaction rates for both precursor nuclides and decontamination processes are presented. Two distinct neutron spectra are considered—one at low energy and another at resonance energy—allowing interaction with a broad range of target nuclides. The production of isotopes such as Sm-153, Lu-177, Re-188, Ho-166, and Yb-169 serves as representative examples of potential radionuclide generation. Additionally, the decontamination of long-lived actinides, including transuranic radionuclides and fission products, through fission or radiative capture reactions, illustrates a promising approach to nuclear waste mitigation. The specific activities and benchmark parameters of various radioisotopes under defined conditions demonstrate the effectiveness of the RSS in both radioisotope production and nuclear decontamination.

7.13 DR. THIAGO LIMA: CURRENT AND FUTURE TRENDS IN NUCLEAR MEDICINE DOSIMETRY

Dr. Thiago holds a BSc in Physics with Medical Physics from UFRJ and a PhD in Medical Physics from University College London. He is the chief diagnostic medical physicist at Lucerne University Hospitals (LUKS) and research physicist at Genolier Hospital. He also lectures at the University of Lucerne. He chairs the education committee of the Swiss Society of Radiobiology and Medical Physics and represents the society in EFOMP working groups. His work focuses on dosimetry, protocol optimization, radioprotection, and research in nuclear medicine and medical imaging.

Description: This lecture delves into the evolution of nuclear medicine dosimetry. It will explore the progression from early rudimentary methods, which relied on basic measurements and empirical estimates, to the sophisticated imaging techniques and computational models employed today. These advancements have significantly enhanced the accuracy of absorbed dose quantification, improving both safety and efficacy in treatments. The lecture will also discuss the future of dosimetry, highlighting the integration of artificial intelligence and personalized medicine. This forward-thinking approach promises even more precise dose calculations tailored to individual patient characteristics, reflecting a continuous effort to optimize therapeutic protocols and improve patient outcomes in nuclear medicine.

7.14 DR. YVONE MASCARENHAS: PERSONAL DOSIMETRY - THE IMPORTANCE OF REGULATION, TECHNOLOGY, AND CULTURE. EVALUATION OF DOSIMETRY SERVICES IN LATIN AMERICA AND THE CARIBBEAN

Dr. Yvone Maria Mascarenhas has a degree in Physics from the University of São Paulo (USP), and a master's, doctorate, and post-doctorate from the USP's Institute of Physics and Chemistry of São Carlos (IFSC), with research in basic physics as well as radiology and dosimetry. She has worked as a coordinator for research projects funded by agencies such as CNPq and FAPESP and has extensive experience in the field of Physics, working primarily in radiological protection, x-ray production, radiodiagnosis, and dosimetry.

Description: Personal dosimetry is a fundamental tool in radiation protection, ensuring that best practices in facilities using ionizing radiation effectively minimize risks to exposed workers. The use of radiation is highly regulated worldwide, and to make personal dosimetry widely accessible, technology plays a key role—not only as a scientific instrument but also as a reliable and reproducible measurement technique. Accurate analysis of population-level dosimetry data is only possible with the correct and appropriate use of personal dosimeters. This can be achieved only when a strong radiation protection culture is promoted and adopted by all stakeholders involved in personal dosimetry. In Latin America and the Caribbean, the evaluation of Individual Monitoring Services (SMIE) varies by country, reflecting differences in regulations, technological infrastructure, and safety culture. This study presents an overview of the current state of personal dosimetry in the region, with emphasis on the 2022 intercomparison exercise organized by REPROLAM. A total of 26 services—both public and private—were evaluated using different irradiation beams and dose levels. The results show that 69.2% of the laboratories complied with ISO 14146:2018. However, some exhibited calibration deviations, particularly with X-ray beams. Notably, OSL (Optically Stimulated Luminescence) dosimetry has seen significant growth, accounting for 34% of the systems evaluated in 2022. In Brazil, the development of Personal Dosimetry Services has been shaped by two decades of intercomparison exercises and over 30 years of a structured certification process. Despite this progress, challenges remain in standardizing methodologies and enhancing laboratory training to improve dosimetry accuracy across the region. This analysis underscores the critical importance of regulation, technological advancement, and the consolidation of a radiological safety culture as pillars for strengthening occupational dosimetry in Latin America and the Caribbean.

7.15 DR. WILSON CALVO: CURRENT STATUS OF STRATEGIC PROJECTS AT CNEN

Dr. Wilson Calvo holds a diploma in Metallic and Ceramic Materials Engineering from the Federal University of São Carlos - UFSCar (1987) and holds a Master's (1997) and a PhD (2005) degrees in Nuclear Technology - Applications, from the University of São Paulo - USP. He began working at the Nuclear and Energy Research Institute (IPEN) in 1988 and went on to pursue a successful career in both administration and technology. He is a Senior Technologist, professor, and postgraduate advisor in the area of Nuclear Technology at IPEN/USP. He published nearly 100 full-length articles in international conferences and indexed magazines, and supervised more than 15 master's, doctoral, and postdoctoral students. He is a leading expert in Nuclear Engineering, with a focus on the applications of Nuclear Techniques in Industry, Health, Agriculture, and the environment. His emphasis is on radioisotope technology (radiotracers and sealed radioactive sources) and ionizing radiation (electron beams, X-rays, and gamma rays). Owing to his great contribution in the nuclear applications field, he was awarded the Nuclear Merit Award from ABDAN (2020). Regarding his administrative career, he served as a Manager of the Radiation Technology Center (2001-2013), Director of Administration and Infrastructure (2014-2016), and Superintendent of IPEN (2017-2023). He was also Vice President of the Board of Directors of the São Paulo Technology-Based Incubator at USP/IPEN-Cietec (2017-2023). He is a Member of the Deliberative Committee of CNEN (2023-), the Advisory Board of ABENDI, and the International Irradiation Association (IIA). Currently, he is the Director of Research and Development at the National Nuclear Energy Commission - CNEN.

Description: The National Nuclear Energy Commission's guidelines for technological innovation focus on three main areas: nuclear technologies—including reactors, instrumentation, materials, and the fuel cycle—applications of ionizing radiation in health, industry, agriculture, and the environment, as well as radioprotection and dosimetry. This lecture will highlight several strategic projects, including CENTENA, GraNioTer, the tracer technique for offshore oil and gas platforms and processing plants, mobile units equipped with electron accelerators, and a multipurpose reactor. The main features of each project, their contributions to addressing major national challenges in developing nuclear technology, and the current progress of each initiative will be discussed. In short, CENTENA aims to design, build, and commission the first nuclear and environmental technology facility in South America for disposing of low-level radioactive waste, thereby establishing a foundation of safety and sustainability for Brazil. GraNioTer's goal is to establish itself as a national technological hub, promoting research, development, and innovation based on graphene, niobium, rare earths, and other materials and minerals with increasing global demand. Another key project is the application of the radiotracer technique to offshore oil and gas platforms and processing plants, which seeks to develop a prototype and method for measuring gas flow in pipelines. Equally important is the Mobile Unit, equipped with an electron beam accelerator, which aims to expand the nation's capacity to treat industrial effluents and improve water quality in urban areas, addressing one of the primary challenges faced by modern society.

7.16 DR. BRUNO MELO MENDES

Dr. Bruno Melo Mendes holds a Bachelor's degree in Biological Sciences and a Master's and PhD in Nuclear Science and Technology from the Federal University of Minas Gerais (UFMG). He is currently a researcher at the Nuclear Technology Development Center (CDTN/CNEN), responsible for the Internal Dosimetry Laboratory, and a professor in the CDTN postgraduate program. He also coordinates the Monte Carlo Modelling Expert Group (MCMEG). His expertise lies in the nuclear field, with a focus on radiopharmacy, computational modeling (Monte Carlo), and internal dosimetry.

7.17 DR. TELMA CRISTINA FERREIRA FONSECA

Dr. Telma C. F. Fonseca is professor at the DEN/UFMG, faculty member of the PCTN/UFMG and member of the Diversity & Inclusion Management Committee (2024–2027) of WinBrasil – Women in Nuclear, and coordinates the Monte Carlo Modelling Expert Group (MCMEG). She holds degrees in the area of computational sciences, a Master's in Radiation Sciences from UFMG, and a PhD in Biomedical Sciences from KU Leuven (Belgium), where her research focused on internal dosimetry. She was a researcher at the Radiation Protection Dosimetry and Calibrations group (RDC) at SCK-CEN in Belgium. Additionally, she completed a postdoctoral fellowship at the Centre for Development of Nuclear Technology (CDTN/CNEN), where she also led the Internal Dosimetry Laboratory. Her expertise and research interests include dose calculation in radiotherapy, radiobiology, nuclear instrumentation, nanodosimetry, computational modeling, and the use of Monte Carlo methods in radiation protection and radiobiology. Website: www.telmafonseca.com.br

CHAPTER 8

Round Tables

8.1 AI'S CHALLENGES AND OPPORTUNITIES: CONNECTING RESEARCH, INDUSTRY, AND SOCIETY

Moderator

- Dr. Telma Fonseca (DEN/UFMG)

Panelists

- Dr. Rita Sebastião (ICEX/UFMG)
- Dr. Ana Carolina (Siemens Healthineers)
- Dr. Andre Henrique Alves Carneiro (USP)

Description: The roundtable is part of the Tech Area on Artificial Intelligence, an innovative space within ISSSD 2025 dedicated to the convergence of technology, healthcare, and innovation. Its main goal is to foster dialogue among startups, researchers, healthcare professionals, and investors, encouraging the use of AI in clinical practice. During the session, strategic topics will be discussed, such as: Challenges and opportunities of AI in clinical radiology Explainability and ethics of diagnostic algorithms Automation and integration of hospital workflows Success stories from innovative startups Trends for the next 5 years in medical AI The roundtable will feature multidisciplinary experts — including radiologists, data scientists, entrepreneurs, and regulators — and will be moderated by professionals with experience in healthcare innovation. There will also be room for audience interaction, with guided questions designed to stimulate debate.

Dr. Telma C. F. Fonseca is professor at the DEN/UFMG, faculty member of the PCTN/UFMG and member of the Diversity & Inclusion Management Committee (2024–2027) of WinBrasil – Women in Nuclear, and coordinates the Monte Carlo Modelling Expert Group (MCMG). She holds degrees in the area of computational sciences, a Master's in Radiation Sciences from UFMG, and a PhD in Biomedical Sciences from KU Leuven (Belgium), where her research focused on internal dosimetry. She was a researcher at the Radiation Protection Dosimetry and Calibrations group (RDC) at SCK-CEN in Belgium. Additionally, she completed a postdoctoral fellowship at the Centre for Development of Nuclear Technology (CDTN/CNEN), where she also led the Internal Dosimetry Laboratory. Her expertise and research interests include dose calculation in radiotherapy, radiobiology, nuclear instrumentation, nanodosimetry, computational modeling, and the use of Monte Carlo methods in radiation protection and radiobiology. Website: www.telmafonseca.com.br



Dr. Rita C. O. Sebastião is a full professor at Universidade Federal de Minas Gerais (UFMG) with experience in Chemistry, working mainly on the topics: inverse problems, artificial neural networks, nuclear magnetic resonance, thermal decomposition, chemical kinetics, combustion processes, and thermodynamics. Extensive experience in education, research, and extension, especially in projects aiming for the popularization of science, educational innovation, and entrepreneurship, connecting the university to basic education. Experience in developing and managing projects on socio-environmental sustainability and education. She is currently a member of the Postgraduate Program in Chemistry (PPGQUI), Postgraduate Program in Technological Innovation (PPGIT) and Professional Master's in Technological Innovation and Intellectual Property (MPITPI) at UFMG.

Dr. André Carneiro is a Senior Data Scientist and Management Consultant with over 6 years of experience working with leading Brazilian companies. Skilled in developing mathematical, statistical, and machine learning models to deliver actionable insights and optimize business decisions. Currently, I am a Senior Data Scientist at Ambev Tech, one of Brazil's largest companies, where I focus on recommendation systems, clustering, predictive modeling, causal inference, and advanced techniques such as double machine learning (e.g., causal forests). In addition to my corporate experience, I have a strong academic background with over 5 years of research experience. I hold a PhD in Production Engineering from Escola Politécnica of the University of São Paulo (Poli-USP), as well as a double degree in Production Engineering from Poli-USP and a Master's degree in Management and Industrial Engineering from the Instituto Superior Técnico of the University of Lisbon. My research focuses on machine learning, statistical control, and project monitoring. I am also a collaborating researcher at LABDAPS (Laboratório de Big Data e Análise Preditiva em Saúde) of the Public Health School of the University of São Paulo, where I pursue postdoctoral research on fairness and social justice in machine learning models, aiming to enhance predictive performance for underrepresented populations.

8.2 RADON IN ENVIRONMENT: TRACER APPLICATIONS AND PUBLIC HEALTH RISKS

Moderator

- Dr. Talita Santos (CDTN/CNEN)

Panelists

- MSc. Elydio Soares (CDTN/CNEN)
- MSc. Laura Takahashi (CDTN/CNEN)
- Dr. Nathan da Silva (CDTN/CNEN)

Description: The two main aspects related to radon will be presented and discussed: the concern regarding the impacts on public health and the applications of radon isotopes as tracers of environmental phenomena.

Dr. Talita de Oliveira Santos holds a degree in Radiology Technology from the Federal Center for Technological Education of Minas Gerais (2007), a master's degree in Nuclear Sciences and Techniques from the Federal University of Minas Gerais (2010) and a PhD in Nuclear Sciences and Techniques from the Federal University of Minas Gerais (2015). She is currently an adjunct professor at the Federal University of Minas Gerais and a collaborator of the National Commission for Nuclear Energy. She has experience in Physics, with an emphasis on Radiation Physics, working mainly on the following topics: natural radioactivity, application of radiation to the environment, nuclear instrumentation and radioprotection.



8.3 LEADERSHIP IN SCIENTIFIC AND TECHNOLOGICAL RESEARCH

Moderator

- Dr. Luciana Carvalheira (IEN/CNEN)

Panelists

- Adriana Marques (Agência Nacional de Água - ANA)
- Anna Letícia Barbosa de Sousa (CNEN)

Description: This roundtable will convene leading experts to explore the pivotal role of leadership in advancing nuclear scientific and technological research. The discussion will address key challenges, including regulatory frameworks, ethical considerations, nuclear applications, diversity and inclusion, effective communication, and the integration of the next generation into the global nuclear sector. By examining how leadership drives innovation, navigates complex challenges, and shapes the future of nuclear science and technology, this session aims to foster meaningful dialogue. Participants will gain insights into emerging technologies, explore strategic leadership approaches, and strengthen collaboration among researchers, policymakers, and industry leaders.

Dr. Luciana Carvalheira is a Black Brazilian scientist and singer from Rio de Janeiro, leading pioneering projects at the Institute of Nuclear Engineering and serving as a professor and vice-coordinator in postgraduate programs for Nuclear Sciences. She co-founded startups like Magtech Brazil and Radíon, driving innovation in radiochemistry and nanomaterials. Her research spans neutron activation, medical polymers, and environmental monitoring, alongside mentoring future scientists through WiN Brazil's Helen Khoury Program. As a member of MunaN, she advocates for the inclusion of young Black women in nuclear sciences. Honored by ALERJ for empowering women and recognized as a leader by ENAP's LideraGOV 4.0, she embodies audacity, joy, and a relentless pursuit of scientific freedom and innovation.

CHAPTER 9

Mentorship Helen Khoury WIN-Brasil

The Helen Khoury Mentorship WinBrasil2025

Program: Advancing Diversity, Inclusion, and Female Leadership in the Nuclear Sector

The Helen Khoury Mentorship Program, promoted by Women in Nuclear Brazil (WiN Brasil), was established to foster the professional development of women in the nuclear sector, integrating diversity in age, educational background, and technical expertise. Over seven months, the program combined one-on-one mentoring sessions with collective workshops, encouraging both academic and professional growth.

Mentors and mentees co-developed final projects that reflected the program's multidisciplinary and inclusive approach. Outcomes included: the design of a critical meaningful learning model for future mentorship cycles; the creation of Radioarretadas, a local initiative to engage women from Pernambuco's countryside in nuclear discussions; research on gender and racial equity within Eletronuclear; studies on hybridization of nuclear and renewable energy; structured career development strategies in nuclear physics; participation in scientific proposal submissions (PPSUS/Innovation 2025); scientific dissemination projects on radiation and radioactivity; and professional milestones such as article submissions, approvals in competitive examinations, and postgraduate advancements.

The results demonstrated progress in strengthening both technical and behavioral competencies, expanding professional networks, and integrating diverse regional and cultural perspectives. By connecting institutions such as CNEN, IPEN, IEN, Eletronuclear, INMETRO, and universities across Brazil, the initiative reinforced the importance of knowledge sharing and inter-institutional collaboration. It is concluded that the Helen Khoury Mentorship has been consolidated as a model for professional development aligned with sectoral demands and individual expectations. The program not only contributes to participants' growth but also reinforces the presence and leadership of women in the nuclear sector, leaving a legacy of diversity, inclusion, and innovation.

Keywords: Mentorship, Diversity, Inclusion, Female Leadership, Nuclear Sector.

CHAPTER 10

Honors Awards

The International Symposium on Solid State Dosimetry (ISSSD) is proud to honor outstanding contributions in the field of radiation science and its applications in medicine, industry, and research. The 2025 Awards recognize excellence, innovation, and leadership that have advanced knowledge, safety, and education in nuclear science.

10.1 PROF. DR. TARCISIO PASSOS RIBEIRO DE CAMPOS



We are honored to recognize the exceptional contributions of Prof. Tarcísio Passos Ribeiro de Campos, a distinguished figure in nuclear engineering and radiation science in Brazil. With a career spanning over four decades, Prof. Campos has significantly advanced the fields of medical dosimetry, radiation protection, and nuclear technology.

As a retired full professor at the Federal University of Minas Gerais (UFMG), he continues to collaborate with the university and the Nuclear Technology Development Center (CDTN). Prof. Campos earned his undergraduate degree in Engineering from UFMG (1983), a master's in Nuclear Engineering from the Federal University of Rio de Janeiro (1985) with a focus on Reactor Physics and Nuclear Resonances, and a Ph.D. in Nuclear Engineering from the University of Illinois at Urbana-Champaign (UIUC, USA, 1993). At UIUC, he conducted research at the Beckman Institute in neuroscience, robotics, and visuo-motor coordination. He completed a postdoctoral fellowship at the Massachusetts Institute of Technology (MIT, USA, 1998) under the MIT-Harvard research program as a Fulbright Fellow, focusing on Boron Neutron Capture Therapy (BNCT) for brain tumors.

Prof. Campos has extensive experience across multiple areas of Engineering and Nuclear Energy, with current emphasis on materials, radiation science, radioisotopes, radiochemistry, and radiobiology. His research spans radioactive biomaterials, ionizing radiation effects, radiotherapy, brachytherapy, experimental and computational

dosimetry, radiopharmaceutical production, development of physical radiology simulators, radiotherapy software, neutron capture therapy, radiobiology, radiochemistry, oncology, nuclear instrumentation, automation, and the design of particle accelerators, isotope transmuters, radioisotope generators, and nuclear reactor cores.

He holds 14 patents and 1 software registered with INPI, with 2 additional invention patents submitted to WIPO, and has developed 4 computational systems. Prof. Campos has published 148 articles in indexed journals, 340 full conference papers, and 34 abstracts, and has supervised 57 master's theses, 21 doctoral dissertations, 2 co-supervisions, and currently has 2 ongoing doctoral supervisions. He has also mentored 13 postdoctoral researchers, including 1 senior postdoc.

Prof. Campos has collaborated with numerous institutions, including departments of Surgery, Biology, Pharmacy, Veterinary, and Medicine at UFMG, as well as radiotherapy centers at Santa Casa de Misericórdia, Hospital Mário Pena, Hospital São Francisco, and CDTN. He has coordinated over 28 research projects funded by government agencies.

He is a member of the editorial boards of the Journal of Radiology (Austin Publishing Group) and the Revista Mineira de Radioterapia. Over 30 years of teaching, he has delivered undergraduate and graduate courses in applied radiobiology, applied radiochemistry, medical imaging concepts, biomedical radiation applications, particle accelerators applied to biomedicine, and radiation protection, within the Radiation Sciences research line focused on engineering and health applications.

Prof. Campos's dedication to education, research, and innovation has left a lasting legacy, inspiring generations of scientists and advancing nuclear science and technology both in Brazil and internationally.

10.2 PROF. DR. TEÓGENES AUGUSTO DA SILVA



We are honored to recognize the outstanding contributions of Prof. Teógenes Augusto da Silva, a distinguished researcher and educator in nuclear physics and medical physics in Brazil. With a career spanning over four decades, Prof. Silva has made significant advances in radiation metrology, radiological protection, and dosimetry.

He graduated in Physics from the Federal University of Rio de Janeiro (1976) and earned his Master's (1981) and Ph.D. in Nuclear Engineering from the Programa de Engenharia Nuclear – COPPE/UF (1996). During his doctoral studies, he was a CNPq sandwich fellow at the National Institute of Standards and Technology (NIST, Gaithersburg, USA) from December 1992 to June 1995.

Prof. Silva served as a senior researcher at the Comissão Nacional de Energia Nuclear (CNEN) from July 1977 until his retirement in September 2023. He was a faculty member in the graduate program at the Nuclear Technology Development Center (CDTN, Belo Horizonte) from March 2015 to September 2023 and served as

an adjunct professor at the Postgraduate Program in Nuclear Science and Techniques, DEN/UFMG from March 2000 to June 2017.

His research has focused on Medical Physics, emphasizing radiation metrology, radiological protection, and dosimetry, particularly in the calibration of dosimeters, medical and dental radiodiagnostics, quality control, and radiation safety. He contributed as a Mineiro Researcher (2011–2019) and as a CNPq Productivity Fellowship (PQ 2) awardee (2009–2022). He also served as Vice-Coordinator of the CNPq INCT Radiation Metrology in Medicine project (2008–2017).

Prof. Silva held leadership positions at CDTN, including Chief of the Nuclear and Radiological Safety Division (1988–2010) and Chief of the Reactor and Radiation Division (2014–2018). After his retirement, he continues as a collaborator at CDTN (from October 2024) and currently serves as Executive Secretary and Administrative Director of the INCT for Nuclear Instrumentation and Applications in Industry and Health (INAIS, 2023–present). Through his extensive career, Prof. Silva has left a profound impact on radiation safety, dosimetry, and medical physics in Brazil, mentoring generations of scientists and promoting excellence in research and education.

10.3 PROF. DR. HELEN JAMIL KHOURY - IN MEMORIAM



Prof. Helen Khoury was a leading figure in radiological protection and scientific outreach in Brazil, dedicating over four decades to teaching, research, and development in the nuclear sector. She served as a full professor at the Department of Nuclear Energy (DEN) at the Federal University of Pernambuco (UFPE) and held a Ph.D. in Nuclear Physics from the Pontifical Catholic University of São Paulo.

Her work focused on nuclear dosimetry and instrumentation, with applications in medical imaging, radiation therapy, and radiation metrology. She led the Occupational Radiological Protection Network for Latin America and the Caribbean (Reprolam), fostering professional training and knowledge exchange, and founded the Nuclear Science Museum at DEN/UFPE, a pioneering initiative for popularizing nuclear science in Brazil.

Prof. Khoury's legacy lives on in the countless professionals she trained, the initiatives she helped establish, and the lives she inspired. Her dedication and vision continue to guide the field of radiological protection and nuclear science.

CHAPTER 11

Helen Khoury Award for Best Works

In recognition of excellence in research and innovation, the XXV International Symposium on Solid State Dosimetry is proud to present the Helen Khoury Award. This award honors the legacy of Professor Helen Khoury, whose significant contributions to the field of dosimetry continue to inspire generations of scientists worldwide.

The award is granted in two categories — Panel and Communication — highlighting outstanding contributions in each format. In addition to the winners, the Scientific Committee has selected the five best-ranked works in each category to receive an Honorable Mention, acknowledging the high quality and impact of their research.

11.1 BEST WORK IN COMMUNICATION

ID work code: G02-011

Title of Work: **Preliminary Evaluation of a Positioning Phantom for Dosimetric Audit in Ir-192 HDR Brachytherapy: Results from an IAEA Collaborative Project**

Authors: *Natasha Briggs e Silva, Alexander Camargo Firmino da Silva, Aneli Silva, Sandro Passos Leite, Alexandre Bernardino Pinto Jorge, Camila Salata, Maria Helena Marechal, Ademir Xavier da Silva, Andrea Mantuano Coelho da Silva.*

Presenters: *Natasha Briggs e Silva*

Mini-bio: Master in Nuclear Engineering, graduated from the Military Institute of Engineering (IME). Bachelor's degree in Physics with an emphasis in Medical Physics from Souza Marques College. Occupational Safety Technician from the Celso Suckow da Fonseca Federal Center for Technological Education (CEFET/RJ). Has experience as a teacher in the field of Occupational Safety Technology. Currently working as a Researcher in Dosimetry and Radioprotection at the Laboratory of Radiological Sciences (LCR/UERJ), and pursuing a Ph.D. in Nuclear Engineering at COPPE/UFRJ, focusing efforts on research and development in the field of Dosimetry for HDR brachytherapy.

More info:

<https://www.linkedin.com/in/natasha-briggs-2b722bb3/>

<http://lattes.cnpq.br/2115738852789471>

11.1.1 Honorable Mentions in Communication

ID work code: G05-023

Title of Work: **Analysis of Natural Radioactivity in Beach Sands Along the Rio de Janeiro State Coast by Gamma Spectrometry**

Authors: *Laisa de Almeida Neves, Wander Nascimento de Araújo Júnior, Jonathan Oliveira, Leandro Barbosa*

da Silva, Ademir Xavier da Silva.

Presenters: *Laisa de Almeida Neves*

ID work code: G02-057

Title of Work: **Generator of Health Optimized Simulation Templates – GHOST a 3D Slicer plugin to voxelized medical images for MCNP**

Authors: *Francisco Harley Dantas Hauradou Xavier, Paula Selvatice Pereira Teles, Ademir Xavier da Silva, Mirta Bárbara Torres Berdeguez.*

Presenters: *Francisco Harley Dantas Hauradou Xavier*

ID work code: G07-002

Title of Work: **Low-Dose Neonatal Ionizing Radiation Induces Modest, Sex-Specific Changes in Adult Mouse Behavior and Hippocampal Oxidative Stress**

Authors: *Jhon Brandon Felix Ferreira, Luciana Penedo de Melo Teixeira, Maria Eduarda Gomes Moreira da Silva, Heitor gabriel de Souza Saraiva, Mauricio Moacir da Silva Borges, Bianca Torres Ciambarella, Carlos Eduardo Veloso de Almeida, Samara Cristina Ferreira Machado.*

Presenters: *Jhon Brandon Felix Ferreira*

ID work code: G02-019

Title of Work: **A new formulation of Fricke gel dosimeter development using salicylic acid**

Authors: *Iasmin Vieira Nishibayaski, Anna Luiza Fraga Silveira, Luiz Cláudio Meira Belo.*

Presenters: *Iasmin Vieira Nishibayaski*

11.2 BEST WORK IN PANEL

ID work code: G02-066

Title of Work: **Modeling and validation of a Whole Body Counter for in vivo measurements using CAD and MCNP6**

Authors: *Daniel de Castro Pacheco, Bruno Melo Mendes, Luiz Cláudio Meira Belo.*

Presenters: *Daniel de Castro Pacheco*

Mini-bio: Graduated in Mechanical Engineering (2020), with over 5 years of experience in the development and coordination of industrial engineering projects for refineries, mining, and steel industries, participating in the preparation of technical and managerial documents, as well as in the coordination and management of people and projects. Holds a Master's degree in Science and Radiation Technology (2023), focused on the development of a computational model for shielding studies in in vivo monitoring, using a CAD tool associated with a Monte Carlo code. Currently a Ph.D. candidate in Science and Radiation Technology at the Nuclear Technology Development Center (CDTN) / National Nuclear Energy Commission (CNEN), working on the development of the dosimetry monitoring and data collection system at CDTN, with an approach based on the Internet of Things (IoT), Internet of Everything, and Digital Twins. Also worked in supporting and developing projects in the field of Information and Communication Technology (ICT), focused on smart cities.

More info:

<https://www.linkedin.com/in/daniel-de-castro-pacheco>

<http://lattes.cnpq.br/0207136434699068>

11.2.1 Honorable Mentions in Panel

ID work code: G02-070

Title of Work: **Monte Carlo Simulation of Depth Dose in Mammography Using Geant4 and 3D-Printed ABS and PETG Phantoms**

Authors: *Greiciane de Jesus Cesário, Crislane de Jesus Cesário, Linda Viola Ehlin Caldas, Edilaine Honório da*



Silva.

Presenters: *Greiciane de Jesus Cesário*

ID work code: G02-035

Title of Work: **Computational Tools for Automatic Conversion of Microscopy Images into Voxelized Cellular Phantoms**

Authors: *Lucas Beloti Silva, Lucas Fabricio de Araujo, Telma Cristina Ferreira Fonseca.*

Presenters: *Lucas Beloti Silva*

ID work code: G04-038

Title of Work: **Validation of an alanine-based postal dosimetry system for quality control in radiotherapy**

Authors: *Aricia Ravane Pereira da Cruz, Lucas D. Albino, Ernesto Roesler, Claudio C. B. Viegas, Josemary Angélica Corrêa Gonçalves, Carmen Cecília Bueno, Vinicius Saito Monteiro de Barros, Viviane Khoury Asfora.*

Presenters: *Aricia Ravane Pereira da Cruz*

ID work code: G02-081

Title of Work: **SMS Notification System for Potential Radiological Risk Scenarios**

Authors: *Felipe Mamed de Souza, Alessandra Vaz dos Anjos Santos, Paulo César Rocha Silveira, Wallace Vallory Nunes.*

Presenters: *Felipe Mamed de Souza*



CHAPTER 12

Symposium Papers

Proposal for the use of thorium as fuel in TRIGA® research reactors

L. N. Welen^{1*}, M. Z. Amir², G. R. Maritza³

¹ Universidade Federal de Minas Gerais/Centro de Desenvolvimento de Tecnologia Nuclear, 30161-970, Belo Horizonte, Brazil

² Universidade Federal de Minas Gerais/Centro de Desenvolvimento de Tecnologia Nuclear, 30161-970, Belo Horizonte, Brazil

³ Amazônia Azul Tecnologias de Defesa S.A. (Amazul)

* welennunes@gmail.com

Abstract

The thorium cycle offers several potential advantages over the uranium fuel cycle. It has better physical and nuclear properties and leads to a lower production of radioactive waste. Brazil also has large reserves of thorium. According to the database of the International Atomic Energy Agency (IAEA), the country has the largest thorium reserves in the world, although Brazil has officially only explored for nuclear minerals on a third of its territory. Thorium is a chemical element that is lighter than uranium and occurs three to four times more frequently on earth and can also be used as fuel in nuclear reactors. This work aims to present a proposal for a technical feasibility study using neutron codes for numerical simulation, which justifies the manufacture of thorium as nuclear fuel and with a zirconium alloy coating, for Triga-type nuclear research reactors. To carry out the study, parameters such as the effective multiplication factor and other nuclear parameters of a Triga-type reactor are evaluated using codes based on the Monte Carlo method. Based on the analysis of the results obtained, it is possible to deduce the feasibility of using Th-232 as a mixed fuel in the Triga IPR-R1 reactor at CDTN. Different percentages and concentrations of the Th-U mixture are simulated to determine the most advantageous composition.

Keywords: Triga IPR-R1; Neutronic Analysis; Monte Carlo; Thorium.

1 Introduction

Thorium was studied in the early stages of nuclear reactor development in the United States and Brazil in the 1960s. Uranium became the preferred fuel because it has the only fissile isotope found in nature in quantities capable of sustaining a chain reaction. Today, there is a real prospect of a possible shortage of uranium, which, among other reasons, such as non-proliferation, has led to thorium once again being the subject of research and development, especially in India. This

country has the world's largest reserve of thorium, and Brazil the second. Our country could become very important in its use.

Among its qualities, thorium stands out for being fertile, stable under irradiation and the possibility of recycling. The adoption of this fuel is associated with the greater sustainability of regenerator reactors and the reduction in the amount of uranium needed to generate nuclear energy. Uranium-238, like thorium-232, are fertile materials. In nature, practically 100% of thorium is Th-232. Th-231, which would be fissile, is only found in traces. Nuclear fuels are generally made up of a mixture of fissile material and fertile material. For example, uranium (U-235 or U-235 + U-238), a mixture of uranium and thorium (Th-232 + U-233) and uranium and plutonium (U-238 + Pu-239), etc., are used as fuel.

1.1. Thorium Group of IPR/UFMG

In any work on the use of thorium in reactors, the Thorium Group cannot fail to be mentioned. The work of this group began in 1965, within the Reactor Engineering Division of the Radioactive Research Institute of the Federal University of Minas Gerais – IPR/UFMG (today the Nuclear Technology Development Center - CDTN). The main task of this group was to study the technical and economic possibilities of using thorium in an autonomous nuclear program. To start a reactor using thorium (fertile material) as fuel, an adequate quantity of fissile material is needed, namely: uranium-235 (enriched uranium) or plutonium-239 (created by the capture of a neutron by the isotope uranium-238). At the time, there were no uranium enrichment facilities in Brazil. Therefore, to design a thorium reactor, it was necessary to design and assemble a reactor using natural uranium and heavy water. To obtain the fissile isotope (Pu-239) for starting the thorium reactor, a heavy water research reactor, Capitu (Thorium-Uranium Heavy Water Circuit), was built. The ultimate goal was to build a regenerator reactor that transforms thorium-232 (a fertile isotope) into uranium-233 (a fissile isotope) (Mesquita, 2016).

The Thorium Group formed a group of engineers and physicists to study research facilities related to power reactors. A Thermohydraulic Laboratory, a Subcritical Unit, a Nuclear Component Testing Laboratory were set up at the IPR (now CDTN), and six tons of heavy water were purchased. The team carried out several reactor physics experiments (Camargo, 2006)

In 1975, with the Brazil-West Germany agreement, the option for a reactor using enriched uranium and light water aborted the ongoing research by the Thorium Group and it was discontinued. However, during the CDTN era, studies were carried out on the use of thorium as a nuclear fuel in PWR reactors, which began to be carried out in cooperation with the Jülich Nuclear Center in Germany. This work indicated the viability of its use in PWR reactors, with substantial gains in fuel economy. (Pinheiro *et al.*, 1988).

It is worth noting that, currently, the concept of breeder reactors, which are the case of reactors that use thorium-232, is once again in the spotlight. Of the six concepts adopted for the new generation of reactors (Generation 4) three are fast reactors. (Mesquita, 2023).

1.2. IPR-R1 Triga Reactor Fuel Elements

In this proposal, the reference research reactor is Triga IPR-R1, located in Belo Horizonte (MG). Triga (or TRIGA®) type reactors are normally installed in universities, research centers,

industries and medical centers in several countries. The Triga IPR-R1 reactor shown in Figure 1 was acquired from General Atomics Co. of San Diego, California by the government of the State of Minas Gerais in 1960, through the American government's "Atoms for Peace" program. It was installed in the former Institute for Radioactive Research (IPR), now the Center for Nuclear Technology Development (CDTN), of the National Nuclear Energy Commission (CNEN), located on the UFMG campus (Pampulha). (Mesquita, 2023).



Source: Mesquita, A. Z. Energia Nuclear: uma introdução, v.1 n92, 2023.

Figure 1 :IPR-R1 TRIGA reactor pool and core.

The current core of the IPR-R1 Triga reactor is loaded with two types of fuel elements. They are cylindrical in shape and between the fuel and the cladding there is a small space (gap) filled with helium. The majority (59) are from the first load of the reactor and have a 1100F aluminum cladding, 0.76 mm thick. In 2002, four new elements with stainless steel cladding (AISI-304) were added, with a wall thickness of 0.5 mm. The total length of the element is 72.24 cm, with its active part measuring 35.56 cm in height (Al) and 38.10 cm (stainless steel), and the diameter is 3.56 cm (Al) and 3.63 cm (stainless steel). The aluminum-clad fuel element has two burnable poison disks made of samarium oxide at both ends of the active part. Completing the element, there is a graphite rod at each end that acts as an axial reflector, having the same diameter as the active part and a height of 10.16 cm (Al) and 8.64 cm (stainless). The stainless-steel fuel element has a 6.35 mm diameter zirconium rod in its center. In addition to the axial reflectors, at each end of the elements there are support terminals in aluminum or stainless steel, depending on the type of fuel (Mesquita *et al.*, 2023).

Each element consists of a homogeneous mixture of metallic uranium (fuel) and zirconium hydride (neutron moderator). The fuel mixture in the aluminum-clad element contains 8.0% by weight of uranium, 91% of zirconium and 1.0% of hydrogen; in the steel-clad element, these percentages are 8.5%, 89.9% and 1.6%. These two compounds, uranium and zirconium hydride, are characterized by the formulas $\text{U-ZrH}_{1.0}$ and $\text{U-ZrH}_{1.6}$, respectively. The enrichment in ^{235}U is 20% in both fuel mixtures. Each fuel-moderator element contains about 37 g of ^{235}U , with a total weight of 3 kg. (Mesquita *et al.*, 2023). Figure 2 shows the current configuration of the Triga IPR-R1 reactor core.

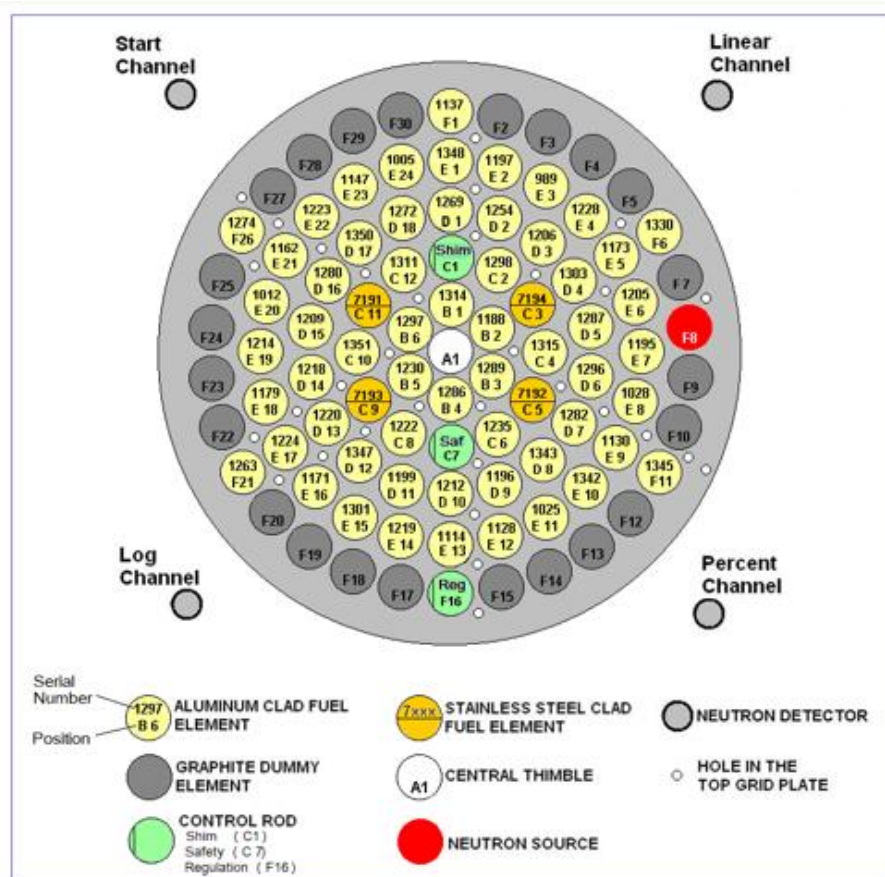


Figure 2: Current configuration of the IPR-R1 Triga reactor core

2 Justification

The license to manufacture fuel for Triga reactors was transferred a few years ago from General Atomics (San Diego, California, USA) to Cerca (Compagnie pour l'Etude et la Realisation de Combustibles Atomiques). Cerca is a subsidiary of Framatome, located in the city of Romans (France). It was recently announced that Cerca will close the Triga manufacturing line, mainly because it would have to invest significant capital to upgrade the facility and comply with the new licensing requirements of the French regulatory authority. Consequently, all Triga reactors worldwide would have to stop operating due to the unavailability of fresh fuel to sustain the operation (Suwarno, 2014).

This is one of the reasons that justify research into the development of fuels for Triga reactors. Below, we will present some characteristics of the importance of using thorium and the Zircaloy coating in Triga fuel elements.

The development of thorium-based fuels was motivated by the nuclear properties of the fuel cycle of this element, when applied in thermal reactors, being superior to those of other elements. In the thorium cycle, the ^{232}Th isotope, instead of ^{238}U , would be the fertile material, and the ^{233}U isotope would be the main fissile actinide, playing the role attributed to ^{239}Pu in the uranium

cycle. Commonly, thorium when used in nuclear reactors is in the form of ThO₂ in ceramic pellets, which is one of the most chemically inert solid substances available and has many advantages over UO₂, such as a higher melting point (one of the highest in existence). It is not predisposed to oxidation and, at usual operating temperatures of fuel elements in reactors, its thermal conductivity is greater than that of UO₂ (Belle and Berman, 1984).

Regarding nuclear data, for the thermal neutron spectrum, the neutron capture cross section of ²³²Th is 7.6 barns, almost three times larger than that of ²³⁸U (2.7 barns). The barn [b] (1 b = 10⁻²⁴ cm²) is the unit that represents the probability of neutron interaction with the nuclei of atoms. The high capture rate in fertile materials implies that in thorium-based reactors there is a need for greater enrichment and, when this need is met, the conversion of ²³²Th (to ²³³U) is higher than that of ²³⁸U (to ²³⁹Pu). Thus, thorium is a better fertile material than ²³⁸U in thermal reactors (IAEA, 2005). ²³³U is produced from the radioactive capture of a neutron by ²³²Th, followed by beta decay, according to the following chain:

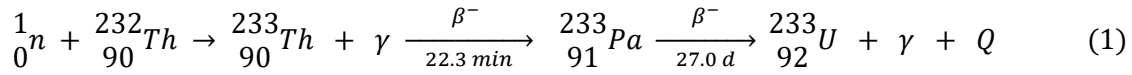


Figure 4 shows the formation diagram of ²³³U. Through neutron capture, ²³²Th produces the radioactive isotope ²³³Th. ²³³Th decays with the emission of a beta particle and forms protactinium (²³³Pa), which in turn decays with the emission of another beta particle and forms the artificial fissile isotope ²³³U. These fissions when it absorbs a low-energy neutron. In other words, when it absorbs a neutron, it can generate more than two neutrons, which allows regeneration (the process by which a fertile isotope, when it captures a neutron, produces more fissile elements than non-fissile elements). In a smaller percentage of cases, two subsequent neutron captures lead to a second fissile isotope, ²³⁵U.

A determining factor in the efficiency of a nuclear reactor, in relation to the use of natural resources, is the average number of thermal neutrons that cause fissions for each neutron absorbed in the fuel, this is the Reproduction Factor, represented by the Greek letter η (eta). Thus, in a fuel containing only ²³⁵U (100% enriched), the capture of the thermal neutron that causes its fission is 2.06 neutrons. In other words, the average number of neutrons released in thermal fission ν (ni) in ²³⁵U is $\nu = 2.43$. However, ~ 0.35 neutrons are captured without causing fission, transforming ²³⁵U into ²³⁶U. In addition, approximately 0.5 of the neutrons are absorbed by the structural materials and other devices present in the reactor core (Mesquita 2023).

²³³U, ²³⁵U, and ²³⁹Pu are the three fissile isotopes that can be used in nuclear reactors. Of the three, ²³³U has the highest Reproduction Factor (η) when fissioned by thermal neutrons. For ²³³U, $\eta = 2.27$, for ²³⁵U, $\eta = 2.06$, and for ²³⁹Pu, $\eta = 2.0$. Of these 2.27 neutrons produced per neutron absorbed in ²³³U, one neutron is sufficient to sustain the fission chain reaction. This leaves 1.27 neutrons available for absorption by additional breeding material and by the other components of the reactor core. This allows the design of breeder reactors that operate in the low neutron energy range (thermal neutrons). In this case, the fertile material inserted into the reactor core is ²³²Th, which will generate ²³³U. This isotope is then removed from the reactor core, chemically separated from ²³²Th, concentrated, and used to make new fuel. In this way, the reactor will

indirectly burning thorium, which is three to four times more abundant than uranium (Vlasov, 2024).

Thorium is a chemical element of the actinide family, widely distributed around the Earth's crust. The thorium cycle offers several potential advantages over a uranium fuel cycle. It has superior physical and nuclear properties and reduced production of radioactive waste. The products of thorium combustion are almost completely free of transuranics. In ^{235}U (fissile) fuels, in which most of the mixture is ^{238}U (fertile), various isotopes of plutonium appear, with ^{239}Pu standing out. However, the use of thorium does not produce plutonium, thus contributing to the non-proliferation of fissile material for non-peaceful purposes.

In the 1960s, many countries showed interest in using thorium in reactors, justified by the fact that ^{233}U (neutron capture in thorium) is a better fuel than ^{239}Pu (neutron capture in ^{238}U) due to the high yield of neutrons produced by that isotope in thermal reactors. Therefore, in the long term, it is expected that thorium reactors will be able to provide a reduction in fuel cycle costs (when compared to uranium reactors, or uranium and plutonium reactors).

Based on its nuclear, physical and mechanical properties, in addition to its stability under irradiation, thorium is a suitable material for use as a fertile element in thermal reactors. Recycling the fuel to extract the ^{233}U -produced with fertile thorium provides an increase in fuel life.

3 Methodology

To achieve the proposed objectives, codes based on the Monte Carlo method will be used to simulate neutron parameters. The reference will be the fuel element with stainless steel cladding used in the CDTN IPR-R1 Triga reactor, that is, metallic uranium fuel enriched to 20%. The weight percentages of each component are: 8.5% uranium, 89.9% zirconium and 1.6% hydrogen. This alloy, uranium and zirconium hydride, is characterized by the formula $\text{U-ZrH}_{1.6}$.

The simulation proposed here will be performed with the Triga core fueled with fuel elements containing a mixture of uranium and thorium (metallic Th-U) in various concentrations (formula $\text{Th-U-ZrH}_{1.6}$). Calculations will be performed with different Th/U ratios, with a percentage of thorium, by weight, always greater than that of uranium. The simulated operating powers will be 100 kW and 250 kW.

Figure 3 shows a diagram of the stainless-steel coated fuel element used in the IPR-R1 reactor. The simulations to be performed will be on an element that will have the same geometry and components as this element. However, the fuel will be changed and will have different concentrations of thorium + uranium, and the coating will be the Zircaloy-4 alloy (Rodrigues, 2018).

They will be simulated using neutron codes to obtain theoretical results such as, neutron multiplication factor, excess reactivity, reactivity/temperature coefficient, control rod reactivity, fuel element reactivity, burnup elements, power distribution, neutron flux in irradiation devices and fuel decay heat after the end of its operational life. An important evaluation will be the

prediction of the resulting production of uranium-233 and the appearance of poisons resulting from the burnup.

Most of these parameters are obtained directly from the effective neutron multiplication factor (k_{eff}) or from the flux distribution. Determining these parameters is fundamental in the design of nuclear reactors and allows the optimization and analysis of various aspects related to their operation. It will also help identify and prevent possible violations of safe operating conditions.

The Monte Carlo transport method will be used to determine the neutron parameters of the Triga IPR-R1 reactor in this new configuration. In many cases, the Monte Carlo code is the only numerical tool available in practice, due to the great geometric flexibility of the Triga reactors. Currently, the use of the Monte Carlo method in the neutron simulation of nuclear reactors is growing and constitutes a global trend.

4 Results and Conclusions

From the analysis of the results to be obtained in the simulations, it will be possible to know whether the use of ^{232}Th as a mixed fuel in a Triga reactor will be advantageous. Several percentages of the Th-U mixture will be simulated to know the most productive composition. According to simulations carried out by Gonçalves *et al.* (2018), in PWR reactors with mixed fuel of ThO_2 and UO_2 , compared to the usual fuels with only UO_2 , there was a lower emergence of fission poisons, such as xenon-135 and samarium-149. The amount of reduction in the generation of these isotopes was, on average, 17% of ^{135}Xe and 21% of ^{149}Sm , for the reactor with 40% ThO_2 in its volume, when compared to the reactor with the original configurations. Similar results are expected with the fuel proposed in this work.

4.1 Simulation with Fuel Elements with Zircaloy Cladding

The following simulation results will be presented regarding the use of zirconium alloys as fuel cladding for the IPR-R1 Triga reactor. Table 1 shows a comparison between the mechanical characteristics of the metal alloys currently used in the IPR-R1 reactor and zirconium and Zircaloy-4. Zirconium and its alloys have intermediate thermal conductivity and density, between aluminum and stainless steel. The heat capacity of zirconium and its alloys is lower than that of the other two materials. Therefore, zirconium transfers heat better.

Table 1

Properties of aluminum, stainless steel, metallic zirconium and Zircaloy-4 alloy.

Material	Tensile Strength [MPa]	Thermal Conductivity [W/m/K]	Density [kg/m ³]	Heat Capacity [J/kg.K]
Aluminum (1100F)	50-90	66.25	2700	900
Stainless steel (AISI 304)	120-220	15.00	7900	460
Zirconium	350-390	22.70	6510	278
Zircaloy-4	413	20.87	6560	350

Source: Kenneth, 2017, Howladera *et al.*, 1998, Rol Metais Aços Finos e Metais, 2017.

Zircaloy-4 is an alloy that contains the most Zr in its composition, among the alloys used in nuclear engineering. The first alloy to be used was Zircaloy-2, which was later improved to become the Zircaloy-4 alloy. Subsequently, the Zirlo alloy was developed, whose physicochemical characteristics make it superior in performance, regarding mechanical and chemical resistance in nuclear environments (Samal et al., 2011). Table 2 provides information on the alloy compositions: Zircaloy-2, Zircaloy-4, and Zirlo

Table 2

Composition of Zircaloy alloys, (the complementary amount constituted of zirconium, Zr)

Alloy	Components					
	Sn	Fe	Cr	Ni	Nb	O
			%			ppm
Zircaloy-4	1.2 ~1.7	0.18~0.24	0.07~0.17	-	-	1000~14000
Zircaloy-2	1.2 ~1.7	0.07~0.2	0.05~0.15	0.03~0.08	-	-
Zirlo	1	0.1	-	-	1.0	1100

Source: Goodfellow, The widest selection of advanced materials, 2025.

The Monte Carlo code MCNPx (Monte Carlo N-Particle eXtended) was used to find the Effective Multiplication Factor (k_{eff}) with two types of fuel elements for the IPR-R1 Triga reactor. The simulation was performed with the current core configuration with 59 aluminum-clad elements and four stainless steel-clad elements. Compared with this same configuration, but replacing all the cladding with Zircaloy-4. In this simulation, the fuel alloy was not yet the thorium alloy, but rather the existing uranium alloy. The results presented in Table 3 showed that the k_{eff} is higher with the fuel elements cladded with the Zircaloy-4 alloy. Thus, leaving more neutrons available for nuclear reactions.

Table 3

Simulation results with the current IPR-R1 Triga reactor core configuration and with the proposed Zircaloy-4 cladding

Cladding	Neutron Multiplication Factor, k_{eff}	Standard Deviation	68% confidence	95% confidence	99% confidence
Current Core: Aluminum, and Stainless steel	1.01874	0.00050	1.01824 to 1.01924	1.01775 to 1.01974	1.01743 to 1.02006
Proposed Core: Zircaloy-4	1.02313	0.00052	1.02261 to 1.02364	1.02210 to 1.02415	1.02177 to 1.02449

Zirconium alloys, in addition to their low neutron capture cross section, also have advantages in terms of resistance to the operating conditions of nuclear reactor cores, which can promote transmutations of the elements. Transmutations modify the atomic and crystalline structure of the metals. This can lead to embrittlement and more pronounced corrosion (Azevedo, (2011).

Acknowledgements

The authors would like to thank the following institutions: Nuclear Technology Development Center (CDTN), Brazilian Nuclear Energy Commission (Cnen), Research Support Foundation of the State of Minas Gerais (Fapemig), and Brazilian Council for Scientific and Technological Development (CNPq).

References

Azevedo, C. R. F. (2011). Selection of fuel cladding material for nuclear fission reactors. *Engineering Failure Analysis*, 18 (8), 1943-1962. DOI: 10.1016/j.engfailanal.2011.06.010.

Belle, J.; Berman, R.M. (1984). *Thorium dioxide: properties and nuclear applications*. Naval Reactors Office, United States Department of Energy.

Camargo, G. O. (2006). *Fogo dos deuses uma história da energia nuclear*. Editora Contraponto, Rio de Janeiro, 344p.

Gonçalves, I. C.; Wichrowski, C. C.; Oliveira, C. L. (2018). Análise de burnup de um reator nuclear PWR com combustível misto de óxido de tório e urânio. *Revista Militar de Ciência e Tecnologia*. 35 (4), 42 – 54.

Goodfellow Advanced Materials. (2025). *The widest selection of advanced materials*. Available at: <https://www.goodfellow.com/usa>. Accessed: on: Feb 02.2025.

Howlade, M. M. R. *et al.* (1998). The electrical conductivity of zircaloy oxide films. *Journal of Nuclear Materials*, 253(1-3), 149-155. DOI: /10.1016/S0022-3115(97)00309-7.

IAEA - International Atomic Energy Agency. (2005). *Thorium fuel cycle: Potential benefits and challenges*. Vienna.

Kenneth, B. (2025). *Periodic Table of Elements - Zirconium - Zr*. Available at: <https://EnvironmentalChemistry.com/yogi/periodic/Zr.html>. Accessed: Jun 18 2025.

Mesquita, A. Z. *Energia Nuclear: uma introdução*. (2023). Editora UFPR - Universidade Federal do Paraná.

Mesquita, A. Z.; Camarano, D. M.; Vilela, J. J. (2023). *Plano de Gestão do Envelhecimento para o Reator Triga® IPR-R1 do CDTN*. No: PGE/IPR-R1/SEURT/CDTN. Centro de Desenvolvimento da Tecnologia Nuclear.

Mesquita, A.Z.; Lameiras, F.S.; Martins, M. D. (2016). O ensino de tecnologia de reatores nucleares em um programa de pós-graduação de Engenharia Multidisciplinar. *Revista Brasileira de Pós-Graduação*, 12(29), 839-855. DOI: 10.21713/2358-2332.2015.v12.866.

Mesquita, A. Z.; Palma, D. A. P; Almeida, V. F. (2024). Experimental study of temperature

distribution and development of a power monitoring channel for the IPR-R1 TRIGA® research nuclear reactor. *Nuclear Engineering and Design*. 417, 112869. DOI: 10.1016/j.nucengdes.2023.112869.

Mesquita, A.Z. (2005). *Investigação Experimental da Distribuição de Temperaturas no Reator Nuclear de Pesquisa Triga IPR-R1*. Universidade de Campinas - Unicamp.

ORNL - Oak Ridge National Laboratory. (2011). *SCALE: a comprehensive modeling and simulation suite for nuclear safety analysis and design*. ORNL/TM-2005/39, Version 6.1.

Pinheiro, R.B. et al. (1988). *Program of Research and Development on Thorium Utilization in PWRs*. Final Report (1979-1988).

Rodrigues, R. (2018). *Sistema para Inspeção de Elementos Combustíveis de Reatores Nucleares de Pesquisa pelo Método de Sipping*. Doctoral Thesis. Centro de Desenvolvimento da Tecnologia Nuclear.

Rol Metais Aços Finos e Metais, S. A. (2017). *Aços Inoxidáveis AISI 304*. São Paulo. 2017.

Samal, M. K.; Sanyal, G.; Chakravartty, J. K. (2011). Investigation of failure behavior of two different types of Zircaloy clad tubes used as nuclear reactor fuel pins. *Engineering Failure Analysis*. 18(8), 2042-2053. DOI: 10.1016/j.engfailanal.2011.06.009.

Suwarno, H. (2014). Development of TRIGA Fuel Fabrication by Powder Technique. *Atom Indonesia*, 40(3), 113-119. DOI: 10.17146/aij.2014.329.

Quality Assurance in ^{90}Y Radioembolization: An Intercomparison Protocol for Activity Optimization

T. C. V. Santos^{1*}, L. Carnevalheira², J. U. Delgado³, L. V. G. Tarelho¹

¹ Instituto Nacional de Metrologia, Qualidade e Tecnologia / INMETRO / DIMIC, 25250-020, Duque de Caxias, Brasil

² Instituto de Engenharia Nuclear / IEN / DINUC, 21941-614, Rio de Janeiro, Brasil

³ Instituto de Radioproteção e Dosimetria / IRD / DIMET / LNMRI, 22780-160, Rio de Janeiro, Brasil

* thamyra11vieira@gmail.com

Abstract

Hepatic radioembolization with ^{90}Y microspheres is a minimally invasive and increasingly adopted therapy for primary and metastatic liver tumors. The efficacy and safety of this procedure are critically dependent on the accuracy of administered activity measurements. However, recent studies have reported discrepancies of up to 44% between manufacturer-declared activities, measurements obtained using dose calibrators, and post-procedural quantification through PET imaging, particularly between glass and resin microspheres. Such variations represent significant clinical and metrological challenges: overdosing may cause hepatic or systemic toxicity, while underdosing may lead to suboptimal tumor control. At present, no standardized protocol exists in Brazil to minimize these discrepancies and ensure the comparability of measurements between centers. To address this gap, the present study proposes a standardized intercomparison protocol for ^{90}Y activity measurements across nuclear medicine services. The proposed methodology includes clear guidelines for the selection and calibration of equipment, homogenization of microspheres by controlled stirring prior to activity measurement, and standardized scatter and attenuation correction procedures in PET imaging. A key component of the protocol is the systematic comparison of activity values measured in dose calibrators with those quantified in PET/CT, expressed as the $A_{\text{PET}} / A_{\text{Calibrator}}$ ratio, enabling the identification and correction of systematic deviations. This approach aims to enhance metrological reliability, reduce inter-center variability, and optimize dosimetric accuracy. The implementation of this protocol represents a significant step toward harmonization of clinical practices, improved patient safety, and optimized therapeutic outcomes. Furthermore, the methodology aligns with international recommendations for personalized dosimetry, contributing to the development of a robust national framework for quality assurance in ^{90}Y hepatic radioembolization.

Keywords: ^{90}Y microspheres; Intercomparison protocol; Activity optimization; Personalized Dosimetry.

1 Introduction

Hepatic radioembolization is a minimally invasive procedure used in the treatment of primary liver tumors, such as hepatocellular carcinoma (HCC), and metastatic liver disease. The technique involves the intra-arterial administration of microspheres loaded with the radioisotope Yttrium-90 (^{90}Y), which emits beta radiation, directly into the vasculature supplying the tumor. Once lodged within the tumor vessels, the microspheres deliver localized radiation, destroying cancer cells while sparing the surrounding healthy tissue. (Carlier, T. *et al.* 2025)

This procedure has emerged as a promising therapeutic alternative; however, its efficacy and safety are critically dependent on precise dosimetry. Variations in the administered activity may lead to hepatic or systemic toxicity in cases of overdose, or therapeutic failure in cases of underdose. Recent studies have revealed significant discrepancies between the activity values stated by manufacturers, measurements performed using dose calibrators, and post-procedural quantification obtained through Positron Emission Tomography (PET), particularly between glass and resin microspheres. These inconsistencies pose both clinical and metrological challenges, compromising the reproducibility and reliability of treatment across different nuclear medicine services. (Gnesin, S. *et al* 2023)

Given this issue, it is essential to establish standardized protocols to ensure the accuracy of ^{90}Y measurements, from equipment calibration to correction of artifacts in PET imaging. This study proposes a unified method for intercomparison of activity measurements among Brazilian centers, incorporating guidelines for instrument selection, microsphere homogenization prior to measurement, and adjustments for attenuation and scatter in PET. By systematically analyzing the ratio between activities quantified by PET and those measured with the dose calibrator (APET/Calibrator), the aim is to identify and correct systematic biases, thereby improving both metrological reliability and clinical outcomes. The implementation of this protocol not only raises patient safety standards but also promotes harmonization with international personalized dosimetry guidelines, reinforcing the role of radioembolization as a precision therapy in the oncological setting.

2 Theoretical Background

2.1 Requirements for Hepatic Radioembolization

According to what was demonstrated in the studies of Weber, M. *et al* (2022) and Busse, N. C. *et al* (2024) To perform hepatic radioembolization with ^{90}Y microspheres, a facility must have appropriate infrastructure, including specialized equipment, qualified human resources, and well-established safety protocols.

First, it is essential to have a trained multidisciplinary team composed of interventional physicians (specialized in radiology or nuclear medicine), medical physicists responsible for dosimetry and radiation safety, nuclear medicine technologists for radiopharmaceutical handling, and nurses skilled in procedures involving radioactive materials.

In terms of equipment, the facility must be equipped with a digital angiography system or cone-beam computed tomography (CBCT) to perform pre-procedural vascular mapping, allowing precise identification of tumor vascularization and selective microsphere delivery. In addition,

hybrid imaging systems such as SPECT/CT or PET/CT are indispensable to evaluate the distribution of ^{99m}Tc -MAA during the pre-procedure phase and to verify the location of ^{90}Y microspheres after treatment, ensuring therapeutic efficacy and patient safety.

Accurate dosimetry is a fundamental pillar, requiring dose calibrators with metrological traceability to measure the administered activity, as well as radiation detectors (such as Geiger–Müller counters and ionization chambers) to monitor potential contamination and assess staff exposure. Radiation protection must be reinforced through appropriate shielding barriers and personal dosimeters for all professionals involved.

Finally, the facility must comply with local and international regulatory requirements, including specific licenses for handling radioactive materials, regular quality control protocols, and ongoing training to minimize operational failures. Standardization of these processes—based on guidelines from the Brazilian Society of Nuclear Medicine (SBNM) and in compliance with regulations from the National Nuclear Energy Commission (CNEN)—is essential to ensure patient safety, therapeutic efficacy, and alignment with international best practices in radioembolization.

2.2 Types of Microspheres

Currently, there are two types of microspheres commercially available worldwide: SIR-Spheres®, produced by Sirtex Medical Inc., indicated for radioembolization of unresectable metastatic liver tumors; and TheraSphere®, manufactured by BTG International Ltd., indicated for selective internal radionuclide therapy of unresectable primary liver tumors (hepatocellular carcinoma).

These two products differ in raw material, production methods, and the specific activity of ^{90}Y , as shown in Table 1. TheraSphere microspheres are composed of glass, with ^{90}Y incorporated into the glass matrix through neutron activation of ^{89}Y . In contrast, SIR-Spheres are composed of resin, with ^{90}Y produced from a $^{90}\text{Sr}/^{90}\text{Y}$ generator and located exclusively on the outer layer of the microsphere.

As a result of these differences, the specific activity of ^{90}Y in TheraSphere is substantially higher than in SIR-Spheres—nominally 2,500 Bq/glass sphere versus 50 Bq/resin sphere at the time of calibration. (Thiam, C. *et al* 2016)

Table 1: Characteristics of ^{90}Y Microspheres

Characteristic	Glass Microspheres	Resin Microspheres
Diameter (μm)	20 – 30	20 - 60
Activity per sphere (Bq)	2.500	50
Number injected per produce (million)	1 – 8	40 - 60
Activation / production method	Neutron activation	Generator $^{90}\text{Sr}/^{90}\text{Y}$

3 Materials and Methods

To establish a standardized protocol for metrological intercomparison among nuclear medicine centers performing hepatic radioembolization with ^{90}Y microspheres, reference sources provided by the National Metrology Laboratory of Ionizing Radiation (LNMRI) will be used to validate the accuracy of activity measurements in dose calibrators and PET/CT systems.

3.1 Reference and Control Standard

The reference sample will consist of a ^{90}Y chloride ($^{90}\text{Y}\text{-Cl}$) solution calibrated by the National Metrology Laboratory of Ionizing Radiation (LNMRI), with certified activity, contained in standardized glass ampoules (25 mm in diameter). Control samples will include glass microspheres (TheraSphere®) and resin microspheres (SIR-Spheres®).

3.2 Participating Centers

Reference centers, duly authorized by the National Nuclear Energy Commission (CNEN), will be selected. These facilities must be equipped with certified dose calibrators capable of measuring ^{90}Y activity, with annual calibration validated. In addition, the centers must have PET/CT scanners with time-of-flight (TOF) technology and technical teams trained in radioembolization procedures.

3.3 Intercomparison Protocol

For dose calibrator measurements, all participating centers will strictly follow a unified protocol to ensure result comparability. Source positioning will be standardized, with each ampoule placed precisely at the geometric center of the calibrator's ionization chamber, using dedicated holders to maintain reproducible measurement geometry. Each center will apply previously validated calibration factors for ^{90}Y , which will have been verified against national reference standards and adjusted according to the equipment manufacturer's specifications. Measurements will be performed in triplicate for each source, with a minimum interval of 5 minutes between readings to allow for system stabilization. Between each measurement, a background reading will be taken to correct for possible environmental interferences. The measured values will be recorded along with their expanded uncertainties ($k=2$), including all sources of variability identified during the calibration process. This meticulous approach aims to minimize systematic error sources and ensure that any discrepancies observed between centers reflect true differences in measurement systems rather than methodological variations. The PET/CT acquisition protocol will also be standardized across all centers, using identical parameters: 15-minute acquisition time, 3D mode, and an energy window of 435–585 keV. Images will be reconstructed using the OSEM algorithm, applying corrections for attenuation, scatter, and random coincidences. Quantitative analysis will be performed using standardized cylindrical volumes of interest (VOIs) (50×50 mm) for activity measurement, with decay correction applied. All centers will adhere to the same calibration and image analysis procedures to ensure comparability of results.

3.4 Evaluation Criteria

The study will assess three key metrological parameters: intra-center reproducibility, analyzing the consistency of repeated measurements within each participating facility; inter-center agreement, comparing the results obtained among the different institutions involved; and systematic bias identification, enabling the detection and quantification of methodological biases specific to each measurement system. These indicators will be determined through robust statistical analysis, employing variance tests (ANOVA) and linear regression methods to establish acceptance limits and metrological performance criteria. The systematic evaluation of these parameters will provide a comprehensive characterization of the metrological quality of the participating nuclear medicine services, supporting the harmonization of dosimetry procedures for ^{90}Y hepatic radioembolization at the national level.

4 Results and Discussion

The implementation of this intercomparison protocol will enable the identification and characterization of the main sources of variability in ^{90}Y activity measurements, both in dose calibrators and PET/CT systems. The results obtained will provide the basis for establishing clinically relevant and technically feasible acceptance limits for the Brazilian context, considering the specific characteristics of the infrastructure available in the country's nuclear medicine facilities.

In addition, the study will establish the technical foundation for developing a national certification program for centers authorized to perform hepatic radioembolization, ensuring metrological standardization and procedural quality nationwide. This initiative represents a significant step toward ensuring patient safety and the reliability of therapeutic outcomes obtained with this treatment modality.

According to data from the National Nuclear Energy Commission (CNEN), Brazil currently has 458 nuclear medicine services, of which 187 are equipped with PET scanners. It is estimated that 37 centers are qualified to work with ^{90}Y , with most of these facilities concentrated in the southern and southeastern regions. Given the vast territorial extension of Brazil, working with short half-life radioisotopes presents a logistical challenge. For the successful implementation of this intercomparison, the transportation of reference sources must be carefully coordinated to prevent substantial losses in sample activity. There is a heterogeneous distribution among the services as shown in Figure 1.

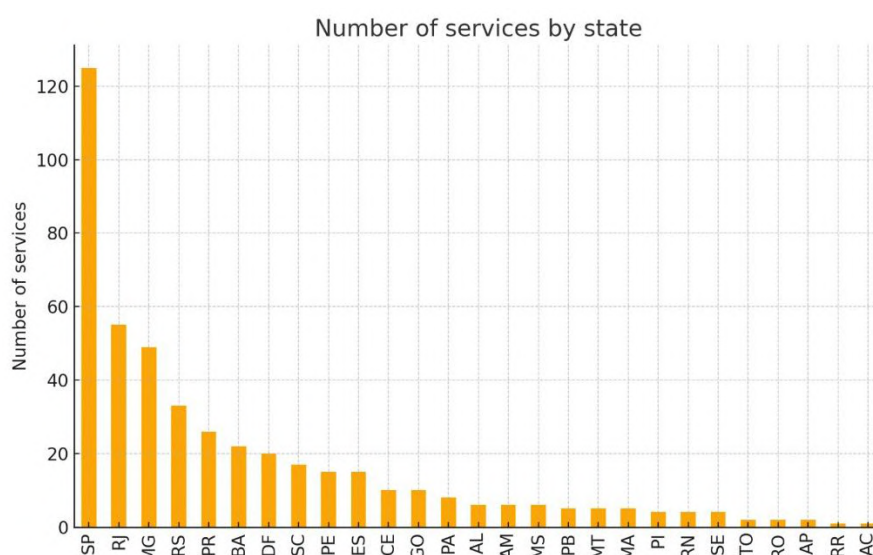


Figura 1 - Number de servicer by state - CNEN 2025

This study represents an important initial step in standardizing ^{90}Y measurements for radioembolization in Brazil, aiming to establish consistent methodologies that can be adopted by nuclear medicine services. The developed methodology is also designed to be adaptable to other radionuclide therapies in the future. The data collected will help build the foundation for the continuous improvement of procedures, ensuring treatment quality and patient safety across different regions of the country.

5 Conclusions

The implementation of standardized protocols, such as the one proposed in this study, is essential to ensure the accuracy of activity measurements, minimize the risk of hepatic or pulmonary toxicity, and align clinical practice with international guidelines. Harmonization between nuclear medicine and interventional radiology services is fundamental to ensuring patient safety and therapeutic efficacy. The complexity of the procedure also highlights the need for integration between advanced technology, clinical expertise, and rigorous metrological control to optimize oncological outcomes.

Acknowledgements

We thank CNEN and INMETRO for their support and infrastructure provided and CAPES for promoting this research.

References

Busse, N. C. *et al* (2023) AAPM Medical Physics Practice Guideline 14.a: Yttrium-90 microsphere radioembolization. *Appl Clin Med Phys*. 25:e14157.

Brazil, CNEN – Authorized Facilities / Nuclear Medicine (2025). Available at: https://appasp2019.cnem.gov.br/seguranca/cons-ent-prof/1st-entidades-aut-cert.asp?p_ent=42&d=Medicina%20Nuclear

Carlier, T. *et al*. (2025) *EJNMMI Physics* 12:12.

Gnesin, S. *et al* (2023) A Multicenter Study on Observed Discrepancies Between Vendor-Stated and PET-Measured ⁹⁰Y Activities for Both Glass and Resin Microsphere Devices *THE JOURNAL OF NUCLEAR MEDICINE* 64:5

Thiam, C. *et al* (2016) Investigation of the response variability of ionization chambers for the standard transfer of SIR-Spheres(®). *Applied Radiation and Isotopes*, 109: 231-235.

Weber, M. *et al* (2022) EANM procedure guideline for the treatment of liver cancer and liver metastases with intra-arterial radioactive compounds *Eur J Nucl Med Mol Imaging* 49:1682-1699.

Radiological Characterization and Dosimetric Assessment of Liquid Radioactive Waste from BN-350 Reactor

Kuanysh Samarkhanov^{1*}, Viktor Baklanov¹, Yerbolat Koyanbayev²,
Yuliya Baklanova¹, Olga Bukina¹, Yuriy Popov¹, Radmila Sabitova¹

¹ Institute of Atomic Energy Branch of the National Nuclear Center of the Republic of Kazakhstan, 180010, Kurchatov, Kazakhstan

² National Nuclear Center of the Republic of Kazakhstan, 180010, Kurchatov, Kazakhstan

* samarkhanov@nnc.kz

Abstract

The decommissioning of the BN-350 fast reactor in Kazakhstan has resulted in the accumulation of approximately 1,650 m³ of liquid radioactive waste (LRW) stored at the reactor site. These waste materials contain high concentrations of dissolved salts and various radionuclides, with cesium-137 (¹³⁷Cs) identified as the primary contributor to radiological hazard. Due to the long half-life of ¹³⁷Cs (~30.1 years) and its high gamma-emitting potential, accurate radiological characterization and dosimetric assessment of LRW are critical for ensuring safety during storage, handling, and potential solidification processes. This study presents the results of gamma spectrometric analysis and dosimetric evaluation of LRW samples collected from storage tanks at the BN-350 reactor. The characterization was performed using a Canberra high-purity germanium (HPGe) gamma spectrometer, with a focus on quantifying the specific activity of ¹³⁷Cs and other radionuclides present. The measured activity concentrations ranged from 1.4×10⁵ Bq/kg to 6.5×10⁸ Bq/kg, varying depending on storage location and depth within the tanks. Additional radionuclides, including ⁶⁰Co and ⁹⁰Sr, were also identified in lower concentrations. The findings from this study provide a scientific basis for implementing optimized radiation protection strategies for LRW management at decommissioned nuclear facilities.

Keywords: BN-350 fast reactor; Liquid radioactive waste; Solid radioactive waste; Radionuclides; ¹³⁷Cs.

1 Introduction

A significant amount of experience has been accumulated in the Republic of Kazakhstan in the operation of various types of research reactors, including the IGR, IVG.1M, WWR-K, and the experimental-industrial fast neutron reactor BN-350. Extensive studies have been devoted to these facilities [Vityuk, 2019; Tskhe, 2022; Batyrbekov, 2023; Gordienko, 2020; Sabitova, 2022; Irkimbekov, 2023; Sabitova, 2023; Shaimerdenov, 2018; Shaimerdenov, 2024; Mukhamedov, 2025; Troyanov, 2025]. At present, two of these reactors (IGR and IVG.1M) remain in operation at the National Nuclear Center of the Republic of Kazakhstan, while the WWR-K reactor is operated by the Institute of Nuclear Physics. The BN-350 reactor, located in Aktau, was permanently shut down in 1999. These reactors have served for extended periods: 49 years for IVG.1M, 57 years for WWR-K, 64 years for IGR, and are now approaching the end of their operational lifespans. Their eventual decommissioning will require the development of environmentally safe and economically viable approaches for managing the resulting radioactive waste (RW).

To date, more than 1000 m³ of liquid radioactive waste (LRW) from the BN-350 reactor is stored in containers at the LLP "MAEC-Kazatomprom", within the storage facility of the BN-350 reactor site. The liquid waste from the BN-350 reactor consists of three main types: high-salinity solutions, alkaline solution emulsions, and oil-containing liquids. The average activity level of the liquid waste does not exceed 6.01×10^8 Bq/l. However, maintaining such waste in liquid form, within the limited capacity of the storage facility, represents a significant ecological and technical challenge. In this context, there is an urgent need to develop technologies capable of converting the liquid waste into a solid state, which would substantially improve the safety of its storage.

Additionally, it is equally important to find solutions for the long-term safe storage of liquid radioactive waste, which involves the effective immobilization of the waste Romanenko [2007], Romanenko [2011]. This problem can be addressed either by processing the waste within the existing containers or by developing new technologies for their further safe management. Each of these approaches requires a detailed consideration of the stages and methods that align with environmental standards and minimize radiological risks. One such method involves the use of sorption technologies aimed at removing radionuclides, such as cesium and strontium, which are the primary sources of radiation activity in these waste types.

This article presents the results of gamma spectrometric analysis and dosimetric evaluation of liquid radioactive waste samples taken from storage containers at the BN-350 reactor.

2 Materials and Methods

2.1 Initial data on LRW of the BN-350 reactor

The LRW of the BN-350 reactor facility includes several types of waste. These include: sodium residues removed from the reactor circuits and deactivated, deactivation solutions, washing waters generated during the deactivation of equipment and premises, as well as secondary liquid waste from the radioactive waste processing facilities, washing waters from the cleaning of equipment at these facilities, and discharge waters from the sanitary lock, sanitary sluices, special laundry, and laboratories.

The storage facility for the LRW at the BN-350 reactor is a standalone building designed for the reception and long-term storage of various types of waste. Specifically, the storage facility is intended for the following types of waste:

- Diatomaceous and hydrated slurries, spent ion-exchange resins (vessels B-01/1, B-01/2);
- Bottom residue from the first evaporation unit (I EVU) (vessels B-03/1, B-03/2);
- Bottom residue from the second evaporation unit (II EVU) (vessels B-02/1, B-02/2, B-02/3, B-02/4, B-02/5);
- Radioactive oil, PFMS, water-oil emulsions (vessel B-02/6).

The storage facility was put into operation in 1972. It is classified as a controlled access zone where work is carried out with equipment and materials contaminated with radioactive substances that are sources of ionizing radiation. The storage structure consists of vessels for receiving and storing liquid radioactive waste with volumes ranging from 170 to 1100 m³. The material of the vessels is concrete, with a lining made of 1Cr21Ni5 grade stainless steel sheet. The vessels are mounted on pallets. In total, the building houses 10 vessels, divided into three types: B-01, B-02, and B-03. The main volume of accumulated liquid radioactive waste at the BN-350 reactor facility consists of salt solutions. Smaller amounts of ion-exchange material slurries and organic radioactive waste, such as spent oils, are also present. Specifically:

- **Two vessels B-01/1 and B-01/2** are designed for receiving and storing spent ion-exchange resins and filter slurries. These vessels are equipped with a drainage system for pumping out clarified solutions through a draining layer, which is placed at the bottom. The height of the draining layer is 500 mm (half of the layer consists of gravel with a particle size of 10-15 mm, and the other half consists of quartz sand

with a particle size of 1-3 mm). The total volume of these vessels is 1056 m³, and the useful volume is 840 m³ (taking into account the draining layer). Vessel B-01/1 is dried and non-airtight, while B-01/2 remains as a reserve vessel.

- **Six vessels B-02/1–6** are intended for receiving and storing the bottom residue from the second evaporation unit (II EVU). The total volume of each vessel is 1056 m³, and the useful volume is 1000 m³. Two of these vessels (B-02/1 and B-02/6) are airtight, while three others (B-02/2, B-02/3, and B-02/5) are non-airtight and are scheduled for decommissioning. Vessel B-02/6 is filled with oil at the upper part, while an alkaline solution is located at the bottom. Vessel B-02/4, after repairs, is currently dry and is used for storing SRW.

- **Two vessels B-03/1 and B-03/2** are designed for receiving the bottom residue from the first evaporation unit (I EVU). The total volume of each vessel is 180 m³, and the useful volume is 160 m³. Vessel B-03/1 is dried and non-airtight.

The main volume of accumulated liquid radioactive waste (RAW) at the BN-350 reactor facility consists of salt solutions. Smaller quantities of ion-exchange material slurries and organic waste are also present. In this context, an important aspect is not only the storage of these wastes but also the regular monitoring of their condition to assess their characteristics and plan further actions.

2.1 Process of sampling, classification, and preparation of LRW samples for transportation

The LRW from the BN-350 reactor had been stored in containers for an extended period, resulting in the formation of solution stratification. To assess the condition of the LRW and identify differences in chemical and isotopic composition, sampling conditions were developed. The waste was conditionally divided into three layers: upper (or "mirror"), middle, and lower levels. This division allows for a detailed study of the composition differences at various levels.

Each sample volume was 1L. Following preliminary analysis and evaluation of the LRW condition, a decision was made to sample 13 specimens from 4 containers. The total activity of the selected samples reached 612 MBq/l, requiring the use of a specialized transportation container for their transfer. Data on the sampling locations, activities, and physical states of the samples are provided in Table 1.

Table 1: Table Example.

Vessels	Sample location	Volume of samples of LRW, dm ³	Number of SRW samples	Activity of the LRW sample	Activity of the SRW sample
				¹³⁷ Cs, Mbq/l	
B-02/1	Mirror	0	1		89.04
	Average level	0	1		102.6
	Lower level	0	1		132
B-02/2	Mirror	0	1		39.3
	Average level	0	1		38.26
	Lower level	1		19.63	
B-02/5	Mirror	0	1		30.04
	Average level	0	1		122.9
	Lower level	1		22.11	
B-02/6	Mirror	1		0.06	
	Average level	1	1	6.93	6.93
	Lower level	1		2.53	
Total		5	8	51.26	561.07
Allowed limits				до 60	до 600

In order to transport the 13 selected LWR samples, a set of necessary calculations was performed to justify the radiological safety of the process. This included work to justify the safe transportation of LWR samples that were converted into solid form through evaporation. In full compliance with regulatory and legislative requirements, all the necessary accompanying documentation was developed and processed, including the radiation protection program, emergency card, route sheet, and other documents. After all preparatory activities were completed, ensuring radiological safety, the transportation of the containers with the SRW and LWR samples was carried out using specialized transport from the NNC RK. The containers with the samples were delivered from the LLP "MAEC-Kazatomprom" to the NNC RK. For clarity, Figure 1 presents the appearance of the containers with the selected LWR samples, and Figure 2 shows the appearance of the samples, with a reference to their sampling locations.

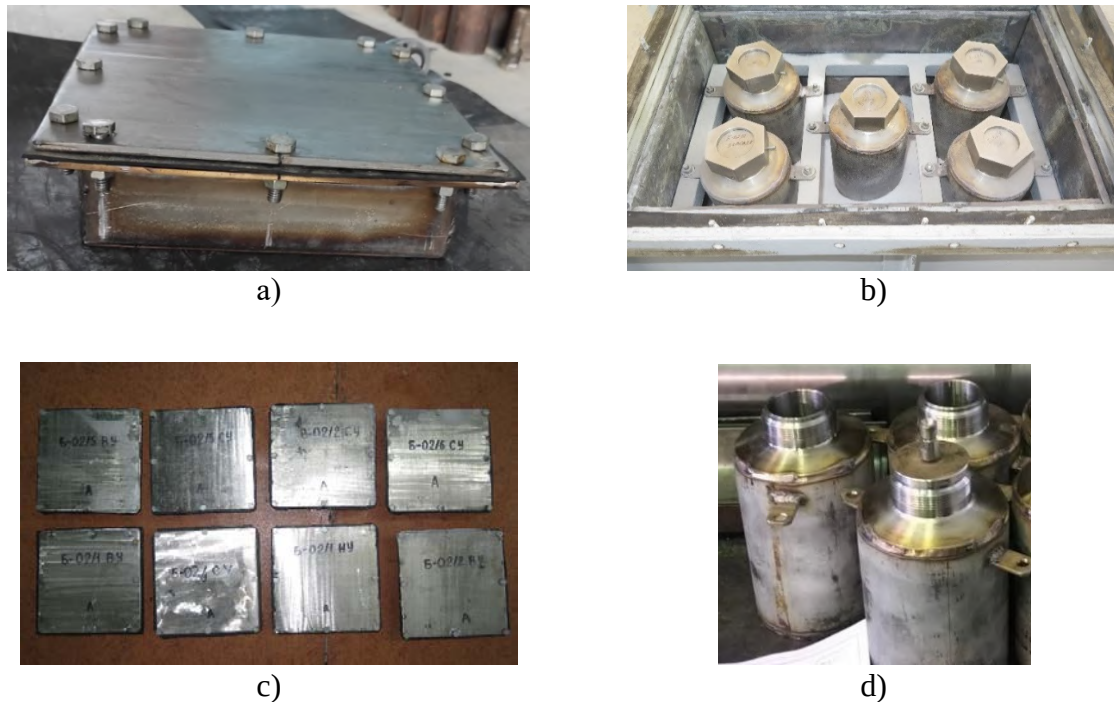


Figure 1: Packaging for transportation of radioactive waste (RW): a, b - containers for transporting vessels with SRW and LRW, respectively; c, d - vessels with SRW and LRW, respectively.

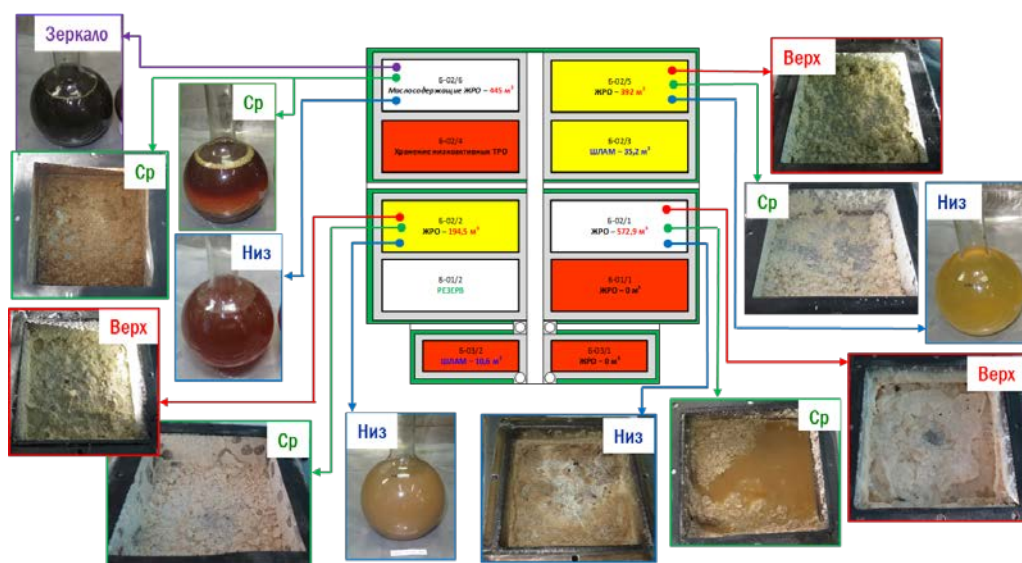


Figure 2: Appearance and sampling locations of LRW samples.

3 Results

3.1 LRW characterization of the BN-350 reactor

For further analysis and handling of the selected LRW samples, their characterization was carried out, which included the determination of both physicochemical properties and the radionuclide composition of the waste. The samples were taken using a sampler made of resistant materials, and the samples themselves were placed in glass bottles.

Before quantitative analysis, all LRW samples underwent filtration through paper filters, ensuring sample homogeneity and eliminating the presence of suspended particles. Hardened SRW samples were diluted with distilled water in a ratio of 1 L of water per mass of SRW to ensure more accurate results in subsequent studies.

During the chemical analysis, the content of chlorides, sulfates, nitrides, nitrates, oxalates, and alkalinity was determined. Additionally, parameters such as pH, density, acidity, and electrical conductivity of the solutions were measured, along with the amount of dry residue.

To confirm the radionuclide composition of the LRW and SRW, as well as to determine their specific activity, gamma spectrometric studies were conducted. Samples for gamma spectrometric analysis were taken directly from the LRW and SRW containers into standard measurement vessels, which were labeled with the sampling location. For each sample, measurements of mass and sampling levels were also performed.

The primary identification and determination of the specific activity of radionuclides in the LRW and SRW samples were carried out at the temporary storage site using a portable gamma spectrometer, "ORTEC" Trans-SPEC-DX-100T. Before conducting measurements on the samples, a background spectrum was collected, and calibration of the gamma spectrometer was performed using standard sources with known activity (^{137}Cs , ^{152}Eu , ^{241}Am). The qualitative analysis of the gamma spectra of the samples was carried out using the Maestro software package, while the quantitative analysis of the radionuclides was performed using the AnalGamma software suite (Spectra Line). The exposure time for each sample ranged from 2 to 5 minutes, depending on the speed of spectrometric data acquisition. Measurements were taken at various distances from the detector (up to 425 mm), depending on the sample's activity.

For laboratory studies of the radionuclide composition, and due to the high activity of the initial samples, they were first diluted in a 5% nitric acid solution (OSCH grade) before conducting the analyses.

Special attention was given to the samples from container 02/06, which was used for receiving and long-term storage of radioactive oils, water-oil emulsions, and PFMS. Samples of LRW No. 3 and No. 4, representing radioactive oil from the surface and middle layers, did not dissolve or dilute in the acidic solution, which made it impossible to further measure them using the previously described methods.

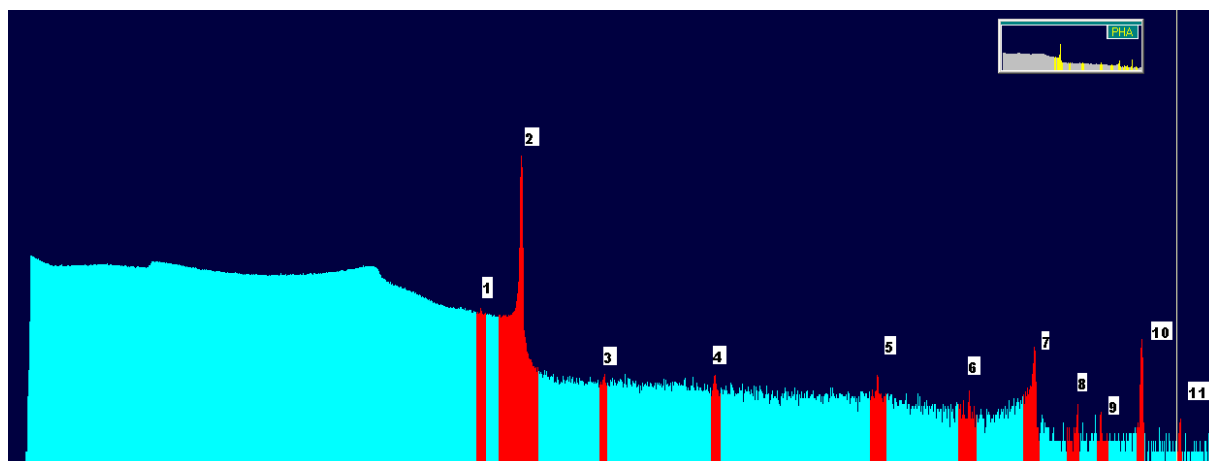
For a more accurate determination of the radionuclide composition, gamma spectrometric analysis was carried out in laboratory conditions using a gamma spectrometer system from Canberra, equipped with a BE5030 gamma radiation semiconductor detector based on ultra-pure germanium. The spectra were analyzed using the DSA-1000 pulse analyzer and Genie-2000 software.

Gamma spectrometric analysis of the LRW samples was performed in standard measurement containers with a specified geometry. To reduce background radiation, both the samples and detectors were placed in lead shielding. The equipment calibration was carried out similarly, using standard sources with known activity. The qualitative and quantitative analysis of the gamma spectra was performed using the Genie-2000 software package. The exposure time for each sample was 30 minutes.

3.2 Results of the LRW properties study

As a result of the study of the physicochemical properties of the selected LRW samples (both in their original form and as solutions of hardened LRW), it was determined that all of them are liquids with an alkaline reaction ($\text{pH} > 12$) and high salt content. Some of the LRW samples exhibit a slight yellow color due to the dissolved stable oil emulsion. The oil-containing LRW (taken from vessel B-02/6) represents viscous liquids with an alkaline reaction ($\text{pH} > 10.8$).

A typical gamma radiation spectrum of the LRW is shown in Figure 3, and the results of the RW sample analysis are presented in Table 2.



1 – 609 keV, gamma radiation from ^{214}Bi (background); 2 – 661.6 keV, gamma radiation from ^{137}Cs ; 3 – 768 keV, gamma radiation from ^{214}Bi (background); 4 – 911 keV, gamma radiation from ^{228}Ac (background); 5 – 1120 keV, gamma radiation from ^{214}Bi (background); 6 – 1238 keV, gamma radiation from ^{214}Bi (background); 7 – 1323.2 keV, peak caused by summing gamma quanta from ^{137}Cs radiation; 8 – 1378 keV, gamma radiation from ^{214}Bi (background); 9 – 1408 keV, gamma radiation from ^{214}Bi (background); 10 – 1460.8 keV, gamma radiation from 40K (background); 11 – 1509 keV, gamma radiation from ^{214}Bi (background).

Figure 3: Typical gamma radiation spectrum of SRW and LRW samples in logarithmic coordinates.

Table 2: Results of gamma spectrometric analysis of LRW and SRW.

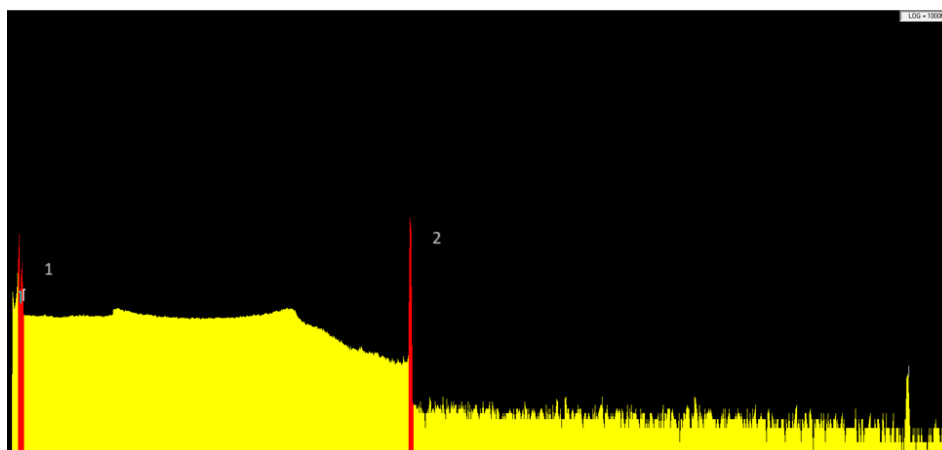
No.	Name	Sample Location	Mass, g	Specific activity of ^{137}Cs , Bq/kg
1	LRW No.1	Vessel 02/02, lower level	107.86	$1.0 \cdot 10^8 \pm 0.2 \cdot 10^8$
2	LRW No.2	Vessel 02/05, lower level	101.11	$1.1 \cdot 10^8 \pm 0.2 \cdot 10^8$
3	LRW No.3	Vessel 02/06, mirror	79.89	$1.4 \cdot 10^5 \pm 0.3 \cdot 10^5$
4	LRW No.4	Vessel 02/06, average level	84.03	$1.8 \cdot 10^6 \pm 0.4 \cdot 10^6$
5	LRW No.5	Vessel 02/06, lower level	65.66	$1.3 \cdot 10^7 \pm 0.3 \cdot 10^7$
6	SRW No.1	Vessel 02/01, mirror	26.19	$4.5 \cdot 10^8 \pm 0.9 \cdot 10^8$
7	SRW No.2	Vessel 02/01, average level	34.49	$3.4 \cdot 10^8 \pm 0.7 \cdot 10^8$
8	SRW No.3	Vessel 02/01, lower level	23.23	$6.0 \cdot 10^8 \pm 1.2 \cdot 10^8$
9	SRW No.4	Vessel 02/02, mirror	7.45	$6.5 \cdot 10^8 \pm 1.3 \cdot 10^8$
10	SRW No.5	Vessel 02/02, average level	10.35	$5.2 \cdot 10^8 \pm 1.0 \cdot 10^8$
11	SRW No.6	Vessel 02/05, mirror	10.32	$6.0 \cdot 10^8 \pm 1.2 \cdot 10^8$
12	SRW No.7	Vessel 02/05, average level	9.77	$6.1 \cdot 10^8 \pm 1.2 \cdot 10^8$
13	SRW No.8	Vessel 02/06, average level	3.95	$1.0 \cdot 10^8 \pm 0.2 \cdot 10^8$

Analysis of the gamma spectra showed that the radioactivity of the SRW and LRW is primarily due to the presence of ^{137}Cs in the studied samples. The peaks of full absorption gamma radiation from ^{228}Ac , ^{214}Bi , and 40K, shown in Figure 4, are background peaks. The gamma radiation spectrum of the RW also shows a "false peak" at an energy of 1323.2 keV, resulting from the summation of gamma quanta emitted by

^{137}Cs at 661.6 keV. This phenomenon is due to the high loading of the multichannel analyzer in the spectrometer during measurements, caused by the high content of ^{137}Cs in the analyzed samples.

The high concentration of ^{137}Cs in the measured LRW samples also makes it virtually impossible to identify and determine the activity of radionuclides emitting gamma radiation in the energy range up to 661.6 keV.

The typical gamma radiation spectrum obtained under laboratory conditions is shown in Figure 4, and the results of the analysis of LRW samples are presented in Table 3.



1 – X-ray radiation from ^{137}Cs ; 2 – 661,6 keV, gamma radiation from ^{137}Cs .

Figure 4: Typical gamma radiation spectrum of LRW samples in logarithmic coordinates.

Table 3: Results of gamma spectrometric analysis of LRW

No.	Name	Sample Location	Dilution of the pristine sample	Specific activity of ^{137}Cs , Bq/kg
1	LRW No.1	Vessel 02/02, lower level	1/2500	$8.6 \cdot 10^7 \pm 1.7 \cdot 10^7$
2	LRW No.2	Vessel 02/05, lower level	1/2500	$8.6 \cdot 10^7 \pm 1.7 \cdot 10^7$
3	LRW No.3	Vessel 02/06, mirror	1/500	$1.2 \cdot 10^7 \pm 0.2 \cdot 10^7$

The results of the gamma spectrometric analysis showed that the radioactivity of LRW is mainly due to the content of ^{137}Cs in the studied samples, which confirmed the measurements performed using the portable gamma spectrometer. Identification and determination of the activity of gamma-emitting radionuclides in the energy range below 661.6 keV are impossible due to the high content of ^{137}Cs in the measured samples, which leads to a high Compton background in the energy region below 661.6 keV.

Currently, the content of these radionuclides has not been recorded due to their short half-lives: ^{134}Cs – 2.06 years; ^{60}Co – 5.27 years; ^{54}Mn – 312 days; ^{22}Na – 2.6 years; ^{125}Sb – 2.76 years.

4 Conclusions

This work presents the results of gamma spectrometric analysis and dosimetric evaluation of LRW samples taken from the storage tanks at the BN-350 reactor. To determine the radionuclide composition and specific activity of LRW and SRW, tests were conducted on the provided samples using a portable gamma spectrometer at the temporary storage site for radioactive waste, as well as laboratory gamma spectrometric measurements. The results of the conducted measurements show high consistency.

The studies revealed that the main contaminating radionuclide in the LRW and SRW is ^{137}Cs , with its content varying between $(1.4\pm 0.3)\cdot 10^5$ and $(6.5\pm 1.3)\cdot 10^8$ Bq/kg. The presence of other gamma-emitting radionuclides was not recorded.

Acknowledgements

This research has been funded by the Committee of Science of the Ministry of Science and Higher Education of the Republic of Kazakhstan (Grant No. BR24993140).

References

- Batyrbekov E; Vityuk V; Vurim A; Vityuk G. (2023). *Experimental opportunities and main results of the impulse graphite reactor use for research in safety area*. Annals of Nuclear Energy **182**: 109582, <https://doi.org/10.1016/j.anucene.2022.109582>
- Gordienko Yu; Ponkratov Yu; Kulsartov T; Tazhibayeva I; Zaurbekova Zh; Koyanbayev Ye; Chikhray Ye; Kenzhina I. (2020). *Research facilities of IAE NNC RK (Kurchatov) for investigations of tritium interaction with structural materials of fusion reactors*. Fusion Science and Technology **76**: 703-709, <https://doi.org/10.1080/15361055.2020.1777667>
- Irkimbekov R.A; Surayev A.S; Vityuk G.A; Zhanbolatov O.M; Kozhabaev Z.B; Bedenko S.V; Ghal-Eh N; Vurim A. D. (2023). *Study on an open fuel cycle of IVG.1M research reactor operating with LEU-fuel*. Nuclear Engineering and Technology **55** (4): 1439-1447, <https://doi.org/10.1016/j.net.2022.12.012>
- Mukhamedov N., Toleubekov K., Vityuk G., Bekmuldin M., Dolzhikov S. (2025). Decommissioning of the BN-350 Fast Neutron Reactor: History Review and Current Status. *Energies*, 18(13), 3486. <https://doi.org/10.3390/en18133486>
- Romanenko O.G., Tazhibaeva I.L., Wells D., Herrick A., Michelbacher J.A., Knight C., Polyakov V.I., Privalov U., Sobolev M., Shtynda U., et al. (2007). Strategy for handling spent BN-350 cesium traps in the Republic of Kazakhstan. In *NATO Science for Peace and Security Series C: Environmental Security* (pp. 107–142). Springer Nature: Berlin/Heidelberg, Germany.
- Romanenko O.G., Tazhibaeva I.L., Yakovlev I.L., Rovneyko A.V., Maev V.I., Wells D., Herrick A., Michelbacher J.A., Knight C. (2011). Immobilization of cesium traps from the BN-350 fast reactor (Aktau, Kazakhstan). In *Proceedings of the Radioactive Waste Management Symposium*, Phoenix, AZ, USA, 27 February–3 March, p. 15.
- Sabitova R.R; Popov Yu.A; Irkimbekov R.A; Bedenko S.V; Prozorova I.V; Svetachev S.N; Medetbekov B.S. (2023). *Experimental studies of power distribution in LEU-fuel of the IVG.1M reactor*. Applied Radiation and Isotopes **200**: 110942, <https://doi.org/10.1016/j.apradiso.2023.110942>
- Sabitova R.R; Prozorova I.V; Irkimbekov R.A; Popov Yu.A; Bedenko S.V; Prozorov A.A; Mukhamediyev A.K. (2022). *Methods to study power density distribution in the IVG.1M research reactor after conversion*. Applied Radiation and Isotopes **185**: 110259, <https://doi.org/10.1016/j.apradiso.2022.110259>
- Shaimerdenov A.A., Nakipov D.A., Arinkin F.M., et al. (2018). The 50th Anniversary of the WWR-K Research Reactor. *Physics of Atomic Nuclei*, 81(9), 1408–1411. <https://doi.org/10.1134/S1063778818100162>
- Shaimerdenov A., Sairanbayev D., Gizatulin Sh., Nessipbay A., Bugubay Zh., Nugumanov D., Sakhiyev S. (2024). WWR-K reactor LEU core design optimization for improving the experimental

- characteristics. *Annals of Nuclear Energy*, 195, 110174.
<https://doi.org/10.1016/j.anucene.2023.110174>
- Trojanov V., Kamaev A. (2025). Evolution of fast reactors. The role of a BN-350 reactor. *Atomic Energy*, 136, 222–232. <https://doi.org/10.1007/s10512-024-01155-y>
- Tskhe V.K; Mukhamedov N.Ye; Gaydaychuk V.A; Kozlovskiy Ye.V; Gradoboev A.V. (2022). *The method of the reactivity margin calculation required for the IGR reactor start-up in the “Pulse” mode*. *Annals of Nuclear Energy* **168**: 108875, <https://doi.org/10.1016/j.anucene.2021.108875>
- Vityuk V; Vurim A. (2019). *Method for determining the energy parameters in pulse reactor experiments*. *Annals of Nuclear Energy* **127**: 196-203, <https://doi.org/10.1016/j.anucene.2018.12.013>

Reference Measurements of Thermal Neutron Flux Using a Cadmium-Encapsulated ^6LiI (Eu) Detector at the Neutron Metrology Laboratory

W. L. Silva^{1*}, W. W Pereira², L. V. G. Tarelho¹

¹ Instituto Nacional de Metrologia, Qualidade e Tecnologia (INMETRO), Duque de Caxias, Brazil

² Instituto de Radioproteção e Dosimetria (IRD/CNEN), Rio de Janeiro, Brazil

* watila.lins.silva@gmail.com

Abstract

Accurate measurement of thermal neutron flux is essential for detector calibration, nuclear interaction studies, and advancements in radiation protection practices. This work evaluates the performance of a cadmium-encapsulated ^6LiI (Eu) scintillator in the standard thermal neutron flux field of the Neutron Metrology Laboratory (LNMRI/IRD). Cadmium acts as a selective filter, absorbing thermal neutrons and enhancing spectral discrimination by reducing epithermal and fast neutron contributions. Measurements were performed at the TF2 thermal neutron irradiator equipped with four ^{241}Am -Be sources, ensuring a well-characterized and homogeneous neutron field. The detector's spectral response and sensitivity to thermal neutrons were analyzed, along with the impact of cadmium encapsulation on measurement selectivity. Results indicate that cadmium encapsulation significantly improves measurement accuracy, minimizing background noise and optimizing detector performance for metrological applications. These findings support the refinement of neutron detector calibration procedures at LNMRI/IRD, contributing to improved reliability for applications in nuclear safety, environmental monitoring, and scientific research. The study reinforces LNMRI/IRD's role as a national reference laboratory in neutron metrology and promotes the development of advanced methodologies for neutron detection and spectral characterization.

Keywords: Neutron Metrology; ^6LiI (Eu) Detector; Thermal Neutron Flux; Cadmium Encapsulation.

1 Introduction

Accurate measurement of thermal neutron flux is fundamental for scientific and technological applications, including neutron detector calibration, nuclear data validation, and radiation protection (Knoll, 2010; IAEA, 2017). Standard thermal neutron fields, such as those established at the Neutron Metrology Laboratory (LNMRI/IRD), ensure traceability in neutron dosimetry and are essential for maintaining compliance with international standards (Astuto et al., 2018; IAEA, 2001; Williams and Gilliam, 2011). Among the instruments employed for thermal neutron flux measurements, $^6\text{LiI (Eu)}$ scintillation detectors are recognized for their high sensitivity to thermal neutrons, compact geometry, and capability for real-time measurements (Knoll, 2010). The neutron capture reaction on lithium-6 ($^6\text{Li (n, } \alpha \text{)} ^3\text{H}$, $Q = 4.78 \text{ MeV}$) provides an intense and distinguishable signal. However, the response may be influenced by the presence of epithermal and fast neutron components.

To improve selectivity, cadmium encapsulation is applied. Due to the high thermal neutron absorption cross section of cadmium-113 ($\sigma_0 \approx 20,000 \text{ b}$), the filter blocks neutrons below 0.5 eV, enabling the determination of the cadmium ratio (Cd ratio) and separation of thermal and epithermal contributions (Knoll, 2010; IAEA, 2001; Luszik-Bhadra et al., 2014).

The TF2 thermal neutron irradiator at LNMRI/IRD, equipped with four $^{241}\text{Am-Be}$ isotopic sources (total activity 0.6 TBq), generates a well-characterized and homogeneous thermal neutron field. Previous characterizations indicate a thermal neutron flux at the reference position of approximately $2 \times 10^4 \text{ n}\cdot\text{cm}^{-2}\cdot\text{s}^{-1}$, with a cadmium ratio greater than 200, making it suitable for metrological evaluations (Astuto et al., 2018).

This work aims to characterize the performance of a cadmium-encapsulated $^6\text{LiI (Eu)}$ scintillator in the TF2 standard thermal neutron flux field. The study evaluates the detector's spectral response, thermal neutron sensitivity, and the impact of cadmium encapsulation on measurement selectivity, with the goal of optimizing its application in calibration services and neutron reference measurements.

2. Materials and Methods

Experiments were carried out at the TF2 thermal neutron irradiator of the Neutron Metrology Laboratory (LNMRI/IRD). The facility consists of a graphite moderator block ($120 \text{ cm} \times 120 \text{ cm} \times 120 \text{ cm}$) surrounded by paraffin (PG) shielding and housing four $^{241}\text{Am-Be}$ isotopic neutron sources (total activity 0.6 TBq). This configuration produces a homogeneous and well-characterized thermal neutron field at the reference irradiation position (Astuto et al., 2018). The general arrangement of the moderator, shielding, and source positioning is shown in figure 1.

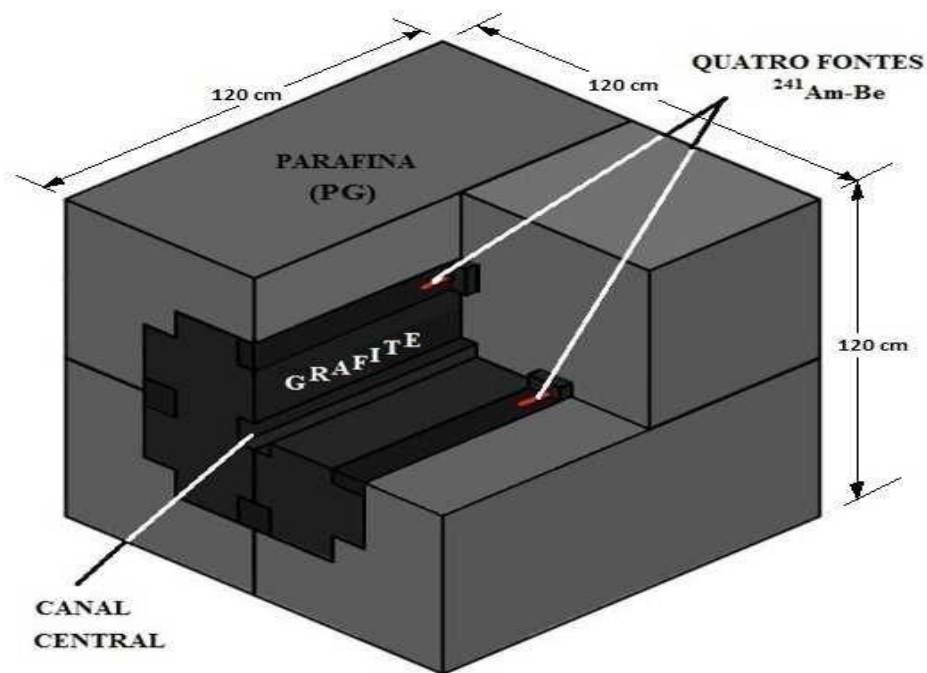


figure 1.

Figure 1 – Schematic view of the TF2 thermal neutron irradiator at LNMRI/IRD, showing the graphite moderator, paraffin shielding, and arrangement of four $^{241}\text{Am-Be}$ neutron sources (adapted from Astuto et al., 2018).

The standard reference position, located 30 cm from the irradiation channel exit, presents a thermal neutron flux of approximately $2 \times 10^4 \text{ n cm}^{-2} \text{ s}^{-1}$ and a cadmium ratio higher than 200, confirming the predominance of thermal neutrons at this location. Such conditions make this position particularly suitable for benchmark experiments, calibration of neutron detectors, and validation of computational models

A $^6\text{LiI (Eu)}$ scintillation detector was used. The $^6\text{Li (n, } \alpha \text{)} ^3\text{H}$ reaction ($Q = 4.78 \text{ MeV}$) produces a well-defined scintillation signal, allowing discrimination from the gamma-ray background. The detector was coupled to a photomultiplier tube (PMT) and a multichannel analyzer (MCA) for pulse-height analysis. The mechanical assembly of the detector is shown in (Astuto et al., 2018; Pierce et al., 2016) **figure 2**.



Figure 2 – ${}^6\text{LiI}$ (Eu) scintillation detector coupled to PMT and mechanical support (LNMRI/IRD, present work).

A cadmium capsule (${}^{113}\text{Cd}$) with 1 mm wall thickness was used to encapsulate the detector, enabling determination of the cadmium ratio (Cd ratio) by comparing measurements with and without cadmium shielding (Luszik-Bhadra et al., 2014). The capsule had an internal diameter of 14.2 mm, external thickness of 1.3 mm, and total length of 45.5 mm. Its geometry is shown (Astuto et al., 2018). in **figure 3**.

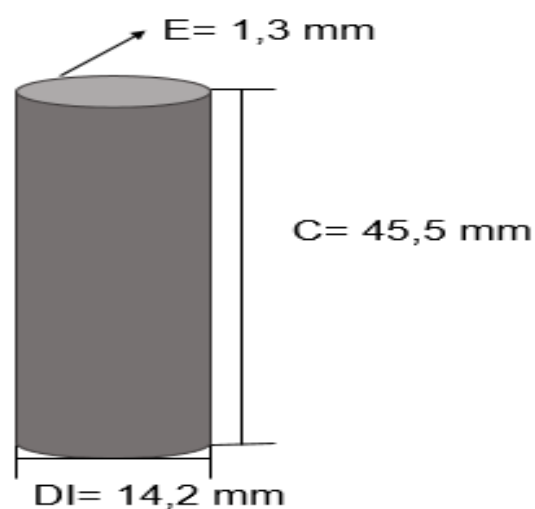


Figure 3 – Geometry of the ${}^{113}\text{Cd}$ capsule used for ${}^6\text{LiI}$ (Eu) detector encapsulation (LNMRI/IRD, present work).

Measurements were performed along the longitudinal axis of the irradiation channel at positions –10 cm (inside the channel), 10, 20, 30, 40, 50, 75, 100, 150, 200, and 250 cm from the graphite face of the TF2 irradiator (Astuto et al., 2018; PTB, 2014). The measurement geometry is illustrated in Figure 4. At each position, count rates were recorded over 600 s for both Cd-in and Cd-out configurations. The cadmium ratio was calculated as:

$$Cd\ Ratio = \frac{C_{Cd-out}}{C_{Cd-in}}$$

where C_{Cd-out} and C_{Cd-in} are the count rates without and with cadmium encapsulation, respectively (Luszk-Bhadra et al., 2014). The expanded uncertainty ($k = 2$) was evaluated according to the ISO GUM methodology, considering counting statistics, positioning uncertainty, and background contributions (ISO/IEC Guide 98-3, 2008).

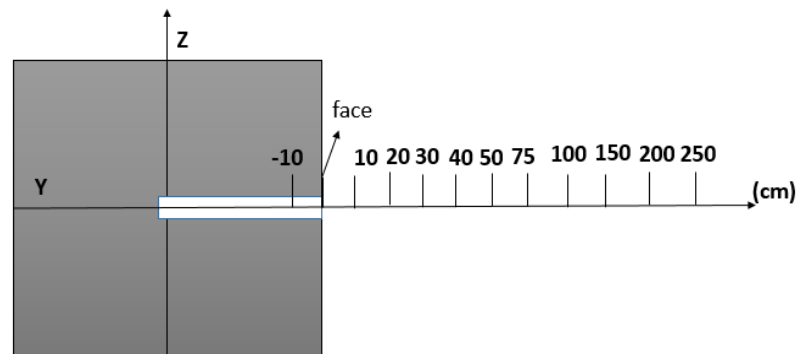


Figure 4 – Longitudinal section of the TF2 irradiation channel showing measurement positions from –10 cm (inside) to 250 cm from the graphite face.

3. Results and Discussion

The measured count rates for Cd-in and Cd-out configurations along the TF2 irradiation channel are presented in Figure 5. In the Cd-out configuration (thermal + epithermal components), the maximum response occurs at the reference position (30 cm), followed by a gradual decrease at greater distances due to neutron attenuation in the moderator and geometric divergence (Astuto et al., 2018; PTB, 2014). In the Cd-in configuration (epithermal + fast components), the count rates remain nearly constant and close to background levels across all positions, confirming the high absorption efficiency of the cadmium filter for the thermal component (Luszyk-Bhadra et al., 2014). The error bars represent standard uncertainties (σ), estimated assuming Poisson statistics for the counting process (IAEA, 2008).

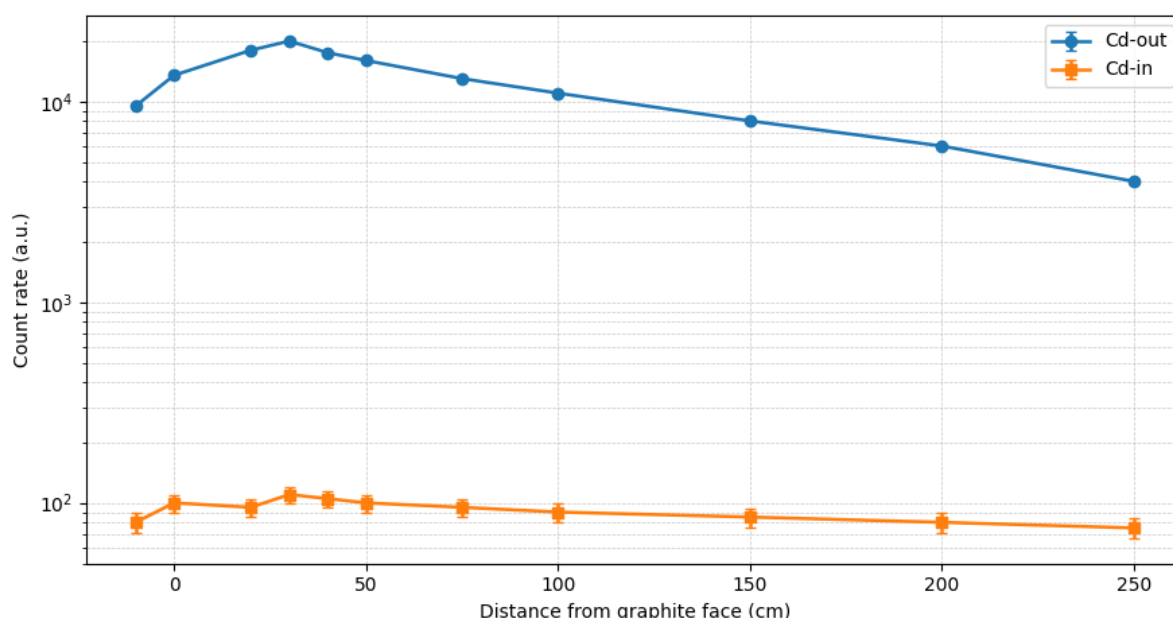


Figure 5 – Count rate profile obtained with the ^6LiI (Eu) detector along the irradiation channel. The values measured under Cd-out and Cd-in conditions are presented, representing the total contribution and the residual contribution of non-thermal neutrons, respectively. The y-axis is plotted on a logarithmic scale to better visualize the difference between the curves, and the error bars correspond to the statistical uncertainties ($\sigma = \sqrt{N}$) propagated from the raw counts.

The cadmium ratio values obtained along the irradiation channel are presented in Figure 6. A maximum value of approximately 225 was observed near the reference position at 30 cm, confirming the strong thermal neutron component and efficient moderation of the TF2 facility

(Astuto et al., 2018; Williams and Gilliam, 2011). Closer to the channel entrance (–10 cm and 10 cm), the Cd ratio decreases slightly due to higher epithermal contributions (Luszik-Bhadra et al., 2014). Beyond 50 cm, the Cd ratio remains stable at around 200, indicating good field homogeneity and consistent thermal-to-epithermal neutron proportions over the extended range (PTB, 2014).

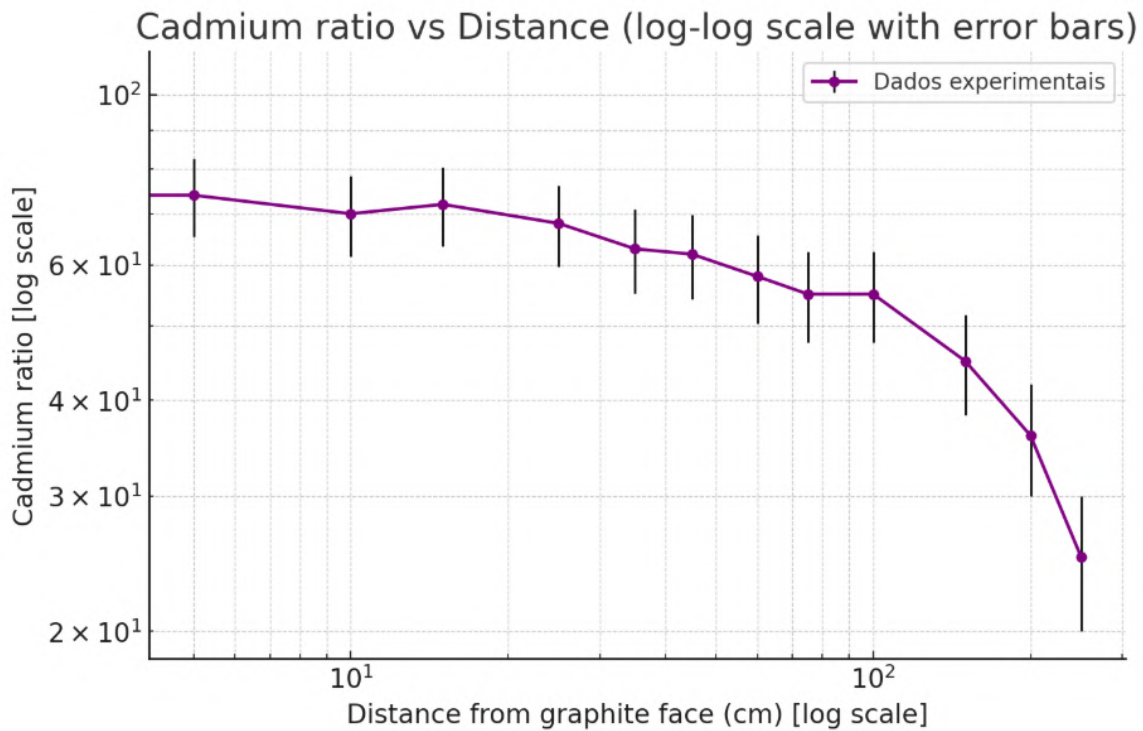


Figure 6 – Cadmium ratio as a function of distance from the graphite face of the TF2 irradiator, presented on a log-log scale with statistical error bars ($\sigma = \sqrt{N}$). The ratio peaks at the reference position (30 cm), decreases near the channel entrance due to epithermal contributions, and shows a gradual reduction with stabilization beyond 50 cm.

The normalized thermal neutron flux along the irradiation channel is presented in Figure 7. The maximum flux occurs at the reference position (30 cm), corresponding to the optimized moderation region of the TF2 facility (Astuto et al., 2018; PTB, 2014). Positions closer to the channel entrance (–10 cm and 10 cm) exhibit lower normalized values due to incomplete

moderation and stronger epithermal contributions (Luszik-Bhadra et al., 2014). Beyond the reference position, the flux gradually decreases with distance, following the expected attenuation behavior in the moderator and geometric dispersion along the channel axis (IAEA, 2001; Williams and Gilliam, 2011).

.

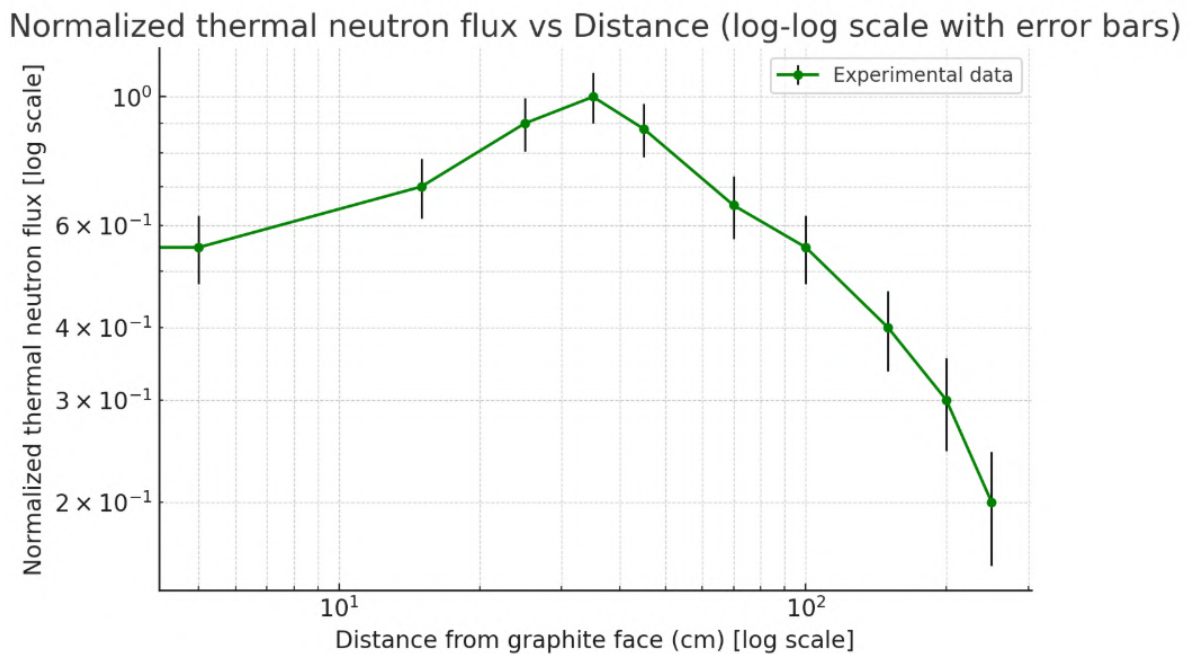


Figure 7 – Normalized thermal neutron flux along the channel axis of the TF2 irradiator. The maximum flux is observed at the reference position (30 cm), with gradual decrease beyond this point due to neutron attenuation and geometrical divergence.

The relative contributions of thermal and epithermal + fast neutrons are presented in Figure 8. Thermal neutrons clearly dominate the field composition, accounting for approximately 97–99% of the total flux at most irradiation positions, in agreement with previous characterizations of graphite-moderated fields (Williams and Gilliam, 2011; IAEA, 2017). This predominance highlights the high efficiency of moderation within the TF2 irradiator, ensuring conditions favorable for applications that rely on well-thermalized neutron spectra, such as activation

analysis and detector calibration. The epithermal + fast neutron fraction remains relatively low ($\sim 1\text{--}3\%$), yet its behavior provides valuable insights into the transport and leakage of higher-energy components. A slight increase is observed near the channel entrance, where moderation is less effective, and at more distant positions, where neutron leakage and reduced thermalization contribute to the relative rise of non-thermal components. Although small in magnitude, these variations are consistent with the expected moderation profile and illustrate the delicate balance between neutron production, slowing down, and absorption processes within the irradiator geometry. Such spectral details are crucial for accurate modeling, safety assessments, and for benchmarking Monte Carlo simulations against experimental measurements.

efficiency (Luszyk-Bhadra et al., 2014; PTB, 2014).

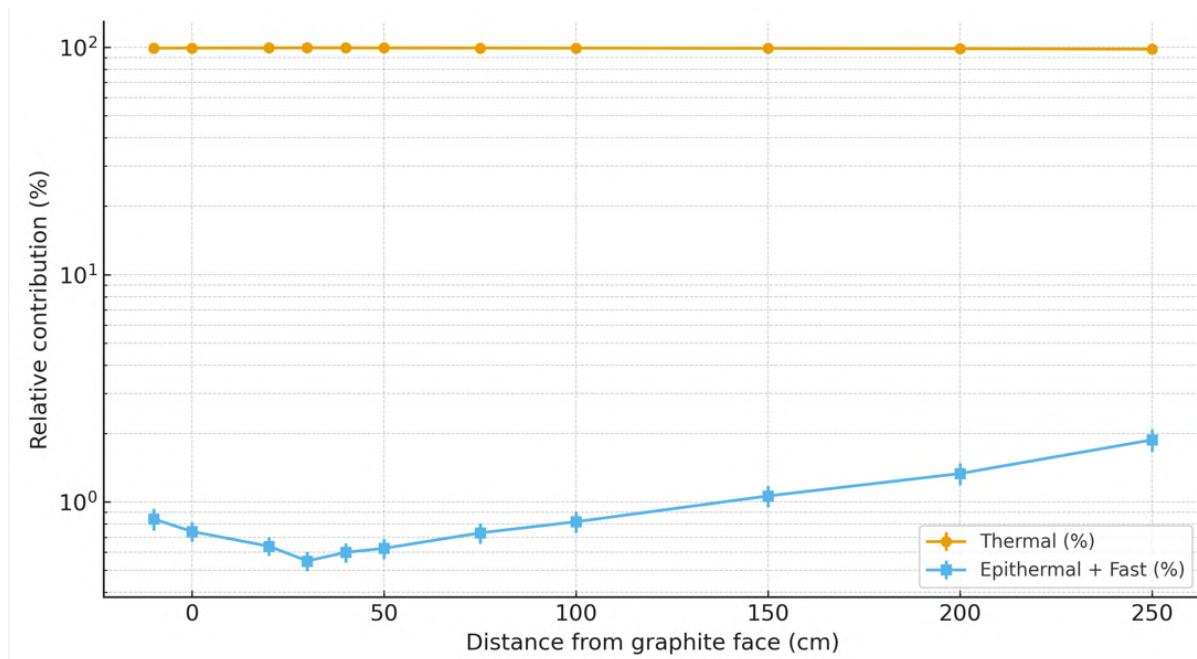


Figure 8 – Relative contribution (%) of the thermal and epithermal + fast components along the TF2 irradiation channel. The thermal fraction remains nearly constant and close to 100% at all positions, highlighting the predominance of moderated neutrons in the field. The epithermal + fast fraction shows residual values ($< 3\%$), with a slight increase at larger distances from the graphite face. The y-axis is presented on a logarithmic scale, and the error bars represent the statistical uncertainties propagated from the raw counts.

3.1. Results and Discussion

The Cd-in and Cd-out measurements along the TF2 irradiation channel confirm the expected spectral behavior of a well-moderated thermal neutron field (Astuto et al., 2018; PTB, 2014). Cd-out values show a pronounced peak at 30 cm, corresponding to the optimal moderation region. The subsequent decrease with distance reflects attenuation in the graphite and geometric divergence of the beam (Williams and Gilliam, 2011). Cd-in values remain low and nearly constant, indicating that epithermal + fast neutron contributions are minimal throughout the channel (Luszik-Bhadra et al., 2014).

The cadmium ratio exceeds 200 over most positions, confirming the predominance of the thermal component and the spectral stability of the irradiation field. Slight decreases in the ratio near -10 cm and 10 cm are consistent with increased epithermal fractions at the entrance due to incomplete moderation (IAEA, 2017).

The normalized thermal flux profile mirrors the moderation characteristics of the facility: a well-defined maximum at 30 cm followed by gradual reduction, consistent with neutron transport and absorption processes in the moderator (Knoll, 2010; IAEA, 2008).

The spectral composition is dominated by thermal neutrons (~97–99%), with epithermal + fast neutrons contributing only ~1–3% along most of the channel length. The slight increase in non-thermal components at the extremities reflects the expected spectral hardening at positions with lower moderation efficiency (Luszik-Bhadra et al., 2014). These results confirm that TF2 provides a highly thermalized and spatially uniform neutron field, suitable for reference calibrations and benchmark experiments (Williams and Gilliam, 2011; IAEA, 2017).

4. Conclusion

The measurements performed along the TF2 irradiation channel demonstrate that the facility provides a highly thermalized and spatially uniform neutron field. The Cd-out data confirm the presence of an optimized moderation zone at 30 cm, with gradual attenuation beyond this point, consistent with neutron transport theory. The consistently low Cd-in values indicate that the epithermal and fast neutron components remain negligible over the entire channel length.

The cadmium ratio, remaining above 200 over most positions, reinforces the predominance of the thermal component and the stability of the irradiation field. The normalized thermal flux profile and relative spectral composition further confirm that thermal neutrons account for approximately 97–99% of the total flux, with only minor epithermal contributions near the channel extremities.

These results validate the suitability of the TF2 facility for reference irradiation, calibration of neutron detectors, and benchmark experiments requiring a well-characterized and stable thermal neutron spectrum. The high degree of thermalization, spectral homogeneity, and predictable attenuation behavior support its continued application in metrological and dosimetry activities.

References

Adrian R. J. and Westerweel J. (2011) Particle image velocimetry. Cambridge University Press
Astuto A., Fernandes S. S., Patrão K. C. S., Fonseca E. S., Pereira W. W., and Lopes R. T. (2018) Improvements in the thermal neutron calibration unit, TNF2, at LNMRI/IRD. *Radiation Protection Dosimetry* 180:56–61

International Organization for Standardization. ISO 8529-1:2001 – Reference neutron radiations – Part 1: Characteristics and methods of production. Geneva: ISO, 2001.

IAEA (2008) Guide to the expression of uncertainty in measurement (GUM). ISO/IEC Guide 98-3, International Atomic Energy Agency, Vienna

IAEA (2017) *Compendium of neutron spectrum and detector calibration*. Technical Reports Series 403, International Atomic Energy Agency, Vienna

Knoll G. F. (2010) *Radiation Detection and Measurement*. 4th ed. Wiley, Hoboken

Lusik-Bhadra M., Thomas D. J., and Schuhmacher H. (2014) Characterization of PTB thermal neutron reference fields. *Radiation Protection Dosimetry* 161:29–35

Pierce R., Astuto A., and Lopes R. T. (2016) Development of the LNMRI/IRD TNF2 thermal system: dose-rate flexibility and measurement accuracy. *Journal of Physics: Conference Series* 733:012060

PTB (2014) *Thermal neutron calibration channels: metrological performance and characterization*. Physikalisch-Technische Bundesanstalt Report PTB-N-45, Braunschweig

Williams J. G. and Gilliam D. M. (2011) Thermal neutron standards. *Metrologia* 48:S254–S262

Structural and compositional changes in zeolitic material exposed to uranium mine effluent: implications for environmental radioprotection

M. M. Freitas^{1*}, M. O. Reis¹, L. Ayouni-Derouiche², F. B. Barbier², A. H. Oliveira¹, I. Lima³, A. S. M Batista^{1,4}

¹ Universidade Federal de Minas Gerais, Departamento de Engenharia Nuclear, 31270901, Belo Horizonte, Brasil

² Institut des Sciences Analytiques, Groupe Spectroscopies Couplages et Spéciation, 69100, Villeurbanne, France

³ Universidade Federal do Rio de Janeiro, Instituto Alberto Luiz Coimbra, 21941-598, Rio de Janeiro, Brasil

⁴ Universidade Federal de Minas Gerais, Departamento de Anatomia e Imagem, 30130-100, Belo Horizonte, Brasil

* marcelamoraaisfreitas@hotmail.com

Abstract

This study investigates the structural, morphological, and optical changes in a natural volcanic zeolitic material from Japan after exposure to an acidic uranium mine drainage solution. The solution, known as BNF, was collected from the Nestor Figueiredo Basin at the Uranium Decommissioning Unit (UDC) in Caldas, Minas Gerais, Brazil. Characterized by high acidity ($\text{pH} < 4$) and elevated concentrations of aluminum, manganese, and uranium, the BNF solution represents a realistic and challenging matrix for evaluating the performance of natural adsorbent materials in environmental remediation. The zeolitic material was exposed to the BNF solution for 24 hours under controlled laboratory conditions. After exposure, the samples were analyzed using Scanning Electron Microscopy with Energy Dispersive X-ray Spectroscopy (SEM-EDS) and Cathodoluminescence (CL). These techniques provided complementary information on the material's surface morphology, elemental composition, and luminescent behavior before and after interaction with the acidic solution. SEM images revealed significant morphological alterations following exposure, including the formation of a silica-rich gel-like layer on the surface of the zeolite. This transformation suggests a dissolution–reprecipitation process driven by the acidic environment. EDS spectra confirmed the presence of silicon, aluminum, and oxygen as major components, while also indicating localized incorporation of other elements, possibly introduced by the BNF solution. Spot-specific EDS analyses highlighted compositional heterogeneities between translucent gel regions and agglomerated volcanic particles, suggesting different roles in contaminant interaction and retention. CL imaging revealed the presence of luminescent centers both before and after exposure, with notable changes in emission patterns following contact with the BNF solution. These changes are likely associated with the formation of new luminescent phases or the activation of existing centers due to structural reorganization or trace element incorporation. The observed structural and optical modifications provide qualitative evidence of remediation mechanisms. These include ion exchange, surface complexation, physical adsorption, and gel-phase entrapment. The findings support the potential application of natural zeolitic materials in the treatment of acidic mine drainage and highlight the value of SEM-EDS and CL as diagnostic tools for evaluating material performance in environmental remediation contexts.

Keywords: Zeolitic Materials; Uranium Mine Effluents; Environmental Radioprotection.

1 Introduction

The environmental consequences of uranium mining have long been a subject of concern due to the persistence, mobility, and toxicity of radionuclides and associated heavy metals in ecosystems. Even after mine closure, residual effluents from uranium extraction processes can continue to leach into surrounding soils and water bodies, posing long-term risks to human health and the environment. These effluents are often acidic and enriched with a complex mixture of radioactive and non-radioactive elements, including uranium (U), thorium (Th), rare earth elements (REEs), and heavy metals such as lead (Pb) and barium (Ba). As such, the development of effective and sustainable remediation strategies is a critical priority in the field of environmental radioprotection (Eddy *et al.*, 2025).

Among the various materials investigated for environmental remediation, zeolites—microporous aluminosilicates of volcanic origin—have emerged as promising candidates due to their high cation-exchange capacity, structural stability, and selectivity for metal ions (Samolej and Franus, 2025). Their ability to adsorb and immobilize contaminants through ion exchange, surface complexation, and physical entrapment makes them particularly suitable for treating acidic mine drainage and radioactive effluents (Dinis and Fiúza, 2021). Furthermore, the structural and optical properties of zeolites can be altered upon interaction with contaminants, offering potential for their use not only in remediation but also in environmental sensing and monitoring applications (Pandya *et al.*, 2024).

This study focuses on the structural, morphological, and compositional changes in a Japanese volcanic zeolitic material after exposure to an acidic drainage solution (BNF) collected from the uranium decommissioning site in Caldas, Minas Gerais, Brazil. The BNF solution, characterized by high acidity and elevated concentrations of aluminum, manganese, and uranium, represents a realistic and challenging matrix for testing the material's remediation potential. To investigate the effects of this interaction, the zeolitic material was exposed to the BNF solution for 24 hours under controlled laboratory conditions. The samples were then analyzed using Scanning Electron Microscopy with Energy Dispersive X-ray Spectroscopy (SEM-EDS) and Cathodoluminescence (CL). These techniques provided detailed information on surface morphology, elemental composition, and the presence of luminescent centers, which may be linked to structural changes or the incorporation of trace elements.

The results offer valuable understanding of the material's behavior under acidic conditions and its potential for application in environmental remediation and optical monitoring of contaminated sites. Importantly, advanced microscopy and spectroscopic techniques not only reveal the extent of structural and compositional changes but also offer valuable understanding of the dominant remediation mechanisms at play. For instance, SEM-EDS can detect ion exchange processes through elemental substitution patterns, while morphological changes observed via SEM may indicate surface restructuring or the formation of silica gels. CL analysis can reveal the presence of luminescent centers associated with specific metal incorporations, suggesting complexation or precipitation phenomena.

Additionally, porosity and surface area measurements—though not the focus of this specific study—are critical for understanding adsorption by pore filling, where contaminants are physically confined within the zeolitic channels. These findings are essential for determining whether the material acts primarily through ion exchange, physical filtration, surface complexation, or a combination of these mechanisms. By integrating these analytical perspectives, this work contributes to a more nuanced understanding of how zeolitic materials interact with complex effluents and how their remediation potential can be optimized. The findings not only support the application of natural zeolites in environmental cleanup but also inform the design of engineered materials tailored for specific contaminant profiles. Ultimately, this research advances the strategic use of zeolitic materials in the sustainable management of radioactive waste and the protection of vulnerable ecosystems.

2 Methodology

This study employed a multi-technique approach to investigate the structural, morphological, and optical changes in a natural volcanic zeolitic material after exposure to an acidic uranium mine drainage solution. The experimental procedure involved exposing the zeolitic material to a drainage solution collected from the Nestor Figueiredo Basin (BNF), located at the Uranium Decommissioning Unit (UDC) in Caldas, Minas Gerais, Brazil (21°58'S, 46°30'W). This basin exhibits high acidity due to water percolation through nearby waste rock piles composed of sulfide minerals. The pH of the solution was measured using a Digimed DM-22 pH meter with a resolution of 0.01 at 20°C.

The characterization of the material before and after exposure was conducted using SEM-EDS and CL. These techniques were selected for their ability to provide detailed information on surface morphology, elemental composition, and luminescent behavior, allowing for the evaluation of structural transformations and potential remediation mechanisms.

2.1 Zeolitic Material and Sample Preparation

The zeolitic material used in this study was a natural volcanic rock provided by IQR Indústria Química (Santa Luzia, MG, Brazil), commercially known as HR17. For the adsorption experiments, samples of HR17 were prepared weighing 1.0 g using a semi-analytical balance and transferred into test tubes with 50 mL of either BNF. The tubes were then placed in a carousel-type tube rotator, which maintained continuous agitation for 24 hours at room temperature (25 °C). This setup ensured uniform contact between the solid and liquid phases throughout the experiment. After the exposure period, the samples were filtered using both standard filter paper to separate the solid and liquid phases.

2.2 Chemical Analysis of BNF Solution

The concentrations of metals and radionuclides in the BNF solution were determined using Tandem Inductively Coupled Plasma Mass Spectrometry (ICP-MS/MS). Analyses were performed using an Agilent 8800 instrument with helium gas employed to mitigate spectral interferences. Nitric acid was used for sample acidification and digestion, and calibration was conducted using a 2A multielement standard solution (SPEX, Horiba Jobin Yvon). These analyses were carried out at the Groupe Spectroscopies Couplages et Spéciation, Institut des Sciences Analytiques (ISA), Lyon, France.

2.3 Characterization of Zeolitic Material

The zeolitic samples were characterized before and after exposure using the following techniques:

- Scanning Electron Microscopy with Energy Dispersive X-ray Spectroscopy (SEM-EDS): Used to assess morphological changes and elemental composition. Analyses were conducted at the Laboratory of Electron Microscopy, CDTN (Centro de Desenvolvimento da Tecnologia Nuclear), Brazil.
- Cathodoluminescence (CL): Employed to identify luminescent centers and mineral phases potentially associated with uranium and other trace elements.

This integrated methodology enabled a comprehensive evaluation of the zeolitic material's behavior in contact with uranium mine effluent, supporting the identification of dominant remediation mechanisms such as ion exchange, surface complexation, and physical adsorption.

3 Results and Discussion

The results presented in this section aim to elucidate the structural, morphological, and optical changes observed in the zeolitic material following exposure to the acidic BNF solution. Through the combined

use of SEM-EDS and CL, it was possible to assess the material's surface transformations, elemental distribution, and the emergence of luminescent features associated with chemical and structural modifications.

3.1 Characterization of BNF Solution

The characterization of the chemical composition and acidity of the solution is presented in Table 1, which shows the concentrations of the main elements detected by ICP-MS/MS, along with the pH value measured for BNF sample.

Table 1: Concentration of elements of interest present in BNF solution, pH value and legislative parameters.

	BNF		Limits of Brazilian legislation¹
pH	3,67 ± 0,05		5 - 9
	<i>Concentration [$\mu\text{g L}^{-1}$]</i>	<i>RSD [%]</i>	<i>Concentration [$\mu\text{g L}^{-1}$]</i>
Al	102498,8	0,9	-
Ca	19986,3	3,1	-
Mn	58816,1	1,3	1000
Fe	242,9	1,4	15000
Zn	8497,4	0,8	5000
Se	72,1	3,9	300
Sr	1396,2	0,7	-
Pb	31,7	1,3	500
Th	103,2	3,1	-
U	4515,2	1,5	30

¹The values were obtained from CONAMA (2011), with the exception of the limit for uranium, which is a provisional recommendation updated in 2012 by the WHO.

Analysis of Table 1 reveals that BNF solution exhibit high concentrations of aluminum (above 100 $\text{mg}\cdot\text{L}^{-1}$) and manganese (greater than 55 $\text{mg}\cdot\text{L}^{-1}$). When compared to the discharge limits established by environmental regulations, the concentrations of manganese, zinc, and uranium are significantly above the permissible thresholds. These findings underscore the urgent need for effective water treatment strategies. Additionally, the high acidity of both acid mine drainage (AMD) solution ($\text{pH} < 4$) falls outside the acceptable pH range for effluent discharge, which is typically between 5 and 9. Although iron is commonly a major concern in AMD scenarios, its concentration in the analyzed sample was relatively low (less than 0.250 $\text{mg}\cdot\text{L}^{-1}$). Thorium was also detected, albeit at a lower concentration, below 0.150 $\text{mg}\cdot\text{L}^{-1}$.

3.2 SEM-EDS Analysis of the Zeolitic Sample Before Exposure

Figure 1 presents a SEM image of the zeolitic material at 132x magnification, alongside a general EDS spectrum.

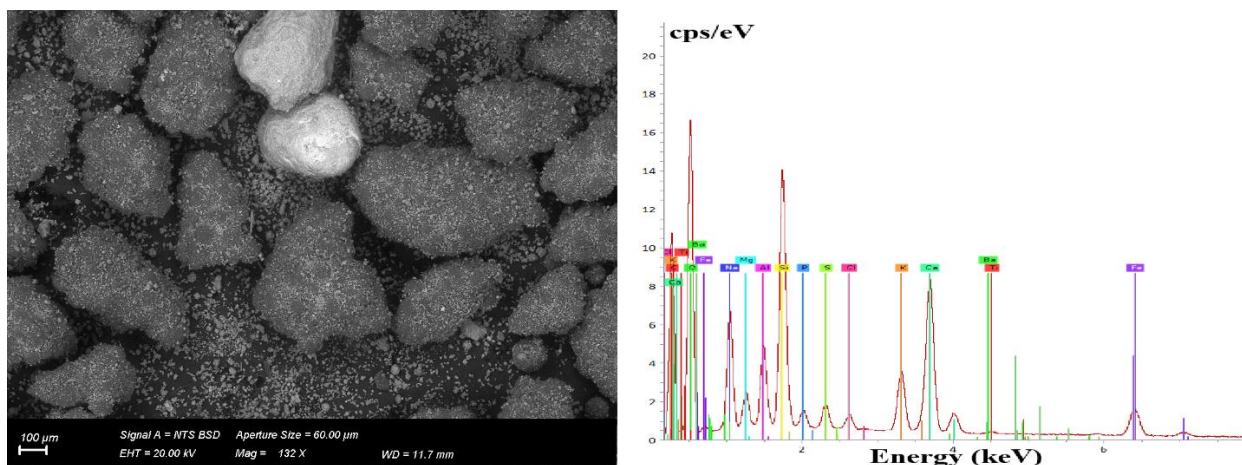


Figure 1: SEM image and EDS spectrum of the zeolitic material before exposure to acidic solutions.

The SEM image reveals a heterogeneous surface morphology with a rough texture and a high degree of porosity. Numerous small and irregularly shaped pores are distributed across the surface, indicating a large specific surface area—an essential feature for adsorption processes. The variation in brightness and contrast across the image suggests differences in either surface topography or material composition. Brighter regions may correspond to denser or more conductive mineral phases, while darker areas likely indicate more porous or less dense zones. These tonal variations motivated a more detailed compositional analysis of specific regions, which is presented in Figure 2.

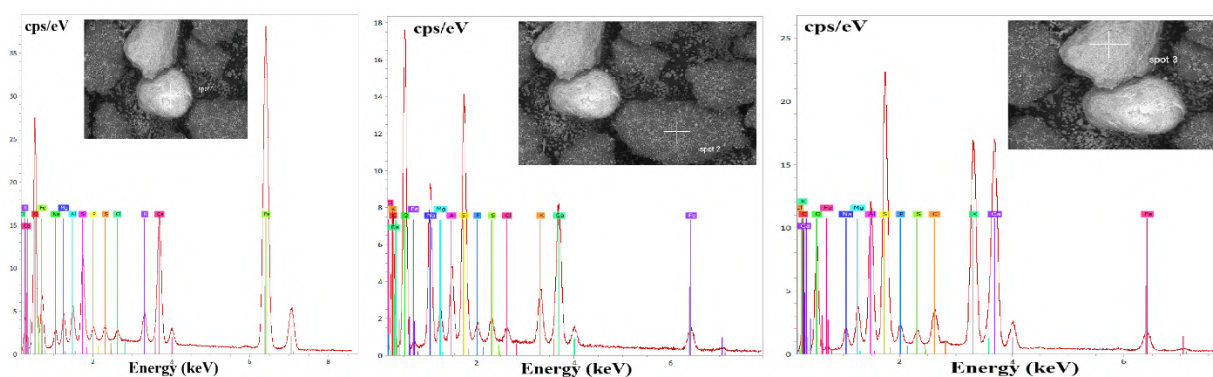


Figure 2: EDS spectra obtained from three distinct spots on the SEM image shown in Figure 1, selected based on tonal variations.

Figure 2 displays three EDS spectra acquired from distinct spots on the same SEM image shown in Figure 1. These spectra were selected based on their differing tonalities, allowing for a localized assessment of elemental composition. The spectrum from Spot 1 shows intense peaks for oxygen (O), silicon (Si), and aluminum (Al), which are characteristic of aluminosilicate frameworks typical of zeolitic materials. This region likely represents a structurally homogeneous area of the sample. In Spot 2, the spectrum also highlights O, Si, and Al as major components, but additional peaks—such as those for potassium (K) and calcium (Ca)—suggest the presence of exchangeable cations or secondary mineral phases. This indicates slight compositional heterogeneity, which may influence the material's ion exchange behavior.

Spot 3 again shows O, Si, and Al as dominant elements, but with variations in peak intensity compared to the other spots. These differences, although subtle, reinforce the presence of micro-scale heterogeneities in the elemental distribution, even within visually similar regions. The combined analysis of Figures 1 and 2 demonstrates that the zeolitic material, while compositionally consistent in its aluminosilicate base, exhibits localized variations in elemental content. These differences are relevant for understanding the material's performance in adsorption and ion exchange applications, as they may affect

the interaction with contaminants in solution. This concludes the SEM-EDS characterization of the zeolitic sample prior to exposure. The next section will present the results of cathodoluminescence (CL) analysis, which aims to investigate the presence of luminescent centers and potential structural changes associated with the material's interaction with metallic elements.

3.3 Cathodoluminescence (CL) Analysis of the Zeolitic Sample Before Exposure

Figure 3 presents a comparative analysis between SEM and CL images of the zeolitic material prior to exposure to acidic solutions. Sub-images (a) and (c) show SEM micrographs of two distinct regions of the sample, while sub-images (b) and (d) correspond to their respective CL images, acquired from the same fields of view. In sub-image (a), the SEM image reveals a heterogeneous surface composed of particles of varying sizes and morphologies. The texture is irregular, with visible porosity and contrast variations that suggest the presence of different mineral phases or densities. Brighter areas may correspond to denser or more conductive regions, while darker zones likely indicate more porous or less compacted areas. The corresponding CL image (b) highlights several luminescent centers distributed across the same region. These emissions are indicative of structural defects, trace elements, or specific mineral phases capable of emitting light under electron excitation. The spatial correlation between the luminescent zones and the morphological features observed in the SEM image suggests that the luminescence may be associated with compositional heterogeneities or the presence of activator elements such as rare earths or uranium.

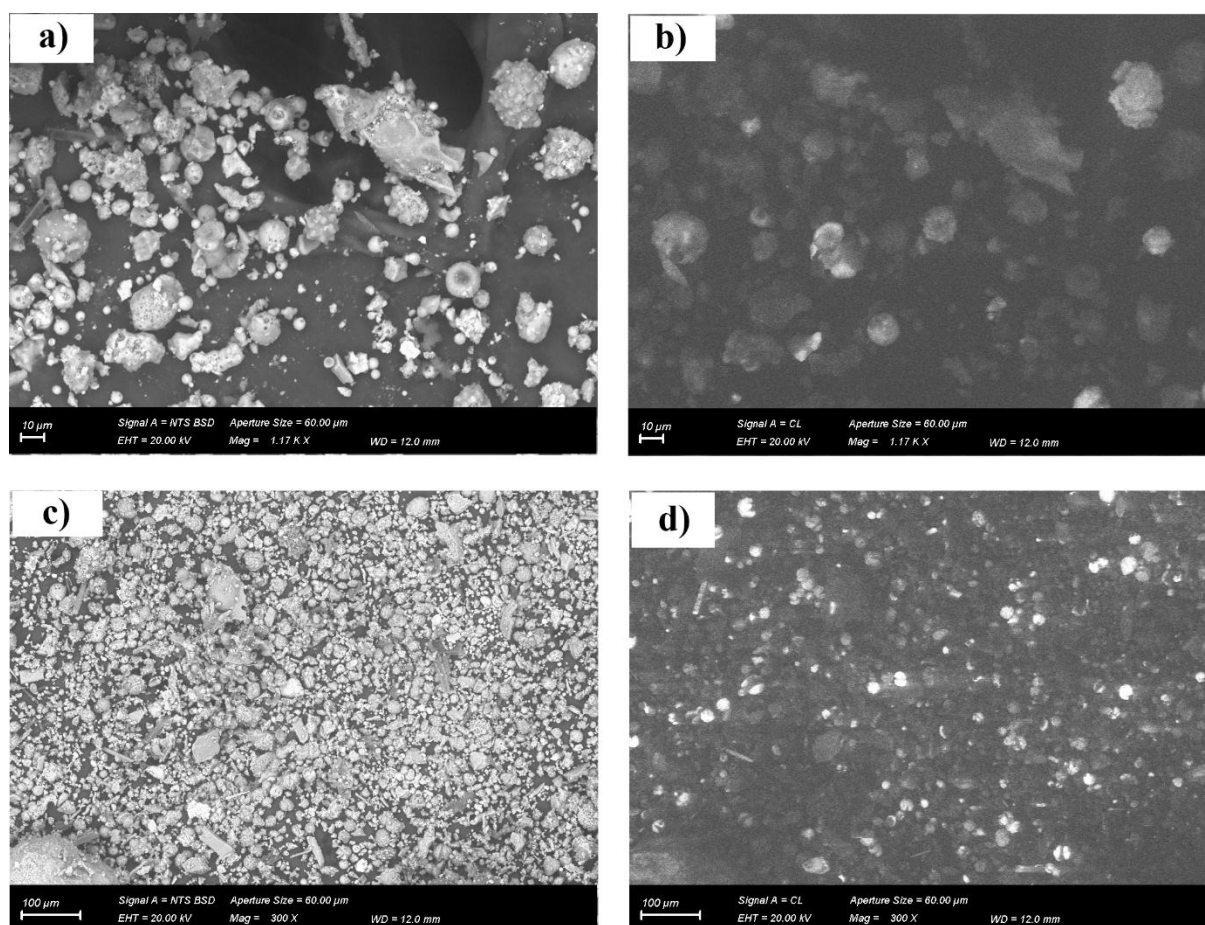


Figure 3: Comparison between Scanning Electron Microscopy (SEM) images and their corresponding Cathodoluminescence (CL) images for zeolitic material samples before exposure to acidic solutions.

Sub-image (c) shows another SEM image of the zeolitic material, this time at a lower magnification (300x), capturing a broader area with a more compact and fine-grained surface. The morphology appears

denser and less porous compared to the region shown in (a), with fewer visible voids and a more uniform texture. The corresponding CL image (d) reveals a distinct luminescence pattern, with numerous bright spots scattered across the surface. These luminescent features suggest the presence of multiple emission centers, possibly linked to different mineral inclusions or structural irregularities within the zeolitic matrix. The distribution and intensity of the luminescence in this region differ from those observed in (b), reinforcing the idea of compositional and structural variability within the material. Together, the SEM and CL analyses provide complementary information about the zeolitic material. While SEM reveals the physical and morphological characteristics, CL contributes to the understanding of the optical and compositional properties, particularly those related to trace elements and structural defects. This dual approach enhances the interpretation of the material's behavior in environmental applications, especially in scenarios involving metal adsorption and optical monitoring.

3.4 SEM, CL, and EDS Analysis After Exposure to BNF Solution

Figure 4 presents the results of the zeolitic material after 24 hours of exposure to the BNF acidic drainage solution. Sub-image (a) shows a Scanning Electron Microscopy (SEM) image, which reveals significant morphological changes compared to the unexposed sample. The surface appears partially coated with a thin, amorphous layer, and the original granular texture of the zeolite is less defined. This transformation is consistent with the formation of a gel-like structure, likely resulting from the dissolution and reprecipitation of silica and other components under acidic conditions.

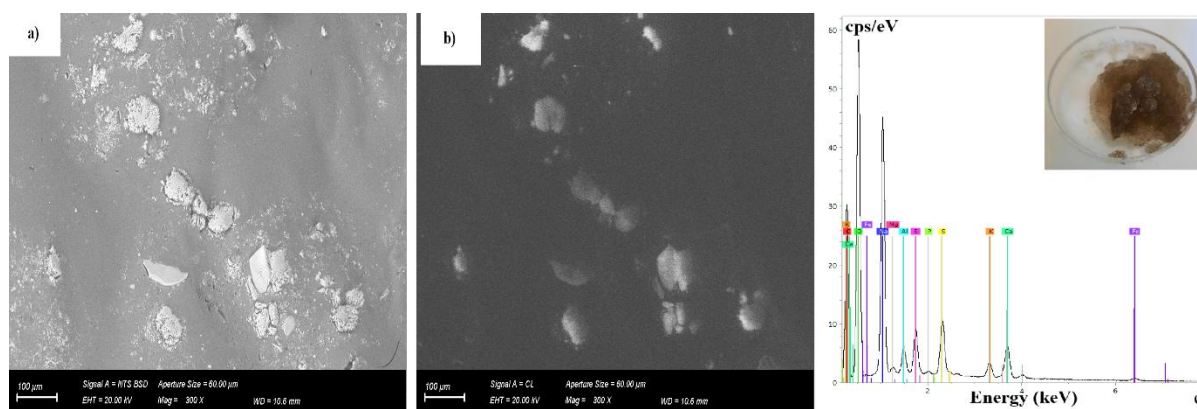


Figure 4: SEM (a), CL (b), and EDS (c) analyses of the zeolitic material after 24 hours of exposure to the BNF acidic solution.

Sub-image (b), the corresponding Cathodoluminescence (CL) image, displays a distinct luminescence pattern with several bright emission centers. These luminescent zones are spatially correlated with the altered surface regions observed in the SEM image, suggesting that the interaction with the BNF solution led to the formation of new luminescent phases or the activation of existing centers through structural modification or incorporation of trace elements. Sub-image (c) presents an EDS spectrum acquired from the altered region, with an inset showing the gel-like structure in greater detail. The spectrum confirms the presence of silicon (Si), aluminum (Al), and oxygen (O) as major components, consistent with the zeolitic matrix. However, additional peaks—potentially corresponding to elements introduced or concentrated by the BNF solution—suggest chemical changes in the surface composition. These may include elements such as sulfur (S), calcium (Ca), or uranium (U), depending on the local chemistry of the gel. To further investigate the origin of the luminescent features observed in the CL image, spot-specific EDS analyses were performed and are presented in Figure 5.

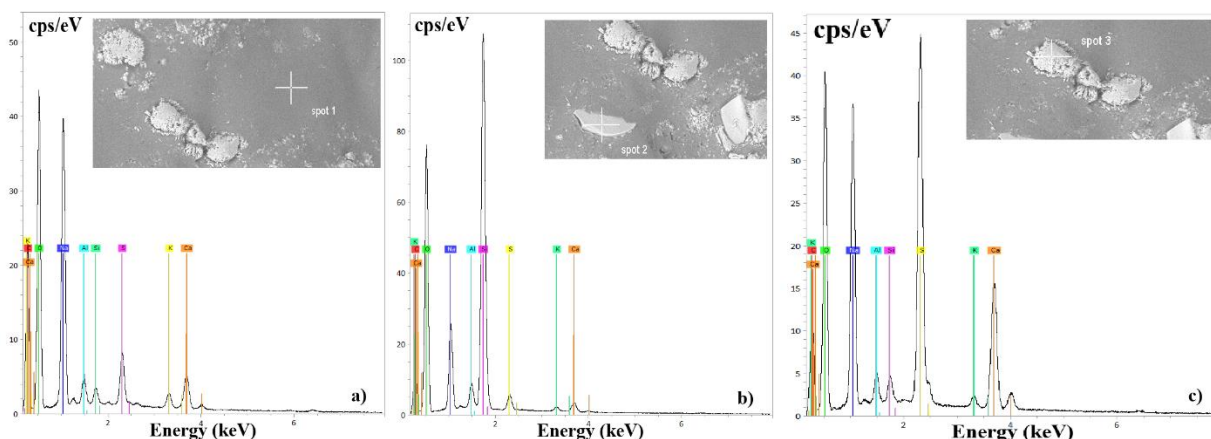


Figure 5: EDS spectra from three distinct spots on the zeolitic sample after exposure to the BNF solution and drying, forming a thin film.

This figure includes three EDS spectra from distinct regions of the sample after drying, which formed a thin film. Spot 1, shown in sub-image (a), corresponds to a translucent film region, while spots 2 and 3, shown in sub-images (b) and (c), are located in areas with agglomerated volcanic rock particles embedded in the final film. The comparison of these spectra will help clarify the elemental composition of the luminescent centers and the role of the gel-like phase in modifying the optical properties of the material.

To further investigate the luminescent centers observed in the CL image of Figure 4 and to understand the chemical nature of the altered surface, Figure 5 presents EDS spectra from three distinct spots on the zeolitic sample after 24 hours of exposure to the BNF solution and subsequent drying. The drying process led to the formation of a thin film, which exhibited both translucent and particulate regions. Sub-image (a) shows the EDS spectrum from Spot 1, located in a translucent region of the film. The corresponding SEM inset reveals a smooth, continuous surface, likely dominated by the gel-like phase observed previously. The EDS spectrum for this spot indicates a composition rich in silicon (Si) and oxygen (O), with lower but detectable levels of aluminum (Al). This suggests that the translucent film is primarily composed of a silica-rich gel, possibly formed by the leaching and reprecipitation of silicate species from the zeolitic matrix under acidic conditions. The absence or low intensity of heavier elements supports the interpretation of this region as a chemically simplified, amorphous phase.

Sub-images (b) and (c) correspond to Spots 2 and 3, respectively, both located in regions where agglomerated volcanic rock particles are embedded within the thin film. The SEM insets show more granular textures, with visible microstructures resembling the original zeolitic material. The EDS spectra for these spots reveal a more complex composition, with prominent peaks for Si, Al, and O, as well as additional elements such as potassium (K), calcium (Ca), and possibly trace metals introduced by the BNF solution. These regions retain more of the original mineralogical complexity of the zeolite and may act as nucleation sites for secondary phases or as reservoirs for adsorbed contaminants. The comparison between the three spots highlights the chemical heterogeneity of the final film. While the translucent region (Spot 1) is dominated by a silica-rich gel, the particulate regions (Spots 2 and 3) preserve the multi-elemental character of the original material. This spatial variation in composition is consistent with the CL emission patterns observed in Figure 4 and supports the hypothesis that both structural reorganization and selective adsorption occurred during the interaction with the BNF solution.

3.5 EDS-Based Assessment of Remediation Pathways in Zeolitic Materials

EDS spectra reveal that, prior to exposure, the zeolitic material exhibited prominent peaks for oxygen (O), silicon (Si), and aluminum (Al), consistent with its aluminosilicate framework. Additionally, minor but distinct peaks for potassium (K) and calcium (Ca) were observed, indicating the presence of these exchangeable cations within the zeolitic structure. After 24 hours of contact with the BNF solution, the

EDS spectrum shows a notable reduction in the intensity of K and Ca peaks, while the major framework elements (Si, O) remain relatively stable. This reduction strongly suggests that ion exchange occurred, with K^+ and Ca^{2+} being released from the zeolite and replaced by cations present in the acidic solution—potentially H^+ , Na^+ , or even trace levels of UO_2^{2+} .

In some spectra, the appearance or enhancement of peaks corresponding to sulfur (S) or other elements from the BNF solution may indicate adsorption or surface complexation processes in addition to ion exchange. These findings support a multi-mechanistic remediation behavior, where ion exchange is predominant, but not exclusive. This observation is supported by previous studies, such as Reis *et al.* (2024), which demonstrated the effectiveness of the same material in removing various metals, including uranium, thorium, cerium, lanthanum, and cobalt, with adsorption rates exceeding 90% in some cases. The adsorption capacity was directly related to the concentration of the adsorbent, with $3.5\text{ g}\cdot\text{L}^{-1}$ identified as the optimal dosage for most of the elements analyzed.

This compositional shift, combined with the morphological evidence of surface restructuring and gel formation, reinforces the interpretation that the zeolitic material undergoes both chemical and physical transformations that enhance its capacity to immobilize contaminants. The loss of exchangeable cations and the formation of a silica-rich gel layer suggest a dynamic interaction between the material and the solution, where structural degradation and reorganization contribute to the overall remediation process.

4 Conclusions

The results of this study demonstrate that the natural volcanic zeolitic material investigated exhibits significant structural and optical transformations upon exposure to acidic uranium mine drainage, specifically the BNF solution. Through detailed SEM-EDS and CL analyses, it was possible to identify key features that support the material's potential for environmental remediation and monitoring applications. Morphological observations revealed the formation of a silica-rich gel layer on the surface of the zeolitic particles after 24 hours of contact with the acidic solution. This gel is likely the result of a dissolution–reprecipitation mechanism, in which the acidic environment promotes the leaching of silicon and aluminum from the zeolitic framework, followed by the reorganization and condensation of silanol species into an amorphous silica network. The resulting gel structure contributes to surface reconfiguration, potentially reducing crystallinity while increasing surface area and modifying porosity. These changes are critical, as they may enhance the material's capacity for physical adsorption and entrapment of contaminants.

The EDS spectra confirmed the persistence of the aluminosilicate matrix while also indicating localized incorporation of elements potentially derived from the solution. The spatial heterogeneity observed in spot analyses suggests that different regions of the material may engage in distinct remediation processes, including ion exchange, surface complexation, and physical adsorption. The translucent gel regions, in particular, appear to act as passive barriers or diffusion media, while the more crystalline zones retain their ion-exchange functionality. Cathodoluminescence imaging revealed the presence of luminescent centers both before and after exposure, with notable changes in emission intensity and distribution. These centers are likely associated with structural defects, trace elements such as rare earths or uranium, or newly formed mineral phases. The persistence and evolution of these luminescent features suggest that CL can serve not only as a diagnostic tool for material characterization but also as a potential method for in situ environmental monitoring, enabling the detection of contaminant interactions through optical signals.

Taken together, the evidence supports a hybrid remediation mechanism for the zeolitic material, combining ion exchange, surface complexation, and gel-phase entrapment. This multifaceted behavior enhances the material's versatility and effectiveness in treating complex effluents such as acid mine drainage. Moreover, the structural integrity of the material, although partially altered, remains sufficient to support its application in engineered systems. These findings position the studied zeolitic material as a promising candidate for integration into remediation engineering projects, such as permeable reactive

barriers, filtration units, or column-based treatment systems. Its dual functionality—remediation and optical sensing—opens new possibilities for the development of smart materials capable of both treating and monitoring contaminated environments in real time.

Acknowledgements

The authors would like to thank the Granioter, the Technological Hub for Advanced Materials and Strategic Minerals, based at CDTN, for supporting the SEM analyses and enabling access to advanced characterization infrastructure. This work was financially supported by Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES) and Fundação de Amparo à Pesquisa do Estado de Minas Gerais (FAPEMIG).

References

- Dinis M. D. L., & Fiúza A. (2021). Mitigation of uranium mining impacts—a review on groundwater remediation technologies. *Geosciences*, 11(6): 250.
- Eddy N. O., Igwe O., Eze I. S., Garg, R., Akpomie K., Timothy C., ... & Paktin, H. (2025). Environmental and public health risk management, remediation and rehabilitation options for impacts of radionuclide mining. *Discover Sustainability*, 6(1): 209.
- Pandya T., Patel, S., Kulkarni M., Singh Y. R., Khodakiya A., Bhattacharya S., & Prajapati B. G. (2024). Zeolite-based nanoparticles drug delivery systems in modern pharmaceutical research and environmental remediation. *Heliyon*, 10(16).
- Reis M. O., Palmieri H. E. L., Souza W.; Sousa R. G.; Batista A. S. M. (2024) Aplicação de adsorvente natural para separação de metais em drenagem ácida de mina de urânio. In *Semana Nacional de Engenharia Nuclear e da Energia e Ciências das Radiações – VII SENCIR, Belo Horizonte, Brasil, Novembro 12-14*.
- Samolej K., & Franus W. (2025). Highly efficient radium removal from mine wastewater with fly ash NaP1 zeolite. *Separation and Purification Technology*, 356: 129900.

Survey of Radiation Exposure from Portable X-Ray Fluorescence

F.S. Dias^{1*}, D.N.S. Giovanni¹, A. J.Potiens Jr. ¹, O. Rodrigues Jr. ¹, M.P.A. Potiens¹ and C. B. Zamboni¹

¹ Instituto de Pesquisas Energéticas e Nucleares – IPEN/USP. Av.Prof. Lineu Prestes, 2242, São Paulo, Brazil

* fabio.sabara.dias@gmail.com

Abstract

This study performed a radiometric survey of an experimental portable X-ray fluorescence (pXRF) system to assess radiation exposure and ensure operational safety. The methodology involved the use of an Amptek® X-123SDD pXRF spectrometer, with a silver anode X-ray tube and a silicon detector, and a Radcal® ionization chamber for measurements. Measurements were performed at different distances and positions to simulate operator exposure, under operating conditions of 50 kV and 200 μ A, and 30 kV and 5 μ A. The results demonstrate the inherent radiological safety of the equipment. Analysis along the primary beam validated the principle of the Inverse Square Law of Distance, with a dose rate reduction to 5.8% of the original value when increasing the distance from 21 cm to 100 cm. Furthermore, the tube head shielding proved highly effective. Leakage radiation rates at the sides and rear were significantly low, ranging from 0.01% to 0.04% of the primary beam intensity. This confirms that the operator is exposed to minimal radiation levels when positioned next to or behind the device. In conclusion, the results attest to the safety of the pXRF system in its routine use configurations and under the tested conditions. The effectiveness of the distance and shielding confirms adherence to the principles of radiation protection optimization. For complete validation, future investigations into the spatial distribution of radiation in other sample geometries and the assessment of accumulated dose in prolonged use scenarios are recommended.

Keywords: Radiometric survey; X-ray fluorescence; radiation metrology

1 Introduction

Energy-dispersive X-ray fluorescence (EDXRF) is a well-established and widely employed analytical technique for materials characterization. Its relevance lies in its ability to identify and quantify elements across diverse matrices, based on the principle of X-ray emission to induce the characteristic fluorescence of each element present in the sample (BECKHOFF *et al*, 2007).

The advent of portable X-ray fluorescence (pXRF) equipment represents a significant advancement in the field of EDXRF, providing a non-destructive and convenient approach to elemental analysis in a vast range of applications (MARGUI & GRIEKEN, 2013; LEMIERE, 2018). These compact devices operate by exciting the atoms of the material via X-ray emission, resulting in the characteristic secondary fluorescence of each element (WEST *et al*, 2010). The versatility of pXRF is evidenced by its broad spectrum of applications, extending from the analysis of metals in industrial scrap to the identification of residues in biological samples, distinguishing itself by its portability and ease of field use, enabling rapid and accurate analyses in various situations (AMPTEK, 2016).

Beyond its versatility, operational safety is a paramount consideration in the use of pXRF equipment. These devices are designed with inherent safeguards to minimize the risks associated with radiation exposure, ensuring the safety of both operators and the general public. Operators undergo training in safe operating practices, including the proper use of protective barriers and regular monitoring of radiation exposure. Strict compliance with radiation safety regulations and the implementation of appropriate control measures ensure that pXRF equipment offers a safe and reliable approach to elemental analysis in a variety of scenarios (WEST *et al*, 2010).

Thus, pXRF equipment represents a valuable tool for elemental analysis across a wide spectrum of applications. Its portability, ease of use, and integrated safety measures make it ideal for field use, providing fast and accurate results without compromising the safety of the operator or the surrounding environment.

Recent studies indicate that the application of pXRF for multielement analysis of blood samples shows great promise for clinical practice (SOUZA *et al*, 2019). Despite the low levels of radiation emitted by these devices, caution is essential in their handling. Therefore, the objective of this study is to perform a radiometric survey of an experimental setup for pXRF analysis of blood samples

2 Materials and Methods

X-ray fluorescence analysis was performed at the Dosimeter Calibration Laboratory of the Institute for Energy and Nuclear Research. This experiment involved the use of an Amptek® X-123 SDD X-ray spectrometer, which has a silicon drift detector (SDD) with an area of 25 mm² and a thickness of 500 µm, equipped with a 12.5 µm beryllium (Be) window, coupled to an X-ray tube with a silver (Ag) anode, as illustrated in Figure 1.



Figure 1: Mini- X-Ray Tube and SDD detector

The experimental setup utilized an Amptek® X-123SDD X-ray spectrometer, equipped with a semiconductor detector, and an acrylic support designed to hold Whatman N.41 papers. Radiometric measurements were performed using a calibrated Radcal® model 10x5 ionization chamber. This specific model was chosen due to its sensitive volume, making it suitable for both beam dosimetry and radiation protection assessments.

The measurements were carried out under two conditions: for the maximum current and voltage values of the equipment, 50 kV and 200 µA, and for the current and voltage values of the blood sample analysis protocol, 30 kV and 5 µA. For each condition, measurements were performed with the ionization chamber positioned at varying distances from the X-ray source's focal spot: 21 cm, 35 cm, 100 cm, and 200 cm. Additionally, the ionization chamber's positioning relative to the source was varied to simulate operator exposure scenarios during equipment use.



Figure 2: Setup for radiometric survey of the pXRF spectrometer

Using the ISO 4037 standard as a reference, corrections were necessary for the environmental conditions, factor $f_{T,P}$, which takes into account temperature and pressure, as shown in equation 1. The relative humidity of the air was not corrected, but is controlled in the range of 20 to 70%.

$$f_{T,P} = \left(\frac{T + 273,15}{293,15} \right) \times \left(\frac{101,325}{P} \right) \quad (1)$$

Subsequently, the electrical charge values were converted into Air Kerma terms, using the following equation (ISO, 1999):

$$K = N_K \times C \times f_{T,P} \quad (2)$$

Where K is the Kerma in the air (air kerma), N_K is the calibration coefficient of the ionization chamber, C is the electrical charge obtained by the electrometer and $f_{T,P}$, temperature and pressure correction factor.

3 Results

The obtained results demonstrate that in the vicinity of the pXRF equipment, with the exception of the direct beam region, the air kerma rate values are consistently low, approaching the detector background level. This finding indicates minimal secondary radiation scatter into the surrounding environment.

Figure 3 schematically illustrates the positions where the dosimeter was placed for the measurements. Complementarily, Table 1 presents the detailed values of the air kerma rates (in Gy/s) recorded at each of these positions, under the operational conditions specified in the Materials and Methods section.

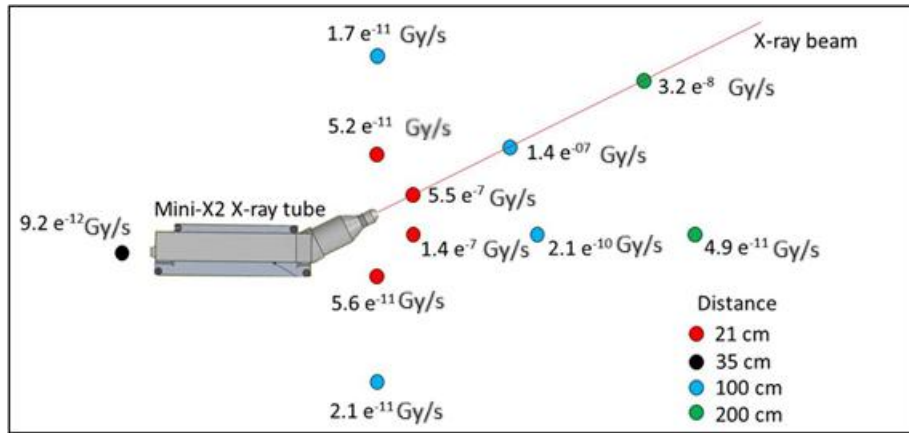


Figure 3: Radiometric survey of the X-ray spectrometer

Table 1: Radiometric survey of the X-ray spectrometer

Position	Air Kerma Rate (Gy/s) for each Detector Source Distance				Uncertainties * (%)
	21 cm	35 cm	100 cm	200 cm	
Direct beam	5.5×10^{-7}	-----	1.4×10^{-7}	3.2×10^{-8}	3.0
Behind pXRf	-----	9.2×10^{-12}	-----	-----	3.0
Front of pXRf	1.4×10^{-7}	-----	2.1×10^{-10}	4.9×10^{-11}	3.0
Next to pXRf (direct beam)	5.2×10^{-11}	-----	1.7×10^{-11}	-----	3.0
Next to pXRf (opposite to direct beam)	5.6×10^{-11}	-----	2.1×10^{-11}	-----	3.0

* The expanded uncertainty (U) of the reported measurement is stated as the standard uncertainty of measurement multiplied by the coverage factor $k = 2.00$.

The radiometric survey of the X-ray spectrometer, using the Mini-X2 tube, allowed fundamental conclusions to be drawn for the safe operation of the equipment. Analysis along the primary beam axis robustly validated the Inverse Square Law of Distance as the predominant factor for dose attenuation. Increasing the distance from 21 cm to 100 cm resulted in a dose rate reduction to 5.8% of the original value, a figure that is in excellent agreement with the theoretically calculated 4.4%. This result confirms that distance is the most effective radiation protection measure against the primary beam.

The remaining points outside the central beam revealed the effectiveness of the tube head shielding. The dose rates measured at the sides and rear of the equipment are orders of magnitude lower than those of the main beam. For example, in the lateral region, the values are on the order of 0.01% of the value recorded in front of the collimator at the same distance, while behind the tube, this value is approximately 0.04%. This significant reduction demonstrates that the shielding is highly effective in containing both escaped and transmitted radiation, directing the beam in a safe and confined manner.

4 Conclusions

Preliminary measurements demonstrated the inherent radiation safety of the X-ray fluorescence spectrometry (pXRf) system under investigation. The measurements validated two fundamental protection principles: the effectiveness of the distance, with a clear attenuation of the primary beam dose following the Inverse Square Law (a reduction of approximately 6% of the value at 21 cm when moving away to 100 cm), and the effectiveness of the head shielding.

Leakage radiation levels to the sides and back were found to be in the order of 0.01% to 0.04% of the primary beam intensity, confirming that the operator is exposed to low radiation levels, approaching the levels found in dosimeter leakage radiation tests, when positioned behind or to the side of the device during operations.

These results attest to the operational safety of the equipment in routine use configurations and under the conditions tested, ensuring compliance with the optimization principles. However, to ensure full system reliability and comprehensive validation under all possible operating conditions, additional testing is recommended. Future investigations should focus on a more detailed analysis of the spatial distribution of radiation across various sample geometries and distances, as well as on assessing the accumulated dose in prolonged and continuous use scenarios, thus closing any remaining gaps in the apparatus's safety characterization.

Acknowledgements

The authors would like to thank the partnership with the National Council for Scientific and Technological Development (CNPq) and the São Paulo Research Foundation (FAPESP) for their financial support.

References

AMPTek. (2016) Mini-X User Manual Rev D2.

Beckhoff, B., Kanngießer, B., Langhoff, N., Wedell, R., & Wolff, H. (Eds.). (2007). *Handbook of practical X-ray fluorescence analysis*. Springer Science & Business Media.

Lemiere, B. (2018). A review of pXRF (field portable X-ray fluorescence) applications for applied geochemistry. *Journal of Geochemical Exploration*, 188, 350-363.

International Organization for Standardization ISO (1999). *X and gamma reference radiation for calibrating dosemeters and dose rate meters and for determining their response as a function of photonenergy*, ISO 4037-3.

Margui, E., & Van Grieken, R. (2013). *X-ray fluorescence spectrometry and related techniques: an introduction*. Momentum press.

Souza, C. S., Zamboni, C. B., Giovanni, D. N., & Metairon, S. (2019, January). Use of X-ray Fluorescence mini-spectrometer as an alternative for dialysate clinical practice. In *Proceedings of the INAC 2019: international nuclear atlantic conference. Nuclear new horizons: fueling our future* (pp. 2829-2835).

West M, Ellis AT, Potts PJ, Streli C, Vanhoof C, Wegrzynek D, et al. *Atomic spectrometry update—X-ray fluorescence spectrometry*. *Journal of Analytical Atomic Spectrometry*. 2010;25(10):1503-45.



Transfer Factors of Natural Radionuclides in Organic Foods from Rio de Janeiro, Brazil

Mirella Maturo da Cruz^{1*}, Leandro Barbosa da Silva², Jonathan Oliveira dos Santos³, Ademir Xavier da Silva⁴

¹ Universidade Federal do Rio de Janeiro, Programa de Engenharia Nuclear, CEP 21941-914, Rio de Janeiro, Brasil

² Universidade Federal do Rio de Janeiro, Programa de Engenharia Nuclear, CEP 21941-914, Rio de Janeiro, Brasil

³ Universidade Federal do Rio de Janeiro, Programa de Engenharia Nuclear, CEP 21941-914, Rio de Janeiro, Brasil

⁴ Universidade Federal do Rio de Janeiro, Programa de Engenharia Nuclear, CEP 21941-914, Rio de Janeiro, Brasil

* mirellamaturo@poli.ufrj.br

Abstract

Assessing the transfer of radionuclides from soil to food is essential for understanding bioaccumulation mechanisms and estimating health risks from ingestion. The transfer factor (TF) is a widely used tool in radiological safety studies, as it empirically synthesizes the dynamic behavior of radionuclides in the soil–plant system. It is strongly influenced by variables such as bioavailability, soil physicochemical characteristics, plant morphology, and local environmental conditions. This study aimed to determine the transfer factors of naturally occurring radionuclides ^{226}Ra , ^{228}Ra , ^{228}Th , and ^{40}K into various edible and non-edible parts of organic fruits and vegetables cultivated in the state of Rio de Janeiro, Brazil. This tropical region, marked by a wide diversity of soils, remains underrepresented in international datasets. Samples of soil and plant tissues were collected from crops including chayote, lemon, papaya, and banana. Gamma spectrometry was performed using a high-resolution HPGe detector, and transfer factors were calculated as the ratio of activity concentration in the sample ($\text{Bq}\cdot\text{kg}^{-1}$ dry weight) to that in the corresponding soil. The highest TF values were observed for ^{40}K , particularly in stems and leaves, with a notable value in yam leaf ($\text{TF} = 16.86$), reflecting the essential physiological role of potassium in plants. In contrast, ^{226}Ra , ^{228}Ra , and ^{228}Th exhibited low mobility, with TFs generally below 1, indicating strong soil retention and limited transfer through the food chain. These results are in line with previous findings reported for other regions. The study highlights the importance of localized assessments in organic production systems, providing critical data for improving radionuclide transport models and for estimating internal doses from ingestion. This study helps fill existing knowledge gaps in tropical environments and supports the development of public policies related to food radiological safety and sustainable agriculture.

Keywords: Transfer Factor; Natural Radionuclides; Gamma Spectrometry; Radiological Risk.

1 Introduction

Naturally occurring radionuclides such as ^{226}Ra , ^{228}Ra , ^{228}Th , and ^{40}K are present in soils due to geogenic processes and can enter the food chain through root uptake by crops. This pathway is of increasing concern in food safety and environmental radioprotection, particularly in tropical agricultural systems where soil diversity and climatic conditions may enhance or inhibit radionuclide mobility International Atomic Energy Agency (2010b); Kabata-Pendias (2010).

The soil-to-plant transfer factor (TF) is a key parameter in radiological risk assessment. It is defined as the ratio between the activity concentration of a radionuclide in plant tissue and that in the corresponding soil (dry weight basis) International Atomic Energy Agency (2010b). TFs are influenced by multiple variables, including soil pH, texture, cation exchange capacity, organic matter content, and the physiological role of the element in plants Carini (2001); da Costa Dantas et al. (2019). For instance, potassium-40 (^{40}K), being an essential nutrient, is typically more mobile than actinides such as ^{226}Ra or ^{228}Th , which tend to bind strongly to soil particles.

Although comprehensive datasets of TFs exist for temperate regions and conventional crops, there is a notable lack of data for tropical environments and organic farming systems, where reduced synthetic input and high organic content may alter radionuclide bioavailability Santos et al. (2002). Moreover, existing literature often generalizes plant tissues without distinguishing between edible and inedible structures, which is crucial for accurately estimating ingestion doses.

This study aimed to quantify the transfer of ^{226}Ra , ^{228}Ra , ^{228}Th , and ^{40}K from soil to different tissues of tropical fruits and vegetables cultivated organically in Rio de Janeiro, Brazil. Specifically, we analyzed edible parts (fruits and pulp) and inedible vegetative parts (leaves, stems, and peels) of crops including banana, papaya, lemon, and chayote. The goal was to generate region-specific data to improve radioecological models and support public health assessments in tropical food systems.

2 Materials and Methods

Composite samples of soil and plant material were collected from organic farming sites in the state of Rio de Janeiro, Brazil. For each crop, different plant parts were sampled, including edible portions (such as fruits or pulp) and non-edible vegetative structures (including peels, stems, and leaves), depending on plant morphology. Soil samples were taken from the root zone (0–20 cm) and transported in sealed containers to the laboratory. A total of 44 plant samples and 10 soil samples were analyzed.

In the laboratory, all samples were oven-dried at 100 °C for 48 hours to remove moisture content. They were then homogenized, ground, and packed in 120 mL or 500 mL polyethylene containers, depending on sample volume. Each container was sealed with PVC film and stored for 45 days to ensure secular equilibrium between radon isotopes and their decay products, following IAEA recommendations, International Atomic Energy Agency (2010a).

2.1 Gamma Spectrometry and Efficiency Calibration

The activity concentrations of ^{226}Ra , ^{228}Ra , ^{228}Th , and ^{40}K were determined by high-resolution gamma spectrometry using a high-purity germanium (HPGe) coaxial N-type detector (Cannberra GC3020), with 30% relative efficiency. The system was housed in a lead shield with inner linings of cadmium and copper to minimize background radiation and X-ray fluorescence. Counting times ranged from 28,800 to 48,000 seconds. Background spectra were acquired using empty containers identical to those used for the samples and subtracted from all measurements.

Efficiency calibration was performed using LabSOCS software (Canberra), which employs Monte Carlo-based modeling that accounts for each sample's geometry and composition. All analytical procedures followed previously validated protocols in accordance with current scientific standards Silva et al. (2021); Garc ez et al. (2019). The Minimum Detectable Activity (MDA) was calculated according to the Currie method, and uncertainties were estimated by standard error propagation involving peak area, efficiency, counting time, and gamma emission probability Barbosa da Silva et al. (2025).

2.2 Activity Concentration Determination

The activity concentration (A) was calculated using the following expression:

$$A = \frac{N}{\epsilon \cdot P_\gamma \cdot t \cdot m} \quad (1)$$

where N is the net peak area (counts), ϵ is the detector efficiency at the corresponding gamma energy, P_γ is the gamma emission probability, t is the counting time (s), and m is the dry mass of the sample (kg).

The gamma energies used to identify the radionuclides were: 609.3 keV and 351.9 keV for ^{226}Ra (via ^{214}Bi and ^{214}Pb), 911.2 keV and 583.2 keV for ^{228}Ra and ^{228}Th (via ^{228}Ac and ^{208}Tl), and 1460.8 keV for ^{40}K (direct emission).

2.3 Transfer Factor Calculation

Transfer factors (TFs) were calculated as the ratio between the activity concentration in plant tissue ($\text{Bq} \cdot \text{kg}^{-1}$ dry weight) and the corresponding soil ($\text{Bq} \cdot \text{kg}^{-1}$ dry weight), according to Eq. 2:

$$TF = \frac{C_{\text{plant}}}{C_{\text{soil}}} \quad (2)$$

where C_{plant} and C_{soil} are the activity concentrations of a given radionuclide in plant and soil samples, respectively.

3 Results and Discussion

The results obtained demonstrate distinct behaviors among the radionuclides investigated, highlighting the importance of jointly considering soil physicochemical characteristics, plant physiology, and the analyzed compartments. Table 1 presents the activity concentrations and corresponding transfer factors (TFs), while Figure 1 summarizes the mean TF values by radionuclide, facilitating direct comparison among elements and providing a clearer visualization of overall trends.

The highest TF values were observed for ^{40}K , particularly in non-edible vegetative structures such as yam leaves ($TF = 16.86$) and stems ($TF = 15.32$), papaya stems ($TF = 16.57$), and banana peels ($TF = 6.07$). The elevated mobility of potassium is related to its essential physiological role in plants, where it regulates osmotic balance, sugar transport, and protein synthesis, thus favoring its root uptake and efficient translocation to various tissues (International Atomic Energy Agency, 2010b; Kabata-Pendias, 2010). Similar behavior has been reported in other tropical environments by Ribeiro et al. (2018) and Haque and Ferdous (2017), where ^{40}K accumulation was especially prominent in leaves and stems.

Table 1: Activity concentrations (Bq·kg⁻¹) and transfer factors (TFs) for natural radionuclides in organic food samples.

Sample	Part	Ra-226	TF Ra-226	Ra-228	TF Ra-228	Th-228	TF Th-228	K-40	TF K-40
AVOCADO	SEED	5.81	0.10	14.05	0.27	5.10	0.10	631.26	2.56
	PEEL	15.95	0.28	49.76	0.97	12.30	0.25	124.79	5.07
	LEAF	10.50	0.19	26.35	0.51	5.79	0.12	443.71	1.80
	PULP	3.86	0.07	15.33	0.30	6.14	0.13	367.33	1.49
	SOIL	56.28		51.37		48.67		246.22	
BANANA	PEEL	7.93	0.22	1.08	0.02	7.86	0.31	1260.28	6.07
	STEM	12.53	0.35	2.75	0.5	17.48	0.68	1353.73	6.52
	PULP	2.60	0.07	0.03	0.00	6.58	0.26	2067.87	9.95
	SOIL	35.51		52.01		25.53		207.78	
CHAYOTE	PEEL	4.66	0.07	3.17	0.04	5.65	0.12	1231.92	4.14
	PULP	5.70	0.09	1.22	0.02	8.04	0.18	1006.49	3.38
	SOIL	65.66		74.01		45.68		297.75	
COCONUT	PEEL	7.13	0.31	14.72	0.29	4.93	0.13	338.54	1.71
	STEM	9.69	0.43	3.66	0.07	25.31	0.67	718.13	3.62
	LEAF	8.90	0.39	24.79	0.49	13.12	0.35	324.84	1.64
	PULP	5.28	0.23	3.99	0.08	3.13	0.08	396.60	2.00
	SOIL	22.68		50.36		37.95		198.40	
LEMON	PEEL	1.21	0.02	11.89	0.21	4.50	0.11	248.79	1.36
	PULP+SEED	2.28	0.05	15.91	0.28	7.35	0.18	311.12	1.71
	SOIL	49.92		57.54		40.29		182.47	
ORANGE	PEEL	2.40	0.03	12.43	0.17	3.39	0.06	218.58	0.81
	STEM	4.64	0.05	16.20	0.22	7.24	0.12	162.39	0.60
	LEAF	7.65	0.09	21.24	0.29	10.03	0.17	282.43	1.05
	PULP	6.54	0.07	16.13	0.22	6.87	0.11	649.90	2.40
	SOIL	88.06		73.55		60.21		270.26	
	JUICE	1.83	0.02	5.51	0.07	4.18	0.07	39.05	0.14
PAPAYA	SEED	0.20	0.01	5.07	0.10	2.42	0.05	440.19	2.44
	PEEL	2.02	0.05	0.68	0.01	6.73	0.15	1440.76	7.98
	STEM	18.50	0.50	3.81	0.08	72.51	1.61	2989.59	16.57
	LEAF	1.84	0.05	1.82	0.04	6.08	0.13	549.44	3.04
	PULP	2.68	0.07	0.05	0.00	7.50	0.17	1208.88	6.70
	SOIL	36.84		50.21		45.04		180.45	
PUMPKIN	STEM	9.69	0.27	41.26	0.80	4.95	0.20	619.45	2.99
	LEAF	6.95	0.20	34.80	0.67	5.15	0.20	714.64	3.44
	PULP	6.23	0.18	27.20	0.52	6.00	0.24	1462.61	7.05
	SOIL	35.43		51.89		25.36		207.48	
TOMATO	STEM	5.04	0.14	40.36	0.81	8.44	0.19	576.85	3.20
	LEAF	8.27	0.23	26.54	0.53	5.86	0.13	416.35	2.31
	PULP	0.26	0.01	12.89	0.26	2.39	0.05	1101.24	6.12
	SOIL	36.59		49.97		44.98		180.06	
YAM	STEM	16.16	0.48	60.30	1.16	13.52	0.53	3186.20	15.32
	LEAF	12.12	0.36	55.51	1.07	4.64	0.18	3506.08	16.86
	PULP	6.72	0.20	20.63	0.40	1.63	0.06	1537.16	7.39
	SOIL	33.97		52.04		25.46		208.01	

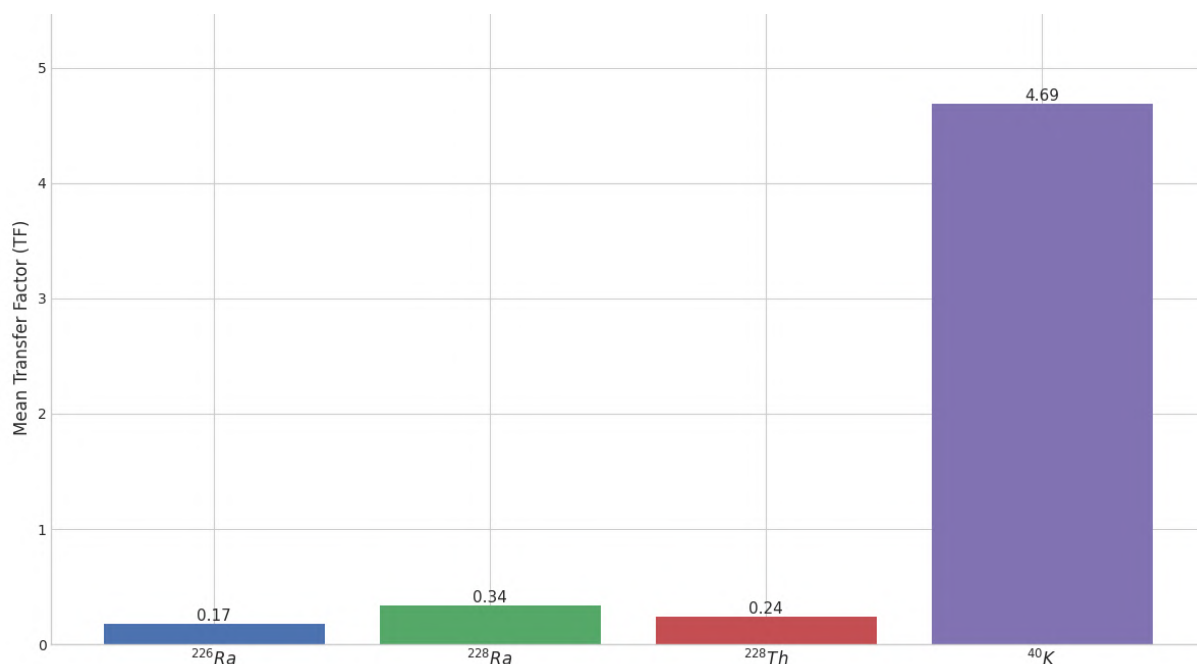


Figure 1: Comparison of Mean Transfer Factor by Radionuclide

In contrast, the actinide-series radionuclides (^{226}Ra , ^{228}Ra , and ^{228}Th) showed low mobility, with TFs generally below 1 for most of the analyzed samples. This limited transfer is attributed to their strong retention in mineral and organic soil fractions, particularly in tropical soils rich in iron oxides and with high cation exchange capacity (Haque and Ferdous, 2017; Ribeiro et al., 2018). Among them, ^{228}Th exhibited the lowest TFs due to its low solubility and strong association with fine particles and organic matter UNSCEAR (2008). In contrast, ^{226}Ra , owing to its chemical similarity to calcium, displayed slightly higher variability, with limited enrichment detected in some vegetative parts. This behavior may be explained by competitive cation exchange processes and local differences in radium availability (Kabata-Pendias, 2010; Sheppard et al., 2005; International Atomic Energy Agency, 2010a).

A clear pattern emerged when comparing edible and non-edible plant parts: leaves, stems, and peels tended to accumulate significantly higher radionuclide concentrations than pulps, juices, and seeds. This distribution reflects physiological compartmentalization, where structural and photosynthetic tissues accumulate more mobile elements, while storage tissues generally show lower radionuclide levels (Kabata-Pendias, 2010; Carini, 2001).

For instance, in yam samples, TF values for ^{40}K were approximately 2.3 times higher in leaves than in pulp, whereas in papaya, stems exhibited TF values almost 2.5 times greater than pulp, reinforcing the need to distinguish plant compartments when assessing ingestion risks.

Figure 1 complements these findings by visually demonstrating the contrast between the behavior of ^{40}K and that of the actinide radionuclides. The pronounced dominance of ^{40}K across all analyzed compartments highlights its critical role as a physiological marker of soil-to-plant dynamics. Conversely, the restricted variability and consistently low TFs for ^{226}Ra , ^{228}Ra , and ^{228}Th emphasize their limited bioavailability, corroborating trends reported in international datasets (International Atomic Energy Agency, 2010b) and regional studies (Ribeiro et al., 2018; Santos et al., 2002; Silva et al., 2021).

Despite these consistent patterns, it is important to note that the present study did not investigate potential

seasonal variations in radionuclide availability and plant uptake. Seasonal dynamics can influence radionuclide mobility and bioavailability due to changes in soil moisture, microbial activity, ionic composition, and plant metabolism (da Costa Dantas et al., 2019). For instance, higher rainfall periods may enhance the leaching of mobile cations such as ^{40}K and ^{226}Ra , whereas drier seasons could favor the retention of less mobile elements such as ^{228}Th .

Although beyond the scope of the current investigation, understanding the potential influence of seasonal variability is essential for refining predictive models and improving ingestion dose assessments. Therefore, future studies should consider multi-seasonal sampling campaigns to evaluate temporal effects on radionuclide transfer and to strengthen the accuracy of radiological risk estimations.

Overall, the TF values obtained in this study are consistent with international reference ranges, such as those provided by the IAEA (International Atomic Energy Agency, 2010b), and align well with regional data reported for the state of Rio de Janeiro (Santos et al., 2002; Ribeiro et al., 2018; Silva et al., 2021). These similarities reinforce the reliability of the measurements and indicate that, despite the distinctive characteristics of tropical soils and organic farming practices, the observed transfer patterns are comparable to those reported in other contexts.

4 Conclusions

This work advances the understanding of radionuclide behavior in organic agricultural systems by presenting transfer factor data for ^{226}Ra , ^{228}Ra , ^{228}Th , and ^{40}K in tropical food crops from Rio de Janeiro, Brazil. The data underscore the differential uptake and distribution of radionuclides across plant compartments, reinforcing the relevance of considering plant physiology and morphology in radioecological assessments.

Beyond reaffirming the well-known mobility of ^{40}K and the low bioavailability of actinide isotopes, the study emphasizes the variability of transfer patterns in organically cultivated species, reflecting both environmental and agronomic influences. These insights are especially valuable for refining dose estimation models and supporting public health strategies in contexts where organic farming practices are expanding but remain insufficiently studied from a radiological perspective.

Overall, the results highlight the need for region-specific datasets to better inform safety standards and predictive models. Continued investigations with broader temporal and spatial coverage, including more detailed risk assessments based on ingestion pathways, will enhance the applicability of TF data in regulatory frameworks and environmental monitoring programs.

Acknowledgements

The authors would like to acknowledge the financial support of government agencies funding research projects in Brazil, National Council for Scientific and Technological Development (NCSTD) and Coordination for the Improvement of Higher Education Personnel (CIHEP).

References

- Barbosa da Silva L. et al. (2025) Radiometric and simulation analysis with mcnp 6.3 of products that expose the skin by absorption. *Radiation Physics and Chemistry* 234:112756
- Carini F. (2001) Radionuclide transfer from soil to fruit. *Journal of Environmental Radioactivity* 52:237–279

- da Costa Dantas R., Navoni J. A., de Alencar F. L. S., da Costa Xavier L. A., and do Amaral V. S. (2019) Natural radioactivity in brazil: a systematic review. *Environmental Science and Pollution Research* 27:143–157
- Garcêz R. W. D., Lopes J. M., Filgueiras R. A., and Silva A. X. (2019) Study of k-40, ra-226, ra-228 and ra-224 activity concentrations in some seasoning and nuts obtained in rio de janeiro city, brazil. *Food Science and Technology* 39:120–126
- Haque M. and Ferdous J. (2017) Transfer of natural radionuclides from soil to plants in savar dhaka. *spanish journal of soil science (sjss)* 7:133–145
- International Atomic Energy Agency (2010a) Analytical methodology for the determination of radium isotopes in environmental solids by gamma spectrometry. Technical report. International Atomic Energy Agency. Vienna
- International Atomic Energy Agency (2010b) *Handbook of Parameter Values for the Prediction of Radionuclide Transfer in Terrestrial and Freshwater Environments*. Technical Reports Series No. 472. IAEA, Vienna
- Kabata-Pendias A. (2010) *Trace Elements in Soils and Plants*. CRC Press. 4 edition
- Ribeiro F., Silva J., Lima E., do Amaral Sobrinho N., Perez D., and Lauria D. (2018) Natural radioactivity in soils of the state of rio de janeiro (brazil): Radiological characterization and relationships to geological formation, soil types and soil properties. *Journal of Environmental Radioactivity* 182:34–43
- Santos E., Lauria D., Amaral E., and Rochedo E. (2002) Daily ingestion of 232th, 238u, 226ra, 228ra and 210pb in vegetables by inhabitants of rio de janeiro city. *Journal of Environmental Radioactivity* 62:75–86
- Sheppard S., Sheppard M., Ilin M., and Thompson P. (2005) Soil-to-plant transfers of uranium series radionuclides in natural and contaminated settings. *Radioprotection* 40
- Silva L. B. et al. (2021) Committed effective dose and lifetime cancer risk due to ingestion of natural radionuclides in grains grown in an area of high background radiation. *Applied Radiation and Isotopes* 172:109656
- UNSCEAR (2008) Sources and effects of ionizing radiation — report to the general assembly. annex B: Exposures of the public and workers from various sources of radiation

Analysis of Natural Radioactivity in Beach Sands Along the Rio de Janeiro State Coast by Gamma Spectrometry

NEVES, A. Laisa^{1*}; JÚNIOR, A. N. Wander¹; SANTOS, O. Jonathan¹; OREJUELA, P. O. Carlos¹; SILVA, B. Leandro¹, SILVA, X. Ademir¹

¹Programa de Engenharia Nuclear (PEN) – COPPE - UFRJ, Rio de Janeiro, BRAZIL.

*sevenlaisa@gmail.com

Abstract

Brazil is a country characterized by a tropical climate, offering numerous opportunities for leisure and contemplation along its extensive coastline, particularly on the beaches of the state of Rio de Janeiro. According to a survey conducted by the Ministry of Tourism (2022), Brazilian beaches emerge as the primary motivational factor for leisure travel in the country, with the combination of sea, sun, and warmth encouraging recreational activities for five out of ten Brazilians. The wide stretches of sand that constitute the coastline of Rio de Janeiro significantly contribute to the intensity of background radiation. Given the high frequency of both residents and tourists visiting these areas throughout the year, it is essential to assess the presence of radionuclides in the beach sands. Such analysis enables the comparison of risk coefficients due to exposure during beach stays, in accordance with the limits established by Standard NN 3.01. This study aims to estimate the potential radiological risks faced by tourists and local residents from exposure to beach sands. To date, forty-six samples have been analyzed, and using gamma spectrometry, activity concentrations were found to range from 1.1 ± 0.1 and 1408.1 ± 32.89 Bq.kg⁻¹ for ²²⁶Ra, from 0.81 ± 0.16 and 6203.57 ± 278.83 Bq.kg⁻¹ for ²³²Th, and from 8.81 ± 10.39 and 1008.36 ± 43.63 Bq.kg⁻¹ for ⁴⁰K. Based on the calculations for absorbed dose, the mean value was 307.15 nGy.h⁻¹, which is over 5% higher than the threshold established by UNSCEAR (2000). The average annual effective dose was 0.1505 mSv.y⁻¹ for local residents and 0.0180 mSv.y⁻¹ for tourists, corresponding to 0.1% and 0.01% of the annual limit set by CNEN's Standard NN 3.01, respectively. The average radiological risk index was 1.94, exceeding the safety threshold of one. Finally, regarding the cancer risk index, only four beaches were classified as negligible risk, twenty-two as intermediate risk, and eight as high-risk areas. Although most samples presented dose values below the recommended limit, fourteen samples exhibited significantly higher values, warranting increased attention from a radiological perspective. Consequently, prolonged exposure to these materials must be justified, and compliance with the ALARA principle should be a prerequisite for the recreational use of these areas.

Keywords: natural radioactivity; beach sands; radiological risk index.

1 Introduction

Brazil stands out for its remarkable ecosystemic richness and topographic diversity. The national territory is predominantly characterized by a tropical climate, which favors recreational and contemplative activities along its extensive coastal zone, with particular emphasis on the beaches of the state of Rio de Janeiro. According to a survey conducted by the Ministry of Tourism (2022), Brazilian beaches are the main attraction for leisure tourism, with the combination of sea, sun, and heat encouraging leisure activities for approximately 50% of the population (ALAM et al., 1999; FREITAS; ALENCAR, 2004).

It is widely recognized that certain geological formations exhibit significant concentrations of naturally occurring radionuclides (NORMs), present since the early formation of the Earth (LIN et al., 2015; VIVES; BATLLE, 2018). Beach sands are composed of a variety of minerals, such as quartz and feldspar, which are noted for their high resistance to weathering caused by wave action. These minerals result from complex geological processes, including rock disintegration, decomposition of constituent elements, and transformations promoted by living organisms—processes known as physical, chemical, and biological weathering (PETTIJOHN et al., 1987; DE MEIJER et al., 2001).

In environmental samples, the concentrations of natural radionuclides are usually low and generally do not pose significant health risks. However, factors such as rock disintegration, cosmic radiation incidence, and industrial processing of raw materials may contribute to increased concentrations. In this context, beach sands may represent a relevant source of elevated background radiation levels. Given the extensive coastal stretch of the state of Rio de Janeiro—especially in its metropolitan region, where beaches are heavily frequented by both residents and tourists throughout the year—it becomes essential to conduct studies aimed at quantifying the presence of radioisotopes in beach sands (DE MEIJER et al., 1988; MOHANTY et al., 2004). According to UNSCEAR (2008), in addition to artificial sources, humans are continuously exposed to ionizing radiation of natural origin, emanating both from the Earth's crust and from cosmic radiation (NEVES et al., 2025). Figure 1 illustrates data related to the global average annual effective dose received by the population (UNSCEAR, 2008).

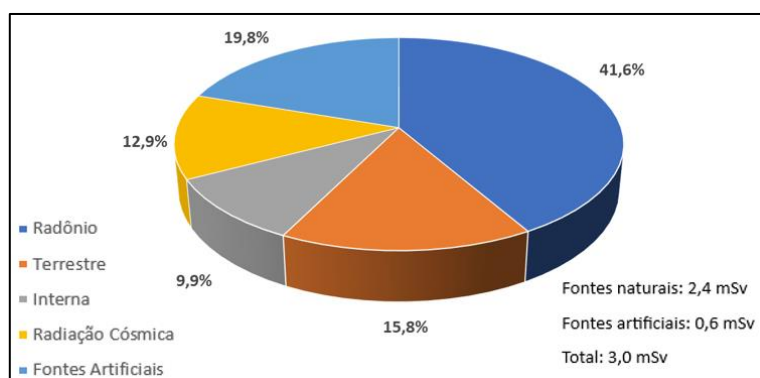


Figure 1: Global distribution of average annual effective dose

In this context, the present study aims to estimate the potential radiological risks associated with exposure to beach sands in the state of Rio de Janeiro, considering both tourists and residents. The study area encompasses the regions of Costa Verde, Costa Azul, the Metropolitan Region, and the Baixada Litorânea. The results obtained will be compared with data from other Brazilian beaches that have also been evaluated, as well as with information available in the scientific literature. This analysis will allow for the assessment of radiation doses received during the time spent on the beaches, in comparison with other regions of the country and with the exposure limits recommended by CNEN Standard NN 3.01. Furthermore, the study seeks to contribute to the understanding of gamma radiation exposure resulting from naturally occurring radionuclides present in beach sands, as well as from the decay products of their radioactive chains.

2 Methodology

2.1 Study Area

The Brazilian coastline extends approximately 7,491 kilometers, ranking as the 15th longest national coastline in the world. The entire coastal strip of the country is bordered by the Atlantic Ocean, which features a variety of geographical formations such as islands, reefs, and bays. It is estimated that the Brazilian coast comprises around 2,095 beaches, of which 380 are located in the state of Rio de Janeiro. Of the 26 Brazilian states, nine have no direct access to the sea, including the Federal District. Among the 17 coastal states, most have their capitals situated near the coastline, with the exception of the

cities of Porto Alegre (RS), Curitiba (PR), São Paulo (SP), Teresina (PI), Belém (PA), and Macapá (AP). This geographical configuration fosters a remarkable diversity of marine and coastal ecosystems, highlighting the environmental and socioeconomic importance of Brazil's coastal zone. The coastal territory of Rio de Janeiro is illustrated in Figure 2 (NOGUEIRA, 2007; IBGE, 1981; DIÁRIO DO RIO, 2020).

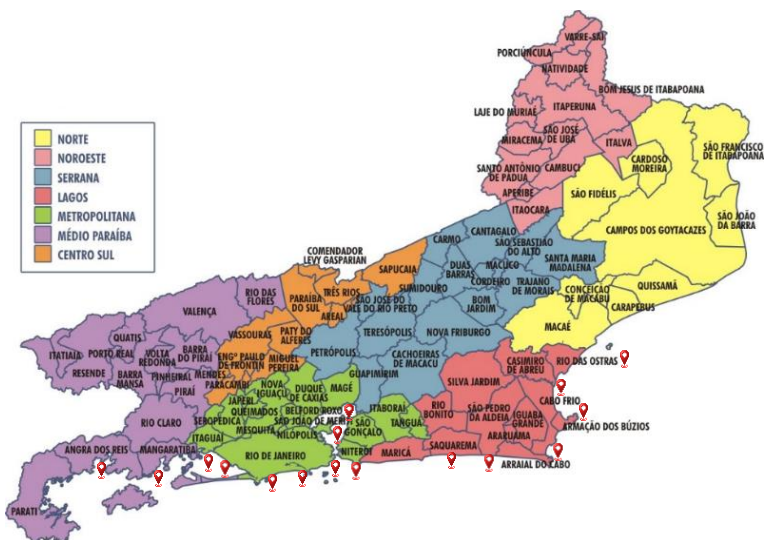


Figure 2: Regional division map of the state of Rio de Janeiro.

2.2 Sample Analysis

A total of 46 sand samples from beaches in the state of Rio de Janeiro were analyzed for this study. The sampling was conducted at an arbitrarily defined point located 3 meters from the shoreline, considering that this approximate stretch of sand typically presents the highest concentration of beachgoers. This choice aims not only to measure radiological risk indices associated with direct exposure but also to evaluate potential anthropogenic contributions in the collected samples. All experimental procedures were carried out at the Laboratory of Environmental Analysis and Computational Simulation (LAASC), affiliated with the Nuclear Engineering Program of the Federal University of Rio de Janeiro (PEN/COPPE/UFRJ). In the laboratory, the samples were stored in 250 mL polypropylene containers, subsequently weighed, and subjected to radioactive counting. For this purpose, a High-Purity Germanium (HPGe) detector was used to analyze the gamma radiation spectrum emitted by the samples. The schematic process employed for sample analysis at LAASC is illustrated in Figure 3.

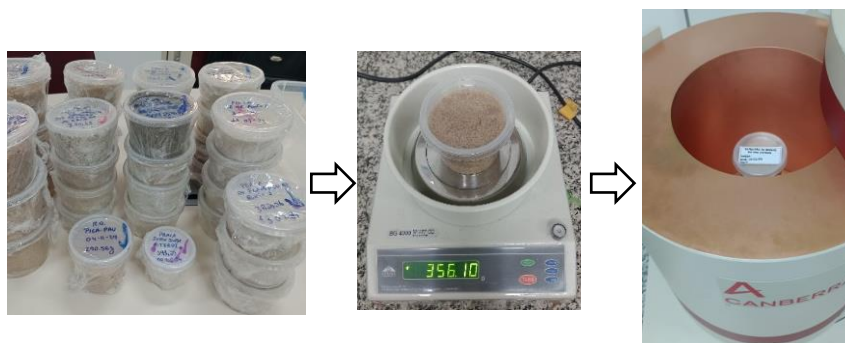


Figure 3: Schematic process used for sample analysis at LAASC.

All analyzed samples are presented in Table 1, along with their respective masses and the locations where they were collected.

Table 1 – List of analyzed samples, indicating corresponding masses and collection sites.

ID	Beach	Location	Mass (kg)
1	Abraão	Ilha Grande – RJ	0.4078
2	Abriçó	Rio das Ostras – RJ	0.2364
3	Areias Negras	Rio das Ostras – RJ	0.5094
4	Barra	Rio de Janeiro – RJ	0.3452
5	Barra de Guaratiba	Barra de Guaratiba – RJ	0.3298
6	Boca da Barra	Rio das Ostras – RJ	0.6539
7	Bosque	Rio das Ostras – RJ	0.3963
8	Botafogo	Rio de Janeiro – RJ	0.4151
9	Canto	Barra de Guaratiba – RJ	0.4832
10	Cemitério	Rio das Ostras – RJ	0.7306
11	Centro	Rio das Ostras – RJ	0.5635
12	Charitas	Niterói – RJ	0.3735
13	Copacabana	Rio de Janeiro – RJ	0.3610
14	Costa Azul	Rio das Ostras – RJ	0.3942
15	Coqueirinhos	Ilha do Governador – RJ	0.3756
16	Dorival Caimmy	Rio das Ostras – RJ	0.4332
17	Ens. Das Gaivotas	Rio das Ostras – RJ	0.3633
18	Extensão do Bosque	Rio das Ostras – RJ	0.3966
19	Flamengo	Rio de Janeiro – RJ	0.3755
20	Forte	Cabo Frio – RJ	0.3609
21	Iriry	Rio das Ostras – RJ	0.4290
22	Itapebussus	Rio das Ostras – RJ	0.4401
23	Itaúna	Saquarema – RJ	0.3550
24	Japariz	Ilha Grande – RJ	0.3593
25	Joana	Rio das Ostras – RJ	0.7448
26	Jurujuba	Niterói – RJ	0.1913
27	Leme	Rio de Janeiro – RJ	0.3415
28	Lopes Mendes	Ilha Grande – RJ	0.4000
29	Mangue	Ilha do Governador – RJ	0.3384
30	Mar do Norte	Rio das Ostras – RJ	0.2824
31	Mauá	Magé – RJ	0.3525
32	Pedrinhas	Rio das Ostras – RJ	0.4114
33	Perigoso	Barra de Guaratiba – RJ	0.2952
34	Pica-Pau	Rio das Ostras – RJ	0.2906
35	Pontal	Arraial do Cabo – RJ	0.4186
36	Praíha	Arraial do Cabo – RJ	0.3780
37	Praíha	Rio de Janeiro – RJ	0.4120
38	Preta	Ilha Grande – RJ	0.7801
39	Remanso	Rio das Ostras – RJ	0.3622
40	Saco	Mangaratiba – RJ	0.3488
41	Sudoeste	São Pedro D’Aldeia – RJ	0.3707
42	Tartaruga	Rio das Ostras – RJ	0.6313
43	Urca	Rio de Janeiro – RJ	0.3998
44	Vermelha	Rio de Janeiro – RJ	0.4249
45	Vila	Saquarema – RJ	0.4212
46	Virgem	Rio das Ostras – RJ	0.6896

2.3 Gamma Spectrometry

Considering the chemical composition of the sands described in the literature—particularly the presence of sulfates – SO_4^{2-} – (KHANDAKER, 2018) gamma radiation spectra of each sample were obtained using an N-type High-Purity Germanium (HPGe) semiconductor detector. The model used was the GC3020, manufactured by Canberra, capable of detecting photons in the energy range from 59.50

keV (^{124}Am) to 1,460.80 keV (^{40}K), with a relative efficiency of 30%. The counting time per sample was 28,800 seconds (equivalent to 8 hours). Spectrometric analysis was performed using the DSA 1000 (Digital Spectrum Analyzer), and energy calibration was carried out with the Gamma Analysis software, included in the Genie 2000 suite. Computational modeling of the experimental system was employed to infer both the physical and chemical characteristics of the samples and the containers used. For this step, the LabSOCS (Laboratory Sourceless Calibration Software) package—also developed by Canberra—was used, allowing the detection efficiency curve to be determined for each specific energy without the need for standard radioactive sources.

2.4 Dosimetric and Radiological Assessment

Several steps were carried out in this study to perform the calculations related to radiation dose and radiological risk indices. Initially, the specific activity concentration of each sample was calculated using Equation 1, where:

- CA is the activity concentration;
- N_L is the net area under the energy peak;
- ϵ is the counting efficiency for a specific energy;
- P_γ is the emission probability of the measured gamma ray;
- m is the sample mass, in kilograms;
- t is the counting time, in seconds.

Considering that approximately 98% of the radiological effects from the ^{238}U decay series are produced by ^{226}Ra and its decay products (UNSCEAR, 2000), the contribution of ^{238}U was determined based on ^{214}Pb and ^{214}Bi . The same approach was adopted for the ^{232}Th decay chain. In this case, the analysis was based on ^{228}Ac and ^{212}Pb , originating from ^{228}Ra and ^{228}Th , respectively. To ensure the establishment of radioactive equilibrium of the radionuclides of interest, the samples were stored and measured 30 days after preparation. Additionally, ^{40}K was directly analyzed. Environmental samples commonly present low levels of radiation; therefore, it is necessary to establish a minimum detection limit for each gamma radiation energy. Equation 2 is used by the Gamma Analysis software to calculate the Minimum Detectable Activity (MDA), where:

- AMD is the minimum detectable activity;
- σ is the standard deviation of the background radiation count.

Subsequently, the absorbed dose rate for each sample was determined using Equation 3, taking into account the necessary conversion factors (calculated using the Monte Carlo method) to convert the activity concentrations of ^{238}U , ^{232}Th , and ^{40}K into absorbed dose at 1 meter above ground level. For validation purposes, a reference value of 55 nGy h^{-1} was adopted, corresponding to the external absorbed dose rate from gamma radiation, as stipulated by UNSCEAR (2000).

To assess and characterize the various biological effects resulting from exposure to ionizing radiation, it is essential to consider the type of radiation involved in the process. This is because, although different types of radiation may release the same amount of energy per unit mass to the body, the biological effects are not necessarily equivalent. Therefore, the effective dose was calculated using Equation 4. The annual effective dose for whole-body irradiation received by the population was estimated for both tourists and local residents, considering the absorbed dose in air (1 meter above the sand surface), the conversion factor, and the occupancy factor (T). In situations of exposure to natural radioactivity, it is possible to calculate the effective dose by considering the mean energy of gamma rays emitted from the ^{238}U , ^{232}Th , and ^{40}K series (ranging from 0.66 to 1.46 MeV). Accordingly, a representative conversion factor of 0.7 is adopted (KEITH et al., 1993; UNSCEAR, 1993). The occupancy factors are estimated, since there are no international recommendations for this specific scenario. Therefore, an occupancy factor of 700 h y^{-1} was established for the local population and 84 h y^{-1} for tourists. As a radiological protection criterion, it is adopted that the annual effective dose (E) should be less than 1 mSv. This value was established based on the recommendations set forth in CNEN Standard NN 3.01, which defines the annual effective dose limit for members of the public.

To ensure that radiation doses remain within permissible limits, the external hazard index was determined using Equation 5. Para manter um risco de radiação insignificante, o valor desse índice deve ser inferior a 1 (UNSCEAR, 2000). This parameter is of significant interest, as it indicates the risk associated with an individual's presence on the beaches where the samples were collected (ICRP-60, 1990; RAVISANKAR, *et al.*, 2015).

In Equation 5, we have:

- $A_{226\text{Ra}}$: specific activity concentration of ^{226}Ra , in Bq kg^{-1} ;
- $A_{232\text{Th}}$: specific activity concentration of ^{232}Th , in Bq kg^{-1} ;
- $A_{40\text{K}}$: specific activity concentration of ^{40}K , in Bq kg^{-1} ;

The constant values of 370, 259, and 4810 are the weighting factors, in Bq kg^{-1} , representing the individual contribution of ^{226}Ra , ^{232}Th , and ^{40}K , respectively, to the specific activity concentration.

Considering the variations and uncertainties inherent to calculations involving radioactivity, the evaluation of the risk of developing radiation-induced cancer is a matter that requires special attention (RAVISANKAR *et al.*, 2015). For this reason, the excess lifetime cancer risk was estimated using Equation 6, where:

- D_{ef} is the annual effective dose, estimated using Equation 4, in mSv y^{-1} ;
- 75,5 is the average life expectancy in Brazil, as reported by IBGE (2022), in years;
- R_f is the fatal cancer risk factor for stochastic effects, as proposed by the U.S. EPA (United States Environmental Protection Agency), in Bq^{-1} .

It is worth noting that, for stochastic effects, the risk factor (R_f) is normalized to 0.05 for members of the public (TASKIN, *et al.* 2009).

The equations used in this study for dosimetric and radiological risk calculations are summarized in Table 2.

Table 2 - Fundamental equations for calculating dosimetric effects on health.

Quantity	Equation	Limit	Reference
Specific Activity Concentration (Bq kg^{-1})	$AC (\text{Bq kg}^{-1}) = \frac{N_L}{\epsilon \cdot P_Y \cdot m \cdot t} \quad (1)$	$^{226}\text{Ra} \leq 35$ $^{232}\text{Th} \leq 30$ $^{40}\text{K} \leq 400$	(UNSCEAR, 2000)
Minimum Detectable Activity (MDA)	$MDA (\text{Bq kg}^{-1}) = \frac{2.71 + 4.66\sigma}{\epsilon \cdot P_Y \cdot m \cdot t} \quad (2)$	[1]	(UNSCEAR, 2000)
Absorbed Dose Rate (nGy h^{-1})	$D_a (\text{nGy} \cdot \text{h}^{-1}) = 0,462 \cdot A_{238\text{U}} + 0,604 \cdot A_{232\text{Th}} + 0,0417 \cdot A_{40\text{K}} \quad (3)$	$D_a \leq 55$	(UNSCEAR, 2000)
Effective Dose (mSv ano^{-1})	$D_{\text{ef}} (\text{mSv} \cdot \text{a}^{-1}) = 10^{-6} \cdot D_a \cdot 0,7 \cdot T \quad (4)$	$D_{\text{ef}} \leq 1$	(CNEN, 2024)
Radiological Risk Index ^[2]	$H_{\text{ex}} = \frac{A_{226\text{Ra}}}{370} + \frac{A_{232\text{Th}}}{259} + \frac{A_{40\text{K}}}{4810} \quad (5)$	$H_{\text{ex}} < 1$	(UNSCEAR, 2000)
Excess Lifetime Cancer Risk ^[2]	$ELCR = D_{\text{ef}} \cdot 75,5 \cdot R_f \quad (6)$	$ELCR \leq 10^{-6}$	(U.S. EPA, 1999)

[1] There are no reference limits for the estimated values.

[2] Dimensional quantities.

It is important to note that, according to the United States Environmental Protection Agency, the excess lifetime cancer risk is considered negligible for $CR \leq 10^{-6}$, moderate for $CR = 10^{-5}$, and a public health concern for $CR \geq 10^{-4}$ (U.S. EPA, 1999).

3 Results and Discussion

As illustrated in Figure 4, some beaches along the coast of Rio de Janeiro stand out due to the distinct characteristics of their sands, particularly those with black sand. These sands generally contain heavy minerals derived from rocks, such as monazite, zircon, ilmenite, rutile, garnet, allanite, and sillimanite. These minerals often contain significant amounts of naturally occurring elements, such as uranium and thorium.



Figure 4: Color variation among some of the collected samples.

During the evaluation, it was observed that samples 3, 5, 17, 22, 25, and 38 exhibited the highest specific activity peaks for most of the radionuclides analyzed in this study (with the exception of ^{40}K). Additionally, it was found that samples 10 and 33 showed specific activity concentration values for ^{232}Th above the global average. Furthermore, sample 29 presented elevated levels of both ^{232}Th and ^{40}K .

Sample 9 exhibited high specific activity concentration levels for all three analyzed radioisotopes (^{226}Ra , ^{232}Th , and ^{40}K), resulting in the highest accumulated specific activity value, approximately $8,152.63 \text{ Bq}\cdot\text{kg}^{-1}$. To highlight the discrepancy between this result and the others obtained for the beach sand samples, reference values from UNSCEAR (2000) for radionuclides in beach sands were used as a comparative parameter. The results were found to be significantly higher—approximately 40 times greater for ^{226}Ra and over 200 times greater for ^{232}Th .

Table 3 presents the individual contribution of each evaluated radionuclide to the specific activity concentration distribution. The data shown in the table are subsequently illustrated in Figure 4.

Table 3 – Activity concentration values obtained from surface sand samples collected at different beaches along the coast of Rio de Janeiro.

Samples		Activity Concentration (Bq kg^{-1})			
ID	Beache	^{226}Ra	^{228}Ra	^{228}Th	^{40}K
1	Abraão	4.14 ± 0.23	8.54 ± 0.75	8.24 ± 0.76	264.66 ± 11.9
2	Abriço	17.16 ± 0.93	27.10 ± 0.83	31.12 ± 2.94	108.93 ± 11.12
3	Areias Negras	128.12 ± 3.03	169 ± 0.98	537.21 ± 42.99	60.42 ± 3.87
4	Barra	1.43 ± 0.12	2.23 ± 0.16	2.12 ± 0.26	9.24 ± 1.61
5	Barra de Guaratiba	66.77 ± 1.96	147.70 ± 0.98	194.59 ± 15.63	150.68 ± 10.84
6	Boca da Barra	5.23 ± 0.33	9.41 ± 0.45	9.07 ± 0.93	250.12 ± 11.89
7	Bosque	16.17 ± 0.7	19.97 ± 0.73	22.26 ± 2.05	181.96 ± 10.82
8	Botafogo	8.37 ± 0.27	23.12 ± 0.61	25.02 ± 2.04	82.89 ± 4.20
9	Canto	1408.1 ± 32.89	5449.93 ± 1.00	6957.21 ± 556.65	540.96 ± 27.82
10	Cemitório	16.6 ± 0.66	29.21 ± 0.76	36.66 ± 3.08	179.66 ± 9.74
11	Centro	6.81 ± 0.47	12.04 ± 0.60	12.6 ± 1.34	148.25 ± 9.42
12	Charitas	2.27 ± 0.14	7.86 ± 0.23	6.05 ± 0.54	86.35 ± 4.41
13	Copacabana	1.99 ± 0.11	3.92 ± 0.10	16.26 ± 1.35	26.42 ± 2.23
14	Costa Azul	1.10 ± 0.10	1.88 ± 0.09	2.00 ± 0.23	36.52 ± 2.43
15	Coqueirinhos	10.46 ± 0.6	18.99 ± 0.73	23.65 ± 2.16	443.60 ± 21.07

16	Dorival Caimmy	2.35 ± 0.18	4.71 ± 0.22	4.84 ± 0.5	54.25 ± 3.76
17	Ens. Das Gaivotas	38.48 ± 1.27	85.14 ± 0.95	99.07 ± 8.10	8.98 ± 9.73
18	Extensão do Bosque	5.56 ± 0.39	7.79 ± 0.33	4.05 ± 0.46	118.48 ± 8.5
19	Flamengo	3.21 ± 0.16	6.90 ± 0.22	6.32 ± 0.56	54.77 ± 3.23
20	Forte	1.20 ± 0.10	2.17 ± 0.18	1.02 ± 0.19	64.48 ± 3.60
21	Iriry	4.74 ± 0.41	9.36 ± 0.5	10.89 ± 1.16	2.07 ± 4.12
22	Itapebussus	39.07 ± 1.22	62.62 ± 0.91	75.65 ± 6.20	110.87 ± 8.11
23	Itaúna	6.63 ± 0.52	11.61 ± 0.61	10.81 ± 1.29	135.01 ± 9.84
24	Japariz	8.4 ± 0.65	13.08 ± 0.70	11.1 ± 1.38	1008.36 ± 43.63
25	Joana	157.22 ± 3.91	408.86 ± 1.00	559.04 ± 44.80	143.33 ± 9.20
26	Jurujuba	3.03 ± 0.22	6.53 ± 0.29	6.24 ± 0.62	61.03 ± 4.55
27	Leme	1.93 ± 0.13	3.22 ± 0.13	3.10 ± 0.33	21.38 ± 2.07
28	Lopes Mendes	1.92 ± 0.17	0.03 ± 0.19	2.85 ± 0.38	80.43 ± 4.85
29	Mangue	13.93 ± 0.77	51.59 ± 0.90	72.97 ± 6.03	545.51 ± 25.48
30	Mar do Norte	7.78 ± 0.62	12.35 ± 0.69	12.90 ± 1.54	216.64 ± 13.75
31	Mauá	7.16 ± 0.51	6.35 ± 0.52	4.85 ± 0.84	8.81 ± 10.39
32	Pedrinhas	5.26 ± 0.44	9.66 ± 0.56	10.87 ± 1.19	136.61 ± 9.02
33	Perigoso	19.27 ± 0.89	32.95 ± 0.85	42.53 ± 3.7	155.68 ± 11.29
34	Pica-Pau	6.92 ± 0.59	10.79 ± 0.67	12.28 ± 1.49	99.86 ± 9.73
35	Pontal	8.99 ± 0.28	19.17 ± 0.66	16.00 ± 1.31	50.24 ± 2.93
36	Prainha	8.08 ± 0.57	15.53 ± 0.77	26.43 ± 2.47	497.52 ± 23.13
37	Prainha	4.54 ± 0.24	10.83 ± 0.34	11.08 ± 0.97	14.73 ± 2.50
38	Preta	113.39 ± 2.68	144.25 ± 0.97	144.69 ± 11.60	12.37 ± 2.36
39	Remanso	1.20 ± 0.11	2.01 ± 0.10	1.70 ± 0.23	42.65 ± 2.77
40	Saco	6.39 ± 0.32	10.23 ± 0.46	12.68 ± 1.12	403.07 ± 17.66
41	Sudoeste	1.17 ± 0.10	1.19 ± 0.15	0.42 ± 0.16	121.41 ± 5.76
42	Tartaruga	22.63 ± 0.83	10.59 ± 0.95	54.00 ± 4.47	104.33 ± 7.48
43	Urca	4.29 ± 0.18	11.56 ± 0.31	12.01 ± 1.00	56.52 ± 3.21
44	Vermelha	1.97 ± 0.12	4.33 ± 0.32	4.44 ± 0.41	16.57 ± 1.65
45	Vila	5.82 ± 0.42	9.84 ± 0.43	10.12 ± 1.14	37.14 ± 5.20
46	Virgem	8.04 ± 0.44	12.72 ± 0.56	13.63 ± 1.33	64.58 ± 563

It is worth noting that the specific activity concentration for ^{40}K in sample 17 (Enseada das Gaivotas Beach) was below the Minimum Detectable Activity (MDA). Therefore, in this case, the MDA value was used as the parameter for calculations related to this sample. Another noteworthy aspect regarding ^{40}K is the high activity concentration observed in samples 15, 24, 36, and 40. It is believed that the contribution of ^{40}K is also directly related to sands with high concentrations of limestone, calcium, coral residues, and shell fragments—as is the case with most light-colored sands.

Table 4 presents the estimated values for Absorbed Dose Rate, Effective Dose, External Radiological Risk Index, and Excess Lifetime Cancer Risk. Subsequently, Figure 5 shows the distribution of the estimated annual effective dose for the local population and tourists, as well as the corresponding contribution to the excess lifetime cancer risk. These figures allow a comparative analysis of annual average exposure and provide a basis for assessing the cancer risk potential induced by radiation as a function of effective dose levels.

Table 4 – Estimated values for Absorbed Dose Rate, Effective Dose, External Radiological Risk Index, and Excess Lifetime Cancer Risk.

ID	Beach	D_a (nGy h ⁻¹)	D_{ef} (mSv ano ⁻¹)		H_{ex}	$ELCR$ ($\leq 10^{-6}$)	
			Locais	Turistas		Locais	Turistas
1	Abraão	23.08	0.0113	0.0014	0.13	42.70	5.12
2	Abriçó	47.64	0.0233	0.0028	0.29	88.11	10.57
3	Areias Negras	488.26	0.2392	0.0287	3.09	903.16	108.38
4	Barra	3.67	0.0018	0.0002	0.02	6.79	0.81
5	Barra de Guaratiba	243.87	0.1195	0.0143	1.53	451.11	54.13
6	Boca da Barra	24.01	0.0118	0.0014	0.14	44.41	5.33
7	Bosque	40.57	0.0199	0.0024	0.24	75.04	9.00

8	Botafogo	36.40	0.0178	0.0021	0.23	67.32	8.08
9	Canto	8167.01	4.0018	0.4802	51.82	15106.93	1812.83
10	Cemitério	54.95	0.0269	0.0032	0.34	101.64	12.20
11	Centro	24.21	0.0119	0.0014	0.14	44.78	5.37
12	Charitas	13.05	0.0064	0.0008	0.08	24.14	2.90
13	Copacabana	11.55	0.0057	0.0007	0.07	21.36	2.56
14	Costa Azul	4.37	0.0021	0.0003	0.03	8.09	0.97
15	Coqueirinhos	49.09	0.0241	0.0029	0.29	90.80	10.90
16	Dorival Caimmy	9.12	0.0045	0.0005	0.05	16.86	2.02
17	Ens. Das Gaivotas	129.42	0.0634	0.0076	0.82	239.39	28.73
18	Extensão do Bosque	14.66	0.0072	0.0009	0.09	27.12	3.25
19	Flamengo	11.75	0.0058	0.0007	0.07	21.73	2.61
20	Forte	5.17	0.0025	0.0003	0.03	9.56	1.15
21	Iriry	14.51	0.0071	0.0009	0.09	26.83	3.22
22	Itapebussus	106.19	0.0520	0.0062	0.66	196.42	23.57
23	Itaúna	22.23	0.0109	0.0013	0.13	41.13	4.94
24	Japariz	60.53	0.0297	0.0036	0.33	111.97	13.44
25	Joana	18.32	0.0090	0.0011	0.11	33.87	4.07
26	Jurujuba	11.66	0.0057	0.0007	0.07	21.56	2.59
27	Leme	5.60	0.0027	0.0003	0.03	10.35	1.24
28	Lopes Mendes	5.98	0.0029	0.0004	0.03	11.06	1.33
29	Mangue	104.42	0.0512	0.0004	0.63	193.15	23.18
30	Mar do Norte	27.88	0.0137	0.0016	0.16	51.57	6.19
31	Mauá	10.51	0.0051	0.0006	0.06	19.43	2.33
32	Pedrinhas	20.53	0.0101	0.0012	0.12	37.97	4.56
33	Perigoso	60.98	0.0299	0.0036	0.38	112.81	13.54
34	Pica-Pau	21.30	0.0104	0.0013	0.13	39.39	4.73
35	Pontal	27.49	0.0135	0.0016	0.17	50.85	6.10
36	Prainha	49.82	0.0244	0.0029	0.29	92.16	11.06
37	Prainha	15.95	0.0078	0.0009	0.10	29.49	3.54
38	Preta	227.42	0.1114	0.0134	1.42	420.67	50.48
39	Remanso	4.57	0.0022	0.0003	0.03	8.46	1.02
40	Saco	33.60	0.0165	0.0020	0.20	62.15	7.46
41	Sudoeste	6.58	0.0032	0.0004	0.03	12.16	1.46
42	Tartaruga	53.82	0.0264	0.0032	0.33	99.55	11.95
43	Urca	18.57	0.0091	0.0011	0.11	34.35	4.12
44	Vermelha	6.90	0.0034	0.0004	0.04	12.75	1.53
45	Vila	16.29	0.0080	0.0010	0.10	30.14	3.62
46	Virgem	22.32	0.0109	0.0013	0.14	41.29	4.96

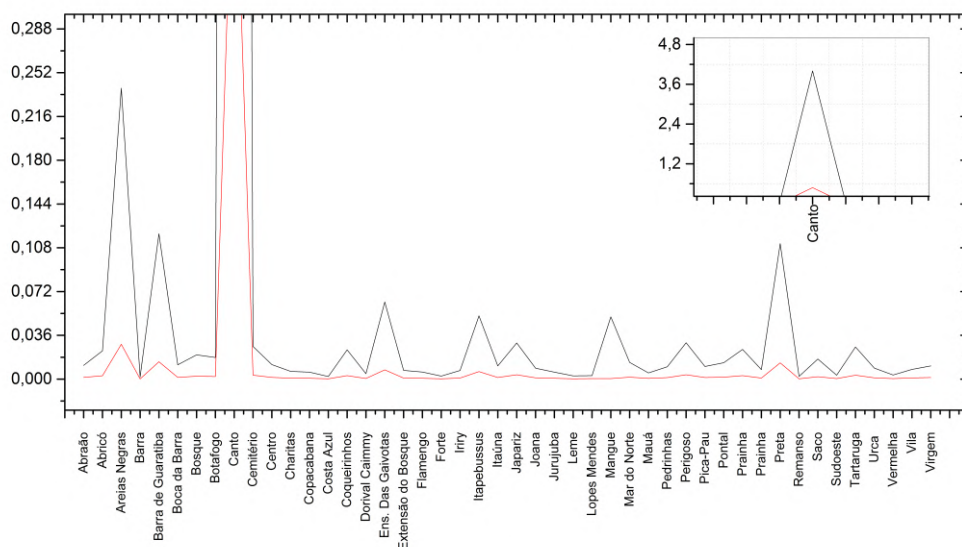


Figure 5: Estimated annual effective dose distribution for the local population and tourists. Corresponding contribution to the excess lifetime cancer risk.

In summary, most of the samples presented absorbed dose values below the recommended limit. However, samples 3, 5, 9, 17, 22, 24, 29, 33, and 38 stood out for exhibiting dose rates above $55 \text{ nGy} \cdot \text{h}^{-1}$. Specifically, sample 9, corresponding to Praia do Canto in Mangaratiba, was the only one to exceed the annual effective dose limit for residents established by CNEN. Regarding radiological risk indices, samples 3, 5, 9, and 38 exceeded the values considered acceptable.

Among the 46 samples analyzed, only four were classified as presenting negligible risk for the development of lifetime cancer for both the local population and tourists. The remaining samples fell into the intermediate or high-risk categories, indicating potential health concerns for these groups in the case of prolonged exposure. These results highlight the need for special attention to regions with elevated radiological values, particularly concerning the frequency and duration of exposure. Therefore, the application of the ALARA principle (As Low As Reasonably Achievable) is reinforced as an essential guideline for the safe and recreational use of these coastal areas.

4 Final Considerations

The results obtained in this study demonstrate that, although the majority of beach sand samples from the state of Rio de Janeiro comply with dose limits established by national and international regulatory agencies, certain locations require special attention. Notable among them are Areias Negras Beach, in Rio das Ostras, which exhibited elevated concentrations of ^{226}Ra and ^{232}Th , and Praia Sul do Oeste, in São Pedro D'Aldeia, with high levels of ^{40}K . In these regions, values significantly exceeded global averages, indicating potential public health risks.

The estimated annual effective dose and excess lifetime cancer risk indices suggest that prolonged or frequent exposure to the sands of these beaches may pose a concern, particularly for the local population. In accordance with radiological protection principles, continuous monitoring of these areas is recommended, along with the application of the ALARA principle, aiming to mitigate risks associated with natural radiation.

In cases where the measured values significantly exceeded global reference indices, the scenario underscores the urgent need for ongoing monitoring and evaluation of these areas—especially regarding the radiological exposure impacts on tourists and local residents. The radiation doses detected—particularly the annual effective dose and radiological risk index—indicate a potentially concerning public health issue, with increased risks of cancer incidence associated with prolonged or frequent exposure to these sands. Therefore, further scientific studies are recommended to explore in greater depth the impacts of radionuclides in black sands, contributing to the development of specific guidelines for land use and occupation in these areas, and consequently, to the preservation of both public and environmental health.

Acknowledgements

This research was funded by CAPES research grants (Coordination for the Improvement of Higher Education Personnel) for graduate studies in the Nuclear Engineering Program at the Federal University of Rio de Janeiro.

References

- ALAM, M. N. et al. The ^{226}Ra , ^{232}Th and ^{40}K activities in beach sand minerals and beach soils of Cox's Bazar, Bangladesh. *Journal of Environmental Radioactivity*, v. 46, p. 243–250, 1999.
- COMISSÃO NACIONAL DE ENERGIA NUCLEAR (Brasil). **Diretrizes básicas de proteção radiológica**. CNEN NN 3.01. Rio de Janeiro: CNEN, 2024.
- DE MEIJER, R. J. et al. Provenance of coastal sediments using natural radioactivity of heavy mineral sands. *Radiation Protection Dosimetry*, v. 24, n. 1/4, p. 55–58, 1988.

DE MEIJER, R. J. et al. Cluster analysis of radionuclide concentrations in beach sand. *Applied Radiation and Isotopes*, v. 54, p. 535–542, 2001.

FREITAS, A. C.; ALENCAR, A. S. Gamma dose rates and distribution of natural radionuclides in sand beaches, Ilha Grande, southeastern Brazil. *Journal of Environmental Radioactivity*, v. 75, p. 211–223, 2004.

FUNDAÇÃO INSTITUTO BRASILEIRO DE GEOGRAFIA E ESTATÍSTICA – IBGE. *IX recenseamento geral do Brasil – 1980: sinopse preliminar do censo demográfico*. Rio de Janeiro: IBGE, 1981.

KHANDAKER, Mayeen Uddin et al. Elevated concentrations of naturally occurring radionuclides in heavy mineral-rich beach sands of Langkawi Island, Malaysia. *Marine Pollution Bulletin*, [S.l.], v. 127, p. 654–663, 2018. Disponível em: <https://doi.org/10.1016/j.marpolbul.2017.12.055>. Acesso em: 24 jul. 2025.

LIN, W.; et al. Radioactivity impacts of the Fukushima nuclear accident on the atmosphere. *Atmospheric Environment*, v. 102, p. 311–322, 2015.

MOHANTY, A. K., et al. Natural radioactivity in the newly discovered high background radiation area on the eastern coast of Orissa, India. *Radiation Measurements*, v. 38, p. 153–165, 2004.

NEVES, Laisa de A. et al. Tissue Reactions (Deterministic Effects) in Diagnostic and Interventional Radiology: An Integrative Review of Recent Literature. *Brazilian Journal of Radiation Sciences*, Rio de Janeiro, v. 13, n. 2, e2802, 2025. DOI: 10.15392/2319-0612.2025.2802.

NOGUEIRA, Kiko. *Guia Quatro Rodas Praias 2007*. São Paulo: Editora Abril, 2007.

PETTIJOHN, F. J.; POTTER, P. E.; SIEVE, R. R. *Sand and sandstones*. New York: Springer, 1987.

RAVISANKAR, R.; et al. Assessments of radio-activity concentration of natural radionuclides and radiological hazard indices in sediment samples from the East coast of Tamilnadu, India with statistical approach. *Marine Pollution Bulletin*, v. 97, n. 1–2, p. 419–430, 2015.

REDAÇÃO. Noroeste Fluminense classificado com bandeira laranja para casos de Covid-19. *Portal ViU!*, 11 dez. 2020. Disponível em: <https://www.portalviu.com.br/cidades/noroeste-fluminense-classificado-com-bandeira-laranja-para-casos-de-covid-19>. Acesso em: 24 jul. 2025.

UNITED NATIONS SCIENTIFIC COMMITTEE ON THE EFFECTS OF ATOMIC RADIATION – UNSCEAR. *Sources and effects of ionizing radiation*. New York: United Nations, 2000.

UNITED NATIONS SCIENTIFIC COMMITTEE ON THE EFFECTS OF ATOMIC RADIATION – UNSCEAR. *UNSCEAR 2008 report on sources and effects of ionizing radiation*. Volume I. New York: United Nations, 2008.

UNITED STATES ENVIRONMENTAL PROTECTION AGENCY – U.S. EPA. *Cancer risk coefficients for environmental exposure to radionuclides*. EPA 402-R-99-001. Washington, D.C.: U.S. EPA, 1999.

VIVES I BATLLE, J.; et al. Marine radioecology after the Fukushima Dai-ichi nuclear accident: are we better positioned to understand the impact of radionuclides in marine ecosystems? *Science of the Total Environment*, v. 618, p. 80–92, 2018.

Estimación del Término Fuente de un irradiador gamma de ^{60}Co utilizando el código MCNP

N. N. Diaz^{1*}, V.M.Vivanco², G.E. Rojas³, J.A. Benavente³

¹ Universidad Nacional del Centro del Perú, Facultad de Ingeniería Química, Huancayo, 120114, Perú

² Instituto Peruano de Energía Nuclear, Laboratorio de Irradiación, 15307, Lima, Perú

³ Instituto Peruano de Energía Nuclear, Laboratorio Secundario de Calibraciones Dosimétricas, 15307, Lima, Perú

*e_2018200698E@uncp.edu.pe

Resumen

El irradiador Gammacell Excel 220 del Laboratorio de Irradiación del Instituto Peruano de Energía Nuclear (LI/IPEN) es utilizado para la esterilización de distintos productos médicos y alimentos, así como la inducción de mutaciones, entre otros. Para realizar una correcta irradiación es importante conocer la distribución de dosis en la cámara de irradiación. El objetivo del presente trabajo es estimar el Término Fuente de un irradiador gamma de ^{60}Co utilizando el código MCNP. Para ello, un modelo computacional de las fuentes de ^{60}Co y del irradiador con sus componentes fueron desarrollados. Para la estimación de la tasa de kerma en aire, se realizaron simulaciones utilizando el Tally F5. También, se obtuvo de manera gráfica la distribución de las curvas de isodosis del campo de radiación en términos de la tasa de kerma en aire. Los resultados obtenidos con las simulaciones computacionales fueron comparados con resultados experimentales obtenidos usando una cámara de ionización PTW Semiflex en el centro de la cámara de irradiación. El resultado en el centro fue de 164,9 Gy/min y 154,8 Gy/min usando cámara de ionización y código MCNP, respectivamente. Para las curvas de isodosis simuladas se obtuvieron resultados que se diferenciaban como máximo en un 9,3% a los reportados por el fabricante. Estas diferencias podrían deberse a factores de corrección de temperatura y presión que, no son contemplados por el código MCNP. Además, la razón de uniformidad de dosis, definida como el valor de dosis máxima sobre el mínimo, fue de 1,8. El modelo computacional servirá de base para una futura recarga del irradiador añadiendo información sobre la actividad, el diseño y material de recubrimiento de las nuevas fuentes. Por lo tanto, se concluye que el modelo computacional del término fuente del irradiador gamma de ^{60}Co utilizando el código MCNP puede ser utilizado para estimar el campo de radiación en próximos proyectos investigación.

Palabras claves: Gammacell Excel 220; kerma en aire; curvas de isodosis; MCNP.

Abstract

The Gammacell Excel 220 irradiator at the Irradiation Laboratory of the Peruvian Institute of Nuclear Energy (LI/IPEN) is used for the sterilization of various medical products and foods, as well as for inducing mutations, among other things. To perform proper irradiation, it is important to know the dose distribution in the irradiation chamber. The objective of this work is to estimate the source term of a ^{60}Co gamma irradiator using the MCNP code. To achieve this, a computational model of the ^{60}Co sources and the irradiator with its components was developed. To estimate the air kerma rate, simulations were performed using Tally F5. Also, the distribution of the isodose curves of the radiation field in terms of the air kerma rate was obtained graphically. The results obtained with the computational simulations were compared with experimental results obtained using a PTW Semiflex ionization chamber in the center of the irradiation chamber. The result in the center was 164.9 Gy/min and 154.8 Gy/min using the ionization

chamber and MCNP code, respectively. For the simulated isodose curves, results were obtained that differed by a maximum of 9.3% from those reported by the manufacturer. These differences could be due to temperature and pressure correction factors that are not taken into account by the MCNP code. In addition, the dose uniformity ratio, defined as the maximum dose value over the minimum, was 1.8. The computational model will serve as the basis for a future reloading of the irradiator, adding information on the activity, design, and shielding material of the new sources. Therefore, it is concluded that the computational model of the source term of the ^{60}Co gamma irradiator using the MCNP code can be used to estimate the radiation field in future research projects.

Keywords: Gammacell Excel 220; kerma in air; isodose curves; MCNP.

1 Introducción

En los últimos años, la radiación gamma ha encontrado diversas aplicaciones entre las que resaltan la inducción de mutaciones (Atak et al., 2004; Pérez et al., 2020), la síntesis de nanopartículas (Ardjoum et al., 2021; Oh et al., 2006), la técnica del insecto estéril (Ahmadi et al., 2018; Ayvaz et al., 2007), la irradiación de alimentos (Jan et al., 2020; Park et al., 2020; Xiong et al., 2024), entre otras. Para garantizar una correcta irradiación de los productos, es importante caracterizar el campo de radiación del irradiator gamma que se utilice (Moradi et al., 2017). Un irradiator comúnmente empleado para la investigación e irradiación de productos es el Gamma Cell Excel 220 que opera con fuentes de ^{60}Co y está diseñado para irradiar volúmenes pequeños con alta precisión.

Es importante conocer la distribución del campo de radiación en el interior de la cámara de irradiación vacía del Gamma Cell Excel 220 (Majer et al., 2019). Esta distribución se puede obtener mediante medidas experimentales y/o cálculos teóricos, como la simulación con códigos que usen el método de Monte Carlo (Edalatkhah & Badiei, 2025; Majer et al., 2019). La forma experimental hace uso de diversos métodos como la dosimetría Fricke (Edalatkhah & Badiei, 2025; P. S. Rodrigues et al., 2024), el uso de dosímetros de alanina (Saputro et al., 2024) o TLD (Moradi et al., 2017), dosimetría con fotodiodo (Gonçalves et al., 2022), así como el uso cámaras de ionización. Rodrigues et al. (2002) desarrollaron una cámara con un volumen sensible de $0,9\text{ cm}^3$, la cual fue empleada para monitorear en tiempo real la magnitud kerma en aire en el rango de $0,6$ a $2,6\text{ kGy/h}$. En el 2019, Majer et al. utilizaron una cámara PTW Farmer con un volumen sensible de $0,6\text{ cm}^3$ para obtener la tasa de dosis en un irradiator Gamma Cell Excel 220.

Las simulaciones computacionales han sido empleadas para estimar valores en irradiadores de ^{60}Co y compararlos con los resultados experimentales por diversos autores. Saputro et al. (2024) tuvo como objetivo validar el modelo computacional del irradiator Gamma Cell Excel 220 utilizando el código PHITS, con el fin de estimar las curvas de isodosis. Para ello, se compararon los resultados simulados con mediciones experimentales obtenidas mediante dosímetros de alanina. Se obtuvo una diferencia de 2.25%, y se identificó la región de mayor uniformidad de dosis dentro de la cámara de irradiación, ubicada entre 7.5 y 12.5 cm del eje z y con un radio de hasta 5 cm desde el centro. Majer et al. (2019) tuvo por objetivo el mapear la distribución de dosis en una instalación panorámica de irradiación gamma con ^{60}Co usando código Geant4 y validarlas con mediciones experimentales con cámara de ionización. La diferencia promedio fue de 2% entre ambas metodologías. Además, se obtuvo la dosis de tránsito que es fundamental para irradiaciones de baja dosis. MCNP permite analizar irradiadores con distintas distribuciones de fuentes y geometrías. Moradi et al. (2017), usaron código MCNP para comparar sus resultados con los obtenidos experimentalmente a través de dosímetros TLD, encontrando una diferencia máxima del 6,7 %. De igual manera, Edalatkhah & Badiei (2025) pudieron mapear la distribución de dosis en un Gamma cell mediante dosímetros de gel Fricke y simulaciones Monte Carlo con MCNP. Los resultados experimentales y simulados coincidieron estrechamente, con razones de uniformidad de dosis (DUR) de 1.53 y 1.48 respectivamente, validando tanto la calibración del sistema como el desempeño de los dosímetros.

En el presente estudio se busca estimar el término fuente del irradiator de ^{60}Co utilizando el código MCNP. Se desarrolló un modelo computacional completo del irradiator Gamma Cell Excel 220 del

Instituto Peruano de Energía Nuclear (IPEN) y se obtuvieron valores de tasa de kerma en aire y curvas de isodosis. Dicho modelo fue validado con la obtención de la tasa de kerma en aire en el centro de la cámara de irradiación mediante una cámara de ionización.

2 Materiales y Métodos

2.1 Gamma Cell Excel 220

El irradiador Gammacell Excel 220 del Laboratorio de irradiación del IPEN emplea como fuente de radiación al radionucleido ^{60}Co , el cual se encuentra rodeada por un blindaje de plomo que garantiza la protección del operador. El irradiador cuenta con un elevador denominado cylindrical drawer, que se desplaza verticalmente y permite que la cámara de irradiación pueda desplazarse desde la posición de carga (load position) hasta la posición de irradiación. La cámara de irradiación presenta un radio interno de 7,7 cm y una altura de 20,32 cm. El tiempo de irradiación puede ser ingresado por un operador en el panel de control. Para medir la temperatura en la cámara de irradiación se implementó una termocupla que tiene su unidad de visualización en la parte exterior del irradiador. La Figura 1 muestra las principales partes del irradiador.

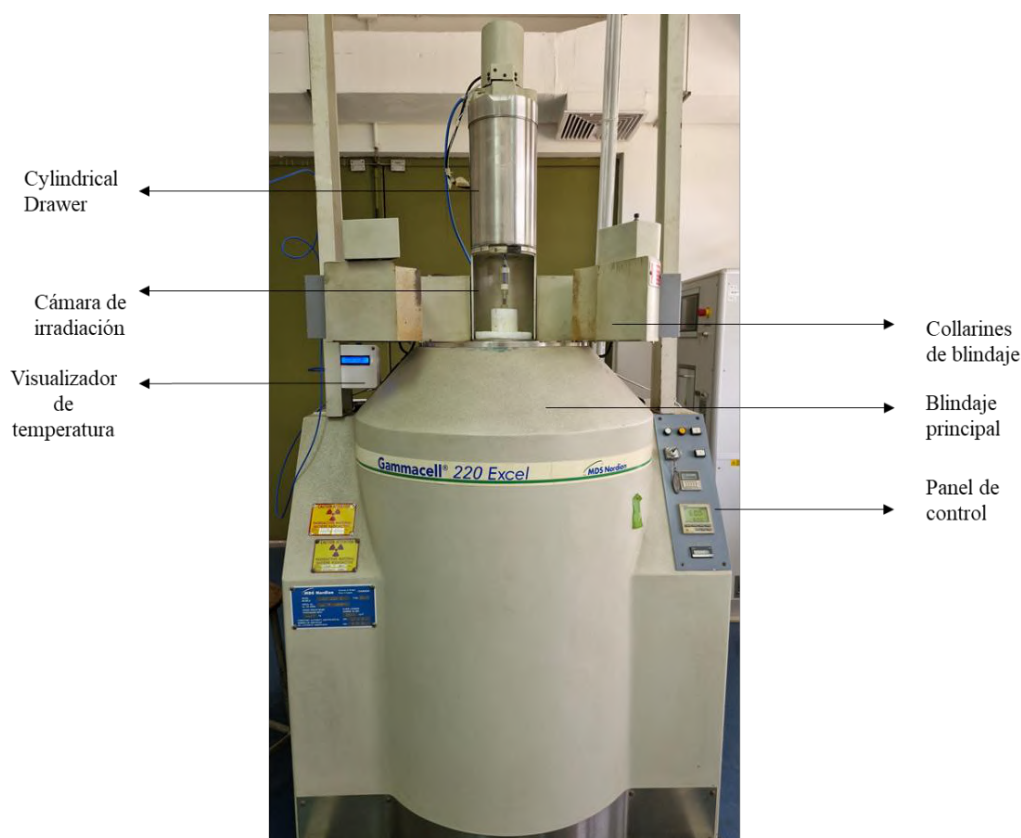


Figura 1: Partes principales del Irradiador Gammacell Excel 220 en la posición de carga (Load position).

Las fuentes radiactivas tienen forma cilíndrica delgada (pencil sources) y se encuentran distribuidas alrededor de la cámara de irradiación cuando está descendiendo a la posición de irradiación. Existen dos tipos de lápices presentes en el irradiador: C-198 y GIK-7-3, las medidas de cada tipo se detallan en la Figura 3. Cada fuente está encapsulada en acero inoxidable tipo 316L y todas están organizadas de manera intercalada y equidistante en 24 de las 48 posiciones disponibles en el carrusel del irradiador, con el fin de permitir una distribución homogénea del campo de radiación. El carrusel tiene un diámetro primitivo (Pitch Circle Diameter, diámetro que coincide con todos los centros de las fuentes) de 20.9 cm como se muestra en la Figura 2.

Las actividades de las fuentes C-198 y GIK-7-3 en la fecha de medición presentaron valores de 54,48 TBq y 412,78 TBq, respectivamente.

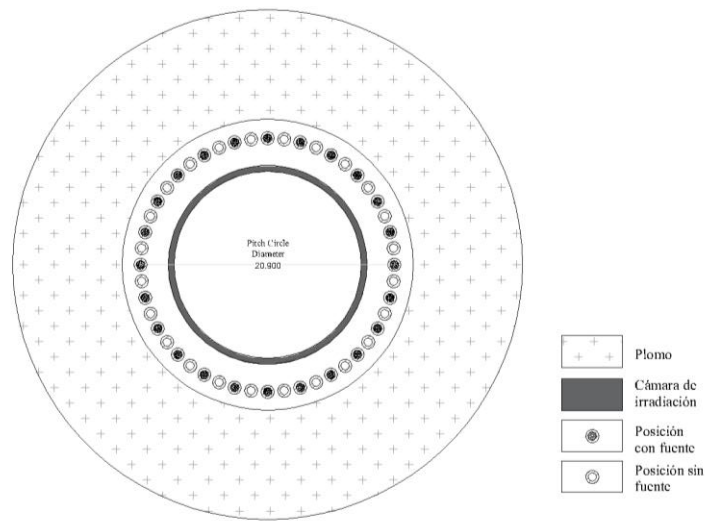


Figura 2: Vista desde arriba (top view) del irradiador.

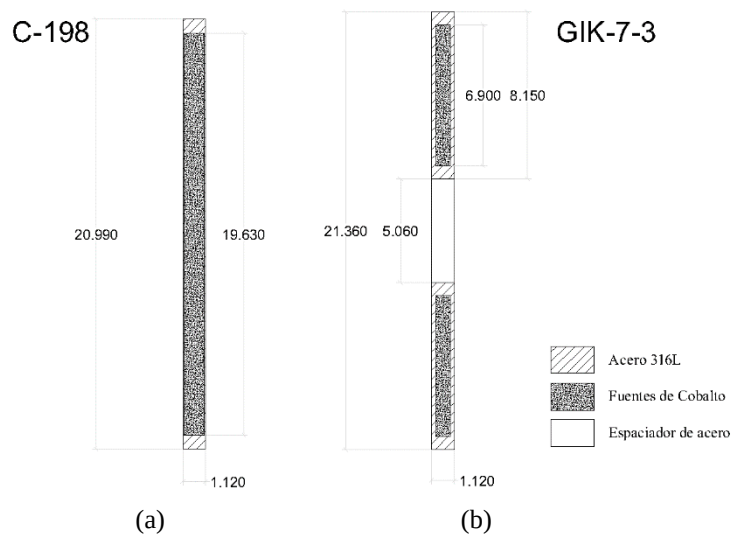


Figura 3: Lámpices de ^{60}Co de tipo C-198 y GIK-7-3. El segundo está compuesto por dos fuentes de ^{60}Co y un espaciador entre ambos.

2.2 Modelo computacional del Irradiador Gammacell Excel 220

El modelo computacional fue desarrollado en el código MCNPX versión 2.5.0. Se creó un archivo de entrada (.i) utilizando Visual Studio Code (VS Code) y VISED como editores. Las fuentes de ^{60}Co , en sus dos presentaciones representadas en la Figura 4, fueron definidas conforme a las medidas especificadas por el fabricante (Figura 3). Se distribuyeron los lámpices de ^{60}Co en 24 posiciones. La cámara de irradiación es de aluminio con una densidad de 2.6989 g/cm^3 . La Figura 5 muestra el arreglo geométrico final de las fuentes. La cámara se desplaza 40.5 cm desde la posición de carga hasta la de irradiación y es esta última posición la que se tuvo en cuenta para la geometría de la simulación MCNP. El arreglo final completo del irradiador Gammacell Excel 220 se encuentra en la Figura 6.

El ^{60}Co emite fotones de 1,33 y 1,17 MeV por lo que se definió el tipo de partículas como fotones (par=2), con una emisión de energía promedio de 1,25 MeV. Las propiedades de los materiales utilizados en la simulación, como la densidad y fracción en peso de los elementos, fueron extraídos del compendio PNNL-15870 (Detwiler et al., 2021), y se resumen en la Tabla 1.

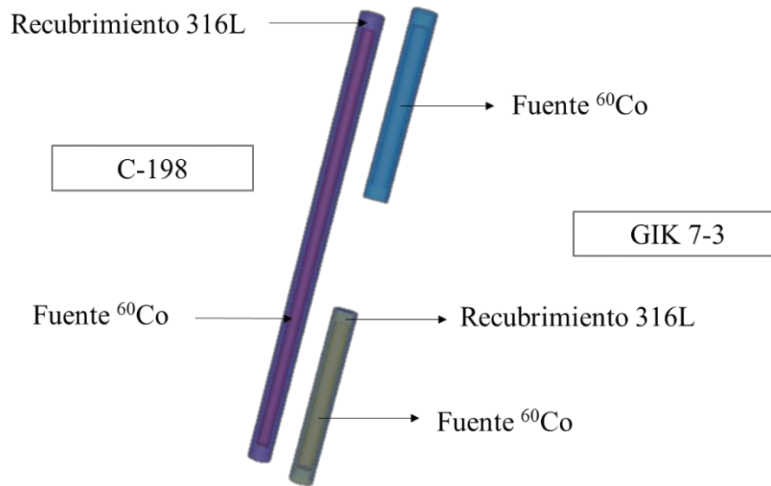


Figura 4: Lápices de ^{60}Co . La imagen fue obtenida con Visual Editor.

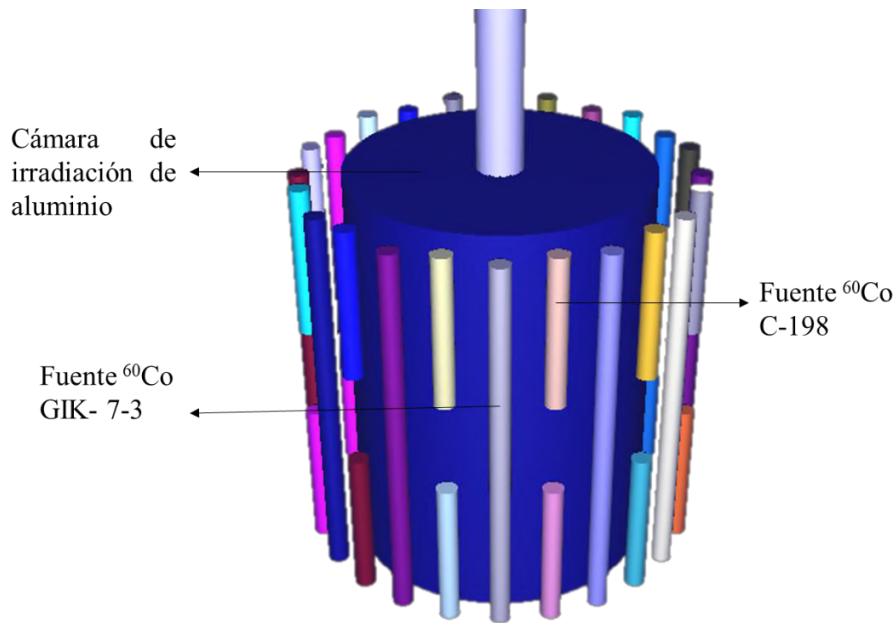


Figura 5: Posiciones de las fuentes tipo lápices en la simulación.

La librería de secciones eficaces utilizada fue MCPLIB04-ENDF/B-VI para el transporte de fotones. El tally F5, un detector puntual, se empleó para obtener la tasa de kerma en aire en el centro de la cámara de irradiación ($x=0$; $y=0$; $z=10,3$ cm). Fueron simuladas 10^{+7} historias, además se obtuvo las curvas de isodosis del campo de radiación al colocar detectores puntuales en términos de tasa kerma en aire. Los resultados se normalizaron respecto al valor central obtenido por simulación (Ecuación 1). Para obtener las curvas de isodosis, los resultados obtenidos se ingresaron en el programa Origin 2025.

$$PL(\%) = 100 + \left(\frac{VO - VC}{VC} \times 100 \right) \quad (1)$$

Donde, PL (%) es el porcentaje de los datos con respecto al centro, VO es el valor obtenido y VC es el valor en el centro de la cámara.

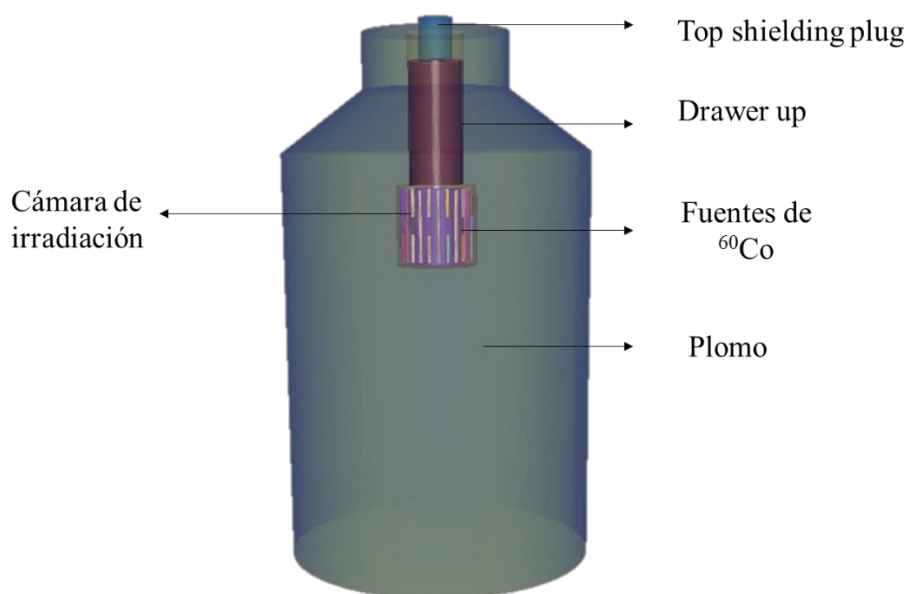


Figura 6: Geometría del Gammacell Excel 220. La cámara de irradiación se encuentra en la posición de irradiación.

Tabla 1: Especificaciones de los materiales incluidos en la simulación

Material	Elemento	Fracción en peso adm.	Densidad g/cm ³
Plomo	Pb	-1	11,35
Aluminio	Al	-1	2,698900
Aire	C	-0,000124	0,001205
	N	-0,755268	
	O	-0,231781	
	Ar	-0,012827	
Acero 316L	C	-0,0003	8,0
	S	-0,01	
	P	-0,00045	
	S	-0,0003	
	Cr	-0,17	
	Mn	-0,02	
	Fe	-0,653950	
	Ni	-0,120000	
	Mb	-0,025000	

2.3 Medición de la tasa de kerma en aire

Se utilizaron una cámara de ionización PTW Freiburg Semiflex modelo 31002 y un Electrómetro PTW Freiburg modelo T10002 para la obtención del kerma en aire. La cámara fue calibrada dosimétricamente en el Laboratorio Secundario de Calibraciones Dosimétricas (LSCD) del Instituto Peruano de Energía Nuclear (IPEN), con factor de calibración (N_k) de $253,1 \text{ mGy} \cdot \text{nC}^{-1}$.

Se empleó una estructura de acrílico, sostenida por un molde de tecnopor, para posicionar el punto efectivo de la cámara de ionización a 10,16 cm de la base de la cámara de irradiación del Gammacell Excel 220 correspondiente al centro geométrico. Las condiciones de temperatura y presión se registraron mediante una termocupla tipo K, fabricada por el IPEN, y un baro-termohigrómetro Kestrel modelo 4000.

La termocupla y la cámara de ionización fueron introducidas a través del tapón de blindaje superior e instaladas en la estructura de acrílico. El arreglo final se encuentra en la Figura 7. Se programó una irradiación de un minuto, verificando el paso de la fuente desde la posición de carga a la de irradiación. Finalizado el proceso, se registraron los valores del electrómetro (nC), la termocupla (°C) y el baro-termohigrómetro (mbar).

La Ecuación 2 fue utilizada para determinar el valor de kerma en aire (K_{air}) en la cámara de irradiación; mientras que la Ecuación 3 permitió calcular el factor de corrección (F_{TP}) para condiciones estándar de 1013,25 mbar y 20°C, asumiendo que el aire se comporta como un gas ideal (Aalbers et al., 2008).

$$K_{air} = (M_1 - M_0) \times F_{TP} \times N_k \quad (2)$$

$$F_{TP} = \left(\frac{1013,25 \text{ mbar}}{P} \right) \times \left(\frac{273,15^\circ\text{C} + T^\circ}{293,15^\circ\text{C}} \right) \quad (3)$$

Donde, M_1 es la medida integrada mostrada por el electrómetro en nC a un minuto de exposición, M_0 es la medida de fondo, F_{TP} es el factor de corrección de presión y temperatura, y N_k es el factor de calibración de la cámara de ionización.



Figura 7: Arreglo experimental donde se observan la termocupla tipo K y la cámara de ionización dentro de la estructura de acrílico.

3 Resultados y Discusiones

3.1 Medición de la tasa de Kerma en aire

El valor promedio experimental obtenido mediante la cámara de ionización, en términos de tasa de kerma en aire en el centro de la cámara de irradiación, fue de $164,9 \pm 2,6$ Gy/min. Por otro lado, el valor estimado por decaimiento para la fecha de la medición, según la información proporcionada por el fabricante de las fuentes, fue de 147,6 Gy/min, lo que representa un error relativo del 11,72 % entre ambos resultados. Esta discrepancia podría deberse a condiciones específicas durante la irradiación, como la temperatura y la presión, las cuales pueden no haber sido completamente consideradas en la estimación proporcionada por el fabricante.

3.2 Modelo computacional del término fuente del irradiador

El resultado de tasa de kerma en aire obtenido por MCNPX fue de 154,8 Gy/min $\pm$ 0,0002 Gy/min en el centro de la cámara de irradiación. Este valor se diferencia en un 6,1% del valor experimental lo que podría estar relacionado con las simplificaciones geométricas del modelo o con las condiciones experimentales no replicadas en la simulación. La tabla 2 muestra los resultados obtenidos por diversos autores que realizaron estudios con simulación de Monte Carlo y que obtuvieron diferencias coherentes con el obtenido en el presente estudio.

Tabla 2: Estudios realizados con simulación de Monte Carlo

Autor	Software	Detector	Diferencia %
Saputro et al. (2024)	PHITS	Alanina	2.2
Moradi et al. (2017)	MCNP	TLD	6.7
Edalatkah & Badiei (2025)	MNCP	Dosímetros gel Fricke	14
Rodrigues et al. (2024)	MCNP y TOPAS	Dosímetros gel xilenol Fricke	1.99 para MCNP y 6.44 para TOPAS
Majer et al. (2024)	PHITS	Cámara de ionización	6
Majer et al. (2019)	GEANT	Cámara de ionización	Rango 5-8

El modelo computacional desarrollado es una herramienta que permite estimar el campo de radiación sin necesidad de recurrir a mediciones experimentales constantes. Es así que la Tabla 3 muestra los valores obtenidos por simulación, normalizados según la Ecuación 1, considerando que el 100 % es el valor correspondiente al centro geométrico de la cámara de irradiación. Para efectos comparativos, se incluyen también los datos reportados por el fabricante. Se observa que la diferencia máxima entre los valores simulados y los proporcionados por el fabricante fue de 9,3 %. Esta discrepancia se encuentra en un rango aceptable considerando las condiciones ideales en la simulación.

Tabla 3: Comparación de curvas de isodosis obtenidas por el fabricante y por MCNP

COORDENADAS		VALORES NORMALIZADO AL CENTRO		DIFERENCIA
EJE X	EJE Z	FABRICANTE	MCNP	%
cm	cm	%	%	
0	0,95	83,8	80,7	3,7
0	2,54	95,1	86,6	8,9
0	6,35	99,4	96,8	2,7
0	10,16	100	100,0	-
3,81	0,95	87,7	85,5	2,5
3,81	2,54	101,3	93,4	7,8
3,81	6,35	104,3	103,7	0,5
3,81	10,16	103,8	104,8	0,9
6,223	0,95	99,4	95,8	3,6
6,223	2,54	119,1	108,1	9,3
6,223	6,35	115,7	118,9	2,7
6,223	10,16	109,9	113,6	3,4

La Figura 8 muestra la distribución de las curvas de isodosis obtenida por MCNPX en un corte transversal XZ. El porcentaje de radiación aumenta en el eje x a medida que se van alejando del centro de la cámara. Los valores mínimos se presentan en los extremos superior e inferior del eje z, debido a la ausencia de fuentes en dichas regiones. Esta tendencia confirma lo mencionado por (Edalatkah & Badiei, 2025). El porcentaje máximo de dosis fue de 133,7% mientras el mínimo valor fue 77,2%. Por lo tanto, la uniformidad de dosis en la cámara de irradiación, definida como el cociente entre la dosis máxima y la mínima, fue de 1,8. Esta relación refleja una distribución no homogénea dentro de la cámara de irradiación lo que puede comprometer la efectividad del proceso de irradiación si no se dispone de manera adecuada

de la muestra. El valor de uniformidad de dosis se ve influenciada por la configuración geométrica del sistema y da lugar a cuatro máximos laterales, influenciados por la disposición y la mayor actividad de las fuentes tipo GIK-7-3, como se aprecia en la Figura 9. No obstante, se identifica una región central de color verde con una distribución relativamente uniforme, lo que sugiere que las muestras que se ubicarían en esta región recibirían una dosis más uniforme.

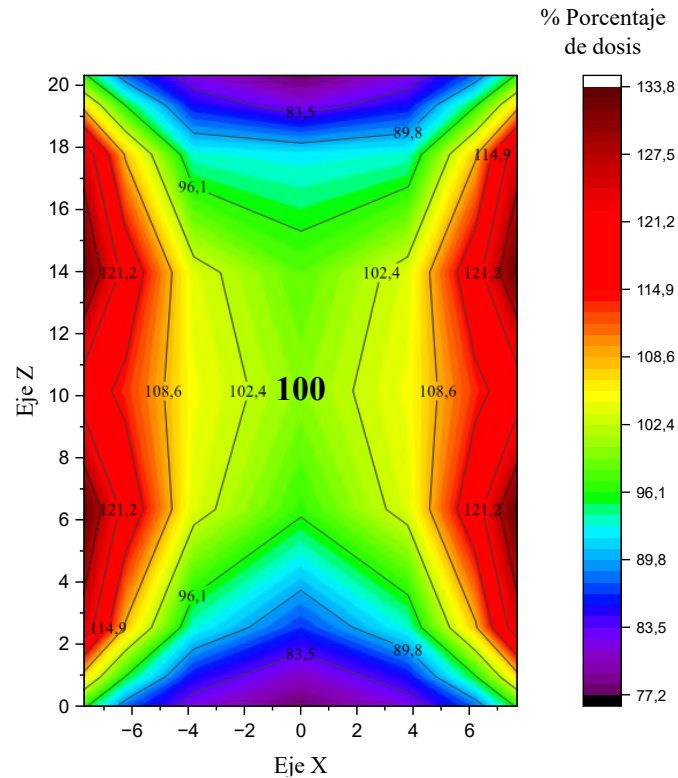


Figura 8: Curvas de isodosis en términos de kerma en aire. Corte transversal en $y=0$. Los valores de la curva están normalizados con respecto al valor central.

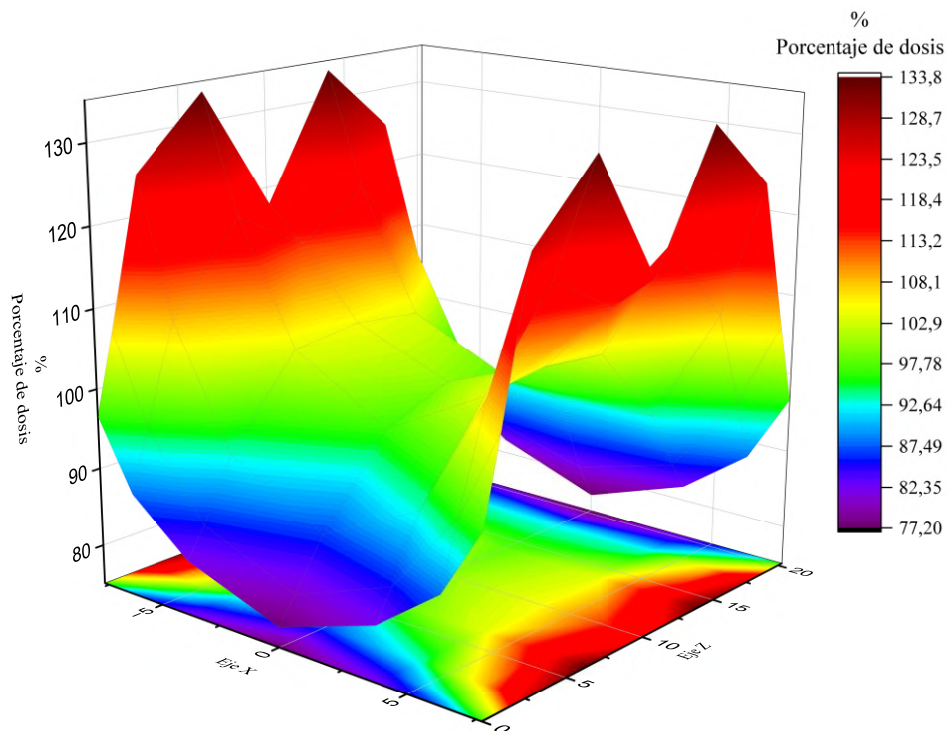
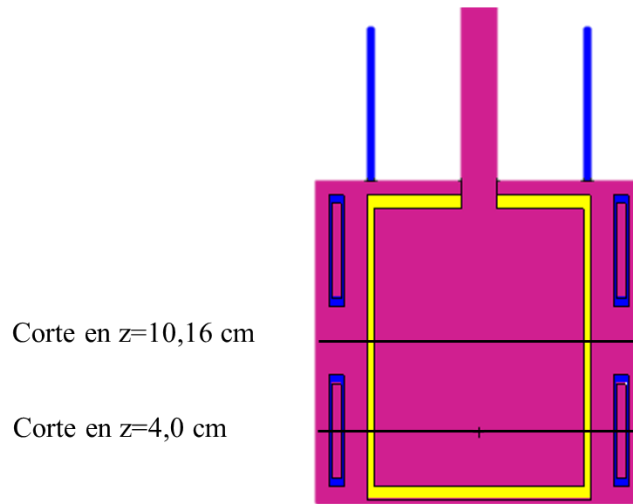
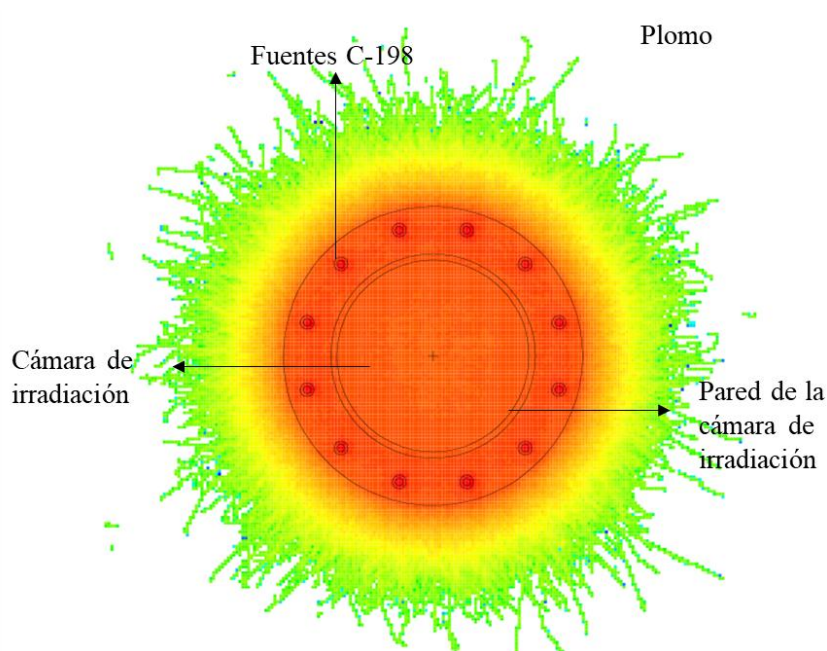


Figura 9: Gráfico tridimensional de la distribución de dosis en la cámara de irradiación.

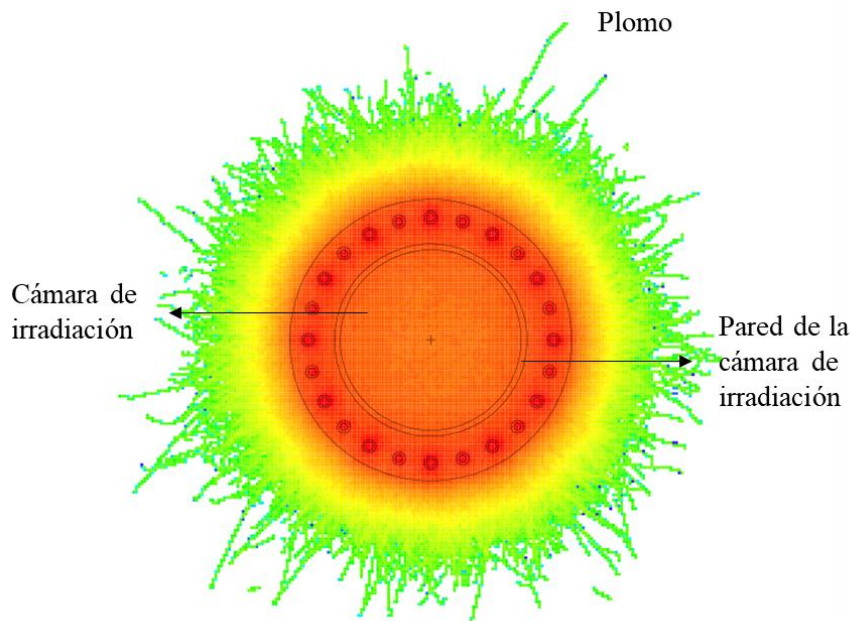
Con el fin de poder apreciar el campo de radiación en distintas áreas de la cámara, se realizaron cortes transversales en los planos $z=10,16\text{cm}$ y $z=4,0\text{cm}$ como se muestra en la Figura 10a. Esto nos permitirá observar las diferencias a distintas alturas. La Figura 10b muestra solo las fuentes tipo C-198 emitiendo radiación ya que desde $z=8,15\text{cm}$ hasta $z=13,21\text{cm}$ las fuentes GIK-7-3 poseen un espaciador impidiendo su vista en este plano. Mientras que la Figura 10c muestra todas las posiciones ocupadas en carrusel del irradiador incluyendo los 2 tipos de fuentes. En ambos casos se observa que la mayor concentración de dosis se encuentra en cámara de irradiación, mientras que la dispersión hacia las zonas periféricas se encuentra atenuada por la presencia del blindaje de plomo, que limita eficazmente la propagación hacia afuera de la cámara.



(a)



(b)



(c)
Figura 10: Cortes transversales de la cámara de irradiación.

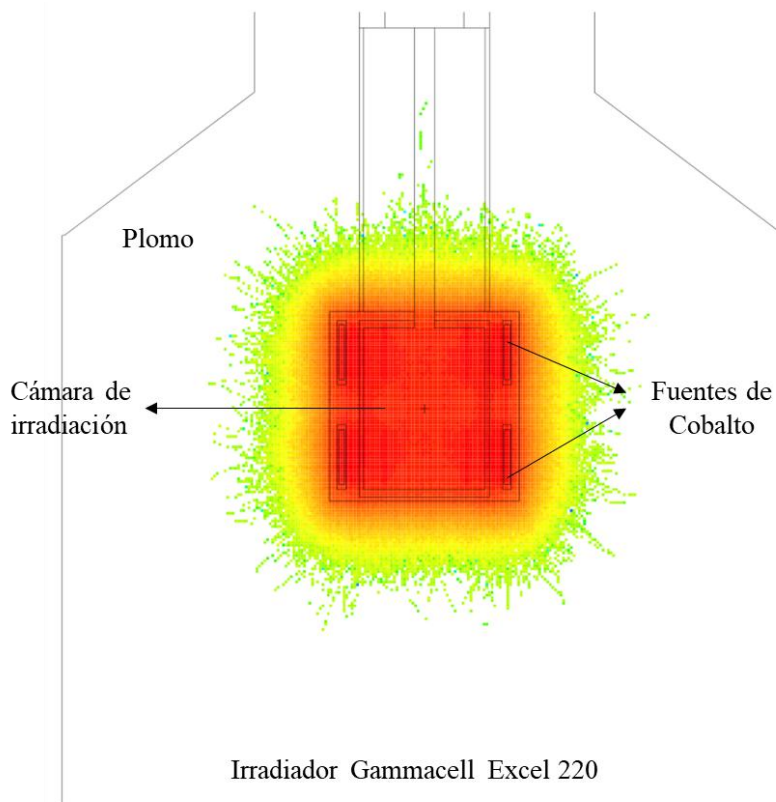


Figura 11: Corte transversal en el plano XZ.

La Figura 11 muestra el campo de radiación en el plano central XZ, simulado con una vista más extendida del irradiador. Se observa una región representada por tonalidades intensas de rojo y anaranjado, correspondiente al interior de la cámara de irradiación, la cual concentra los niveles más altos de radiación. La intensidad observada en esta área se debe a la disposición geométrica de las fuentes radiactivas y al blindaje de plomo que rodea la cámara, el cual atenúa eficazmente la propagación de la radiación,

confinando el campo de radiación dentro del volumen útil. El diseño del sistema GammaCell Excel 220 ha sido desarrollado para lograr una alta eficiencia en el proceso de irradiación dentro de la región de interés, asegurando a su vez una contención efectiva de la radiación.

4 Conclusiones

Se estimó el Término Fuente de un irradiador gamma de ^{60}Co mediante el empleo del código MCNP. La tasa de dosis, en términos de kerma en aire, fue determinada experimentalmente en el centro de la cámara de irradiación utilizando una cámara de ionización y comparada con los resultados obtenidos por simulación, obteniéndose una diferencia del 6,1 %. Adicionalmente, se generaron curvas de isodosis y se analizaron los campos de irradiación en distintos planos de la cámara, lo que permitió visualizar la distribución espacial de la radiación y evaluar la uniformidad de la dosis dentro del volumen útil. En tal sentido, se recomienda colocar las muestras en la región central donde la dosis es más uniforme, ya que la configuración actual genera cuatro picos de dosis elevadas. El modelo desarrollado demuestra ser una herramienta eficaz para caracterizar el campo de radiación y puede ser empleado en futuras evaluaciones que impliquen modificaciones en el irradiador, como la recarga o el reemplazo de fuentes radiactivas.

Agradecimientos

Los autores agradecen al Laboratorio de Irradiación y al Laboratorio Secundario de Calibración Dosimétrica (LSCD) del Instituto Peruano de Energía Nuclear (IPEN) por facilitar el acceso a sus instalaciones y por el soporte técnico proporcionado durante el desarrollo del proyecto de investigación.

Referencias

- Aalbers, T., Duane, S., Kapsch, R.-P., & Meghzifene, A. (2008). Measurement Uncertainty. *Measurement Uncertainty*, 1–62. <https://www.iaea.org/publications/7913/measurement-uncertainty>
- Ahmadi, M., Salehi, B., Abd-Alla, A. M. M., & Babaie, M. (2018). Feasibility of using the radiation-based sterile insect technique (SIT) to control the olive fruit fly, *Bactrocera oleae* Gmelin (Diptera: Tephritidae) in Iran. *Applied Radiation and Isotopes*, 139, 279–284. <https://doi.org/10.1016/J.APRADISO.2018.05.032>
- Ardjoum, N., Shankar, S., Chibani, N., Salmieri, S., & Lacroix, M. (2021). In situ synthesis of silver nanoparticles in pectin matrix using gamma irradiation for the preparation of antibacterial pectin/silver nanoparticles composite films. *Food Hydrocolloids*, 121, 107000. <https://doi.org/10.1016/J.FOODHYD.2021.107000>
- Atak, Ç., Alikamanoğlu, S., Açıık, L., & Canbolat, Y. (2004). Induced of plastid mutations in soybean plant (*Glycine max* L. Merrill) with gamma radiation and determination with RAPD. *Mutation Research/Fundamental and Molecular Mechanisms of Mutagenesis*, 556(1–2), 35–44. <https://doi.org/10.1016/J.MRFMMM.2004.06.037>
- Ayvaz, A., Albayrak, S., & Tunçbilek, A. Ş. (2007). Inherited sterility in Mediterranean flour moth *Ephestia kuehniella* Zeller (Lepidoptera: Pyralidae): Effect of gamma radiation on insect fecundity, fertility and developmental period. *Journal of Stored Products Research*, 43(3), 234–239. <https://doi.org/10.1016/J.JSPR.2006.06.003>
- Detwiler, R., McConn, R., Grimes, T., Upton, S., & Engel, E. (2021). *Compendio de datos de composición de materiales para el modelado del transporte de radiación*. Richland, WA: Laboratorio Nacional del Pacífico Noroeste. <https://www.pnnl.gov/publications/compendium-material-composition-data-radiation-transport-modeling-1>
- Edalatkhah, E., & Badiei, S. (2025). Dose mapping of an irradiation chamber of a gamma cell unit using Fricke gel dosimeters and Monte Carlo simulation. *Applied Radiation and Isotopes*, 217, 111631. <https://doi.org/10.1016/J.APRADISO.2024.111631>

- Gonçalves, J. A. C., Mangiarotti, A., & Bueno, C. C. (2022). Dose rate mapping of an industrial ^{60}Co irradiator using an online photodiode-based dosimetry system. *Radiation Physics and Chemistry*, 200. <https://doi.org/10.1016/j.radphyschem.2022.110387>
- Jan, A., Sood, M., Younis, K., & Islam, R. U. (2020). Brown rice based weaning food treated with gamma irradiation evaluated during storage. *Radiation Physics and Chemistry*, 177, 109158. <https://doi.org/10.1016/J.RADPHYSICHEM.2020.109158>
- Majer, M., Pasariček, L., & Knežević, Ž. (2024). Dose mapping of the ^{60}Co gamma irradiation facility and a real irradiated product – Measurements and Monte Carlo simulation. *Radiation Physics and Chemistry*, 214, 111280. <https://doi.org/10.1016/J.RADPHYSICHEM.2023.111280>
- Majer, M., Roguljić, M., Knežević, Ž., Starodumov, A., Ferenček, D., Brigljević, V., & Mihaljević, B. (2019). Dose mapping of the panoramic ^{60}Co gamma irradiation facility at the Ruđer Bošković Institute – Geant4 simulation and measurements. *Applied Radiation and Isotopes*, 154, 108824. <https://doi.org/10.1016/J.APRADISO.2019.108824>
- Moradi, F., Khandaker, M. U., Mahdiraji, G. A., Ung, N. M., & Bradley, D. A. (2017). Dose mapping inside a gamma irradiator measured with doped silica fibre dosimetry and Monte Carlo simulation. *Radiation Physics and Chemistry*, 140, 107–111. <https://doi.org/10.1016/J.RADPHYSICHEM.2017.01.032>
- Oh, S. D., Lee, S., Choi, S. H., Lee, I. S., Lee, Y. M., Chun, J. H., & Park, H. J. (2006). Synthesis of Ag and Ag– SiO_2 nanoparticles by γ -irradiation and their antibacterial and antifungal efficiency against *Salmonella enterica* serovar Typhimurium and *Botrytis cinerea*. *Colloids and Surfaces A: Physicochemical and Engineering Aspects*, 275(1–3), 228–233. <https://doi.org/10.1016/J.COLSURFA.2005.11.039>
- Park, H. Y., Kim, Y. N., Lee, Y. J., Song, B. S., Park, J. H., Kim, J. K., & Son, E. J. (2020). Effects of gamma irradiation on the quality of manila clam porridge and in vivo immunomodulatory activity of hesperidin as immunopotentiator for sterilized patient food. *Radiation Physics and Chemistry*, 176, 109015. <https://doi.org/10.1016/J.RADPHYSICHEM.2020.109015>
- Pérez, M., Tallón, C., & Pérez, O. (2020). Inducing mutations in Citrus spp.: Sensitivity of different sources of plant material to gamma radiation. *Applied Radiation and Isotopes*, 157, 109030. <https://doi.org/10.1016/J.APRADISO.2019.109030>
- Rodrigues, A., Vieira, J. M., & Hamada, M. M. (2002). Development of the Mini Ionization Chamber for High Dose Monitoring on Line Inside a Gamma Irradiation Facility. *IEEE Nuclear Science Symposium and Medical Imaging Conference*, 1, 406–410. <https://doi.org/10.1109/TNS.2003.815114>
- Rodrigues, P. S., Burin, A. L., Talacimon, C. F., Medeiros, I. M. M. A., Pereira, J. J. N., Teodoro, L. E. H., Rigo, M. E. Z., Gesserame, M. M. F., Tavares, P. V. S., Martins, A. C., Souza, C. D., Zeituni, C. A., Rodrigues, O., & Rostelato, M. E. C. M. (2024). Investigation for Monte Carlo simulation with geometric simplification of the GammaCell 220 irradiator with experimental measurements using Fricke xylene gel (FXG). *Radiation Physics and Chemistry*, 224, 112000. <https://doi.org/10.1016/J.RADPHYSICHEM.2024.112000>
- Saputro, B., Saputro, A. H., Nuraeni, N., Prasetyo, H., Firmansyah, O. A., nugroho, F., & Mayditia, H. (2024). Monte Carlo Simulation As Precision Predictive Tools to Find Isodose Curve of Gamma Irradiator: A Preliminary Study. *INDONESIAN JOURNAL OF APPLIED PHYSICS*, 14(2), 386–398. <https://doi.org/10.13057/IJAP.V14I2.93092>
- Xiong, C., Zou, X., Phan, C. W., Huang, W., & Zhu, Y. (2024). Enhancing the potential of rapeseed cake as protein-source food by γ -irradiation. *Bioscience Reports*, 44(3). <https://doi.org/10.1042/BSR20231807>

Alpha Particle Spectrometry and Monte Carlo Evaluation of ^{241}Am Sources for Use in Elastic Recoil Detection Analysis (ERDA)

P. A. L. Cruz^{1*}, I. T. Vicente², E. M. Oliveira¹, R. S. F. Dam¹, C. M.
Salgado¹, R. C. Nunes¹, W. L. Salgado¹, J. C. Suita¹

¹ Instituto de Engenharia Nuclear/Divisão de Radiofármacos, 21941-614, Rio de Janeiro, Brasil

² Universidade Federal do Rio de Janeiro, Escola Politécnica/Departamento de Engenharia Nuclear,
21941-909, Rio de Janeiro, Brasil

* paulo.cruz@ien.gov.br

Abstract

Alpha spectrometry involves determining the energy distribution of alpha particles emitted by radioactive sources or produced through nuclear reactions. By employing a surface barrier semiconductor detector, coupled with pulse-processing electronic modules, a multichannel analyzer, and data acquisition and analysis software, the detection system can be calibrated for both energy and efficiency. The high measurement precision of this system is primarily attributed to the near 100% intrinsic detection efficiency of semiconductor detectors for alpha particles and their negligible background typically associated with alpha radiation. Due to these characteristics, alpha spectrometry is widely used in Nuclear Physics for determining half-lives, identifying new nuclides, and analyzing nuclear decay schemes, as well as in Radioprotection and Dosimetry, for assessing biological materials and surface contamination. The goal of this work is to implement an alpha spectrometry system for the characterization of an electrodeposited ^{241}Am source, which will be obtained through the radiochemical processing of this nuclide recovered from decommissioned lightning rods. This nuclide will serve as an alpha particle emitter in Elastic Recoil Detection Analysis (ERDA), a non-destructive analytical technique suited for measuring light elements in heavy matrices. To complement the experimental setup, alpha particle transport through a metallic target was performed using a Monte Carlo simulation with the FLUKA/FLAIR codes to establish the alpha particle energy that avoids nuclear reactions in the ERDA technique. The alpha spectrometry system was implemented and calibrated for both energy and efficiency using individual and mixed alpha-emitting standards. Energy calibration included the determination of peak positions and full-width at half-maximum (FWHM) to assess energy resolution. Efficiency calibration was performed across a range of energies using well-characterized standard sources. The simulations with FLUKA/FLAIR codes were performed to reveal the energy threshold for initiating nuclear reactions with alpha particles in niobium. The results of the efficiency and resolution detection, and FLUKA/FLAIR simulations were discussed.

Keywords: Alpha Particle Spectrometry; Elastic Recoil Detection Analysis (ERDA); FLUKA/FLAIR Monte Carlo simulation codes; Electrodeposited americium-241 (^{241}Am)-standard source; electrodeposited mixed (^{233}U , ^{241}Am , and ^{244}Cm)-standard source.

1 Introduction

Alpha spectrometry is a technique used to determine the energy distribution of alpha particles emitted by radioactive sources or generated through nuclear reactions. Heavy atomic nuclei are often energetically unstable and can spontaneously emit alpha particles—helium-4 nuclei consisting of two protons and two

neutrons. These particles are typically emitted with discrete, well-defined energies, resulting in monoenergetic peaks within the energy spectrum. Each alpha decay corresponds to a specific nuclear transition, most commonly from the ground state of the parent nucleus to the ground state of the daughter nucleus. However, if the daughter is produced in an excited state, multiple transition pathways may occur, each characterized by a distinct Q-value. Consequently, the energy spectrum can display several discrete alpha groups, each associated with a particular transition and exhibiting its own relative intensity (Knoll, 2010).

In practice, the effectiveness of an alpha-emitting source is determined by a careful balance between its emission energy and half-life. Most alpha particles emitted by usable sources fall within the energy range of approximately 4 to 6 MeV. Among these, Americium-241 (^{241}Am) is one of the most widely utilized, especially for calibrating silicon-based solid-state detectors. It emits a dominant alpha line at approximately 5.486 MeV and has a half-life of 432.2 years, providing both excellent stability and sustained activity for long-term applications. Its highly monoenergetic emission makes it particularly well-suited for energy calibration and detector characterization purposes (Vajda et al., 2012).

The objective of this work is to implement an alpha spectrometry system for the characterization of a high-activity, electrodeposited ^{241}Am source. This source will be obtained through the radiochemical processing of americium recovered from decommissioned lightning rods. The resulting nuclide will serve as an alpha particle emitter in Elastic Recoil Detection Analysis (ERDA), a non-destructive analytical technique particularly suited for the quantitative analysis of light elements embedded in heavy matrices (Verma, 2007, Romanenko and Goncharova, 2011), a research effort in progress.

The high measurement precision of alpha spectrometry is largely attributed to the near 100% intrinsic detection efficiency of semiconductor detectors for alpha particles, combined with the typically negligible background associated with alpha radiation. These characteristics make alpha spectrometry a widely employed technique in Nuclear Physics, where it is used to determine half-lives, identify new nuclides, and analyze nuclear decay schemes. Additionally, it plays a crucial role in Radioprotection and Dosimetry, particularly for evaluating biological samples and detecting surface contamination.

Alpha particle transport through a niobium foil target was modeled using a FLUKA/FLAIR Monte Carlo simulation (Ahdida et al., 2022, Donadon et al., 2024), in order to determine the alpha particle energy range that avoids inducing nuclear reactions during the ERDA technique. These simulations aimed to determine the reaction threshold energy, i.e., the minimum energy at which undesired nuclear reactions are initiated in niobium. This threshold is of critical importance in ERDA applications, as the presence of nuclear reactions can produce secondary radiation (such as neutrons, protons, or gamma rays) that compromise the spectrometric purity of the measurement. By identifying the energy range free of nuclear reaction channels, it is possible to optimize the alpha beam energy for purely elastic interactions, ensuring accurate recoil detection without contamination from inelastic or nuclear processes. The results of the simulation provide a predictive model for selecting appropriate operational alpha energies that remain below the nuclear reaction thresholds in niobium, preserving the integrity and specificity of ERDA measurements.

FLUKA is a general-purpose Monte Carlo simulation code designed to model the interaction and transport of particles, such as neutrons, protons, heavy ions, and alpha particles, through matter. It enables highly detailed modeling of complex geometries and physical processes, including nuclear reactions, energy loss, multiple scattering, and residual nuclei production.

To facilitate the use of FLUKA, the FLAIR (FLUKA Advanced Interface) graphical user interface was developed. FLAIR provides an integrated environment for building and visualizing geometries, managing input and output files, executing simulations, and analyzing results (Donadon, A., 2024).

Niobium (Nb), a transition metal, is known for its high melting point, excellent corrosion resistance, and broad range of technological applications. In its pure monoatomic form, niobium can be obtained with a

high degree of purity, typically 99.99%, making it well-suited for advanced research and industrial use. With a density of 8.57 g/cm³, niobium is widely employed in the production of superconducting materials, high-performance alloys, and components for nuclear and aerospace systems. Its monoatomic structure and high purity help minimize interference from impurities, enabling precise characterization in both physical and nuclear experiments.

In practice, niobium foil was readily available in the laboratory and was therefore selected as the target material to prospect irradiation. The data obtained from the alpha irradiation simulations of the niobium foil will also support another ongoing research project. Overall, the combined use of alpha particle spectrometry and Monte Carlo simulation plays a critical role in the characterization of ²⁴¹Am sources intended for application in the ERDA technique.

2 Material and Methods

To implement the alpha spectrometry system, the following electronic modules and equipment were used: an 4001A standard NIM-BIN, an model 807 Vacuum Chamber, an model ED 020-100-100 and Serial No. 15-024F surface barrier detector, an A576 APAD pulse-processing electronic module, an ADC-928 analog-to-digital converter and multichannel analyzer, and the MAESTRO data acquisition and analysis software, all supplied by ORTEC. Additionally, an EDWARDS RV5 vacuum pump was used.

To ensure accurate measurements, a high voltage of 95 volts was applied to the surface barrier detector to optimize charge collection. The alpha spectrometry system was subsequently calibrated in terms of energy and efficiency using certified electrodeposited alpha-emitting standards, both individual and mixed, supplied by the National Metrology Laboratory (LNMRI, Brazil). These standards enabled precise calibration across a range of alpha energies, ensuring reliable identification and quantification of peaks in subsequent measurements.

Energy calibration is essential for establishing the relationship between the channel number recorded by the multichannel analyzer (MCA) and the corresponding alpha particle energy (in MeV), thereby enabling the accurate identification of alpha-emitting radionuclides based on their characteristic energies. The procedure involves the determination of peak positions and the full width at half maximum (FWHM) of the peaks to evaluate the system's energy resolution.

It was carried out using a well-characterized electrodeposited mixed alpha source containing ²³³U, ²⁴¹Am, and ²⁴⁴Cm. The energies of the most intense alpha peaks emitted by these nuclides, were used to construct the energy calibration curve. A polynomial adjustment was applied to the relationship between the measured channel numbers and the corresponding alpha particle energies. This energy calibration was then loaded into the MAESTRO software to enable accurate identification of the alpha particle energies in the acquired spectra.

To evaluate the contribution of electronic noise to the overall energy resolution of the alpha spectrometry system, a test pulse measurement was performed. A precision pulse generator was connected to the preamplifier test input, and the resulting pulse height distribution was acquired using the same signal processing chain employed for actual alpha events. The width of the test pulse peak (typically measured as full width at half maximum, FWHM) reflects the intrinsic electronic noise and allows for distinguishing it from contributions due to statistical fluctuations and charge collection processes in the detector.

The detection efficiency calibration was carried out using a certified ²⁴¹Am standard source (Emission rate = $1766.9 \pm 19.1 \text{ } \alpha \cdot \text{s}^{-1} 2\pi$, on 2025/07/28) by comparing the recorded alpha count rate to the known emission rate of the source, taking into account the measurement geometry and acquisition time. The results of the both energy calibration and efficiency are presented in Table 2 and Figures 1 and 2.

Monte Carlos simulation using FLUKA/FLAIR Monte Carlo simulations were conducted with 5 million primary number to evaluate the interaction of alpha radiation in a bulk a niobium (Nb) foil with a thickness of 125 μm . The primary goal was to investigate potential nuclear reactions that could be induced in niobium within the alpha energy range of 5 to 6 MeV, typical of ^{241}Am sources used in Elastic Recoil Detection Analysis (ERDA). The results are presented in table 3.

3 Results

Table 1 presents the the nuclides peak energies and the respective channels obtained in the MAESTRO from measuring of mixed (^{233}U , ^{241}Am , and ^{244}Cm)-standard source used for calibration of the alpha spectrometry system.

Table 1: Main alpha energies and corresponding channels of the electrodeposited mixed (^{233}U , ^{241}Am , and ^{244}Cm)-standard source used for calibration of the alpha spectrometry system.

Mixed-Nuclide Source	Energy (keV)	Intensity (%)	Channel
^{233}U	4824,2 (12)	84,3 6	173.74
^{241}Am	5485.56 (12)	84.8 (5)	293.18
^{244}Cm	5804.77 (5)	76.9	347.9

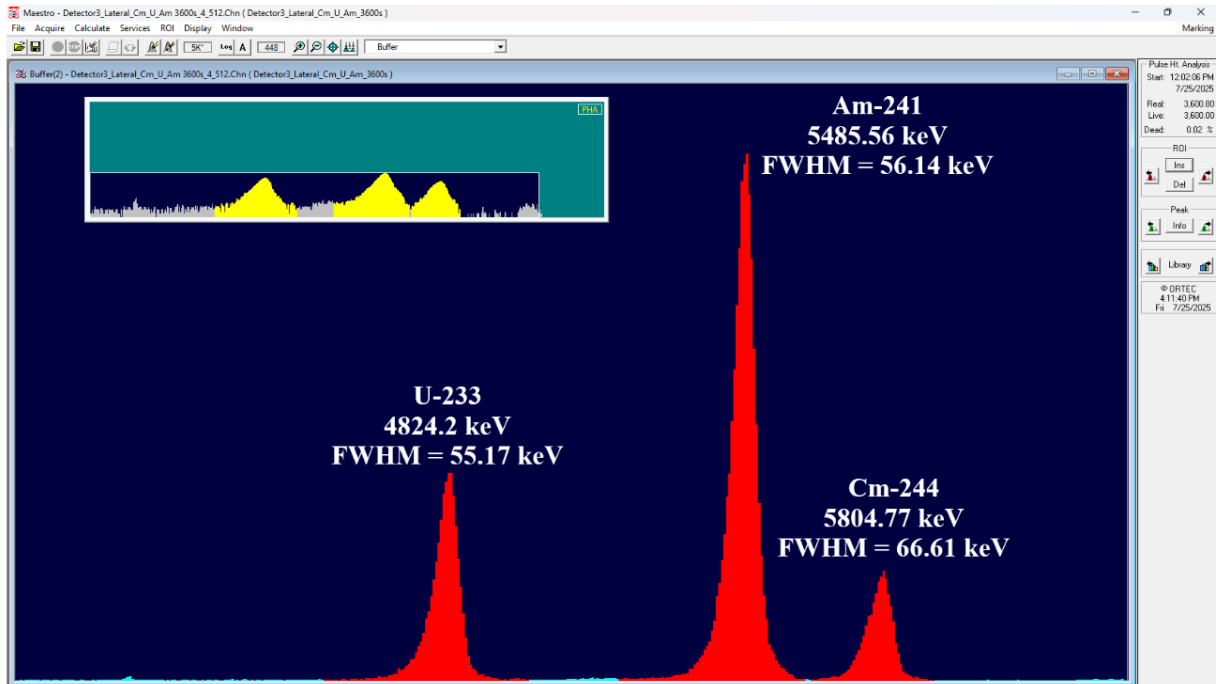


Figure 1: Nuclides spectra obtained by measurement a mixed (^{233}U , ^{241}Am , and ^{244}Cm)-standard source in the alpha spectrometry system.

Figure 1 presents the spectra of the nuclides obtained from measurements used for the calibration of the alpha spectrometry system. The excerpt highlights a measured energy resolution of 56.14 keV for the ^{241}Am peak, which contrasts with the detector's nominal resolution of 20 keV. However, the observed peaks exhibited sufficient resolution for accurate identification and energy calibration of the alpha-emitting radionuclides.

The spectra were analyzed to evaluate the contributions of various factors to the degradation of the energy resolution observed in Figure 1. In particular, a pulser signal was applied to the electronic chain to verify the extent of its contribution to the overall resolution. This procedure allows for the separation of electronic noise from other resolution-broadening effects such as source thickness, self-absorption, and energy straggling. Figure 2 shows the spectrum obtained using a pulser applied to the electronic chain, representing the contribution of electronic noise to the system's resolution. The pulser peak exhibited a resolution of 30 keV, which is significant for evaluating the overall resolution observed in the alpha spectrometry system.

The age of the surface barrier detector also contributed to the degradation of the peak signal, as the detector had remained unused for an extended period prior to the measurements. The prolonged storage and age of the surface barrier detector likely affected its performance, contributing to signal degradation and a broader energy resolution. Detectors that remain unused for long periods may experience issues such as increased leakage current or reduced charge collection efficiency.

Additionally, the prolonged storage and age of the surface barrier detector likely affected its performance, contributing to signal degradation and a broader energy resolution. Detectors that remain unused for extended periods may suffer from issues such as increased leakage current or reduced charge collection efficiency. Despite the resolution being broader than expected for the alpha spectrometry system, it was still sufficient to resolve the peaks of the mixed nuclide source, allowing for successful energy calibration. A study using a certified mixed standards of ^{239}Pu , ^{241}Am , and ^{244}Cm presented similar spectra (Bramha and Chandrasekaran, 2024).

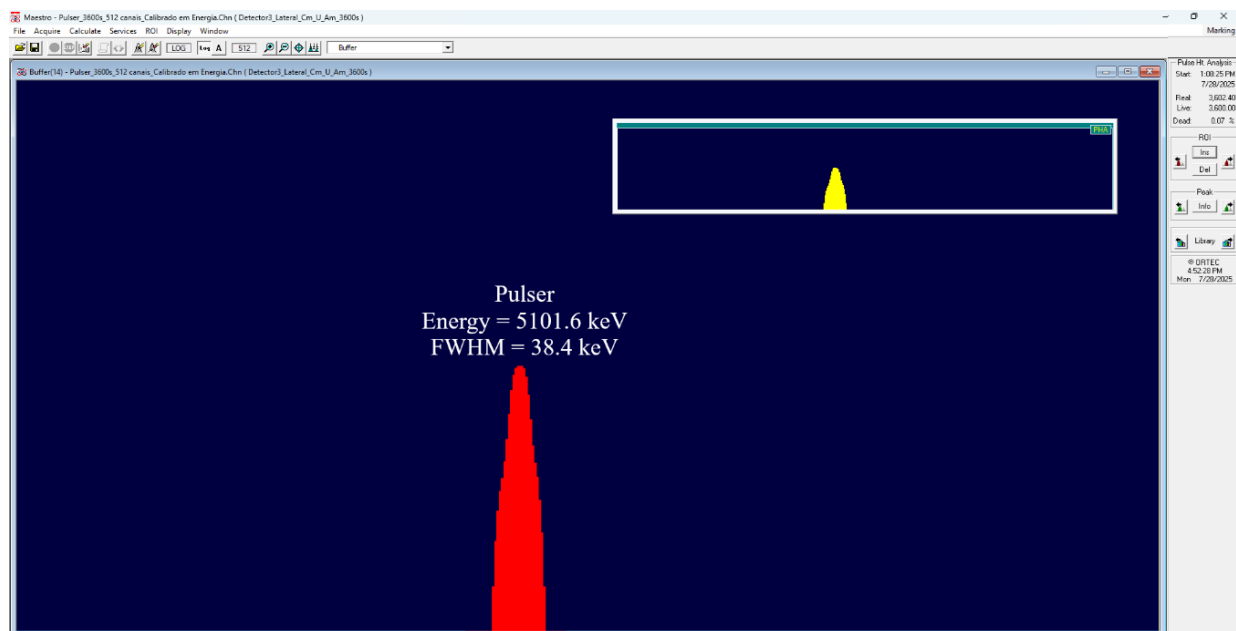


Figure 2: Pulser signal applied to the electronic chain to assess the electronic noise contribution to the energy resolution of the alpha spectrometry system.

The energy calibration curve, presented in Figure 3, exhibited a good correlation, as a second-order polynomial fit was applied to the measured relationship between the channel numbers and the corresponding alpha particle energies, ensuring accurate identification of spectral peaks.

The results of the measurements performed to obtain the counting efficiency using the electrodeposited ^{241}Am standard are presented in Table 2. The measured efficiency values demonstrate excellent repeatability, with only minimal variation among replicates.

This consistency indicates a stable and reliable measurement process, confirming both the instrumental stability and the reproducibility of the experimental setup. Such results validate the integrity of the source preparation, detector response, and overall system calibration, ensuring that the efficiency values can be confidently used for subsequent analyses and quantifications.

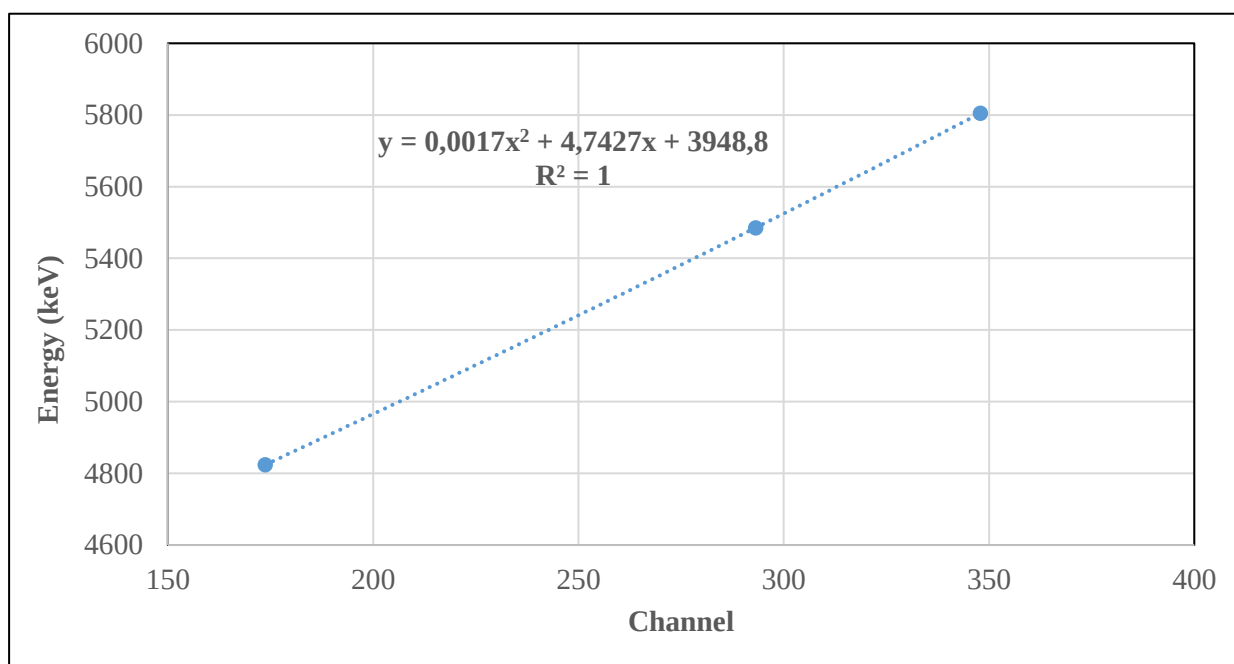


Figure 3: Calibration curve obtained by count a mixed (^{233}U , ^{241}Am , and ^{244}Cm)-standard souce in the alpha spectrometry system.

Table 2: Results of the measurements performed to determine the counting efficiency of the alpha spectrometry system using an electrodeposited ^{241}Am standard source.

Nuclide	Emission rate (k = 1)	Counts	Mean value	Count Efficiency	Mean value
	$\alpha.s-1\ 2\pi$ (2025/07/28)	(keV)	$\alpha.s-1\ 2\pi$	(%)	(%)
^{241}Am	1736.6 (19.1)	141.38 (25)		8.14	
		141.51 (23)	141.53 (16)	8.15	8.15 (1)
		141.69 (23)		8.16	

In the aforementioned study (Bramha and Chandrasekaran, 2024) the α -particle detection efficiency was evaluated using certified mixed standards of ^{239}Pu , ^{241}Am , and ^{244}Cm with 7 mm active diameter. For a source–detector distance of 10 mm and an ORTEC ULTRA-AS detector, model BU-020-450-AS (450 mm² active area), a typical ^{241}Am (5 mm active diameter) peak efficiency of ~11 % was reported. In the present work, a similar ^{241}Am efficiency of ~8 % was obtained at the same 10 mm distance using a detector model ED-020-100-100 with a smaller active area (100 mm² active area). Despite the geometric and

model differences between detectors, the agreement suggests that the effective acceptance is limited by the measurement geometry, showing that the results obtained are consistent.

Table 3: Output data from the FLUKA/FLAIR Monte Carlo simulation describing the interaction of alpha particles with a niobium (Nb) foil.

Alpha Energy (MeV)	A/Z Isotopes	A/Z Isotopes_Activity (Bq)	A/Z Isomers	A/Z Isomers_Activity (Bq)
5.6	-	-	-	-
5.8	⁹⁷ Mo	6.09E-06	^{97m} Mo	6.41E+01
6	⁹⁷ Mo	6.40E-06	⁹⁷ Mo	6.07E+01

The results presented in Table 3 show that the energy threshold for initiating the nuclear reaction with alpha particles is approximately 5.8 MeV. This indicates that ²⁴¹Am, which emits alpha particles with an energy of 5.486 MeV, can be used in the ERDA (Elastic Recoil Detection Analysis) technique for surface analysis or with lighter target elements, and may not have sufficient energy to induce nuclear reactions in heavier elements such as niobium.

4 Conclusions

This work successfully implemented and validated an alpha spectrometry system for the characterization of a high-activity electrodeposited ²⁴¹Am source, to be produced from americium recovered from decommissioned lightning rods. The system was calibrated in energy and efficiency using a mixed (²³³U, ²⁴¹Am, and ²⁴⁴Cm) and ²⁴¹Am standard sources, achieving sufficient resolution and excellent counting efficiency repeatability. Despite the detector's age and contributions from electronic noise, the system demonstrated reliable performance for peak identification and energy calibration. Monte Carlo simulations indicated that the 5.486 MeV alpha particles from ²⁴¹Am are suitable for surface-level ERDA applications, particularly for the analysis of light elements in heavy matrices.

Acknowledgements

Our thanks to the Fundação de Amparo à Pesquisa do Estado do Rio de Janeiro (FAPERJ) for supporting this research work with resources provided under project E-26/211.667/2021 (269606) – APQ1.

References

Ahdida, C., Bozzato, D., Calzolari, D., Cerutti, F., Charitonidis, N., Cimmino, A., Coronetti, A., D'Alessandro, G. L., Donadon Servede, A., Esposito, L. S., Froeschl, R., García Alía, R., Gerbershagen, A., Gilarioni, S., Horváth, D., Hugo, G., Infantino, A., Kouskoura, V., Lechner, A., Lefebvre, B., Lerner, G., Magistris, M., Manousos, A., Moryc, G., Ogallar Ruiz, F., Pozzi, F., Prelipcean, D., Roesler, S., Rossi, R., Sabaté Gilarte, M., Salvat Pujol, F., Schoofs, P., Stránský, V., Theis, C., Tsinganis, A., Versaci, R., Vlachoudis, V., Waets, A., Widorski, M., (2022). New Capabilities of the FLUKA Multi-Purpose Code, *Frontiers in Physics* 9, 788253.

Bramha, S. and Chandrasekaran, S. (2024). Evaluation and assessments of efficiency calibration and resolution of an alpha spectrometry system using standard sources. *Radiation Protection and Environment*, 47(3):p 205-210.

Donadon, A., Hugo, G., Theis, C., Vlachoudis, V. (2024). FLAIR3 – recasting simulation experiences with the Advanced Interface for FLUKA and other Monte Carlo codes, *EPJ Web Conf. Volume 302*, (SNA + MC 2024).

Knoll, G. (2010). *Radiation Detection and Measurement*, John Wiley & Sons, 4th Edition.

Romanenko, A. and Goncharova, L. V. (2011). Elastic recoil detection studies of near-surface hydrogen in cavity-grade niobium. *Supercond. Sci. Technol.* 24 105017.

Vajda, N., Martin, P., Kim, C-K. (2012). Alpha Spectrometry. In: L'Annunziata MF (ed) *Handbook of radioactivity analysis*, 3rd edn. Academic Press, San Diego.

Verma, H. R. *Atomic and Nuclear Analytical Methods* (2007). Springer-Verlag Berlin Heidelberg 123.

Axial neutron flux mapping in the Argonauta Research Reactor at IEN-CNEN using neutron activation and γ -ray spectrometry techniques

R. C. Nunes^{1*}, G. de O. Barreto², M. de L. Moreira¹, P. A. L. da Cruz¹, E. M. de Oliveira¹, C. M. Salgado¹, L. Carvalheira¹, A. P. Junior¹, A. L. N. Barbosa¹

¹ Comissão Nacional de Energia Nuclear, Instituto de Engenharia Nuclear/Divisão de Radiofármacos, 21941-614, Rio de Janeiro, Brasil

² Comissão Nacional de Energia Nuclear, Instituto de Engenharia Nuclear/Programa de Pós-graduação do Instituto de Engenharia Nuclear, 21941-614, Rio de Janeiro, Brasil

* rogerio.chaffin@ien.gov.br

Abstract

The density and spatial distribution of neutron flux inside a nuclear reactor exhibit non-uniform behavior, which results from the interactions that occur through moderation, multiplication, and neutron absorption. These interactions are influenced by the materials of the fuel, moderator, and reflector present in the system. Neutrons, after multiple interactions and collisions, tend to concentrate in the central region of the nucleus and have a greater probability of dispersing to areas of lower density, which are its extremities (Duderstadt & Hamilton, 1976). Understanding the spatial distribution of neutron flux allows for improvements in fuel element design, optimization of the core geometry, ensuring operational safety of the reactor under different operating regimes, and optimizing experimental conditions. Additionally, these data can be used to validate numerical methods. The objective of this research is to experimentally observe, through indirect measurement, the behavior of the curvature of the neutron flux in the axial axis of the Argonauta Research Reactor at the Instituto de Engenharia Nuclear (IEN/CNEN), under routine operational conditions, with a power of 340 W and a neutron flux on the order of 10^9 n/cm²s. The methodology employed in this work combined Neutron Activation (Kruger, 1971) and Gamma-Ray Spectrometry (Knoll, 2000) techniques. In the first technique, radioactive sources were produced by irradiating thirteen small dysprosium (¹⁶⁴Dy) wires distributed on one of the legs of a U-shaped Lucite ruler, sequentially and equidistant from each other, and positioned axially between plates 2 and 3 of fuel element number 6. In the second technique, the acquisition and analysis of experimental data of the radioisotope ¹⁶⁵Dy, formed by the (n, γ) reaction, are performed. An ORTEC Hyperpure Germanium (HPGe) detector with 25% efficiency and 1.8 keV resolution at the 1332 keV photopeak of ⁶⁰Co, Maestro v7.0 Software and DSPEC jr 2.0 electronic system were used. The result of this experiment confirmed the theoretical hypothesis presented, since, through qualitative analysis, it is possible to graphically visualize the distribution of the relative and normalized neutron flux in the fuel element as a function of position

Keywords: Neutron Activation; Gamma-Ray Spectrometry; Neutron Flux Mapping; Argonauta Research Reactor.

1 Introduction

The dynamics of neutrons in a nuclear reactor are broadly described by transport theory, which takes into account spatial, angular, energy, and temporal dependencies. Although this approach allows predicting

the reactor's response to changes in core composition or geometry, the simplifications adopted in the calculations, combined with uncertainties in nuclear data and limitations in the models, result in approximate solutions. Therefore, experimental measurements are essential both to validate the models and to adjust them to the observed physical reality (Bitelli, 1988).

According to theory, both the density and the spatial distribution of neutron flux inside a nuclear reactor are, therefore, non-uniform. This non-uniformity is a consequence of moderation, multiplication, and absorption interactions, which directly depend on the properties of the materials that make up the core: fuel, moderator, and reflector. After successive collisions, neutrons tend to concentrate in the central region of the core, spreading more likely to the peripheral regions (Duderstadt & Hamilton, 1976).

The distribution of neutrons in the core of a reactor is directly related to the location of the neutron source, that is, to the arrangement of the fuel elements. In the case of fission, this distribution is influenced by the very presence of the neutrons and the interactions that occur with atomic nuclei as the neutrons move away from their origin (Stacey, 2003).

From the perspective of diffusion theory, which is an approximation of the transport equation based on Fick's Law, the net neutron flux occurs only when there are spatial gradients in the flux. Thus, neutrons tend to migrate from regions of higher to regions of lower flux density, analogously to the diffusion of solutes in continuous media (Lamarsh & Baratta, 2001).

Therefore, understanding the spatial distribution of neutrons in the core of a reactor is essential to optimize the design of fuel elements, core geometry, operational safety, and experimental conditions (Bowen and Busch, 2023). Abstracting the complexity and mathematical formalism of neutron transport theory, this work — purely experimental in nature — aimed to observe, through indirect measurement, the curvature of the neutron flux along the axial axis in the Argonauta Research Reactor (IEN/CNEN), routinely operating at 340 W, with flux on the order of 10^9 n/cm²·s. The methodology employed combined Neutron Activation techniques (Kruger, 1971), (Bitelli, 1988) and (IAEA-TECDOC-564, 1990) and Gamma Spectrometry (Knoll, 2000) and (Gilmore and Joss, 2024). Dysprosium wires (¹⁶⁴Dy) were irradiated, positioned in fuel element 6. The analysis of the ¹⁶⁵Dy radioisotope formed by the (n, γ) reaction was performed with an HPGe detector (25% efficiency, 1.8 keV resolution at 1332 keV).

2 Materials and Methods

As mentioned in the introduction, the results of this experiment were obtained through a combination of Neutron Activation Analysis and Gamma Spectrometry techniques. The details regarding the materials, equipment, and procedures used are presented in the following subsections of this section.

2.1 General Overview of the Argonauta Research Reactor at IEN/CNEN

The Argonauta research reactor, shown in Figure 1, was developed at Argonne National Laboratory (USA) and built in Brazil by CBV Ltda. for the IEN/CNEN. It reached criticality on February 20, 1965. Designed for research, training, and experimental testing, the reactor is notable for its inherent safety, low operational cost, and easy experimental access, making it ideal for educational use and low-power nuclear research.



Figure 1: Argonauta research reactor – IEN/CNEN (RJ/Brazil).

The core of the Argonauta reactor features a concentric cylindrical configuration, consisting of two aluminum cylinders. The fuel is a cermet of U_3O_8 dispersed in an aluminum matrix, enriched to 19.91% in ^{235}U , supplied by the United States and manufactured by IPEN/CNEN. The fuel elements (FE) are arranged in a “blade-type” configuration, meaning they are distributed as a ring segment, as illustrated in Figure 2.

The current core loading includes four FEs with 17 plates each (~ 20 g of ^{235}U per element), two FEs with 11 plates plus 6 additional plates (~ 30 g of ^{235}U in total), and two FEs with 7 plates each (~ 10 g of ^{235}U per element). The configuration is complemented by a half graphite prism, which functions as a neutron reflector (Albuquerque, 2020).

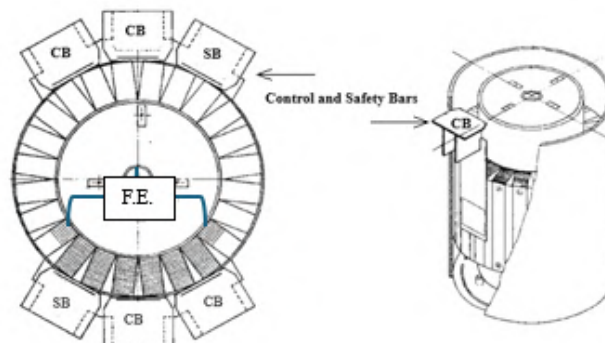


Figure 2: Argonauta core - a “Blade Type” configuration.

The nominal power of the reactor is 5 kW, and it is licensed by CNEN, the Brazilian nuclear regulatory authority, to operate continuously at 500 W, with the possibility of reaching up to 1 kW for a maximum duration of one hour. However, its routine operational power is typically limited to 340 W.

The reactor uses deionized water as a moderator, pure graphite as a reflector, and biological shielding made of concrete. The control and safety rods are made of cadmium and operate by gravity, ensuring a passive and inherently safe response.

Figure 3 shows an identical replica of the original fuel element, which consists of 17 plates.



Figure 3: Identical replica of the original FE.

2.2 Configuration of Dy Wire Samples for Axial Neutron Flux Mapping

Dysprosium was chosen as the activation material due to its particularly favorable nuclear properties for the detection of thermal neutrons. The isotope ^{164}Dy , with a natural abundance of 28.26%, has a high thermal neutron capture cross section — $(2,650 \pm 25)$ barns — making it highly sensitive to the presence of these neutrons. Upon irradiation, the radionuclide ^{165}Dy is formed, which is a gamma emitter with a suitable half-life ($T_{1/2} = 8,391.6$ s), allowing measurements to be taken with good temporal resolution and without the need for long waiting periods after irradiation (IAEA, 2025).

Additionally, dysprosium is available in metallic or wire form, facilitating its insertion into specific regions of the reactor core with minimal disturbance to the neutron flux field. Finally, the characteristic gamma energy peaks emitted by ^{165}Dy — especially the most intense one at 94.7 keV — are well defined and easily detectable by gamma spectrometry systems, contributing to the accuracy and reliability of the induced activity analysis.

To measure the axial neutron flux profile in the core of the Argonauta Reactor, thirteen dysprosium (Dy) wires were used as neutron activation detectors. The wires were evenly spaced and fixed with adhesive tape along a polymethylmethacrylate (PMMA, Lucite) ruler with cavity holders, each wire positioned in an equally spaced cavity. The assembly was then inserted between plates 2 and 3 of fuel element number 6 (FE-No.6), as illustrated in Figure 4(a), Figure 4(b), and Figure 5.

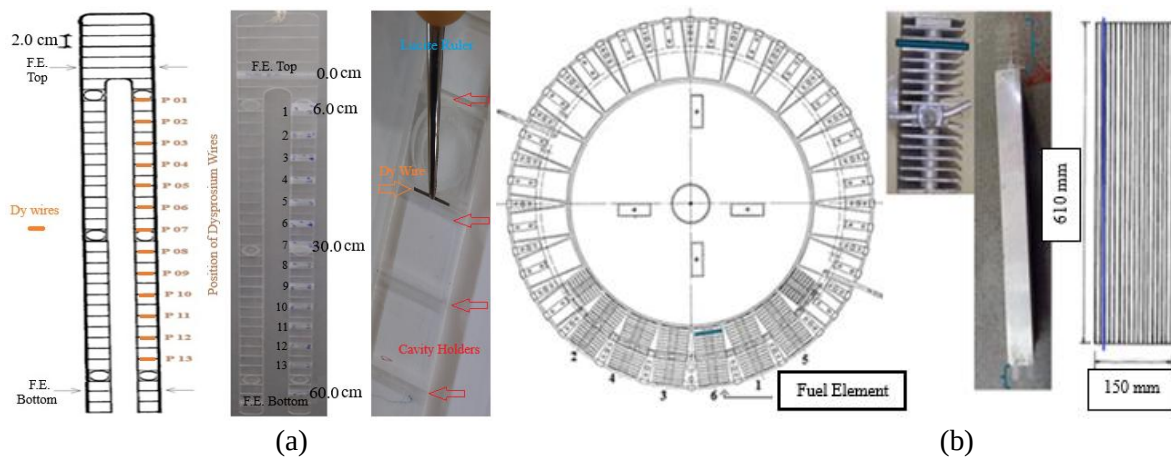


Figure 4: Configuration of dysprosium wires. (a) Positioning of the Dy wires on the Lucite ruler with emphasis on the cavity holders. (b) Insertion position of the Lucite ruler within the reactor core highlighted in blue.

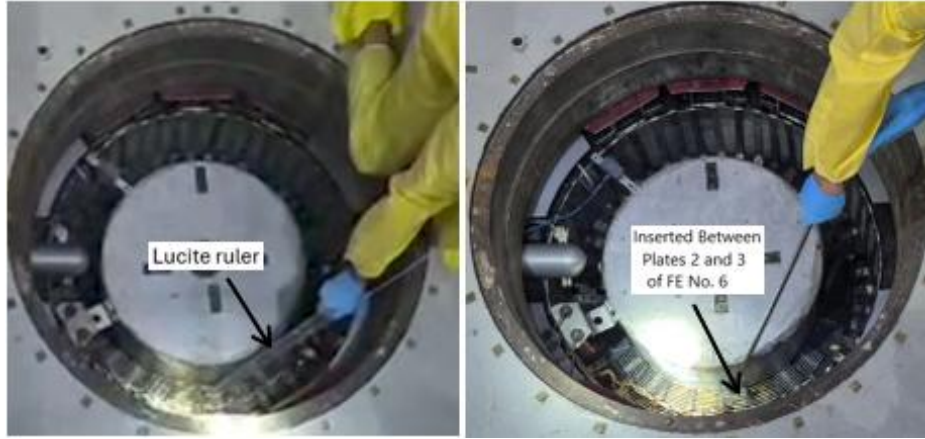


Figure 5: Insertion of the Lucite ruler between fuel plates of FE No.6.

Each wire, with a diameter of 0.711 cm and precisely cut to a length of 1.1 cm, was made of a 10% Dy–Al alloy supplied by Reactor Experiments, Inc. (USA), providing suitable sensitivity for neutron activation. Their masses were then determined using a Shimadzu ATY224 balance, which provides four-decimal precision and an associated uncertainty of 0.00005 g, with each wire containing (0.0013 ± 0.0001) g of Dy.

2.3 Flux Mapping via Neutron Activation and Gamma-Ray Techniques

Neutron detection is associated with the emission of ionizing particles, since neutrons, being electrically neutral, cannot be detected directly. To achieve detection, nuclear reactions are induced that result in the emission of detectable secondary radiation. Neutron flux measurement is based on the induced activity in certain materials exposed to the neutron field; these materials are known as activation detectors. Because capture cross sections are higher for thermal neutrons, activation detectors are widely used for measuring this type of radiation. To minimize disturbance of the neutron flux field, these detectors are usually manufactured in small geometries, such as wires or thin foils (Bitelli, 1988).

There is a proportionality relationship between Activation and Flux, expressed by Equation 1 and Equation 2 (Kruger, 1971) and (Kimeridze and Nguyen, 2023), as follows.

$$A_0^i = \phi * Const. \quad (1)$$

$$\Sigma_{act} * (1 - e^{-\lambda t_i}) = Const. \quad (2)$$

Where (A_0^i) represent the activity of the activation detector at the end of the irradiation and (ϕ) the neutron flux, respectively. (Σ_{act}) is the macroscopic activation cross section, (λ) the decay constant, and (t_i) the irradiation time, respectively — all of which are known and constant values, allowing the neutron flux to be determined from the measured activity.

This research does not aim to calculate the neutron flux quantitatively but rather to conceptually use the flux–activation relationship defined in Equation 1 to perform a simple qualitative analysis focused on mapping the relative neutron flux. For this purpose, the activity of each wire must be calculated at the reference time (A_0^i) , corresponding to the exact end of irradiation. To ensure accuracy, dead-time

corrections (DTC) were applied to the net peak areas ($a_{n(DTC)}^i$) of each spectrum. According to the literature, the dead time (DT^i) in each spectrum should remain below 30% to ensure reliable qualitative results (Gilmore and Joss, 2024).

The method for obtaining dead-time corrected net areas is presented in Equation 3, along with its associated uncertainty in Equation 4:

$$a_{n(DTC)}^i = \left(\frac{a_n^i}{1-DT^i} \right) \quad (3)$$

$$\frac{\delta a_{n(DTC)}^i}{a_{n(DTC)}^i} = \frac{\delta a_n^i}{a_n^i} \quad (4)$$

Following the dead-time correction, the activity at the reference time is denoted as ($A_{0(DTC)}^i$), and its uncertainty is given in Equation 6:

$$A_{0(DTC)}^i = \left(\frac{a_{n(DTC)}^i / t_L^i}{I_\gamma * \epsilon_\gamma} \right) * e^{\lambda * t_w^i} \quad (5)$$

$$\frac{\delta A_{0(DTC)}^i}{A_{0(DTC)}^i} = \frac{\delta a_{n(DTC)}^i}{a_{n(DTC)}^i} \quad (6)$$

Next, the ratio between the dead-time corrected activity and the mass of each wire (m^i) is calculated, determining the dead-time corrected specific activity of each Dy wire ($A_{0(DTC)}^{i,m^i}$), as shown in Equation 7, with its associated uncertainty provided in Equation 8:

$$A_{0(DTC)}^{i,m^i} = \frac{A_{0(DTC)}^i}{m^i} \quad (7)$$

$$\frac{\delta A_{0(DTC)}^{i,m^i}}{A_{0(DTC)}^{i,m^i}} = \sqrt{\left(\frac{\delta A_{0(DTC)}^i}{A_{0(DTC)}^i} \right)^2 + \left(\frac{\delta m_i}{m_i} \right)^2} \quad (8)$$

Where (i) represents the index identifying each irradiated wire; (a_n^i) and $a_{n(DTC)}^i$ are the net count area and the dead-time corrected net count area of the characteristic gamma energy peak of ^{165}Dy , obtained individually for each wire through gamma spectrometry; (t_L^i) corresponds to the live time of spectrum acquisition; (I_γ) is the relative intensity of the selected gamma energy peak; (ϵ_γ) represents the efficiency of the HPGe detector for the considered gamma energy in the adopted counting geometry; (t_w^i) is the waiting time, defined as the interval between the end of irradiation and the start of spectrometric acquisition; (λ) refers to the decay constant of the ^{165}Dy radionuclide; and (δ) represents the uncertainties associated with each of the quantities involved in the calculation.

In general, to obtain the relative flux distribution curve (mapping) in a reactor region, it is not necessary to calculate the flux at any point. It is sufficient to distribute the activation detectors in the area of interest while keeping the following parameters constant: the detector material, its mass and dimensions, the distance between the detectors, the same irradiation time, and the same counting/data acquisition position in the spectrometry system. Under these conditions, the induced specific activity measured in each detector directly reflects the relative variation of the neutron flux in each position of the analyzed region.

The graphical visualization of the mapping is directly obtained by plotting a graph of the dead-time corrected specific activity ($A_{0(DTC)}^{i,m^i}$) as a function of the wire positions within the reactor core, allowing for a clear spatial interpretation of the relative flux distribution.

In this research, the relative neutron flux mapping was carried out through gamma spectrometry analysis (Knoll, 2000) of dysprosium (Dy) wires irradiated in the core of the Argonauta reactor, positioned between plates 2 and 3 of fuel element No. 6. The arrangement of the wires was chosen to allow the assessment of the variation in neutron flux along the axial direction of the core.

For the acquisition of gamma spectra, a system consisting of a High-Purity Germanium (HPGe) detector manufactured by ORTEC was used. This detector features 25% relative efficiency and a resolution of 1.8 keV at the 1332 keV photopeak of the isotope ^{60}Co . The detector was coupled to a DSPEC jr 2.0 digital electronics system and is located at the LMN/SEREA/IEN laboratory.

The collection, processing, and storage of the spectrometric data were performed using Maestro v7.0 software, installed on a dedicated computer. This setup enabled visualization, energy calibration, and detailed analysis of the acquired spectra. Additionally, the system includes lead shielding to protect the detector from background radiation.

3 Results and Discussions

The Dy wires were irradiated under the operational conditions presented in Table 1. The samples were fixed to a Lucite ruler and inserted into the core of the Argonauta reactor, as shown in Figure 4(a), Figure 4(b), and Figure 5.

Table 1: Operational Data of the Irradiation.

Date	Time	Power	Location	Control Rod Position (%)
May 13, 2025	30 min	340 W	Core (FE No.6)	>80 (withdrawn)

For the proper application of the gamma spectrometry technique in this work, regarding counting efficiency in the treatment of wire-shaped sources, the following counting geometry was adopted: each Dy wire (1.1 cm) was placed flat on a spacer, at a distance of 6 cm from the HPGe detector cap, as illustrated in Figure 6. In the analysis of this source–detector system, considering the ratio between the spacer–detector distance (d) and the Dy wire length (L) ($d/L \approx 5.45$), the sources could be treated as point-like, since the distance exceeds three times the sample dimension (Bevelacqua, 2004), making any additional efficiency correction unnecessary.

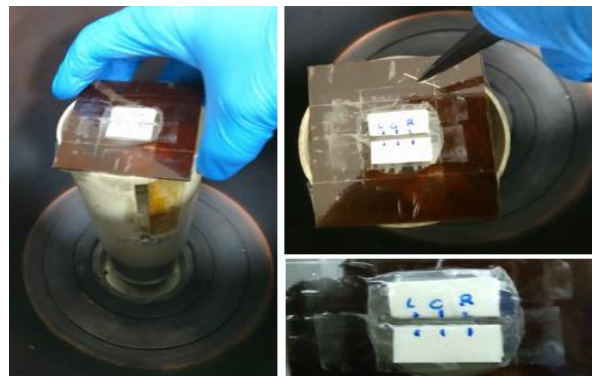


Figure 6: Source–detector geometry showing the spacer and the sample holder.

Figure 7 shows the spectrum of the ^{165}Dy wire at position 8, obtained with the ORTEC HPGe detector, highlighting the 94.7 keV peak—the most intense—which was used in all calculations. The live time counting for each spectrum was set to 180 s.

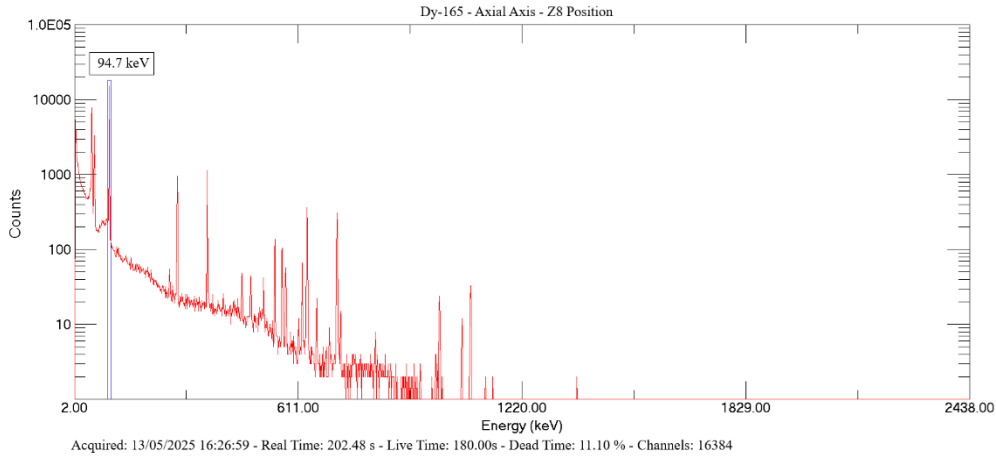


Figure 7: Gamma spectrum of the ^{165}Dy wire at position 8.

The distribution of the specific activity measured for ^{165}Dy along the wires revealed a relative neutron flux profile with maximum concentration in the central region, decreasing symmetrically toward the extremities. This distribution is consistent with theoretical expectations for graphite-moderated cores and reflects the reactor’s geometry and materials (Duderstadt and Hamilton, 1976) and (Lamarsh and Baratta, 1972).

The relative neutron flux and the relative neutron flux with dead-time correction were normalized by the highest specific activity value and are presented in Table 2, as well as graphically in Figure 8, allowing for a qualitative observation of the axial flux profile. The observed “parabolic” shape is consistent with the solutions of the diffusion equation for homogeneous thermal reactors.

It should be noted that the fuel element measures approximately 61 cm, and the first Dy wire is positioned about 6.0 cm below the top of the fuel element, while the last wire is located approximately 7.0 cm above the bottom, as illustrated in Figure 4(a).

Table 2: Normalized Relative Neutron Flux (NRNF) along FE no.6. with Dead-Time Correction (DTC) and Dead-Time (DT).

Dy Wire (No.)	F.E. Position (cm)	NRNF	NRNF (DTC)	DT (%)
1	6.0	0.56±0.01	0.54±0.01	7.37
2	10.0	0.69±0.01	0.67±0.01	8.87
3	14.0	0.78±0.01	0.76±0.01	9.58
4	18.0	0.86±0.01	0.85±0.01	10.34
5	22.0	0.99±0.01	1.00±0.01	11.64
6	26.0	0.98±0.01	0.98±0.01	11.25
7	30.0	0.99±0.01	0.99±0.01	11.08
8	34.0	1.00±0.01	1.00±0.01	11.10
9	38.0	0.95±0.01	0.95±0.01	10.45
10	42.0	0.93±0.01	0.91±0.01	9.88
11	46.0	0.77±0.01	0.74±0.01	8.05
12	50.0	0.75±0.01	0.72±0.01	7.64
13	54.0	0.60±0.01	0.57±0.01	6.07

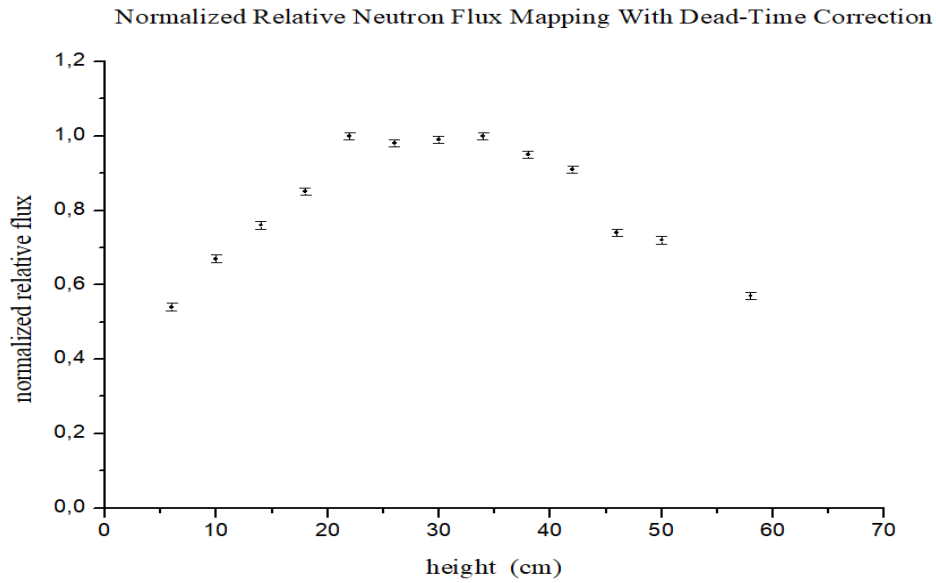


Figure 8: Normalized Neutron Flux Mapping with Dead-Time Correction and Error Bars.

4 Conclusions

The experimental methodology adopted in this research enabled the qualitative mapping of the axial neutron flux distribution between plates 2 and 3 of Fuel Element No. 6 in the core of the Argonauta Research Reactor at IEN/CNEN, under routine operational conditions at a power of 340 W, with control rods 80% withdrawn, over a 30-minute period. The combination of dysprosium wire neutron activation with analysis via gamma spectrometry using a high-resolution HPGe detector allowed for the indirect observation of the neutron flux behavior along the axial axis of the referenced fuel element. The results obtained corroborate theoretical predictions described by neutron transport and diffusion theories, particularly the tendency for neutron concentration in the central region of the core and dispersion toward the extremities. The relative and normalized flux profile generated from the measurements demonstrates the feasibility of the proposed method for the qualitative evaluation of the flux distribution. This type of experimental validation is essential for improving the understanding of neutron behavior in research reactors and for supporting the refinement of numerical models and core optimization strategies.

However, the study presents certain limitations, such as the spatial restriction of sampling to a single axial column within the core. For future work, it is recommended to carry out mappings in other regions of the core, including the radial plane. Comparing the experimental data with computational simulations using transport and diffusion codes also appears promising to validate and refine models applicable to the Argonauta Reactor.

Acknowledgements

The authors would like to thank the Instituto de Engenharia Nuclear (IEN/CNEN) for the technical support and the laboratory infrastructure provided, as well as the Argonauta Reactor Service Team for the operational assistance during the irradiations.

References

- Albuquerque, B. F. (2020) *Técnicas analíticas para a análise qualitativa e quantitativa em amostras de cimento Portland*. Rio de Janeiro: CNEN/IEN
- Bevelacqua, J. J. (2004) Point source approximations in Health Physics. *Radiation Protection Management*. Vol. 21 (5): 09–13
- Bitelli, U. d'U. (1988) *Medida e cálculo da distribuição espacial e energética de neutrons no núcleo do reator IEA-R1*. São Paulo: IPEN/CNEN
- Bowen, D. G. and Busch, R. D. (2023) *Hand calculation methods for nuclear criticality safety*. ORNL
- Duderstadt, J. J., & Hamilton, L. J. (1976) *Nuclear reactor analysis*. John Wiley & Sons
- Gilmore, G.; Joss, D. (2024) *Practical gamma-ray spectrometry*. John Wiley & Sons
- IAEA. (2025.) *Live chart of nuclides*. IAEA Nuclear Data Section
- IAEA-TECDOC-564. (1990) *Practical aspects of operating a neutron activation analysis laboratory*. Austria: IAEA
- Kimeridze, T., and Nguyen, T. C. H. (2023). Neutron Flux Measurement in TRIGA Reactor using Gold Foil Activation Method. *Georgian Scientists*. Vol. 5 Issue 3:189-205
- Knoll, G. F. (2000) *Radiation detection and measurement*. John Wiley & Sons
- Kruger, P. (1971) *Principles of activation analysis*. John Wiley & Sons
- Lamarsh, J.R. and Baratta, A. J. (1972) *Introduction to nuclear reactor theory*. Addison-Wesley
- Stacey, W. M. (2007) *Nuclear reactor physics*, WILEY-VCH Verlag GmbH & Co .KGaA

Comparative Study of Gamma Efficiency Curves in HPGe Detector with Different Calibration Standards

E. M. de Oliveira^{1*}, A. G. M. dos Santos^{1,2}, P. A. L. da Cruz¹, R. C. Nunes¹,
R. S. F. Dam¹, C. M. Salgado¹, W. L. Salgado¹

¹ Comissão Nacional de Energia Nuclear, Instituto de Engenharia Nuclear, Divisão de Radiofármacos, CEP: 21941-614, Rio de Janeiro, Brasil

² Universidade Federal do Rio de Janeiro, Centro de Tecnologia, Escola Politécnica, Departamento de Engenharia Metalúrgica e de Materiais, CEP 21941-901, Rio de Janeiro, Brasil

* corresponding.author: estela.oliveira@ien.gov.br

Abstract

High-purity germanium (HPGe) detectors are widely used in high-resolution gamma spectrometry due to their excellent response and ability to accurately and precisely discriminate gamma-ray energies. To ensure reliable quantitative analyses, it is essential to determine the absolute efficiency curve, which relates the detection efficiency to the energy of the incident gamma photons.

This work presents the determination of the absolute efficiency curve of a high-purity Germanium (HPGe) detector through calibration using two certified standards provided by the National Laboratory for Metrology of Ionizing Radiation (LNMRI/IRD), containing different gamma-emitting radionuclides. Standard 20L24 contains ²⁴¹Am and ¹⁵²Eu, while standard 23L24 contains ²⁴¹Am, ¹³³Ba, ¹³⁴Cs, ¹³⁷Cs, ⁶⁰Co, and ¹⁰⁶Ru. These selected standards cover a wide gamma energy range, from approximately 59 keV to 1408 keV, enabling a precise adjustment of the detector's response.

The objective of this work is to determine the efficiency calibration curve, with the estimate of the associated uncertainties, for each standard and select the most appropriate curve for the counting geometry used, based on a Gamma sample from the 2024 round of the Programa Nacional de Intercomparação (PNI - Proficiency Test in environmental samples).

Measurements were performed in a fixed geometry, with the standards positioned directly on top of the detector. A counting time of 259,200 seconds was used to ensure high statistical quality. Peak areas were determined through spectral analysis using the Genie 2000 software, and efficiencies were calculated based on the certified activity of the standards, counting time, and gamma emission probabilities of the corresponding radionuclide energies. The results are presented along with their uncertainty estimates for each efficiency curve.

Among the efficiency curves obtained, the curve corresponding to the 20L24 standard exhibited lower uncertainty compared to that of the 23L24 standard. Furthermore, the results obtained for the PNI reference sample showed smaller deviations from the reference values provided by the Program. Consequently, the efficiency curve derived from the 20L24 standard was validated for use in this specific geometry.

Keywords: Gamma spectrometry; efficiency curve; counting geometry

1 Introduction

Accurately determining radionuclide activity in environmental samples is essential for reliable environmental impact assessments, especially in areas near facilities that use radioactive materials. The accuracy and traceability of these measurements are fundamental requirements in radiological monitoring and regulatory compliance studies.

High-purity germanium (HPGe) detectors are widely used in high-resolution gamma spectrometry due to their excellent energy response and ability to discriminate gamma peaks with high precision (Semkov et al., 2015).

Gamma spectrometry is a highly versatile nuclear technique used to identify and quantify gamma-emitting radionuclides in different types of matrices—solid, liquid, and gaseous—without requiring prior radiochemical treatment (Rodrigues *et al.*, 2015).

Efficiency calibration of HPGe detectors is commonly performed using standards containing reference radionuclides, the selection of which directly impacts the accuracy, energy range, and robustness of the generated efficiency curve. Radionuclides such as ^{152}Eu , which exhibits multiple gamma emissions across a wide energy range, are widely used because they allow for a more complete curve fit. In contrast, monoenergetic sources, such as ^{137}Cs or ^{65}Zn , are valuable for the punctual validation of efficiency at specific energies (Knoll, 2010). To ensure the quality of analytical results and establish metrological traceability of measurements, the Nuclear and Physical Measurements Laboratory of the Institute of Nuclear Engineering has participated in the National Intercomparison Program (PNI), a proficiency test promoted by the National Laboratory of Ionizing Radiation Metrology (LNMRI/IRD) since 2024.

In this work, the absolute efficiency curves of an HPGe detector obtained from two standards containing different radionuclides are determined and compared: a multienergy ^{152}Eu standard and a standard containing multiple radionuclides (such as ^{137}Cs , ^{60}Co , ^{133}Ba and others). The objective is to evaluate the differences between the curve fits generated, verify their applicability in different spectrometric analysis contexts, and discuss the implications of standard selection in the calibration of HPGe systems.

2 Methods

2.1 Efficiency curve

The efficiency of a detector is defined as the ratio between the number of radiation events detected and the number of radiation events emitted by the source (Nascimento, 2024). By employing standard sources with well-characterized energies and known relative intensities, and maintaining consistent source geometry and counting conditions, a set of efficiency data points can be established.

The efficiency $\varepsilon(E)$ for each energy was calculated by Equation 1:

$$\varepsilon(E) = \frac{N}{A \cdot I_\gamma \cdot t} \quad (1)$$

where:

- N is the net count at the gamma peak;
- A is the activity of the source (decays/s);
- I_γ is the emission intensity of the gamma photon (in fraction);
- T is the counting time (s).

The experimental efficiency points were adjusted using an empirical log-polynomial function as Equation 2:

$$\ln(\varepsilon) = a_0 + a_1 \ln(E) + a_2 [\ln(E)]^2 + \dots + a_n [\ln(E)]^n \quad (2)$$

The curves obtained for each standard were compared in terms of shape, amplitude and range of applicability, allowing a critical evaluation of the advantages and limitations of each radionuclide in the calibration of the HPGe system.

2.2 Sample Activity

The efficiency curve selection and validation step was performed using the GAMA EM ÁGUA sample from the 2024 National Intercomparison Program (PNI), sample no. 508. This sample consists of an aqueous matrix with gamma-emitting radionuclides (^{133}Ba , ^{134}Cs , ^{106}Ru , ^{65}Zn , ^{137}Cs and ^{60}Co) in concentrations determined by the program. Sample measurements were performed under the same geometric and operational conditions as those used for the detector's efficiency calibration standards.

To determine the activity, it is necessary to use the efficiency curve, which relates the number of counts recorded at a given energy to the number of radiations emitted by the sample at that energy per unit time; in other words, it establishes the connection between measured counts and the true emission rate. Sample Activity is determined using Equation 3.

$$A_i = \frac{\text{area}}{\varepsilon_i \cdot I_{\gamma_i}} \quad (3)$$

where:

- A_i is the activity of the source (Bq);
- Area count rate (cps), proportional to the area of the photoelectric peak, determined by the spectrum analysis program, divided by the counting time;
- ε_i is the detector efficiency for energy radiation E_i (obtained by the efficiency curve equation)
- I_{γ} is the relative intensity of radiation emission of energy E_i (LNHB, 2025).

The activities of the radionuclides present in the sample were then experimentally determined and compared with the PNI reference values, allowing the accuracy and applicability of each generated efficiency curve.

The standard deviation D , defined in accordance with the official PNI report, was used as a criterion for evaluating the analytical performance of the measurements. This parameter is calculated using Equation 4.

$$D = \frac{X - U}{su} \quad (4)$$

where:

- X is the average value obtained by the laboratory;
- U is the reference value assigned by the program to the sample (considered as true);
- su is the standard deviation associated with the reference value.

The D value quantifies the difference between the obtained result and the reference value, normalized by the associated standard deviation. Values of $|D| < 2$ indicate good performance (“GOOD”); between 2 and 3, acceptable performance (“ACCEPTABLE”); and values of $|D| \geq 3$ indicate that the analytical system performed does not meet the acceptance criteria and requires corrective action, (“OUT OF CONTROL”), (Tauhata et al., 2006). This metric constitutes the validation methodology adopted by PNI, being systematically applied in all proficiency test rounds. Therefore, the analysis based on D enables consistent comparison among participating laboratories and ensures the metrological reliability of the reported results.

2.3 Uncertainties

When evaluating the uncertainties associated with the calculated efficiency values, the main contributing sources include: certified activity of the standard sources, gamma emission intensity, radionuclide half-life, the determination of the photopeak area and correction of decay during counting.

The total uncertainty (u_t) is obtained by propagating the individual uncertainties through Equation 5.

$$u_t = (u_A^2 + u_{I_\gamma}^2 + u_{T_{1/2}}^2 + u_{area}^2)^{1/2} \quad (5)$$

where:

- u_A is the uncertainty certified activity of the standard sources;
- u_{I_γ} is the uncertainty in gamma emission intensity;
- $u_{T_{1/2}}$ is the uncertainty in radionuclide half-life;
- u_{area} is the uncertainty in determination of photopeak area.

2.4 Experimental

For the experimental determination of the efficiency curve of the hyperpure germanium (HPGe) detector, with setup shown in Figure 1, two distinct certified radionuclide standards provided by the National Laboratory of Metrology of Ionizing Radiation (LNMRI/IRD) were used. The first standard (20L24) is a multienergy source composed of ^{215}Eu and ^{241}Am , while the second (23L24) contains monoenergetic radionuclides: ^{241}Am , ^{133}Ba , ^{134}Cs , ^{137}Cs , ^{60}Co , and ^{106}Ru . The activities, uncertainties, and half-lives of the radionuclides are presented in Tables 1 and 2. The standards are packaged in a 1 L polyethylene bottle with a screw cap.



Figure 1: Experimental arrangement - HPGe.

Source: own authorship.

Table 1: Data from standard 20L24. Reference date: 05/16/2024 – 12 h (Brasília time).

Radionuclide	Total Activity (Bq)	Uncertainty (% k=2)	Half Life (y)
^{241}Am	293.763	1.7	(432.6 ± 0.6)
^{152}Eu	527.087	0.94	(13.522 ± 0.016)

*Subtitle: y- years.

Table 2: Data from standard 23L24. Reference date: 05/16/2024 – 12 h (Brasília time).

Radionuclide	Total Activity (Bq)	Uncertainty (% k=2)	Half Life (*)
^{41}Am	283.360	1.7	(432.6 ± 0.6) y
^{133}Ba	544.802	0.98	(10.539 ± 0.006) y
^{106}Ru	480.564	3.2	(371.5 ± 2.1) d
^{134}Cs	647.298	1.3	(2.0644 ± 0.0014) y
^{137}Cs	494.093	2.7	(30.05 ± 0.08) y
^{60}Co	495.954	1.4	(5.2711 ± 0.0008) y

*Subtitle: y- years; d- days.

Measurements were performed using a fixed geometry, with the source base positioned flush with the active face of the HPGe detector (Canberra, model GC 2520, 34.60 mm thickness, 62.00 mm active diameter). The counting time was 259,200 seconds for each measurement to ensure good statistical accuracy for the gamma peak counters. The detector is coupled to an electronic system consisting of a preamplifier, pulse amplifier, and multichannel. The peak area evaluation were performed using the data acquisition software Genie 2000 - 3.3 version (Canberra, 2013).

3 Results and Discussion

The efficiency calibration curves of the 20L24 and 23L24 standards are shown in Figure 2, and the equations obtained from the polynomial fit are shown in Table 3.

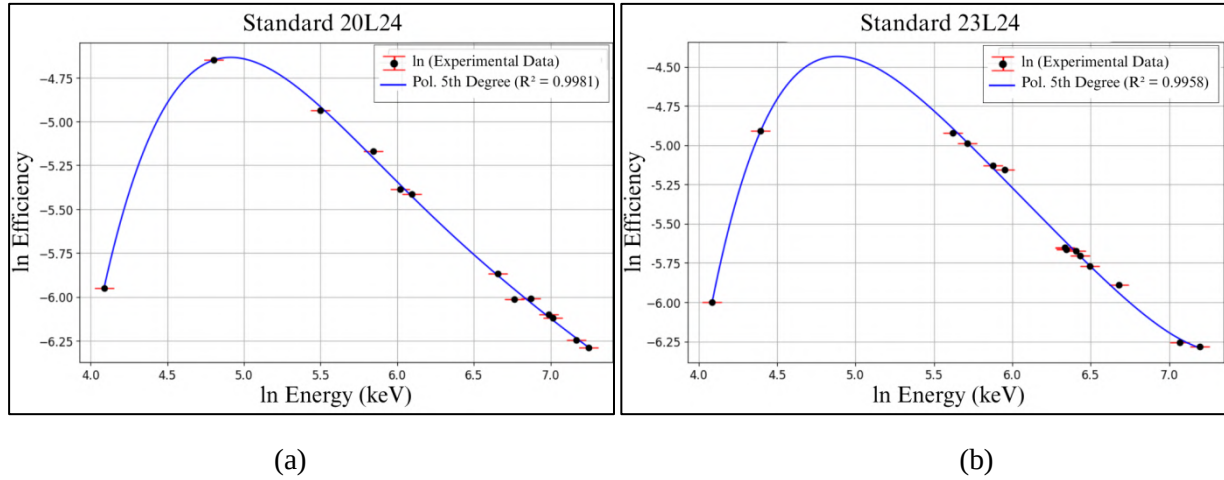


Figure 2: a) Efficiency curve for standard 20L24. (b) Efficiency curve for standard 23L24.
Source: own authorship.

Table 3: Efficiency curve fitting based on radioactive standards.

Standard	Polinomial Adjustment (5th Degree)	R ²
20L24	$\ln(\epsilon) = 0.0204 \ln(\epsilon)^5 - 0.6972 \ln(\epsilon)^4 + 9.5 \ln(\epsilon)^3 - 64.42 \ln(\epsilon)^2 + 216.4 \ln(\epsilon) - 291.5$	0.9981
23L24	$\ln(\epsilon) = 0.0591 \ln(\epsilon)^5 - 1.809 \ln(\epsilon)^4 + 22.24 \ln(\epsilon)^3 - 137.2 \ln(\epsilon)^2 + 423.5 \ln(\epsilon) - 526$	0.9958

Based on the efficiency curves obtained from the two certified standards, the activities of the radionuclides present in the sample provided by the PNI were determined. Three measurements (C1, C2 e C3) were taken of sample no. 508 of the 2024 PNI under the same conditions as the counts of the LNMRI 20L24 and 23L24 Standards. Tables 4 and 5 below present the experimental values obtained for the three counts of Sample No. 508 using the efficiency curve of standards 20L24 and 23L24, the respective reference and uncertainty values provided by the PNI program, as well as the normalized deviations (D) calculated for each radionuclide, allowing the performance of the applied method to be evaluated. The reference date of the PNI results used for validation is April 1, 2024.

Table 4: Determination of activity (Bq) using the efficiency curve of standard 20L24. Reference Date: April 1, 2024.

Radionuclide	Activity Bq (C1)	Activity Bq (C2)	Activity Bq (C3)	Activity Average Bq	PNI Value Bq	su PNI Value Bq	D (*)
¹³³ Ba	3.208	3.192	3.211	3.204	2.949	0.442	0.998
¹³⁴ Cs	1.652	1.735	1.635	1.674	1.692	0.254	-0.124
¹⁰⁶ Ru	6.142	5.641	5.823	5.869	5.72	0.858	0.300
⁶⁵ Zn	1.007	1.047	1.048	1.034	0.923	0.138	1.398
¹³⁷ Cs	4.687	4.523	4.498	4.569	4.28	0.642	0.781
⁶⁰ Co	1.432	1.426	1.404	1.421	1.415	0.212	0.046

*Subtitle: ■ Good; ■ Acceptable; ■ Out of control.

Table 5: Determination of activity (Bq) using the efficiency curve of standard 23L24. Reference Date: April 1, 2024 – 12 h (Brasília time).

Radionuclide	Activity Bq (C1)	Activity Bq (C2)	Activity Bq (C3)	Activity Average Bq	PNI Value Bq	su PNI Value Bq	D (*)
¹³³ Ba	3.237	3.222	3.241	3.233	2.949	0.442	1.114
¹³⁴ Cs	1.798	1.888	1.780	1.822	1.692	0.254	0.886
¹⁰⁶ Ru	6.690	6.144	6.342	6.392	5.72	0.858	1.356
⁶⁵ Zn	1.104	1.148	1.149	1.133	0.923	0.138	2.639
¹³⁷ Cs	5.078	4.900	4.873	4.950	4.28	0.642	1.809
⁶⁰ Co	1.492	1.485	1.463	1.480	1.415	0.212	0.533

*Subtitle: ■ Good; ■ Acceptable; ■ Out of control.

4 Conclusions

This study enabled a comparative analysis of two calibration curves for the efficiency of an HPGe detector using two certified standards provided by LNMRI/IRD. The experimental procedure for obtaining the curves was based on a fixed counting geometry and a prolonged acquisition time of 259.200 seconds to ensure high statistical reliability.

The efficiency curves were fitted to the log-polynomial model on a logarithmic scale, and both standards resulted in curves with good fits, covering a wide energy range. The activity determined for sample #508 using the efficiency curve of Standard 20L24 showed the best agreement with the reference values of the National Intercomparison Program (PNI), used to validate this study.

Validation using measurements of the PNI reference sample confirms the better precision and accuracy of the 20L24 standard curve for the geometry and energy range of this study, resulting in a "GOOD" result for all activity values found. The maximum D was 1.398 for ¹³⁷Cs, a result considered good by the program parameters. The good agreement between the results obtained and the reference values confirms the adequacy of the methodology adopted to select the standard to be used for this geometry and matrix.

Acknowledgements

The authors thank the Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq) for granting the scholarship, the Instituto de Engenharia Nuclear (IEN) for all the support and the Laboratório Nacional de Metrologia das Radiações Ionizantes (LNMRI/IRD) for providing the standards.

References

- Canberra Industries (2013). *Genie 2000 Spectroscopy System – Operations Manual, Version 3.3*. Meriden, CT, USA.
- Knoll, G. F. (2010). *Radiation Detection and Measurement* (4th ed.). John Wiley & Sons, Inc.
- Laboratoire National Henri Becquerel, Table of Atomic and Nuclear data recommended - <http://www.lnhb.fr/home/nuclear-data/nuclear-data-table/> (2025, May)
- Nascimento, D. S.; Chierici, A.; Ciolini, R.; Chiappini, S.; d'Enrico, F.; Chiappini, M. (2024). Calibration HPGe detector using IAEA-U source for CBRNe. In *The European Physical Journal Plus* 139: 654.

Rodrigues, J. L.; Kastner, G. F.; Ferreira, A. V. (2011). Determination of efficiency curves for HPGE detector in different counting geometries. in *International. Nuclear Atlantic Conference (INAC 2011)*, Belo Horizonte, MG (Brazil).

Semkov, T. M.; Bradt, C. J.; Beach, S.E.; Haines D. K.; Khan, A. J.; Bari. A. (2015). *Nuclear Instruments and Methods Research A*. 799: 105-113.

Tauhata, L.; Vianna, M. E. C.; Oliveira, A. E; Ferreira, A. C.; Bragança, M.J.C.; Clain, A. F.; de Faria, R. Q. (2006). The Brazilian National Intercomparison Program (PNI/IRD/CNEN): evaluation of 15 years of data. In *Journal of Environmental Radioactivity* 86, 384-390.

Determination of the mass attenuation coefficient of polymers using Monte Carlo N-Particle simulation

J.P.C. Andrade^{23*}, R.S.F Dam¹, C.M. Salgado¹, W.L. Salgado¹

¹ Instituto de Engenharia Nuclear (IEN), Divisão do Radiofármacos Rua Hélio de Almeida, 75, 21941-906, Cidade Universitária, RJ, Brazil

² Instituto de Engenharia Nuclear (IEN), Divisão do ensino, Rua Hélio de Almeida, 75, 21941-906, Cidade Universitária, RJ, Brazil

³ Universidade Federal do Rio de Janeiro (UFRJ), Avenida Brigadeiro Trompowsky, Cidade Universitária Ilha do Fundão, RJ, Brazil

* jp.vc9999@gmail.com

Abstract

The increasing use of 3D printing has increased the relevance of polymers such as acrylonitrile butadiene styrene (ABS), polyethylene terephthalate glycol (PETG), and polylactic acid (PLA), which are widely employed in industry, prototyping, and consumer products, and to the nuclear field the shielding of gamma rays. In gamma densitometry applications, these materials are widely used to establish fixed geometries for detector calibration, thus, the accurate determination of their mass attenuation coefficients is essential. In this study, the MCNPX code was used to simulate a geometry using circular plated samples of these polymers (ABS, PETG, PLA), radiation sources of ²⁴¹Am (59.54 keV), ¹³⁷Cs (661.66 keV), and ⁶⁰Co (1332.49 and 1173.23 keV) and one 2"x2" NaI(Tl) detector. For the simulation was used 0.60 mm thickens for each material, and the source-detector distance was fixed at 150 mm. The aim of the simulation was to evaluate the radiation interaction with the polymers throughout the mass attenuation coefficient. The simulation setup replicated real conditions, with the radiation's source collimated in one direction, the photon beam striking the sample of the polymers, and the attenuated beam is collimated and then reaches the detector. Thus, the simulations included collimated detectors and non-collimated geometries to reduce scattering effects. It is important to highlight that in addition to polymers, aluminum was used as a pure reference material. The results were compared with reference data from the NIST XCOM database and showed excellent agreement, for the energy range used (59.54 keV to 1132.49 keV), the maximum relative error was 6.44% without collimation and 0.51% with collimation, validating the accuracy and robustness of the simulation methodology. Furthermore, the results show that these polymers have little influence on the measurements and can be used as support for detectors.

Keywords: Mass attenuation coefficient; Gamma-rays; Polymers; MCNPX code

1 Introduction

The increasing use and adoption of 3D printing technologies has emphasized the relevance of polymers such as acrylonitrile butadiene styrene (ABS), polyethylene terephthalate glycol

(PETG), and polylactic acid (PLA), which are widely employed in industry, prototyping, and consumer products. In recent years, these materials have also attracted growing due to their resistance, durability and flexibility [10,2], and their interest isn't different in the nuclear field their potential application in gamma radiation shielding [1,5] and in support for detectors [13]. To assess their shielding capabilities, it is essential to determine the mass attenuation coefficient (MAC), a fundamental parameter that describes the interaction between photons and matter. In this study, we employed the MCNPX Monte Carlo simulation code [11] to calculate the MAC for selected 3D-printed polymer samples. The simulations modeled the interaction of photons emitted by standard gamma-ray sources, ^{241}Am (59.54 keV), ^{137}Cs (661.66 keV), and ^{60}Co (1173.23 and 1332.49 keV), with the material samples. Photons were directed toward the samples, and those that passed through were detected using a 2"x2" NaI(Tl) scintillation detector. The number of photons reaching the detector was used to calculate the attenuation coefficient via the Beer–Lambert law.

For many decades, studies of attenuation and absorption properties of materials are conducted [6,7] due to the growing interest in nuclear techniques. Simulation and experiments of these properties include several material types, such as wood [14], building materials [12,4], stainless steel [3], among others

The linear coefficient of a material varies with the energy of the radiation and its physical state. To avoid the dependency, the coefficient is divided by the density of the material, thus being called mass attenuation coefficient [16], this coefficient is possible to compare between different materials.

To validate the methodology, an aluminum sample was used as a control. The simulated results were compared with reference data from NIST XCOM database [9], allowing for an evaluation of the accuracy and reliability of the MCNPX modeling approach in estimating attenuation parameters for potential shielding materials.

Additionally, to minimize the contribution of low-energy photon scattering, particularly Rayleigh scattering [15] at 59.54 keV (^{241}Am), lead collimators were employed in the simulation setup. This ensured a more defined photon beam geometry and reduced off-axis scattering effects, contributing to a more accurate estimation of the mass attenuation coefficient under narrow-beam conditions.

2 Methodology

To determine the mass attenuation coefficient (MAC) of the investigated 3D-printed polymers, simulations were performed using the MCNPX Monte Carlo code. The simulations modeled a typical narrow-beam transmission experiment, where a collimated photon beam passes through the sample and the transmitted intensity is measured by a detector placed behind the material.

2.1 Theoretical Background

The attenuation of gamma radiation in matter follows the Beer–Lambert law, which describes the exponential decrease in intensity I of a monoenergetic photon beam as it traverses a material of thickness x , as shown in Equation 1:

$$I = I_0 \exp(-\mu x) \quad (1)$$

Where:

I_0 is the incident photon intensity,

I is the transmitted intensity,

μ is the linear attenuation coefficient (cm^{-1}),

x is the material thickness (cm).

To obtain the mass attenuation coefficient μ/ρ , Equation (2) is normalized by the material's density ρ (g/cm^3), yielding:

$$\frac{\mu}{\rho} = \frac{1}{x\rho} \ln \left(\frac{I_0}{I} \right) \quad (2)$$

This coefficient allows for a direct comparison between materials of different densities and is widely used in radiation shielding studies.

2.2 Simulation Geometry

The simulation geometry illustrated in Figure 1, where two setups are considered: one with a lead collimator in front of the detector (to reduce low-energy scattering effects such as Rayleigh scattering), a one without the collimator.

The radiation source emits a monoenergetic pencil beam directed perpendicularly toward the center of a circular polymer plate (diameter = 6.0 cm, thickness = 0.6 cm), for every simulation it was used only plates using this dimension. The detector, modeled as a 2"x2" NaI(Tl) scintillation crystal, is positioned 15 cm from source.

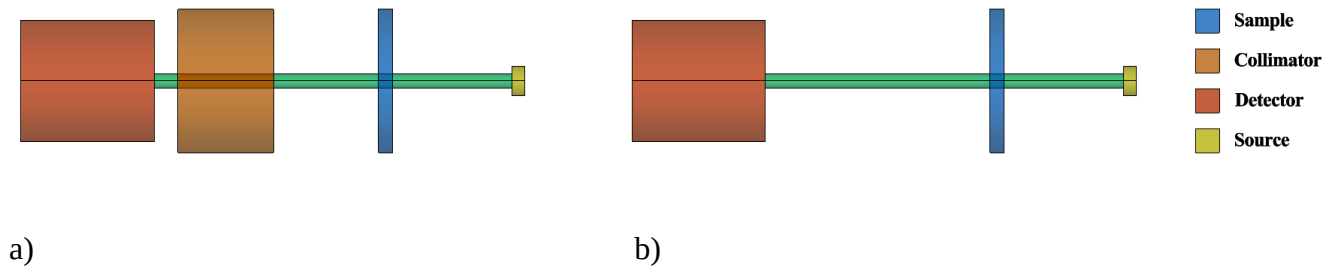


Figure 1: Geometry implemented in the MCNPX code: a) with lead collimator before the detector. b) without a collimator.

2.3 Photon Sources and Tallies

Simulations were performed for three gamma-emitting radionuclides commonly used in attenuation studies:

- ^{241}Am – 59.54 keV
- ^{137}Cs – 661.66 keV
- ^{60}Co – 1173.23 and 1332.49 keV

In all cases, the number of photons reaching the detector was recorded using the F8 pulse height tally, normalized per source particle, and used to calculate the transmitted intensity I . Separate runs were conducted to determine I_0 , with the sample removed from the geometry. All the simulations were terminated when 100 million particles histories were done, and the relative error of the number of photons counted were below 1%.

2.4 Material Composition and Density

For accurate simulation, the elemental mass fractions and densities of the three polymers were defined in the MCNPX input. The compositions were as follows:

- PLA and PETG: elemental compositions calculated from their chemical formulas (PLA: $C_3H_4O_2$, PETG: $C_{14}H_{20}O_5S$) [8].
- ABS: composition and density obtained from the study by Moayyad Alssabbagh [1], due to its nature as a complex copolymer.

The elemental mass fractions and density values are presented in Table 1.

Table 1: Elemental composition and densities of simulated polymers.

Polymer	C (%)	H (%)	O (%)	S (%)	N (%)	Density (g/cm^3)
PLA	50.0025	5.5942	44.4032	-	-	1.29
PETG	55.9821	6.7106	26.6321	10.0675	-	1.30
ABS	85.51	5.26	1.65	0.08	5.26	1.09

With these same elemental compositions was used to obtain the values of mass attenuation coefficient from NIST platform.

3 Results

Tables 3 and 4 present the mass attenuation coefficients simulated using the MCNPX code without and with a collimator, respectively.

Table 3: Mass attenuation coefficient from the simulation without collimator.

Source	Energy (keV)	Mass attenuation coefficient (μ/ρ) (cm^2/g)			
		ABS	PLA	PETG	Al
^{241}Am	59.6	1.82E-01	1.84E-01	2.07E-01	2.62E-01
^{137}Cs	662	8.23E-02	8.09E-02	8.17E-02	7.39E-02
^{60}Co	1173.2	6.23E-02	6.13E-02	6.19E-02	5.60E-02
	1332.5	5.85E-02	5.75E-02	5.81E-02	5.27E-02

Table 4: Mass attenuation coefficient from the simulation with collimator.

Source	Energy (keV)	Mass attenuation coefficient (μ/ρ) (cm^2/g)			
		ABS	PLA	PETG	Al

²⁴¹ Am	59.6	1.87E-01	1.91E-01	2.15E-01	2.80E-01
¹³⁷ Cs	662	8.26E-02	8.13E-02	8.21E-02	7.44E-02
⁶⁰ Co	1173.2	6.28E-02	6.17E-02	6.23E-02	5.65E-02
	1332.5	5.88E-02	5.78E-02	5.84E-02	5.29E-02

Table 5 presents the reference values from the NIST XCOM platform, calculated based on the mass fraction of the elements comprising the materials.

Table 5: Reference mass attenuation coefficients from NIST.

Source	Energy (keV)	Mass attenuation coefficient (μ/ρ) (cm ² /g)			
		ABS	PLA	PETG	AL
²⁴¹ Am	59.6	1.88E-01	1.91E-01	2.15E-01	2.80E-01
¹³⁷ Cs	662	8.29E-02	8.15E-02	8.22E-02	7.47E-02
⁶⁰ Co	1173.2	6.31E-02	6.21E-02	6.27E-02	5.68E-02
	1332.5	5.91E-02	5.81E-02	5.87E-02	5.22E-02

The relative error was calculated to compare the simulation results with the reference values. These are presented in Table 6.

Table 6: Relative error (%) between simulation and NIST values.

Source	Energy (keV)	Relative error (%)							
		Without collimator				With collimator			
		ABS	PLA	PETG	Al	ABS	PLA	PETG	Al
²⁴¹ Am	59.6	3.18	3.77	3.88	6.44	0.20	0.21	0.15	0.04
¹³⁷ Cs	662	0.72	0.68	1.27	0.99	0.29	0.24	0.08	0.36
⁶⁰ Co	1173.2	1.28	1.27	1.25	1.26	0.51	0.51	0.48	0.47
	1332.5	1.01	0.99	0.99	1.03	0.49	0.49	0.47	0.49

The results obtained demonstrate good agreement with the NIST reference values, validating the reliability of the simulation methodology. In the initial configuration, without the use of a collimator, the highest relative error observed was 6.44%, which occurred for the Am-241 source. This discrepancy is primarily attributed to the increased contribution of Rayleigh scattering, especially relevant for low-energy photons. To mitigate this effect, an additional simulation was performed with a collimator positioned in front of the detector. The presence of the collimator effectively limited the acceptance angle of incoming photons, thereby reducing the contribution of scattered radiation. This refinement led to a marked improvement in the results: for photons emitted by the Cs-137 and Co-60 sources, the relative errors were consistently reduced to below 1%, indicating excellent agreement with the NIST data.

The Am-241 source, which initially exhibited the highest deviation, showed the most significant improvement. With the introduction of the collimator, the relative error dropped dramatically from 6.44% to just 0.21 highlighting the collimator's critical role in minimizing scattering effects at lower photon energies.

Furthermore, when analyzing the attenuation in different polymers, the relative errors remained below 0.52% with the collimator in place, reinforcing the method's robustness across diverse materials. The control material, aluminum, also yielded relative errors below 0.50%, further confirming the accuracy and consistency of the simulation approach. These findings underscore the effectiveness of the proposed methodology and support its applicability for precise determination of attenuation coefficients in photon interaction studies.

4 Conclusions

The simulation results demonstrated the effectiveness and accuracy of the methodology employed using the MCNPX code for determining the mass attenuation coefficients of polymers commonly used 3D printing. The simulated values exhibited excellent agreement with the NIST XCOM reference data, particularly when a collimator was incorporated into the setup, which led to a significant reduction in relative errors, down to a maximum of only 0.51%.

The pronounced decrease in error for low-energy photons, especially those emitted by the ^{241}Am source, underscores the critical role of collimation in minimizing scattering effects and improving the fidelity of photon transport simulations. These findings emphasize the necessity of accounting for geometrical and scattering factors in computational dosimetry and materials characterization.

Moreover, the observed low interaction of the evaluated polymers with gamma radiation across the studied energy range confirms their suitability for use as structural materials in detector assemblies and calibration phantoms for gamma densitometry applications. The proposed methodology proved to be a reliable tool for generating accurate attenuation data, reinforcing the potential of 3D-printed polymers in the design of nuclear instrumentation and shielding systems.

However, a notable limitation of the present approach is its reliance on idealized geometries and monoenergetic sources, which may not fully capture the complexities of real experimental conditions, such as beam divergence, source energy spectra, and environmental scattering.

Future studies should aim to validate the simulation results through experimental measurements, as well as to explore the behavior of other emerging 3D printing materials under a broader range of photon energies. Additionally, expanding the simulations to more complex geometries may further enhance the applicability of this methodology to practical scenarios in nuclear medicine, industrial radiography, and radiation protection.

Acknowledgements

The authors gratefully acknowledge scholarships from Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq).

References

- [1] ALSSABBAGH, Moayyad et al. *Evaluation of 3D printing materials for fabrication of a novel multifunctional 3D thyroid phantom for medical dosimetry and image quality*. Radiation Physics and Chemistry, v. 135, p. 106–112, 2017.
- [2] Barros, A. de F. 2024. *Preparação, caracterização físico-química e avaliação do comportamento térmico e mecânico de blendas de poli (tereftalato de etileno) (PET) e poli (éterimidia) (PEI), PET/PEI*. 254 f. Tese (Doutorado em Ciências Exatas e da Terra) - Universidade Federal de São Carlos, São Carlos.
- [3] BUYUK, B. *Gamma attenuation behavior of some stainless and boron steels*. Acta Physica Polonica A, v. 127, n. 4, 2015.
- [4] DEMIR, Nilgun; TARIM, Urkiye Akar; POPOVICI, Maria-Ana; DEMIRCI, Zehra Nur; GURLER, Orhan; AKKURT, Iskender. *Investigation of mass attenuation coefficients of water, concrete and bakelite at different energies using the FLUKA Monte Carlo code*. Journal of Radioanalytical and Nuclear Chemistry, 2013.
- [5] GULTEKIN, Batuhan; BULUT, Fatih; YILDIZ, Hatice; US, Hakan; OGUL, Hasan. *Production and investigation of 3D printer ABS filaments filled with some rare-earth elements for gamma-ray shielding*. Nuclear Engineering and Technology, v. 55, n. 12, p. 4664–4670, 2023.
- [6] Hubbell, J.H., Seltzer, S.M., 1995. *Tables of X-Ray Mass Attenuation Coefficients and Mass Energy-Absorption Coefficients 1 keV to 20 MeV for Elements Z = 1 to 92 and 48 Additional Substances of Dosimetric Interest*. National Institute of Standards and Technology.
- [7] Jackson, D.F., Hawkes, D.J., 1981. *X-ray attenuation coefficients of elements and mixtures*. Phys. Rep. 70 (3), 169–233.
- [8] McConn Jr., R.J., Gesh, C.J., Pagh, R.T., Rucker, R.A., Williams III, R.G., 2011. *Compendium of material composition data for radiation transport modeling*. PNNL-15870 Rev. 1. <https://doi.org/10.2172/1023125>
- [9] NIST National Institute of Standards and Technology. Available online. <https://physics.nist.gov/PhysRefData/Xcom/html/xcom1.html>.
- [10] Ogul, H., 2022. *Radiation attenuation properties of polymer composites mixed with tantalum carbide*. Radiation Effects and Defects in Solids, 177(5–6), 531–544.
- [11] Pelowitz, D.B., 2005. *MCNP-X Tm User's Manual*, 2.5 ed. Los Alamos National Laboratory, New Mexico, USA.
- [12] SALINAS, I. C. P.; CONTI, C. C.; LOPES, R. T. Effective density and mass attenuation coefficient for building material in Brazil. *Applied Radiation and Isotopes*, [S.l.], v. 64, p. 13–18, 2006.
- [13] Santos, A.G.M., Dam, R.S.F., Salgado, W.L., Schirru, R., Salgado, C.M., 2020. *Determination of mass attenuation coefficient of polylactic acid using gamma densitometry in 50-1000 keV energy range*. Radiation Chemistry and Physics, 177, 109097.
- [14] SARITHA, B.; RAO, A. S. Nageswara. *A study on photon attenuation coefficients of different wood materials with different densities*. Journal of Physics: Conference Series, v. 662, 2015.
- [15] TAUHATA, L.; SALATI, I. P. A.; DI PRINZIO, R.; DI PRINZIO, M. A. R. R. *Radioproteção e dosimetria: fundamentos*. 10. rev. abr. 2014. Rio de Janeiro: IRD/CNEN, 2014.

[16] Tsoulfanidis, N., 1983. *Measurement and Detection of Radiation*. McGraw-Hill. New York & USA.

Identification and Quantification of K, Na and Mn in Textured Soy Protein Sold in the City of Rio de Janeiro Using the Argonauta Research Reactor of IEN/CNEN and the Techniques of Instrumental Neutron Activation Analysis and γ -ray Spectrometry

R. C. Nunes^{1*}, G. T. Echeverria², P. A. L. da Cruz¹, E. M. de Oliveira¹, C. M. Salgado¹, L. Carvalheira¹, A. P. Junior¹, A. L. N. Barbosa¹

¹ CNEN, Instituto de Engenharia Nuclear/DIRAD, 21941-614, Rio de Janeiro, Brasil

² UFRJ, POLI/Departamento de Engenharia Mecânica, 21941-914, Rio de Janeiro, Brasil

* rogerio.chaffin@ien.gov.br

Abstract

Glycine max., popularly known as soybean, is one of the legumes cultivated in Brazil. In general, it has a high nutritional value for human consumption, being an abundant source of proteins, oil and fiber, in addition to essential minerals for maintaining good health (Silva et al., 2022). In the form of textured soy protein, TSP, it constitutes a valuable low-cost food alternative (Hamdy, 1974). The objective of this study was to identify and quantify the chemical elements potassium, K, sodium, Na, and manganese, Mn, present in samples of TSP, purchased in popular markets in the city of Rio de Janeiro. The analysis methodology used in this research associates the techniques of Instrumental Neutron Activation Analysis (Kruger, 1971), INAA, by the comparative method, with Gamma Ray Spectrometry (Knoll, 2000), GS. The samples were positioned inside the J9 irradiation channel of the Argonauta research reactor of the Instituto de Engenharia Nuclear, IEN/CNEN, under operating conditions with a power of 34 W and a neutron flux of 10^8 n/cm²s for 30 min. In the qualitative and quantitative tests, an ORTEC Hyperpure Germanium, HPGe, detector with 25% efficiency and 1.8 keV resolution at the 1332 keV photopeak of ⁶⁰Co, the Maestro v7.0 Software and the DSPEC jr 2.0 electronic system were used. The results obtained, through spectrum, tables and calculations, demonstrate the identification of the elements by their Gamma energies and half-lives, as well as the concentration values (mg/100g) of the elements K (1520.7 ± 93.3), Na (4.3 ± 1.4) and Mn (2.3 ± 0.1), respectively. In conclusion, the INAA and GS techniques proved to be effective in both the identification and quantification of the elements, and the results show that this food constitutes an excellent source of potassium, presenting, at the same time, low sodium content. These characteristics make it a nutritious and beneficial option for a balanced diet, also highlighting the nutritional potential of TSP as a substitute for proteins of animal origin. This information has great potential to assist and guide healthier dietary decisions, contributing to the promotion of public health in the city of Rio de Janeiro.

Keywords: Textured soy protein (TSP), Neutron activation, γ -ray spectrometry, Argonauta research reactor.

1 Introduction

Soybean (*Glycine max*), belonging to the Fabaceae family, is a legume widely cultivated in Brazil (Silva, 2022), standing out for its high nutritional value (Riaz, 2001). The largest soybean producers in the world are the USA, Brazil, Argentina, China, and India, with global production shares of 35 %, 28 %, 17 %, 4

%, and 3 %, respectively (Rizzo and Baroni, 2018). It is recognized as an excellent source of proteins, lipids, amino acids, fibers, and essential minerals for human health (Hamdy, 1974).

In the soybean industry, oil is the first commercial product to be extracted and, at the end of this stage, by-products are separated, such as forage, intended for livestock, and flour, which is used in the food industry (Rizzo and Baroni, 2018). When this flour is subjected to an industrial thermoplastic extrusion process, it is transformed into a textured food product with the appearance of animal protein, known as textured soy protein, TSP (Zhang et al., 2018).

TSP is an excellent alternative to animal-derived proteins (Alamu and Busie, 2019), as it has a low cost and a high concentration of nutrients. However, its consumption is little encouraged, and although it is economically accessible, it is generally still underutilized by the Brazilian population.

Still in the Brazilian context, the national regulatory agency linked to the Ministry of Health, responsible for promoting public health through sanitary control of the production and consumption of products and services that may pose health risks, the Agência Nacional de Vigilância Sanitária, ANVISA, recommends a minimum daily intake of 25 g (ANVISA, 2016) to obtain benefits for maintaining good health (Nogueira-de-Almeida et al., 2020).

Despite all its potential, studies focused on quantifying the essential chemical elements present in this product in Brazil are rare. According to the available literature, it is known that each cultivar absorbs properties from the soil, as well as from fertilizers and added manures (Paulilo et al., 2015), and in this way, they become foods rich in essential minerals with nutritional relevance such as potassium (K), sodium (Na), and manganese (Mn).

A diet rich in potassium contributes to the control of blood pressure, the prevention of cardiovascular diseases and kidney disorders, and the improvement of overall health (He & MacGregor, 2008). Although a high intake of potassium is generally encouraged, there is no universal consensus regarding the exact recommended amount (D'Elia, 2024). However, the Institute of Medicine (IOM, 2005) suggests an adequate intake of 4700 mg/ day for adults.

Sodium represents about 2% of the total mineral content in the body and plays an essential role in various human body functions. It is important for maintaining acid-base balance, ensuring the body's pH remains within a healthy range. Imbalances in sodium levels can lead to health problems (Kodintsev et al., 2022). The World Health Organization (WHO, 2012) recommends a daily intake of 2 g (Jaques and Ponte, 2023) and (IOM, 2005).

Manganese, when consumed in controlled doses, supports various physiological functions; however, excessive exposure can lead to neurological disorders. Therefore, it is essential to observe safe exposure limits to maintain good health. The adequate intake for adults is estimated to range from 1.8 mg/ day for women to 2.3 mg/ day for men (IOM, 2001). Manganese deficiency may increase calcium and phosphorus concentrations in the blood and cause dermatitis (Santamaria, 2008).

In this context, the present study aimed to identify and quantify the concentrations of these elements in a commercial sample of TSP, purchased from a large natural products retail chain in the city of Rio de Janeiro. For this purpose, the techniques of Instrumental Neutron Activation Analysis, INAA (Kruger, 1971) and (IAEA-TECDOC-564, 1990) and Gamma-Ray Spectrometry, GS (Knoll, 2000) and (Gilmore and Joss, 2024) were employed, using a high-purity germanium detector, HPGe, from ORTEC with 25% efficiency and a resolution of 1.8 keV at the 1332 keV photopeak of ^{60}Co , the Maestro v7.0 software, and the DSPEC jr 2.0 electronic system. The neutron source used was the Argonauta research reactor, located at the Instituto de Engenharia Nuclear, IEN/CNEN, operating under conditions of 34 W power and a thermal neutron flux of $10^8 \text{ n/cm}^2\cdot\text{s}$ for 30 minutes.

2 Materials and Methods

As already mentioned in the introduction, the results of the elemental identifications and their respective quantifications in the TSP samples were obtained through the combination of the INAA and GS techniques and the materials, equipment and methods used will be described in detail in the subtopics of this section, below.

2.1 Samples of TSP and Certified Chemical Reference Reagents

The neutron activation process of samples requires preparatory steps before irradiation in the reactor. The granular TSP samples were processed in a manual mini-processor, converted into a fine powder, aliquoted, and placed into 5 ml polyethylene vials, which were duly sealed and labeled. Their masses were then determined on a Shimadzu ATY224 balance with four-decimal precision and associated uncertainty of 0.00005 g.

The reference reagent samples (KCl, NaCl, MnO₂), in powder form, did not undergo processing, but were simply aliquoted, placed in individual vials, and weighed sequentially on the same balance.

Figure 1 illustrate one of the sample arrangements, and the Figure 3 shows the placement location for insertion into the J9 channel of the Argonauta reactor.



Figure 1: TSP and Reference Samples in Powder Form

Since completely pure and isolated elements are not available to work with, it is necessary to proceed with the calculation of the elemental mass fraction as shown in Equation 1 (Albuquerque, 2020). The corresponding results are presented in Table 1, which also lists the reagents used as reference materials along with their manufacturer-specified purity grades.

$$EMF_e = \left[\frac{MM_e}{MM_R} \right] \times M_R \quad (1)$$

Where EMF_e , MM_e , MM_R and M_R denote the elemental mass fraction, the elemental molar mass, the molar mass of the reference compound used as reagent, and the mass of that reference compound used as reagent, respectively; in this table, EMF_e has been set to 1 g.

Table 1: Elemental Mass Fraction per Gram of K, Na, and Mn in Reference Reagents.

Certified Chemical Reference Reagents (R)	Chemical Purity Grade (CPG)	Company Name	Chemical Elements (e)	Elemental Mass Fraction per 1 g (EMF)
KCl	0.966	Pro Analyti	K	0.393
NaCl	0.999	Merck	Na	0.524
MnO ₂	0.900	Vetec	Mn	0.632

The mass measurement procedure was as follows: first, all vials were measured empty, and then TSP and the reagents were added, each to its own vial, with the mass of each compound obtained by the difference between these two measurements. Since the reagents (KCl, NaCl, and MnO₂) did not weigh exactly 1 g, Equation 1 was adjusted to calculate the masses of the reference elements (K, Na, and Mn) and their associated uncertainties. Equation 2 and Equation 3 describe the calculations for the TSP samples, while Equation 4 and Equation 5 apply to the reference elements.

$$M_{TSP} = M_{TSP}^{vial} - M_{empty}^{vial} \quad (2)$$

$$\delta_{M_{TSP}} = M_{TSP} * \left[\left(\frac{\delta_{M_{TSP}^{vial}}}{M_{TSP}^{vial}} \right)^2 + \left(\frac{\delta_{M_{empty}^{vial}}}{M_{empty}^{vial}} \right)^2 \right]^{1/2} \quad (3)$$

$$M_R^e = (M_R^{vial} - M_{empty}^{vial}) * CPG_e * EMF_e \quad (4)$$

$$\delta_{M_R^e} = M_R^e * \left[\left(\frac{\delta_{M_R^{vial}}}{M_R^{vial}} \right)^2 + \left(\frac{\delta_{M_{empty}^{vial}}}{M_{empty}^{vial}} \right)^2 \right]^{1/2} \quad (5)$$

Where index *e*, M_{TSP}^{vial} , M_{empty}^{vial} , M_R^{vial} , δ , CPG_e and EMF_e represents elements (K, Na, Mn), the total mass of TSP without discounting the mass of the vial, the mass of the empty vial, the mass of the reference chemical compound without discounting the mass of the empty vial, the uncertainties associated with each measurement, the chemical purity grade of each reference compound presented by the manufacturer and the elemental mass fraction per 1g related to each chemical compound.

2.2 Argonauta Reactor - The Neutron Source

Among various neutron sources, nuclear reactors provide the highest analytical sensitivity for most elements due to their intense neutron fluxes. Neutron energy distributions and flux intensities may vary depending on reactor type and irradiation position, primarily influenced by the moderator materials used.

Neutron spectra generally consist of three components: thermal, epithermal, and fast neutrons. Thermal neutrons ($E < 0.5$ eV), in equilibrium with the reactor moderator, represent up to 95 % of the total incident neutron flux on a sample during irradiation. These low-energy neutrons are primarily responsible for inducing radiative capture reactions (n, γ) in target-nuclei, which form the basis of the activation techniques employed in this study (Glascok, 2003).

The Argonauta reactor at IEN was the neutron source used in this research to activate the samples. This reactor has a licensed power of 1 kW for up to 1 h, and 500 W continuously, with its usual power being approximately 340 W. It uses uranium fuel enriched to 19.91 % in ²³⁵U, with a current critical mass of about 2113.44 g of ²³⁵U and an operational mass of approximately 2136.21 g of ²³⁵U.

The standard loading includes 4 fuel elements, FE, with 17 plates each, totaling about 20.0 g of ²³⁵U per element, in addition to 4 other FE with different plate configurations and uranium quantities, totaling 8 FE. The core arrangement involves two concentric aluminum cylinders, forming a cylindrical ring, with internal diameters of 603.2 mm and 904.4 mm, and a height of 1165 mm. Between the plates, there is deionized water and graphite wedges, which help control the reaction.

In the peripheral region, there is graphite, while in the upper and lower parts, there is deionized water. Regarding safety, the reactor is protected by concrete blocks of various sizes, forming a biological shield

that prevents the formation of unwanted neutron and gamma beams. The top of the reactor has a concrete shield lined with a steel plate, with a circular hatch that allows access to the core (Albuquerque, 2020).

Its main irradiation channel is called J9, its access is external to the reactor core, so there is no direct contact with the moderator or the fuel element, allowing a mutual process of preservation for both the sample and the core in case of leaks. Figures 2 (a) and (b) illustrate, on the left side, the access to the core through J9 channel, as well as the configuration of the fuel elements in the core, and on the right side, an external photograph of the Argonauta highlighting the location of J9 channel.

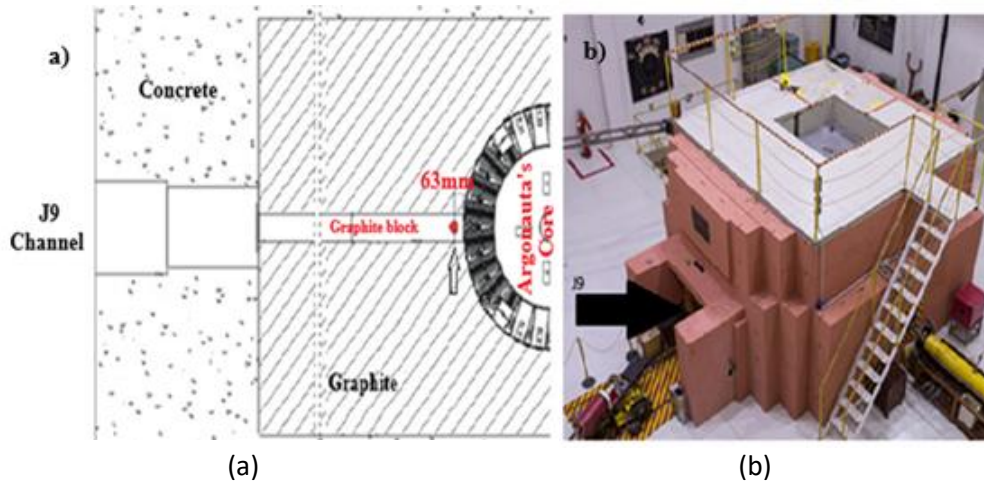


Figure 2: (a) J9 channel with graphite blocks inside. b) Argonauta research reactor.

The samples inserted into the Argonauta through J9 channel are housed in a solid graphite block called the optional inner segment and are placed close to the core at about 63 mm from the outer cylinder, as can be seen in Figure 3. The height of this channel relative to the axial position of the core is exactly at the middle, where the neutron flux is at its maximum (Duderstadt, 1976), ensuring the best activation conditions for the samples

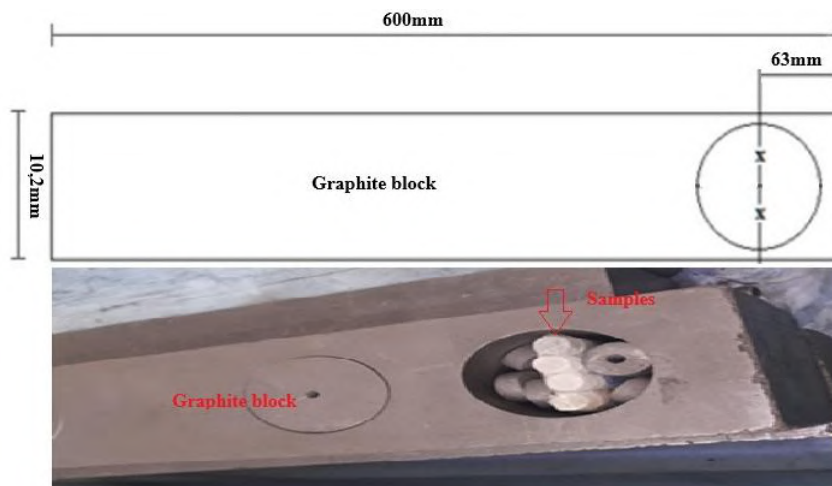


Figure 3: Solid graphite block used in sample irradiation - called optional inner segment.

2.3 Instrumental Neutron Activation Analysis Technique – Comparative Method

Neutron Activation Analysis, NAA, is a nuclear analytical technique for elemental quantification based predominantly on radioactive capture (n, γ), in which a stable nucleus interacts with a thermal neutron,

forming an excited nucleus that emits prompt gamma radiation and frequently delayed gamma radiation upon decay.

In the Instrumental modality, INAA, the analysis is performed directly on the irradiated sample without subsequent chemical steps, using high-resolution detectors such as High Purity Germanium, HPGe. It is a precise, fast, non-destructive, and highly sensitive method, widely applied in various scientific fields (De Bruin, 1983).

The comparative method is the simplest way to apply INAA for the quantitative analysis of an element and involves simultaneous irradiation of samples under identical conditions and neutron flux. For each analysis, the required number of reference samples, duly certified for the purity of the elements to be quantified, is used. The quantification of elements in the target samples is performed by direct comparison of gamma spectrometry results between the reference and the target sample, analyzing the net area of the spectral gamma energy peaks, which allows estimation of the mass of the element present in the target sample (Glascok, 2003), (Kruger, 1971), (IAEA-TECDOC-564, 1990) and (Albuquerque, 2020).

Following the same counting geometry of the samples in the gamma spectrometry system, Equation 6 and Equation 7 can be written for calculating the elemental mass of the target sample and its respective uncertainty.

$$\frac{M_{TSP}^e}{M_R^e} = \frac{A_{TSP}^{n,e}/t_{TSP}^{l,e}}{A_R^{n,e}/t_R^{l,e}} * \frac{Exp^{(\lambda_e * t_{TSP}^{w,e})}}{Exp^{(\lambda_e * t_R^{w,e})}} \quad (6)$$

$$\frac{\delta M_{TSP}^e}{M_{TSP}^e} = \left[\left(\frac{\delta A_{TSP}^{n,e}}{A_{TSP}^{n,e}} \right)^2 + \left(\frac{\delta A_R^{n,e}}{A_R^{n,e}} \right)^2 + \left(\frac{\delta M_R^e}{M_R^e} \right)^2 \right]^{1/2} \quad (7)$$

Where,

$M_{TSP}^e = (M_{TSP}^K, M_{TSP}^{Na}, M_{TSP}^{Mn})$ and $M_R^e = (M_R^K, M_R^{Na}, M_R^{Mn})$ are the masses of the elements (e) K, Na and Mn in the TSP and reference samples;

$A_{TSP}^{n,e} = (A_{TSP}^{n,K}, A_{TSP}^{n,Na}, A_{TSP}^{n,Mn})$ and $A_R^{n,e} = (A_R^{n,K}, A_R^{n,Na}, A_R^{n,Mn})$ are the net peak areas (n) corresponding to the gamma-ray energies of the elements (e) K, Na, and Mn in the TSP and the reference samples;

$t_{TSP}^{l,e} = (t_{TSP}^{l,K}, t_{TSP}^{l,Na}, t_{TSP}^{l,Mn})$ and $t_R^{l,e} = (t_R^{l,K}, t_R^{l,Na}, t_R^{l,Mn})$ are the values corresponding to the live time (t^l) of counting in the gamma-ray energy spectrum of the TSP and the reference samples;

$t_{TSP}^{w,e} = (t_{TSP}^{w,K}, t_{TSP}^{w,Na}, t_{TSP}^{w,Mn})$ and $t_R^{w,e} = (t_R^{w,K}, t_R^{w,Na}, t_R^{w,Mn})$ correspond to the waiting time (t^w), defined as the interval between the end of neutron irradiation and the start of gamma spectrum acquisition for the TSP and the reference sample in the spectrometry system;

λ_e is the radioactive decay constant, which in this study refers to the radioisotopes ^{42}K , ^{24}Na e ^{56}Mn ;

δ the uncertainties associated with each measurement.

For the TSP samples, the total mass was determined using Equation 2, and the corresponding elemental masses were calculated using Equation 6. Consequently, the concentrations (mg/100 g) and associated uncertainties for the elements K, Na, and Mn in these samples were derived according to Equation 8 and Equation 9, as presented below.

$$C_{PTS}^e = M_{PTS}^e / M_{PTS} \quad (8)$$

$$\delta_{C_{PTS}^e} = C_{PTS}^e * \left[\left(\delta_{M_{PTS}^e} / M_{PTS}^e \right)^2 + \left(\delta_{M_{PTS}} / M_{PTS} \right)^2 \right]^{1/2} \quad (9)$$

2.4 Gamma-ray Spectrometry and Half-life Test - Elemental Identification Technique

Gamma spectrometry, GS, is a technique used for the identification and quantification of radionuclides through the detection of emitted gamma radiation. Among the various types of available detectors, the High-Purity Germanium, HPGe, semiconductor stands out, employed in this study due to its high energy resolution.

The GS technique enables the generation of energy spectra, which are graphs that relate the number of events (counts) as a function of the energy of the detected photons (keV). These graphs show well-defined peaks at specific energies, corresponding to emissions from the radiation source. By analyzing these peaks, it is possible to identify which radionuclides are present in a sample (Knoll, 2000), since all radioisotopes are characterized by their half-lives ($T_{1/2}$), as well as by the nature and characteristic energy (keV) of the radiation they emit (ANVISA, 2019).

For the acquisition of gamma spectra, a system composed of an HPGe detector from ORTEC was used, with 25 % relative efficiency and 1.8 keV resolution at the 1332 keV photopeak of the isotope ^{60}Co , coupled to the DSPEC jr 2.0 electronic system, located in the LMN/SEREA/IEN laboratory. The collection, processing, and storage of spectrometric data were carried out using Maestro v7.0 software, installed on a computer dedicated to the system, allowing the visualization, energy calibration, and analysis of the obtained spectra. The system also includes a lead shielding, used to protect the detector from background radiation.

Table 2 presents the gamma-ray emitting radionuclides, resulting from the most probable neutron capture reactions $^A\text{X}(n,\gamma)^{A+1}\text{X}$ for the elements K, Na, and Mn, as well as their characteristic energies and half-lives.

Table 2: Gamma Energy and Half-Life of the Activate Elements K, Na and Mn – (IAEA Data Source)

TSP	Gamma Energy (keV)	($T_{1/2}$) (s)
^{42}K	1524.86	44,478.0
^{24}Na	1368.62	53,841.6
^{56}Mn	846.76	9,284.0

In order to avoid false diagnoses that may lead to the incorrect identification of radionuclides after spectrum acquisition, a qualitative test is necessary to prove that the obtained characteristic energy peaks correspond to the indicated elements, with the half-life test being an irrefutable proof of such identifications.

The half-life test ($T_{1/2}$) of an element by means of GS is relatively simple. It is sufficient to perform successive acquisitions of the spectra from the same radioactive source, with time intervals that allow observing the decrease in the areas of the peaks of interest in each stored spectrum, according to the radioactive decay law. After subtracting the background radiation spectrum, the remaining peaks can be analyzed.

Selecting the peaks to be identified, the count rate, CR, of each is calculated by dividing the area of each peak by the corresponding acquisition time. Repeating this procedure for the other spectra yields the necessary values for constructing graphs of the count rates of the elements X time. When the ordinate axis is represented on a logarithmic scale, the curve takes on a linear form (ANVISA, 2019).

The slope of this fitted line corresponds exactly to the decay constant, λ , of the radionuclide under analysis, which is inversely related to the half-life, as described in Equation 10, Equation 11, Equation 12 and Equation 13.

$$y = y_0 - a * x \quad (10)$$

$$CR = A_{TSP}^{n,e} / t_{TSP}^{l,e} \quad (11)$$

$$\ln(CR) = \ln(CR)_0 - \lambda * t \quad (12)$$

$$T_{1/2} = \ln 2 / \lambda \quad (13)$$

3 Results and Discussions

The first TSP sample, with a mass of (1.6 ± 0.00005) g, was irradiated for 30 minutes at a power of 340 W. After the GS analysis, the peaks related to the elements under study were identified in the spectrum, as highlighted in Figure 4. The confirmatory half-life test was performed, and the results, presented in Table 3, certify the presence of Mn, Na, and K in the sample.

Table 3: Half-Lives Measured and Relative Error for the K, Na and Mn of the TSP Sample

TSP	Gamma Energy (keV)	$T_{1/2}$ (IAEA) (s)	$T_{1/2}$ (this work) (s)	Relative error (%)
^{42}K	1524.86	44,478.0	$(45,123.2 \pm 720.5)$	1.5
^{24}Na	1368.62	53,841.6	$(51,233.4 \pm 3196.8)$	4.5
^{56}Mn	846.76	9,284.0	$(9,430.6 \pm 187.9)$	1.6

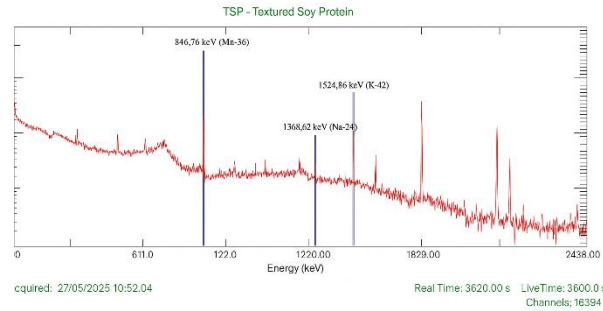


Figura 4: TSP spectrum [log(counts) x energy] with Mn^{56} , Na^{24} , and K^{42} peaks marked from left to right side.

Five additional irradiations were carried out at a power of 34 W for 30 minutes each. Table 4 presents the results of the mean elemental concentrations (mg/100 g) obtained by the INAA technique using the comparative method, and the result is compared with that obtained by Hamdy (1974).

Table 4: Concentrations (mg/100 g) of K, Na and Mn in the TSP Sample and Dietary Reference Intakes

TSP	C_{PTS} (Hamdy, 1974) (mg/100 g)	$\overline{C_{PTS}}$ (this work) (mg/100 g)	Dietary Reference Intakes (mg/ day)
K	2090.0	(1520.7 ± 93.3)	4700.0 (IOM, 2005)
Na	0.9	(4.3 ± 1.4)	2000.0 (WHO, 2012)
Mn	3.8	(2.3 ± 0.1)	{ 2.3 – Men 1.8 – Women (IOM, 2001)

4 Conclusions

The results obtained through spectra, tables, and calculations demonstrate the identification of the elements based on their gamma energies and half-lives, as well as the concentration values (mg/100 g) of the elements K (1520.7 ± 93.3), Na (4.3 ± 1.4) and Mn (2.3 ± 0.1), respectively.

Regarding the comparison between the results of this study and those reported by Hamdy (1974), as well as the Dietary Reference Intakes for potassium (IOM, 2005), sodium (WHO, 2012), and manganese (IOM, 2001), several factors should be taken into consideration. These include the geographic origin of the samples, soil composition, agricultural practices, including the use of different modern inputs, and the extrusion process, which involves the addition of liquids that may contain varying concentrations of these elements. Such variables can significantly influence the elemental composition of soy-based products.

It is concluded that the INAA and GS techniques proved to be effective both in the identification and quantification of the elements present in the analyzed sample. This confirmation was made based on the analysis of the obtained spectra, considering the characteristic energy peaks and the half-life tests for each radionuclide, which ensured accuracy in the assignment of the elements. The combined use of these techniques offers significant advantages, including non-destructive analysis, high sensitivity, multi-element capability, and minimal sample preparation, enabling efficient utilization of the infrastructure available at the Argonauta research reactor (IAEA-TECDOC-564, 1990).

The results indicate that the TSP analyzed is an excellent source of potassium (IOM, 2005), an essential mineral for muscle function, blood pressure regulation, and maintenance of hydroelectrolytic balance. Furthermore, the low sodium content found is a positive factor since excessive sodium intake is associated with cardiovascular diseases (WHO, 2012). The presence of manganese (IOM, 2001) in adequate concentrations is also relevant, given that this element participates in several metabolic reactions and is essential for the proper functioning of antioxidant enzymes.

Thus, the data obtained reinforce the nutritional value of TSP, not only as a plant-based protein source but also as a functional food that can significantly contribute to a balanced diet. This is particularly important in urban contexts such as the city of Rio de Janeiro, where the search for healthy and sustainable alternatives to animal-based proteins has intensified. Therefore, this study provides valuable information that can guide public policies, school feeding programs, and nutritional awareness campaigns, promoting the health of the population.

Acknowledgements

The authors thank IEN/CNEN for technical and infrastructural support, and CNPq for financial support through the Scientific Initiation (IC) scholarship.

References

- Alamu, E. O., & Busie, M. D. (2019) Effect of textured soy protein (TSP) inclusion on the sensory characteristics and acceptability of local dishes in Nigeria. *Cogent Food & Agriculture*, 5(1)
- Albuquerque, B. F. (2020) *Técnicas analíticas para a análise qualitativa e quantitativa em amostras de cimento Portland*. Rio de Janeiro: CNEN/IEN
- ANVISA. (2019) *Farmacopeia brasileira*. Vol. 1. 6. ed. Brasília: ANVISA
- ANVISA. (2016) *Alimentos com alegações de propriedades funcionais e ou de saúde*. Brasília: ANVISA
- De Bruin, M. (1983) *Instrumental neutron activation analysis – a routine method*. Delft University Press

- D'Elia, L. (2024) Potassium intake and human health. *Nutrients*, 16(6), 833
- Duderstadt, J. J., & Hamilton, L. J. (1976) *Nuclear reactor analysis*. John Wiley & Sons
- Gilmore, G.; Joss, D. (2024) *Practical gamma-ray spectrometry*. John Wiley & Sons
- Glascocock, M. (2003) *An overview of neutron activation Analysis*. University of Missouri Research Reactor
- Hamdy, M.M. (1974) Nutritional aspects in textured soy proteins. *J Am Oil Chem Soc*, Vol.51: 85A-90A
- He F.J, MacGregor GA. (2008) Beneficial effects of potassium on human health. *Physiol Plant*. Aug;133(4):725-735
- IAEA-TECDOC-564. (1990) *Practical Aspects of Operating a Neutron Activation Analysis Laboratory*. Austria: IAEA
- IOM. (2005) *Dietary Reference Intakes for Water, Potassium, Sodium, Chloride, and Sulfate*. National Academies Press
- IOM. (2001) *Panel on Micronutrients. Dietary Reference Intakes for Vitamin A, Vitamin K, Arsenic, Boron, Chromium, Copper, Iodine, Iron, Manganese, Molybdenum, Nickel, Silicon, Vanadium, and Zinc*. Washington: National Academies Press
- Jaques, D. A., & Ponte, B. (2023) Dietary sodium and human health. *Nutrients*, 15(17), 3696
- Knoll, G. F. (2000) *Radiation detection and measurement*. John Wiley & Sons
- Kodintsev V.V., Lenda I.V., Ponomarev A.V., Naumov N.A., Salatov Ya.S. (2022) Sodium as one of the basic elements in human life, *European Journal of Natural History*. N. 5: 4-7
- Kruger, P. (1971) *Principles of activation analysis*. John Wiley & Sons
- Nogueira-de-Almeida, C. A., Ferraz, I. S., Ued, F. V., Almeida, A. C. F., & Del Ciampo, L. A. (2020) Impact of soy consumption on human health: integrative review. *Brazilian Journal of Food Technology*, 23, e2019129
- Paulilo, M. T. S.; Viana, A. M.; Randi, Á. M. (2015) *Fisiologia vegetal*. 1a. ed. Florianópolis: Universidade Federal de Santa Catarina. p. 182
- Riaz, M. N. (2001) Textured soy protein and its uses. *Agro Food Industry Hi-Tech*. 12(5): 28–31.
- Rizzo, G., and Baroni, L. (2018) Soy, Soy Foods and Their Role in Vegetarian Diets. *Nutrients*, 10(1), 43
- Santamaria, A. B. (2008) Manganese exposure, essentiality & toxicity. *Indian J Med Res*. Oct;128(4):484-500. PMID: 19106442
- Silva, F., Câmara, G., Sediyaama, T., Borém, A. (2022) *Soja: do plantio à colheita*. 2. ed. Oficina de Textos
- WHO. (2012) *Guideline: Sodium intake for adults and children*. World Health Organization
- Zhang, J., Liu, L., Liu, H., Yoon, A., Rizvi, S. S. H., & Wang, Q. (2018) Changes in conformation and quality of vegetable protein during texturization process by extrusion. *Critical Reviews in Food Science and Nutrition*, 59(20), 3267–3280

Neutron Source Applied for Eccentric Scale Prediction in Oil Pipelines using Deep Learning

C.M. Salgado¹, W.L. Salgado¹, E.M. de Oliveira¹, P.A.L. da Cruz¹, R.S.F. Dam^{1*}

¹ Instituto de Engenharia Nuclear (IEN), Divisão de Radiofármacos (DIRAD), Zip Code 21941-906, Rio de Janeiro, Brazil.

* rdam@coppe.ufrj.br

Abstract

Asymmetric scale buildup, such as barium sulfate deposits, remains a critical issue in oil and gas pipelines due to its potential to restrict flow and reduce system performance. This work introduces an innovative method for estimating the maximum thickness of calcium carbonate (CaCO_3) eccentric scale using a combination of neutron-based nuclear techniques and deep neural networks (DNNs). The approach employs a $^{241}\text{Am-Be}$ neutron source and 72 NaI(Tl) scintillation detectors, modeled through MCNP6 Monte Carlo simulations, to analyze a three-phase annular flow containing crude oil, saltwater, natural gas, and asymmetric scale. The neutron source was collimated (4.0 cm^2 emission surface), with energy spectrum standardized by ISO 8529-1. The MCNP6 F1 tally computed integrated gamma flux across a spherical surface, complemented by a theoretical 4π Spherical Detector as a performance benchmark. Gamma rays resulting from neutron interactions with hydrogen, oxygen, and barium are detected and processed. In total, 2,880 cases were generated from 40 thickness levels (26–164 mm) and 72 angular positions (5° steps), capturing a wide range of asymmetric scale deposition patterns. Specific energy window counts from the gamma spectra serve as input features for training the DNN model. Hyperparameter tuning using the Optuna library enhanced model performance and generalization. The optimized network achieved 99.51% accuracy in predicting the maximum scale thickness, with a relative error below 5% and a mean absolute percentage error (MAPE) of 0.85% on the test set. The results support the feasibility of this hybrid approach for non-invasive, real-time scale monitoring in multiphase industrial flow systems.

Keywords: Eccentric scale; Deep neural network; Neutron, Multiphase flow, MCNP6 code.

1 Introduction

In Brazilian pre-salt oil and gas production, CO_2 concentrations in associated gas can surpass 20% in some fields (Cezar et al., 2015). When combined with high-salinity environments containing calcium ions up to 10,000 ppm, this promotes inorganic scale formation, severely affecting flow assurance (Bezerra et al., 2013). Given the region's vast reserves and the complexity of deep offshore operations, addressing scale buildup is essential to maintain system efficiency (Santana and Manzela, 2016).

Scale originates from thermodynamic and chemical instabilities during production, forming precipitates like BaSO_4 , SrSO_4 , CaSO_4 , and CaCO_3 (Kamal et al., 2018), with CaCO_3 being the most prevalent in subsea and surface pipelines (Lakshmi et al., 2013). CO_2 and H_2S increase fluid acidity, enhancing mineral solubility, while microbial activity further contributes to CaCO_3 precipitation. These deposits form under specific pH, temperature, and pressure conditions, leading to reduced flow and requiring costly interventions (Kamal et al., 2018). Adhering to surfaces throughout reservoirs and pipelines, scale morphology depends on factors such as flow rate, supersaturation, and surface roughness (Liu et al., 2022). While removal techniques vary, prevention remains the most effective strategy (Vazirian et al., 2016). Conventional prediction models, based on solubility and saturation indices, often lack precision

under dynamic multiphase conditions (Olajire, 2015). Nuclear-based methods, like gamma densitometry, neutron activation, and X-ray analysis, are emerging as viable alternatives for non-invasive monitoring (Roshani et al., 2021; Osman and Abdel-monem, 2021; Salgado et al., 2021).

Artificial Neural Networks (ANNs) enhance these techniques by interpreting complex data, especially in eccentric scale scenarios where thickness varies spatially. Using MCNP6 simulations (Goorley et al., 2016), recent studies have demonstrated the successful integration of ANNs with gamma densitometry (Dam et al., 2023; Carvalho et al., 2024) and neutron activation (Salgado et al., 2023, 2025; Dam et al., 2024). For instance, Salgado et al. (2025) reported a mean relative error of 3.01% and 92.68% of predictions below 5% when estimating concentric scale thickness with a ^{241}Am -Be source and DNNs. Building on this, the present study aims to predict eccentric CaCO_3 scale thickness in three-phase flow (oil, saline water, gas) using deep learning and neutron interactions, targeting real-time industrial applications.

2 Methodology

This study proposes a computational framework for analyzing eccentric calcium carbonate (CaCO_3) scale deposition in annular three-phase flows using neutron-gamma transport simulations via the MCNP6 code (Goorley et al., 2016). The approach relies on prompt gamma emissions from $^1\text{H}(n,\gamma)$, $^{16}\text{O}(n,\gamma)$, and $^{56}\text{Fe}(n,\gamma)$ reactions, induced by a ^{241}Am -Be neutron source, combined with nuclear data from ENDF/B-VI.8 and MCPLIB04 libraries, from mathematical simulations. Taking advantage, the contrasting interaction behavior of neutrons and photons, light elements like hydrogen being neutron-sensitive, while gamma attenuation is intense in dense materials like iron, the method enables non-invasive distinction of fluids and scale, even in heterogeneous environments.

2.1 Geometry Simulation

The simulations modeled a 316L stainless steel pipe containing oil, gas, brine, and CaCO_3 scale deposits, based on mass weight and densities from McConn Jr. et al. (2011). The neutron source was collimated (4.0 cm^2 emission surface), with energy spectrum standardized by ISO 8529-1 (Kluge and Weise, 1982). The long half-life of ^{241}Am (432 years) ensures operational longevity (Sohrabpour et al., 1999; Nunes et al., 2002). A total of 72 detectors were positioned circumferentially around the pipe to capture omnidirectional photon flux, supporting detailed gamma spectral analysis for estimating maximum thickness of eccentric scale (ScaleMax). To reduce scattering artifacts and maintain computational efficiency, collimated geometries were applied (Nunes et al., 2002). The MCNP6 F1 tally computed integrated gamma flux across a surface (4π Detector Surface as a performance benchmark). Gamma spectra were segmented into 190 energy bins (10 keV from 10 keV–1 MeV; 100 keV from 1–10 MeV). The simulations modeled annular three-phase flow (80% oil, 10% brine, 10% gas) with eccentric CaCO_3 scale (see Figure 1). In total, 2,880 cases were generated from 40 thickness levels (26–164 mm) and 72 angular positions (5° steps), capturing a wide range of asymmetric scale deposition patterns.

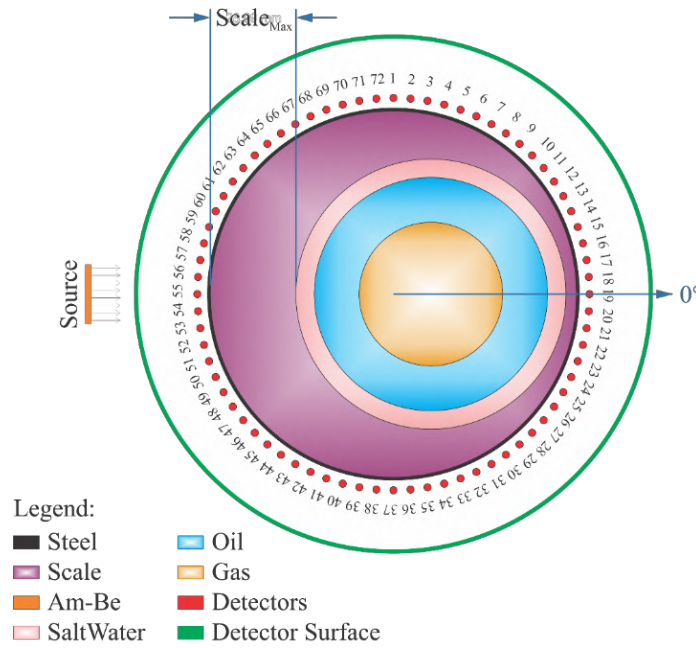


Figure 1: Geometry model for three-phase in the annular flow simulations using MCNP6.

2.2 Deep Neural Network

Deep Neural Networks (DNNs), a class of Artificial Neural Networks (ANNs), are designed to model complex, non-linear relationships through multiple hidden layers that perform hierarchical feature learning (Schmidhuber, 2015). Training was performed using backpropagation, with adaptive training rate, (Adam optimizer) (Kingma and Ba, 2015), with early stopping applied to avoid overfitting. The dataset (2,880 patterns) was split into training (70%), validation (20%), and test (10%) subsets (Haykin, 1999). Input data were normalized using z-score transformation (mean = 0, standard deviation = 1) to improve convergence stability. The Optuna library (Akiba et al., 2019) automated hyperparameter optimization, exploring:

- Hidden layers: 1–10
- Neurons per layer: 100–1000
- Activation functions: ReLU, ELU, SELU, Swish (Glorot and Bengio, 2010; Clevert et al., 2016; Klambauer et al., 2017; Ramachandran et al., 2017)

Fixed parameters included the MSE loss function, and batch size = 256. The DNN's input layer consisted of 13,680 features (190 energy bins $\times$ 72 detectors), with a single output neuron predicting eccentric CaCO_3 scale thickness.

Performance was assessed using MSE, RMSE, MAPE, and R2. The model was implemented in Python using scikit-learn (Pedregosa et al., 2011). Two configurations were compared:

- ALL: Full 13,680-variable dataset from all detectors.
- SURF: Spherical surface (reference) detector.

3 Results

To improve eccentric scale thickness prediction in annular multiphase systems, various Deep Neural Network (DNN) architectures were designed and optimized using the Optuna library (Akiba et al., 2019), which enables automated hyperparameter tuning. The optimization involved adjusting the number of hidden layers, neurons per layer, activation functions, learning rates, and dropout rates to balance performance and overfitting risk. Two optimized architectures were selected:

- ALL: Incorporating spectral data from 72 detectors.
- SURF: Using a spherical detector response as a benchmark reference.

Table 1 presents the detailed characteristics of the specific DNN architectures adopted in this study, with a focus on balancing generalization capacity and overfitting risk.

Table 1: Summarize the architectures.

Configuration	#Hidden Layers	L0 (U/A)	L1 (U/A)
SURF	2	662/Swish	835/ReLU
ALL	1	996/ELU	–

Ln (U/A): layer n – number of units (U) / activation function (A)
 “–” indicates the absence of a second hidden layer.

The SURF setup is a design based on an integrated gamma signal, it lacks the spatial resolution provided by multiple detectors, which limits its accuracy. Nevertheless, it serves as a valuable theoretical benchmark. In contrast, the ALL model, using full spatial and spectral information (72 detectors $\times$ 190 energy bins), exhibited enhanced sensitivity to spatial variations in scale deposition. Despite being a shallower input architecture, its compact single-layer configuration (996 neurons, ELU activation) provided strong generalization and robustness. These findings confirm that architecture leveraging richer input features, deliver superior predictive accuracy for eccentric scale thickness estimation in multiphase flow systems.

Model performance was evaluated using four standard metrics: mean squared error (MSE), mean absolute percentage error (MAPE), root mean squared error (RMSE), and the coefficient of determination (R²). These indicators collectively offer a robust framework to assess predictive accuracy, generalization capability, and error magnitude. As shown in Figure 2, the comparison across architecture reveals how model complexity and input dimensionality influence overall performance. The joint interpretation of these metrics ensures a comprehensive and reliable evaluation of each DNN model’s robustness and predictive consistency.

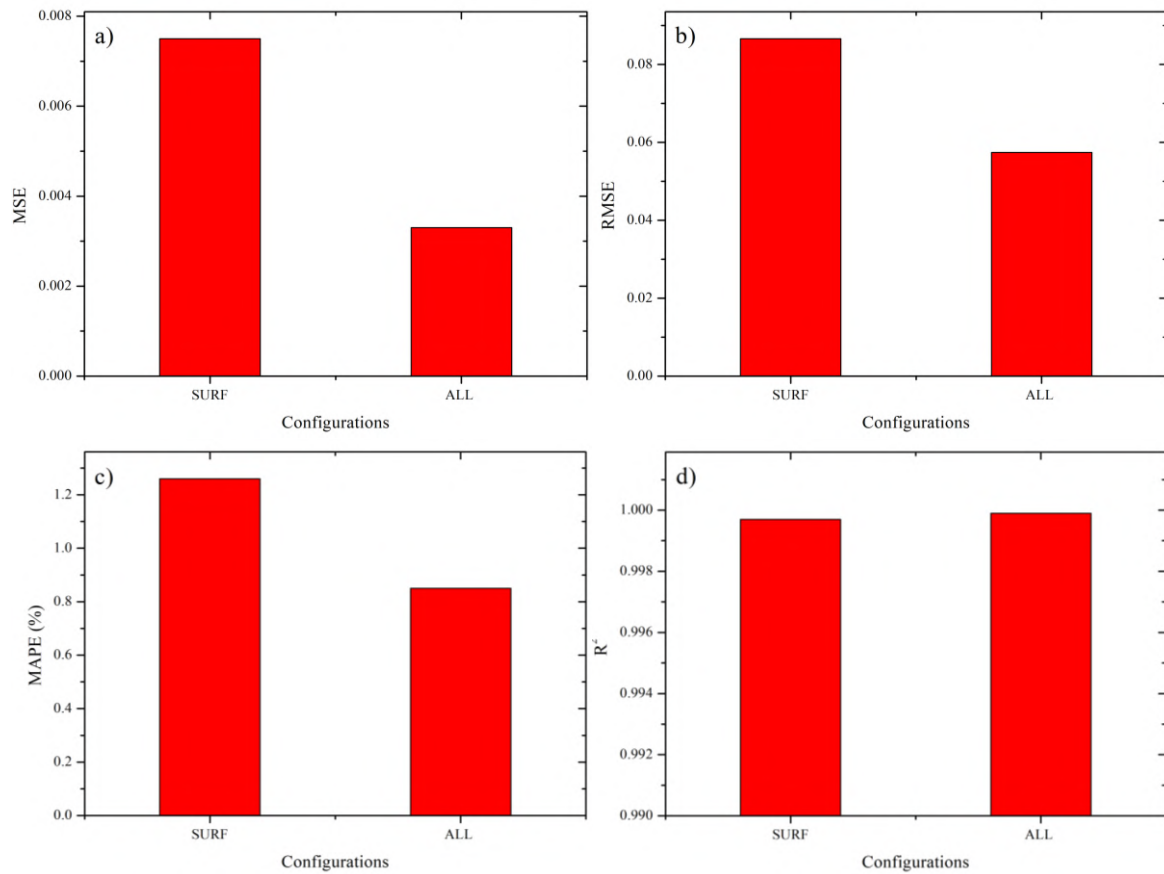


Figure 2: Comparison of SURF and ALL models in terms of the following metrics: a) MAPE (%); b) MSE; c) RMSE; d) R^2 .

The analysis of performance metrics (MSE, RMSE, MAPE, and R^2) demonstrates that the model utilizing all detectors (ALL) achieved the best overall performance, with the lowest MSE and RMSE values and an R^2 close to 1. These results highlight its robustness and strong predictive capability, particularly in complex scenarios. The SURF model, based on a single integrated reading of total radiation over the system's surface, showed a good performance too. Despite its lower accuracy compared to the ALL model, SURF is notable for its simple implementation and value as a theoretical benchmark. Its condensed structure, which aggregates radiation into a single spectrum, offers a conceptually elegant and computationally lightweight approach. However, results indicate that the lack of spatial information limits its predictive capability. In contrast, the ALL model leverages the spatial and spectral distribution of radiation across 72 detectors and 190 energy channels, providing a more detailed physical representation and, consequently, superior performance. Overall, models with richer input information, such as ALL, deliver better predictive results.

In addition to traditional evaluation metrics, a complementary analysis was conducted based on relative error range classification, aiming to assess not only the average prediction error but also its distribution across configurations. Table 2 presents the distribution of sample counts (and corresponding percentages) within different relative error intervals. Configurations that concentrate most of their errors within the (0%, 5%] range are considered more robust and reliable, particularly for applications requiring high predictive accuracy.

The selection of the best configuration was based on two main criteria: the highest concentration of errors within the (0%, 5%] range and the lowest number of occurrences with errors exceeding 10%. A detailed analysis reveals the differences in robustness and reliability between the tested models.

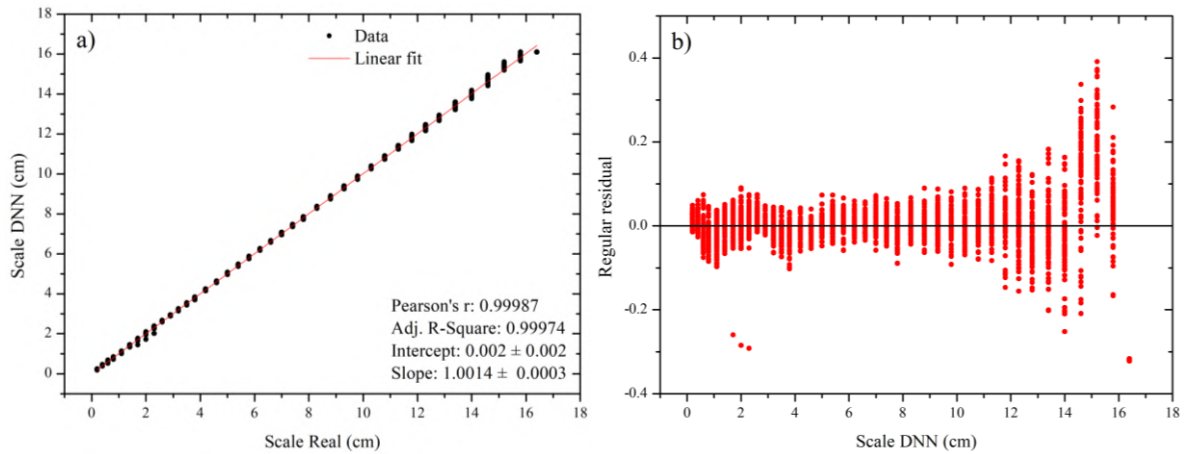
Table 2: Distribution of Relative Error Ranges for Different Detector Configurations.

Configuration	(0, 5]	(5, 10]	(10, 15]	(15, 20]	(20, 25]
SURF	94.55 % (2723)	3.68 % (106)	1.35 % (39)	0.31 % (9)	0.10 % (3)
ALL	99.51 % (2866)	0.45 % (13)	0.03 % (1)	0 % (0)	0 % (0)

- ALL exhibited the best overall performance, with 99.51% of predictions having a relative error $\leq 5\%$ and only 0.03% with errors exceeding 10%, demonstrating high precision and robustness.
- SURF achieved 94.55% of predictions with error $\leq 5\%$, but displayed higher error dispersion, with 1.76% of predictions having errors above 10%, indicating greater variability.

The SURF configuration, based on a 4π detector surface, is conceptually simple and serves as a theoretical reference. However, because it does not account for spatial variations in radiation distribution, it lacks practical applicability in real-world scenarios. In contrast, the ALL configuration, which includes 72 detectors with their respective energy distributions, achieved the best performance in the evaluated statistical metrics. Nevertheless, its practical implementation is significantly constrained by the system's complexity, high installation costs, and the challenges associated with calibrating and maintaining multiple sensors and energy channels.

Regression analysis is a key tool for evaluating the relationship between dependent and independent variables. Linear regression enables estimation, assessment of variable influence, and evaluation of model fit. Figure 3 shows the regression and residual plots for the Test subset, providing insight into the model's accuracy and the validity of linear assumptions.



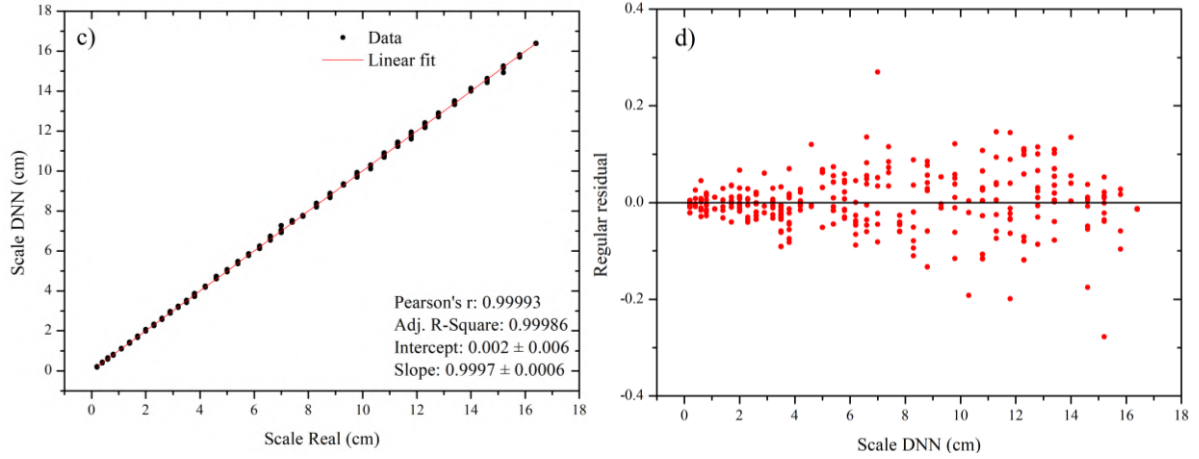


Figure 3: Scatter plots with adjusted regression lines and residual plots, considering the configurations for a,b) SURF; c,d) ALL.

The analysis of the linear regression plots demonstrates the high performance of the deep neural network (DNN) architecture in predicting scale thickness. The adjusted coefficients of determination ($R^2 > 0.999$), slopes close to unity, and intercepts near zero indicate an excellent linear agreement between predicted and actual values, with no significant systematic bias. The close alignment of data points with the identity line and the low dispersion observed reinforce the model's accuracy and the absence of significant outliers. This consistent behavior, particularly in the ALL configuration, illustrates the robustness and stability of the model across different combinations of input features. These results validate the effectiveness of the proposed DNN architecture, which exhibits high accuracy, and generalization capability, essential attributes for industrial applications exposed to varying operational conditions.

Following the regression analysis, residual plots were examined to assess model adequacy and to verify whether key regression assumptions, such as homoscedasticity, normality, and independence of residuals, were satisfied. Most residuals are tightly clustered around zero, suggesting a high level of prediction accuracy. For thickness values below approximately 10 cm, the residuals remain consistently low and centered, indicating a good fit and absence of bias in this range. However, beyond 12 cm, an increase in residual dispersion is observed, indicating reduced predictive performance, mainly in SURF configuration.

Overall, the reduction in input features negatively impact prediction reliability for larger thickness values, introducing non-constant error variance that may undermine the model's robustness. Thus, an appropriate balance between model complexity and predictive power must be achieved. Future improvements should consider expanding the dataset, implementing stronger regularization strategies, and performing careful hyperparameter optimization to improve accuracy, particularly for higher scale thickness values. Figure 4 shows the errors in histogram produced by the deep neural network, highlighting the distribution of residuals relative to the predicted values.

The histogram for the SURF configuration (Figure 4a) shows an error distribution centered around zero, indicating that the model predictions are generally unbiased. However, it exhibits slightly greater dispersion compared to the ALL configuration, with a higher frequency of errors in the -0.1 to $+0.2$ range. This suggests increased variability in predictions and a reduced precision. Additionally, a subtle positive skewness is observed, implying a tendency for overestimation in some predictions. In contrast, the ALL configuration (Figure 4b) presents a narrower and more symmetric error distribution centered at zero, indicating higher precision and a lower frequency of large errors. The symmetry of the distribution suggests that the model maintains a balanced prediction behavior without systematic over- or underestimation.

These results imply that while both models achieve acceptable accuracy, the ALL configuration, by incorporating a more comprehensive set of features, enhances consistency and reduces prediction

variance. Thus, merging multiple input variables to mitigate the limitations observed in simpler configurations such as SURF, resulting in increased robustness and generalizability.

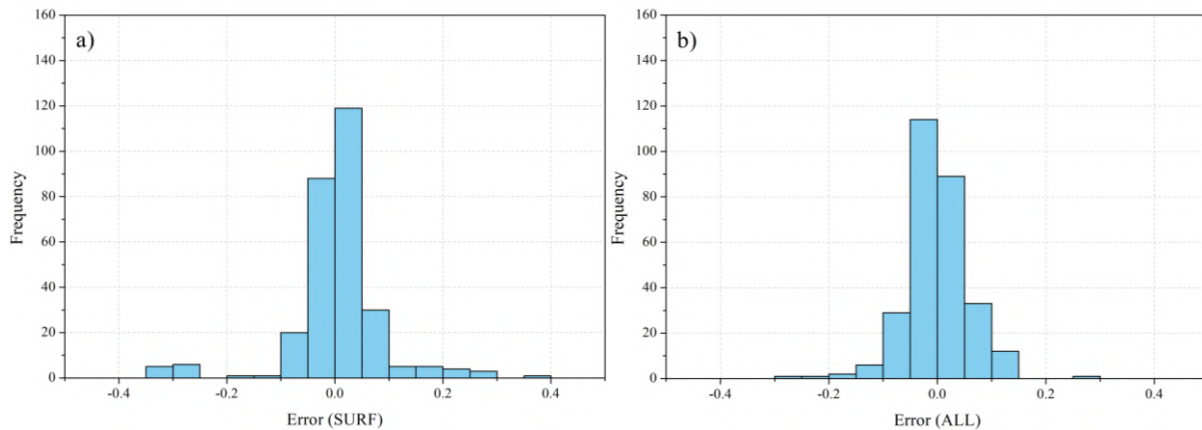


Figure 4: Histogram of error distribution for the configurations for: a) SURF; b) ALL.

4 Conclusion

This study proposes an innovative methodology for predicting the eccentric thickness of calcium carbonate (CaCO_3) scale in oil pipelines by combining radiative transport simulations using a ^{241}Am -Be neutron source with deep neural networks. The most effective configuration, involving a full-detector setup (ALL), demonstrated high predictive accuracy, with over 99.51% of predictions exhibiting absolute errors below 5% and a MAPE of 0.85% on the test set. Analyses of regression coefficients and residuals confirmed the models' strong generalization ability, accurately capturing scale thickness trends even in the presence of spatial heterogeneity. Error distributions further demonstrated that both SURF and ALL models achieved acceptable accuracy, though increased dispersion at higher thicknesses indicated a reduction in predictive precision under those conditions. The findings underscore the potential of combining nuclear techniques with artificial intelligence as a robust and non-invasive approach to scaling monitoring in multiphase systems. Nonetheless, the models' reduced performance for higher scale thicknesses suggests the need for further improvements. Future work should prioritize the optimization of regularization and feature extraction strategies to enhance model robustness, reduce configuration complexity, and improve performance across a wider range of scale thicknesses. This is particularly important, as the practical implementation of the ALL configuration may be constrained by system complexity and costs associated with sensor integration and calibration.

Acknowledgements

The authors gratefully acknowledge financial support from Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq) and Comissão Nacional de Energia Nuclear (CNEN). Thanks also to Instituto de Engenharia Nuclear (IEN).

References

- Akiba, T., Sano, S., Yanase, T., Ohta, T., Koyama, M., 2019. Optuna: A Next-generation Hyperparameter Optimization Framework, in: Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining. pp. 2623–2631. <https://doi.org/10.1145/3292500.3330701>.
- Bezerra, M. C. M.; Rosário, F. F.; Rosa, K. R. S. A., 2013. Scale management in deep and ultra-deep water fields. In: Offshore Technology Conference Brazil, 2013, Rio de Janeiro. Proceedings [...]. Rio de Janeiro: Offshore Technology Conference.

- Carvalho, A.C.L., Salgado, W.L., Dam, R.S.F., Conti, C.C., Salgado, C.M., 2024. Calcium carbonate scale thickness prediction in annular three-phase flow using gamma-ray densitometry and artificial neural networks, *Radiation Physics and Chemistry*, v.223. <https://doi.org/10.1016/j.radphyschem.2024.112018>.
- Cezar, A.S.P. Pereira, A. R., Daniel. L.L., Oazen, E.V., Trovoado, L.C., Almeida, M.C., 2015. Subsea solutions in the pre-salt development projects. In: *Offshore Technology Conference*, 2015, Houston. Proceedings [...]. Houston: Offshore Technology Conference.
- Clevert, D.A., Unterthiner, T., Hochreiter, S., 2016. Fast and accurate deep network learning by exponential linear units (ELUs), *ICLR*, arXiv:1511.07289.
- Dam, R.S.F., Salgado, W.L., Conti, C.C., Schirru, R., Salgado, C.M., 2024. Volume fraction detection in multiphase systems using neutron activation analysis and artificial neural network, *Applied Radiation and Isotopes*, v.214. <https://doi.org/10.1016/j.apradiso.2024.111504>.
- Dam, R.S.F., Santos, M.C., Salgado, W.L., Cruz, B.L., Schirru, R., Salgado, C.M., 2023. Prediction of fluids volume fraction and barium sulfate scale in a multiphase system using gamma radiation and deep neural network, *Applied Radiation and Isotopes*, v.201. <https://doi.org/10.1016/j.apradiso.2023.111021>.
- Glorot, X., Bengio, Y., 2010. Understanding the difficulty of training deep feedforward neural networks. In: *Proceedings AISTATS*, 9, pp.249 – 256.
- Goorley, T., James, M., Booth, T., Brown, F., Bull, J., Cox, L.J., Durkee, J., Elson, J., Fensin, M., Forster, R.A., Hendricks, J., Hughes, H.G., Johns, R., Kiedrowski, B., Martz, R., Mashnik, S., McKinney, G., Pelowitz, D., Prael, R., Sweezy, J., Waters, L., Wilcox, T., Zukaitis, T., 2016. Initial MCNP6 release overview. *Nucl. Technol.* 180 (3), 298–315. <https://doi.org/10.13182/NT11-135>.
- Haykin, S., 1999. *Neural networks: A comprehensive Foundation*. 9^a. ed. [S. l.]: Pearson Education, 2005. ISBN 81-7808-300-0.
- Kamal, M. S., Hussein, I., Mahmoud, M., Sultan, A., Saad, M.A.S., 2018. Oilfield scale formation and chemical removal: A review. *Journal of Petroleum Science and Engineering*, v. 171, pp. 127-139.
- Kingma D.P., Ba J.L., 2015, Adam: A method for stochastic optimization. *Proceedings of 3rd International Conference for Learning Representations*, San Diego. arXiv:1412.6980.
- Klambauer G., Unterthiner T., Mayr A., Hochreiter S., 2017. Self-Normalizing Neural Networks. *Advances in Neural Information Processing Systems*, 30, pp.972–981.
- Kluge, H., Weise, K., 1982. The neutron energy spectrum of a ^{241}Am -Be(γ ,n) source and resulting mean fluence to dose equivalent conversion factors. *Radiation Protection Dosimetry*. 2, pp.85–93.
- Lakshmi, D.S., Senthilmurugan, B., Drioli, E., Figoli, A., 2013. Application of ionic liquid polymeric microsphere in oil field scale control process. *Journal of Petroleum Science and Engineering*, v. 112, pp.69–77.
- Liu, B., Zhang, X., Zhou, X., Hashimoto, T., Wang j., 2017. The corrosion behavior of machined AA7150-T651 aluminum alloy. *Corrosion Science*, v. 126, pp. 265–271.

- McConn Jr., R.J., Gesh, C.J., Pagh, R.T., Rucker, R.A., Williams III, R.G., 2011. Compendium of Material Composition Data for Radiation Transport Modeling. Radiation Portal Monitor Project. National Laboratory, Pacific Northwest. PIET- 43741-TM-963 PNNL-15870 Rev. 1.
- Nunes, W.V., Silva, A.X., Crispim, V.R., Schirru, R., 2002. Explosives detection using prompt-gamma neutron activation and neural networks. *Applied Radiation and Isotopes*, 56, pp. 937–943.
- Olajire, A.A., 2015. A review of oilfield scale management technology for oil and gas production. *Journal of Petroleum Science and Engineering*, v. 135, pp. 723–737.
- Osman, A.M., Abdel-monem, A.M., 2021. Measurement of void fraction in pipes by nuclear transmission based techniques. *Annals Nuclear Energy* 152. [https://doi.org/ 10.1016/j.anucene.2020.107994](https://doi.org/10.1016/j.anucene.2020.107994).
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., Duchesnay, É. 2011. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, pp.2825–2830.
- Ramachandran P., Zoph B., Le Q.V., 2017. Searching for Activation Functions. arXiv:1710.05941.
- Roshani, M., Phan, G., Roshani, G.H., Hanus, R., Nazemi, B., Corniani, E., Nazemi, E., 2021. Combination of X-ray tube and GMDH neural network as a nondestructive and potential technique for measuring characteristics of gas-oil–water three phase flows. *Measurement*, 168. <https://doi.org/10.1016/j.measurement.2020.108427>.
- Salgado, C.M., Salgado, W.L., Dam, R.S.F., Conti, C.C., 2021. Calculation of scales in oil pipeline using gamma-ray scattering and artificial intelligence, *Measurement*, v.179. <https://doi.org/10.1016/j.measurement.2021.109455>.
- Salgado, W.L., Dam, R.S.F., Puertas, E.J.A, Salgado, C.M., 2025. Predicting scale thickness in three-phase flow using neutron activation analysis and deep learning, *Measurement*, v.242, Part A. <https://doi.org/10.1016/j.measurement.2024.115880>.
- Salgado, W.L., Dam, R.S.F., Salgado, C.M., 2021. Optimization of a flow regime identification system and prediction of volume fractions in three-phase systems using gamma-rays and artificial neural network. *Appl. Radiat. Isot.* 169. <https://doi.org/10.1016/j.apradiso.2020.109552>.
- Salgado, W.L., Dam, R.S.F., Silva, A.X., Salgado, C.M., 2023. Void fraction prediction using prompt gamma neutron activation analysis and artificial intelligence. *Radiation Physics and Chemistry*, 213 <https://doi.org/10.1016/j.radphyschem.2023.111212>.
- Santana, C.J., Manzela, A.A., 2016. Incrustações Inorgânicas em Campos do Pré-Sal. *Revista de Engenharias da Faculdade Salesiana*, v. 4, pp. 22–31.
- Schmidhuber, J., 2015. Deep learning in neural networks: An overview, *Neural Networks*, 61, pp.85-117. <https://doi.org/10.1016/j.neunet.2014.09.003>.
- Sohrabpour, M., Shahriari, M., Zarifian, V., Moghadam, K.K., 1999. Borehole prompt gamma neutron activation and comparison with Monte Carlo simulation using MCNP code: Borehole PGNA experiment comparison with MCNP. *Applied Radiation and Isotopes*, 50, pp.805–810.

Vazirian, M.M., Charpentier, T.V.J., Penna, M.O., Neville, A., 2016. Surface inorganic scale formation in oil and gas industry: As adhesion and deposition processes. *Journal of Petroleum Science and Engineering*, v. 137, pp. 22–32.

Response Function Validation of HPGe Detector Using GADRAS Simulation

Jonathan O. dos Santos^{1*}, Alexander C. F. da Silva¹, Lorena C. Fernandes¹,
Gabriel N. Freitas¹, Leandro B. Silva¹, Ademir X. da Silva¹

¹ Federal University Of Rio de Janeiro, Nuclear Engineering Program, 21941-914, Rio de Janeiro, Brazil.

* jonathan.oliveira.physics@gmail.com

Abstract

In the nuclear field, the automation of gamma spectrum analysis offers significant advantages, such as minimizing human errors in the identification of radioisotopes and reducing analysis time. In this context, the Gamma Detector Response and Analysis Software (GADRAS) stands out due to several features, including modeling of source-detector configurations and automated analysis of gamma spectra. This study aimed to validate the accuracy of GADRAS in identifying radioactive sources and estimating their activity, using a methodology that included experimental tests with a high-purity germanium (HPGe) detector to acquire reference gamma spectra for standard sources (²⁴¹Am, ¹³³Ba, ⁶⁰Co, and ¹³⁷Cs). The validation process involved detector calibration within GADRAS, modeling of the experimental setup, and systematic comparison between predicted and experimental spectra. The results showed agreement between simulated and experimental values, confirming the accuracy of GADRAS in spectrometric analysis. The radionuclides ²⁴¹Am and ¹³⁷Cs exhibited χ^2 values close to the ideal reference value of 1, with 0.37 and 1.53, respectively. Although the radionuclides ⁶⁰Co and ¹³³Ba presented higher χ^2 values (3.77 and 5.63, respectively), these values remain within acceptable limits and are consistent with results reported in the literature. The percentage differences between the activities determined using the HPGe detector and those obtained through the GADRAS software were approximately 3% for ⁶⁰Co, 4% for ¹³³Ba, 7% for ¹³⁷Cs, and 13% for ²⁴¹Am. Therefore, the automation of spectrum analysis using GADRAS constitutes a valuable tool for reliable and efficient nuclear analyses.

Keywords: GADRAS; Gamma spectrometry; HPGe Detector; Automation of radiation spectrum analysis.

1 Introduction

High-resolution gamma spectrometry has been widely employed in the analysis of radioactive samples. It is a non-destructive technique that enables both qualitative and quantitative evaluation of gamma-emitting radionuclides directly within the samples, without the need for chemical separation procedures. Among the various instruments available for gamma radiation detection and measurement, semiconductor detectors particularly high-purity germanium (HPGe) detectors stand out due to their intrinsic advantages. These include excellent energy resolution, linear response over a broad energy range, short pulse rise time, and low sensitivity to magnetic fields (Semkova et al., 2023; Prozorova et al., 2019).

The detector response function is essential for the accurate interpretation of experimental data and is fundamental for the characterization of nuclear instruments under development. However, the experimental determination of this function typically requires the use of several monoenergetic sources, long acquisition times, and a carefully designed experimental setup (Prozorova et al., 2019). The Gamma Detector Response and Analysis Software (GADRAS), developed by Sandia National Laboratories (SNL), is a computational tool designed for detailed analysis of gamma radiation spectra. Its interface incorporates a robust set of features that make it effective for interpreting spectra obtained, for instance, from HPGe detectors. Additionally, GADRAS is capable of identifying and quantifying radionuclides in unknown sources, as well as modeling the geometry of the radiation source-detector system (Jeffcoat et al., 2010).

The present study aims to validate and apply GADRAS as an alternative tool for identifying and quantifying radioisotopes in radioactive samples. To this end, experimental tests were conducted using an HPGe detector, in order to generate gamma radiation spectra corresponding to the radionuclides ^{241}Am , ^{133}Ba , ^{60}Co , and ^{137}Cs . The results were compared to evaluate the effectiveness of the proposed methodology.

2 Materials and Methods

2.1 Experimental Trials

Analogous to the computational simulations performed using GADRAS, all experimental procedures involving the HPGe detector were conducted at the Laboratory of Environmental Analysis and Computational Simulation (LAASC) of the Federal University of Rio de Janeiro (UFRJ). The detector model used was the GC3020, manufactured by Canberra, which features a relative efficiency of 30%, as illustrated in Figure 1.

The experimental evaluations were carried out using four standard radioactive sources (^{241}Am , ^{133}Ba , ^{137}Cs and ^{60}Co), duly certified by the Institute for Radiation Protection and Dosimetry (IRD). Each radiation spectrum was acquired until reaching a total of 10,000 counts, with detection times ranging from 48.27 seconds for ^{137}Cs to 461.38 seconds for ^{133}Ba .

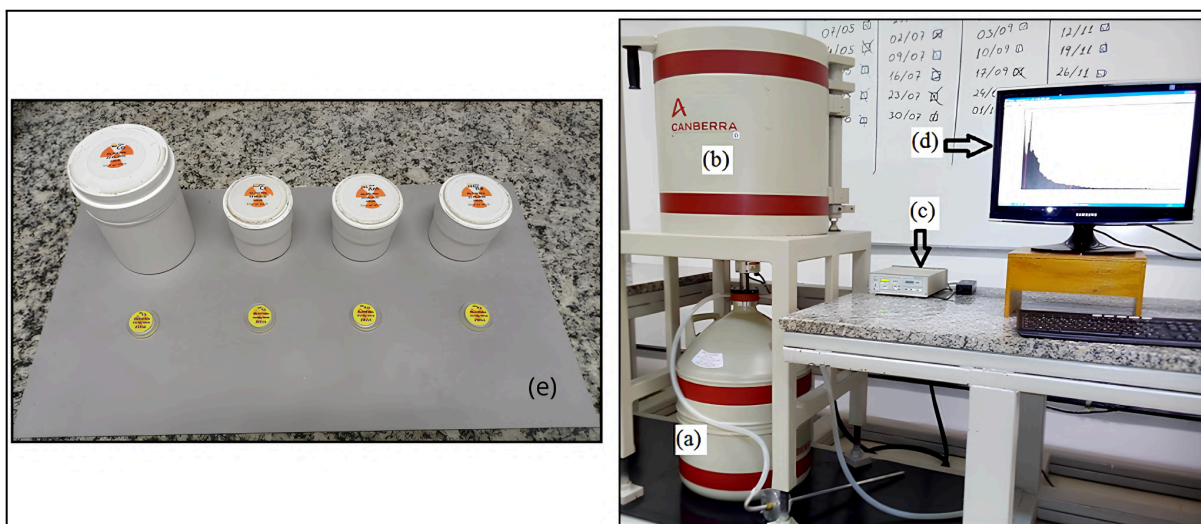


Figure 1: (a) Dewar. (b) Lead shielding. (c) Multichannel analyzer. (d) CPU. (e) Radioactive sources.

To calculate detector efficiency, computer modeling was required. This stage allows you to model the experimental environment by inferring chemical and physical properties of the samples and beakers (Figure 2). Following the geometric modeling, Canberra's LabSOCS (Laboratory Sourceless Calibration Software) discretizes the source region, which includes the sample holder and the sample itself, into 1024 voxels. From this division, a point is randomly chosen in each voxel, and the detection efficiency is determined using the values associated with these specific points. The software then doubles the amount of voxels and repeats the process, comparing the results based on convergence criteria. If the simulated efficiency values do not converge, the number of voxels is doubled until adequate convergence is obtained (Canberra, 2012). Figure 3 presents a typical detection efficiency curve obtained from the analysis of the samples.

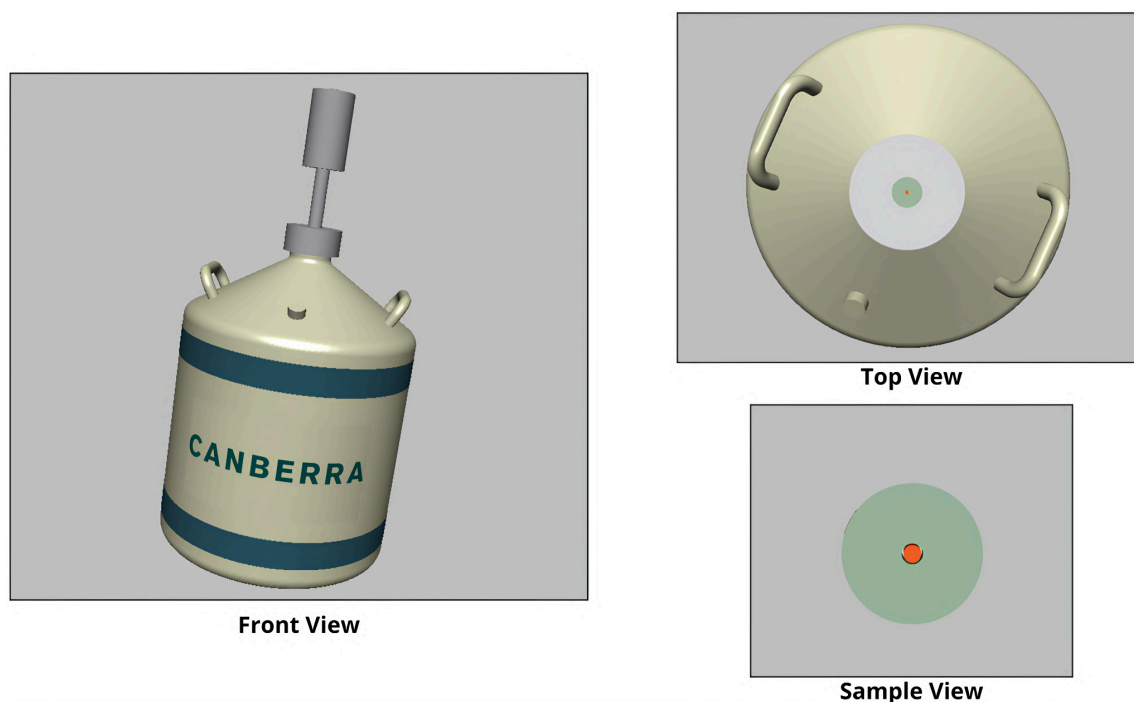


Figure 2: The system is made up of the detector and the sample after modeling.

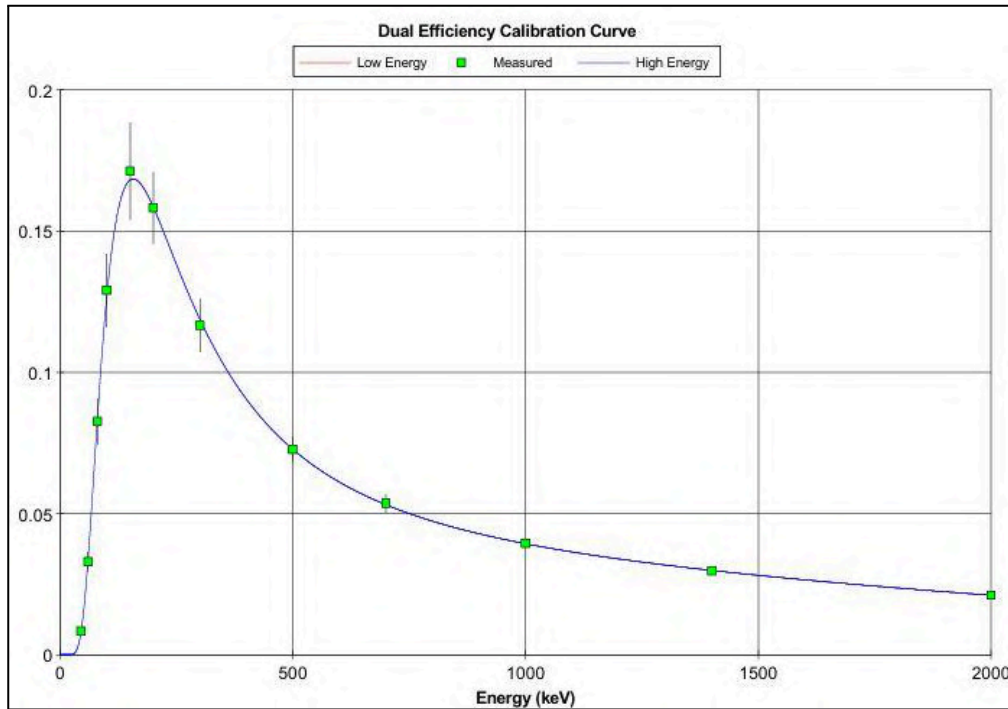


Figure 3: Typical detection efficiency curve obtained from the analysis of the samples.

The activity concentration (AC) of each sample was computed, which was then estimated by evaluating the energy peaks for each radionuclide. Equation 1 quantifies the AC and Equation 2 quantifies the uncertainty of activity concentration (σ_{AC}). Table 1 shows the formulae used to calculate the activity concentrations of samples.

Table 1: Formulas for determining activity concentration and uncertainty.

Calculations	Formula	Reference
(AC)	$AC \text{ (Bq kg}^{-1}\text{)} = \frac{N_L}{\varepsilon \times P_{\gamma} \times m \times t} \quad (1)$	(IAEA, 1989)
σ_{AC}	$\sigma_{AC} = \sqrt{\left(\frac{1}{\varepsilon \cdot P_{\gamma} \cdot m \cdot t}\right)^2 \cdot \delta N_L^2 + \left(\frac{N_L}{\varepsilon^2 \cdot P_{\gamma} \cdot m \cdot t}\right)^2 \cdot \delta \varepsilon^2 + \left(\frac{N_L}{\varepsilon \cdot P_{\gamma}^2 \cdot m \cdot t}\right)^2 \cdot \delta P_{\gamma}^2} \quad (2)$	(IAEA, 1989)

2.2 Detector and Sample Characterization

The computational modeling of samples and the HPGe detector was based on the physical characteristics of the equipment used in the experimental procedures conducted in this study. To ensure high accuracy in the simulation results, it is essential that the geometry of the system be represented with the highest possible level of detail, as variations in the dimensions of the semiconductor crystal can significantly influence detection efficiency (Sima, 1990). For this purpose,

technical manuals provided by the manufacturer were consulted and used as references for the modeling and characterization of the detection system.

The high-purity germanium crystal has a diameter of 6.2 cm and a length of 4.0 cm. This crystal is mounted on an aluminum holder with an internal diameter matching that of the sensitive volume, having an external diameter of 6.74 cm and a length of 6.75 cm. The entire assembly is encapsulated in an aluminum end cap with internal dimensions of 7.3 cm in diameter, external dimensions of 7.6 cm in diameter, and a length of 13.5 cm (Canberra, 2012).

Each source consists of an acrylic cylinder measuring 2.54 cm in diameter and 0.5 cm in height, with a central hole of 0.4 cm in diameter. A solution containing the desired radionuclide is dropwise deposited into the internal area of the disk. After sample application, an acrylic cover with a thickness of 0.1 cm is placed over the assembly, sealing the source.

The acquired data were incorporated into the GADRAS software, allowing for an accurate representation of the experimental setup in the simulations, as illustrated in Figure 4

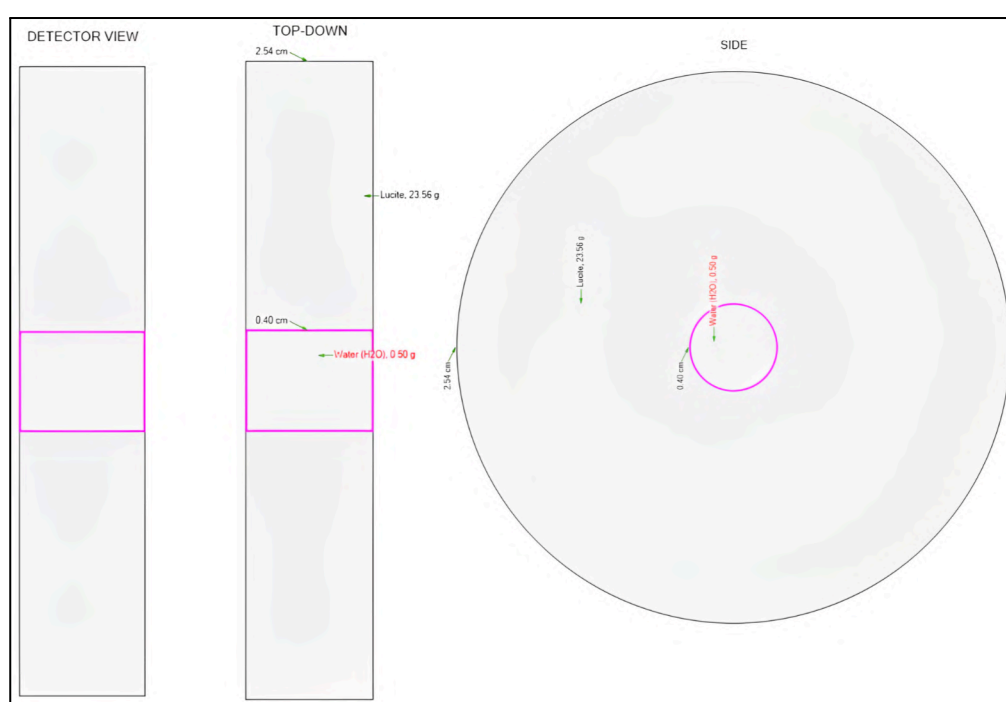


Figure 4: Sample geometry modeled using the GADRAS software.

2.2 Analysis

GADRAS is capable of processing spectral data acquired at varying distances between the source and the detector. The software supports the import of multiple spectral file formats from different manufacturers, which can be internally converted into the processing format used by the program, the PCF file.

Gamma radiation spectrum analysis using GADRAS required the input of two distinct spectra: one corresponding to the measurement in the presence of the radioactive source, and the other corresponding to the background spectrum of the experimental environment. The initial evaluation phase was conducted using the software's automatic radionuclide identification functionality, enabling the qualitative determination of present emitters. This was followed by the quantification phase, aimed at estimating the activity of the radioactive material.

GADRAS performs a simple regression analysis after the radionuclide has been inferred. This regression is based on the statistical analysis of the chi-square (χ^2) value, which quantifies the

deviation between the experimental data and the fitted model. When the spectral data is of high quality, the energy calibration is accurate, and the detector response function is well-defined, the χ^2 value is expected to be close to 1, indicating a good fit (Horne et al., 2019). The ideal curve generated by the model is overlaid onto the measured spectrum, allowing for a visual assessment of the fit quality.

3 Results and Discussions

The count spectra of the radionuclides ^{241}Am , ^{133}Ba , ^{137}Cs , and ^{60}Co , obtained using the HPGe detector and the GADRAS software, are presented in Figures 5, 6, 7, and 8, respectively. It is observed that the software correctly and uniquely identified all radionuclides present in the experimental spectra.

The χ^2 coefficient values obtained from the curve fitting performed by GADRAS for the analyzed radionuclides provided a quantitative assessment of the software's accuracy in identifying the emitters present in the spectra. The radionuclides ^{241}Am and ^{137}Cs exhibited χ^2 values close to the ideal reference value of 1, with 0.37 and 1.53, respectively, indicating that the computational model of the detector and the analyzed samples is adequate and that the experimental data acquisition was performed with quality. These results suggest that the detector was properly calibrated, with uncertainties minimized as much as possible. Although the radionuclides ^{60}Co and ^{133}Ba presented higher χ^2 values (3.77 and 5.63, respectively), these values remain within acceptable limits and are consistent with results reported in the literature, such as in Horne et al. (2019).

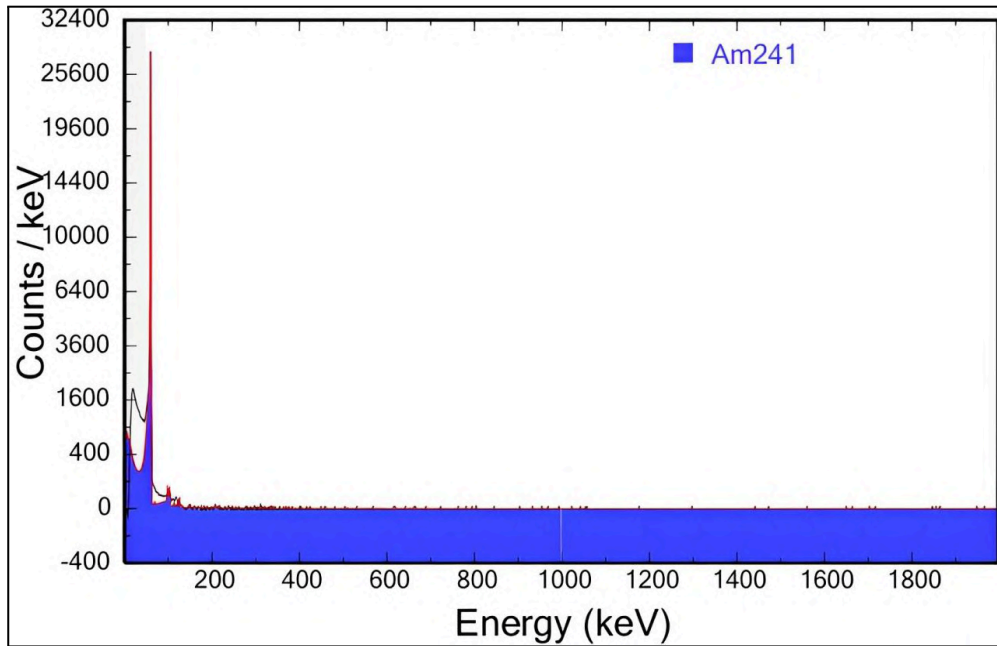


Figure 5: Count spectrum of the radionuclide ^{241}Am obtained experimentally using the HPGe detector (black line) and by simulation in the GADRAS software (red line).

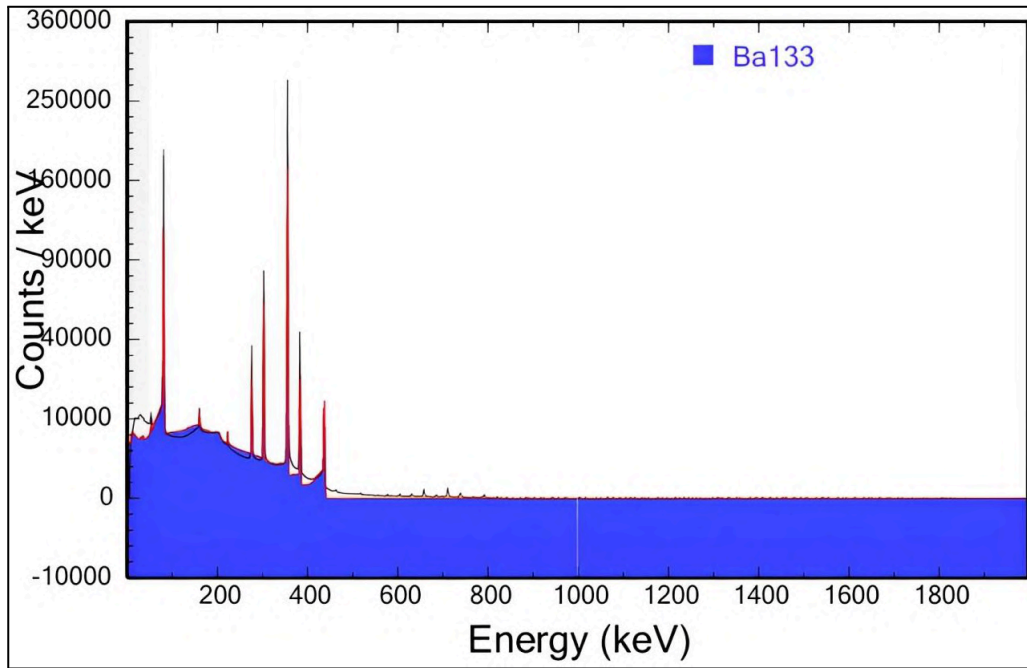


Figure 6: Count spectrum of the radionuclide ^{133}Ba obtained experimentally using the HPGe detector (black line) and by simulation in the GADRAS software (red line).

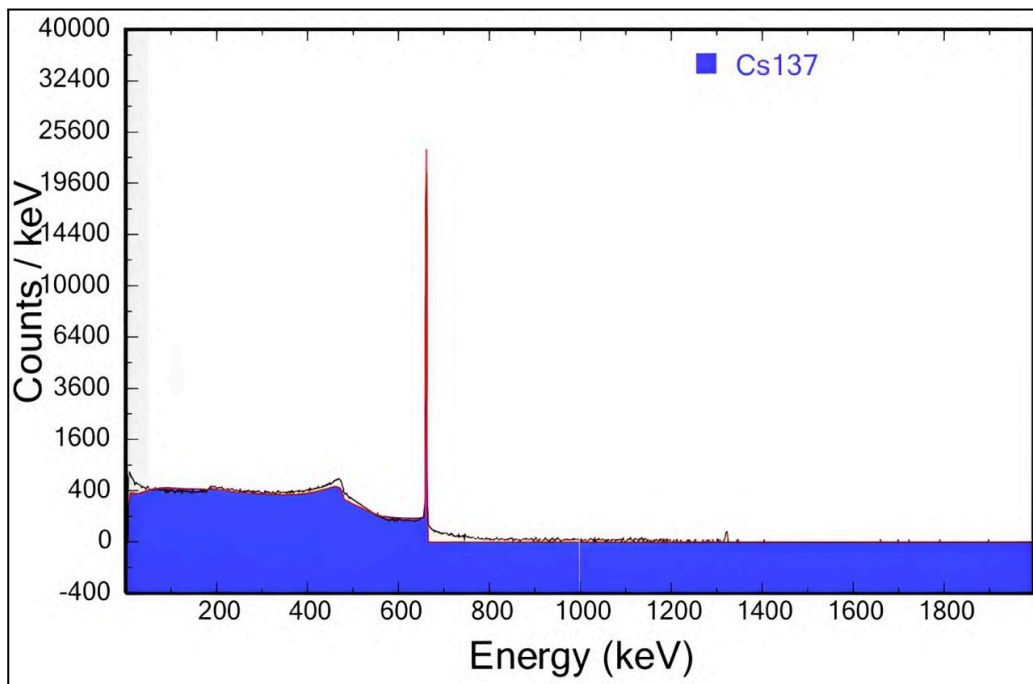


Figure 7: Count spectrum of the radionuclide ^{137}Cs obtained experimentally using the HPGe detector (black line) and by simulation in the GADRAS software (red line).

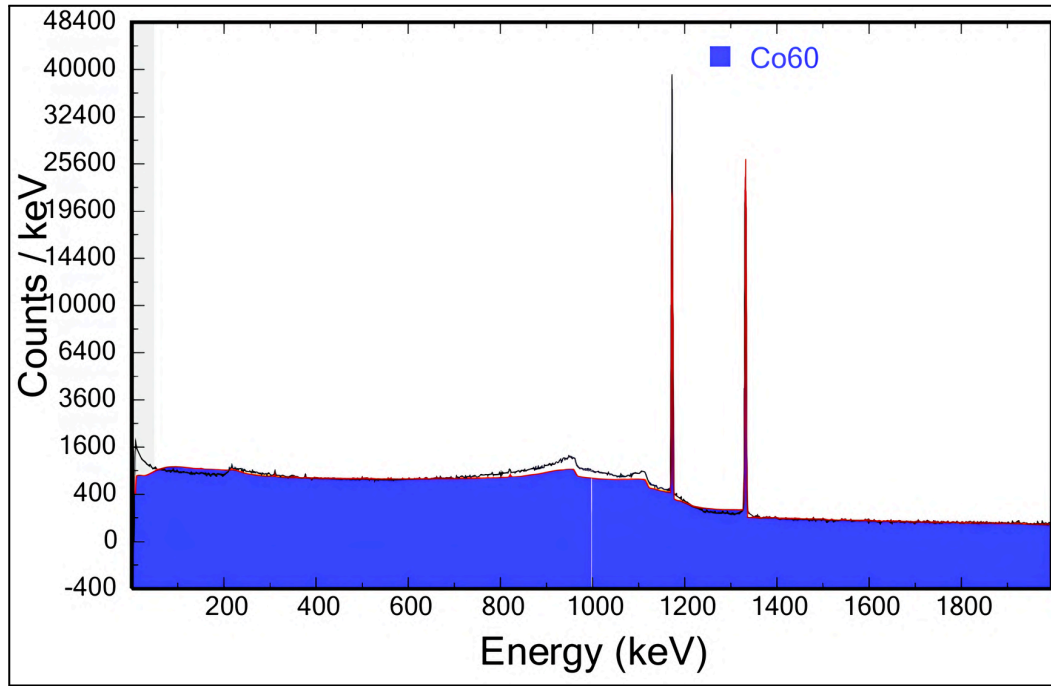


Figure 8: Count spectrum of the radionuclide ^{60}Co obtained experimentally using the HPGe detector (black line) and by simulation in the GADRAS software (red line).

Table 2 presents the activities of the radionuclides ^{241}Am , ^{133}Ba , ^{60}Co , and ^{137}Cs determined using the HPGe detector and the GADRAS software. Figure 9 displays the percentage difference between the activities obtained by both methods. A comparative analysis of Table 2 and Figure 9 reveals that the largest percentage difference was observed for ^{241}Am , which exhibited the highest activity among the analyzed isotopes. Conversely, ^{60}Co , which had the lowest activity, was the radionuclide with the smallest percentage difference.

It is noteworthy that the photon energies emitted by these radionuclides vary significantly: ^{241}Am emits radiation in the low-energy range, ^{133}Ba emits in the low- to medium-energy ranges, ^{137}Cs emits at medium energy, and ^{60}Co emits at high energies. This distribution may directly influence the observed discrepancies, as larger percentage differences tend to occur at lower energies, while smaller discrepancies are more common at higher energies.

Table 2: Activity values obtained experimentally using the HPGe detector and theoretically through the GADRAS software, expressed in nCi.

Isotope	HPGe Activity	GADRAS Activity
^{241}Am	1591.4	1300
^{133}Ba	872.36	908.5
^{137}Cs	885.2	951.02
^{60}Co	262.36	269.59

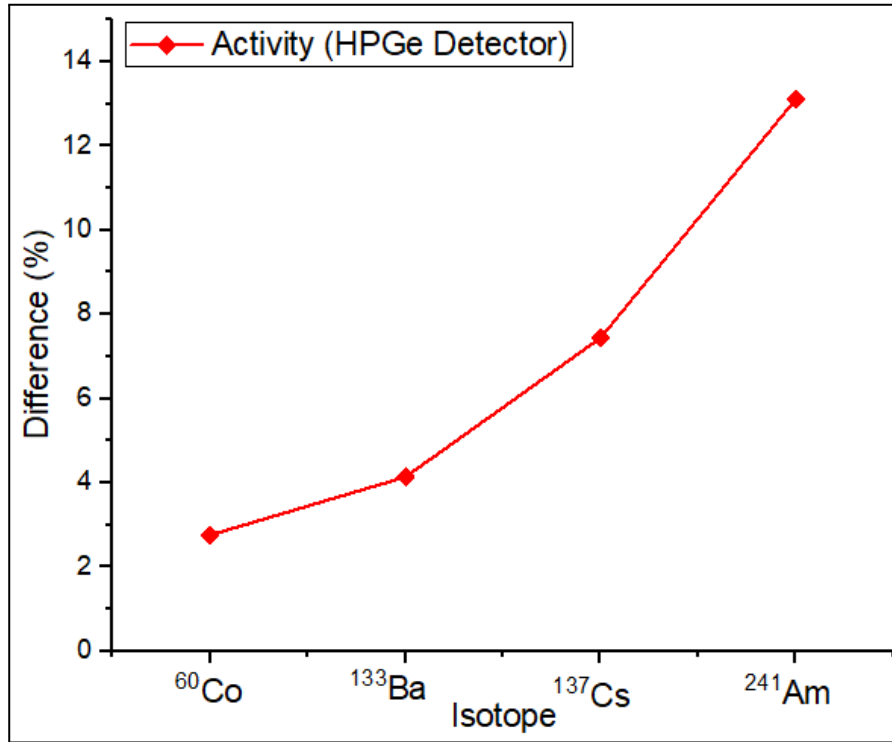


Figure 9: Percentage difference between the activities obtained by the HPGe detector and the GADRAS software.

4 Conclusions

The results obtained with the GADRAS software for the radionuclides ^{241}Am , ^{133}Ba , ^{137}Cs , and ^{60}Co showed variable fitting quality, with ^{137}Cs exhibiting the best fit. The ^{241}Am results may require a reassessment of uncertainties, as GADRAS did not perform well in the low-energy range, while ^{133}Ba and ^{60}Co indicate the need for further investigation.

Regarding activity estimation, GADRAS demonstrated good agreement for radionuclides with high-energy emissions. For ^{137}Cs , the percentage difference was approximately 2.80%, indicating good accuracy of the software within this energy range. It is noteworthy that discrepancies in low-energy counts directly affect the activity estimation of the sources.

The differences observed in the spectra may be related to limitations in modeling the detector response and the simplification of experimental conditions in the simulated model.

Despite these limitations, the results demonstrate that GADRAS is a promising tool for spectral analysis and activity estimation, particularly in the higher energy range. Improvement of input models and proper system calibration can contribute to better agreement between simulation and experiment, thereby enhancing the applicability of the software in nuclear security, safeguards, and radioactive material characterization contexts.

References

CANBERRA, Genie™ 2000 3.3: Operations Manual, EUA, 2012.

Horne S. M., Thoreson G. G., Theisen L. A., Mitchell D. J., Harding L. T., O'Brien S. E.
GADRAS-DRF Version 18 User's Manual. Sandia National Laboratories, Printed December 2019,
Sandia Report: [SAND2019-14655](#).

International Atomic Energy Agency. Naturally Occurring Radioactive Material (NORM VI),
Technical Reports Series No. 295, IAEA, Vienna (1989). [Measurement of Radionuclides in Food and the Environment | IAEA](#).

I. V. Prozorova et al., Modeling an HPGe detector response to gamma-rays using MCNP5 code.
International Journal of Modern Physics C, doi: [10.1142/S0129183119500992](#).

Jeffcoat R., Salaymeh S., Clare A. A Comparison of GADRAS-Simulated and Measured Gamma-ray
Spectra. 51st Annual Meeting of the Institute of Nuclear Materials Management 2010, INMM 51,
Volume 1 of 5, Baltimore, Maryland, USA 11-15 July 2010, ISBN: [978-1-61782-192-9](#).

Semkova, V., Otuka, N., & Plompen, A. J. M. (2023). Characterization of HPGe Detector Response
for Activation Cross-Section Measurements: Regression Method versus Monte Carlo Method. Journal
of Nuclear Science and Technology, 61(2), 151–160, doi:
<https://doi.org/10.1080/00223131.2023.2278598>.

Sima O. Monte Carlo simulation versus semiempirical calculation of autoabsorption factors for
semiconductor detector calibration in complex geometries. Progress in Nuclear Energy, Volume 24,
Issues 1–3, 1990, Pages 327-336, ISSN 0149-1970, doi:
[https://doi.org/10.1016/0149-1970\(90\)90052-7](https://doi.org/10.1016/0149-1970(90)90052-7).

Pulsed X-ray-Induced Nuclear Fragmentation in a Radioresistant Ovarian Cancer Cell Line

F. Castañeda¹, J. Moreno², J. Jain², C. Letechipia¹, L. Soto²,
E.V. Velásquez^{3*}.

¹ Universidad Autónoma de Zacatecas, Unidad Académica de Estudios Nucleares, 98000, Zacatecas, México.

² Comisión Chilena de Energía Nuclear, División de Investigación y Aplicaciones Nucleares, Centro de Investigación en la Intersección de Física de Plasmas, Materia y Complejidad, 7600713, Santiago, Chile.

³ Comisión Chilena de Energía Nuclear, División de Investigación y Aplicaciones Nucleares, Centro de Investigaciones Nucleares para Aplicaciones en Salud y Biomedicina, 7600713, Santiago, Chile.

* ethel.velasquez@cchen.cl

Abstract

Pulsed radiation is a promising alternative to conventional radiotherapy regimens, especially for the treatment of neoplasms with radioresistant phenotypes. In the present study, nuclear fragmentation was evaluated in the HEY cell line — a human cell line derived from ovarian cancer and reported in the literature as radioresistant — after exposure to pulsed X-rays generated by an experimental Plasma Focus device. This type of source emits high-intensity, ultra-short duration (tens of nanoseconds) radiation pulses, which could promote cellular damage mechanisms that differ from those induced by continuous radiation used in conventional radiotherapy. The experimental protocol involved the administration of 40 X-ray pulses, equivalent to an estimated cumulative dose of 0.8 Gy, at an instantaneous dose rate of 10^5 Gy/s. Nuclear morphology was analyzed using fluorescence microscopy and specific staining. Significant nuclear fragmentation was observed in the irradiated cells compared to the unexposed controls. The preliminary results indicate that pulsed radiation can induce early nuclear damage, even in cells with high repair capacity such as the HEY line, suggesting a potential pathway to overcome radioresistance and providing a basis for future research aimed at exploring new possibilities for its therapeutic application.

Keywords: Nuclear fragmentation, Pulsed X-rays, Radioresistance, Ovarian cancer, Plasma focus.

1 Introduction

Ionizing radiation has been used for decades as a highly effective tool in the treatment of various cancers and remains a basis of modern oncology. Its therapeutic efficacy is primarily based on the inducement of DNA damage, leading to cell cycle arrest, senescence, or programmed cell death (Joiner & van der Kogel, 2018). However, the optimization of these technologies continues to be an active area of research, particularly in terms of preserving adjacent healthy tissues and designing more precise, personalized treatment regimens. Likewise, the clinical success of radiotherapy may be compromised by the presence of radioresistant tumor phenotypes, which exhibit an enhanced capacity to repair radiation-induced damage and to survive exposure to therapeutic radiation doses (Oike et al., 2016).

Ovarian cancer presents a significant therapeutic challenge due to late diagnosis, high recurrence rates, and intrinsic resistance to multiple treatments (Choi et al., 2019). Among the *in vitro* models used to study this neoplasm, the HEY cell line is distinguished for its serous epithelial origin and well-documented radioresistant phenotype, linked to p53 mutations, altered expression of DNA repair proteins, and apoptosis avoidance (Concin et al., 2000; Choi et al., 2019; Sanij et al., 2020). Due to these features, the HEY line serves as a valuable experimental model for evaluating radiosensitization strategies.

In recent years, there has been growing interest in non-conventional irradiation schemes, such as pulsed low-dose radiotherapy (PLDR), which differs from traditional approaches by delivering the total dose in brief fractions or pulses, separated by definite intervals (Ma, 2022). This delivery pattern has been shown to induce differential responses in tumor cells, including hyper-radiosensitivity (HRS), a phenomenon whereby small doses exert disproportionately high cytotoxic effects, particularly in cells in the G2 phase of the cell cycle (Marples et al., 2004; Short et al., 2003). However, even as PLDR employs a pulsed administration pattern, the radiation itself remains continuous. The recent development of experimental devices such as the Plasma Focus (PF) — capable of producing nanosecond-duration X-ray pulses with extremely high dose rates (Soto, 2015) — has opened new possibilities for studying the biological effects of ultra-short pulses on tumor cell lines *in vitro*, as demonstrated in previous studies on various cancer models (Jain et al., 2021; Virelli et al., 2015).

Nuclear fragmentation is one of the most sensitive cellular responses to radiation exposure and can be detected by fluorescence microscopy as a morphological marker of apoptosis or irreversible lethal damage (Louis & Siegel, 2011). This phenomenon may occur even before complete loss of cell viability, making it a widely used early indicator of ionizing radiation-induced damage (Feng et al., 2015; Iwakawa et al., 2020). Its assessment in radioresistant cell lines may provide critical insights into the efficacy of non-conventional irradiation schemes and guide the development of more effective therapeutic strategies.

The present work aims to explore the effects of pulsed X-ray exposure on nuclear fragmentation in a radioresistant human ovarian cancer cell line, using an experimental Plasma Focus device. Specific fluorescent staining and microscopy techniques were employed to evaluate nuclear morphology and early post-irradiation damage. The findings are expected to contribute to a better understanding of the cellular mechanisms triggered by pulsed radiation and to the identification of novel therapeutic options for overcoming radioresistant phenotypes.

2 Methodology

A comparative *in vitro* study was designed to assess the differential effects of two ionizing radiation modalities—continuous gamma radiation and pulsed X-rays—on the human HEY cell line, derived from serous ovarian carcinoma and recognized for its radioresistant phenotype. Two primary parameters were analyzed: nuclear fragmentation and cell viability. Evaluations were performed 24 hours post-irradiation, with three independent biological replicates for each analysis ($n = 3$). The experimental conditions, incubation times, and specific analytical methods are described in the following sections. Figure 1 presents a flow diagram summarizing the overall experimental design and the methodological sequence applied.

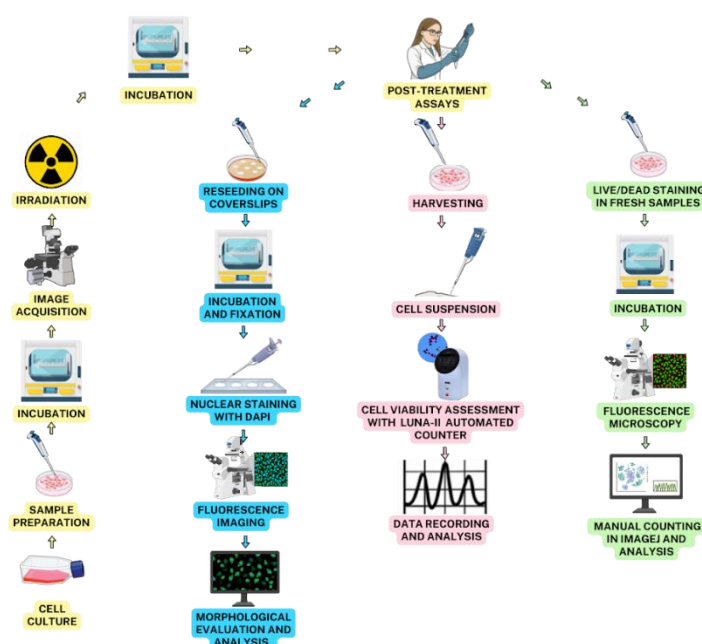


Figure 1: General outline of the experimental design. Flow chart summarizing cell culture, treatments with continuous gamma radiation or pulsed X-rays, and analyses performed 24 hours post-irradiation: nuclear fragmentation and cell viability. Each assay was performed with three independent biological replicates (n=3).

2.1 Cell culture and experimental preparation

The HEY human cell line was cultured in DMEM/F12 medium supplemented with 10% fetal bovine serum and antibiotics (penicillin 100 U/mL, streptomycin 100 µg/mL), maintained under standard conditions at 37 °C in a humidified atmosphere with 5% CO₂. For the experiments, the cells were seeded at a density of 2.5×10^5 cells per plate and allowed to adhere for at least 24 hours prior to treatment.

2.2 Gamma Irradiation and Evaluation with Trypan Blue

Cells were exposed to 2 Gy of gamma radiation using a Gamma Cell 220R self-shielded irradiator (MDS Nordion) equipped with a Cobalt-60 source. The dose rate was 5.9 Gy/min, verified by Fricke chemical dosimetry in accordance with ASTM E-1026 standards (Rivera et al., 2021). Non-irradiated controls were maintained under identical conditions. Twenty-four hours after exposure, cell viability was assessed using 0.4% trypan blue staining. Cells were gently dissociated, resuspended in serum-free medium, and mixed with the dye. Live and dead cells were counted using a LUNA-II Automated Cell Counter, following a protocol previously optimized for the HEY cell line.

2.3 Pulsed X-ray Irradiation and Fresh Viability Assessment

Pulsed irradiation was performed using an experimental Plasma Focus PF-2kJ device, developed at the Chilean Nuclear Energy Commission. The system operates through a ~17 kV discharge in a hydrogen-filled vacuum chamber (13 mbar), generating nanosecond X-ray pulses in the 8–10 keV range (Jain, 2021). A total of 40 pulses were applied, corresponding to an estimated absorbed dose of ~0.8 Gy, delivered at a dose rate in the order of 10^7 Gy/min, as determined with TLD-100 dosimeters (Díaz, 2018).

Cell viability was assessed 24 hours after irradiation using the ReadyProbes™ Cell Viability Imaging Kit (Blue/Green) (Thermo Fisher). This assay enables discrimination between total nuclei (Hoechst 33342) and nuclei of non-viable cells (NucGreen). Reagents were added directly to the culture medium, followed

by incubation for 60 min at 37 °C. Live-cell imaging was performed without fixation, acquiring fluorescence in the blue and green channels. Five random fields per sample were captured, and manual counting was performed using ImageJ software. Quantitative data are presented as mean $\pm$ standard deviation (SD).

2.4 Nuclear Fragmentation Analysis

Following irradiation, cells were recovered and seeded onto sterile coverslips, allowing them to adhere for 24 hours. Samples were then fixed with 4% paraformaldehyde for 20 minutes, permeabilized with 0.1% Triton X-100, blocked with 1% bovine serum albumin (BSA), and stained with DAPI (1 μ g/mL) for nuclear visualization. Finally, coverslips were mounted using antifade medium containing TRITC–phalloidin for cytoskeletal observation. Nuclear morphological features were assessed, including the identification of fragmented nuclei (FN), apoptotic bodies (AB), dividing cells (DC), viable cells (VC), and limited cell confluence (LC). Images from multiple fields were captured, with at least five fields analyzed per sample.

2.5 Image Acquisition

Samples were examined using an inverted fluorescence microscope (ZEISS AxioObserver 7) with ZEN 3.0 Blue Edition software. Imaging was performed with 40X and 100X oil-immersion (for fixed samples) objectives. The green fluorescence channel was used to visualize GFP and NucGreen stains, the red channel for TRITC–phalloidin, and the blue channel for DAPI and Hoechst 33342.

3 Results

3.1 Morphological Assessment of Nuclear Damage Following Pulsed X-ray Exposure

Nuclear morphological alterations were assessed by DAPI staining in HEY Green cells cultured for 24 hours following exposure to pulsed radiation. This transformed cell line stably expresses a green fluorescent protein, allowing direct examination of overall cellular morphology without additional staining.

In non-irradiated samples (control group), well-defined nuclei and homogeneous chromatin distribution were observed. Overall cell morphology showed no alterations and there were no signs of disorganization in F-actin distribution; cells grew as an adherent monolayer in which mitotic cells could be identified (Figure 2). By contrast, cells exposed to pulsed radiation (40 X-ray pulses, approximately equivalent to a total dose of $\sim$ 0.8 Gy) exhibited reduced confluence alongside morphological changes suggestive of nuclear damage. Irregularly shaped nuclei, nuclear fragmentation, and apoptotic bodies were observed, accompanied by rounded cellular morphology and F-actin reorganization (Figure 3). These features are indicative of severe damage and are consistent with morphological alterations reported during programmed cell death processes.

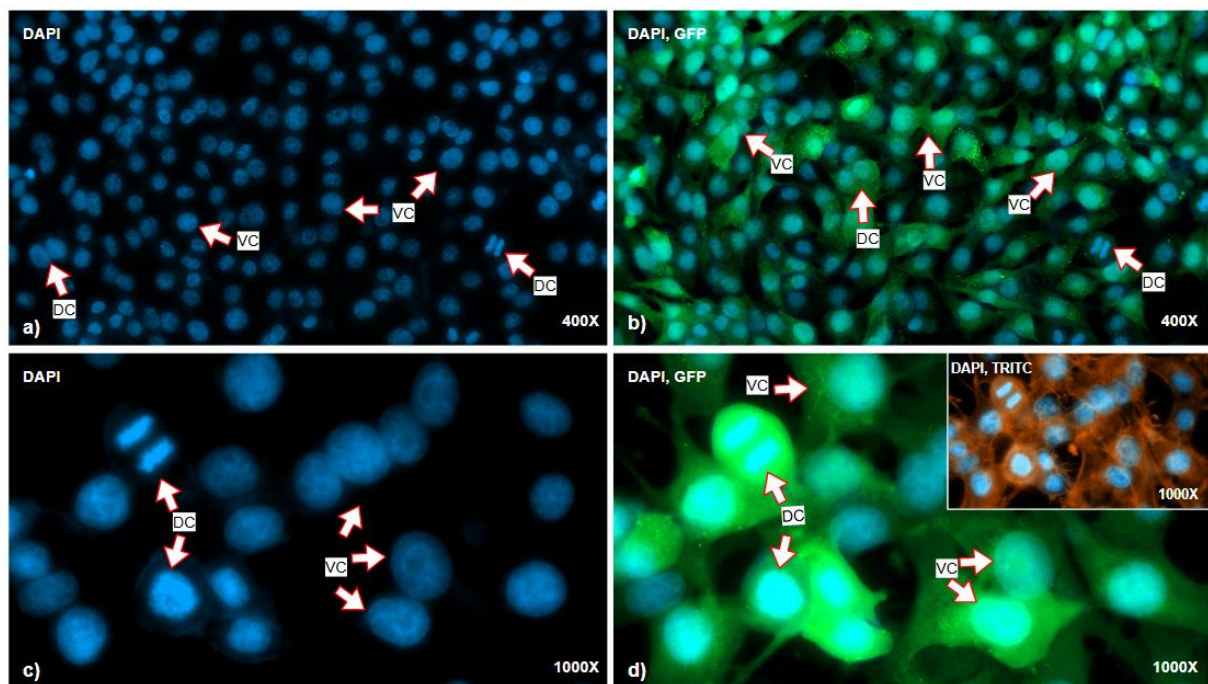


Figure 2: Characteristic morphology in HEY Green cell line without irradiation after 24 h culture. The nuclei show regular morphology, with viable adherent cells (VC), abundant F-actin fibres, dividing cells (DC) and homogeneous nuclear organization. The green fluorescence corresponds to the recombinant protein expressed by the HEY Green line. Panels: (a) DAPI channel 400X; (b) merge DAPI + green fluorescence 400X; (c) DAPI 1000X; (d) merge DAPI + green fluorescence, DAPI + Phalloidin-TRITC (insert) 1000X.

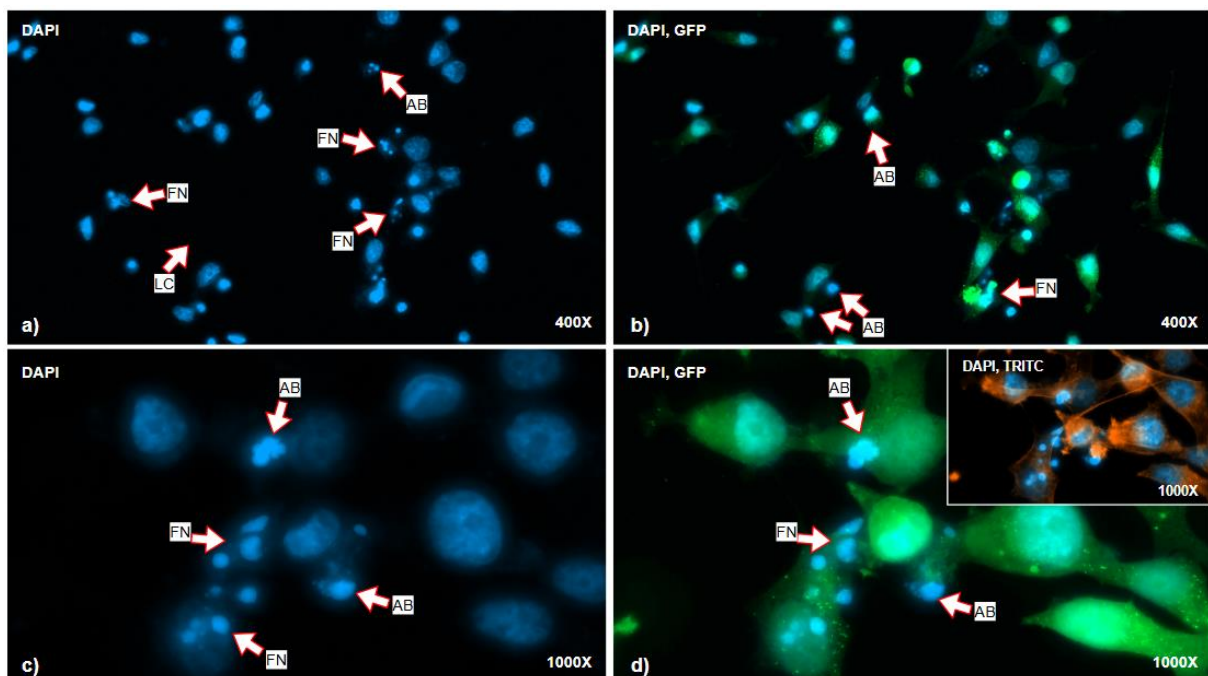


Figure 3: Nuclear fragmentation in HEY Green cell line 24 hours after pulsed X-ray exposure (~0.8 Gy). Pyknotic, fragmented nuclei (FN), apoptotic bodies (AB), low cell confluence (LC) and irregular or rounded morphology are evident, indicative of severe cell damage. Green fluorescence corresponds to the recombinant protein expressed by the HEY Green line. Panels: (a) DAPI channel 400X; (b) merge

DAPI + green fluorescence 400X; (c) DAPI 1000X; (d) merge DAPI + green fluorescence, DAPI + Phalloidin-TRITC (insert) 1000X.

3.2 Effect of Continuous Gamma Radiation on Cell Viability

To compare the effects of pulsed and continuous radiation, HEY cells were exposed to continuous gamma radiation (2 Gy) using a self-shielded irradiator. Cell viability was assessed 24 hours post-irradiation by trypan blue stain exclusion. No significant morphological alterations were observed compared to the non-irradiated control group. The proportion of viable cells remained at approximately 98%, similar to the control group; there was no evidence of detachment or reduced confluence (Figure 4).

These results indicate that the HEY cell line maintains high viability following exposure to this dose of continuous radiation, which is consistent with its radioresistant phenotype previously reported at the literature (Sanij et al., 2020; Iwakawa et al., 2020).

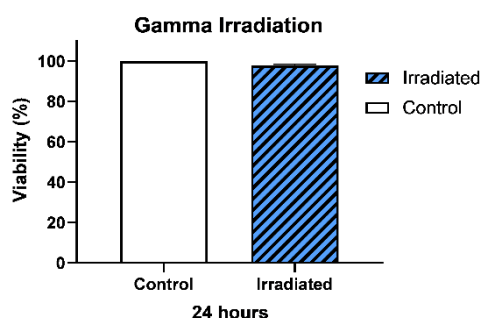


Figure 4: Analysis of cell viability in HEY cells 24 hours after gamma irradiation. The evaluation was performed by staining with trypan blue in an automated cell counter. No significant differences were observed in the percentage of viability between the samples in control conditions and those irradiated with 2 Gy. Irradiated cells retained an adherent monolayer distribution, achieving similar confluence with no signs of cell death.

3.3 Effect of Pulsed X-rays on Cell Viability

Following exposure to pulsed X-rays (~0.8 Gy), cell viability was assessed at 24 hours using fresh staining with the ReadyProbes™ kit (Hoechst 33342/NucGreen). Compared to the control group, irradiated cells exhibited a substantial decrease in cell density and an increased proportion of NucGreen-stained nuclei, which was used as a marker of plasma membrane integrity loss. Additionally, partial loss of cell adhesion and morphological alterations consistent with early stages of cell death were observed.

Quantitative analysis was performed on five representative fields for each sample. On average, irradiated samples showed a 73% reduction in the total number of nuclei per field (control: 713 ± 32 nuclei/field; irradiated: 191 ± 60 nuclei/field). Likewise, an 18% decrease in relative cell viability was also estimated (Figure 5). These findings indicate that, despite the low total dose administered, the pulsed modality was able to induce a significant cytotoxic effect in a radioresistant cell line, highlighting its differential potential compared to conventional modalities.

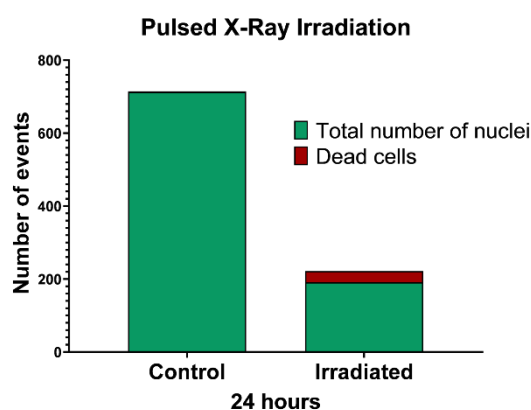


Figure 5: Cell count in the HEY line 24 hours after pulsed X-ray irradiation. A 73% reduction in the total number of nuclei per field is observed. Analysis performed by manual counting in ImageJ from fluorescence images captured at 200X magnification.

4 Discussion

This study showed that pulsed X-ray irradiation (~ 0.8 Gy) generated by the PF-2kJ device induced early nuclear fragmentation and a reduction in cell density in HEY cells, a human cell line previously described as being resistant to conventional continuous radiation schemes (Concin et al., 2000; Choi et al., 2019). Consistent with the literature, our results showed that exposure of HEY cells to continuous gamma rays did not result in any obvious morphological changes nor a significant decrease in cell viability, highlighting the importance of dose delivery modality in the biological response (Marples et al., 2004; Ma, 2022).

The observed alterations, such as nuclear fragmentation, apoptotic bodies, and loss of adhesion, suggest that exposure to pulsed X-rays triggered an apoptotic process that could be associated with acute genotoxic stress (Mah et al., 2010). The morphological characteristics of irradiated cells have been linked to the generation of DNA double-strand breaks, which in turn activate DNA damage responses, including the phosphorylation of histone H2AX (γ -H2AX), a widely used marker for assessing ionizing radiation-induced damage (Wilson et al., 2011; Takatori et al., 2013). Although the present study did not include molecular markers assessment, previous research in irradiated models has demonstrated that γ -H2AX expression correlates with morphologically evident nuclear damage and activation of cell death pathways such as apoptosis (Cook et al., 2009; Shichijo et al., 2023).

The phenomenon of hyper-radiosensitivity, characterized by an amplified response to low doses, has been described as a result of incomplete activation of cell cycle checkpoints, especially during the G2/M phase (Joiner & van der Kogel, 2018; Short et al., 2003). This mechanism could explain the 73% decrease in nuclear density observed after 24 hours in cells irradiated with high dose rate pulses (Jain et al., 2021; Ma, 2022), an effect that was not replicated following continuous gamma radiation exposure. This finding supports the hypothesis that pulsed regimens induce a qualitatively distinct cellular response, with the delivery modality playing a critical role on the severity of the induced damage (Virelli et al., 2015).

Despite the presence of mutations in the HEY cell line that confer p53-dependent radioresistance (Concin et al., 2000), our results suggest that this resistance can be overcome by pulsed radiation schemes. Recent studies have reported similar effects using an RNA polymerase I inhibitor (CX-5461), which induces cell death in high-grade serous lines, including HEY (Sanij et al., 2020). This suggests that alternative mechanisms of apoptotic activation may be involved, which could potentially be exploited for therapeutic purposes through pulsed irradiation.

The use of the modified HEY Green line, which constitutively expresses GFP, allowed for live analysis of nuclear morphology and cell viability without requiring additional staining (Louis & Siegel, 2011; Iwakawa et al., 2020). This technical advantage facilitated monitoring during sample preparation and assay follow-up.

Although pulsed radiation remains within the research domain, there is growing interest in developing new technologies to find therapeutic alternatives for radioresistant cancers. Modalities such as ultra-high dose rate radiotherapy (FLASH) and novel fractionation schemes are being reconsidered as strategies for the treatment of resistant tumors (Lee et al., 2025). The results obtained in this study show that pulsed X-rays can induce nuclear damage in an ovarian-origin cell line described as radioresistant. While further research is needed to elucidate the cellular and molecular mechanisms associated with this response, our findings suggest that the pulsed modality could have potential clinical applicability in the future.

5 Conclusions

The results obtained in this study show that pulsed X-ray irradiation generated by a Plasma Focus device effectively induces nuclear morphological alterations in HEY Green cells, including pyknosis, nuclear fragmentation, and the appearance of apoptotic bodies. These observations coincide with reduced cell confluence and decreased cell viability. Remarkably, these effects were observed at an estimated total dose of 0.8 Gy, in contrast to the cellular response following exposure to continuous gamma radiation at a 2.5-fold higher dose (2 Gy), which did not produce visible morphological changes or a significant reduction in cell viability. This difference suggests that the pulsed modality is capable of eliciting early cellular damage even in cell lines known for their radioresistant phenotype.

Although preliminary, these findings suggest that low-dose pulsed regimens may represent a promising strategy to overcome radioresistance mechanisms in cancer. Detailed morphological characterization using nuclear staining and live-cell microscopy revealed early nuclear damage that was not apparent in conventional viability assays. Future research should explore the correlation of these morphological alterations with DNA damage and apoptosis markers, such as γ H2AX and TUNEL, as well as evaluate the impact on the proliferative capacity of irradiated cells. Such studies would enhance understanding of the mechanisms underlying radiosensitization induced by ultra-short radiation pulses and support the development of more effective and selective therapeutic approaches.

Acknowledgements

This work was partially funded by the project

ANID-FONDECYT Regular 1240375

CONAHCYT 4036525.

UAZ Ago-Dic 2024.

COZCyT 2024.

References

- Choi Y., Lee Y. J., Park Y. (2019) HEY ovarian cancer cells and DNA repair gene alterations. *Cancer Cell International* 19:254.
- Concin N., Zeillinger C., Stimpfl M., et al. (2000) p53-dependent radioresistance in ovarian carcinoma cell lines. *Cancer Letters* 150:191–199.
- Cook P. J., Ju B. G., Telese F., et al. (2009) Tyrosine dephosphorylation of H2AX modulates apoptosis and survival decisions. *Nature* 458(7238):591–596
- Díaz Contreras, D. A. (2018) Characterization of the ionizing radiation emission of a plasma focus device. Tesis de Magíster, Pontificia Universidad Católica de Chile.
- Feng L., Ma R., Liu S., et al. (2015) Cold plasma-activated media suppresses ovarian cancer cell growth and induces apoptosis. *Scientific Reports* 6:18459.
- Iwakawa M., Shibata A., Oike T., et al. (2020) Radiation therapy targeting cancer stem cells: Current perspectives. *Cancer Science* 111(5):1442–1450.
- Jain J., Araya H., Moreno J., et al. (2021) Hyper-radiosensitivity in tumor cells following exposure to low dose pulsed X-rays emitted from a kilojoule plasma focus device. *Journal of Applied Physics* 130:164902.
- Joiner M. C., van der Kogel A. J. (2018) *Basic Clinical Radiobiology*. CRC Press.
- Kim S. I., Kim H. S. (2025) Is it time to reassess the role of radiotherapy treatment in ovarian cancer? *Journal of Gynecologic Oncology* 36(1):e72.
- Louis K. S., Siegel A. C. (2011) Tissue culture techniques used in radiobiology. *Tissue and Cell* 43(1):61–69.
- Ma C. M. C. (2022) Pulsed low dose-rate radiotherapy: radiobiology and dosimetry. *Physics in Medicine and Biology* 67(3):301.
- Mah L. J., El-Osta A., Karagiannis T. C. (2010) γ H2AX: a sensitive molecular marker of DNA damage and repair. *Leukemia* 24:679–686.
- Marples B., Wouters B. G., Collis S. J., et al. (2004) Low-dose hyper-radiosensitivity: a consequence of ineffective cell cycle arrest. *Radiation Research* 161:247–255.
- Oike T., Sato H., Noda S., Nakano T. (2016) Translational research to improve the efficacy of carbon ion radiotherapy. *Frontiers in Oncology* 6:139.
- Sanij E., Hannan K. M., Yip K. Y., et al. (2020) CX-5461 activates the DNA damage response and demonstrates therapeutic efficacy in high grade serous ovarian cancer. *Nature Communications* 11:2641.
- Shichijo K., Takatsuji T., Uzbekov D., et al. (2023) Radiation makes cells select the form of death dependent on external or internal exposure: apoptosis or pyroptosis. *Scientific Reports* 13:12002.

- Short S. C., Woodcock M., Marples B., et al. (2003) Effects of cell cycle phase on low-dose hyper-radiosensitivity. *International Journal of Radiation Biology* 79(2):99–105.
- Soto L. (2005) New trends and future perspectives on plasma focus research. *IEEE Transactions on Plasma Science* 43(9):2910–2916.
- Takatori E., Shoji T., Takada N., et al. (2013) Analysis of gemcitabine and carboplatin activity in ovarian clear-cell carcinoma using γ H2AX. *OncoTargets and Therapy* 6:901–907.
- Virelli A., Zironi I., Pasi F., et al. (2015) Early effects comparison of X-rays delivered at high-dose-rate pulses and at low dose rate on human tumour cells. *Radiation Protection Dosimetry* 166(1–4): 383–387.

Low-Dose Neonatal Ionizing Radiation Induces Modest, Sex-Specific Changes in Adult Mouse Behavior and Hippocampal Oxidative Stress

Ferreira, J. B. F^{1*}, Penedo, L. S. T, Saraiva, H. G. S.¹, SILVA, M. E. G. M¹., Borges, M.M.S.¹, Lau, C. S. C¹, Ciambarella, B.T.¹, Pacífico, L. C. P.¹, Almeida, C. E.¹, Ferreira-Machado S. C¹.

¹ University of State of the Rio de Janeiro, Institute of Biology Roberto Alcântara Gomes/Department of Radiological Sciences / RJ 20550-900, Rio de Janeiro, Brazil

* jhon.cg52@gmail.com

Abstract

Ionizing radiation is known for its harmful effects on biological tissues, including DNA damage, oxidative stress, and functional alterations in sensitive organs such as the brain. However, while the impacts of high-dose exposure are well established, the effects of low-dose radiation, particularly during critical developmental windows such as the neonatal period, remain poorly understood. Swiss Webster mice of both sexes, at 10 days of age (PND10), were exposed to low doses of radiation within a range comparable to that used in diagnostic imaging procedures. Four groups were evaluated: a control group (no irradiation) and groups exposed to 100 mGy, 60 mGy, and 25 mGy. At PND85, an age equivalent to human adulthood, the mice underwent behavioral assessments to evaluate anxiety-like behavior and short-term memory, followed by molecular and biochemical analyses of hippocampal tissue, including SOD, TBARS, and RT-qPCR for pro-inflammatory cytokines (IL-6 and TNF- α). The results indicated dose- and sex-specific variations, with differences in anxiety-like behavior in males at 100 mGy and alterations in oxidative stress markers (SOD and TBARS) in males at 60 mGy. These effects were modest and restricted to males, with no differences in the expression of the pro-inflammatory cytokines analyzed in either sex. The present study suggests that, for the low doses analyzed, neonatal exposure to ionizing radiation induces subtle and sex-specific behavioral and biochemical changes, without evidence of significant neuroinflammatory responses.

Keywords: Ionizing Radiation; Brain; Low Doses; Oxidative Stress; Neonatal.

1 Introduction

Ionizing radiation (IR) plays a fundamental role in modern medicine, serving as the main physical agent in key imaging diagnostic techniques and cancer therapies. However, procedures involving IR have the potential to cause unintended alterations in biological systems due to their high capacity for cellular interaction and ionization, leading to the generation of free radicals (Takehiko Katsurada, 2017). Computed Tomography (CT), widely used for diagnostic purposes, raises particular concern when applied to the cranial region of pediatric patients, given the increased sensitivity of their tissues to IR (Loganovsky, 2009). Unlike radiotherapy procedures, in which the absorbed doses are higher and well-documented with established tissue effects and dose thresholds, the effects of the lower doses used in radiodiagnostic procedures remain poorly characterized, with dose-response relationships often complex or controversial (Abu Bakar et al., 2019).

Although diagnostic imaging techniques employing IR use relatively low doses, several studies have shown that even these doses can induce changes in the expression of genes and proteins involved in essential metabolic pathways (Marquette et al., 2003). In addition to fulfilling their primary purpose, be it treatment or image acquisition, such procedures can often trigger inflammatory processes in the affected tissues (Loganovsky, 2009). Regarding nervous tissue, particularly the hippocampus, any disruption of homeostasis in this region may impact behavior and memory (Hladik et al., 2020).

Exposure of neural tissue to moderate and high doses of radiation typically results in neuroinflammation, capillary rarefaction, and reduced neurogenesis (Kempf et al., 2013). The neuroinflammatory response acts as a protective mechanism, aiming to eliminate harmful agents and promote tissue repair, as seen in radiation-induced injuries (Abu Bakar et al., 2019). In diagnostic imaging procedures, where absorbed doses are low and not sufficient to cause significant cell death, the inflammatory response tends to be less pronounced but may still involve cellular activation and alterations in the expression of inflammatory cytokines (Lee et al., 2019). The lack of longitudinal studies on tissue responses to such low-dose exposures leaves unanswered questions about the long-term recovery of neural tissue and potential behavioral implications. This is particularly relevant in the context of the hippocampus, a brain structure essential for processing various forms of memory, especially spatial memory, and a critical component in cognitive function (Szabo & Hennerici, 2014).

While studies on higher doses of IR have demonstrated the hippocampus' notable sensitivity and its relevance to cognitive performance, research focused on the effects of low doses, similar to those used in CT scans and conventional X-rays or fluoroscopy, remains limited and inconclusive. To date, no studies have firmly established the significance of low-dose radiation exposure for pediatric patients, especially regarding long-term outcomes (Goodman, 2019; Jánošíková, 2019). In this study, neonatal mice were exposed to low-dose X-ray irradiation using an experimental approach designed to mimic, from a translational perspective, clinical exposures comparable to those encountered during pediatric computed tomography procedures. The same energy range was maintained, and irradiation was restricted to the cranial region to assess the potential impact of this radiation on the hippocampus. The doses of interest, considered low and below 100 mGy, include those commonly used in diagnostic radiology, as reported in the literature (Lee et al., 2019). To investigate long-term effects, the same animals underwent behavioral testing in adulthood. Subsequently, hippocampal samples were analyzed for the expression of genes involved in inflammatory processes (TNF- α and IL-6) using RT-qPCR, along with oxidative stress assessment through different methodologies aimed at quantifying both damage and antioxidant response in hippocampal tissue. TBARS (thiobarbituric acid reactive substances) assays were conducted to quantify lipid peroxidation, while the SOD (superoxide dismutase) was measured to evaluate the antioxidant capacity of the tissue in response to radiation exposure.

2. Methodology

2.1 Organization of Experimental Groups

All procedures involving animals in this study were approved by the Institutional Animal Care and Use Committee (CEUA-UERJ) under protocol number 011/2023. Final gestation-stage pregnant Swiss Webster mice (*Mus musculus*) were housed in the vivarium located in the Department of Radiological Sciences (DCR/IBRAG/UERJ) under controlled temperature and a 12-hour light/dark cycle (light: 6:00 AM–6:00 PM; dark: 6:00 PM–6:00 AM). The animals were kept in standard plastic cages with wood shavings, drinking water, and food *ad libitum*. The birth date was recorded daily, with the day of delivery designated as PND0 (Postnatal Day 0). A total of 20 mice (10 males and 10 females) were used for each of the four radiation doses tested: 0, 25, 60, and 100 mGy. Experimental groups were defined as follows:

- Males: M0 (non-irradiated control), M25 (25 mGy), M60 (60 mGy), M100 (100 mGy);
- Females: F0 (non-irradiated control), F25 (25 mGy), F60 (60 mGy), F100 (100 mGy).

Animals in the M0 and F0 groups (controls) were not irradiated but underwent the same anesthesia and handling procedures as the other groups.

2.2 Radiation Exposure and Dosimetry

The mice were irradiated on postnatal day 10 (PND10) using the SARRP irradiator (LCR/UERJ), with dosimetry performed by the PTW NOMEX system (IEC 61674). The fixed parameters included a 0.30 mm copper filter, a 50 cm source-to-target distance, 90 kV, and 30 mA, while the exposure time (7, 16, and 27 seconds) was varied to achieve doses of 25, 60, and 100 mGy (air kerma). The irradiation area ($\sim 1 \text{ cm}^2$) was verified using GAFCHROMIC™ films.

The animals were positioned in the equipment without sedation, with control groups undergoing the same procedure without activation of the irradiator. The selected doses were based on international guidelines: 100 mGy (the upper limit for neonatal computed tomography scans – UNSCEAR 2020/2021, ICRP 2007) and lower doses (25 and 60 mGy) following the ALARA principle (ICRP 2021). After irradiation, the pups remained with their mothers until weaning (PND21) and were then separated by sex and dose, forming eight experimental groups (four per sex).

2.3 Behavioral Testing Paradigms

On the 81st postnatal day (PND81), behavioral testing was conducted in the afternoon (13:00–16:00). First, the Open Field Test was performed in an acrylic arena ($40 \times 40 \times 40 \text{ cm}$) divided into 16 quadrants. Each animal was individually assessed for 10 minutes, with recording of time spent in the central and peripheral zones (an indicator of anxiety) and total distance traveled (locomotor activity), using the ANY-maze™ system for automated analysis.

Subsequently, the Object Location Test was administered to assess spatial memory. The protocol consisted of two 5-minute sessions with a 60–70-minute interval. During the first exposure, the animals explored two identical objects; in the second session, one object was displaced. Exploration time for each object was recorded, with increased attention to the moved object indicating preserved memory (Lueptown, 2017; Cloke et al., 2015).

2.4 Euthanasia and Tissue Harvesting

Twenty-four hours after behavioral testing, the animals were euthanized in accordance with ethical guidelines (RN37/CONSEA). Immediately following euthanasia, bilateral hippocampi were dissected for collection. The left hemisphere was allocated for molecular analysis, while the right hemisphere was reserved for enzymatic assays.

2.5 Molecular and Biochemical Assays

The left hippocampus was processed for RNA extraction and subsequent RT-qPCR analysis, focusing on the expression of pro-inflammatory cytokines IL-6 and TNF- α . The right hippocampus was used to assess the activity of redox-balance-related enzymes. This approach enabled correlation of behavioral parameters with molecular markers of neuroinflammation and oxidative stress within the same animal.

2.6 Statistical analysis

The data were tested for normality using the Shapiro-Wilk test, followed by one-way ANOVA analysis in GraphPad Prism 8. Statistical significance was set at $*p* < 0.05$. Results are presented as mean $\pm$ standard deviation (SD)

3. Results

3.1 Behavioral Outcomes

The assessment of anxiety-related parameters was performed by analyzing the time spent in the central quadrants of the open field, expressed as a percentage of the total test duration. The results

revealed distinct patterns among experimental groups. While female subjects exhibited no significant alterations in the proportion of time spent in the center area at any tested dose ($F(3,38) = 0.6964$; $p = 0.560$; Figure 1.A), male groups demonstrated a differential response to radiation exposure ($F(3,38) = 4.883$; $p = 0.0057$; Figure 1.B). Specifically, males in the M100 group (100 mGy) showed a significant increase in time spent in the central area compared to the non-irradiated control group (M0), suggesting a potential reduction in anxiety-related behaviors.

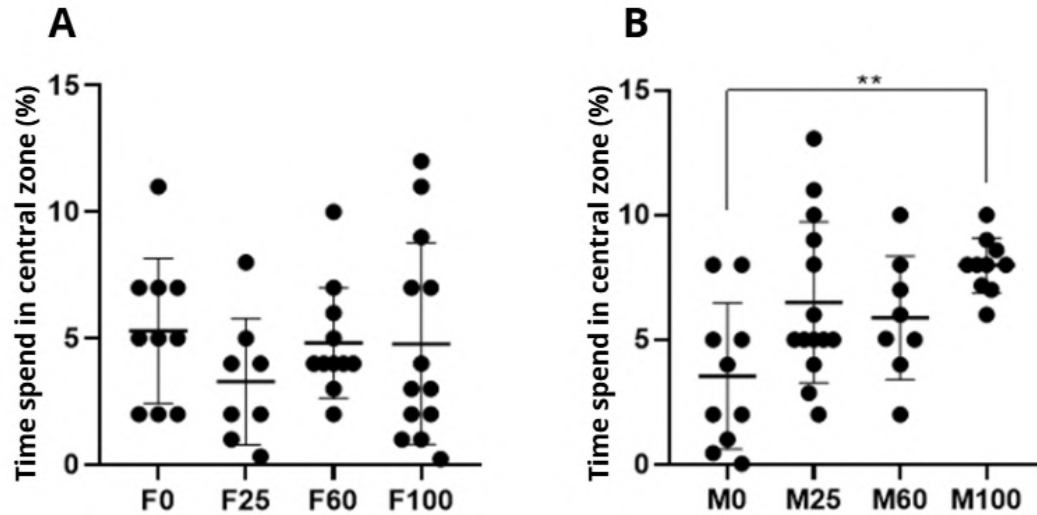


Figure 1: Open field test performance: Central quadrant dwell time (%) (A) Female groups (B) Male groups. Data presented as mean $\pm$ SD; statistical significance set at $p < 0.05$

It is noteworthy that this pattern was not maintained in the object location recognition test, where exploration times of the displaced object showed no significant differences among experimental groups in either sex (ANOVA: females - $F(3,40)=0.5740$, $p=0.6354$; males - $F(3,35)=0.5269$, $p=0.6667$).

While the open field test suggests potential sex-dependent sensitivity, particularly in males at the highest tested dose, the absence of effects in the location recognition test implies that these alterations may not extend to all aspects of exploratory behavior and spatial memory. The lack of changes in the location recognition test, even at the dose that showed effects in the open field (100 mGy), suggests that radiation effects may be specific to certain neural circuits or behavioral domains, or that different tests possess varying sensitivities for detecting subtle alterations.

3.2 Hippocampal Gene Expression Profiling

Gene expression of the inflammatory cytokines IL-6 and TNF- α in the hippocampus of mice was assessed by RT-qPCR. For IL-6 (Figure 2A: females; Figure 2B: males), the data indicated no statistically significant differences between experimental groups in either females ($F(3,18) = 1.456$; $p = 0.2599$) or males ($F(3,15) = 2.318$; $p = 0.1169$), suggesting that the treatments did not substantially alter the expression of this pro-inflammatory cytokine in the hippocampus. Similarly, for TNF- α (Figure 2C: females; Figure 2D: males), the analyses showed that although males exhibited a p-value below the significance threshold ($F(3,21) = 4.200$; $p = 0.0178$), post hoc multiple comparisons revealed no significant differences between groups, indicating that the result may be attributable to intragroup variability. In females, no relevant changes were observed ($F(3,20) = 2.095$; $p = 0.1330$). These findings suggest that, within the evaluated period, neonatal cranial irradiation did not induce detectable modulation of the transcription of these cytokines at

measurable levels. This may reflect either an absence of active inflammation at the time of sampling or an already resolved adaptive response.

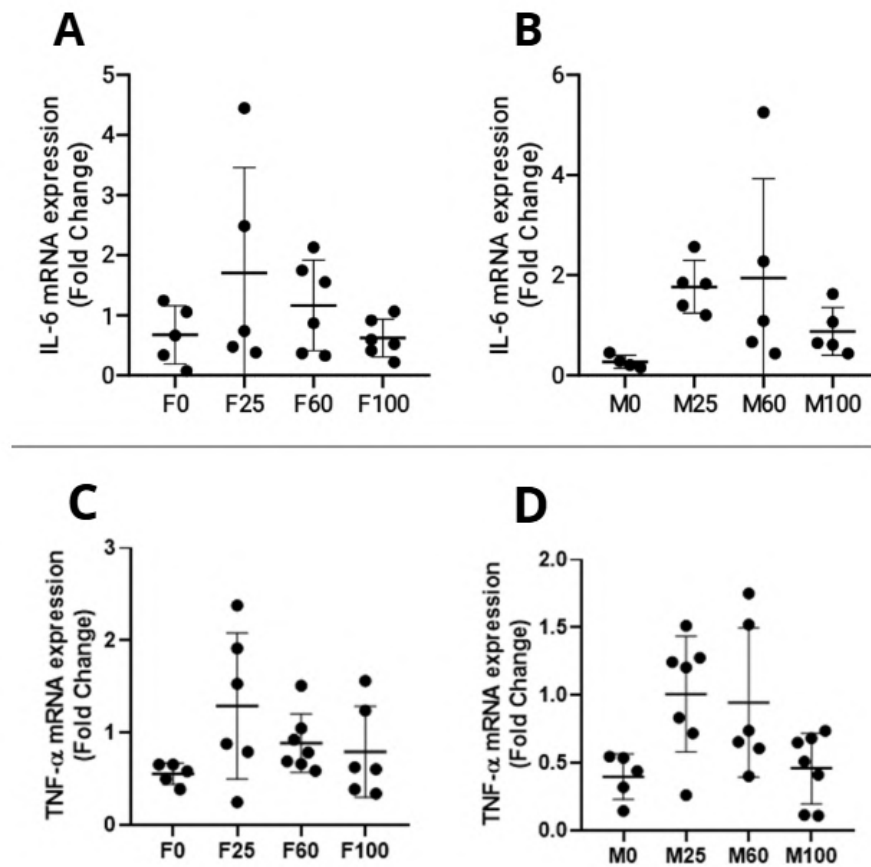


Figure 2: Hippocampal cytokine mRNA expression following neonatal irradiation. (A-B) IL-6 expression levels in (A) females and (B) males (C-D) TNF- α expression levels in (C) females and (D) males. Data presented as mean $\pm$ SD; statistical significance set at $p < 0.05$

3.3 Enzymatic Activity Analysis

Assessment of lipid peroxidation through the TBARS assay revealed distinct sex-specific patterns, as shown in Figures 3A-B. In males, a significant increase in malondialdehyde (MDA) levels was observed in the 60 mGy irradiated group ($F(3,20) = 4.486$; $p = 0.0145$), with statistically significant differences compared to both M0 and M100 groups. This non-linear dose-response pattern suggests either a specific window of vulnerability or threshold-dependent activation of oxidative damage mechanisms.

Conversely, female subjects ($F(3,18) = 2.079$; $p = 0.1388$) showed no statistically significant alterations, indicating either greater stability against radiation-induced oxidative stress or more efficient endogenous antioxidant defense mechanisms. These sex-dimorphic responses highlight the importance of considering biological sex as a critical variable when evaluating radiation-induced oxidative damage in neural tissue.

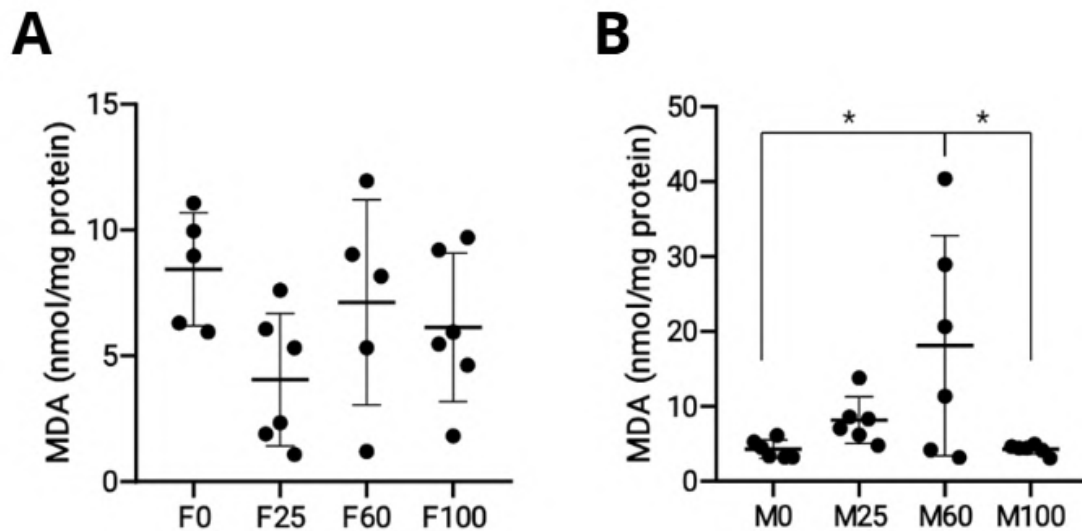


Figure 3: Radiation-induced lipid peroxidation measured by TBARS assay. (A) Female hippocampal MDA (B) Male hippocampal MDA levels. Data presented as malondialdehyde (MDA) concentration (nmol/mg protein) $\pm$ SD; * $p < 0.05$.

Analysis of superoxide dismutase (SOD) enzyme activity, presented in Figures 4A-B, demonstrated sexually dimorphic response patterns. In females (Figure 4A), statistical analysis revealed no significant differences among groups ($F(3,19) = 1.460$; $p = 0.2570$). However, male subjects (Figure 4B) exhibited significantly increased SOD activity in the M60 group ($F(3,19) = 3.690$; $p = 0.0301$), with Bonferroni post-hoc testing confirming differences relative to controls (M0).

This enzymatic upregulation likely represents a compensatory mechanism against the oxidative damage detected by TBARS in the same experimental group, suggesting an adaptive activation of antioxidant defenses following low-dose radiation exposure. The absence of this response in female animals reinforces the existence of sexually dimorphic regulation in cerebral redox homeostasis, highlighting fundamental sex-based differences in neurobiological responses to oxidative stress.

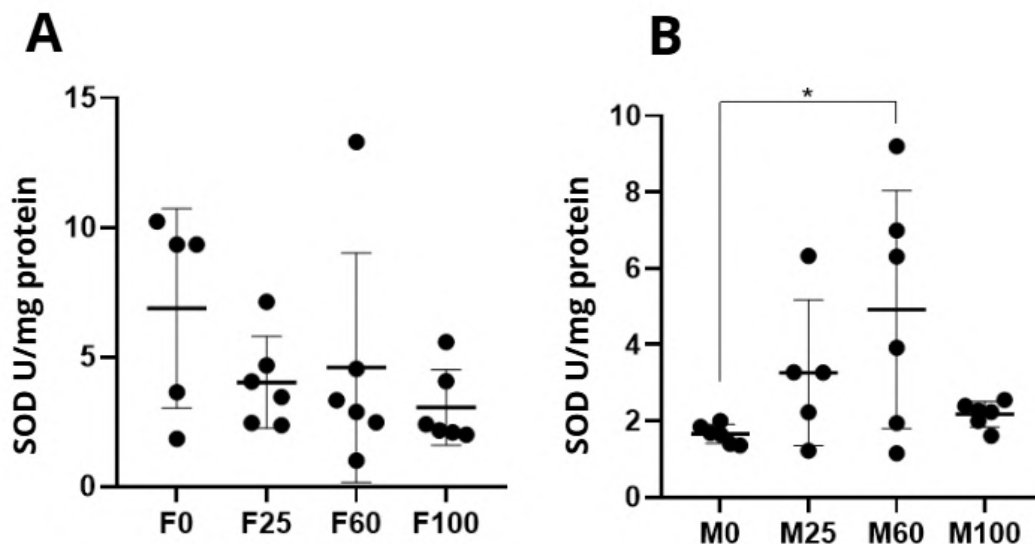


Figure 4: Hippocampal SOD activity following radiation exposure. (A) Female groups (B) Male groups. Data presented as superoxide dismutase (SOD) activity (U/mg protein) $\pm$ SD; * $p < 0.05$.

4. Conclusions

The findings of this study indicate that neonatal cranial exposure to low-dose ionizing radiation resulted in dose- and sex-specific alterations, including alterations in anxiety-like behavior in males exposed to 100 mGy and modulation of oxidative stress markers (SOD activity and lipid peroxidation) in males exposed to 60 mGy. Although the observed effects were modest and not associated with potential inflammation (no expression of the analyzed pro-inflammatory cytokines), they were not absent in males. The lack of alterations in females suggests a possible influence of sex-dependent biological factors, including hormonal regulation, in modulating the long-term response to radiation exposure. The variability observed in behavioral and biochemical outcomes, characterized by notable data dispersion, may be partially related to individual differences in response to low-dose exposure.

Furthermore, the non-linear patterns detected in some endpoints raise important considerations regarding the universal applicability of the linear no-threshold (LNT) model in radioprotection, highlighting the biological complexity of radiation effects on the developing nervous system. Taken together, these results emphasize the need for further research exploring both genetic background and hormonal influences as determinants of susceptibility to low-dose irradiation, contributing to evidence-based approaches that balance the clinical benefits and potential risks of diagnostic imaging in pediatric populations.

Acknowledgements

We gratefully acknowledge the dedicated efforts of all researchers, technicians, and interns whose tireless work made this project possible. Special thanks to University of State of Rio de Janeiro for providing the infrastructure and intellectual environment essential to this study. This research was supported by funding from FAPERJ and CNPQ, whose financial and structural support was instrumental to our work.

References

- Abu Bakar, N. F., Amira Othman, S., Amirah Nor Azman, N. F., & Saqinah Jasrin, N. (2019). Effect of ionizing radiation towards human health: A review. *IOP Conference Series: Earth and Environmental Science*, 268(1). <https://doi.org/10.1088/1755-1315/268/1/012005>
- Cloke, J. M., Jacklin, D. L., & Winters, B. D. (2015). The neural bases of crossmodal object recognition in non-human primates and rodents: A review. *Behavioural Brain Research*, 285, 118–130. <https://doi.org/10.1016/J.BBR.2014.09.039>
- Goodman, T. R., Mustafa, A., & Rowe, E. (2019). Pediatric CT radiation exposure: where we were, and where we are now. *Pediatric Radiology*, 49(4), 469–478. <https://doi.org/10.1007/S00247-018-4281-Y/METRICS>
- Hladik, D., Dalke, C., von Toerne, C., Hauck, S. M., Azimzadeh, O., Philipp, J., Ung, M. C., Schlattl, H., Rößler, U., Graw, J., Atkinson, M. J., & Tapio, S. (2020). CREB Signaling Mediates Dose-Dependent Radiation Response in the Murine Hippocampus Two Years after Total Body Exposure. *Journal of Proteome Research*, 19(1), 337–345. <https://doi.org/10.1021/ACS.JPROTEOME.9B00552>

Jánošíková, L., Juričková, M., Horváthová, M., Nikodemová, D., Klepanec, A., & Šalát, D. (2019). RISK EVALUATION IN THE LOW-DOSE RANGE CT FOR RADIATION-EXPOSED CHILDREN, BASED ON DNA DAMAGE. *Radiation Protection Dosimetry*, 186(2–3), 163–167. <https://doi.org/10.1093/RPD/NCZ195>

Kempf, S. J., Azimzadeh, O., Atkinson, M. J., & Tapio, S. (2013). Long-term effects of ionising radiation on the brain: Cause for concern? *Radiation and Environmental Biophysics*, 52(1), 5–16. <https://doi.org/10.1007/s00411-012-0436-7>

Lee, C., Journy, N., Moroz, B. E., Little, M., Harbron, R., McHugh, K., Pearce, M., & Berrington De Gonzalez, A. (2019). ORGAN DOSE ESTIMATION ACCOUNTING FOR UNCERTAINTY FOR PEDIATRIC AND YOUNG ADULT CT SCANS IN THE UNITED KINGDOM. *Radiation Protection Dosimetry*, 184(1), 44. <https://doi.org/10.1093/RPD/NCY184>

Loganovsky, K. (2009). Do Low Doses of Ionizing Radiation Affect the Human Brain? *Data Science Journal*, 8, BR13–BR35. <https://doi.org/10.2481/DSJ.BR-04>

Marquette, C., Linard, C., Galonnier, M., van Uye, A., Mathieu, J., Gourmelon, P., & Clarençon, D. (2003). IL-1 β , TNF α and IL-6 induction in the rat brain after partial-body irradiation: Role of vagal afferents. *International Journal of Radiation Biology*, 79(10), 777–785. <https://doi.org/10.1080/09553000310001610998>

Sources, effects and risks of ionizing radiation.: United Nations Scientific Committee on the Effects of Atomic Radiation: UNSCEAR 2013 report to the General Assembly with scientific annexes. Volume II, scientific annex B. (2013). United Nations.

Szabo, K., & Hennerici, M. G. (2014). The hippocampus in clinical neuroscience. *The Hippocampus in Clinical Neuroscience*, 34, 1–161. <https://doi.org/10.1159/ISBN.978-3-318-02568-2>

Takehiko Katsurada, Mutsumi Nishida, & Naoya Sakamoto. (2017). Imaging (X-ray - CT - MRI - ultrasound). *Nihon Rinsho*, 75(3), 387–391.

The 2007 Recommendations of the International Commission on Radiological Protection. ICRP publication 103. (2007). *Annals of the ICRP*, 37(2–4), 1–332. <https://doi.org/10.1016/J.ICRP.2007.10.003>

Germination response of “Micro-Tom” to controlled magnetization by using a Rodin coil

E. Romero Muñoz¹ D. Sánchez Guzmán¹, J.C. Martínez Espinosa¹, T. Córdova Fraga², A. Hernández Rayas², H. A. Pérez Olivas², L. Herrera Isidró^{1*}

¹ Instituto Politécnico Nacional - UPIIG, Ave. Mineral de Valenciana N 200, Fraccionamiento Industrial Puerto Interior, 36275 Silao de la Victoria, Gto., Mexico.

² División de Ciencias e Ingeniería, Universidad de Guanajuato campus León, Loma del Bosque N 103, Lomas del campestre, 37150 León, Gto., Mexico.

*corresponding.author: lherrerai@ipn.mx

Abstract

Exposure of *Solanum lycopersicum* L. “Micro-Tom” seeds to variable magnetic fields generated by a Rodin coil significantly influenced physiological processes associated with early germination. Magnetostimulation has been proposed as a biophysical strategy to enhance plant development by modulating cellular metabolism, including ion transport, membrane permeability, and reactive oxygen species (ROS) production. In this study, the effect of magnetization on “Micro-Tom” germination was evaluated under *in vitro* culture. Seeds were exposed to a low-frequency (10 Hz) alternating magnetic field at intensities of 0.3, 0.7, and 1.9 mT for 48, 96, and 144 hours and subsequently cultured alongside a non-magnetized control. The initial seed viability was confirmed at 100% using the tetrazolium test (TTC). Results indicated that seeds treated at 1.9 mT achieved full germination after six days, compared to 83.33% in control. A positive correlation was observed between magnetic field intensity and the germination rate. Moreover, seedlings from treated seeds exhibited greater root elongation, with the most pronounced effect in those exposed to 1.9 mT for six days. These findings suggest that variable magnetic fields stimulate growth-related physiological processes, possibly through redox signaling mechanisms associated with controlled oxidative stress. Therefore, magnetostimulation represents a promising biotechnological tool to optimize germination and early seedling development in cultivated species.

Keywords: Micro-Tom; magnetic field; Rodin coil; germination rate.

1 Introduction

Solanum lycopersicum is among the most consumed and economically valuable vegetables worldwide, with Mexico ranking tenth in global production (FAO, 2012). The “Micro-Tom” cultivar originates from the Andean region, spanning southern Colombia to northern Chile, and was later domesticated in Central America and Mexico. Tomato has been used as an experimental model for the introduction of agronomically relevant genes into dicotyledonous plants (Paduchuri et al., 2010). “Micro-Tom”, belonging to the *Solanaceae* family, is a dicotyledonous, herbaceous, and perennial plant. It is characterized by being one of the tomato varieties with a maximum height of 20 cm, producing fruits between 1 and 2 cm in length and 1 and 3 cm in diameter (Cestoni et al., 2006). Owing to its small size,

short lifecycle and compact genome of 950 Mpb, “Micro-Tom” has become an established model system in plant biology (Dan et al., 2007).

The tomato seeds are lenticular in shape, consisting of an embryo, endosperm, and seed coat, and undergo germination in three phases: imbibition, metabolic activation, and growth, with the radicle emerging first (Nuez, 2001; Woolhouse et al., 1980). Germination may be epigeal or hypogeal, depending on hypocotyl elongation and the cotyledon position relative to the substrate (Alzugaray et al., 2007). To enhance germination and crop performance, physical treatments such as exposure to magnetic fields exceeding the Earth's geomagnetic field have been employed, as they can accelerate seedling development and promote more vigorous growth (Shimazaki & Shizuoka, 1986).

Magnetic fields, generated by moving charges, act on moving charges (Young & Freedman, 2009). This study dates back about 2,500 years to the discovery of magnetite in Magnesia, Greece (Roy, 2023). The Earth's magnetic field, produced by the motion of liquid metal in its core, resembles a giant magnet that influences the living organisms. Its intensity varies with latitude, being stronger near the poles (67–70 μT) and weaker at the equator (33–35 μT). It is generally considered homogeneous, static, and of low intensity (Maffei, 2014; Galland & Pazur, 2005).

As mentioned by Sears and Zemansky (2016), Lorentz's law states that a charged particle q moving with a velocity $\vec{v}$ through a magnetic field $\vec{B}$ will experience a force $\vec{F}$, known as the Lorentz force. This force is proportional to the values of q , $\vec{v}$, and $\vec{B}$, and is given by a specific formula (Equation 1).

$$\vec{F} = q\vec{v} \times \vec{B} \text{ [Lorentz's force]} \quad (1)$$

Where:

$\vec{F}$ = Magnetic force on a moving charged particle [N]

q = Charge [C]

$\vec{v}$ = Velocity vector [m/s]

$\vec{B}$ = Magnetic field [T] = [N / (A·m)]

The Rodin coil is a toroidal device with a design based on numerical and geometric patterns proposed by Marko Rodin, characterized by a special winding of 36 poles arranged at 150° above its midplane, which seeks to optimize the interaction between the electric and magnetic fields, although its effectiveness and applications are still under study (Figure 1) (Galindo et al., 2024).

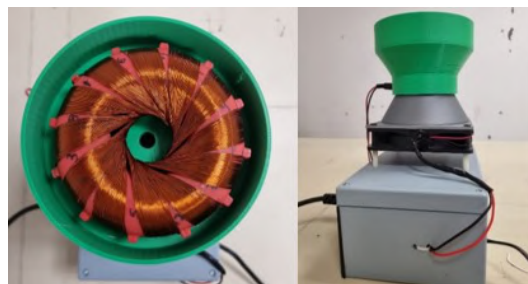


Figure 1: Rodin coil: (A) top view, (B) Side view.

The Earth's magnetic field, due to its low intensity, can easily penetrate cellular tissues and influence physiological and biochemical processes in plants, especially in tissues with high metabolic activity.

These effects are partly explained by ion cyclotron resonance, which alters the behavior of ions such as Ca^{2+} and K^{+} , affecting intracellular signaling and magnetoreception. At physiological and molecular levels, magnetic fields can improve water and nutrient uptake, stimulate enzymes and hormones, promote germination and photosynthesis, alter membranes and gene expression, and induce oxidative stress through ROS. The intensity, frequency, duration of the field, and the state of the plant determine the magnitude of these effects (Saletnik et al., 2022; Smith et al., 1995).

2 Methodology

2.1 Evaluation of “Micro-Tom” seeds viability using the triphenyl tetrazolium (TTC) test.

To determine seed viability, the triphenyl tetrazolium (TTC) test was applied, following the protocol described by Salazar et al. (2020). Prior to conducting the test, a 0.25% TTC solution was prepared in 100 mL of a phosphate buffer solution (1.1% potassium diphosphate and 0.9% sodium monophosphate). To prepare the phosphate buffer, 0.9 g of sodium monophosphate was weighed and dissolved in distilled water before diluting to 100 mL, thus obtaining Solution A. Similarly, 1.1 g of potassium diphosphate was weighed and dissolved in distilled water to reach a final volume of 100 mL, obtaining Solution B. Finally, 50 mL of Solution A and 50 mL of Solution B were mixed.

The seeds were partially dissected to expose the embryo and covered with the 0.25% TTC solution and kept in the dark for 24 hours at room temperature. After this time, they were rinsed and analyzed under a stereoscope. An intense red color indicates viability.

2.2 Characterization of the magnetic field of a Rodin coil.

A "magnetic induction vortex" device, which presents a spiral pattern to produce magnetic field vortices, was used. This toroidal coil was built by researchers from the Division of Science and Engineering at the University of Guanajuato, León Campus (Galindo et al., 2024).

A controller was used to supply sine wave frequencies to the coil. Frequencies from 5 Hz to 50 Hz were tested, in 5 Hz intervals, to determine which frequency produced the greatest magnetic field intensity. Once the optimal frequency (10 Hz) was determined, a grid was created with a series of points to be measured using the HFBTE TD8620 teslameter. The maximum values obtained by the teslameter were measured and recorded to obtain a magnetic field profile on the surface of the Rodin coil (Figure 2).

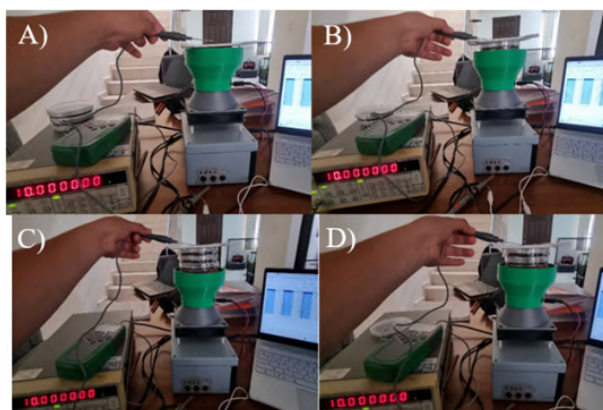


Figure 2. Schematic representation of the measurements on the Rodin coil: (A) surface, (B) level 1 at 1.8 cm, (C) level 2 at 3.6 cm, and (D) level 3 at 5.4 cm.

2.3 Magnetization treatments on seeds of *S. lycopersicum* L. “Micro-Tom” with variable magnetic fields.

The seeds were subjected to variable magnetic fields of 0.3 mT, 0.7 mT, and 1.9 mT at a frequency of 10 Hz for periods of 48, 96, and 144 hours. In addition, a control of non-magnetized seeds was included for each treatment. Thirty seeds were magnetized in each treatment. MINITAB 16 statistical software (ANOVA $p < 0.05$ and Tukey’s pairwise comparison test was used to examine the data. Once the magnetic field exposure time was over, both the magnetized and control seeds were disinfected and cultured under *in vitro*.

2.4 Disinfection of magnetized seeds and evaluation under *in vitro* germination.

The previously magnetized seeds were immersed in 70% ethanol and left to stir for 5 minutes. The ethanol solution was then removed, and a 20% sodium hypochlorite solution with 2% Tween was added and stirred for 20 minutes. The liquid was then decanted until only the seeds were left, and three washes were performed with sterile distilled water until the sodium hypochlorite was eliminated.

The disinfected seeds were sown in Petri dishes with Murashige and Skoog (MS) medium, supplemented with 30 g/L of sucrose, 1 g/L of activated carbon, and 0.5% of smoked water, at a pH of 5.8. Ten seeds were sown in each Petri dish. This experiment was performed in triplicate. The number of germinated seeds was recorded over a 10-day period.

3 Results and discussion

3.1 Evaluation of *Solanum lycopersicum* seed viability with triphenyl tetrazolium (TTC).

To evaluate seed viability, 0.25% 2,3,5-triphenyltetrazolium chloride (TTC) was used on dissected seeds of *Solanum lycopersicum* L. “Micro-Tom”. After 24 hours of incubation in the reagent, viable embryos developed an intense reddish coloration, whereas untreated control seeds remained white (Figure 3). All evaluated seeds exhibit staining, confirming 100% viability and indicating high physiological quality.

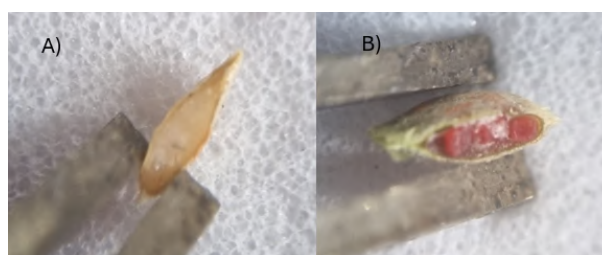


Figure 3. *Solanum lycopersicum* seed before (A) and after staining with 2,3,5-triphenyltetrazolium (TTC) (B).

3.2 Characterization of the electromagnetic field by the Rodin coil.

To characterize the distribution and intensity of the magnetic field generated by the Rodin coil, different frequencies between 5 and 50 Hz were tested to identify the one producing the highest field strength. The optimal frequency was determined to be 10 Hz, which generated a maximum intensity of approximately 4 mT at the center of the coil. Both frequency and wave type were controlled using a sinusoidal function generator.

A flat plate was positioned to provide a homogeneous surface for measurements. Seventeen measurement points were recorded: one at the center and 16 arranged in cardinal (north, south, east, west) and intercardinal (northeast, southeast, northwest, southwest) direction. Measurements were taken at distances of 2.25 cm and 4.5 cm from the center in each direction (Figure 4).

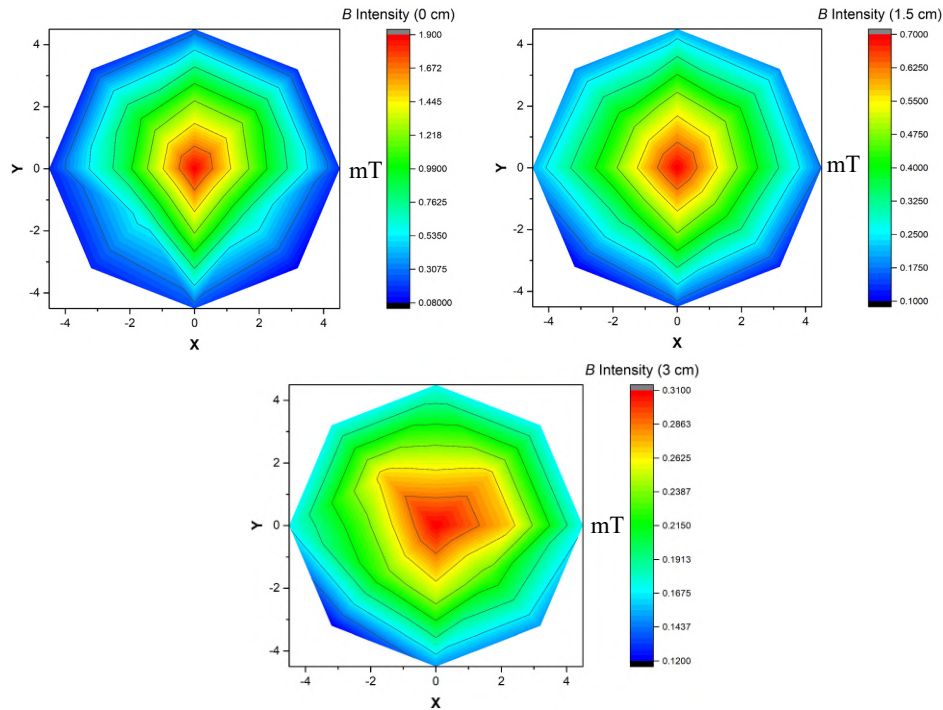


Figure 4. Spatial distribution of the intensity of the electromagnetic field generated by a Rodin coil at 10 Hz, evaluated in cardinal directions and 0, 1.5, and 3 cm from the origin.

3.3 Magnetization treatments for *Solanum lycopersicum* seeds.

Prior to disinfection and the *in vitro* culture, “Micro-Tom” seeds were magnetized using a Rodin coil at intensities of 0.3, 0.7, and 1.9 mT for 48, 96, and 144 hours. Germination of both magnetized and non-magnetized (control) seeds was monitored daily. The first radicles emerged six days after sowing. Except for seeds exposed to 0.3 mT for 48 and 96 hours, which showed germination on day 6, all magnetic treatments displayed higher gemination rates compared to the control, and by 10 days the cumulative germination had reached 100% across all treatments (Figure 5).

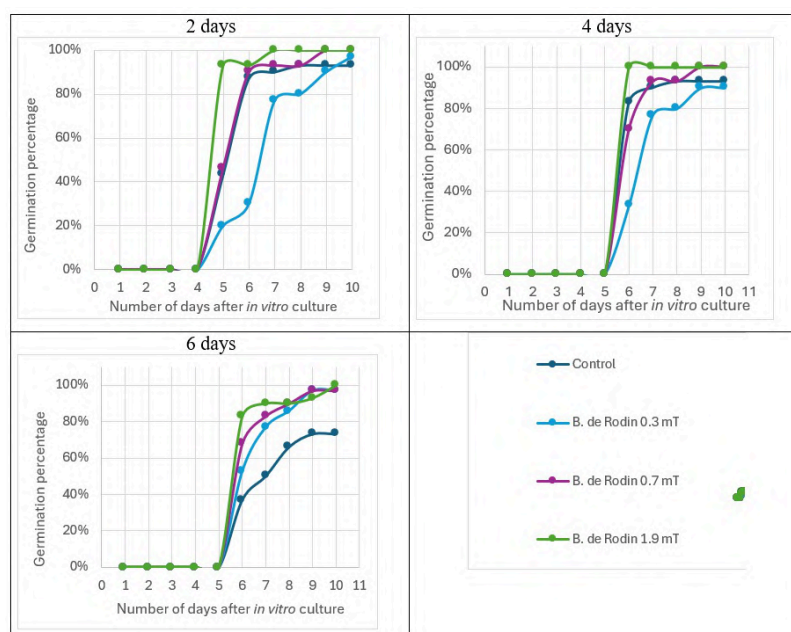


Figure 5. Germination percentage of *S. lycopersicum* L. seeds under magnetic treatments (48, 96 and 144 hours) over 10 days under *in vitro* culture.

This positive effect is consistent with that reported by Naz et al. (2012) in *Abelmoschus esculentus* (99 mT, 3 and 11 min) where increases of 32.8% and 33% were observed. Similarly, Vashisth and Nagarajan (2008, 2010) found increases of 9% to 15% in *Cicer arietinum* (100 mT, 1 h; 150 mT, 4 h) and *Helianthus annuus* (50 mT, 1 h; 200 mT, 2 h). This suggests that the effect of the field on seed germination and development depends not only on the field strength but also on the stimulus duration and the type of magnetic signal applied.

On day 6, seed germination showed a dependence on magnetic field intensity. The control reached 83% germination, while 0.3 mT resulted in only 33%. Seeds exposed to 0.7 mT reached 70%, and those treated 1.9 mT achieved 100% germination, the highest response observed (Figure 6).

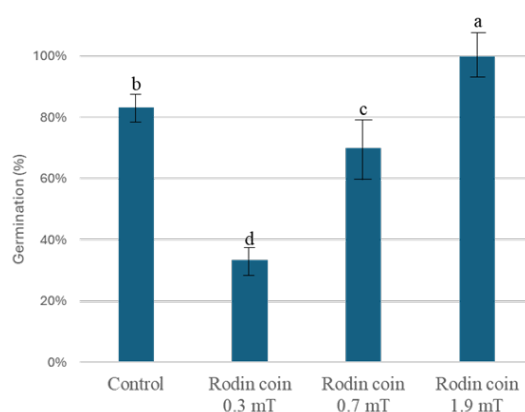


Figure 6. Germination percentage of *S. lycopersicum* “Micro-Tom” seeds on day 6 under different magnetic field intensities for 144 hours.

Another key criterion was germination synchrony. Unlike the control seeds, which germinated gradually over time, seeds exposed to magnetic treatments generally synchronized the day of germination. Exposure of “Micro-Tom” seeds to magnetic fields generated by the Rodin coil promoted not only a higher germination rate but also greater synchrony, as most treated seeds germinated simultaneously around days 5-6, in contrast to the gradual germination in the control group. This effect was most evident at 1.9 mT, which consistently accelerated and concentrated the germination; 0.7 mT produced a moderate effect, whereas 0.3 mT had little or even a delaying influence. These findings indicate that magnetic field intensity is more decisive than exposure duration in promoting uniform synchronous germination (Figure 7).

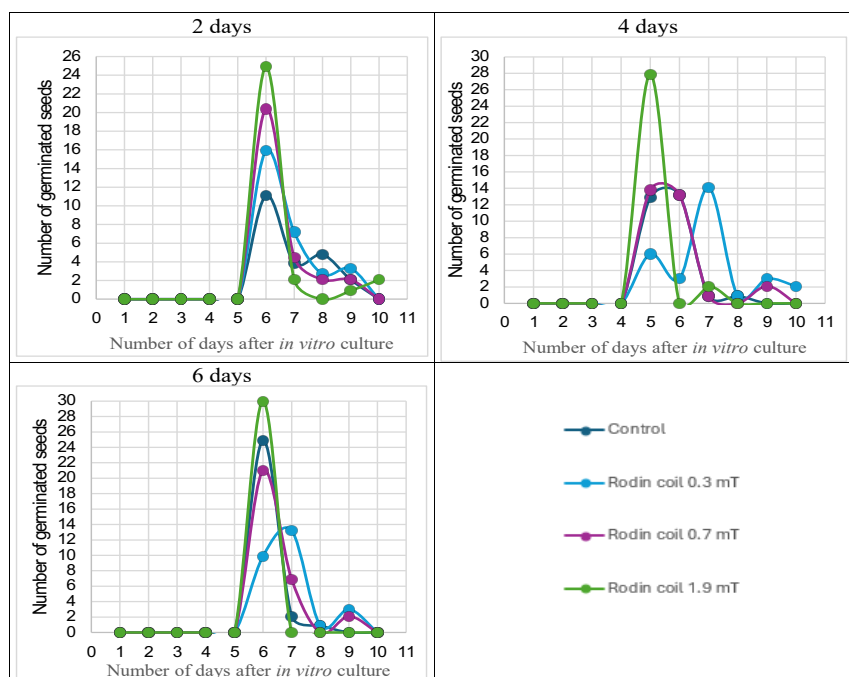


Figure 7. Number of germinated seeds of *S. lycopersicum* L. under magnetic treatments (48, 92 and 144 hours) over 10 days under *in vitro* culture.

This synchrony is highly desirable for agricultural and biotechnological applications, as it facilitates the standardization of seedling development. Overall, these results suggest that exposure of “Micro-Tom” seeds to magnetic fields exerts a positive effect on germination and promotes synchronous germination of seeds emergency.

Root length was measured to evaluate the effectiveness of magnetic field treatments on early seedling development. On day 6, root length increases with field intensity (Figures 8 and 9). Control seedlings average 1.6 cm while treatments with 0.3, 0.7, and 1.9 mT reach approximately 1.8, 2.2, and 2.7 cm, respectively, with 1.9 showing the greatest stimulation. These results coincide with those reported by Selim et al. (2022), who observed a 19.52% increase in root length under similar conditions.

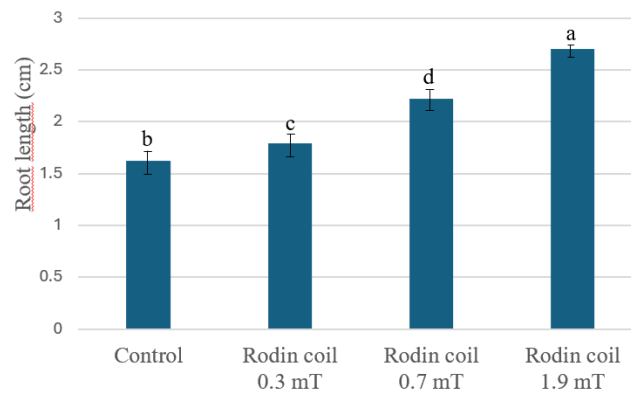


Figure 8. Root length of “Micro-Tom” seedlings treated for 144 hours with the Rodin coil and control, measured on day 6 under *in vitro* culture.

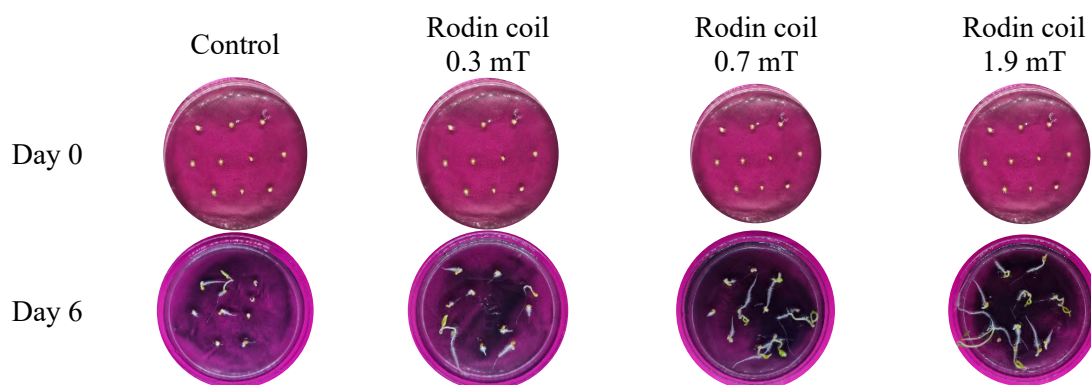


Figure 9. Germination of *S. lycopersicum* “Micro-Tom” seeds under magnetic field treatments (0.3, 0.7 and 1.9 mT) and control at day 6 under *in vitro* culture.

4 Conclusions

Rodin coil treatments enhanced both the rate and synchronization of germination, promoting more uniform seedling development and offering a potential agronomic benefit. Seedlings from magnetized seeds exhibited greater root elongation, with the pronounced effect at the highest intensities (1.9 mT). This suggests that a minimum threshold of magnetic field intensity is required to elicit significant physiological responses.

Acknowledgements

The authors thank the Research and Graduate Studies Secretariat of the Instituto Politécnico Nacional for financial support through SIP project 20231512.

References

- Abhary, M., y Akhkha, A. (2023). Effects of neodymium magneto-priming on seed germination and salinity tolerance in tomato. *Bioelectromagnetics*, 44(1-2), 47-56. <https://doi.org/10.1002/bem.22438>.
- Alzugaray, C., Carnevale, J., Salinas, A., y Pioli, R. (2007). Factores bióticos y abióticos que afectan la calidad de las semillas de *Schinopsis balansae* Engl. y *Aspidosperma quebracho-blanco* Schltdl. *Revista Iberoamericana de Micología*, 24(2), 142-147. [https://doi.org/10.1016/S1130-1406\(07\)70030-X](https://doi.org/10.1016/S1130-1406(07)70030-X).
- Aranzazu, J. (2019). Análisis del efecto del tratamiento magnético sobre la actividad enzimática en semillas de maíz (*Zea mays* L.). <https://repositorio.unal.edu.co/handle/unal/75611>.
- Argüello, L. (2002). Tratamientos para la prolongar la conservación de la semilla de cocona (*Solanum sessiliflorum* Dunal) (Doctoral dissertation, Zamorano: Escuela Agrícola Panamericana, 2016).
- Cecerska-Heryć, E., Gliźniewicz, M., Grygorcewicz, B., Serwin, N., Stodolak, P., Słodzińska, W., ... & Dołęgowska, B. (2024). The effect of a rotating magnetic field on the antioxidant system in healthy volunteers-preliminary study. *Scientific Reports*, 14(1), 8677.
- Cestoni, F., De Jovel, G. y Urquilla, A. (2006). Perfil de negocios de tomate cherry o cereza hacia el mercado de los Estados Unidos. El Salvador. 73 p.4
- Dan, Y., Fei, Z., y Rothan, C. (2007). MicroTom—a new model system for plant genomics. *Genes, Genomes and Genomics*, 1(2), 167-179.
- Erazo, A. (2022). Análisis del efecto del tratamiento magnético sobre la concentración de proteínas totales en semillas de tomate (*Solanum lycopersicum* L.).
- FAO, Food Agricultural Organization. (2012). Organización de las Naciones Unidas para la Agricultura y la Alimentación. www.apps.fao.org
- Galindo, Á., Olivas, H., Islas, G., Rayas, A., López, F., y Fraga, T. (2024). Vortex magnetic induction: Mathematical, geometric and experimental characterization. *Revista Mexicana de Física*, 70(3 May-Jun), 030901-1.
- Galland, P. y Pazur, A. (2005). Magnetoreception in plants. *Journal of plant research*, 118, 371-389.
- Hozayn, M., Gouda, A., El-Mahdy, A. A., & Al Askar, n. M. (2024). Impact of Magnetic Technology on Germination, Seedling vigour, Growth, Yield, and Nutritional value of Groundnut (*Arachis Hypogaea* L) under Water Salinity Stress Conditions in Sandy Soil. *Egyptian Journal of Chemistry*.
- León, E., Fernández, R., y De Souza Torres, A. (1999). Efecto del tratamiento magnético de semillas de tomate (*Lycopersicon esculentum* Mill) sobre la germinación y el crecimiento de las plántulas. *Investigación Agraria. Producción y Protección Vegetales*, 14(3), 437-444.
- Maffei, M. (2014). Magnetic field effects on plant growth, development, and evolution. *Frontiers in plant science*, 5, 445.

- Martínez, E. (2015). Bobinas de helmholtz. *Mexico, México DF*.
- Murashige, T., y Skoog, F. (1962). A revised medium for rapid growth and bio assays with tobacco tissue cultures. *Physiologia plantarum*, 15(3). <https://doi.org/10.1111/j.1399-3054.1962.tb08052.x>
- Naz, A., Jamil, Y., Iqbal, M., Ahmad, M. R., Ashraf, M. I., y Ahmad, R. (2012). Enhancement in the germination, growth and yield of Okra (*Abelmoschus esculentus*) using pre-sowing magnetic treatment of seeds.
- Nuez, F. (2001). El cultivo del tomate. Ediciones Mundi-Prensa Madrid, Barcelona, México.
- ONU. (2022). Una población en crecimiento. <https://www.un.org/es/global-issues/population#:~:text=Una%20poblaci%C3%B3n%20en%20crecimiento,y%202000%20millones%20desde%201998>.
- Paduchuri, P., Gohokar, S., Thamke, B., y Subhas, M. (2010). Transgenic tomatoes-a review. *Int J Adv Biotechnol Res*, 1, 69-72.
- Pérez-Alcázar, G. (2016). Imanes permanentes: características, aplicaciones y futuro. *Revista de la Academia Colombiana de Ciencias Exactas, Físicas y Naturales*, 40(155), 221-233.
- Radhakrishnan, R. (2019). Magnetic field regulates plant functions, growth and enhances tolerance against environmental stresses. *Physiology and Molecular Biology of Plants*, 25(5), 1107-1119.
- Romero, J., y Anuardo, E. (2018). Desarrollo de instrumentación de Resonancia Magnética Nuclear con Ciclado Rápido de Campo magnético para imágenes y relaxometría localizada (Doctoral dissertation, Tesis de Doctorado, Universidad Nacional de Córdoba).
- Roy, S. B. (2023). A Short History of Magnetism and Magnetic Materials. En *Experimental Techniques in Magnetism and Magnetic Materials* (pp. 3–12). Capítulo 1, Cambridge: Cambridge University Press. 10.1017/9781108784948.003.
- Salazar, S., Quintero, J. y Botello, E. (2020). Optimización de la prueba de tetrazolio para evaluar la viabilidad en semillas de *Solanum lycopersicum* L. *Corpoica Ciencia y Tecnología Agropecuaria*, 21(3 (2020)), 1-12.
- Salazar, S., Quintero, J., y Bustos, V. (2020). Implementación de la prueba de tetrazolio en las semillas de *Raphanus sativus* L. *Revista Facultad De Ciencias Básicas*, 15(2 (2019)), 7-15.
- Saletnik, B., Saletnik, A., Słysz, E., Zaguła, G., Bajcar, M., Puchalska-Sarna, A., & Puchalski, C. (2022). The static magnetic field regulates the structure, biochemical activity, and gene expression of plants. *Molecules*, 27(18), 5823.
- Shimazaki, F., & Shizuoka, H. (1986). Verfahren zur Herstellung von Magnitisiertem. Patente P36138916.

- Shokrollahi, S., Ghanati, F., Sajedi, R. H., & Sharifi, M. (2018). Possible role of iron containing proteins in physiological responses of soybean to static magnetic field. *Journal of plant physiology*, 226, 163–171. <https://doi.org/10.1016/j.jplph.2018.04.018>
- Skomski, R. y Coey, J. (1998). Permanent Magnetism, Institute of Physics, Bristol.
- Smith, S., McLeod, B., y Liboff, A. (1995). Testing the ion cyclotron resonance theory of electromagnetic field interaction with odd and even harmonic tuning for cations. *Bioelectrochemistry and bioenergetics*, 38(1), 161-167.
- Torres, J., Muñoz, B., y Ramirez, W. (2007). Caracterización y calibración automática de bobinas de Helmholtz en DC. *Scientia et technica*, 13(34), 535-539.
- Vashisth, A., & Nagarajan, S. (2008). Exposure of seeds to static magnetic field enhances germination and early growth characteristics in chickpea (*Cicer arietinum* L.). *Bioelectromagnetics: Journal of the Bioelectromagnetics Society, The Society for Physical Regulation in Biology and Medicine, The European Bioelectromagnetics Association*, 29(7), 571-578.
- Vashisth, A., & Nagarajan, S. (2010). Effect on germination and early growth characteristics in sunflower (*Helianthus annuus*) seeds exposed to static magnetic field. *Journal of plant physiology*, 167(2), 149-156.
- Woolhouse, H., Bewley, J., y Black, M. (1980). Physiology and Biochemistry of Seeds in Relation to Germination. Vol. 1: Development, Germination and Growth. *Journal of Ecology*, 68(1), 315. <https://doi.org/10.2307/2259257>.
- Young, H. y Freedman, R. (2009). Física Universitaria, Decimosegunda Edición, vol. Volumen, 2, 780.
- Young, H. y Freedman, R. (2016). Física Universitaria (Vol. 2) (13ª ed.). Pearson Educación.

Impact of physical exercise on irisin expression in the brain of irradiated mice: effects on behavior and memory

Teixeira, L.P.M^{1*}, Ferreira, J.B.F¹, Saraiva, H.G¹, Moreira, M.E¹, Silva, I.S, Silva, J.F.R², Andrade, C.B.V², Nagagata, B.A¹, Borges, M.M.S¹, Ferreira-Machado, S.C¹

¹ Rio de Janeiro State University / Roberto Alcântara Gomes Institute of Biology / Department of Radiological Sciences, Rio de Janeiro, RJ, 20550-090, Brazil

² Rio de Janeiro State University / Roberto Alcântara Gomes Institute of Biology / Department of Histology and Embryology, Rio de Janeiro, RJ, 20550-090, Brazil

* lulupenedo@gmail.com

Abstract

Brain radiotherapy can induce cognitive deficits through neuroinflammation, capillary rarefaction, and neuronal apoptosis. Physical exercise, via irisin release, acts as a neuroprotective intervention. This study assessed the effects of exercise on behavior and hippocampal expression of irisin, brain-derived neurotrophic factor (BDNF), tumor necrosis factor- α (TNF- α), and synaptophysin in irradiated mice (10 Gy). After 90 days of treadmill training, memory tests, ELISA, and immunohistochemistry were performed. Exercise attenuated behavioral deficits, increased irisin, BDNF, and synaptophysin, and decreased TNF- α , highlighting its neuroprotective potential against adverse effects of brain irradiation.

Keywords: Hippocampus; Irisin; Radiation; Exercise; TNF- α

1 Introduction

Radiotherapy remains a key strategy in treating tumors of the brain, head, and neck, including brain metastases. While it improves survival, whole-brain radiation (WBR), one of the most used techniques, often causes collateral damage to healthy tissue, leading to long-term cognitive impairments and neurological dysfunctions (Witzmann et al., 2021). Neuroinflammation is central to radiation-induced damage, as ionizing radiation activates glial cells and promotes the release of pro-inflammatory mediators, including tumor necrosis factor- α (TNF- α). Elevated TNF- α levels contribute to blood–brain barrier disruption, oxidative stress, and neuronal apoptosis (Wang et al., 2024).

Conversely, regular physical exercise exerts antioxidant, anti-inflammatory, and neuroprotective effects. One of the molecules associated with these effects is irisin, a peptide hormone (myokine) produced by the proteolytic cleavage of fibronectin type III domain-containing protein 5 (FNDC5), a transmembrane protein highly expressed in skeletal muscle. Once released into the circulation, irisin can cross the blood–brain barrier and modulate neuroinflammatory pathways, as well as promote neurogenesis (Askari et al., 2018; Pignataro et al., 2021). In the present study, we focused on the measurement of endogenous irisin produced by the animals, rather than any exogenous

administration, in order to evaluate the natural response to exercise and irradiation. Irisin is also linked to increased brain-derived neurotrophic factor (BDNF) expression, which plays a critical role in synaptic plasticity, learning, and memory (Miranda et al., 2019). Synaptophysin, a synaptic vesicle protein, serves as a marker of synaptic integrity and is enhanced by exercise-induced neuroplasticity (Zheng et al., 2021). Understanding whether the induced increase in endogenous irisin expression (a product of FND5 protein cleavage) is associated with elevated BDNF, synaptophysin, and TNF- α in the hippocampus of irradiated animals subjected to physical exercise is essential to assess the therapeutic potential of exercise in mitigating radiation-induced brain injury.

2 Materials and Methods

2.1 Animals and ethics statement

A total of 60 male Swiss Webster mice (*Mus musculus*), aged between 70 and 90 days and weighing between 30 and 45 grams, were housed in the animal facility of the Department of Radiological Sciences. The animals were maintained under controlled environmental conditions, including a 12-hour light/dark cycle (lights on at 6:00 AM) and regulated temperature. Mice were housed in appropriate cages containing wood shavings, with food and water provided *ad libitum*. This study was conducted in accordance with ethical guidelines and approved by the Animal Use Ethics Committee (CEUA) of the Roberto Alcântara Gomes Institute of Biology (protocol no. 003/2022).

2.2 Experimental design

The animals were divided into four groups, with $n = 15$ per group: sedentary control (SC group); sedentary irradiated (SI group); exercise control (EC group); and exercise irradiated (EI group).

2.3 Dosimetry

The absorbed dose to water (Dw) was obtained through chemical dosimetry using the Fricke dosimeter, which is based on the oxidation of ferrous ions (Fe^{2+}) to ferric ions (Fe^{3+}) (Pickler et al., 2021). The standard solution was irradiated under conditions equivalent to those used for the animals to obtain the absorbed dose to water in a similar condition that the animals were irradiated. The analysis is performed in a UV-Vis spectrophotometry at 304 nm. The absorbed dose calculated was $10,0 \pm 0,2$ Gy.

2.4 Irradiation Procedure

X-ray photon brain irradiation for the SI and EI groups was conducted using the Small Animal Radiation Research Platform (SARRP 200). Irradiation parameters were set to 220 kV, 13 mA, with a 0.15 mm copper filter, delivering a single dose of 10 Gy. A collimator defined a 10 mm $\times$ 10 mm irradiation field, encompassing the entire cranial region. Individual mice from the male SI and EI groups were anesthetized intraperitoneally with a ketamine (100 mg/kg) and xylazine (10 mg/kg) cocktail, placed in the prone position, and subjected to irradiation. Control groups (SC and EC) underwent anesthesia and were transported to the irradiation facility but were not exposed to radiation.

2.5 Maximal Exercise Test

Twenty-four hours prior to the initiation of aerobic exercise sessions, the EC and EI groups underwent a maximal exercise test (MET) to determine the appropriate training speed, corresponding to 60% of the maximal velocity achieved during the aerobic exercise protocol. All animal groups reached an average maximal speed of 28 m/min. To maintain moderate exercise intensity (60% of MET), a training speed of 17 m/min was established. No significant differences were observed in maximal speed between the EC and EI groups.

2.6 Treadmill exercise

The groups subjected to physical exercise underwent a protocol using an automated treadmill (Bonther V.1000), characterized as forced exercise. Animals were exercised for a period of 3 months, always at the same time of day, between 2:00 PM and 5:00 PM, five days per week, for 30 minutes per day, at a speed corresponding to 60% of their maximal exercise test.

2.7 Object location memory task

Ninety-four days after irradiation, each animal was individually placed in a corner of a square acrylic arena (40 cm X 40 cm) (Figure 1A), with the base marked into 16 squares measuring 10 cm each, containing two identical objects (Figure 1B). The animals remained in the arena for five minutes before being returned to their home cages. After one hour, they were reintroduced to the arena, where one object (object 1) remained in the same position while the other (object 2) was relocated. The mice were allowed to explore both objects for five minutes. Behavioral interactions were later analyzed using the Anymaze software, quantifying the percentage of time spent exploring the relocated object (object 2).

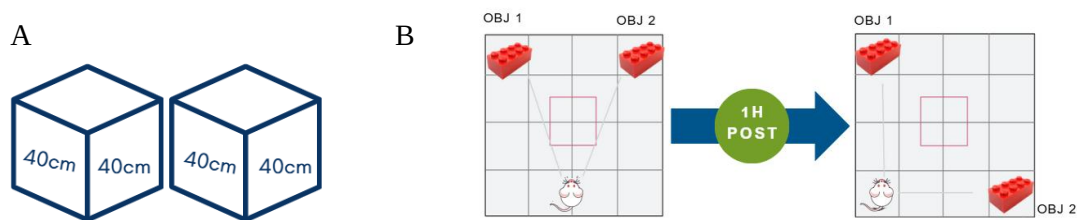


Figure 1. Object location memory task. (A) Two acrylic box arenas measuring 40 × 40 cm. (B) Test: two identical objects in parallel fixed positions (session 1). One hour later: one object was relocated to assess short-term spatial memory.

2.8 Immunofluorescence assay

Twenty-four hours after the behavioral test, six animals from each group were anesthetized via intraperitoneal injection with a ketamine (100 mg/kg) and xylazine (10 mg/kg) mixture for brain fixation by transcardiac perfusion with 0.9% saline for 5 minutes, followed by 4% paraformaldehyde for 15 minutes. The brains were then extracted and cryoprotected by immersion in increasing concentrations of sucrose solutions (10%, 20%, and 30%) in PBS buffer. The brains, embedded in OCT compound, were stored at -20°C until processing. Samples were sectioned on a cryostat at -30°C (LEICA® CM 1850),

and the sections mounted on slides for immunolabeling procedures targeting FNDC5 (fibronectin type III domain-containing protein 5) (BS8486R - Invitrogen, 1:500), synaptophysin (MA514532 - Invitrogen, 1:500), and the pro-inflammatory cytokine TNF- α (PA5-46945 - Invitrogen, MA, 1:1000) in the hippocampal regions (DG and CA1). Images were captured using a fluorescence microscope (EVOS) at 40x magnification with a 100 μ m scale and analyzed using Image Pro-Plus® software.

2.8 ELISA

Eight animals from each group underwent the same transcardiac perfusion protocol used in the immunofluorescence assay. After perfusion with 0.9% saline solution, the brain was removed, and the hippocampus was dissected for subsequent analyses. The hippocampal samples were stored at -80°C until analysis by ELISA, performed according to the manufacturers' instructions using the Elabscience® Mouse BDNF (Brain-Derived Neurotrophic Factor) ELISA Kit and the Irisin (Mouse) ELISA Kit (Catalog No. EK-067-29) with a detection range of 0.1–1000 ng/mL.

2.9 Statistical analyses

Data were expressed as means $\pm$ standard deviations and tested for normality using the Shapiro-Wilk test, appropriate for small sample sizes. Subsequently, a two-way analysis of variance (two-way ANOVA) was performed, followed by Tukey's post hoc test for multiple comparisons. A p-value < 0.05 was considered statistically significant. Analyses were conducted using GraphPad Prism software, version 10 for Windows (GraphPad Software, San Diego, CA, USA).

3 Results

The irradiation was well tolerated, with no signs of distress or suffering observed. All irradiated animals and/or those subjected to exercise survived the study period. Some irradiated animals exhibited mild skin reactions and hair loss, without symptoms such as nausea or ataxia. Body weight gain was similar across all groups, with no statistically significant differences ($p > 0.05$, data not shown), indicating that neither irradiation nor exercise affected the growth of the mice.

3.1 Object location memory task

The analysis of the location recognition test in male mice revealed significant differences among the experimental groups: SC, SI, EC, and EI. As shown in Figure 2, the SI group exhibited a 31% reduction in the percentage of exploration of object 2 compared to the SC group ($p < 0.001$), indicating the presence of cognitive deficit. Conversely, the EI group showed a significant increase compared to the SI group (+43%; $p < 0.01$), suggesting a potential neuroprotective effect of physical exercise under irradiation conditions. The results from the EI group were comparable to those observed in the SC and EC groups.

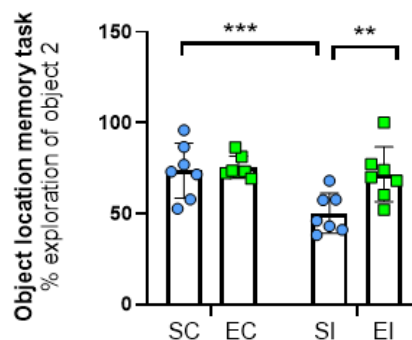


Figure 2. Effect of physical exercise on behavior of irradiated male mice. Percentage of exploration of object 2, an indicator of short-term spatial memory. Groups: SC, Sedentary Control; SI, Sedentary Irradiated; EC, Exercise Control; and EI, Exercise Irradiated. Statistical comparisons: (**) $p < 0.01$; (***) $p < 0.001$, $n = 7$ per group.

3.2 Immunofluorescence assay

Immunofluorescence analysis in the GD and CA1 brain regions of male mice revealed significant differences in immunodensity levels among the experimental groups. In the GD region, the FNDC5 marker (Fig. 3A) showed no differences between the SC, SI, and EC groups. However, the EI group was the only one to exhibit a significant increase compared to the others. In the CA1 region (Fig. 3D), no differences were observed between the SC and SI groups, suggesting that irradiation alone does not affect the expression of this marker. Conversely, a significant increase in FNDC5 expression was observed in the exercise groups (EC and EI), indicating that exercise induces a 486% upregulation of this protein above the basal levels seen in the SC group. Regarding synaptophysin, increased expression was noted in both GD and CA1 regions in animals subjected to exercise (EC and EI; Fig. 3B and 3E). These findings indicate that physical exercise promotes synaptophysin expression in both analyzed regions. In contrast, radiation exposure did not significantly affect this marker, as no differences were detected between the SC and SI groups. Concerning the TNF- α marker (Fig. 3C), a significant increase in expression levels was observed in the SI group compared to the SC group (+2736%, $p < 0.001$) in the GD region, suggesting an exacerbated inflammatory response induced by irradiation. Conversely, the EI group showed a significant reduction in TNF- α levels compared to the SI group ($p < 0.01$), indicating an anti-inflammatory effect mediated by physical exercise. TNF- α expression levels in the EI group were comparable to those in the SC and EC groups. In the CA1 region (Fig. 3D), the immunolabeling profile of TNF- α followed a similar pattern to that observed in the GD region.

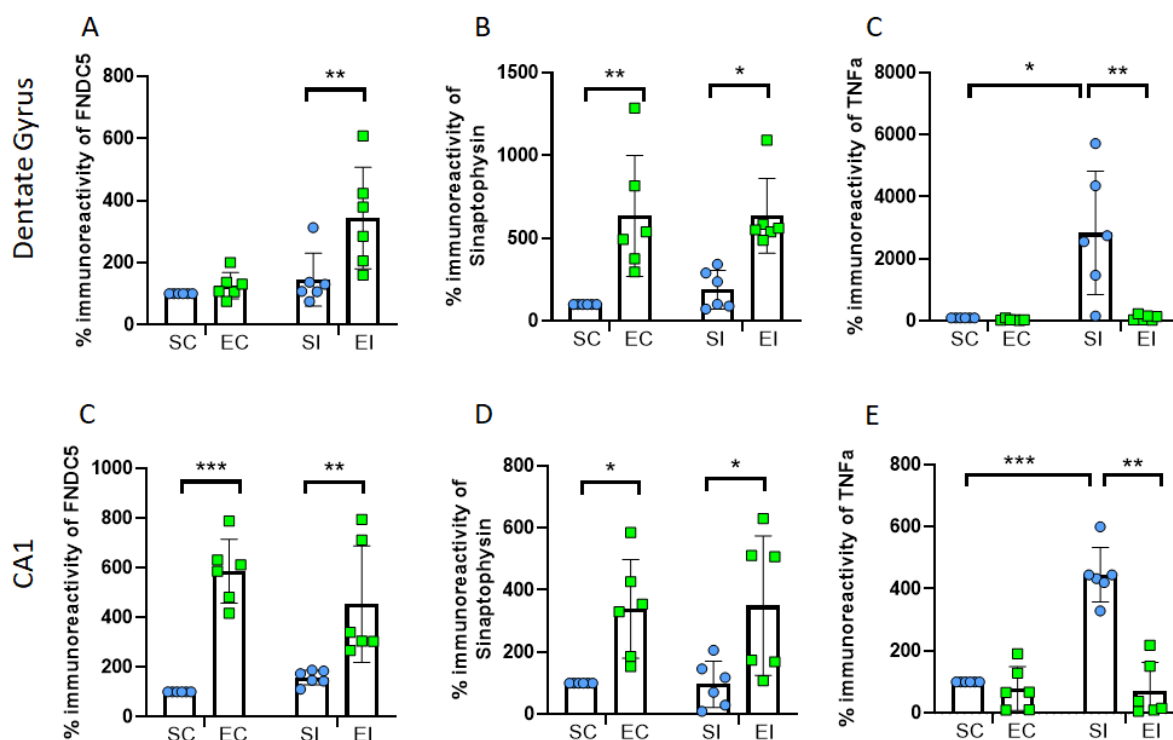


Figure 3. Immunoreactivity (%) of FNDC5, Synaptophysin, and TNF- α markers in the GD and CA1 regions of male mice. The analyzed groups were SC, SI, EC, and EI, with statistical differences indicated by (*) $p < 0.05$, (**) $p < 0.01$, and (***) $p < 0.001$. Data are expressed as mean $\pm$ SD.

3.2 ELISA

ELISA analysis evaluated the levels of Irisin (Fig. 4A) and BDNF (Fig. 4B) in the hippocampus of mice from the experimental groups SC, SI, EC, and EI. Results indicated that Irisin levels were significantly higher in the EI group compared to the SI group ($p < 0.001$), demonstrating a robust neuroprotective effect of physical exercise under irradiation conditions. Irisin levels in the EI group were also significantly elevated compared to the EC group ($p < 0.05$), suggesting an additional increase in this marker's concentration in response to exercise combined with radiation exposure. The SC and SI groups exhibited similar Irisin levels, indicating that irradiation alone does not directly impact this marker's hippocampal expression in the absence of exercise. Regarding BDNF, levels were significantly higher in the EI group compared to the SI group ($p < 0.001$), further supporting the neuroprotective role of exercise under irradiation. BDNF levels in the EI group were also significantly greater than those in the EC group ($p < 0.01$), indicating an additional upregulation in response to the combination of exercise and radiation. The SC and SI groups showed comparable BDNF levels, suggesting that irradiation without exercise does not significantly affect hippocampal BDNF expression.

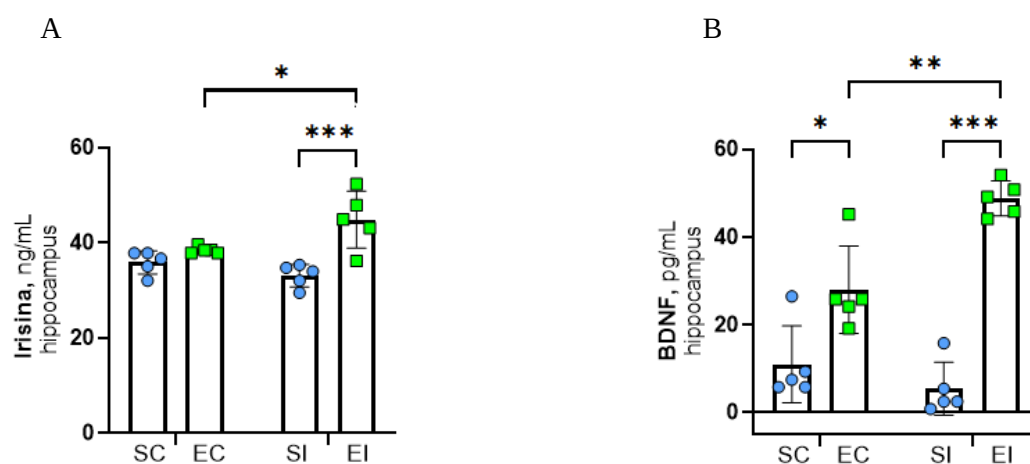


Figure 4. Levels of Irisin (ng/mL) (A) and BDNF (pg/mL) (B) in the hippocampus of male mice. Experimental groups analyzed were SC, SI, EC, and EI. Statistical differences are indicated by (*) $p < 0.05$, (**) $p < 0.01$, and (***) $p < 0.001$. Data are presented as mean $\pm$ SD.

4 Conclusion

The results of this study demonstrate that brain irradiation induces cognitive deficits and a significant increase in TNF- α expression, evidencing a neuroinflammatory profile associated with ionizing radiation exposure. However, the findings indicate that regular physical exercise can attenuate these deleterious effects by reducing TNF- α expression and preventing cognitive performance impairment. Furthermore, exercise increased synaptophysin levels in the dentate gyrus (DG) and CA1 regions of the hippocampus, suggesting a potential enhancement of synaptic plasticity under adverse conditions.

Additionally, the significant elevation of irisin and BDNF in exercised groups under irradiation conditions reinforces the role of these molecules as key mediators of exercise-induced neuroprotective effects. The upregulation of FNDC5, together with increased synaptophysin expression and modulation of TNF- α , highlights a relevant molecular axis involved in neural tissue protection against radiation-induced damage. Collectively, these data support physical exercise as an effective non-pharmacological strategy to prevent neuroinflammatory and neurodegenerative damage associated with radiotherapy, paving the way for future integrative therapeutic approaches in clinical settings.

Acknowledgments

Special thanks are extended to Professor Verônica Morandi, from the Laboratory of Endothelial Cell Biology and Angiogenesis (LabAngio), as well as to the laboratory technicians Francisco Vardiero, Marcos Temperini e Marcella Tavares. This work was made possible by the support of the funding agencies CAPES, CNPq, and FAPERJ.

References

- Askari, H., et al. (2018). A glance at the therapeutic potential of irisin against diseases involving inflammation, oxidative stress, and apoptosis: An introductory review. *Peptides*, 110, 1–10. Elsevier.
- Miranda, M., et al. (2019). Brain-derived neurotrophic factor: A key molecule for memory in the healthy and the pathological brain. *Frontiers in Cellular Neuroscience*.
- Pickler, A., Mantuano, A., Salata, C., Mota, C. L., David, M. G., de Amorim, G. J., Magalhães, L. A. G., & deAlmeida, C. E. (2021). *Analysis of the Fricke solution-PMMA interaction and potential effect on the accuracy of dose measurements*. *Radiation Measurements*, 140, 106513.
- Pignataro, P., et al. (2021). FNDC5/irisin system in neuroinflammation and neurodegenerative diseases: Update and novel perspective. *International Journal of Molecular Sciences*, 22(4), 1605.
- Witzmann, K., Raskche, F., & Toost, E. G. C. (2021). MR image changes of normal-appearing brain tissue after radiotherapy. *Cancers*, 13, 1573.
- Zheng, H., Kang, J., Chen, X., Yang, M., & Zhang, Z. (2021). Exercise-induced neuroplasticity: Synaptophysin expression and cognitive improvement in animal models. *Neurobiology of Learning and Memory*, 178, 107374.

Magnetic stimulation as a promoting factor for germination and development in *Guazuma ulmifolia* Seeds

R. K. Delgado Velázquez¹, L. Herrera Isidró¹, A. Hernández Rayas², T. Córdova Fraga², D. Sánchez Guzmán¹, M. A. Sosa Aquino², J. C. Martínez Espinosa^{1*},

¹ Instituto Politécnico Nacional - UPIIG, Ave. Mineral de Valenciana N 200, Fraccionamiento Industrial Puerto Interior, 36275 Silao de la Victoria, GTO, Mx.

² División de Ciencias e Ingeniería, Universidad de Guanajuato campus León, Loma del Bosque N 103, Lomas del campestre, 37150 León, GTO, Mx.

* jcmartineze@ipn.mx

Abstract

Static magnetic fields influence physiological processes in plants, such as germination and growth. previous studies indicate that their application can improve water absorption, ion transport, and enzyme activity in seeds. However, little is known about their effects on *Guazuma ulmifolia*, a species of ecological and medicinal interest. This study evaluated the impact of these fields on the germination and early development of *Guazuma ulmifolia*. in MS medium. Twelve dissected seeds were used, evenly distributed in three Petri dishes. One was exposed to a static magnetic field of 100 mT for 14 days, another to 300 mT, and the third was kept as a control without magnetic treatment. The results showed that seeds treated with 300 mT reached 100% germination, while the control achieved only 14.29%. Additionally, seedlings in this group had an average length of 3.5 ± 0.35 cm, compared to 2.1 ± 0.13 cm in the control group. Seeds treated with 100 mT also showed favorable growth, with an average length of 3.28 ± 0.61 cm. These findings indicate that exposure to magnetic fields, especially at 300 mT, significantly improves both germination and early growth of *Guazuma ulmifolia*. It is concluded that this treatment could be a useful tool to optimize propagation of this species. Further studies are recommended to better understand the physiological mechanisms involved and to define the optimal application conditions

Keywords: Magnetism, germination, *Guazuma ulmifolia*, early development

1 Introduction

Guazuma ulmifolia L. It is a plant belonging to the Malvaceae family (Martínez, 2018; Villa et al., 2009). In Mexico, it is found in tropical areas, especially in Veracruz, where this species is quite abundant (Román, 2001, cited by Villa et al., 2009). The guásima (*Guazuma ulmifolia*) bears fruits in indehiscent capsules, dark in color when ripe and with a warty surface (Martínez, 2018). This species has numerous uses, among which Villa et al. (2009) mention that guásima is used for firewood and charcoal; its wood is used to make furniture and packing boxes; the shoots are used as stakes or

to support agricultural crops; while the leaves and fruits are edible for livestock. The fruits are also used to treat inflammation, skin rashes, diarrhea, and other conditions.

Dormancy is a common phenomenon, occurring primarily in seeds of certain vegetables and forage crops, some fruit trees, and ornamental trees. These seeds fail to germinate after harvest due to internal mechanisms, which may be physical or physiological, that block germination (Mérola & Díaz, 2012). In the case of *Guazuma ulmifolia*, its fruits are rounded capsules with a large number of physically dormant seeds (Muñoz et al., 2004). Furthermore, the seeds are highly impermeable to water, therefore, pre-germination treatments are required for the embryo to hydrate and germinate (Muñoz et al., 2012). Despite its ecological and economic importance, the propagation of *Guazuma ulmifolia* faces challenges due to the dormancy of its seeds, which reduces its germination rate. Therefore, in the search for alternative propagation strategies, one possible option is *in vitro* cultivation. *In vitro* cultivation of plants consists of reproducing numerous specimens from small parts of tissue from a mother plant. Each fragment forms a plant identical to the mother plant, obtaining homogeneous plants in large quantities, free of pathogens, and at any time of year (Amaya & Torres, 2012).

Magnetic fields affect all living beings, which is why they are the subject of worldwide research. The magnetic field that protects the Earth from solar radiation and winds formed 3.45 billion years ago, promoting the emergence of biological activity on the planet. The intensity of the Earth's magnetic field ranges between 0.4 and 0.6 Gauss, depending on the latitude (Tarduno et al., 2010). A magnetic field is a vector field, that is, a vector quantity associated with each point in space where forces are exerted on magnetic materials or moving electrical charges. This field is generated by magnets or electrical currents and is represented by field lines that indicate its direction and intensity (Arias, 2001). Research has been conducted to determine whether a magnetic field changes the rate of germination and development of plants; cases of both stimulation and inhibition of growth have been found when magnetic fields are applied to seeds of various families. Due to the wide range of conditions under which the magnetic field can be applied, depending on whether it is stationary or variable, its intensity, exposure time, or the plant species treated (Maqueda, 2005). In general, studies on the use of low-intensity magnetic fields yield mostly positive results in terms of germination, growth, and production yield in different crops. Fields between 50–200 mT are usually beneficial, but fields above 350–500 mT can be inhibited, depending on the exposure time (Carbonell et al., 2017).

In this research study, the impact of a permanent magnetic field on the germination and early development of *Guazuma ulmifolia* seeds in MS medium was evaluated. Magnetic field intensities of 100 mT and 300 mT were used for 14 continuous days. Cotyledon measurements were obtained and compared with the control (without magnetic stimulation).

2 Materials, methods and procedure

Collection and processing of Guazuma ulmifolia seeds

Guazuma ulmifolia fruits were obtained from Veracruz, de-seeded, and surface-sterilized. The disinfection protocol involved sequential treatments: a 30-minute wash with 2% Extran, a 30-second rinse in 70% ethanol, a 10-minute immersion in 1% sodium hypochlorite, followed by rinsing with sterile water. The sterilized seeds were then germinated in Petri dishes containing Murashige and Skoog (MS) medium supplemented with 30 g/L sucrose and 3 g/L Gelrite. The medium was adjusted

to a pH of 5.7 and maintained at 25 °C. The outline of a permanent magnet was marked on each dish: three whole seeds were positioned just above the magnet's perimeter, three whole seeds just below it, and one seed bisected in half was placed at the center with a slight gap between the halves (Figure 1).

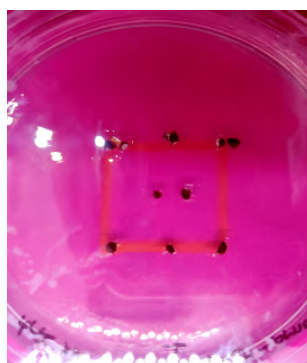


Figure 1. Distribution of the dissected and whole seeds that were placed on the permanent magnet

Magnetic stimulation setup

To magnetize the seeds, a 2.5×2.5 cm neodymium magnet was used. Two magnetic field intensities were applied: approximately 300 mT (seeds located ~ 0.3 cm from the magnet surface), and approximately 100 mT (seeds positioned ~ 1.8 cm above the magnet surface). The seeds were continuously exposed to the magnetic fields for two weeks, during which plant responses were recorded. Figure 2 illustrates the placement of the Petri dishes: the first tier receives ~ 300 mT, the second tier ~ 100 mT, and dishes not placed on the magnets served as the non-magnetized control. The experiment was performed in duplicate.

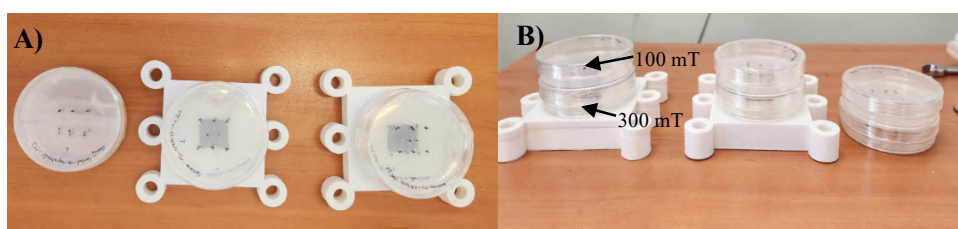


Figure 2. Distribution of the Petri dishes that were magnetized on the permanent magnet and the control boxes, where A) top view and B) side view.

Measurement of Guazuma ulmifolia cotyledons

The length of seedlings exposed to magnetic fields of 300 mT and 100 mT, as well as non-magnetized controls, was measured using the software ImageJ. This program enabled the establishment of a scale based on a reference object of known length within each photograph. Using this calibrated scale, the total length of each seedling from the radicle to the cotyledons was accurately traced and recorded to facilitate comparative analysis of growth differences among the treatment groups.

3 Results and discussions

To visualize the influence of magnetism on the seeds of *Guazuma ulmifolia*, the percentage of seed germination was observed daily, where on the second day 100% of the seeds magnetized at 300 mT and 62.5% of the seeds magnetized at 100 mT had already germinated, while only 14.29% of the seeds that had not been magnetized had germinated, in figure 3, you can see the percentage of germination through the days for each intensity and the control. It is worth mentioning that only dissected seeds were considered, because the whole ones did not germinate.

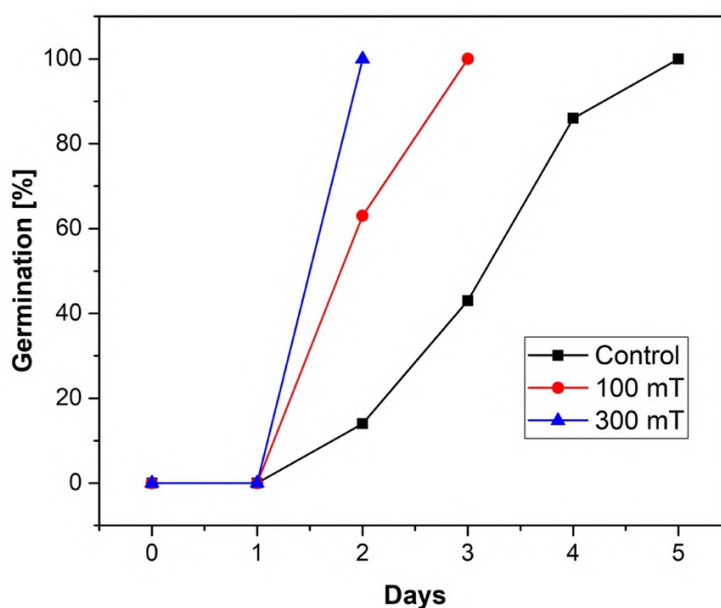


Figure 3. Germination percentage of magnetized and non-magnetized *Guazuma ulmifolia* seeds over the days.

The seeds were exposed to magnetic fields for two weeks, during which observable changes were detected in the seedlings. Specifically, magnetized seedlings exhibited greater growth compared to non-magnetized controls, as illustrated in Figure 4.

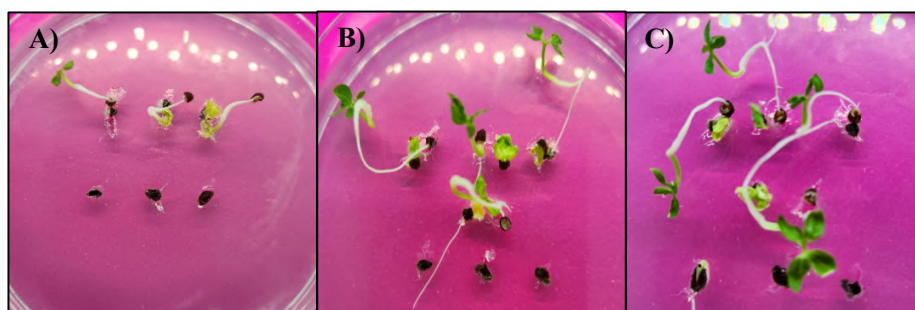


Figure 4. *Guazuma ulmifolia* seedlings. A) non-magnetized seedlings, B) magnetized seedlings at 100 mT, and C) magnetized seedlings at 300 mT.

To quantify differences in seedling size, the average length from the radicle to the cotyledons was measured using ImageJ software. As shown in Figure 5, seedlings exposed to a 300 mT magnetic field exhibited a significantly greater average length (3.5 ± 0.55 cm), while those exposed to 100 mT measured 3.24 ± 0.79 cm, compared to the non-magnetized control group (2.1 ± 0.13 cm).

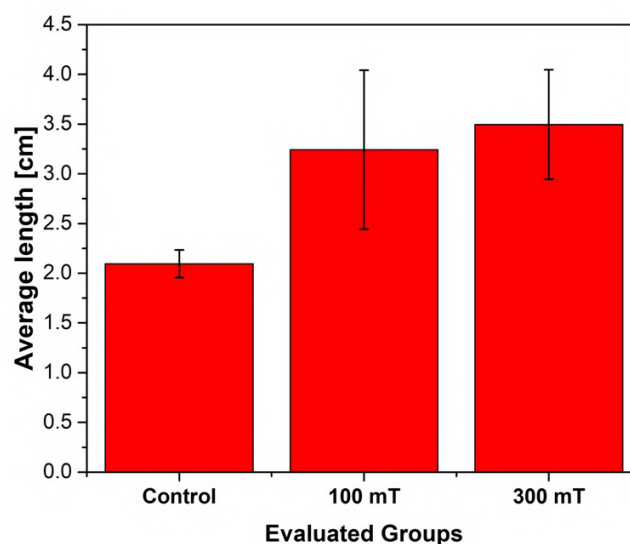


Figure 5. Average length of magnetized and non-magnetized *Guazuma ulmifolia* seedlings.

Whole seeds of *Guazuma ulmifolia* failed to germinate due to their thick, impermeable seed coat, which restricts the passage of water and oxygen to the embryo, thereby inhibiting the initiation of germination even under favorable conditions (Muñoz et al., 2012). In contrast, dissected seeds successfully germinated, as mechanical scarification removed this physical barrier, allowing the embryo to access water and oxygen and thus triggering the germination process. In a study by Ferroni et al. (2023), maize seeds were exposed to static magnetic fields ranging from 50 to 350 mT for one hour. The results showed that all intensities, except 350 mT, significantly stimulated seedling growth. Notably, the 150 mT treatment resulted in a 108.9% increase in total seedling length by the tenth day post-germination compared to the control. Similarly, in *Guazuma ulmifolia*, seeds treated with a 300 mT magnetic field exhibited a 40% increase in total seedling length 15 days after germination relative to the control, indicating a comparable, albeit less pronounced, stimulatory effect.

Ćalić et al. (2023) reported that exposure of flax seeds, previously stored under unfavorable conditions, to static magnetic fields ranging from 50 to 350 mT resulted in 100% germination and significant increases in vigor indices, seedling length, and catalase activity compared to the control group. These findings are consistent with the present results, where higher magnetic field intensity led to 100% germination in a shorter time than the control, along with increased seedling length. Faster germination in magnetized seeds may be attributed to the earlier or more intense activation of enzymes such as α -amylase. This enzyme facilitates the rapid conversion of stored starch into simple sugars, providing energy for embryonic development and accelerating seedling growth (Vashisth & Nagarajan, 2010). Additionally, since water absorption is the initial step in germination, increased membrane permeability induced by magnetic fields may enhance water uptake, thereby triggering internal biochemical processes earlier (Shabrangi & Majd, 2009). Magnetic fields also influence ion

transport (e.g., Na^+ , K^+ , Ca^{2+}) by altering the membrane's electrical potential. This change affects the electrochemical gradient and activates specific ion channels and transporters, resulting in more efficient ion flux and, consequently, greater water absorption (Vashisth & Nagarajan, 2010; Shabrangi & Majd, 2009).

4 Conclusions

Magnetic field stimulation accelerates seed germination and seedling growth. The percentage reached 100% germination after 2 days in seeds magnetized at 300 mT and an average length of 3.5 cm after 15 days. Control seeds germinated completely after 5 days and reached an average length of 2.1 cm after 15 days.

Acknowledgements

We thank the Secretariat for Research and Graduate Studies of the Instituto Politécnico Nacional for financial support through project SIP 20231512. We also thank the Laboratory of Medical Physics and Biological Materials at the University of Guanajuato for technical support and access to biomagnetic instrumentation.

References

- Amaya, J. E. R., & Torres, C. O. P. (2012). Advances in *in vitro* micropropagation. National Open and Distance Learning University (UNAD), Bogotá.
- Arias, A. G. (2001). What is magnetism? University of Salamanca.
- Ćalić, D., Ristić-Djurović, J. L., Ćirković, S., Milojević, J., Belić, M., Stanišić, M., & Zdravković-Korać, S. (2023). Overcoming Low Germination and Low Quality of Flax Seeds (*Linum usitatissimum* L.) in Unfavorable Storage Using Static Magnetic Fields. *Agriculture*, 13(11). <https://doi.org/10.3390/agriculture13112120>.
- Carbonell, M. V., Flórez, M., Martínez, E., & Álvarez, J. (2017). Contributions to the magnetic field: History and influence on biological systems. *Intrópica*, 12(2), 143-159.
- Ferroni, L. M., Dolz, M. I., Guerra, M. F., & Makinistian, L. (2023). Static magnetic field stimulates growth of maize seeds. 7-10. <https://doi.org/10.48550/arXiv.2303.00512>
- Maqueda, M. R. (2005). Effect of exposure to 125 mT and 250 mT magnetic fields on germination and early growth stages of pea. School of Agricultural Engineering, Polytechnic University of Madrid, Spain.
- Martínez, J. (2018). Seed hardening to improve germination of *Guazuma ulmifolia* (Malvaceae) under stress. Comparison among pioneer tropical trees, University of Havana, Faculty of Biology.

Mérola, R., & Díaz, S. (2012). Methods, techniques, and treatments to inhibit dormancy in forage plant seeds.

Muñoz García, B. C., Sánchez Rendón, J. A., Montejo Valdés, L. A., Herrera Oliver, P., & Gamboa Valerino, A. (2012). Technical guide for the reproduction of pioneer tree species (Beatriz Rodriguez Elias).

Muñoz, B., Sánchez, J. A., & Almaguer, W. (2004). Germination, dormancy, and potential longevity of *Guazuma ulmifolia* seeds. ResearchGate, 27, 25–33.

Shabrangi, A., & Majd. (2009). Effect of Magnetic Fields on Growth and Antioxidant Systems in Agricultural Plants Effect of Magnetic Fields on Growth and Antioxidant Systems in Agricultural Plants. ResearchGate, 1142-1147.

Tarduno, J. A., Cottrell, R. D., Watkeys, M. K., Hofmann, A., Doubrovine, P. V., Mamajek, E. E., Liu, D., Sibeck, D. G., Neukirch, L. P., & Usui, Y. (2010). Geodynamo, solar wind, and magnetopause 3.4 to 3.45 billion years ago. Science, 3(27). 1238-1240.

Vashisth, A., & Nagarajan, S. (2010). Effect on germination and early growth characteristics in sunflower (*Helianthus annuus*) seeds exposed to static magnetic field. Journal of Plant Physiology, 167(2), 149-156. <https://doi.org/10.1016/j.jplph.2009.08.011>.

Villa-Herrera, A., Nava-Tablada, M. E., López-Ortiz, S., Vargas-López, S., Ortega-Jimenez, E., & López, F.-G. (2009). Use of guácimo (*Guazuma ulmifolia* Lam.) as a forage source in extensive cattle ranching in the Mexican tropics. 10(2), 253-261.

Radiometric analysis of sediments and water from the Lagoons of East Fluminense, Rio de Janeiro.

Y. S. Pereira^{1,2*}, L. M. Freire¹, A. Kelecom², I. B. M. Ferreira², G. N. P. de Oliveira², M. D. Barreto², C. O. P. Orejuela⁴, A. X. Silva^{3,4}, J. L. Thalhofer^{2,3}, M. R. Costa¹

¹ Fluminense Federal University, Laboratory of Nekton Biology and Fisheries Ecology, 24210-201, Niterói, Brazil

² Fluminense Federal University, Radiobiology and Radiometry Laboratory, 24210-201, Niterói, Brazil

³ Federal University of Rio de Janeiro, Nuclear Engineering Program, 21941-914, Rio de Janeiro, Brazil

⁴ Federal University of Rio de Janeiro, Laboratory of Environmental Analysis and Computational Simulation, 21941-598, Rio de Janeiro, Brazil.

* ysarmento@id.uff.br

Abstract

Radionuclides are atoms with unstable nuclei that emit radiation in the form of particles and electromagnetic waves. Naturally, radionuclides are found in the environment, especially ⁴⁰K and elements from the ²³⁸U and ²³²Th series. Artificial radionuclides are introduced into the environment mainly through human activities. Radionuclides reach lagoons through the process of leaching. In an aquatic environment, besides the organisms consumed by humans, the concentration of radionuclides in sediments is also an important topic, as soils tend to accumulate radionuclides from both natural processes and anthropogenic activities. Another direct pathway of exposure in these environments is the water itself, through the consumption of aquatic organisms and activities such as fishing and swimming. This study was conducted in lagoon systems on the eastern coast of the state of Rio de Janeiro, Brazil (Piratininga-Itaipu, Maricá-Guarapina, Saquarema, and Araruama). These are highly productive environments connected to the sea by tidal channels, offering a wide variety of protein sources, particularly through artisanal fishing. Water and sediment samples were collected from the lagoons during August and September 2024, processed in the laboratory, and left in secular equilibrium for thirty days. Afterward, the samples were analyzed using gamma-ray spectrometry with an HPGe detector, evaluating 13 energy peaks to identify radionuclides (²²⁶Ra, ²²⁸Ra, ²³²Th, ²³⁸U, ⁴⁰K) and calculate activity concentration (AC) in Bq·kg⁻¹. The results from sediment samples were within reference values and were compared with other studies, showing values that were either lower or higher than those reported by other authors. Among the lagoons studied in August, Saquarema showed the highest radionuclide concentrations, potentially related to pollution. In this lagoon system, Saquarema has the lowest rate of households connected to a sewage system (27.36%), suggesting that wastewater from unconnected homes may be discharged into the lagoon. Radionuclide levels are influenced by anthropogenic activities such as pollution, discharge of industrial and agricultural effluents, among others, and the lagoons are affected by their proximity to urban areas. The water samples showed results below the minimum detectable activity (MDA) or presented low levels, which is expected. The findings of this study are essential for the health of humans, the environment, and the organisms living there. Monitoring and understanding these levels are fundamental for effective environmental management and ensuring public health protection.

Keywords: Radionuclides, Environmental monitoring, Gamma spectrometry

1 Introduction

Besides aquatic organisms used for food, it is also necessary to consider the concentration of radionuclides in sediments, since soil tends to accumulate radionuclides originating from natural processes such as weathering, erosion, leaching, and surface runoff (Jibiri et al., 2023). In the case of radionuclides, it is possible to differentiate those of natural origin, known as Naturally Occurring Radioactive Materials (NORM), whose presence is related to the mineral composition of the environment, from those derived from human activities or industrial processes. NORM stands for “Naturally Occurring Radioactive Materials,” referring to radioactive materials naturally present in the environment. These materials are derived from radioactive nuclei, such as the Uranium-238 (^{238}U) and Thorium-232 (^{232}Th) series, and have not undergone any anthropogenic interaction (Lince, 2025). The most important natural radionuclides that can pose radiological protection concerns are uranium (^{238}U and ^{235}U), thorium (^{232}Th), their decay products, and potassium (^{40}K). It is important to highlight that the presence of natural radioactivity in the environment, as well as external exposure to gamma radiation, is directly related to the geological and geographical characteristics of each location, resulting in variations in levels across different regions of the world (UNSCEAR, 2000). The natural terrestrial gamma dose rate is a significant contributor to the average dose received by the population, and estimating the distribution of radiation doses is essential for assessing health risks to living organisms and serves as a reference for detecting changes in environmental radioactivity caused by human activities (Agbalagba & Onoja, 2011).

Another pathway of contact with radionuclides is through water. Water is a potential source of direct exposure, through direct consumption, ingestion of food from these environments, and through activities such as fishing and swimming (JRC, 2019). Fish and crustaceans absorb radionuclides directly from the water and through their diet, accumulating them in their tissues and transferring them along the food chains. Thus, they play an important role in the transfer of radionuclides to humans, exposing organisms to ionizing radiation. This exposure can cause adverse biological effects, impacting individuals, populations, and ecosystems (UNSCEAR, 2000; Carvalho et al., 2011; Mársico et al., 2014).

Monitoring radioactive materials is extremely important for all living organisms and for environmental protection. Since everything in the environment is interconnected, it is necessary to conduct studies that monitor radiation levels and human exposure.

2 Material and Methods

2.1 Study Area

The study was conducted in the lagoon systems of the East Fluminense coast – RJ (Piratininga-Itaipu, Maricá-Guarapina, Saquarema, and Araruama; 22°57'S - 43°03'W and 22°57'S - 42°03'W), oriented in a west-east direction (Figure 1). These environments have high productivity, provide various sources of protein through artisanal fishing, are connected to the sea by tidal channels, cover an area of about 220 km², and have an average depth between 0.7 and 3 meters (Knoppers & Kjerfve, 1999).

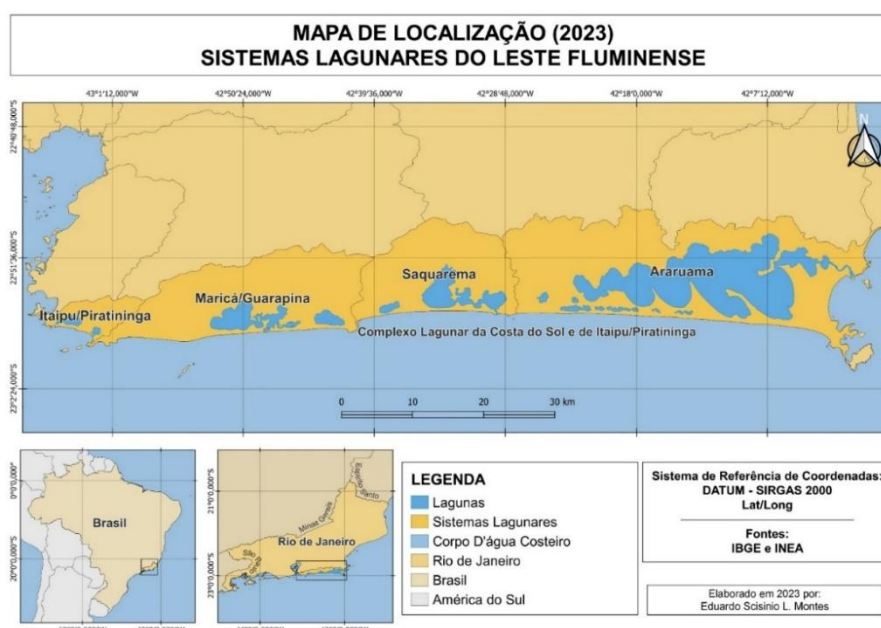


Figure 1: Location map of the Lagoons of the East Fluminense Lagoon System.

2.2 Sampling, analysis and sample preparation

The analyses and procedures were carried out through a collaborative effort involving the Nekton Biology and Fisheries Ecology Laboratory (ECOPESCA) and the Radiobiology and Radiometry Laboratory (LARARA) at Fluminense Federal University (UFF), the Environmental Analysis and Computational Simulation Laboratory (LAASC) at the Federal University of Rio de Janeiro (UFRJ), and the Nuclear Engineering Section of the Military Institute of Engineering (IME).

Water and sediment samples were collected from the lagoons, following a pattern of three sampling points each month (August and September) (Figure 2). Water samples were collected in 1 L bottles, and sediments were collected using a scoop designated exclusively for this purpose and stored in ziplock bags, with approximately 1 kg of sample. After transportation to the laboratory, the water samples were filtered to remove traces of organic matter and sediment, multiparameters were measured again (previously measured at the sampling site (Table 1)), the samples were acidified, and then stored in 500 ml containers, sealed with plastic wrap, for 30 days to reach secular equilibrium before analysis. The sediments undergo a homogenization process, after which they are stored wet in 500 ml containers and sealed until the 30-day secular equilibrium is reached.

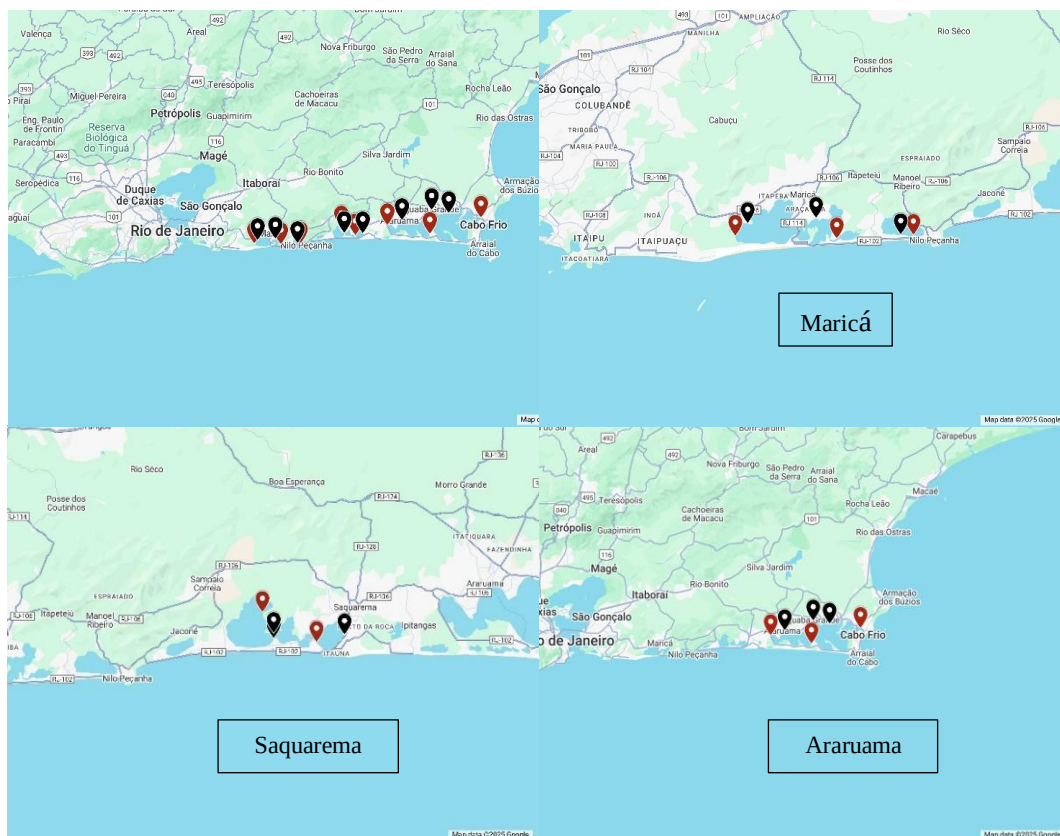


Figure 2: Sampling points in the lagoons of Maricá, Saquarema and Araruama.

Table 1: Abiotic data measured by multiparameter at lagoon collection sites.

Maricá										
Date:	Coordinate:	Temp (°C):	Depth (m):	Salinity:	TDS (mg/L):	pH:	Conductivity:	OD (mg/l):	Transparency (m):	Weather condition:
Point 1	28/08/2024 -22.952775, -42.884913	21,6	0,5	7,9	10,3	7,6	14,5	8,98	total	sunny
Point 2	28/08/2024 -22.955000, -42.785895	24,1	0,35	16,6	19,9	8,3	28,3	10,54	total	sunny
Point 3	28/08/2024 -22.951959, -42.711139	23,1	0,45	21,4	25,6	8,8	35,5	18,47	0,35	sunny
Point 1	24/09/2024 -22.941636, -42.872896	28,1	0,35	0,34 (E)	10,7	7,7	15,1	7,2	0,15	sunny
Point 2	24/09/2024 -22.937044, -42.806101	30,9	0,45	11,9	14,9	7,7	20,9	7,54	0,2	sunny
Point 3	24/09/2024 -22.951512, -42.723372	31,2	0,25	21,5	25,9	8,3	36,5	11,3	total	sunny
Saquarema										
Date:	Coordinate:	Temp (°C):	Depth (m):	Salinity:	TDS (mg/L):	pH:	Conductivity:	OD (mg/l):	Transparency (m):	Weather condition:
Point 1	30/08/2024 -22.924739, -42.505947	21,6	0,4	27,4	32,2	5,3	45,4	8,13	0,3	sunny
Point 2	30/08/2024 -22.921458, -42.547881	24,5	0,4	24,5	28,9	6,6	40,4	7,18	0,3	sunny
Point 3	30/08/2024 -22.898709, -42.558886	24,7	0,3	24,7	29,3	7,1	41,2	10,88	0,3	sunny
Point 1	25/09/2024 -22.918265, -42.478646	27,1	0,3	28,3	33,1	7,7	46,3	8,4	total	sunny
Point 2	25/09/2024 -22.921458, -42.547881	31	0,2	26,4	31,3	8	44	14,01	0,2	sunny
Point 3	25/09/2024 -22.916983, -42.547864	29,5	0,4	27,1	32	7,8	44,9	8,71	0,2	sunny with wind
Araruama										
Date:	Coordinate:	Temp (°C):	Depth (m):	Salinity:	TDS (mg/L):	pH:	Conductivity:	OD (mg/l):	Transparency (m):	Weather condition:
Point 1	29/08/2024 -22.891244, -42.386561	22,5	0,45	41,1	46,5	6,6	65,5	6,76	total	sunny
Point 2	29/08/2024 -22.920600, -42.227352	22,3	0,45	42,5	47,9	8	67,5	8,05	total	sunny
Point 3	29/08/2024 -22.866215, -42.035504	23,4	0,35	34,8	40	7,3	56,4	12,49	total	sunny
Point 1	26/09/2024 -22.874154, -42.331681	25,5	0,6	42,7	48,1	7,5	67,6	6,66	total	sunny
Point 2	26/09/2024 -22.839857, -42.220193	25,7	0,55	41,2	47,1	7,5	66,2	8,2	total	sunny
Point 3	26/09/2024 -22.851146, -42.156911	26,6	0,55	41,5	47	7,3	65,6	8,7	total	sunny with wind

2.2.1 Gamma Ray Spectrometry

To obtain the gamma radiation spectrum of natural radionuclides, a high-purity germanium coaxial semiconductor detector with a relative efficiency of 30% (HPGe, model GC3020 from Canberra Industries) was used. This detector was equipped with a low-noise preamplifier (model 2002 CSL) and connected to a Digital Spectrum Analyzer (DSA 1000) with 8192 channels, operated using the Genie 2000 software. The analyzed energy range varied from 50 keV to 2 MeV, as recommended by the

manufacturer. The spectrum acquisition time for each sample was 28,800 seconds. For energy calibration, sealed radioactive sources of ^{137}Cs , ^{60}Co , and a mixed source containing ^{155}Eu and ^{22}Na were used to establish a linear channel-energy relationship.

To ensure the quality of the analyses, a certified reference soil material (089/ERA) traced to the National Institute of Standards and Technology (NIST) was used. This material was also analyzed by the Gamma Spectrometry Laboratory (LGS) of IRD/CNEN to verify any discrepancies between the certified values and the material used, as well as to validate the LAASC detection system. Specific activities were calculated following standard methodology for reference materials. LAASC is an accredited laboratory and regularly participates in analytical performance tests through the National Intercomparison Program (PNI), coordinated by IRD-CNEN/RJ (PNI – IRD, 1997–2011; Gabriel et al., 2022), which is conducted by the National Nuclear Energy Commission (CNEN) under the supervision of the National Laboratory for Ionizing Radiation Metrology (LNMRI). LNMRI is overseen by the Bureau International des Poids et Mesures (BIPM), ensuring traceability and accuracy of measurements.

Activity concentrations and their uncertainties were determined based on the statistical uncertainties of the peak areas provided by the Genie2000 software. The formulas used to calculate the activity concentration (AC) and the minimum detectable activity (MDA) follow the works of Currie (1968) and Knoll (2000).

$$A_c = \frac{N_L}{\epsilon * P_\gamma * m * t}$$

$$MDA = \frac{2.71 + 4.66 * \sigma}{\epsilon * P_\gamma * m * t}$$

Where AC is the activity concentration ($\text{Bq}\cdot\text{kg}^{-1}$), N_L is the net count area, σ is the standard deviation of the background spectrum count, ϵ is the counting efficiency for a specific gamma energy, P_γ is the emission probability of the measured gamma energy, m is the sample mass (kg), and t is the counting time (s). The LabSOCS software (Laboratory Sourceless Calibration Software) from Canberra was used to determine ϵ , with a spectrum acquisition time of 28,800 s.

The energies 238 keV from ^{212}Pb , 338.3 keV, 911.6 keV, and 969.1 keV from ^{228}Ac , 583 keV from ^{208}Tl , and 727.2 keV from ^{212}Bi from the ^{232}Th series; 186.2 keV from ^{226}Ra , 295.2 keV and 351.9 keV from ^{214}Pb , and 609.3 keV, 1120 keV, and 1764.5 keV from ^{214}Bi from the ^{238}U series; and 1460.8 keV from ^{40}K were analyzed. For the identification of radionuclide ^{226}Ra , the 609.3 keV peak from ^{214}Bi was used; for ^{228}Ra , the 969.1 keV peak from ^{228}Ac ; for ^{232}Th , the 583 keV peak from ^{208}Tl ; for ^{238}U , the 1764.5 keV peak from ^{214}Bi ; and the 1460.8 keV peak for ^{40}K . The calculation of activity concentration (AC) took into account the emission probability of $25.8 \pm 0.4\%$ for ^{228}Ac , $45.4 \pm 0.2\%$ for ^{214}Bi , and $10.6 \pm 0.1\%$ for ^{40}K . A value of 1.96σ was estimated for the dataset used in the MDA calculation.

3 Results and discussions

The activity concentrations (AC) of the radionuclides ^{226}Ra , ^{228}Ra , ^{238}U , ^{232}Th and ^{40}K in the sediments and water of the lagoons are shown in Table 3.

Comparing the lagoons, the graph shows the difference in radionuclide concentrations present in the sediments, keeping in mind that only the Point 2 locations are repeated from one month to the next, while all the others correspond to different locations within the lagoon (Figure 3).

Table 2: Activity concentration, in Bq kg⁻¹, of radionuclides found in sediment and water in August and September (MDA: Minimum Detectable Activity)

Sediment August						
		²²⁶ Ra	²²⁸ Ra	²³⁸ U	²³² Th	⁴⁰ K
Maricá	Point 1	11,12 ± 0,87	19,70 ± 1,58	7,42 ± 1,06	6,93 ± 0,67	6,03 ± 1,76
	Point 2	8,27 ± 0,78	6,3 ± 1,52	3,9 ± 1,02	4,75 ± 0,46	14,74 ± 2,17
	Point 3	1,8 ± 0,25	0,27 ± 0,50	0,69 ± 0,35	0,96 ± 0,13	5,66 ± 0,78
Saquarema	Point 1	9,94 ± 0,82	15,49 ± 1,56	4,98 ± 1,02	7,82 ± 0,53	29,85 ± 2,60
	Point 2	6,37 ± 0,66	3,07 ± 1,26	3,54 ± 0,94	5,22 ± 0,35	7,18 ± 1,66
	Point 3	44,91 ± 2,17	44,17 ± 2,89	23,41 ± 1,80	27,64 ± 1,22	177,23 ± 8,28
Araruama	Point 1	4,92 ± 2,06	2,77 ± 1,69	2,02 ± 1,19	3,13 ± 1,11	31,11 ± 12,64
	Point 2	8,48 ± 0,69	13,32 ± 1,32	5,1 ± 0,82	5,99 ± 0,48	4,68 ± 1,46
	Point 3	5,9 ± 0,71	4,92 ± 1,44	4,1 ± 0,99	3,72 ± 0,41	39,9 ± 2,95
Sediment September						
		²²⁶ Ra	²²⁸ Ra	²³⁸ U	²³² Th	⁴⁰ K
Maricá	Point 1	16,66 ± 1,02	27,95 ± 2,04	8,65 ± 1,11	10,63 ± 0,84	19,25 ± 2,08
	Point 2	17,77 ± 1,11	22,34 ± 1,87	8,07 ± 1,15	13,64 ± 0,69	29,11 ± 2,63
	Point 3	10,32 ± 0,70	19,16 ± 1,42	4,99 ± 0,75	7,48 ± 0,59	14,39 ± 1,54
Saquarema	Point 1	10,61 ± 0,82	10,67 ± 1,46	4,26 ± 0,95	5,81 ± 0,46	244,85 ± 10,75
	Point 2	3,65 ± 0,50	1,88 ± 1,01	1,57 ± 0,69	2,03 ± 0,27	3,66 ± 1,36
	Point 3	8,86 ± 0,62	4,54 ± 1,07	3,63 ± 0,68	5,74 ± 0,32	85,78 ± 4,26
Araruama	Point 1	4,02 ± 0,55	3,76 ± 1,15	3,08 ± 0,72	2,07 ± 0,32	169,63 ± 7,67
	Point 2	2,52 ± 0,41	1,28 ± 0,88	0,96 ± 0,57	1,92 ± 0,23	75,48 ± 3,84
	Point 3	4,01 ± 0,58	2,54 ± 1,21	2,67 ± 0,76	2,49 ± 0,32	117,29 ± 5,69
Water August						
		²²⁶ Ra	²²⁸ Ra	²³⁸ U	²³² Th	⁴⁰ K
Maricá	Ponto 1:	< MDA	1,58 ± 25,38	1,31 ± 21,06	1,19 ± 17,02	3,01 ± 48,33
	Ponto 2:	< MDA	< MDA	1,04 ± 15,45	1,05 ± 8,53	4,56 ± 67,31
	Ponto 3:	0,24 ± 3,80	0,37 ± 5,75	0,04 ± 0,72	< MDA	1,66 ± 25,66
Saquarema	Ponto 1:	< MDA	< MDA	< MDA	< MDA	2,54 ± 37,02
	Ponto 2:	0,02 ± 0,53	1,49 ± 2,46	0,38 ± 1,02	< MDA	3,76 ± 5,87
	Ponto 3:	< MDA	< MDA	0,27 ± 4,32	< MDA	2,51 ± 39,21
Araruama	Ponto 1:	< MDA	0,04 ± 1,26	0,09 ± 1,60	< MDA	6,78 ± 102,83
	Ponto 2:	< MDA	0,2 ± 2,95	0,12 ± 1,70	< MDA	3,19 ± 45,57
	Ponto 3:	< MDA	0,28 ± 4,19	0,74 ± 10,80	0,54 ± 3,44	2,5 ± 36,72

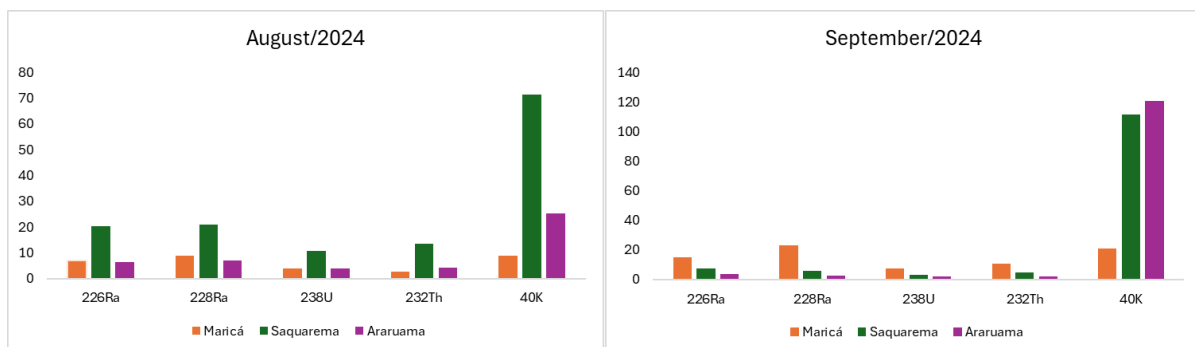


Figure 3: Graphs comparing the average concentration of Bq kg⁻¹ found in the sediments of the lagoons in August and September.

The radionuclide concentrations were compared with other studies (Table 3). A study conducted in 2007 by Povinec et al. on sediments from Fangataufa Lagoon in French Polynesia, intended for use as a reference material for analyses, reported average concentrations lower than those found in this study, with

the exception of ^{238}U . It is important to note that these are distinct environments. The lagoon system of eastern Rio de Janeiro consists of coastal restinga lagoons with variable salinity, where sediments can be sandy, muddy, or contain shell-derived carbonates, among other compositions. The primary sources of radionuclide concentration variations in these lagoons are domestic sewage, agricultural pesticides, and other anthropogenic interactions. In contrast, Fangataufa Lagoon is an oceanic atoll lagoon (coral reef) with high salinity due to its marine setting. Its sediment is nearly 100% calcium carbonate (CaCO_3), and the site has a history of nuclear weapons testing conducted by France between 1966 and 1996. The observed differences in ^{238}U concentrations are directly linked to these nuclear tests, as the explosions released non-fissioned uranium from the weapons and fragmented the carbonate-rich substrate (Povinec et al., 1999).

In Bramha et al. (2019), Chilika Lagoon in India, the largest brackish water lagoon in Asia, was studied. This environment is influenced by agricultural activities, domestic sewage, and the input of continental sediments. Its sediment is dominated by mud (60%), with textural variations (sand, silt, clay) and a high organic matter content. The study conducted in India found elevated activities of natural radionuclides (^{226}Ra , ^{232}Th , ^{40}K), significantly higher than the values reported in the lagoons of the eastern Rio de Janeiro lagoon system. This difference is mainly due to geological factors (rocks rich in thorium and uranium in India), sedimentological factors (predominance of muddy sediments in Chilika versus sandy sediments in the Brazilian lagoons), and anthropogenic factors (intensive use of agricultural fertilizers in India increasing ^{40}K levels).

A 2018 lake study in Northwestern Greece by Noli & Tsamos presented results for sediment and radionuclides. The water-related results were reported in mBq/L ; when converted to Bq kg^{-1} (assuming a freshwater density of 1.00 kg/L), the values found in the four lakes were significantly lower compared to coastal lagoons. The lakes in the study showed ^{238}U levels influenced by mining activities, whereas coastal lagoons may exhibit higher concentrations due to natural processes like evaporation, interaction with marine sediments, and direct anthropogenic sources. For sediments, overall concentrations were higher in the lakes, though it is important to note that the elevated levels in lakes Zazari and Cheimaditis are likely due to increased dry and wet deposition of fly ash, as they are located near the large Amyntaio lignite mine and coal-fired power plants (Noli & Tsamos, 2018). In contrast, lakes Vegoritis and Petron, which showed lower activity levels compared to the other lakes in the study, are likely linked to the region's geological characteristics. Studies suggest that carbonate rock formations in these areas, combined with high water conductivity (particularly in Lake Petron), promote uranium dissolution from sediments and its transfer into the water column (Gavrilescu et al., 2009; Wu et al., 2014).

According to UNSCEAR (2000), terrestrial soils in Argentina naturally present an average of 650 Bq kg^{-1} of ^{40}K . In the analyzed lagoons, including the Araruama hypersaline lagoon, where, due to the known positive correlation between salinity and ^{40}K enrichment in aquatic environments (Ramadan et al., 2014), higher concentrations were expected, the values were still below those reported for Argentina. However, it should be noted that the dynamics of radioactive elements in aquatic environments may differ from those observed in soils due to factors such as solubility, water transport, and others; therefore, conclusions should be interpreted with caution. Compared to the study by Uosif (2007), conducted on the Nile River in Upper Egypt, the average concentrations of ^{226}Ra were higher than those found in our study, except for Point 3 in Saquarema (August), which showed a higher value than any sample from the Nile. This sampling site was located within a farm with abundant manure residues and fertilizer use. Regarding ^{228}Ra , the average concentrations in the Nile study were lower than those in the lagoons (Point 3 - Saquarema - August; Maricá - September), but when analyzing the complete results, elevated levels were also found in comparison. In the 2007 study, values up to $76.2 \pm 6.2 \text{ Bq kg}^{-1}$ were recorded, while the highest value in the lagoons was $27.9 \pm 2.04 \text{ Bq kg}^{-1}$. When comparing the last radionuclide assessed in the Egypt study, ^{40}K concentrations were higher than those found in the lagoons, except for Saquarema Point 1 (August). It is important to highlight that these comparisons involve different environments: a river with greater

water flow (in the study, a location unconnected to the sea), while the lagoons have lower water exchange but are connected to the ocean.

Another riverine study, focusing on the riverbank soils of the Chao Phraya River, Thailand's primary fluvial system, which flows through densely populated regions (Santawamaitre et al., 2014), evaluated activity concentrations of radionuclides in sediments from its four major tributaries (Ping, Wang, Yom, and Nan) and the main river (Chao Phraya). All results reported in the study were higher than those found in lagoons. The authors cite natural geology and human activities (agriculture/industry) as key drivers of these variations.

Table 3: Values found in other studies for sediments, soils from lagoons, lakes and rivers.

	Concentration in sediment (Bq kg ⁻¹)					
	²²⁶ Ra	²²⁸ Ra	²³⁸ U	²³² Th	⁴⁰ K	
UNSCEAR 2000					540 – 750	Range from Argentina (South America)
Povinec et. al, 2007	2,5 ± 0,6	1,5 ± 0,5	35,3 ± 1,7*	0,018 ± 0,008	6,8 ± 0,4*	Mean r std. Dev (* = certified activities)
Uosif 2007	20,0 ± 2,5	24,8 ± 1,5		183,9 ± 9,2		Mean of samples concentrations
Santawamaitre et. al, 2014			29,19 ± 0,10	30,43 ± 0,10	309,0 ± 1,6	Total mean values of activity concentrations
Noli & Tsamos (2018)			19,2–28,7	11,0–38,7	183,0–585	Mass activities (Bq/kg)
			9,28–24,2	0,43–12,7	1,11–145	
Bramha, et al. (2019)	26,27			81,61	620,09	Averages
	Concentration in water					
	²²⁶ Ra	²²⁸ Ra	²³⁸ U	²³² Th	⁴⁰ K	
Noli & Tsamos (2018)			26,55 ± 1,49			Average seasonal results (mBq/L)
			45,1 ± 2,75			
			56,2 ± 3,63			
			61,78 ± 4,49			

Most of the water samples showed values below the minimum detectable activity (MDA), which is expected. Water has a high dilution capacity, meaning that the amount of radionuclides present tends to become dispersed. Additionally, these radionuclides tend to adsorb onto sediments and accumulate there, where higher concentrations are typically found (UNSCEAR, 2000; IAEA, 2003).

4 Conclusions

This study evaluated natural radionuclide concentrations (²²⁶Ra, ²²⁸Ra, ²³²Th, ²³⁸U, and ⁴⁰K) in sediments and waters of coastal lagoons in eastern Rio de Janeiro State (Maricá-Guarapina, Saquarema, and Araruama). Results indicated that sediment concentrations fall within reference limits reported in the literature, though significant variations were observed among lagoons—particularly in Saquarema, which showed the highest levels. This disparity may be linked to inadequate sanitation coverage (only 27.36% of households connected to sewage networks), suggesting potential anthropogenic influences from domestic and agricultural effluents. Water samples exhibited concentrations below the minimum detectable activity (MDA), as expected due to the high dilution capacity of water bodies and radionuclides' tendency to adsorb to sediments. Comparisons with other aquatic ecosystems show that the measured levels align with typical ranges for coastal environments, though regional variations may occur due to geological, sedimentological, and land-use differences.

In summary, these findings provide critical data for environmental management and radiological protection of lagoon ecosystems, highlighting the need for public policies to reduce anthropogenic contamination and promote sustainability. Future studies should include long-term temporal assessments, aquatic organism analyses, and deeper investigations into radionuclide-environment interactions amid increasing urbanization around the lagoons.

References

Agbalagba, E. O., & Onoja, R. A. (2011). Evaluation of natural radioactivity in soil, sediment and water samples of Niger Delta (Biseni) flood plain lakes, Nigeria. *Journal of Environmental Radioactivity*, 102(7), 667-671.

Bramha S., Sahoo S. K., Joel E. S., Venkatraman B., Mohanty P. K., Rath, P. (2019). Influence of geochemical properties on natural radionuclides in the sediment of Asia's largest brackish water lagoon, Chilika-East Coast of India: evaluation through geo-statistical applications. *Environmental Earth Sciences*, 78(652).

Carvalho F.P., Oliveira J.M., Malta M. (2011) Radionuclides in deep-sea fish and other organisms from the North Atlantic Ocean. *ICES Journal of Marine Science*, 68(2), 333–340.

Chowdhury C.R., Khijmatgar S., Chowdhury A., Kumari P. D., Lynch E., Gootvel M. (2020) Radionuclide activity concentration in soil, granites and water in a fluorosis endemic area of India: An oral health perspective. *Journal of Oral Biology and Craniofacial Research* 10: 259-262.

European Commission, Joint Research Centre (2019) – Cinelli, G., De Cort, M. & Tollefsen, T. (Eds.), European Atlas of Natural Radiation, Publication Office of the European Union, Luxembourg.

Gavrilescu M, Pavel LV, Cretescu I (2009) Characterization and remediation of soils contaminated with uranium. *J Hazard Mater* 163:475–510

International Atomic Energy Agency (IAEA). (2003). Radiation protection of the public and the environment (Technical Reports Series No. 1363). IAEA. https://www-pub.iaea.org/MTCD/Publications/PDF/te_1363_web.pdf

Jibiri N.N., Ugbechie A., Sowunmi A.A., Akomolafe I.R., (2023). Evaluation of Radiological Risks From Radionuclides in Fish and Sediment of Eleyele Reservoir, Ibadan, Nigeria. *Pollut. Bull.* 192, 114992.

Knoppers B., Kjerfve B. (1999) Coastal Lagoons of Southeastern Brazil: Physical and Biogeochemical Characteristics. *Estuaries of South America. Environmental Science*.

Lince – Instrumentos e Radioproteção. (2025) NORM e TENORM – O que são, quais os riscos e como fazer sua gestão? <https://lincebrasil.com/norm-e-tenorm-o-que-sao-quais-os-riscos-e-como-fazer-sua-gestao/>

Linsley G., Sjöblom K.L., Cabianca T., (2004) Chapter 4 Overview of point sources of anthropogenic radionuclides in the oceans. *Marine Radioactivity*, cap 4.

Mársico E. T. (2005). Bioaccumulation of ²¹⁰polonium and ²¹⁰lead in fish. [Doctoral Dissertation]. Niterói (RJ): Fluminense Federal University.

Milenkovic B., Stajic J.M., Stojic N., Pucarevic M., Strbac S. (2019). Evaluation of heavy metals and radionuclides in fish and seafood products. *Chemosphere* 229: 324e331, 2019.

Noli F., Tsamos P. (2018) Seasonal variations of natural radionuclides, minor and trace elements in lake sediments and water in a lignite mining area of North-Western Greece. *Environ Sci Pollut Res* 25:12222–12233

Povinec P. P., Woodhead D., Blowers P., Bonfield R., Cooper M., Che, Q., Dahlgaard H., Dovlete C., Fox, V. Froehlich K., Gastaud J., Groning M., Hamilton T., Ikeuchi Y., Kanisch G., Kruger A., Kwong L. L. W., Matthews M., Morgenstern U., ... Tinker R. (1999). Marine radioactivity assessment of Mururoa and Fangataufa atolls. *The Science of the Total Environment* 237r238 1999 249 Ž . j267

Povinec P. P., Pham M. K., Sanchez-Cabeza J. A., Barci-Funel G., Bojanowski R., Boshkova T., Burnett W. C., Carvalho F., Chapeyron B., Cunha I. L., Dahlgaard H., Galabov N., Fifield L. K., Gastaud J., Geering J.-J., Gomez I. F., Green N., Hamilton T., Ibanez F. L., ... Wyse E. (2007). Reference material

for radionuclides in sediment IAEA-384 (Fangataufa Lagoon sediment). *Journal of Radioanalytical and Nuclear Chemistry*, 273(2), 383–393.

Ramadan K. A., Seddeek M. K., Sharshar T., Elnimr T., Badran H. M. (2014) Effect of salinity on the concentrations of radioisotopes in the aquatic environment of a hypersaline coastal lagoon. *Isotopes in Environmental and Health Studies*, Vol. 50, No. 2, 235–256

Santawamaitre T., Malain D., Al-Sulaiti H. A., Bradley D. A., Matthews M. C., Regan P. H. (2014) Determination of ^{238}U , ^{232}Th and ^{40}K activity concentrations in riverbank soil along the Chao Phraya river basin in Thailand. *Journal of Environmental Radioactivity* 138: 80-86

UNSCEAR - United Nations Scientific Committee on the Effects Atomic Radiation (1993). Sources, effects and risks of ionizing radiation.

UNSCEAR - United Nations Scientific Committee on the Effects Atomic Radiation (2000). Sources and effects of ionizing radiation - Annex B.

Uosif M.A. (2007) Gamma-ray spectroscopic analysis of selected samples from Nile River sediments in Upper Egypt. *Radiation Protection Dosimetry*, Vol. 123, No. 2, pp. 215–220.

Wu Y, Wang Y, Xie X (2014) Occurrence, behavior and distribution of high levels of uranium in shallow groundwater at Datong basin, northern China. *Sci Total Environ* 472(15):809–817

Radiometric analysis of the bivalve mollusk *Perna perna*

I.B. F. Martins^{1,2,4*}, G. A. S. Rodé², L. S. L. Neto², R. L. Silva², T. A. Lima^{2,5}, C. O. P. Orejuela⁵, M. A. F. Lima³, A. X. Silva⁵, A. G. A. C. Kelecom², A. S. Gomes^{1,4}, J. L. Thalhoffer^{1,2,5}

¹Pós-Graduação em Dinâmica dos Oceanos e da Terra (PPG-DOT/UFF) - Niterói - Brasil

²Laboratório de Radiobiologia e Radiometria (LARARA/UFF) - Niterói - Brasil

³Laboratório Central Radiométrica (LCR/UFF) - Universidade Federal Fluminense - Niterói - Brasil

⁴Laboratório de Ecologia de Sedimentos (ECOSD/UFF) - Niterói - Brasil

⁵Programa de Engenharia Nuclear (PEN/COPPE/UFRJ) - Rio de Janeiro – Brasil

*irismartins@id.uff.br

Abstract

Bivalve organisms have long been recognized as effective biomonitors in environmental monitoring programs, with the mussel *Perna perna* serving as a prominent example. These organisms play a crucial role in assessing a wide range of conditions in aquatic ecosystems, including the presence of chemical compounds and heavy metals, which can be detected either directly or indirectly in their tissues. A combination of biological, physiological, and behavioral traits such as sessile lifestyle, ease of collection, and low maintenance requirements make them particularly suitable for this purpose. The present study aimed to perform radiometric analysis by gamma spectrometry on different compartments shell, soft tissue, and byssus of *Perna perna* in order to evaluate its potential as a biomonitor species through the determination of activity concentrations (ACs). Samples of *P. perna* were collected from Guanabara Bay and categorized by sex, as well as by tissue type. The mussels were dissected and separated by sex. Fresh soft tissue was placed in sealed containers and stored at – 4 °C. Shell and byssus samples were cleaned, oven-dried at 80 °C for approximately 3h until a constant weight was reached, and then the shells were ground, weighed, and stored in sealed containers. All samples were stored for at least 28 days to allow for secular equilibrium before gamma spectra were obtained using a high-purity germanium (HPGe) detector. Results showed that for ²²⁶Ra and ²²⁸Ra, most samples had activity concentrations below the minimum detectable activity (MDA). In contrast, for the radioisotope ⁴⁰K, activity concentrations in female soft tissue, byssus, male shells, and female shells were 32.24 Bq/kg, 4.75 Bq/kg, 3.39 Bq/kg, and 2.12 Bq/kg, respectively. Although Guanabara Bay is currently impacted by pollution, the radiological assessment indicated no significant activity concentrations for ²²⁶Ra and ²²⁸Ra at the study site. The ⁴⁰K values observed likely reflect species-specific metabolic requirements, as potassium plays an essential role in the overall functioning of the organism.

Keywords: Mussels, *Perna perna*, Gama Spectrometry

1 Introduction

Bivalves play an important role in assessing a wide range of conditions present in the marine ecosystem and are widely recognized for their role as biomonitors of elements found in this environment, such as metals, metalloids, organic substances, pharmaceutical components, microplastics, and radionuclides (Lima et al., 2022; Muncaster et al., 1990; Brenner et al., 2000). Bivalves have proven to be suitable organisms for evaluating contamination in aquatic ecosystems due to specific characteristics that favor their use as environmental indicators. Among these characteristics, factors such as ease of collection, seasonality, physiological aspects, and natural habits make these organisms ideal for biomonitoring studies (Rodrigues, 2012).

Bioaccumulation is a process by which organisms absorb and accumulate radioactive material/contaminants available in the environment, contributing to biomagnification at higher trophic levels (Lokas et al., 2014). The bioaccumulation of these contaminants can cause adverse health effects in organisms and, consequently, affect the ecological dynamics of aquatic environments and the quality of water resources, reinforcing the importance of monitoring and assessing these compounds in biotic compartments (Eça et al., 2012). Considering the potential of bivalves as biomonitors, it is essential to employ analytical techniques that enable the precise and reliable detection and quantification of contaminants, including radionuclides.

Among these techniques, gamma spectrometry stands out as a method that involves measuring the energy of photons emitted from the radioactive decay of nuclides that may be present in the analyzed sample. Each photon has a characteristic energy, allowing for its identification (Patricio, 2020). Among environmentally relevant radionuclides, naturally occurring ones that can be found in marine environments and accumulate in organisms are notable, such as Radium-226 (^{226}Ra), Radium-228 (^{228}Ra), and Potassium-40 (^{40}K). The accumulation of these radioisotopes can represent a significant pathway for transfer to higher trophic levels in both aquatic and terrestrial food chains (Brenner et al., 2007).

Understanding the behavior and distribution of these radionuclides in the environment is of utmost importance in coastal and estuarine ecosystems, especially those near large urban centers, such as Guanabara Bay a site surrounded by approximately 10 million people and a large number of industries, marine terminals, commercial ports, shipyards, among others. Furthermore, the surroundings of Guanabara Bay encompass a complex network of transportation and industrial products, factors that contribute to maintaining its current polluted state (Filho, 2020).

In this context, it becomes necessary to select organisms that are efficient for monitoring the environmental conditions of the region, such as the bivalve mollusk *Perna perna*. Considering the characteristics presented and the environmental scenario of Guanabara Bay, this study aims to evaluate the presence of radionuclides ^{226}Ra , ^{228}Ra , and ^{40}K in the different biological compartments of the organism *Perna perna* through gamma spectrometry.

2 Methodology

2.1 Study Area and Sampling

Guanabara Bay (Figure 1) constitutes an important estuarine system located in the metropolitan region of Rio de Janeiro, also representing a significant source of income for the local population, mainly through commercial and fishing activities. However, due to rapid urbanization, Guanabara Bay is currently polluted, primarily by waste originating from domestic sewage (Coppe, 2017).

Surrounding the bay are several municipalities, among which the municipality of Niterói stands out. Niterói is subdivided into the following regions: East, Pendotiba, Oceânica, North, and Bay Beaches, where Jurujuba, a historic neighborhood, is located (SiGeo, 2025)



Figure 1: Municipality of Niterói, Jurujuba, sample collection point 22°42'54"S 42°39'37"W. (SiGeo, Mapas, 2025)

2.2 Sampling procedure

Specimens of *Perna perna* were collected in the field and subsequently taken to the laboratory, where they underwent a cleaning process aimed at removing encrusting organisms and particulate material adhered to the shell, such as barnacles, algae, and sediments. The individuals were then properly prepared for the next step, which consisted of dissecting the samples.

Initially, a scalpel blade number 12 was used to make an incision in the membrane that joins the shells. Once sectioned, the shells were easily separated, exposing the internal tissues and muscles (anterior muscle, retractor muscle, and posterior muscle). The removal of the soft tissue was performed by cutting these muscles, thereby separating the visceral mass from the shell. During this step, male and female individuals were distinguished and stored separately for analysis. Once the visceral mass was exposed, the byssus was also removed and separated for analysis.

A total of five samples were prepared, as described in the table below (Table 1). For the byssus sample, it was necessary to combine both sexes to obtain a sufficient mass for analysis. Both the shells and the byssus were placed in an oven at approximately 80°C for about 3 hours to eliminate any moisture from the samples, preventing microorganism growth and decomposition.

Next, the containers were sealed with plastic wrap and properly stored. Containers with shell and byssus samples were kept at a controlled temperature of 27°C, while containers with soft tissue samples were stored at -4°C. All samples underwent a minimum storage period of 28 days; this procedure is necessary to allow the samples to reach the secular equilibrium required for gamma spectrometry analysis.

Table 1- Mussel samples

	Female	Male
Soft Tissue	369,111g	452,205g
Shell	401,326g	662g
Byssus	32,522g	

2.3 Gama Spectrometry (HPGe)

To obtain the gamma radiation spectrum of naturally occurring radionuclides, a coaxial high-purity germanium (HPGe) semiconductor detector with a relative efficiency of 30% (model GC3020 from Canberra Industries) was used. A low-noise preamplifier (model 2002 CSL) coupled to a Digital Spectrum Analyzer (DSA 1000) with 8192 channels was connected to the detector, along with Genie 2000 software. The scanned energy range was from 50 keV to 2 MeV. The counting time for sample spectrum acquisition was 28,800 seconds.

For energy calibration, sealed radioactive sources of ^{137}Cs , ^{60}Co , and a mixed source of ^{155}Eu and ^{22}Na were used to establish a linear channel-energy relationship. To ensure the quality of the analyses, a certified reference soil material (089/ERA) traceable to the National Institute of Standards and Technology (NIST) was employed.

The LAASC is accredited and regularly participates in analytical performance proficiency tests through the National Intercomparison Program (PNI), coordinated by IRD-CNEN/RJ (Gabriel et al., 2022), conducted by the National Nuclear Energy Commission (CNEN) under the supervision of the National Laboratory of Ionizing Radiation Metrology (LNMRI). The LNMRI is monitored by the Bureau International des Poids et Mesures (BIPM), ensuring traceability and accuracy in measurements.

Activity concentrations and their uncertainties were determined based on the area under the photopeak and their respective statistical uncertainties, as provided by the Genie2000 software. The following equations were used to calculate the activity concentration (AC) and the minimum detectable activity (MDA) (Curie, 1968; Knoll, 2000).

$$A_C = \frac{N_L}{\varepsilon \cdot P_\gamma \cdot m \cdot t} \quad (1)$$

$$MDA = \frac{2.71 + 4.66 \cdot \sigma}{\varepsilon \cdot P_\gamma \cdot m \cdot t} \quad (2)$$

Where AC is the activity concentration ($\text{Bq} \cdot \text{kg}^{-1}$), NL is the net count area, σ is the standard deviation of the background spectrum counts, ε is the counting efficiency for a specific γ energy, P_γ is the emission probability of the measured γ energy, m is the sample mass (kg), and t is the counting time (s).

Canberra's LabSOCS (Laboratory Sourceless Calibration Software) was used to determine ϵ , and the counting time for spectrum acquisition was 28,800 seconds. The energies and radionuclides used to evaluate AC were as follows: 238 keV from ^{212}Pb ; 338.3 keV, 911.6 keV, and 969.1 keV from ^{228}Ac ; 583 keV from ^{208}Tl ; and 727.2 keV from ^{212}Bi of the ^{232}Th series; 186.2 keV from ^{226}Ra ; 295.2 keV and 351.9 keV from ^{214}Pb ; and 609.3 keV, 1120 keV, and 1764.5 keV from ^{214}Bi of the ^{238}U series; and 1460.8 keV from ^{40}K .

The activity concentration (AC) of each radionuclide was calculated using energy and standard deviation. A 1.96σ limit was applied to the dataset in the calculation of the minimum detectable activity (MDA).

3. Results and discussion

Table 2 presents the activity concentrations of the radionuclides ^{226}Ra , ^{228}Ra , and ^{40}K determined by gamma spectrometry in different biological fractions of the bivalve mollusk *Perna perna*, including shells, soft tissue, and byssus.

Table 2- Activity in Becquerel per Kilo (Bq/Kg) of the analyzed samples

Table 2- Activity (Bq/Kg)			
Sample	^{226}Ra	^{228}Ra	^{40}K
Female shell	<MDA	<MDA	2,12
Male shell	<MDA	<MDA	3,89
Female Soft Tissue	<MDA	0,93	32,24
Byssus	<MDA	<MDA	4,75

In all analyzed samples, the activity concentrations of the radionuclide ^{226}Ra remained below the minimum detectable activity, indicating that the presence of this radionuclide in the analyzed biological fractions is very low or insignificant. Radium tends to preferentially accumulate in hard tissues rich in calcium (National Research Council, 1988). Due to chemical similarities, both ^{226}Ra and ^{228}Ra behave like calcium ions and can replace calcium in plant tissues, shells, bones, and muscle tissue of organisms when bioaccumulated. Therefore, when this radioisotope is present in organisms, it is likely bioavailable in the environment, either naturally or through contamination (Brenner et al., 2000).

According to the results, the amount of ^{226}Ra in the analyzed samples is below the detection limit, which may indicate that, although Guanabara Bay is a polluted water body, the presence of this radioisotope was not detected.

Similarly, ^{228}Ra was also found to be below the detection limit for shell samples from both males and females, as well as for the byssus sample. However, ^{228}Ra was detected in the female soft tissue sample, with an activity concentration of 0.93 Bq/kg. This finding suggests a selective bioaccumulation of this radionuclide in female soft tissue, which may indicate differences in the dynamics of incorporation and retention among the analyzed fractions, possibly associated with physiological or metabolic factors specific to females.

^{40}K was detected in all samples, with the highest activity observed in female soft tissue (32.24 Bq/kg), followed by the byssus (4.75 Bq/kg), and lower concentrations in male and female shells, with activities of 3.39 Bq/kg and 2.12 Bq/kg, respectively. Potassium-40 is a naturally occurring radionuclide present in all living organisms, as it is associated with stable potassium, which is essential for maintaining and ensuring the proper functioning of biological processes, particularly in metabolically active tissues (Risal et al. 2024).

4 Conclusion

The present study aimed to evaluate the presence of the radionuclides ^{226}Ra , ^{228}Ra , and ^{40}K in different biological fractions of the bivalve mollusk *Perna perna* collected from Guanabara Bay. The results showed that ^{226}Ra and ^{228}Ra concentrations were, in most samples, below the minimum detectable activity, with the exception of the female soft tissue sample, in which ^{228}Ra activity of 0.93 Bq/kg was observed. This suggests that, despite the pollution in Guanabara Bay, no significant evidence of radiological contamination by these radionuclides was found during the period and under the conditions analyzed.

^{40}K was detected in all samples, with the highest concentration in female soft tissue, followed by the byssus and shells. This result is consistent with expectations, since potassium is a crucial element for the maintenance and proper functioning of organisms and is associated with metabolically active tissues. Thus, the data contribute to a radiometric profile analysis of Guanabara Bay through the assessment of *Perna perna* mussels, reinforcing the role of this species as a biomonitor. However, further studies are necessary to expand the database, consider seasonal variations, and evaluate possible physiological influences among the organisms studied, such as sex, age, and reproductive period.

Acknowledgements

The authors thank FAPERJ, CNPQ, CAPES and MCTI for the financial support provided; without the help of these institutions, this work would not have been possible.

References

- Brenner, M., Smoak, J. M., Allen, M. S., Schelske, C. L., & Leeper, D. A. (2000). Biological accumulation of ^{226}Ra in a groundwater-augmented Florida lake. *Limnology and Oceanography*, 45(4), 710–715. <https://doi.org/10.4319/lo.2000.45.4.0710>
- Brenner, M., Smoak, J. M., Leeper, D. A., Streubert, M., & Baker, S. M. (2007). Radium-226 accumulation in Florida freshwater mussels. *Limnology and Oceanography*, 52(4), 1614–1623. <https://doi.org/10.4319/lo.2007.52.4.1614>
- COPPE-UFRJ. (2017). *Baía de Guanabara: Relatório de monitoramento e análise*. Universidade Federal do Rio de Janeiro. Disponível em: https://www.baiasdobrasil.coppe.ufrj.br/assets/relatorios/rel_baia_guanabara.html (Acesso em: 28 ago. 2024).
- Currie, L. A. (1968). Limits for qualitative detection and quantitative determination. Application to radiochemistry. *Analytical Chemistry*, 40(3), 586–593. <https://doi.org/10.1021/ac60259a007>

Eça, G. F., Mendonça Filho, C. V. S., Hatje, V., & Santos, W. P. C. (2012). Corais como organismos biomonitores: Aplicação, pré-tratamento e determinação de elementos majoritários e minoritários. *Química Nova*, 35(3), 1–9.

Filho, J. T. de S., Mello, S. C. R. P., Faria, A. S., & Souza, L. L. (2020). Análise socioambiental da poluição por esgoto da Baía de Guanabara do Rio de Janeiro. *Valore*, 5(5022).

Gabriel, L. H., Rodrigues, C. E. C., Nogueira, P. R., Damatto, S. R., Isiki, V. L. K., Teixeira, L. F. L., Máduar, M. F., & Alencar, M. M. (2022). Monitoração radiológica dos efluentes gasosos do reator IEA-R1 do IPEN. In P. S. C. da Silva, G. S. Zahn, & F. de Assis Souza (Orgs.), *Contribuições do reator IEA-R1 para a pesquisa nuclear: II Workshop anual do reator de pesquisas – WARP 2* (pp. 331–338). Blucher Open Access. ISBN 978-65-5550-148-3.

Knoll, G. F. (2000). *Radiation detection and measurement* (3rd ed.). John Wiley & Sons.

Lima, D. F., di Benedetto, A. P. M., & Franco, R. W. de A. (2022). Bivalves como biomonitores ambientais: Uma revisão de literatura. *Conjecturas*, 22(2), 1142–1156. <https://doi.org/10.53660/conj-817-f09>

Łokas, E. L., Bartmiński, P., Wachniew, P., Mietelski, J. W., Kawiak, T., & Srodoń, J. (2014). Sources and pathways of artificial radionuclides to soils at a High Arctic site. *Environmental Science and Pollution Research*, 21(21), 12479–12493. <https://doi.org/10.1007/s11356-014-3163-6>

Muncaster, B. W., Hebert, P. D. N., & Lazar, R. (1990). Biological and physical factors affecting the body burden of organic contaminants in freshwater mussels. *Archives of Environmental Contamination and Toxicology*, 19, 25–34.

National Research Council (US). Committee on the Biological Effects of Ionizing Radiations (BEIR IV). (1988). *Health risks of radon and other internally deposited alpha-emitters*. National Academies Press. Disponível em: <https://www.ncbi.nlm.nih.gov/books/NBK218126/> (Acesso em: 29 jul. 2025).

Patricio, M. L. V. (2020). *Espectrometria gama de alta resolução aplicada na análise de radioatividade em águas devido aos isótopos naturais de rádio*.

Risal, S. Y., Islam, S., Ferdous, J., Siddik, M. N. A., & Bakshi, P. K. (2024). Natural radioactivity and heavy metal contamination in edible fish, shellfish and mollusks at the Bay of Bengal, Kuakata, Bangladesh. *Heliyon*, 10(18). <https://doi.org/10.1016/j.heliyon.2024.e37787>

Rodrigues, V., Abessa, D. M. S., Hortellani, M. A., Sarkis, J. E. S., & Sígolo, J. B. (2012). Uso do bivalve límico *Anodontites tenebricosus* (Lea, 1834) no biomonitoramento de metais do Rio Ribeira de Iguape. *Química Nova*, 35(3).

SIGeo – Prefeitura Municipal de Niterói. (2025). *Mapas – HUB SIGeo*. Disponível em: <https://www.sigeo.niteroi.rj.gov.br/pages/mapas> (Acesso em: 23 jul. 2025).

Study of Drinking Water Quality in Rio de Janeiro: Assessment of Gamma Radiation Levels and Physicochemical Parameters

Leite, R. S.^{1*}, Rodé, G. A. S.¹, Martins, I. B. F.¹, Ferreira, C. A.²,
Nascimento, G. P. O.¹, Orejuela, C. O. P.², Nepomuceno, A. M. J.⁴,
Kelecom, A. G. A. C.¹, Silva, A. X.², Souza, E. M.³, Diniz, M. B.¹,
Thalhofer, J. L.^{1,2 *}

¹ Radiobiology and Radiometry Laboratory (LARARA/UFF), 24210-201, Niterói, Brazil

² Nuclear Engineering Program (PEN/COPPE/UF RJ), 21941-598, Rio de Janeiro, Brazil

³ Reduced Models Laboratory (LMR), State University of Rio de Janeiro (UERJ), West Zone Campus,
23070-200, Rio de Janeiro, Brazil

⁴ Aquatic Environments Biogeochemistry Laboratory (UFF), 24210-201, Niterói, Brazil

* raylineleite@id.uff.br

* jardellt@id.uff.br

Abstract

Drinking water naturally contains radioisotopes such as potassium-40 (⁴⁰K) and decay products from the uranium and thorium series, originating from geological and environmental processes. However, anthropogenic activities and increasing pollution of water resources may intensify the presence of gamma-emitting radionuclides, posing a potential risk. When present at concentrations above recommended limits, these radionuclides may significantly impact human health, as prolonged exposure to gamma radiation can induce DNA mutations, thereby increasing the likelihood of genetic disorders and neoplastic diseases, such as cancer. This study aims to evaluate gamma radiation levels and physicochemical parameters in potable water samples collected from the metropolitan region of the state of Rio de Janeiro. Twenty sampling points were analyzed over the course of one year, covering six different municipalities. Physicochemical characteristics were determined using the AKSO multiparameter meter AK88, and gamma spectrometry was conducted using a high-purity germanium (HPGe) coaxial semiconductor detector (Canberra, model GC3020) with 30% relative efficiency. Samples were stored for 30 days prior to counting to ensure secular equilibrium. Spectra acquisition lasted 28,800 seconds, and the data were processed using Genie 2000 software. Preliminary results indicate that none of the sampling points analyzed so far exceed the recommended limits established by specialized organizations, such as ICRP. The findings, although not indicating critical levels so far, highlight the importance of continuous radiological monitoring to ensure the safety of drinking water and to support preventive public health measures.

Keywords: Gamma radiation; Drinking water; Gamma spectrometry; Water quality; Public health.

1 Introduction

Ionizing radiation is a natural phenomenon present in the environment, although it often goes unnoticed. It exists in the soil, in the air we breathe, in the water we consume, and in the cosmic radiation that continuously reaches the Earth's surface. Constant exposure to this radiation has been part of the evolutionary history of life on Earth and, at natural levels, generally does not pose significant risks to human health. However, when elevated concentrations of radionuclides are present, the biological effects of radiation become a concern, especially due to its mutagenic and carcinogenic potential.

In drinking water, it is common to find natural radionuclides such as potassium-40 (^{40}K) and those from the uranium and thorium decay series, which are introduced into the environment through geological and physicochemical processes. Gamma radiation, in turn, is frequently emitted alongside alpha and beta particles and is characteristic of several natural radionuclides, such as ^{208}Tl , ^{214}Bi , and ^{40}K itself. Gamma radiation differs from the others due to its high penetrating power and indirect ionizing nature, allowing it to traverse large thicknesses of materials and reach internal human tissues even without ingestion or inhalation of the radioactive material (WHO, 2017).

Studies such as Alharbi et al. have shown that increased water pollution, particularly from industrial effluents and mining activities, can intensify the mobilization of radionuclides from the soil to water bodies, raising the levels of gamma radiation detected in environmental samples. Additionally, sectors such as phosphate fertilizer production naturally concentrate radioactive elements, contributing to the radiological enrichment of water (Giovanardi et al., 2012).

Water contamination has become an increasingly critical environmental and public health issue, threatening entire ecosystems, compromising the efficiency of Water Treatment Plants (WTPs), and directly impacting human health. Although many studies focus on chemical and biological contaminants — such as heavy metals (e.g., lead and arsenic) and pathogenic microorganisms responsible for diseases like hepatitis A, typhoid fever, and cholera (Riviere, J.E., 2006; CCD/SES-SP, 2009) — the presence of gamma-emitting isotopes must also be considered a critical factor. Due to their high penetration capacity, even relatively low concentrations of gamma activity may pose health risks, especially in cases of chronic and cumulative exposure.

In recent years, studies have increasingly highlighted the importance of monitoring ionizing radiation in drinking water. While much of the research focuses on risks associated with alpha and beta radiation, it is essential to consider that gamma radiation also contributes to the total exposure dose and can have significant public health implications. The World Health Organization (WHO) emphasizes that approximately 80% of cancer cases are related to environmental factors and that monitoring and controlling these factors are essential for prevention strategies (WHO, 2002).

Given this context, it becomes crucial to broaden the scope of water quality assessments by also incorporating gamma radiation evaluation. The main objective of the present study is to determine the levels of gamma radiation in drinking water samples collected from different municipalities in the state of Rio de Janeiro, thereby identifying potential areas of radiological risk and providing relevant data to support water safety and public health.

2 Methodology

2.1 Sampling, storage and sample preparation

Drinking water samples were collected from 20 different sites (Table 1), distributed across the state of Rio de Janeiro, covering the following cities: Rio de Janeiro, Niterói, São Gonçalo, Itaboraí, Petrópolis, and Nova Friburgo, as shown in Figure 1. The collected samples correspond

to the water consumed by residents in their homes. Some of these samples originate from water treatment plants (WTPs) and are distributed by the companies listed in Table 1, while others are obtained from wells (groundwater) and are untreated.

Sampling was carried out during different months. For each sample, a volume of 1 liter of water was collected from a flowing system and stored in pre-cleaned containers suitable for sample collection. Subsequently, in the Laboratory of Radiobiology and Radiometry (LARARA), the samples underwent physicochemical analysis and were then sealed and refrigerated for at least 28 days to allow for secular equilibrium.

Table 1. Specifications of collection points and water origin.

Sampling Points	Location (city)	Water Source Type	Water Supplier
Point 01	Complexo da Maré (Rio de Janeiro)	WTP	Águas do Rio
Point 02	Jacarepaguá (Rio de Janeiro)	WTP	Iguá
Point 03	Catete (Rio de Janeiro)	WTP	Águas do Rio
Point 04	Santa Bárbara (Niterói)	WTP	Águas de Niterói
Point 05	Itaipu (Niterói)	WTP	Águas de Niterói
Point 06	São Domingos (Niterói)	WTP	Águas de Niterói
Point 07	Pacheco (São Gonçalo)	Groundwater	-
Point 08	Pacheco (São Gonçalo)	WTP	Águas do Rio
Point 09	Pacheco (São Gonçalo)	Groundwater	-
Point 10	Portão do Rosa (São Gonçalo)	WTP	Águas do Rio
Point 11	Mutuá (São Gonçalo)	WTP	Águas do Rio
Point 12	Nova Cidade (São Gonçalo)	WTP	Águas do Rio
Point 13	Lindo Parque (São Gonçalo)	WTP	Águas do Rio

Point 14	Caluge (Itaboraí)	Groundwater	-
Point 15	Apolo II (Itaboraí)	Groundwater	-
Point 16	Apolo II (Itaboraí)	Groundwater	-
Point 17	Alto da Serra (Petrópolis)	WTP	Águas do Imperador
Point 18	Alto da Serra (Petrópolis)	Groundwater	-
Point 19	Downtown (Petrópolis)	Groundwater	-
Point 20	Lumiar (Nova Friburgo)	Groundwater	-

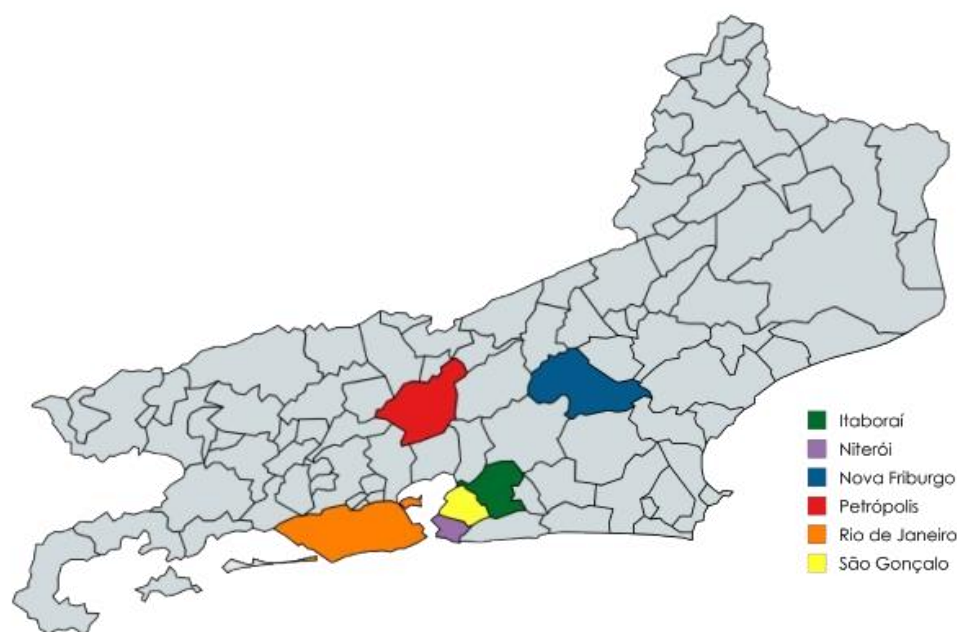


Figure 1: Map of the state of Rio de Janeiro, highlighting the cities where water sampling was conducted.

2.2 Physicochemical Parameters

The physicochemical parameters of the collected water samples were analyzed using the AK88 multiparameter meter (Akso brand). The following parameters were measured: temperature, pH, conductivity, dissolved oxygen, salinity, and total dissolved solids (TDS).

All analyses were conducted in the laboratory. The samples were poured into storage containers, and the meter's probes were rinsed with distilled water before being immersed in the sample. Measurements were taken after one minute of stabilization to ensure reliable readings. The multiparameter meter was previously calibrated according to the manufacturer's

specifications using standard solutions, as detailed in the device manual (AKSO, 2003).

2.3 Gamma Spectrometry

In this study, gamma spectrometry was employed to analyze naturally occurring radionuclides in the water samples. Measurements and analyses were conducted at the Laboratory of Environmental Analysis and Computational Simulation (LAASC), affiliated with the Federal University of Rio de Janeiro (UFRJ).

For activity concentration quantification, a high-purity germanium (HPGe) coaxial semiconductor detector (model GC3020, Canberra Industries) was used. The detector is coupled with a low-noise preamplifier (model 2002 CSL) and operated with the Genie 2000 software, connected to a 8192-channel Digital Spectrum Analyzer (DSA 1000). The voltage applied to form the depletion region was 4.5 kV, according to the manufacturer's specifications.

Energy calibration of the HPGe detector was performed using sealed sources of ^{137}Cs , ^{60}Co , and a mixed $^{155}\text{Eu}/^{22}\text{Na}$ source, to establish a linear relationship between channel numbers and corresponding energies, thus enabling the identification of radionuclides present in the analyzed samples.

The activity concentrations and their uncertainties were determined based on the net area under the photopeak, considering the statistical uncertainties provided by the Genie 2000 software. Equations 1 and 2 were used to calculate the activity concentration (AC) and the Minimum Detectable Activity (MDA), respectively:

$$AC = \frac{NL}{\varepsilon \cdot P\gamma \cdot m \cdot t} \quad (1)$$

$$MDA = \frac{2.71 + 4.66\sigma}{\varepsilon \cdot P\gamma \cdot m \cdot t} \quad (2)$$

Where:

AC = Activity concentration (in Bq/L);

N_L = The net area under the photoelectric peak;

ε = Counting efficiency for a specific energy (γ);

m = Sample volume (in L);

t = Counting time (in seconds);

Pγ = Emission probability of the corresponding gamma radiation;

σ = Standard deviation of the background count.

The detection efficiency (ε) was determined using LabSOCS (Laboratory Sourceless Calibration Software), developed by Canberra. The total spectral acquisition time for each sample was 28,800 seconds.

The activity concentration of ^{228}Ra was determined based on the 911.1 keV energy peak of ^{228}Ac (emission probability of $25.8 \pm 0.4\%$). For ^{226}Ra , the 609.3 keV line of ^{214}Bi was used (emission probability of $45.5 \pm 0.2\%$). The activity of ^{40}K was determined from the 1460.8 keV line (emission probability of $10.66 \pm 0.13\%$). The MDA value considered in this study corresponds to a 95% confidence level.

3 Perspectives and Final Considerations

As this study is still in its initial phase, only a small portion of the samples has been preliminarily analyzed by gamma spectrometry. In the samples analyzed so far, no activity concentrations exceeding the reference limits established by international bodies — such as the International Commission on Radiological Protection (ICRP) — have been detected. Although these initial results suggest a low probability of radiological risk associated with water ingestion at the sampled locations, it is important to note that the complete dataset is still being acquired and validated.

As the analyses progress and considering the adopted methodology, it is expected that radionuclides emitting gamma radiation, such as ^{40}K , ^{226}Ra , and ^{228}Ra , will be detected at concentrations consistent with those reported in the literature for regions with similar geological characteristics. Groundwater samples, due to their increased contact with rocks and sediments, tend to present higher gamma activity concentrations when compared to surface or treated waters.

Furthermore, it is expected that there will be an association between the previously analyzed physicochemical parameters, such as electrical conductivity, salinity, and total dissolved solids, and gamma radiation levels, since these variables may reflect natural processes of radionuclide solubilization. Confirming this hypothesis could contribute to the identification of indirect markers of radiological risk in aquatic environments.

Although the experimental results are still under analysis, the purpose of this work is to contribute to the radiological characterization of drinking water, providing a more comprehensive understanding of population exposure to environmental ionizing radiation.

The final dataset is expected to support future investigations into the health impacts of natural radioactivity and to inform public policies aimed at water quality monitoring and safety, particularly in regions with potential geological risks or a history of contamination.

Acknowledgements

The authors thank FAPERJ, CNPq, CAPES and MCTI for the financial support provided. Without the help of these institutions, this work would not have been possible.

References

- AKSO. (2003). *Multiparameter meter - AK88*. Akso Produtos Eletrônicos Ltda. Available at <https://akso.com.br>
- Alharbi, T.; Aboulnasr, A.; Alzahrani, A.; Alzahrani, M. Assessment of water quality and radioactivity levels in Ismailia Canal, Egypt. *Environmental Science and Pollution Research*, [s.l.], 2024. Available at: <https://link.springer.com/article/10.1007/s11356-024-32672-9>
- Brazil. Ministry of Health. (2017). *Consolidation Ordinance No. 5, of September 28, 2017. Consolidation of standards on health actions and services of the Unified Health System*. Official Gazette of the Union, Brasília, DF, October 3, 2017. Supplement, pp. 360-568. Available at <https://bvsms.saude.gov.br>
- Cardoso, E. *Educational booklet: Radioactivity*. National Nuclear Energy Commission (CNEN). Available at <http://radioatividade.cnen.gov.br>
- Comissão Nacional de Energia Nuclear (CNEN). Radiation Protection and Safety Requirements for Nuclear Facilities (Standard CNEN-NN-3.01). Brasília: CNEN, 2012. Available at: https://www.gov.br/cnen/pt-br/acao-rapido/normas/grupo-3/grupo3-pr301_12.pdf. Accessed on: Apr. 13, 2025.

- National Nuclear Energy Commission (CNEN). (2020). *Regulatory Position 3.01/012:2020 - Investigation and Reference Levels for Radioactivity in Drinking Water*. Rio de Janeiro: CNEN. Available at <http://cnen.gov.br>
- Division of Waterborne and Foodborne Diseases (DDTHA), Center for Epidemiological Surveillance (CVE), CCD/SES-SP. (2009). *Waterborne or water-transmitted diseases*. Available at <https://www.cve.saude.sp.gov.br>
- Giovanardini, C. M. A.; Lemos, D. S.; Ferraz, E. S.; Lopes, R. T. Natural radioactivity in water and the influence of anthropogenic activities. *Archives of Industrial Hygiene and Toxicology*, Zagreb, v. 63, n. 4, p. 362–373, 2012. Available at: <https://scielosp.org/article/aiss/2012.v48n4/362-373>
- Instituto Adolfo Lutz (São Paulo). (2008). *Physical-chemical methods for food analysis* (4th ed.). São Paulo: Instituto Adolfo Lutz. Available at <http://analisedealimentosial.com>
- Kannel, P. R.; Lee, S.; Lee, Y.-S.; Kanel, S. R.; Khan, S. P. Application of Water Quality Indices and Dissolved Oxygen as Indicators for River Water Classification and Urban Impact Assessment. *Environmental Monitoring and Assessment*, v. 132, n. 1–3, p. 93–110, 2007. DOI: 10.1007/s10661-006-9505-1. [PubMed+2ResearchGate+2SpringerLink+2](#)
- Mateo-Sagasta, J., Marjani Zadeh, S., & Turrall, H. (2018). *More people, more food, worse water?* In *A Global Review of Water Pollution from Agriculture*. FAO, Rome. Available at <https://www.fao.org>
- Picetti, R. (2022). *Nitrate and nitrite contamination in drinking water and cancer risk: A systematic review with meta-analysis*. *Environmental Research*, 210. Available at <https://www.ed.ac.uk>
- Quiroz Damian, B. G. (2018). *Characterization of radionuclides present in hot springs in the Chimborazo province and estimation of the equivalent dose rate for the population*. Graduation Thesis (BIOPHYSICS) – Escuela Superior Politécnica de Chimborazo, Riobamba.
- Radiation. Oswaldo Cruz Foundation (Fiocruz). Available at <https://www.fiocruz.br>
- Riviere, J. E. (2006). *Biological Concepts and Techniques in Toxicology* (1st ed.). New York: Taylor & Francis.
- Sang, C., et al. (2021). *Gross alpha and beta measurements in drinkable water from seven major geographical regions of China and the associated cancer risks*. *Ecotoxicology and Environmental Safety*. Available at <https://www.sciencedirect.com>
- Shah, T.; Sharma, A.; Kumar, R. Sources and Leaching of Nitrate Contamination in Groundwater. *ResearchGate*, 2020. Available at: https://www.researchgate.net/publication/340115969_Sources_and_Leaching_of_Nitrate_Contamination_in_Groundwater. Accessed on: Apr. 13, 2025.
- United States Environmental Protection Agency (EPA). Radionuclide Basics: Radium. Washington, D.C.: EPA, 2023. Available at: <https://www.epa.gov/radiation/radionuclide-basics-radium>. Accessed on: Apr. 13, 2025.
- United States Environmental Protection Agency. (2002). *Radionuclides in Drinking Water: A Small Entity Compliance Guide*. Available at <https://www.epa.gov>

World Health Organization. (2002). *National cancer control programmes: policies and managerial guidelines* (2nd ed.). Geneva: World Health Organization. Available at <http://www.who.int>

World Health Organization. (2017). *Guidelines for Drinking-water Quality*. Available at <https://www.who.int>



Growth Inhibition of *S. cerevisiae* BY4741 Exposed to Gold Nanoparticles and Low-Energy Radiation

B. Gallotti¹, L. F. Araújo¹, A. F. Oliveira², P. L. Square², C. Engler², E. M. B. Sousa², M. J. Neves², J. Ferreira³, M. G. L. Rocha⁴, F. S. Martins⁵, T. C. F. Fonseca^{1*}

¹ Departamento de Engenharia Nuclear, Universidade Federal de Minas Gerais (UFMG), Av. Antônio Carlos, 6627, Pampulha, 31270-901, Belo Horizonte, MG, Brazil

² Centro de Desenvolvimento da Tecnologia Nuclear (CDTN) Av. Antônio Carlos, 6627, Campus UFMG – Pampulha, 31270-901, Belo Horizonte, MG, Brazil

³ Instituto Mário Penna, R. Joaquim Cândido Filho, 91 - Luxemburgo, 30380-420, Belo Horizonte, MG, Brazil

⁴ Departamento de Análises Clínicas e Toxicológicas, Faculdade de Farmácia, Universidade Federal de Minas Gerais (UFMG), Av. Antônio Carlos, 6627, Pampulha, 31270-901, Belo Horizonte, MG, Brazil

⁵ Departamento de Microbiologia, Instituto de Ciências Biológicas, Universidade Federal de Minas Gerais (UFMG), Av. Antônio Carlos, 6627, Pampulha, 31270-901, Belo Horizonte, MG, Brazil

* tcff01@gmail.com

Abstract

Radiotherapy is an effective treatment for solid tumors, but the side effects associated with it negatively impact the quality of life of patients. Strategies to optimize and reduce the adverse effects of treatment are needed and the use of nanoparticles, for example gold nanoparticles (AuNPs), might be an alternative. Once submitted to ionizing radiation, AuNPs increase the radiation dosage directed to neoplastic cells and minimize the damage to adjacent healthy tissue and improve the outcome of treatment. Although there are many studies associating nanoparticles and radiation, how nanoparticles work remain unclear. The Monte Carlo Modelling Expert Group has been developed a study simulating the use of AuNPs with different energetic beams with different protocols. Now, the group is interested in perform a experimental study to validate the computational models using yeasts. So, in the present study, we aim to investigate the effect of radiosensitization of AuNPs in low-energy X-ray beam using *Saccharomyces cerevisiae* BY4741 strain. Yeasts were grown in Yeast, Glucose and Peptone (YPG) broth containing different concentrations of AuNPs to evaluate the maximum concentration of nanoparticles that do not compromise the microorganism growth. Next, yeast with nanoparticles were irradiated in a clinical X-ray device, which is a low-energy X-ray beam, in order to obtain a curve of viable cells in function of AuNPs concentration to evaluate the radiosensitizing effect of nanoparticles. This work contributes to improve the radiation therapy protocols and, therefore, reduce the side effects of the neoplastic treatment.

Keywords: *Saccharomyces cerevisiae*, nanoparticles, radiation, radiosensitization.

1 Introduction

Radiotherapy is a widely used therapeutic modality in cancer treatment, based on the application of high doses of ionizing radiation to destroy tumor cells or inhibit their proliferation. By damaging the DNA of neoplastic cells, radiation prevents their multiplication, leading to cell death and subsequent removal by the body. This approach can be used alone or in combination with other therapies such as surgery, chemotherapy, and immunotherapy, thereby increasing treatment efficacy. In addition to its curative role, radiotherapy is also employed palliatively to relieve symptoms in patients with advanced tumors. Despite its well-established benefits, the technique still faces significant limitations, including the low selectivity of radiation, intrinsic or acquired tumor cell resistance, and adverse effects on surrounding healthy tissues (National Cancer Institute (NIH), 2025).

In this context, the use of nanoparticles has emerged as a promising strategy to enhance the efficacy and selectivity of radiotherapy. Due to their unique physicochemical properties, such as high surface-to-volume ratio, functionalization capacity, and selective accumulation in tumor tissues, nanoparticles act as radiosensitizers, intensifying the effects of radiation on cancer cells while reducing damage to normal tissues. Mechanisms such as the generation of reactive oxygen species (ROS), inhibition of DNA repair systems, interference with the cell cycle, and modulation of the tumor microenvironment help explain their synergistic action with radiotherapy (Haque et al., 2023; Parwaie et al., 2025).

Among the various types of nanoparticles studied, gold nanoparticles (AuNPs) stand out due to their high atomic number (Z), which favors processes like the photoelectric effect and Auger electron emission, thereby increasing the locally absorbed radiation dose. Preclinical studies have demonstrated that AuNPs amplify DNA damage, promote ROS accumulation, and reduce tumor volume in animal models, reinforcing their potential as radiosensitizers. However, the detailed mechanisms of interaction between these nanoparticles and radiation are not yet fully understood, and experimental validation of these effects is essential to enable their clinical application (Singh et al., 2018; Cazzanelli et al., 2018).

In this regard, the experimental model *Saccharomyces cerevisiae* has proven to be an effective tool for studying the biological effects of nanoparticles in combination with radiation Oliveira et al., 2023. Despite being a unicellular organism, *S. cerevisiae* shares functional conservation with human genes involved in cell cycle regulation, DNA repair, apoptosis, and oncogenic signaling, allowing the investigation of fundamental tumorigenesis mechanisms in a simple, reproducible, and low-cost system. Yeast also allows for high-throughput experiments and functional analysis of human genes through heterologous expression, in addition to enabling initial toxicity and therapeutic response testing (Mohammadi et al., 2015).

Thus, the present study aimed to evaluate the effects of exposing *S. cerevisiae* to low-energy radiation in the presence of gold nanoparticles, to investigate the potential radiosensitizing impact of these particles in an accessible and highly controllable experimental model. The data generated from this well-characterized and straightforward system will not only contribute to the validation of computational models developed by the research group but will also provide preliminary experimental evidence for the improvement of diagnostic and therapeutic strategies in radiotherapy.

2 Material and Methods

2.1 Microorganism

The yeast *Saccharomyces cerevisiae* BY4741, strain was stored in a solution composed of 1% yeast extract, 2% peptone, and 20% glycerol, and kept at -80°C in 2 mL cryovials. For all assays, the yeast cells were grown in YPG broth (1% yeast extract, 2% peptone, and 2% glucose) at room temperature for 24 hours, with constant shaking at 150 rpm using an orbital shaker (OrbiCultTM Shaker, Esco, Singapore). A growing

curve was obtained from experiments being performed measuring the sample absorbance every hour during 24 h.

2.2 Nanoparticle Characterization

Gold nanoparticles (AuNPs) were synthesized using the coprecipitation method at CDTN/CNEN (Meireles et al., 2021, 2024; Nikoobakht and El-Sayed, 2003). The synthesis process involved a 25 kGy irradiation dose, resulting in nanoparticles with an average size of 26.9 ± 6.0 nm and a Zeta potential of +56.8 mV. The nanoparticle size and shape were assessed using FEG-SEM model SIGMA VP, Carl Zeiss Microscopy (Germany). Nanoparticle spectra was measured using UV-Vis spectrophotometer UV-2450, Shimadzu (Japan). The nanoparticles were prepared at a concentration of 3 mg/mL and were diluted to the following test concentrations: 0.25, 0.5, 1, 3, 5, 25, 50, and 100 $\mu\text{g/mL}$.

2.3 Cell Viability Assay

To evaluate the potential toxic effects of gold nanoparticles (AuNPs) on *Saccharomyces cerevisiae* BY4741, it was performed a growth inhibition assay. Initially, yeast cells were cultured in YPG medium (yeast extract, peptone, and glucose) for 24 hours at room temperature under constant agitation (150 rpm, KS 260 Basics, IKA) in the presence of various AuNP concentrations. Following this incubation, the cultures were centrifuged at 6000 rpm for 10 minutes, the supernatant was discarded, and the cell pellet was resuspended in fresh YPG medium. The suspension was homogenized and incubated for an additional 4 hours under the same conditions.

Subsequently, 2% of this yeast suspension was inoculated into fresh YPG medium containing increasing concentrations of AuNPs (0.25, 0.5, 1, 3, 5, 25, 50, and 100 $\mu\text{g/mL}$). At designated time points (0, 4, 8, 16, and 24 hours), aliquots were collected and stained with trypan blue and yeast concentration was calculated using a Neubauer chamber. In summary, 100 μL of yeast culture was collected and mixed with 100 μL of 0.4% trypan blue solution to distinguish viable from non-viable cells. A 20 μL aliquot of the mixture was loaded onto the chamber, and cells were counted under a light microscope at 40 $\times$ magnification.

Cell counts were performed in four large squares (each subdivided into 25 smaller squares) of the Neubauer chamber. Only viable (unstained) cells were counted. The average number of cells per square was calculated and used to determine the concentration of viable yeast cells per milliliter (cells/mL) using the Equation 1:

$$Concentration_{\text{cells/mL}} = \text{Average count per square} \times \text{Dilution factor} \times 10^4 \quad (1)$$

2.4 Effect of AuNPs and X-ray Irradiation on Yeast Viability

Scenarios of Irradiation

For the irradiation two main scenarios were proposed. Figure 2 shows the setup of the experimental scenarios. The distance between the radiation source and the samples was 50 cm. All groups were Cell viability was assessed at 0, 24, 48, and 72 hours post-irradiation. At each time point, aliquots from the cultures were serially diluted in 0.9% saline solution and plated onto YPG agar. The plates were incubated at room temperature for 48 hours. Following incubation, colony-forming units (CFU) were counted, and the yeast concentration was calculated using the Equation 2:

$$\text{CFU/mL} = \frac{\text{Number of colonies} \times \text{Dilution factor}}{\text{Inoculated volume (mL)}} \quad (2)$$

The two irradiation protocols are described as follow:

- First experiment: a total of four samples of *Saccharomyces cerevisiae* BY4741 with different AuNPs concentrations: CT (control), 0.25, 0.50, 1.00 $\mu\text{g/mL}$, were exposed to X-ray irradiation at a dose of 100 mGy and an beam energy of 70 kVp.
- Second experiment: ten samples (five CT and five samples) of BY4741 with fixed AuNPs concentrations, CT and 1.00 $\mu\text{g/mL}$ of AuNPs and different irradiation doses were applied, 20, 40, 60, 80 and 100 mGy. The nanoparticle-treated groups were compared to their respective controls that received the same dose of radiation (for example, the CT 20 mGy group was compared to the 1 $\mu\text{g/mL}$ 20 mGy group).

During irradiation, samples were placed in T-25 flask and a analysis was conducted to examine the impact of attenuation in the T-25 flask wall on measurement outcomes. This analysis was performed under a configuration of 70 kVp and 40 mAs, utilizing a detector with an empty flask and without the flask.

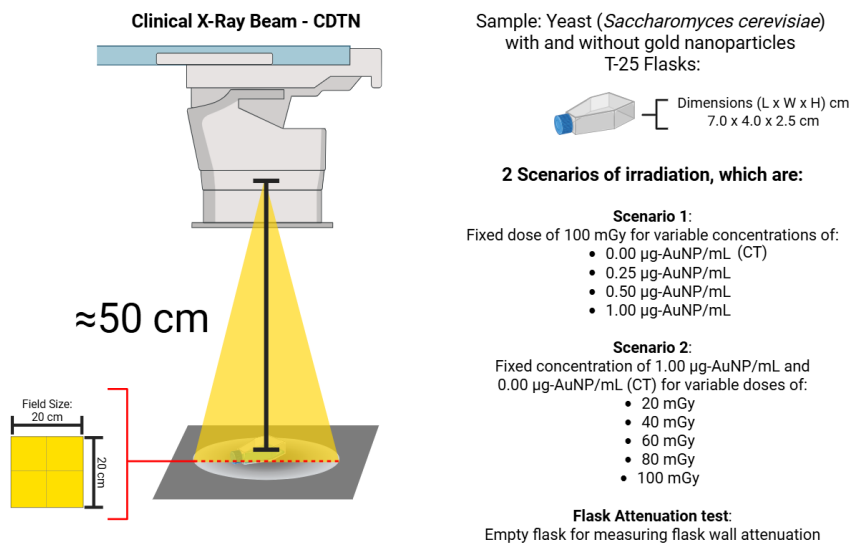


Figure 1: Scheme with the two irradiation scenarios performed in this study

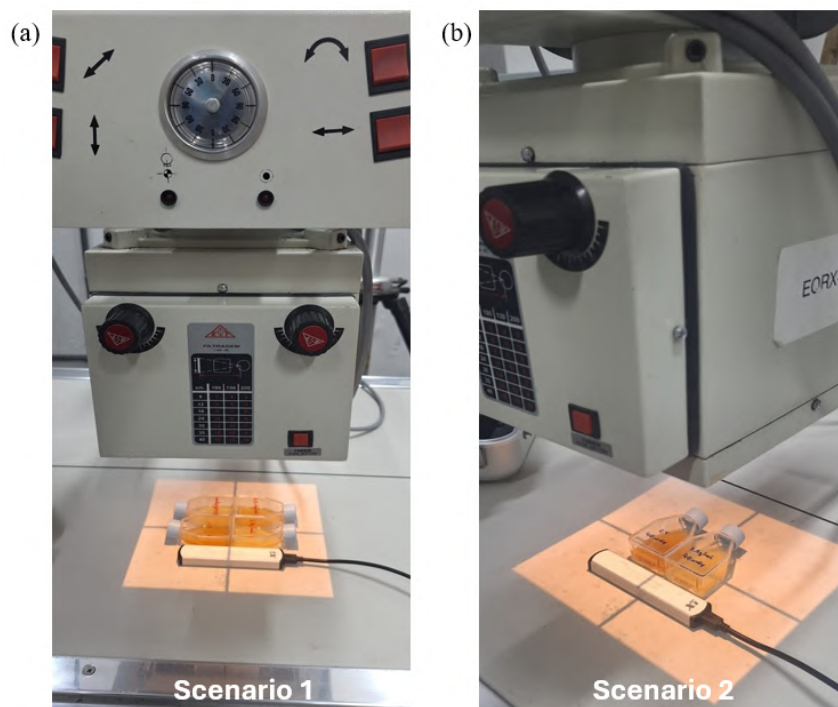


Figure 2: Clinical X-ray unit VMI (Model Pulsar 800 Plus) from CDTN/CNEN and the samples being irradiated in Scenario 1 (a) and Scenario 2 (b) along with a detector device to characterize the X-ray beam

Samples were irradiated using the clinical X-ray unit VMI (Model Pulsar 800 Plus) from CDTN/CNEN. Table 1 summarizes the irradiation protocol parameters used for sample exposure. The technical settings and corresponding air kerma values for each experimental design are presented. For each configuration, the tube voltage (kVp), tube loading (mAs), and number of radiation pulses are provided. The total air kerma refers to the cumulative value over all pulses, while the mean air kerma per pulse corresponds to the average dose delivered in a single pulse. The half-value layer (HVL), expressed in mmAl, indicates the beam quality. All parameters were measured simultaneously with the sample during irradiation using an solid state detector Unfors RaySafe Xi (Model 8202031-H, Unfors RaySafe, Inc., Billdal, Sweden).

Table 1: Technical parameters and corresponding air kerma values for the X-ray exposure in the experimental designs.

Scenario	kVp	mAs	Number of Pulses	Total Air Kerma (mGy)	Mean Air Kerma per Pulse (mGy)	HVL (mmAl)
1	70	40	5	112.65	22.53	1.77
	100	18	1	20.66	20.66	—
	100	32	1	39.47	39.47	—
2	100	26	2	61.03	30.52	—
	100	32	2	74.67	37.335	—
	100	40	2	96.85	48.425	2.90

3 Results

3.1 Growth Curve of *S. cerevisiae* BY4741

Figure 3 represents the growth curve for *S. cerevisiae* BY4741 starting from a 1% or 2% inoculum. It is possible to see that the yeast is in the stationary phase between 13 and 15 h after inoculation. In this phase, the yeast concentration is at its highest, and the experiments were performed when the yeast entered this phase.

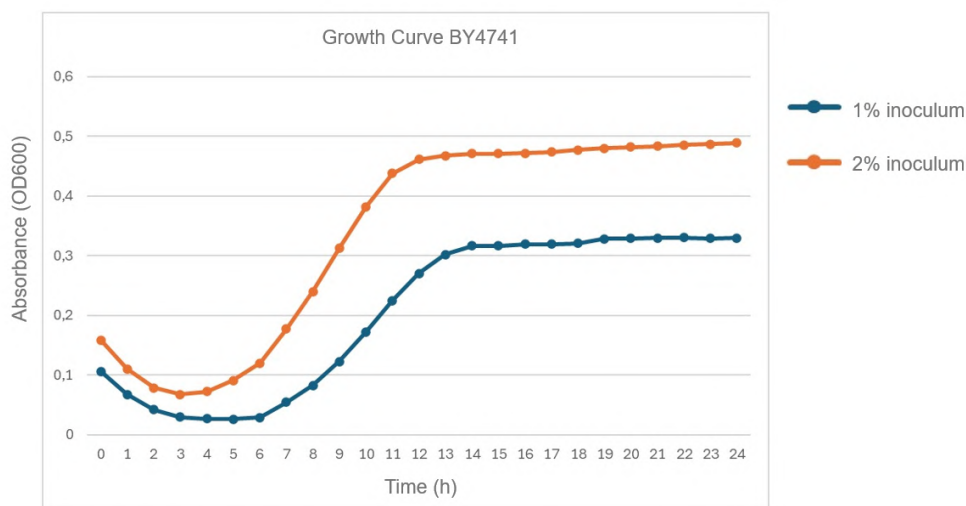


Figure 3: Growth curve of *S. cerevisiae* BY4741 from a 1% inoculum (blue) and 2% inoculum (red)

3.2 Nanoparticle characterization

Transmission electron microscopy (TEM) analysis revealed that the synthesized AuNPs were predominantly spherical, with an average diameter of 26.98 nm (Figure 4 (a)). The UV-Vis absorbance spectrum of the nanoparticles is shown in Figure 4 (b), exhibiting a surface plasmon resonance (SPR) peak at approximately 530 nm, which is characteristic of gold nanoparticles. The zeta potential of the colloidal suspension was measured as +56.8 mV, indicating high colloidal stability due to strong electrostatic repulsion between particles which can be seeing in Figure 4 (c).

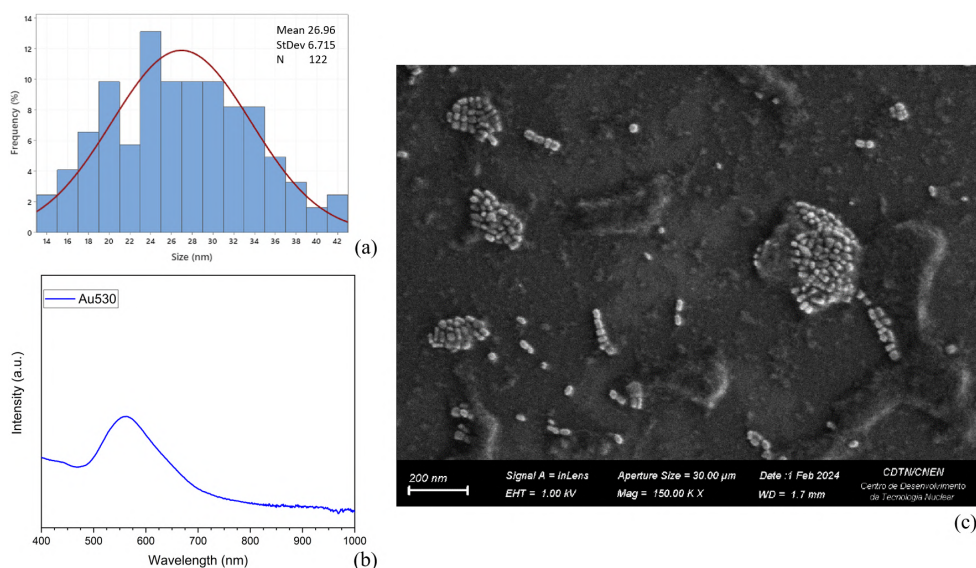


Figure 4: Nanoparticle characterization: (a) Nanoparticle size; (b) UV-Vis spectra; (c) Transmission electronic microscopy (TEM) image.

3.3 Cell Viability Assay: *Saccharomyces cerevisiae* BY4741 + AuNPs

To evaluate the toxicity of AuNPs toward *Saccharomyces cerevisiae* BY4741, yeast cells were exposed to different concentrations of AuNPs: 0.25, 0.5, 1, 3, 5, 25, 50, and 100 μ g/mL. Cell growth was monitored over time by counting viable cells using a Neubauer chamber at 0, 4, 8, 16, and 24 hours.

Table 2 shows the results obtained for the different concentrations of cells using Equation 1. It is possible to see that, no viable cells were detected at concentrations of 25, 50, and 100 μ g/mL, indicating that these concentrations were toxic to the yeast. Concentrations of 0.25, 0.5, and 1 μ g/mL exhibited growth rates comparable to the control group (CT), suggesting that these levels of AuNPs did not affect yeast proliferation. In contrast, exposure to 3 and 5 μ g/mL resulted in a reduced growth rate relative to the control, as evidenced by lower cell counts over time. Based on these findings, the concentration of 1 μ g/mL were identified as the highest non-toxic dose for the BY4741 and it was the selected AuNP concentration for subsequent experiments.

Table 2: Cell counting of *S. cerevisiae* BY4741 after incubation with different concentrations of AuNPs.

	Viable cells								
	CT	0.25 μ g/mL	0.5 μ g/mL	1 μ g/mL	3 μ g/mL	5 μ g/mL	25 μ g/mL	50 μ g/mL	100 μ g/mL
Time 0	0	0	0	0	0	0	0	0	0
Time 4h	1.00E+07	2.04E+07	1.90E+07	1.60E+07	1.92E+06	7.06E+06	0	0	0
Time 8h	8.20E+07	6.52E+07	6.88E+07	1.58E+07	8.50E+06	9.80E+06	0	0	0
Time 20h	1.16E+08	1.17E+08	1.26E+08	1.56E+08	4.60E+06	3.08E+07	0	0	0
Time 24h	1.52E+08	1.96E+08	1.70E+08	1.37E+08	1.04E+07	1.41E+07	0	0	0

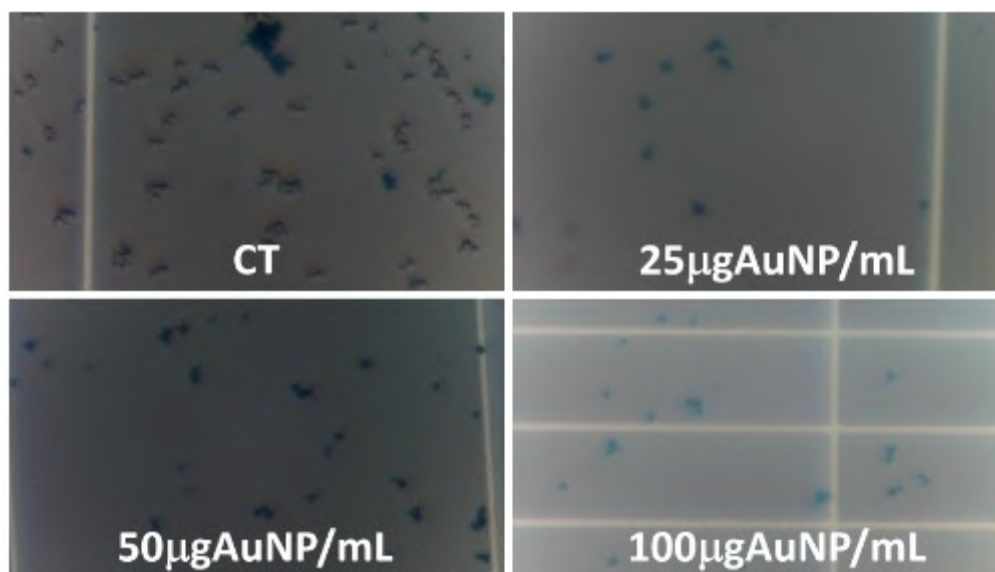


Figure 5: Microscope image of yeast *S. cerevisiae* BY4741 stained with tripan blue after irradiation with clinical X-ray unit VMI (Model Pulsar 800 Plus) from CDTN/CNEN. Viable and nonviable cell are not stained and blue-stained, respectively. 40x magnification

3.4 Irradiation of *S. cerevisiae* BY4741

During irradiation, samples were placed in a T-25 flask, and an analysis was conducted to evaluate the impact of attenuation by the flask wall on measurement outcomes. This analysis was performed using a configuration of 70 kVp and 40 mAs, employing a detector both with an empty flask and without the flask. The results indicated a variation of 2.9% (equivalent to 0.69 mGy) in the measured dose. This discrepancy is considered negligible, suggesting that the influence of the flask wall on the dose measurement is minimal.

The two irradiation scenarios were proposed for the irradiation protocol performed. The first experiment, a total of four samples of *S. cerevisiae* BY4741, placed in T-25 flask, with different AuNPs concentrations: CT (control), 0.25, 0.50, 1.00 µg/mL, were exposed to X-ray irradiation at a dose of 100 mGy and an beam energy of 70 kVp. It was observed that for all groups, the higher the concentration of AuNPs used, the lower the yeast concentration measured compared to the control group.

The second experiment, ten samples (five CT and five samples) of BY4741 with fixed AuNPs concentrations, CT and 1.00 µg/mL of AuNPs and different irradiation doses were applied, 20, 40, 60, 80 and 100 mGy. The nanoparticle-treated groups were compared to their respective controls that received the same dose of radiation (for example, the CT 20 mGy group was compared to the 1 µg/mL 20 mGy group). The results showed that, as in scenario 1, the nanoparticle affects yeast concentration compared to the control group. The group with the nanoparticle showed a lower yeast concentration than the control group. However, there was no significant difference in yeast concentration between the different radiation doses applied, neither within the control groups nor the nanoparticle groups.

Figure 6 shows the results obtained for both experimental scenarios using the clinical X-ray unit VMI (Model Pulsar 800 Plus) from CDTN/CNEN.

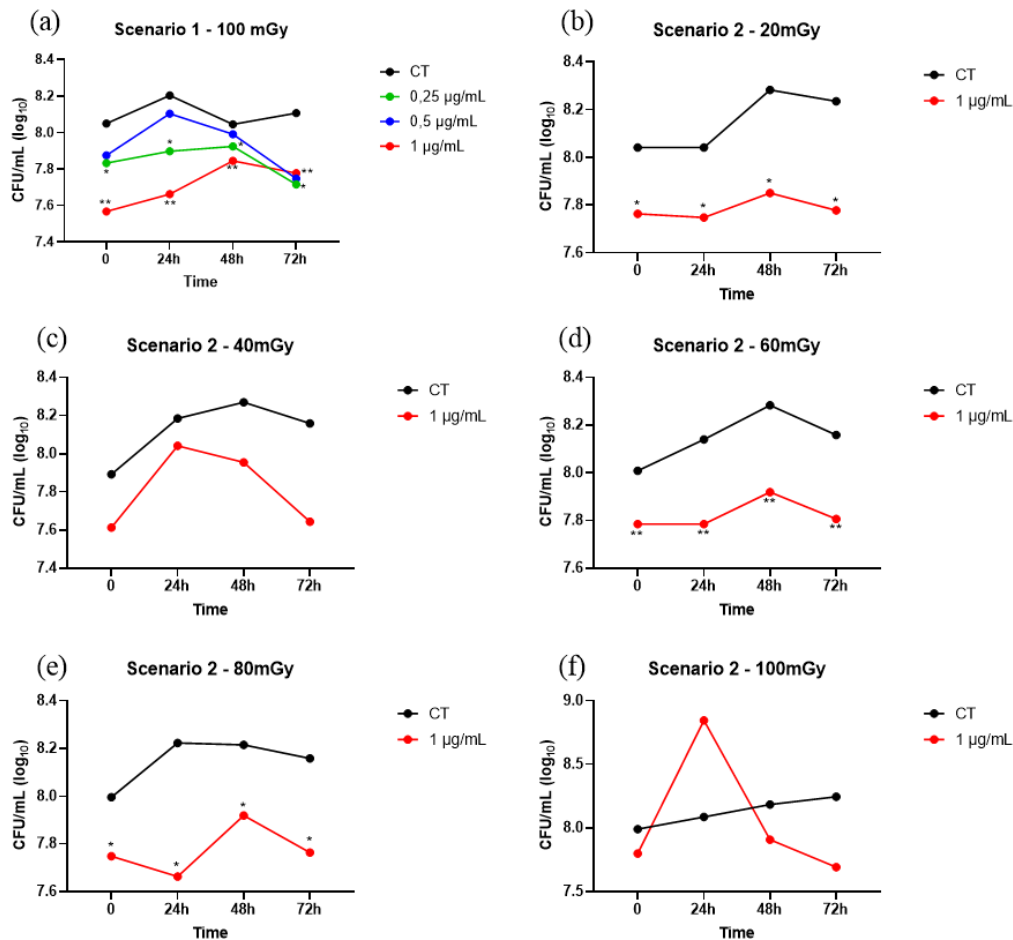


Figure 6: Irradiation Scenarios: (a) Scenario 1: constant dose (100mGy) and energy (70kVp) with different concentration of AuNPs (0, 0.25, 0.5 and 1 µg/mL); (b)-(f) Scenario 2: Constant AuNP concentration (1 µg/mL) with different X-ray doses (20, 40, 60, 80, and 100 mGy).

The radiosensitizer effect of AuNPs was evaluated in two different scenarios. The results presented in Figure 6 demonstrate the effect of gold nanoparticles (AuNPs) at different concentrations and under varying X-ray irradiation doses on the viability of *Saccharomyces cerevisiae* BY4741, assessed through CFU/mL (log scale) at different time points post-irradiation (0, 24, 48, and 72 hours).

Figure 6(a) shows the results from scenario 1 which it was fixed the dose of 100 mGy and it was varying the AuNP Concentrations. In scenario 1, yeast cells with different concentrations of AuNPs underwent a low-energy X-ray beam with a dose of 100 mGy and 70 kVp. Here, we observed that increasing concentrations of AuNPs (0.25, 0.5, and 1 µg/mL) led to a decrease in yeast cell concentration following irradiation, compared to the non-treated control group. Although these AuNP concentrations were previously shown to be non-toxic in the absence of radiation, their combination with low-energy X-rays resulted in reduced cell viability. This outcome supports the expected radiosensitizing role of gold nanoparticles; the observed reduction in yeast growth suggests that AuNPs can enhance the biological impact of radiation, allowing for a more localized and effective response even at lower radiation doses. The control group (CT) showed stable CFU levels over time. The presence of AuNPs resulted in a dose-dependent reduction in CFU/mL, most notably with the highest concentration (1.00 µg/mL), indicating a radiosensitizing effect. A clear decrease in cell viability at 48 h and 72 h was observed for cells treated with 1.00 µg/mL AuNPs. Lower concentrations

(0.25 and 0.50 $\mu\text{g/mL}$) also showed a reduction in viability but to a lesser extent. These results suggest that AuNPs enhance radiation-induced cytotoxicity in a concentration-dependent manner, supporting their potential use as radiosensitizers.

Figure 6(b to f) show the results of scenario 2. In scenario 2, *S. cerevisiae* BY4741 was exposed to varying doses of low-energy X-ray radiation (20, 40, 60, 80, and 100 mGy at 100 kVp) to assess the dose-dependent effects of radiation in the presence of gold nanoparticles (AuNPs). Consistent with the results obtained in the first experiment, a reduction in yeast concentration was observed in the groups treated with AuNPs compared to their respective control that was irradiated with the same dose but without the AuNP, reinforcing the hypothesis that AuNPs exert a radiosensitizing effect. This effect is likely due to enhanced local energy deposition around the nanoparticles during irradiation, leading to increased cellular stress or damage. However, no significant differences in yeast concentration were found among the groups exposed to different radiation doses, either in the presence or absence of AuNPs. This may be explained by the relatively low range of radiation doses applied in this study. It is possible that, while AuNPs are capable of amplifying the effects of radiation, the incremental increases in dose (from 20 to 100 mGy) were not sufficient to produce a measurable biological gradient in this model. The control groups (black curves) consistently showed higher CFU values than the AuNP-treated groups (red curves) across all doses and time points, indicating that AuNPs potentiate the effects of X-ray irradiation. At 20 mGy (b) and 40 mGy (c), slight reductions in CFU/mL were observed in treated groups compared to controls, with minimal but consistent differences, suggesting initial signs of radiosensitization at these lower doses. At 60 mGy (d), a more pronounced reduction in CFU/mL was observed in the AuNP-treated group, especially at 48 h, indicating a stronger synergistic effect between radiation and AuNPs at this dose. At 80 mGy (e) a significant reduction in CFU/mL was observed at 24 h, followed by a partial recovery at 48 h and another decline at 72 h, indicating a possible biphasic response to the combined treatment. At 100 mGy (f) a sharp reduction in CFU/mL at 24 h in the AuNP-treated group compared to control suggests an acute cytotoxic effect. Interestingly, CFU levels increased at 48 h and then dropped again at 72 h, indicating dynamic cellular responses post-treatment. These findings confirm a synergistic effect between AuNPs and X-ray exposure, with the greatest reduction in cell viability occurring at higher doses (60–100 mGy). The temporal profiles suggest a delayed cellular response, with maximal effects generally observed at 48–72 h post-irradiation.

Importantly, the observed reduction in cell concentration in the AuNP-treated groups highlights the potential of gold nanoparticles to act as radiosensitizers by enhancing the biological impact of ionizing radiation. In the context of cancer therapy, this property is highly desirable, as it may allow for effective tumor targeting using lower radiation doses, thereby minimizing damage to surrounding healthy tissue. Future studies should explore higher radiation doses or different exposure protocols to determine whether a dose-dependent radiosensitizing effect can be more clearly delineated in this experimental model.

4 Conclusion

This study demonstrated that gold nanoparticles exert dose-dependent cytotoxicity on *Saccharomyces cerevisiae* BY4741, with concentrations of 25 $\mu\text{g/mL}$ and above proving lethal, while concentrations up to 1 $\mu\text{g/mL}$ did not affect yeast growth. Based on this threshold, further analyses were conducted using low, non-toxic concentrations of AuNPs in combination with low-energy X-ray irradiation. Although no statistically significant differences in yeast viability were observed among the irradiated groups, a reduced cell concentration was noted when compared irradiated groups with control groups, suggesting a potential radiosensitizing effect. These findings contribute to the preliminary steps in validating computational models developed by our research group, aimed at optimizing radiation therapy protocols. The use of yeast as a model organism provides a cost-effective and scalable approach to studying the interactions between nanoparticles and ionizing radiation. Ultimately, this research supports the broader goal of improving radiotherapy outcomes by enhancing the precision of radiation delivery while minimizing adverse effects on

healthy tissue.

References

- Cazzanelli G., Pereira F., Alves S., Francisco R., Azevedo L., Dias Carvalho P., Almeida A., Côrte-Real M., Oliveira M., Lucas C., Sousa M., and Preto A. (2018) The yeast *saccharomyces cerevisiae* as a model for understanding ras proteins and their role in human tumorigenesis. *Cells* 7:14
- Haque M., Shakil M., and Mahmud K. (2023) The promise of nanoparticles-based radiotherapy in cancer treatment. *Cancers (Basel)* 15:1892
- Meireles I. B. d. C. J., Oliveira A. F., Rodrigues M. A., and Sousa E. M. B. d. (2024) A platform of gold nanoparticles coated with silica as controlled drug delivery for application in cancer treatment. *Journal of Materials Science* 59:22181–22205
- Meireles I. B. d. C. J. et al. (2021) Synthesis and characterization of gold nanorods coated by mesoporous silica mcm-41 as a platform for bioapplication in photohyperthermia. *Nanotechnology* 32:505720
- Mohammadi S., Saberidokht B., Subramaniam S., and Grama A. (2015) Scope and limitations of yeast as a model organism for studying human tissue-specific pathways. *BMC Systems Biology* 9:96
- National Cancer Institute (NIH) (2025) Radiation therapy to treat cancer. <https://www.cancer.gov/about-cancer/treatment/types/radiation-therapy>. updated 2025 May 15 [cited 2025 Jul 15]
- Nikoobakht B. and El-Sayed M. A. (2003) Preparation and growth mechanism of gold nanorods (nrs) using seed-mediated growth method. *Chemistry of Materials* 15:1957–1962
- Oliveira S., Campos L., Amaral M. et al. (2023) Evaluation of a functional craft wheat beer fermented with *saccharomyces cerevisiae* ufmg a-905 to treat salmonella typhimurium infection in mice. *Probiotics and Antimicrobial Proteins* 15:1180–1192
- Parwaie W., Molazadeh M., and Mortezaazadeh T. (2025) Evaluation of the impact of energy, radiation type, and concentration on dose enhancement by gold nanoparticles. *Cancer Treatment and Research Communications* 43:100933. epub 2025 Apr 28
- Singh P., Pandit S., Mokkapati V., Garg A., Ravikumar V., and Mijakovic I. (2018) Gold nanoparticles in diagnostics and therapeutics for human cancer. *International Journal of Molecular Sciences* 19:1979

Cathodoluminescence behavior of PVDF films doped with CuSO₄ under high-dose gamma irradiation

G. P. Cardoso^{1*}, C. P. Ricardo¹, J. M. Geraldo^{2,3,4}, A. S. M. Batista^{1,2}

¹ Department of Nuclear Engineering - Federal University of Minas Gerais, 31270-901, Belo Horizonte, MG, Brazil.

² Department of Anatomy and Imaging - Federal University of Minas Gerais, 30130-100, Belo Horizonte, MG, Brazil.

³ Luxemburgo Hospital - Mario Penna Institute, 30380-420, Belo Horizonte, MG, Brazil.

⁴ Alberto Cavalcanti Hospital - FHEMIG, 30730-540, Belo Horizonte, MG, Brazil.

* gabrielapontesc@gmail.com

Abstract

Polymer-based luminescent materials have emerged as promising candidates for applications in optoelectronics, sensing, and radiation detection due to their flexibility, low cost, and tunable properties. In this study, the luminescent behavior of polyvinylidene fluoride (PVDF) films—both pure and doped with copper sulfate pentahydrate (CuSO₄·5H₂O)—was investigated under electron beam excitation, before and after exposure to high-dose gamma irradiation (100 kGy). The films were prepared via solvent casting using N,N-dimethylformamide (DMF), with doped samples containing 50 mg of CuSO₄ per 100 mg of PVDF. After drying at 60 °C for 24 hours, the films were irradiated using a Cobalt-60 panoramic irradiator at the Gamma Irradiation Laboratory (LIG) of CDTN. Characterization was performed exclusively using a ZEISS Sigma field emission scanning electron microscope (FE-SEM) equipped with multiple detectors, including a variable pressure secondary electron detector (VPSE G3), a backscattered electron detector (NTS BSD), and a cathodoluminescence (CL) detector. This configuration enabled simultaneous assessment of surface morphology, compositional contrast, and luminescent emission. Results showed that pure PVDF exhibits a stable morphology and weak luminescent response, even after irradiation. In contrast, CuSO₄-doped samples displayed enhanced luminescence following irradiation, suggesting the formation of new emissive species. Elemental analysis using energy-dispersive X-ray spectroscopy (EDS) confirmed the presence of copper and sulfur in the doped samples, consistent with the incorporation of copper sulfate. Additionally, heat maps were generated from the cathodoluminescence images using ImageJ® software to enhance the spatial analysis of emission intensity. This complementary approach provided a more detailed visualization of luminescent behavior across different sample conditions, reinforcing the interpretation of SEM-CL results. This study demonstrates that, despite relying on a single characterization technique, the strategic use of multiple SEM detectors enables a comprehensive evaluation of both structural and optical properties of polymer-based materials. These findings highlight the potential of PVDF/CuSO₄ composites as radiation-responsive materials and support their application in the development of sensors or optoelectronic devices with tunable emission properties.

Keywords: Cathodoluminescence; Gamma irradiation; PVDF–Copper Sulfate.

1 Introduction

Polymer-based luminescent materials have gained significant attention in recent years due to their potential applications in optoelectronics, sensing technologies, and radiation detection. These materials offer several advantages over traditional inorganic systems, including mechanical flexibility, low processing costs, and the ability to tailor their properties through chemical modification or the incorporation of functional additives (Lv *et al.*, 2020). Among the various polymers studied,

polyvinylidene fluoride (PVDF) stands out for its excellent chemical resistance, thermal stability, and semicrystalline nature, which contributes to its functional versatility (Dallaev *et al.*, 2022).

One promising strategy for enhancing the optical properties of polymer matrices involves the incorporation of inorganic dopants (Kumar and Singh, 2023). Copper sulfate pentahydrate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$) is a compound of particular interest due to its potential to form luminescent species upon thermal or radiation-induced transformation. Under certain conditions, CuSO_4 can partially convert into copper oxides such as CuO or Cu_2O , which are known for their characteristic emission properties (Chaudhary *et al.*, 2019). However, the luminescent behavior of PVDF/ CuSO_4 composites, especially under high-energy excitation and ionizing radiation, remains poorly understood.

In this study, a single characterization technique—scanning electron microscopy (SEM)—was employed in a multimodal approach using different detectors to extract complementary information. This strategy demonstrates that, even with a single instrument, it is possible to obtain a comprehensive understanding of both morphological and luminescent properties of polymer-based materials. The effectiveness of this approach is supported by recent literature. For example, Lisak *et al.* (2022) demonstrated the use of SEM to characterize composites materials incorporating conductive and luminescent components, showing that a single technique, when applied strategically, can provide meaningful information about both structure and function. Their work reinforces the value of SEM as a versatile platform for investigating complex hybrid materials.

Cathodoluminescence (CL), the emission of light from a material under electron beam excitation, is a powerful technique for probing the optical response and defect-related emission in materials at the microscale. When applied to polymer-inorganic composites, CL can reveal how structural or chemical changes—such as those induced by gamma irradiation—affect luminescent behavior (Bugrov *et al.*, 2018). The objective of this work is to investigate the cathodoluminescence behavior of PVDF films, both pure and doped with $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, before and after exposure to high-dose gamma irradiation. The study aims to evaluate the influence of irradiation on the luminescent properties of the composites and to explore their potential for use in radiation-responsive optical devices. To support the identification of dopant-related features, energy-dispersive X-ray spectroscopy (EDS) was also employed to verify the elemental composition of the samples, particularly the presence of copper and sulfur associated with CuSO_4 .

2 Materials and Methods

PVDF homopolymers were supplied by ATOCHEM (France). PVDF films were prepared in two configurations: pure and doped with copper sulfate pentahydrate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$). The preparation method was based on solvent casting. For each sample, 100 mg of PVDF granules were dissolved in 3 mL of N,N-dimethylformamide (DMF) oven at 60 °C for one hour until complete homogenization. For the doped samples, 50 mg of analytical-grade $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (molar mass: 249.68 g/mol, Synth) was added to the solution and stirred until fully dispersed. The resulting mixtures were poured into Petri dishes and dried in a laboratory oven at 60 °C for 24 hours, forming uniform thin films.

Gamma irradiation was performed at the Gamma Irradiation Laboratory (LIG) of the Nuclear Technology Development Center (CDTN), Brazil. The facility uses a panoramic irradiator equipped with a Cobalt-60 source, capable of delivering high-dose exposures with uniform distribution. In this study, the samples were irradiated at a total dose of 100 kGy under ambient conditions.

Characterization was performed exclusively using scanning electron microscopy (SEM) with a ZEISS Sigma field emission microscope. Three detectors were employed to extract complementary information:

- The VPSE G3 detector (Variable Pressure Secondary Electrons) was used to analyze surface morphology and topographic features, particularly in pure PVDF samples.
- The NTS BSD detector (Backscattered Electrons) was used to highlight compositional contrast, especially useful for identifying regions enriched with CuSO_4 in the doped samples.

- The CL detector (Cathodoluminescence) was used to evaluate the spatial distribution and intensity of luminescent emission under electron beam excitation.

All images were acquired at 5,000 $\times$ magnification, with working distances of approximately 9.0 mm. CL images were obtained at 20 kV to enhance excitation, while morphological images were captured at 10 kV. To complement the SEM imaging, EDS analysis was conducted using the same microscope to identify the elemental composition of selected regions. Image analysis was performed using grayscale intensity measurements to compare luminescent responses across different sample conditions. In addition, cathodoluminescence images were further processed using ImageJ® software to generate heat maps. These maps allowed for enhanced visualization and comparative assessment of emission intensity distributions across the different sample types, providing a complementary layer of analysis to support the interpretation of CL behavior.

3 Results and Discussion

This section presents a comprehensive analysis of the morphological and luminescent behavior of PVDF-based materials under different conditions. Initially, the surface morphology and cathodoluminescence (CL) response of pure, non-irradiated PVDF films are examined using SEM imaging with the VPSE G3 detector, which provides detailed topographic information. For the CuSO₄-doped composites, the NTS BSD detector was employed to enhance compositional contrast and identify dopant-rich regions. CL images were acquired for all samples to assess their luminescent behavior before and after gamma irradiation at 100 kGy. At the end of the section, a complementary analysis is presented using heat maps generated from the CL images by ImageJ® software, allowing a more detailed comparison of emission intensity distributions across the different sample types.

3.1 Morphological and Luminescent Characterization of Pure, Non-Irradiated PVDF

The morphological and luminescent properties of pure, non-irradiated PVDF films were initially evaluated using SEM, as shown in Figure 1. Two distinct detection modes were employed: the VPSE G3 detector (left image), which captures secondary electrons under variable pressure conditions and provides detailed surface topography, and the cathodoluminescence (CL) detector (right image), which records light emission induced by the electron beam, revealing the spatial distribution of luminescent centers.

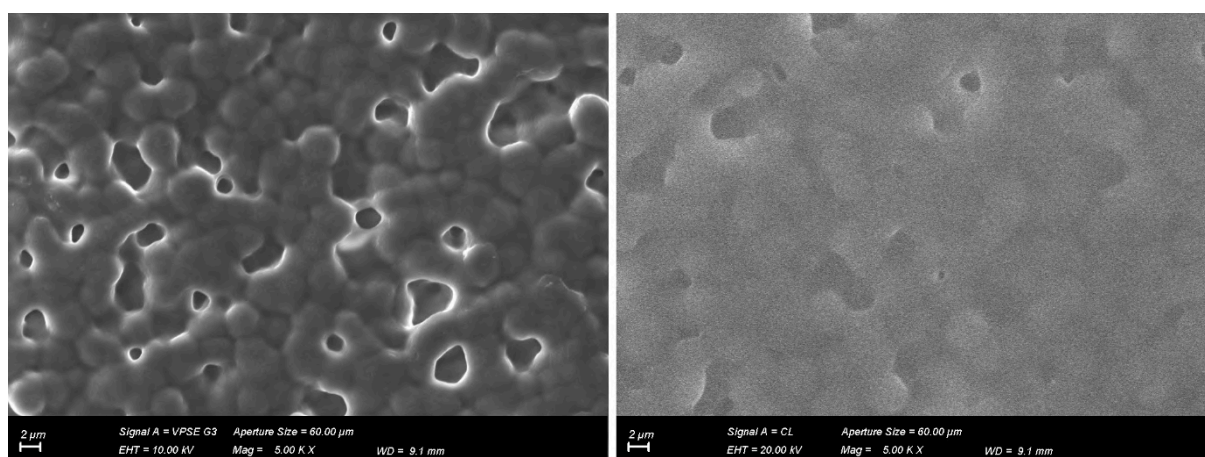


Figure 1: SEM images of pure, non-irradiated PVDF film acquired using two different detectors. Left: VPSE G3 detector. Right: CL detector.

The VPSE G3 image revealed a relatively homogeneous surface morphology, with well-defined spherulitic structures distributed across the film. These rounded formations, characterized by distinct boundaries, are indicative of crystalline domains formed during the solvent evaporation and film solidification process. The presence of spherulites is consistent with the semicrystalline nature of PVDF and reflects the organization of polymer chains into radially growing crystalline lamellae. Their

visibility under SEM confirms that the casting and drying conditions (60 °C for 24 hours) were adequate to promote partial crystallization of the polymer matrix.

In contrast, the CL image of the same sample, acquired under identical magnification but with a higher accelerating voltage (20 kV), exhibited a very weak luminescent response. The surface appeared mostly uniform and lacked distinct bright regions typically associated with strong emissive centers. However, subtle variations in brightness suggest the presence of low-intensity luminescence, possibly arising from minor structural imperfections or localized ordering within the crystalline regions. This behavior aligns with the expected optical inactivity of pure PVDF under electron beam excitation, as the polymer does not inherently possess efficient radiative recombination centers.

These observations establish a baseline for comparison with doped and irradiated samples, where changes in morphology and luminescent behavior can be attributed to the presence of copper sulfate and the effects of gamma irradiation.

3.2 Morphological and Luminescent Characterization of Pure, Non-Irradiated PVDF/CuSO₄ Composite

Figure 2 presents SEM images of the PVDF composite doped with copper sulfate pentahydrate (CuSO₄·5H₂O), acquired using two different detection modes. The left image, obtained with the NTS BSD detector (Backscattered Electron Detector), emphasizes compositional contrast, while the right image, captured with the CL detector, reveals the luminescent response of the material under electron beam excitation.

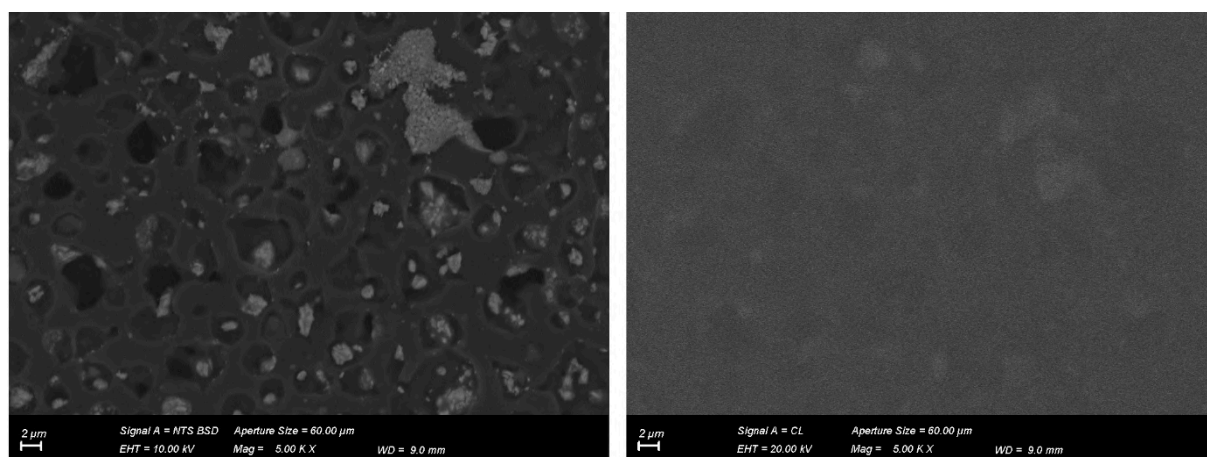


Figure 2: SEM images of the non-irradiated PVDF/CuSO₄ composite. Left: Image acquired using the NTS BSD detector. Right: CL image showing a weak but distinguishable luminescent response.

The use of the BSD detector in this case is particularly appropriate, as it enhances the visibility of regions with higher atomic number elements—such as copper (Cu) and sulfur (S)—present in the dopant. In contrast to the pure PVDF sample, the composite image shows heterogeneous contrast, with brighter regions likely corresponding to areas enriched with CuSO₄. This suggests a non-uniform distribution of the dopant within the polymer matrix. Additionally, the spherulitic structures observed in the pure PVDF are less defined in the composite, indicating that the presence of CuSO₄ may interfere with the crystallization process, either by acting as nucleating agents or by disrupting the regular growth of crystalline domains.

In the CL image (right), a noticeable increase in luminescent signal is observed compared to the pure PVDF sample. Although the emission is still relatively weak, the presence of distinguishable bright regions suggests that the dopant introduces or enhances emissive centers within the material. This behavior is consistent with the known potential of CuSO₄ to form luminescent species, even in the absence of irradiation, possibly due to partial hydration loss or local interactions with the polymer matrix. These findings indicate that the incorporation of CuSO₄ not only alters the morphological

features of PVDF but also contributes to the emergence of luminescent activity, setting the stage for further enhancement upon gamma irradiation.

To further support the identification of the dopant within the composite, energy-dispersive X-ray spectroscopy (EDS) was performed on the same region shown in Figure 2. The resulting spectrum (Figure 3) revealed distinct peaks corresponding to copper (Cu) and sulfur (S), confirming the presence of copper sulfate in the analyzed area. A minor peak associated with aluminum (Al) was also detected, which is attributed to the metallic sample holder used during SEM analysis. Additionally, the EDS spectrum exhibited prominent signals for fluorine (F) and carbon (C), which are intrinsic to the PVDF matrix. These elements originate from the polymer's repeating unit $-\text{CH}_2-\text{CF}_2-$, and their detection further validates the integrity of the polymeric structure in the analyzed region. These results validate the compositional contrast observed in the BSD image and reinforce the interpretation that the brighter regions correspond to CuSO_4 -rich domains embedded within the PVDF matrix.

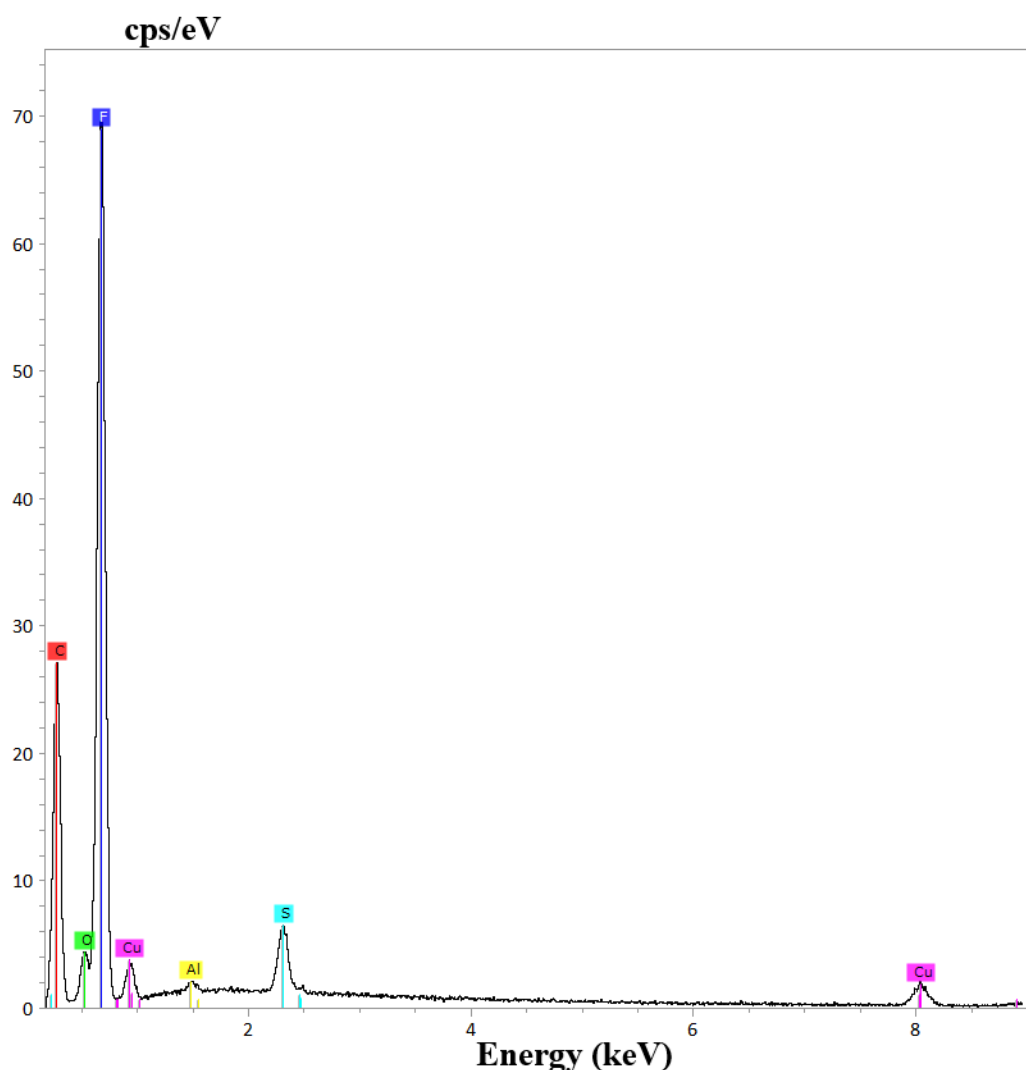


Figure 3: Energy-dispersive X-ray spectroscopy (EDS) spectrum of the non-irradiated PVDF/ CuSO_4 composite, corresponding to the region shown in Figure 2.

3.3 Comparative Analysis of Irradiated and Non-Irradiated Samples

Figure 4 presents a comparative overview of the morphological and luminescent characteristics of pure PVDF and PVDF/ CuSO_4 composites, both before and after gamma irradiation at 100 kGy. The images were acquired using different SEM detectors—VPSE G3, NTS BSD, and CL—each selected to highlight specific features of the materials.

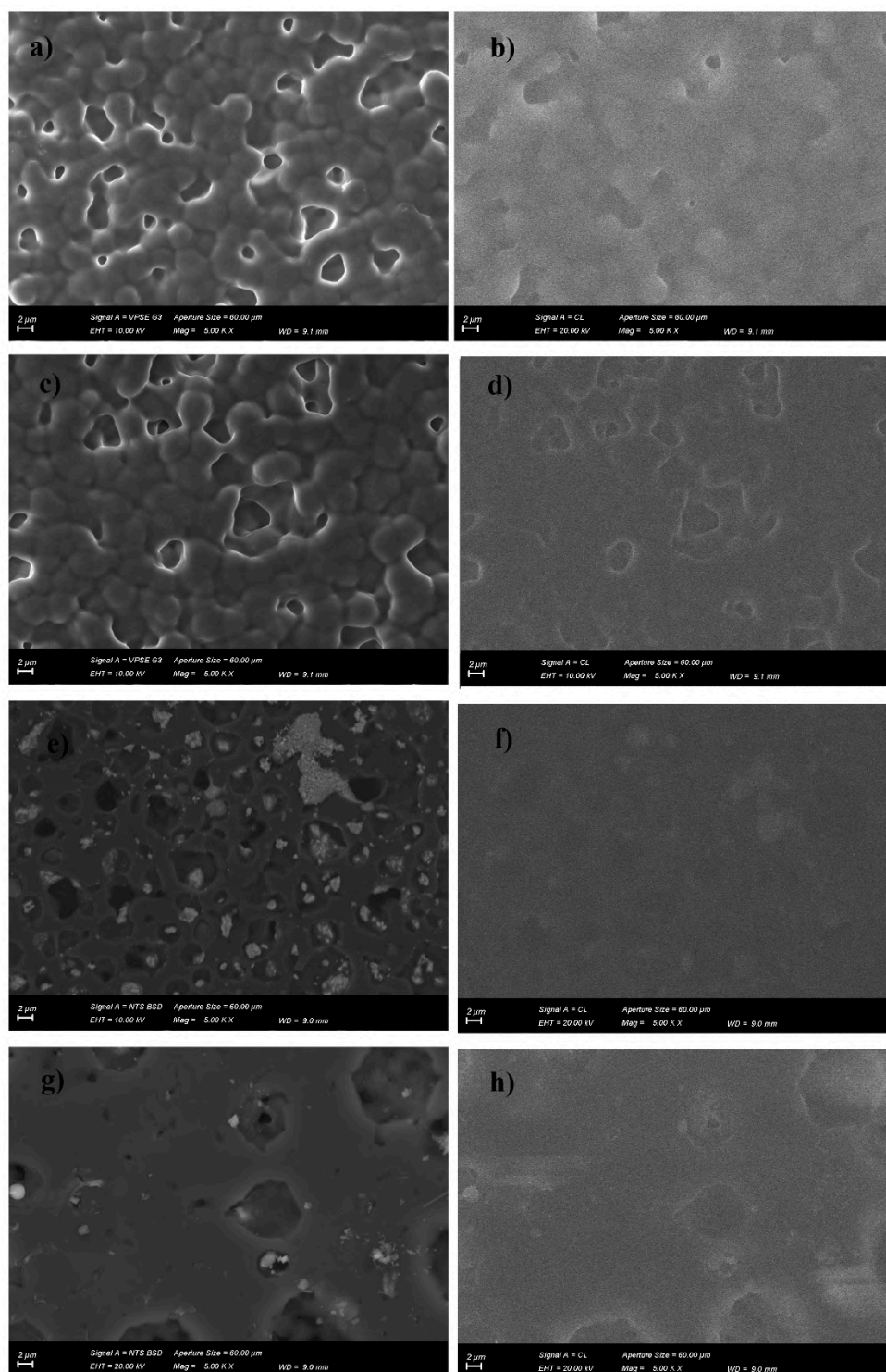


Figure 4: Comparative SEM and CL images of pure PVDF and PVDF/CuSO₄ composites, before and after gamma irradiation at 100 kGy.

Figure 4 presents a comparative overview of the morphological and luminescent characteristics of pure PVDF and PVDF/CuSO₄ composites, both before and after gamma irradiation at 100 kGy. The images were acquired using different SEM detectors—VPSE G3, NTS BSD, and CL—each selected to highlight specific features of the materials. In images a and b, corresponding to pure, non-irradiated PVDF, the VPSE G3 detector (a) reveals well-defined spherulitic structures, indicative of the polymer's semicrystalline morphology. The CL image (b), however, shows minimal luminescent activity, consistent with the expected optical inactivity of undoped PVDF.

After irradiation (images c and d), the VPSE G3 image (c) shows no significant morphological degradation, suggesting that the polymer maintains its structural integrity under high-dose gamma exposure. Interestingly, the CL image (d) reveals a slight increase in luminescent signal compared to the non-irradiated sample, possibly due to the formation of radiation-induced defects or localized structural rearrangements that act as weak emissive centers. For the PVDF/CuSO₄ composite, the NTS BSD image of the non-irradiated sample (e) displays heterogeneous contrast, with brighter regions likely corresponding to CuSO₄-rich domains. The CL image (f) shows a modest luminescent response, indicating the presence of emissive centers introduced by the dopant.

Upon irradiation (images g and h), the BSD image (g) suggests a more diffuse distribution of the dopant, potentially due to radiation-induced migration or partial decomposition. The CL image (h) exhibits a marked increase in luminescent intensity, supporting the hypothesis that gamma irradiation promotes the formation of new emissive species, such as CuO or Cu₂O, through partial transformation of the copper sulfate. Overall, the comparative analysis highlights the stability of PVDF under irradiation and the significant enhancement of luminescent properties in the doped composite, particularly after exposure to gamma radiation. These results reinforce the potential of PVDF/CuSO₄ systems for applications in radiation sensing and tunable optoelectronic devices.

3.4 ImageJ®-Based Heat Map Processing of CL Micrographs

Heat maps were generated from the cathodoluminescence (CL) images using ImageJ® software to enhance the spatial interpretation of emission intensity. This processing step allowed for a clearer visualization of luminescent variation across the sample surfaces, particularly in irradiated and doped regions where emission was more heterogeneous. The conversion of grayscale CL micrographs into color-coded intensity maps made it easier to distinguish subtle differences in emission behavior that were not readily apparent in the original images.

In the Figure 5 the samples analyzed included pure PVDF (non-irradiated and irradiated with 100 kGy of gamma radiation) and PVDF doped with copper sulfate (also in non-irradiated and irradiated conditions). The SEM-CL images (a, c, e, g) and their respective heat maps (b, d, f, h) revealed distinct emission behaviors. The non-irradiated pure PVDF sample (a) exhibited a relatively homogeneous CL emission, which was confirmed by the corresponding heat map (b) showing predominantly cool to intermediate tones. In contrast, the irradiated pure PVDF sample (c) displayed increased contrast and brighter regions, with the heat map (d) indicating higher emission intensities through warmer color zones. This suggests that gamma irradiation induces structural defects or changes in crystallinity that enhance luminescence.

For the PVDF samples containing copper sulfate, the non-irradiated sample (e) showed localized emission zones, with the heat map (f) highlighting distinct hot spots, likely due to the heterogeneous distribution of the salt or its interaction with the polymer matrix. The irradiated PVDF/CuSO₄ sample (g) exhibited the most intense and complex emission patterns, with the heat map (h) showing extensive high-intensity regions. This indicates a synergistic effect between gamma irradiation and the presence of copper sulfate, potentially due to the formation of new emissive centers or increased electronic excitation states. These results demonstrate that both gamma irradiation and copper sulfate incorporation significantly influence the cathodoluminescence behavior of PVDF-based materials.

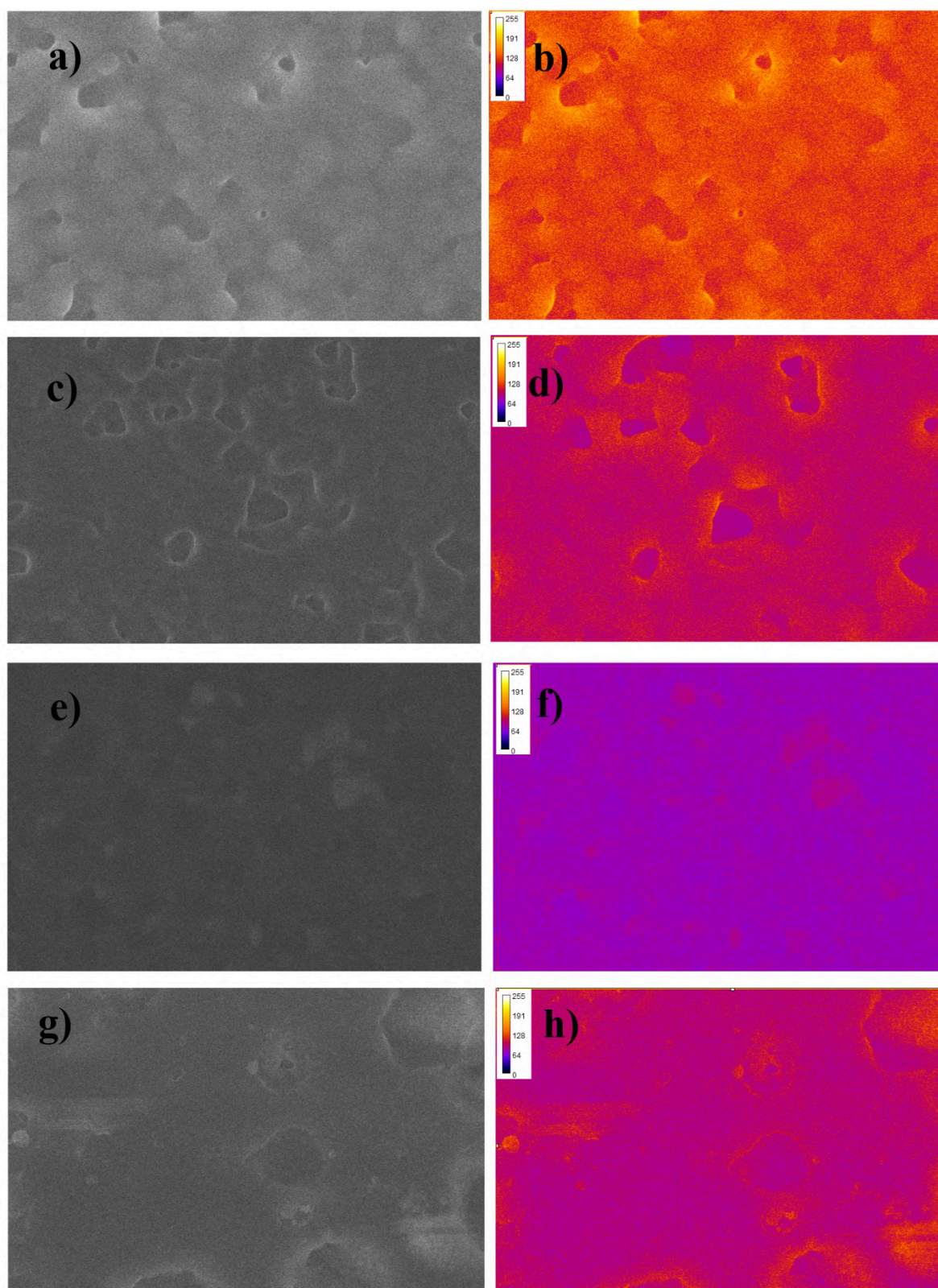


Figure 5: Heat maps generated from cathodoluminescence (CL) images of PVDF-based films using ImageJ® software. Images a), c), e), and g) correspond to CL micrographs of pure PVDF (a and c) and PVDF/CuSO₄ composites (e and g), with c) and g) representing samples irradiated with 100 kGy of gamma radiation. Images b), d), f), and h) are the respective heat maps derived from the CL images, highlighting spatial variations in emission intensity. Warmer colors indicate regions of higher luminescent activity.

Table 1 presents a schematic summary of the cathodoluminescence behavior observed in the PVDF-based samples, based on the heat map analysis performed using ImageJ® software.

Table 1: Summary of Cathodoluminescence Behavior Based on Heat Map Analysis Using ImageJ

Sample	Irradiated	CuSO ₄ Doping	CL Emission
a/b	No	No	Low to moderate
c/d	Yes (100 kGy)	No	Increased
e/f	No	Yes	Localized
g/hd	Yes (100 kGy)	Yes	High

This table consolidates the visual interpretation of emission intensity distributions derived from the processed CL images, highlighting the effects of gamma irradiation and copper sulfate doping on luminescent performance. The classification considers both the qualitative features of the heat maps and the relative intensity variations among the different sample conditions.

4 Conclusions

This study demonstrated the effectiveness of using scanning electron microscopy (SEM) with multiple detectors—VPSE G3, NTS BSD, and CL—for the comprehensive characterization of polymer-based luminescent materials. Despite relying on a single technique, the strategic use of these detectors enabled simultaneous evaluation of morphological, compositional, and optical properties of PVDF films, both pure and doped with copper sulfate pentahydrate (CuSO₄·5H₂O), before and after gamma irradiation.

The pure PVDF films exhibited well-defined spherulitic morphology and a weak, stable luminescent response, even after exposure to a 100 kGy gamma dose, indicating high structural and optical stability. In contrast, the PVDF/CuSO₄ composites showed altered morphology with less defined spherulitic structures and heterogeneous dopant distribution, as revealed by the BSD detector. Notably, the CL images revealed a significant enhancement in luminescent emission after irradiation, suggesting the formation of new emissive species, likely copper oxides, induced by the radiation.

As an additional analytical layer, heat maps generated from CL images using ImageJ software proved to be a valuable tool for visualizing and comparing emission intensity across the samples. This approach highlighted subtle spatial variations in luminescence that were not readily apparent in the raw SEM-CL images, particularly in irradiated and doped samples. The methodology adopted here also reinforces the value of SEM as a versatile and powerful platform for advanced materials characterization. These findings confirm that gamma irradiation can be used as a tool to modulate the optical properties of polymer-inorganic composites. Moreover, the results highlight the potential of PVDF/CuSO₄ systems for applications in radiation sensing and the development of tunable luminescent materials.

In future studies, additional characterization techniques will be employed to complement and compare with the SEM-based findings presented here, aiming to further validate and expand the understanding of the luminescent behavior of these materials.

Acknowledgements

The authors would like to thank the Gamma Irradiation Laboratory (LIG) at the Nuclear Technology Development Center (CDTN) for providing the gamma irradiation services used in this study. Special thanks are also extended to Granioter, the Technological Hub for Advanced Materials and Strategic Minerals, based at CDTN, for supporting the SEM analyses and enabling access to advanced characterization infrastructure. This work was financially supported by Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES) and Fundação de Amparo à Pesquisa do Estado de Minas Gerais (FAPEMIG).

References

- Bugrov A. N. *et al.* (2018) Luminescence of Eu³⁺ ions in hybrid polymer-inorganic composites based on poly (methyl methacrylate) and zirconia nanoparticles. *Luminescence* 33(5):837-849.
- Chaudhary S. *et al.* (2019) Synthesis and characterizations of luminescent copper oxide nanoparticles: toxicological profiling and sensing applications. *Ceramics International* 45(12):15025-15035.
- Dallaev R. *et al.* (2022) Brief review of PVDF properties and applications potential. *Polymers* 14(22):4793.
- Kumar A. and Singh G (2023) Recent advances and future perspectives of photoluminescent liquid crystals and their nanocomposites for emissive displays and other tunable photonic devices. *Journal of Molecular Liquids*, 386:122607.
- Lv Q. *et al.* (2020) Analysis of dominant non-radiative recombination mechanisms in InGaN green LEDs grown on silicon substrates. *Journal of Luminescence* 222: 117186.
- Lysak A. *et al.* (2022) Effect of rapid thermal annealing on short period {CdO/ZnO} m SLs grown on m-Al₂O₃. *Materials Science in Semiconductor Processing* 142: 106493.

Properties of nanoDot OSL Detectors for Radiotherapy Quality Control: An Initial Characterization and Uncertainty Analysis

D. S. Machado^{1*}, C. C. B. Viegas², L. A. R. da Rosa³, S. C. Cardoso¹

¹Rio de Janeiro Federal University, Physics Institute, 21941-909, Rio de Janeiro, Brazil

²National Cancer Institute, Radiotherapy Quality Assurance Program, 20231-092, Rio de Janeiro, Brazil

³Institute of Radiation Protection and Dosimetry (IRD/ANSN), Graduate Program in Radiological Protection and Dosimetry, 22780-160, Rio de Janeiro, Brazil

*daniele.machadofisica@gmail.com

Abstract

OSLDs have the ability to store energy from ionizing radiation and release it as light upon optical stimulation, allowing absorbed dose measurement when calibrated for this purpose. Their characterization is crucial for safe and effective radiotherapy application. To this end, a commercial OSLD system is being evaluated for the measurement of quality control parameters. This study proposes the development of a protocol for the characterization of OSLD nanoDots, providing consistent results supported by uncertainty analysis, in which several dosimetric properties of $\text{Al}_2\text{O}_3:\text{C}$ nanoDots irradiated with a ^{60}Co beam were investigated. The combined total uncertainty of the OSL signal is primarily influenced by the uncertainty in the Bg. The most appropriate approach was found to be the adoption of a Type B uncertainty for the Bg, estimated at 1.93% based on 30 readings. Optical bleaching of the dosimeters, previously irradiated with an absorbed dose of 100 cGy, resulted in a 99% signal reduction after 30 minutes of exposure, thus enabling the reuse of OSLDs. The transient signal immediately post-irradiation stabilizes after approximately 10 minutes in the dark. Between 24 hours and 7 days post-irradiation, natural signal fading is negligible, and the main source of signal reduction is depletion, associated with the accumulation of repeated readouts, with a reduction factor of 0.9995 after 108 readings, corresponding to a 0.05% decrease in signal per readout. Repeatability tests to select the most stable detectors have found 75% of the batch within $\pm 1.5\%$ signal variation, and an average individual calibration coefficient of 2.31×10^{-3} cGy/OSL. All results demonstrate that OSL dosimetry is a viable alternative for use in quality control procedures, particularly in radiotherapy.

Keywords: OSLD; dosimetry; radiotherapy.

1 Introduction

Quality control in radiotherapy is fundamental to ensuring the level of accuracy required for optimal treatment outcomes, enhancing tumor control while minimizing damage to healthy tissues. Achieving this level of quality demands a collaborative effort from a multidisciplinary team, focused on reducing errors and uncertainties throughout every stage of the radiotherapy workflow (Mayles, 1999).

Optically Stimulated Luminescence Dosimeters (OSLDs) are materials that, after exposure to ionizing radiation, have the ability to store energy and subsequently release it as luminescent light under optical stimulation, enabling the measurement of absorbed dose when properly calibrated to this quantity. Due to their high sensitivity, small size, and the ability to be read multiple times without significant signal degradation (Jursinic, 2007), OSLDs – particularly $\text{Al}_2\text{O}_3:\text{C}$ nanoDots – have gained

widespread application in various areas of radiotherapy, including *in vivo* dosimetry, radiotherapy auditing, and individual dose monitoring (Lye, 2014; Mrcela, 2011).

Furthermore, (Mrcela, 2011) showed that OSL dots irradiated with ^{60}Co beams exhibited acceptable reproducibility and sufficient accuracy for entrance dose measurements, with no significant dependence on radiation parameters typically encountered in clinical settings. The commercial InLight™ OSLD system (Landauer Inc.) has been shown to be effective for a variety of dosimetric applications. As reported by (Viamonte, 2008), this system provides consistent readings across multiple measurement types and shows good agreement with both ion chamber and diode measurements.

Accurate characterization of OSLDs is essential for ensuring their safe and reliable use in clinical radiotherapy. Several factors can influence their response, such as optical bleaching efficiency, signal stability, repeatability, and dose-response linearity. Therefore, a calibration protocol is required, along with a detailed evaluation of the uncertainties inherent to each stage of the measurement process.

This study proposes the development of a protocol for the characterization of $\text{Al}_2\text{O}_3:\text{C}$ nanoDot OSLDs using a commercial OSLD system (Landauer Inc.). Dosimetric properties were assessed under ^{60}Co beam irradiation conditions, based on detailed studies with consistent results, and accompanied by a comprehensive analysis of the uncertainties involved, for radiotherapy quality control application.

2 Materials and Methods

2.1. Equipment

All irradiations were performed using a ^{60}Co unit (Theratron 780C) with a nominal dose rate of 70 cGy/min at the Radiotherapy Quality Program (INCA). Irradiation parameters were adjusted to account for radioactive decay, with a source-to-surface distance (SSD) of 60 cm. Two types of phantoms were employed: virtual water slabs (Standard Imaging, GR200 – Virtual Water Phantom), measuring $(40 \times 40 \times 5) \text{ cm}^3$, and a water phantom consisting of an acrylic cube $(40 \times 40 \times 40) \text{ cm}^3$ filled with water.

The OSL system used in this study was the InLight™ OSL (Landauer Inc., Glenwood, IL, USA), comprising nanoDot dosimeters, a microStar reader, and external reading software. Each nanoDot contains an $\text{Al}_2\text{O}_3:\text{C}$ OSL active element encapsulated in a light-tight plastic casing with dimension of $(1 \times 1 \times 0.1) \text{ cm}^3$. During readout, the capsule opens automatically, but it can also be opened manually for optical bleaching, i.e., exposure of the active element to light. The microStar reader uses green LEDs with a peak wavelength of 530 nm, operating in continuous wave stimulation mode. The reading time is approximately 1 second, during which only a small fraction of the trapped charge is released, enabling multiple readings and reuse of the dosimeters (Mrcela, 2011).

Prior to each irradiation, dosimeters were optically bleached to empty charge traps, and background signal measurements were performed.

2.2. Basic Characteristics of OSLD

Tests were conducted to evaluate trap emptying through optical bleaching, early fading, signal stability under repeated readouts to assess fading and depletion, repeatability and non-linearity at different absorbed dose levels.

Bleaching

To determine the required bleaching time for effective trap emptying, an experiment was performed to monitor signal reduction as a function of light exposure time. The setup consisted of a home-made device with four 9 W LED lamps, as illustrated in Figure 1.

For this analysis, the nanoDot OSLDs were irradiated with an absorbed dose of 100 cGy. The active detector element was removed from its plastic holder to allow direct exposure of the sensitive material to light, and reinserted at defined intervals for signal readout. Prior to each reading, dosimeters were kept in the dark for 3 minutes and then subjected to three consecutive readouts. After each set of readings, they were returned to light exposure until the next measurement point. The total exposure time was 1.5 h.



Figure 1: Home-made device for bleaching.

Early Fading

To investigate the presence of the transient signal, each dosimeter was individually irradiated with an absorbed dose of 200 cGy at 5 cm depth in a solid water phantom. Readouts began immediately after irradiation and were repeated for approximately 30 minutes to assess signal decay over time.

Fading and Depletion

Three groups of five OSLDs, each composed of the same five batches – provided by the manufacturer – were irradiated with an absorbed dose of 100 cGy at 5 cm depth in a solid water phantom. An experimental design was implemented to distinguish fading and depletion effects by comparing these groups with different numbers of readouts and reading times, according to the protocol shown in Table 1.

Table 1: Readout protocol for three groups of OSLDs irradiated with an absorbed dose of 100 cGy, indicating the post-irradiation readout time points and the number of readings at each point.

Group	Readout Time Points	Total Readouts
1	10 min, 24 h, 7 days	108
2	24 h, 7 days	72
3	7 days only	36

Individual Calibration Coefficient

To identify the most stable OSLDs, a repeatability test was conducted for 200 nanoDots under identical conditions. Irradiations were performed with 100 cGy, (20 × 20) cm² field size, at 5 cm depth in a solid water phantom. Each dosimeter underwent five complete cycles of bleaching, background reading, irradiation and readout. For each cycle, five consecutive readings were taken. The individual calibration coefficient (CC) was calculated as the mean ratio between the measured signal and the reference absorbed dose.

$$CC = \frac{1}{n} \sum_{i=1}^n \frac{M_i}{D} \quad (1)$$

Non-Linearity

The non-linearity dose-response was also assessed. Five sets of three dosimeters were irradiated with absorbed doses from 150 cGy to 250 cGy, in water phantom. The dose-response correction factor f_{lin} is defined as the ratio of the OSL response M_{D_0} per unit of absorbed dose measured at the absorbed dose $D_0 = 200$ cGy to the OSL response M_D per unit of absorbed dose measured at the absorbed dose D (IAEA, 2023):

$$f_{lin} = \frac{M_{D_0}/D_0}{M_D/D} \quad (2)$$

The factor $f_{lin}(D)$ was determined by performing a linear fit to the experimental data from the OSL response per absorbed dose as a function of absorbed dose. The fit was normalized so that the response per absorbed dose equals 1 at 200 cGy, as given in Equation 3.

$$f_{lin}(D) = a + b \cdot D \quad (3)$$

where a is the intercept of the least squares fit, b is the slope of the linearity correction ($a + 2b = 1$).

2.3. Estimation of Measurement Uncertainties

In this study, both Type A – based on statistical analysis of repeated measurements – and Type B – based on non-statistical sources such as prior knowledge – approaches were assessed to estimate the background uncertainty, aiming to determine which method is more appropriate based on its contribution to the total combined uncertainty of the net OSL signal.

To measure the Bg under the Type A approach, three pre-irradiation readouts are performed; the associated uncertainty is estimated from the experimental standard deviation of the mean and multiplied by the Student's t -factor corresponding to the desired confidence level. For the Type B approach, 30 readings were performed using three OSLDs to obtain a representative estimate of the Bg uncertainty.

To estimate the average OSL signal and its associated uncertainty, five post-irradiation readouts are performed. The standard uncertainty is obtained from the experimental standard deviation of the mean s :

$$s = \frac{s(\overline{M_i})}{\sqrt{n}} \quad (4)$$

In OSL dosimetry, it is often necessary to assess measurement uncertainty based on a single reading. When only a single measurement is available, the uncertainty cannot be directly calculated. In such cases, as noted by (Knoll, 2010), it can be estimated by assuming a known statistical distribution, where the variance is approximated by the measured value itself, i.e., $s^2 \approx \sigma^2 \approx M$. Therefore, the best estimate of the deviation from the true value can be expressed by the standard deviation σ , given by:

$$\sigma = \sqrt{M} \quad (5)$$

The OSL dosimeter reading is obtained using a digital reader, whose uncertainty may be influenced by the resolution of the display device. Even when repeated readings yield identical values, the

resolution introduces an inherent uncertainty (INMETRO, 2008). For a resolution ΔM , the associated standard uncertainty u is given by:

$$u = \frac{\Delta M}{\sqrt{12}} \quad (6)$$

The total combined uncertainty was obtained by propagating the uncertainties for all evaluations in this work.

3 Results and Discussion

3.1. Basic Characteristics of OSLD

Bleaching

OSL dosimeters can be reused through optical bleaching. Successive readouts were performed during the bleaching process to evaluate the signal decay as a function of light exposure time, considering three nanoDots analyzed. A mathematical model describing the signal decay is given by:

$$N(t) = A + B \cdot e^{-kt} \quad (7)$$

where $N(t)$ is the OSLD signal at time t , A and B are constants, and k is the decay constant. The analysis of the measurements shown in Figure 2 is presented in Table 2.

The decay parameters vary slightly among the OSLDs. However, under optical stimulation using four 9 W LED lamps, a bleaching duration equivalent to twelve half-lives (approximately 30 minutes) results in the removal of 99% of the OSL signal. Complete elimination of the signal is not achievable without thermal treatment, due to the presence of deep traps (Yukihara, 2004). Although the residual absorbed dose is typically negligible or can be subtracted when measuring relatively high absorbed doses, as is radiotherapy applications.

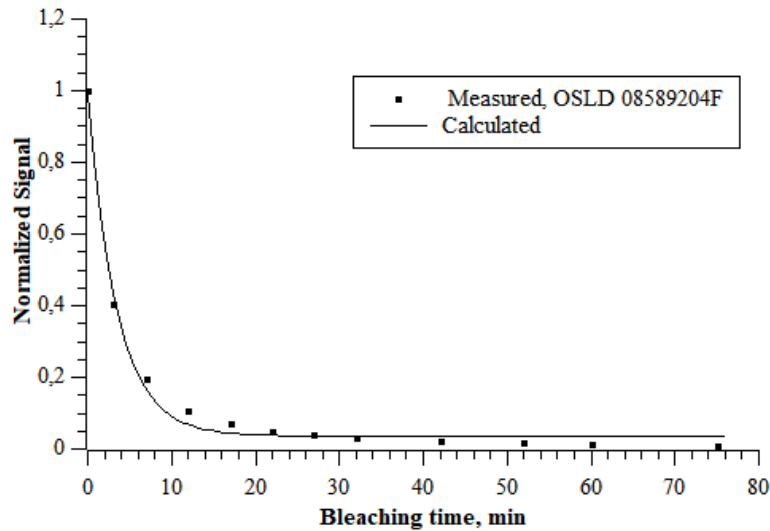


Figure 2: Repeated measurements of an OSLD vs the time of light exposure. OSLDs irradiated with an absorbed dose of 100 cGy were normalized to the first signal measured. The solid lines are calculations based on Equation 7 with parameters shown in Table 2. Error bars represent the uncertainty of each reading.

Table 2: Analysis of the decay of the OSLD signal with light exposure for three OSLDs. These OSLDs were irradiated with an absorbed dose of 100 cGy. The reading data were analyzed by nonlinear regression, according to Equation 7.

OSLD	A	B	k (min ⁻¹)	$T_{1/2}$ (min)
08589204F	0.0360 ± 0.0094	0.954 ± 0.028	0.284 ± 0.020	2.45 ± 0.17
090055930	0.0193 ± 0.0049	0.977 ± 0.014	0.271 ± 0.010	2.562 ± 0.092
097010929	0.0280 ± 0.0070	0.965 ± 0.021	0.270 ± 0.014	2.56 ± 0.13

Early Fading

OSLDs contain shallow energy traps that are unstable, leading to a transient signal immediately after irradiation. However, it was observed that this transient signal stabilizes within a few minutes post-irradiation. The decay parameters vary for each OSLD, and the signal decay is also described by the mathematical model in Equation 7. A dark waiting time corresponding to 8 half-lives (approximately 8 minutes post-irradiation) is adequate to avoid this transient signal in all of the OSLDs, as also reported by (Jursinic, 2007). The analysis of the measurements shown in Figure 3 is also presented in Table 3. The uncertainty of each reading was 2.8%.

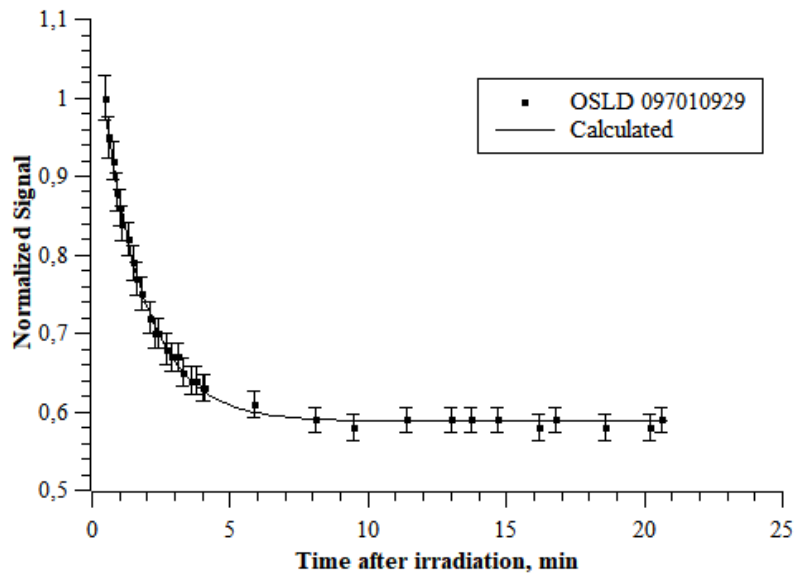


Figure 3: Repeated measurements of an OSLD vs the time after irradiation. OSLDs irradiated with an absorbed dose of 200 cGy were normalized to the first signal measured. The solid lines are calculations based on Equation 7 with parameters shown in Table 3. Error bars represent the uncertainty of each reading.

Table 3: Analysis of the decay of the OSLD signal after irradiation for two OSLDs. The OSLDs were irradiated with an absorbed dose of 200 cGy. The data were analyzed by nonlinear regression, according to Equation 7.

OSLD	A	B	k (min ⁻¹)	$T_{1/2}$ (min)
097010929	0.5877 ± 0.0017	0.541 ± 0.010	0.655 ± 0.015	1.059 ± 0.025
093306041	0.5496 ± 0.0023	0.505 ± 0.017	0.671 ± 0.025	1.033 ± 0.039

Fading and Depletion

Each reading of an OSL dosimeter partially discharges the trapped charges in the material, progressively reducing the emitted signal. The effects of fading (signal decay over time) and depletion (signal reduction due to repeated readings) were analyzed by comparing groups subjected to different reading intervals and frequencies, Table 1. The mathematical model of the decreasing signal, shown in (Jursinic, 2007), is given by:

$$N(i) = N_0 \cdot f^i \quad (8)$$

where $N(i)$ is the signal at reading i , N_0 is the assumed signal before any depletion, and f is the fraction by which the signal is depleted per reading. Equation 8 can be linearized as follows:

$$a = 1 - f,$$

$$f^i = (1 - a)^i \approx 1 - i \cdot a \text{ when } a \ll 1.$$

Substituting into Equation 8,

$$N(i) = N_0 - N_0 \cdot a \cdot i \quad (9)$$

where $N_0 \cdot a$ represents the slope. To evaluate signal reduction, a total of 15 OSLDs were selected and divided into three readout groups, each containing five dosimeters. One representative OSLD from each group (Groups 1, 2, and 3), all from batch 088, was selected to illustrate the results. The data are presented in Figure and Table 4. The uncertainty of each reading was 2.8%.

The results suggest that depletion does not depend on the time interval between irradiation and readout. The main contribution to signal reduction is associated with the number of readouts. Although the fraction of signal loss per readout varies slightly among OSLDs, it is consistently close to 0.9995, meaning each readout reduces the OSLD signal by approximately 0.05%. This finding is also reported by (Jursinic, 2007).

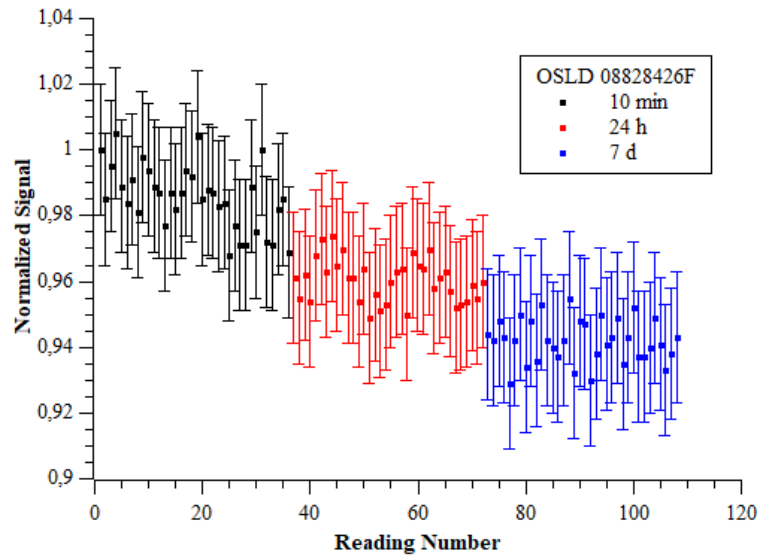


Figure 4: Signal of the OSLD over 108 readings taken at intervals of 10 min, 24 h, and 7 d (36 readings per interval), normalized by the first reading. Error bars represent the measurement uncertainty.

Table 4: Analysis of the decay in OSLD signal reduction for one representative among five OSLDs in each readout group. These OSLDs were irradiated with an absorbed dose of 100 cGy. The readout data were analyzed using linear regression.

	Time post-irradiation	Intercept	Slope	Reduction factor (f)
Group 1: 08828426F				
	10 min	0.995 ± 0.007	-0.0005 ± 0.0004	0.9995 ± 0.0004
	24 h	1.02 ± 0.02	-0.0003 ± 0.0003	0.9998 ± 0.0003
	7 d	1.01 ± 0.03	-0.0001 ± 0.0003	0.9999 ± 0.0003
	Total	0.997 ± 0.004	-0.0006 ± 0.0001	0.9994 ± 0.0001
Group 2: 08828436E				
	24 h	1.000 ± 0.007	-0.0005 ± 0.0003	0.9995 ± 0.0003
	7 d	1.010 ± 0.020	-0.0002 ± 0.0003	0.9998 ± 0.0003
	Total	0.998 ± 0.005	-0.0004 ± 0.0001	0.9996 ± 0.0001
Group 3: 08828470M				
	7 d	0.989 ± 0.007	-0.0005 ± 0.0003	0.9995 ± 0.0003

To investigate a possible correlation between the manufacturer batch and the sensitivity, the readings were compared and analyzed. Figure 5 presents the comparison among five different batches values, each represented by three nanoDots. The data reflect the mean signal obtained from 108 readings, with an uncertainty of 1.94%. However, no clear correlation was identified. Higher batch values did not necessarily correspond to higher signal readings, and the same was observed for lower batches. This highlights the importance of individually calibrating each dosimeter.

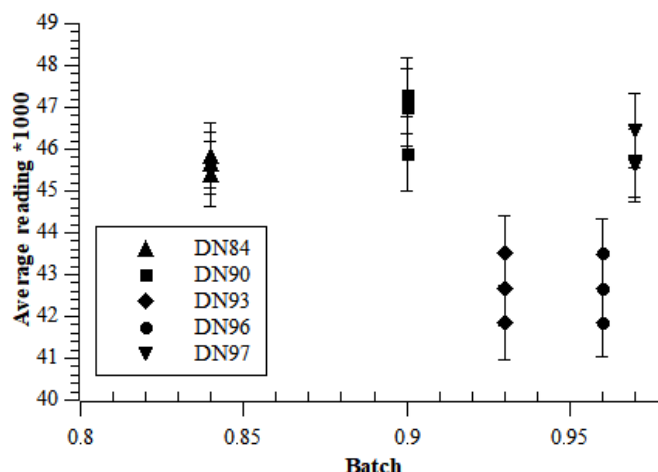


Figure 5: Average signal readings of nanoDots with different batches indicated by the manufacturer. Each data point represents the mean signal of three nanoDots for a given batch, measured over 108 readings. Error bars correspond to the standard uncertainty.

Individual Calibration Coefficient

The repeatability of the OSLD response was evaluated over five irradiation cycles. Based on the signal standard deviation, approximately 75% of the 200 nanoDots tested exhibited deviations within $\pm 1.5\%$, allowing the identification of the most stable detectors. The results are presented in Figure 6. An individual calibration coefficient (CC) was determined to estimate the absorbed dose. It was calculated using Equation 1, resulting in an average value of 2.31×10^{-3} cGy/OSL with a mean uncertainty of 2.3%.

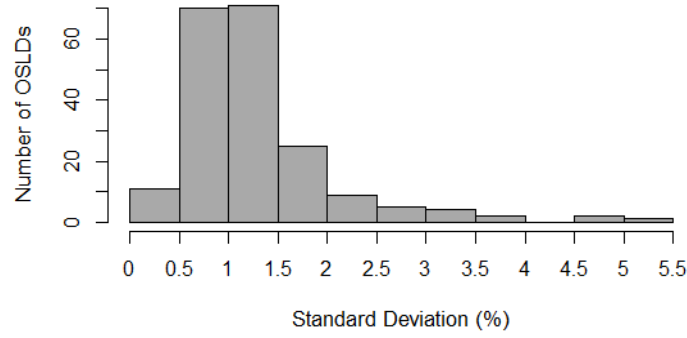


Figure 6: Distribution of the standard deviation values obtained from the repeatability test of 200 nanoDots irradiated with an absorbed dose of 100 cGy across five irradiation cycles under identical conditions. The standard deviation was calculated based on the response of each detector to assess signal stability.

Non-Linearity

The usefulness of a luminescence signal in dosimetry relies on its dose-dependent response, which should be reproducible and preferably linear within the relevant absorbed dose range. Figure 7 indicates that the OSLD dose-response correction factor is linear between 150 and 250 cGy, as also reported by (Viamonte, 2008). A variation of ± 50 cGy results in a correction factor $f_{lin}(D)$ of approximately 2%.

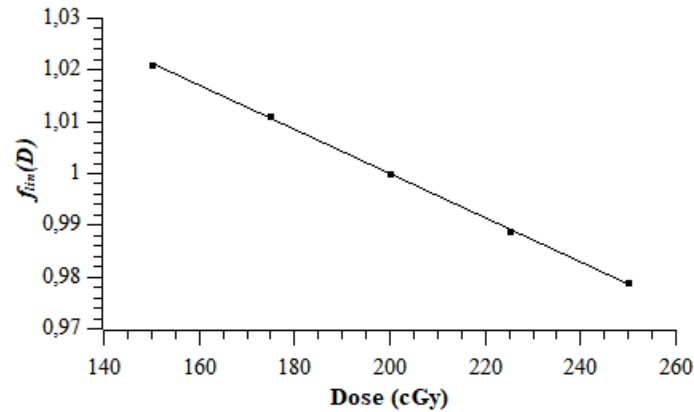


Figure 7: OSLD dose-response correction factor $f_{lin}(D)$ vs absorbed dose for ^{60}Co irradiations. Three different detectors were used at each absorbed dose value. For each detector, the response is the average value of three readings. The solid line shows a linear fit.

A linear fit was applied to the data in Figure 7 and the resulting equation was:

$$f_{lin}(D) = -4.240 \times 10^{-4} \cdot D + 1.0848$$

where D is the absorbed dose in cGy, with $(\text{Chi}^2/\text{dof})$ of 1.33×10^{-7} .

3.2. Estimation of Measurement Uncertainties

According to (INMETRO, 2008), measurement uncertainty comprises different components, classified based on how they are estimated. Type A uncertainties arise from statistical analysis of experimental data, while Type B uncertainties rely on other sources, such as prior knowledge or expert judgment. A complete uncertainty statement must include all relevant components and specify the

estimation method used. Considering this, the contribution of the Bg uncertainty to the total combined uncertainty was evaluated to determine the most appropriate estimation approach.

The Type B uncertainty of the Bg was estimated based on the average relative uncertainty obtained from 30 readings performed on three distinct OSLDs. As shown in Figure 8, the relative uncertainty decreases as the number of readings increases. The Bg signal is the main contributor to the total combined uncertainty. Hence, the Type B uncertainty for the Bg was used in the results presented in this work.

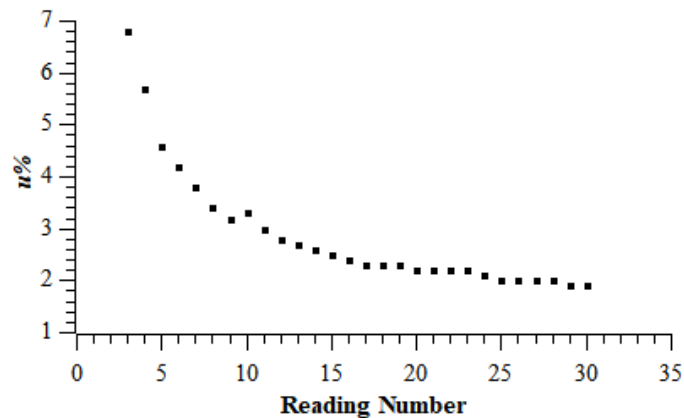


Figure 8: Variation of the average uncertainty as a function of the number of readouts.

4 Conclusions

The results presented in this study demonstrate that nanoDot dosimetry is a viable alternative for quality control applications, particularly in radiotherapy. OSL dosimeters offer several advantages: they are easily bleached and reused, and can be safely read within minutes after irradiation. Additionally, they exhibit acceptable reproducibility in absorbed dose measurements using ^{60}Co beams. The dose-response correction factor is linear within the absorbed dose range typically used in conventional radiotherapy. Although a slight signal depletion occurs with each readout, it remains negligible within the clinical absorbed dose levels. Furthermore, the descriptive analysis of uncertainties played a key role in ensuring a more robust and reliable interpretation of the results.

Acknowledgements

I would like to thank the Programa de Qualidade em Radioterapia from Instituto Nacional de Câncer (PQRT/INCA) for providing access to the OSL dosimeters, the reader, and a collaborative environment for research and discussion. I am also grateful to the Programa Institucional de Bolsas de Iniciação Científica (PIBIC) for supporting this work through a research grant. Finally, I sincerely thank the research group for all the valuable discussions and for guiding me in the process of learning how to conduct scientific research.

References

- Botter-Jensen L., McKeever S. W. S., Wintle A. G. (2003) *Optically Stimulated Luminescence Dosimetry*. Amsterdam: Elsevier.
- IAEA (2024) *Absorbed Dose Determination in External Beam Radiotherapy: An International Code of Practice for Dosimetry Based on Standards of Absorbed Dose to Water*. Vienna: IAEA, Chapter 5.
- IAEA (2023) *Radiotherapy Dosimetry Audits: Structure, Methodology, Scientific Procedures – Appendix I*. International Atomic Energy Agency, Vienna.

INMETRO (2008) Guia para a expressão da incerteza de medição. 3ª ed., Duque de Caxias, RJ: INMETRO.

Jursinic P. A. (2007) Characterization of optically stimulated luminescent dosimeters. *Medical Physics* 34(12):4594.

Knoll G. F. (2010) *Radiation Detection and Measurement*. 4th ed., Hoboken: Wiley, p. 84, Chapter 3.

Lye J., Dunn L., Kenny J. (2014) The use of optically stimulated luminescent dosimeters in radiotherapy. *Medical Physics* 41:032102.

MAYLES, W. P. M. et al. (1999) *Physics aspects of quality control in radiotherapy*. York: Institute of Physics and Engineering in Medicine. ISBN 0 904181 91 X.

McKeever S. W. S. (2001) Optically stimulated luminescence dosimetry. *Nuclear Instruments and Methods in Physics Research B* 184:29–54.

Mrcela I., Bokulic T., Izewska J. (2011) Analysis of calibration methods in radiotherapy dosimetry audits. *Physics in Medicine and Biology* 56:6065–6082.

Viamonte A., da Rosa L. A. R., Buckley L. A., Cherpak A., Cygler J. E. (2008) Radiotherapy dosimetry using a commercial OSL system. *Medical Physics* 35:1261–1266.

Yukihara E G and McKeever S W S 2008 Optically stimulated luminescence (OSL) dosimetry in medicine *Phys.Med. Biol.* 53 R351–79.

Investigation of the Luminescent Properties of Zirconia-Enriched Dental Resins

Paula Thays Santos Malachias^{1,*}, Luiz Cláudio Meira Belo¹

¹Centro de Desenvolvimento da Tecnologia Nuclear - CDTN, Belo Horizonte, Minas Gerais, Brasil

* paula.malachias@cdtn.br

Abstract

Zirconia has been widely incorporated into dental composite resins to enhance their mechanical and optical properties, particularly in applications requiring high wear resistance, optical stability, and longevity under masticatory stress. Beyond its established role in restorative dentistry, zirconia has attracted attention in the field of solid-state dosimetry due to its capacity to trap and subsequently release electric charges upon optical stimulation. This property enables the application of zirconia-containing materials in optically stimulated luminescence (OSL) dosimetry, a technique that measures the luminescence emitted from a material when it is stimulated by light after exposure to ionizing radiation. The present study investigated the OSL properties of zirconia-doped dental composite resins with the aim of determining their response to radiation and their potential use as passive dosimeters in retrospective dosimetry. Disk-shaped samples were prepared and irradiated using the beta radiation source of a $^{90}\text{Sr}/^{90}\text{Y}$ system, covering a dose range from 50 mGy to 3,500 mGy. OSL measurements were performed with a continuous-wave stimulation protocol, using blue LEDs as the stimulation source and a photomultiplier tube for signal detection. The measurement sequence included preheating to eliminate unstable signal components, followed by optical stimulation and signal acquisition. The results showed a well-defined and reproducible OSL signal, with a linear dose–response relationship extending up to 3,500 mGy ($R^2 > 0.99$). The short-term signal stability was confirmed by negligible fading within 72 hours after irradiation, under laboratory storage conditions. The material demonstrated low measurement uncertainty, with relative standard deviations typically below 5% for repeated irradiations at the same dose level. These properties indicate that zirconia-doped dental composite resins can provide accurate and reliable dose estimates when subjected to OSL readout. Although this study focused on experimental evaluation under controlled laboratory conditions, the findings suggest promising applicability of these materials in scenarios involving accidental or unplanned exposures to ionizing radiation. In such cases, retrospective dosimetry can be performed by retrieving dental restorations and analyzing their OSL signal to estimate the absorbed dose. This could be particularly valuable in nuclear or radiological events where conventional dosimeters are absent, complementing biological dosimetry and other retrospective assessment techniques. In conclusion, zirconia-doped dental composite resins exhibit favorable OSL characteristics, including linear dose response, high reproducibility, and minimal fading, supporting their potential as passive dosimeters for radiation dose reconstruction in emergency contexts. Further research will focus on long-term signal stability, sensitivity optimization, and the effects of clinical variables such as resin aging, intraoral conditions, and light exposure.

Keywords: Composite resin; Zirconia; Optically stimulated luminescence; Radiation dosimetry; Aesthetic dentistry; Restorative materials;

1 Introduction

The smile is an essential element of social interaction, influenced by beauty standards that value white, aligned, and proportional teeth, contributing to greater social acceptance (Reis et al., 2018). Modern Dentistry offers solutions such as aesthetic composite resin restorations, which use minimally invasive techniques to restore teeth's color, shape, and function, promoting confidence and social inclusion (Muraro et al., 2016).

Composite resins have undergone significant advancements, including a reduction in particle size and an increase in filler load, resulting in greater strength and durability (Ruivo et al., 2019). Replacing inorganic particles with silica and silica-zirconia nanoclusters has improved properties such as elastic modulus, compressive strength, and hardness due to greater interaction with the organic matrix. Moreover, the shape of the particles influences viscosity and wear resistance, optimizing the material's performance (Usha et al., 2018). Zirconia is a restorative material of choice in this context due to its proven biological, functional, and aesthetic benefits.

Additionally, due to its crystalline nature, zirconia has shown potential in applications beyond dentistry, such as optically stimulated luminescence (OSL) dosimetry. This technique, which uses crystals to emit light when irradiated and stimulated by light, is widely studied for applications in personal dosimetry. Studies on the behavior of zirconia under different radiation absorbed dose levels are essential to understanding its properties better and expanding its use in various contexts (Pereira et al., 2023).

In this study, an investigation was conducted into the luminescent properties of dental resin samples enriched with zirconia, proposing that luminescent activation effects and exposure to ionizing radiation fields, whether of natural origin or not, may contribute to the filling of electron traps within the crystalline structures. Understanding this phenomenon may support retrospective dosimetry, using dental restorations removed from patients or simulated ones, especially those enriched with zirconia, by analyzing their potential as passive dosimetric materials.

Optically stimulated luminescence (OSL) applied to sediment dating was first reported by Huntley et al. in 1985 and has since become a fundamental technique in retrospective and accident dosimetry, as well as in geochronology (Huntley et al., 1985). Conventional OSL materials—such as $\text{Al}_2\text{O}_3\text{:C}$, BeO , and natural quartz—combine high sensitivity with well-known trap structures, allowing accurate dose reconstruction over a wide range of radiation fields (Yukihara and McKeever, 2011); (West et al., 2006); (Sommer et al., 2007).

Despite their proven performance, these materials are exogenous to the human body, limiting their forensic value in scenarios where in-situ or post-extraction dose estimation is desirable. Dental tissues and restorative composites, on the other hand, are permanently present in the oral cavity and can be ethically removed after an exposure event. Exploratory studies on dental enamel (DeWitt et al., 2010), feldspathic ceramics (Bailiff et al., 2002), and silica-based restorations (Nakauchi et al., 2016) already indicate measurable luminescent signals.

Zirconia-enriched composites stand out as desirable candidates: (i) zirconia is mechanically and biologically robust, (ii) its crystalline phase can host a variety of electron traps, and (iii) its high atomic number increases the radiation interaction cross sections (Geber-Bergstrand et al., 2012; Roberts, 2014). Investigation of the OSL behavior of these composites, therefore, addresses a dual gap: advancing esthetic dentistry and broadening the portfolio of passive dosimeters suitable for incidental or forensic applications.

2 Methodology

To achieve the proposed objectives, composite resin from the brand Forma Ultradent enriched with zirconia was used. For sample preparation, a 0.0536 g portion was molded into a shape similar to a dosimeter and photopolymerized for 40 seconds.

After preparation, the sample was placed in the RISØ TL/OSL reader, Model DA-20, produced by DTU – Technical University of Denmark, using a clinical tweezer. The reader was programmed to perform optically stimulated luminescence (OSL) readings at 20°C and under a constant flow of high-purity nitrogen.

Initially, a spontaneous sample reading was performed following the standards described. Then, irradiations were carried out using a $^{90}\text{Sr}/^{90}\text{Y}$ source installed in the reader. Four doses were applied: 684 mGy, 1,140 mGy, 2,280 mGy, and 3,420 mGy. After each irradiation, a new reading was taken to analyze the sample's behavior.

3 Results

Luminescent signals were observed in the zirconia-enriched resin sample investigated in this study. An initial spontaneous readout was performed on the sample without prior radiation exposure to assess the feasibility and reproducibility of such luminescent phenomena. Subsequently, the sample underwent controlled beta irradiation at gradually increasing doses of 684 mGy, 1,140 mGy, 2,280 mGy, and 3,420 mGy, aiming to evaluate its luminescent response under varying exposure conditions.

To ensure that the observed effects were solely attributable to the material's intrinsic properties, environmental conditions were meticulously controlled throughout all experimental procedures, maintaining a constant temperature of 20 °C and ensuring an inert atmosphere through a continuous flow of high-purity nitrogen. This approach minimized potential external interferences, such as humidity or oxygen quenching, thereby improving signal stability and reproducibility.

The luminescence emission profiles obtained after each irradiation stage were plotted as glow curves, which depict the emitted light intensity as a function of the stimulation time or temperature. These curves are a key diagnostic tool in optically stimulated luminescence (OSL) analysis, allowing for comparing signal amplitude, shape, and dose-response trends. Figure 1 presents the glow curves acquired for the zirconia-enriched resin at each irradiation dose, enabling direct visual comparison of changes in signal intensity and profile with increasing absorbed dose.

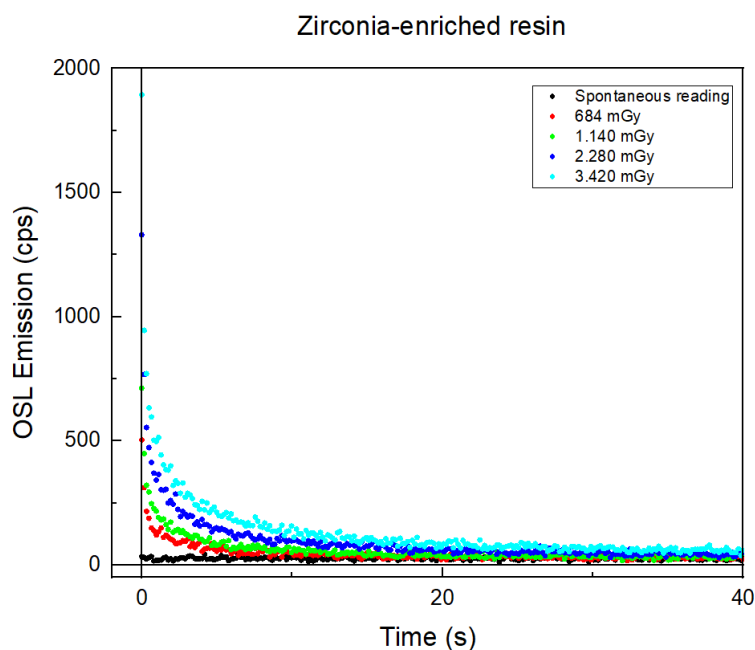


Figure 1: Glow curves from the spontaneous reading and the readings after each irradiation.

The analysis of the optically stimulated luminescence (OSL) response as a function of the absorbed beta radiation dose in the zirconia-enriched resin sample revealed an approximately linear behavior up to a limit of 3,500 mGy. This linearity indicates that, within this dose range, the intensity of the luminescent signal is proportional to the amount of absorbed radiation—an essential characteristic for materials with potential application in dosimetry.

The linearity suggests that the trapping centers in the zirconia-enriched resin matrix respond uniformly to the charge accumulation induced by irradiation, with no significant saturation observed up to the applied dose. Such a consistent dose-response behavior is crucial to ensure the accuracy and reliability of dose measurements in radiological monitoring systems employing OSL technology.

To illustrate this relationship, the dose-response data were plotted to generate a calibration curve representing luminescence intensity as a function of absorbed beta dose. Figure 2 shows this experimentally obtained calibration curve, demonstrating the linear correlation between dose and emission intensity in the evaluated range. This graphical representation reinforces the potential of the zirconia-enriched resin sample as a sensitive and reliable material for dosimetric applications in controlled environments exposed to ionizing radiation.

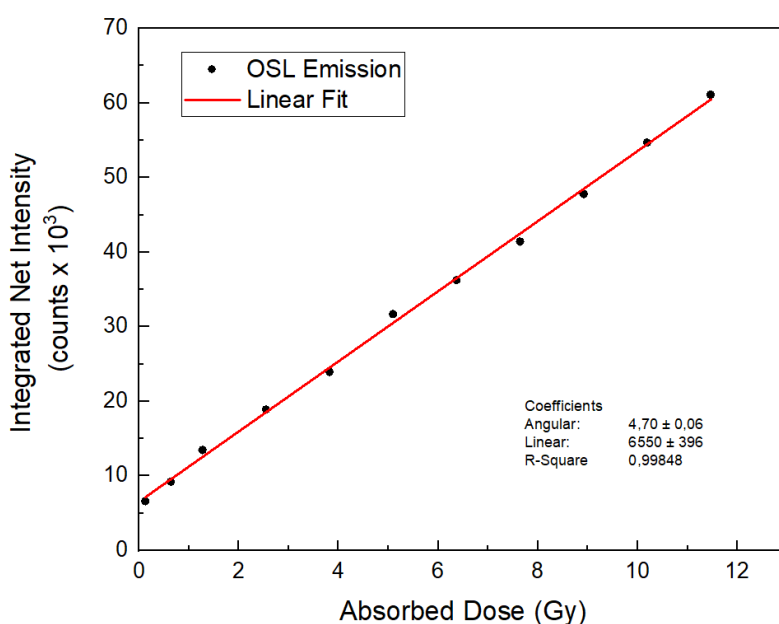


Figure 2: OSL response as a function of the absorbed dose in the resin.

4 Conclusions

The findings of the present study demonstrate that the zirconia-modified composite resin exhibits detectable luminescent properties, as well as an approximately linear optically stimulated luminescence

(OSL) response concerning the applied beta radiation doses up to a limit of 3,500 mGy. These results highlight the promising potential of this material not only for conventional applications in the field of Dentistry but also for its use in dosimetry systems based on optically stimulated luminescence.

Furthermore, the physicochemical characteristics of zirconia significantly contributed to the stability of the observed responses and the repeatability of the measurements. These critical factors reinforce the feasibility of employing this composite as a subject of investigation in dosimetry. Based on these results, future research may focus on optimizing the material's formulation and assessing its performance under varying environmental conditions. In this context, the range of applications of the zirconia-modified composite resin may be expanded, indicating its potential relevance in emerging scientific and technological fields.

Acknowledgements

The authors gratefully acknowledge the financial support provided by the Brazilian National Council for Scientific and Technological Development (CNPq), the National Institute of Science and Technology – INCT–INAIS (Instrumentation for Nuclear Applications with Quality in Industry and Health), and the Foundation for Research Support of the State of Minas Gerais (FAPEMIG).

References

- Bailiff, I.K., Correcher, V., Delgado, A., *et al.*, 2002. *Luminescence characteristics of dental ceramics for retrospective dosimetry: a preliminary study*. Radiat. Prot. Dosim. **101**(1–4), 519–524. <https://doi.org/10.1093/oxfordjournals.rpd.a006040>
- DeWitt, R., Klein, D.M., Yukihara, E.G., *et al.*, 2010. *Optically stimulated luminescence of tooth enamel and its potential use in post-radiation-exposure triage*. Health Phys. **98**(2), 432–439. <https://doi.org/10.1097/01.HP.0000347997.57654.17>
- Geber-Bergstrand, T., Bernhardsson, C., Mattsson, S., Rääf, C.L., 2012. *Retrospective dosimetry using OSL of tooth enamel and dental repair materials irradiated under wet and dry conditions*. Radiat. Environ. Biophys. **51**(4), 443–449. <https://doi.org/10.1007/s00411-012-0434-9>
- Huntley, D.J., Godfrey-Smith, D.I., Berger, G.W., 1985. *Optical dating of sediments*. Nature **313**, 105–107. <https://doi.org/10.1038/313105a0>
- Muraro, D.F., *et al.*, 2016. *Resinas composta de preenchimento único – relato de caso*. Int. J. Braz. Dent. **12**(2), 180–185.
- Nakauchi, D., Okada, G., Yanagida, T., 2016. *Scintillation, OSL and TSL properties of yttria-stabilized zirconia crystal*. J. Lumin. **172**, 61–64. <https://doi.org/10.1016/j.jlumin.2016.07.006>
- Pereira, B. H., *et al.*, (2023) *Characterization of an optically stimulated luminescence reader using BeO dosimeters* (undergraduate thesis). Universidade Federal de Uberlândia, Brazil.
- Reis, G.R., *et al.*, 2018. *Mock-up: previsibilidade e facilitador das restaurações estéticas em resina composta*. Rev. Odontol. Bras. Central **27**(81), 105–111.
- Roberts, H.M., 2014. *Recent developments in optically stimulated luminescence materials and techniques for dosimetry and dating*. Radiat. Meas. **71**, 16–23. <https://doi.org/10.1016/j.radmeas.2014.05.020>

- Ruivo, M.A., *et al.*, 2019. *Surface roughness and filler particles characterization of resin-based composites*. *Microsc. Res. Tech.* **82**(10), 1756–1767.
- Sommer, M., Freudenberg, R., Henniger, J., 2007. *New aspects of a BeO-based optically stimulated luminescence dosimeter*. *Radiat. Meas.* **42**(4–5), 617–620.
<https://doi.org/10.1016/j.radmeas.2007.01.052>
- Usha, C., Rao, S., George, G., 2018. *A comparative evaluation of the staining capacity of microhybrid and nanohybrid resin-based composite to Indian spices and food colorants: An in vitro study*. *Indian J. Dent. Res.* **29**(2), 201–205.
- West, W.G., Kearfott, K.J., Bernal, S.M., 2006. *The sunlight OSL response of a commercially available α -Al₂O₃:C personnel dosimetry material*. *Radiat. Prot. Dosim.* **119**(1–4), 344–349.
<https://doi.org/10.1093/rpd/nci691>
- Yukihara, E.G., McKeever, S.W.S., 2011. *Optically Stimulated Luminescence: Fundamentals and Applications*. Wiley. <https://onlinelibrary.wiley.com/doi/book/10.1002/9780470977064>

Radiation Detection of Lead-Free BaZrO₃ Perovskite Scintillators

D. G. Mesquita^{1*}, G. F. C. Bispo², M. E. G. Valério³, Z. S. Macedo³, S. España⁴, T. Kurtukian-Nieto⁵, P. Silva^{1,5}, M. L. Moreira¹

¹ University of Pelotas, 96160-000, Capão do Leão, Brazil

² Brazilian Center for Research in Energy and Materials, 1303-970, Campinas, Brazil

³ Federal University of Sergipe, 49100-000, São Cristóvão, Brazil

⁴ Consejo Superior de Investigaciones Científicas, 2800, Madrid, Spain

⁵ Instituto Venezolano de Investigaciones Científicas, 1020-A, Caracas, Venezuela

*mesquitagdani@gmail.com

Abstract

Barium zirconate (BaZrO₃, BZO) is a chemically stable, lead-free oxide perovskite with a wide band gap (~5 eV). Despite its structural and environmental advantages, its potential as a scintillating material remains largely unexplored. This study investigates the scintillation response of BZO-based materials under different types of ionizing radiation, contributing new and relevant experimental data to the emerging application field. BZO powders were synthesized via a microwave-assisted hydrothermal method in both pristine and Ce-doped forms. Under X-ray excitation, undoped BZO exhibited radioluminescence centered around 450 nm. Ce-doping, however, significantly suppressed the emission intensity, suggesting that the scintillation mechanism is governed primarily by intrinsic defect-related emission centers rather than by extrinsic dopant states. BZO thin films fabricated by blade coating were evaluated under exposure to various radioactive sources (¹³⁷Cs, ²²Na, and ⁹⁰Sr). No detectable scintillation was observed under gamma excitation ¹³⁷Cs excitation, likely due to low source activity combined with limited film thickness. In contrast, both doped and undoped samples responded under beta and positron exposure, with undoped films consistently showing detectable counts. A pronounced response to ²²Na positrons provided strong evidence of the material's capability to detect charged particles directly. These findings highlight the influence of excitation energy, dopant concentration, and film geometry on scintillation behavior. The measures made in this research should be interpreted as relative indicators of light output rather than absolute yield. Overall, the results support the feasibility of using BZO as a non-toxic scintillator in radiation detection systems, including compact dosimetric devices and nuclear instrumentation, while also reinforcing the need for further material optimization.

Keywords: Scintillator; Perovskite; X-rays; Positron;

1 Introduction

The development of new scintillating materials is essential for improving both the performance and environmental sustainability of detectors used in radiation dosimetry and high-energy applications. Traditional scintillators such as Bi₄Ge₃O₁₂ (BGO), CdWO₄, and CsI:Tl are becoming increasingly inadequate to meet the demands of modern technologies, particularly in medical imaging and high-resolution detection systems (Liu et al., 2024). These materials suffer from limitations such as low light yield, slow response times, limited X-ray stopping power, and unsatisfactory energy resolution. Similarly, conventional semiconductor detectors often fail to meet the stringent requirements of advanced X-ray and gamma-ray detection, prompting the search for alternative materials with superior performance.

In this context, perovskite materials have attracted significant attention due to their unique optoelectronic properties and compositional flexibility. Lead halide perovskites, in particular, have demonstrated outstanding potential in photovoltaic devices, owing to their high efficiency, low fabrication costs, and rapid crystal growth (Moseley et al., 2021). These same features, combined with fast scintillation decay times, make them promising candidates for ionizing radiation detection, especially in time-resolved applications such as positron emission tomography (PET) using Time-of-Flight (TOF) techniques (Pagano et al., 2024).

However, the toxicity and environmental risks associated with lead have intensified the search for stable, lead-free alternatives. In this regard, copper-halide perovskites have shown higher light yield compared to CsI:Tl (Cheng et al., 2021) and promising sensitivity to charged particles under radiation exposure (Hunyadi et al., 2022). Despite these advantages, issues such as slow decay times and complex crystal growth processes still hinder their applicability as fast scintillators.

Among the emerging lead-free options, oxide perovskites have shown great potential for scintillation applications. Notably, barium zirconate (BaZrO_3 , BZO) has gained attention due to its high structural stability under gamma irradiation (Mahadik et al., 2017), wide band gap (~ 5 eV), chemical durability, and low-cost synthesis. BZO also exhibits potential for ultraviolet (UV) detection, particularly when doped with rare earth elements to enhance luminescence through the formation of additional emissive centers (Fujishiro et al., 2024). Studies on the microwave-assisted hydrothermal synthesis of BZO have revealed its radioluminescence properties under X-ray irradiation, as well as the role of intrinsic defect centers formed during the process, suggesting its potential as a scintillator for ionizing radiation detection (Moreira et al., 2011). However, since then, this material has not been further explored for such applications. In particular, the incorporation of rare-earth elements especially cerium (Ce^{3+}) remains largely uninvestigated in BZO. Ce^{3+} is one of the most widely used activators in state-of-the-art scintillators, such as $\text{Gd}_3(\text{Ga},\text{Al})_5\text{O}_{12}$ (GGAG:Ce), Lu_2SiO_5 (LSO:Ce), $\text{Lu}_{1.8}\text{Y}_{0.2}\text{SiO}_5$ (LYSO:Ce), and $\text{LaBr}_3\text{:Ce}$, owing to its allowed 5d–4f transitions, which lead to fast scintillation decay times, high light yield, and excellent energy resolution. Given that intrinsic defect centers have already been identified in BZO and are known to influence its luminescence behavior, investigating the incorporation of Ce^{3+} becomes relevant for understanding the interaction between dopants and native defect states. This approach allows the evaluation of whether rare-earth doping can enhance, or possibly suppress the scintillation performance in oxide perovskites, as well as expand their applicability beyond X-ray detection to other types of ionizing radiation.

Accordingly, the present study investigates BZO-based materials both in pure form and doped with cerium (Ce) synthesized via the microwave-assisted hydrothermal (MAH) method, as potential detectors for ionizing radiation. Preliminary experimental results are presented regarding their detection capabilities under exposure to X-rays and various radioactive sources.

2 Material and methods

2.1 Materials Synthesis

The powders of Barium zirconate (BaZrO_3 , BZO) were synthesized by microwave-assisted hydrothermal (MAH) method, using barium chloride ($\text{BaCl}_2 \cdot 8\text{H}_2\text{O}$), zirconium chloride ($\text{ZrCl}_4 \cdot 8\text{H}_2\text{O}$), potassium hydroxide (KOH), and cerium nitrate ($\text{Ce}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$) as precursors. For a pure sample, in a 50 mL of distilled water was dissolved 0.01 mol of zirconium precursor and added the same quantity of barium precursor and mixed until the completed dissolution. In this solution 6 M of KOH solution was added and stirred under a nitrogen atmosphere at room temperature until complete dissolution. The final solution was transferred to a reaction vessel and placed in a hydrothermal microwave system and heated to 140 °C in 1 minute for 40 minutes. The precipitated powders were centrifuged and washed four times with distilled water until a neutral pH and dried at 80 °C for approximately 12 hours. To obtain the Cerium doped samples the synthesis was following the same procedure, with additions of cerium nitrate in the first mix with the zirconium precursor solution. According to the mass substitution method the The doping concentrations followed the formula $\text{BaZr}_{1-x}\text{Ce}_x\text{O}_3$, with $x = 0.5\%$, 1%, 2%, and 4% Ce.

2.2 Materials Characterization

X-ray diffraction (XRD) analysis of undoped and Ce-doped BZO powder samples was performed using a Shimadzu XRD-6100 diffractometer with Cu K α radiation ($\lambda = 1.5406 \text{ \AA}$). Measurements were conducted at room temperature over a 2θ range of 20° to 80° , with a step size of 0.02° and a scanning speed of 0.02° per second, under an operating voltage of 40 kV.

Radioluminescence (RL) measurements were carried out using an X-ray tube operated at 40 kV. The visible light emitted by the irradiated BZO powder sample was collected via an optical fiber and analyzed using an Ocean Optics QE65000 Pro spectrometer (spectral range: 350–1140 nm). Figure 1 illustrates the experimental setup, designed to detect scintillation light scattered from the sample surface during X-ray excitation

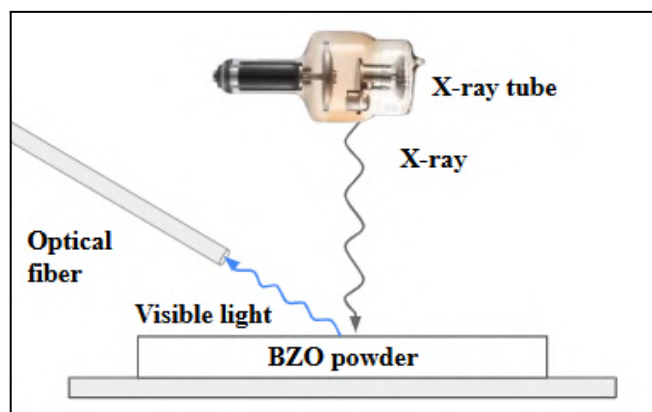


Figure 1: Schematic illustration of the radioluminescence experimental setup used for X-ray excitation of BZO powder samples. (Illustration prepared by the author.)

BZO thin films, both undoped and doped with rare-earth elements, were fabricated on polyvinyl substrates via the blade coating technique. The deposition solution consisted of BZO powder dispersed in a mixture of acrylic varnish and ethanol, yielding a stable suspension. The resulting films exhibited an average thickness of $\sim 50 \text{ }\mu\text{m}$ and an area of $\sim 20 \text{ mm}^2$ (see the picture in Figure 2-a). These films functioned as scintillating layers in the detection setup for all subsequent measurements under radioactive excitation.

The scintillation response of the BZO films to different radioactive sources, was evaluated using a silicon photomultiplier (SiPM). The BZO films were optically coupled to the Broadcom SiPM (AFBR-S4E001) with an active area of $4 \times 4 \text{ mm}^2$ (Figure 2-b), using optical grease to enhance light collection. The SiPM output was connected to an amplifier circuit and a digital oscilloscope (PicoScope), which acquired and processed the signal on a computer. The recorded signal, expressed in counts per second (cps), was used as a relative indicator of light yield. Radioactive excitation was provided by sealed disk sources with a standard diameter of 25 mm with a central active region, commonly used in didactic and laboratory environments.

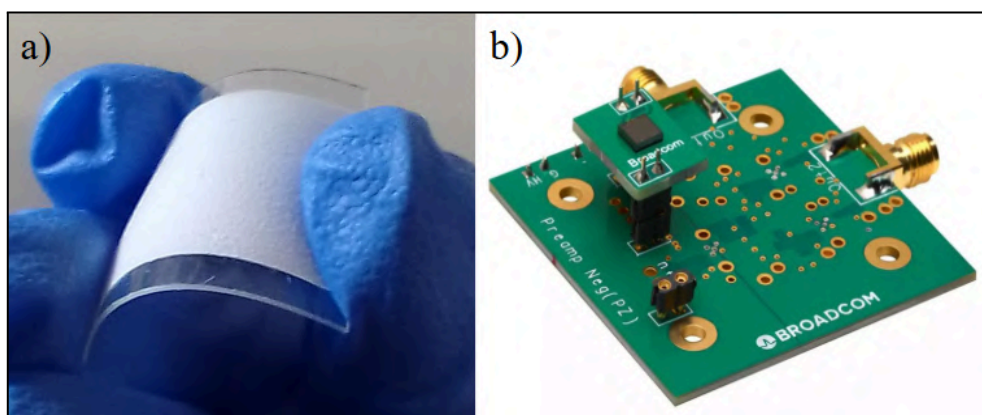


Figure 2: Images from BZO thin film obtained by blade coating (a), and the SiPM module AFBR-S4E001 by Broadcom (2024).

Measurements were carried out using encapsulated radioactive sources of ^{137}Cs , ^{22}Na and ^{90}Sr . The sources were positioned in very close proximity to the BZO films ensuring that the active region of each source was axially aligned with the center of the scintillating film and the SiPM detection area. This configuration formed a layered, parallel “sandwich” structure consisting of source, film, and SiPM, as schematically illustrated in Figure 3. Background signals were also recorded for each configuration to ensure accurate relative comparison.

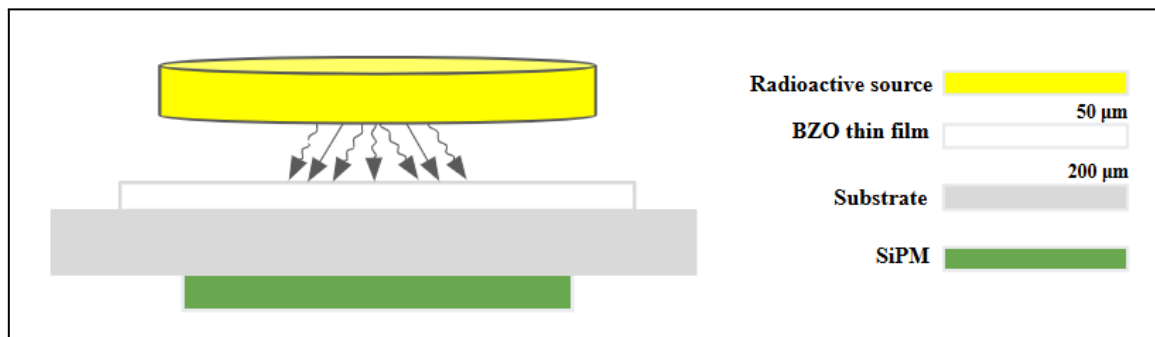


Figure 3: Schematic representation of the experimental arrangement used for count rate measurements. The drawing shows the coupling of the BZO scintillating film to the SiPM, including the polyvinyl substrate and the direction of incoming radiation. (Illustration prepared by the author.)

To provide context for the type and energy of radiation interacting with the materials, Table 1 summarizes the key decay characteristics of the three radioisotopes sources employed, including their activity, decay mode, emission type and typical energy values.

Table 1: Radioactive Source Decay Properties.

Source	Activity (μCi)	Primary Decay Mode	Radiation Emitted	Energy (Typical)
^{137}Cs	0.1	β^- decay $\rightarrow \gamma$ ($^{137\text{m}}\text{Ba}$)	$\beta^- + \gamma$ (662 keV)	β^- : ~ 512 (máx), 174 keV (avg), γ : 662 keV ($^{137\text{m}}\text{Ba}$)
^{90}Sr	0.1	β^- decay ($\rightarrow ^{90}\text{Y}$) $\rightarrow \beta^-$	Pure β^- (no γ)	^{90}Sr β^- : 546 keV (max) ^{90}Y β^- : 2.28 MeV (max), 933 keV (avg)
^{22}Na	10	β^+ decay $\rightarrow$ annihilation	$\beta^+ + 511 \text{ keV } \gamma$ (2x)	β^+ : 546 keV (max); γ : 1275 keV (from $^{22}\text{Ne}^*$), + 511 keV $\times 2$ (annihilation)

These radioisotopes exhibit distinct half-life characteristics, which are crucial for interpreting the experimental results. ^{137}Cs possesses a 30.17-year half-life and undergoes β^- decay to metastable $^{137\text{m}}\text{Ba}$, which subsequently emits a characteristic 662 keV γ photon. Similarly, ^{90}Sr presents a comparable half-life of 28.8 years and decays via β^- emission to ^{90}Y , which itself undergoes further β^- decay with a much shorter half-life of 64.1 hours. Although the initial decay of ^{90}Sr releases low-energy electrons (maximum energy ~ 0.546 MeV), the decay of ^{90}Y dominates the total energy output, emitting high-energy electrons with a maximum energy of 2.28 MeV. In contrast, ^{22}Na has the shortest half-life among the three (2.6 years) and decays via positron emission (β^+), producing two 511 keV annihilation photons. In approximately 90% of these decays, the daughter nucleus ^{22}Ne is left in an excited state, which promptly de-excites by emitting a 1275 keV gamma photon.

3 Results and discussion

3.1 Phase identification

Figure 4 presents the X-ray diffraction (XRD) patterns of $\text{BaZr}_{1-x}\text{Ce}_x\text{O}_3$ ($x = 0.0, 0.5, 1.0, 2.0$ and 4.0 %) alongside the reference pattern (ICSD No. 90049). The XRD results confirm the formation of the desired perovskite phase with a cubic structure, indexed to the typical space group $Pm\bar{3}m$. A secondary phase is evidenced by small peaks around $2\theta = 24^\circ, 34^\circ, 42^\circ$, and 45° , indicating the presence of a minor amount of barium carbonate (BaCO_3). These peaks are associated with the orthorhombic phase of BaCO_3 , indexed in ICSD No. 15196 and corresponding to the $Pm\bar{c}n$ space group. The presence of this secondary BaCO_3 (BCO) phase is likely due to the diffusion of atmospheric carbon dioxide (CO_2) into the solution during synthesis, despite the use of a nitrogen flux intended to minimize such contamination. This carbonate phase does not exhibit luminescence and is considered irrelevant for the current scintillation investigation. Its formation may be favored in regions where the BaZrO_3 matrix is locally non-stoichiometric or structurally unstable, potentially associated with the presence of vacancies or unreacted barium precursors.

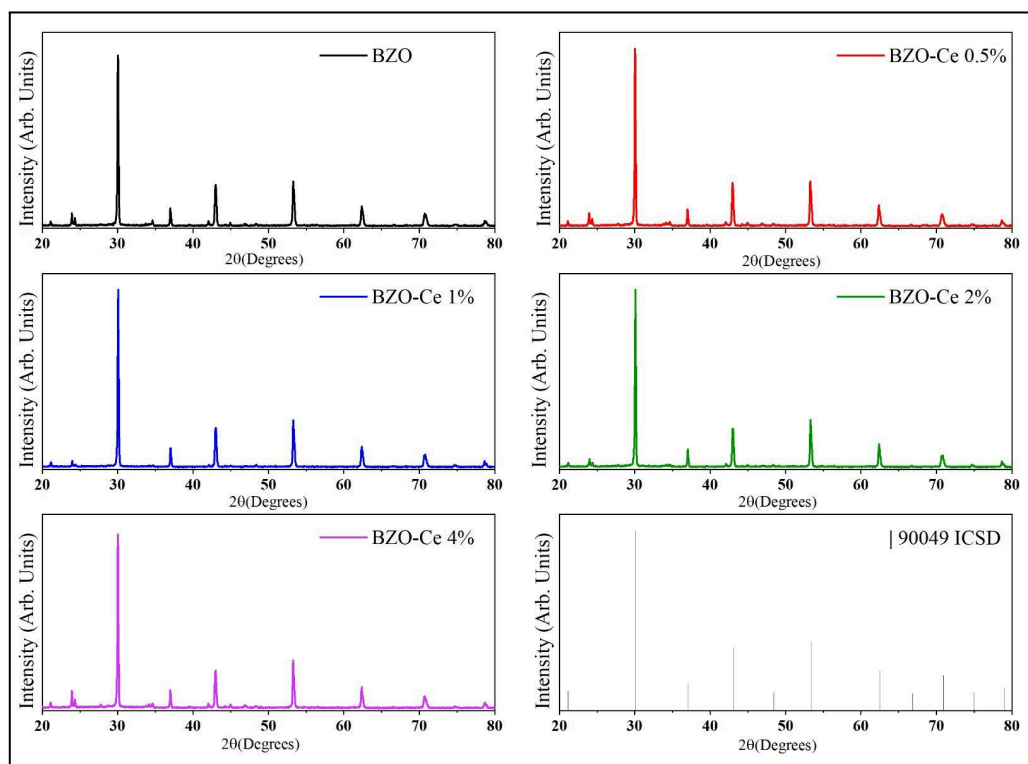


Figure 4: XRD patterns of $\text{BZO}:x\%\text{Ce}$.

The intensity and scattering displacements do not present a significant modification with the Ce-doped, suggesting that the modifications with the incorporations of the rare-earth elements doped does not affect the crystallinity, difference that was observed in studies related with this material under the same growth method (Fassbender et al., 2021).

3.3 Radioluminescence emission

The radioluminescent process involves several consecutive steps. First, high-energy X-ray photons are absorbed by the material, creating electron-hole pairs. Next, these pairs undergo thermalization, where electrons move to the bottom of the conduction band and holes to the top of the valence band. From this point, there are two main routes: either the electrons and holes recombine directly, emitting high-energy photons in the UV range, or they become trapped at defect sites such as vacancies. The trapped charge carriers can later recombine, either through detrapping and recombination with each other or via tunneling, resulting in visible light emission (radioluminescence). This emission typically occurs in the blue-green region and is influenced by the presence and population of interband states associated with defects like oxygen vacancies.

Figure 5 shows the dependence of radioluminescence emission intensity on Ce-doped concentration. The emission maxima were centered around 467 nm to BZO pure sample. The radioluminescence spectra from pure BZO exhibits a broad emission band ranging from 350 nm to nearly 600 nm, as observed by Moreira et al. (2011), with maximum peak reported before. The radioluminescence is attributed to F-centers formed during the synthesis process, associated with intrinsic defects such as oxygen vacancies, which function as electron traps, and barium vacancies, which act as traps for holes. As described by Moreira et al. (2011), F-centers play a crucial role in the radioluminescence mechanism of BZO. These centers trap electrons, introducing intra-gap energy levels within the material's band gap. Upon irradiation, electrons are generated in the matrix and can be captured by these F-centers, creating electron trap states. These electrons remain temporarily stored until they are released through detrapping processes. When these liberated electrons recombine with holes also trapped at other defect sites, this recombination results in photon emission in the visible range, producing the observed luminescence. To Ce-doped samples as shown in Figure 5, a noticeable suppression of the luminescence is observed.

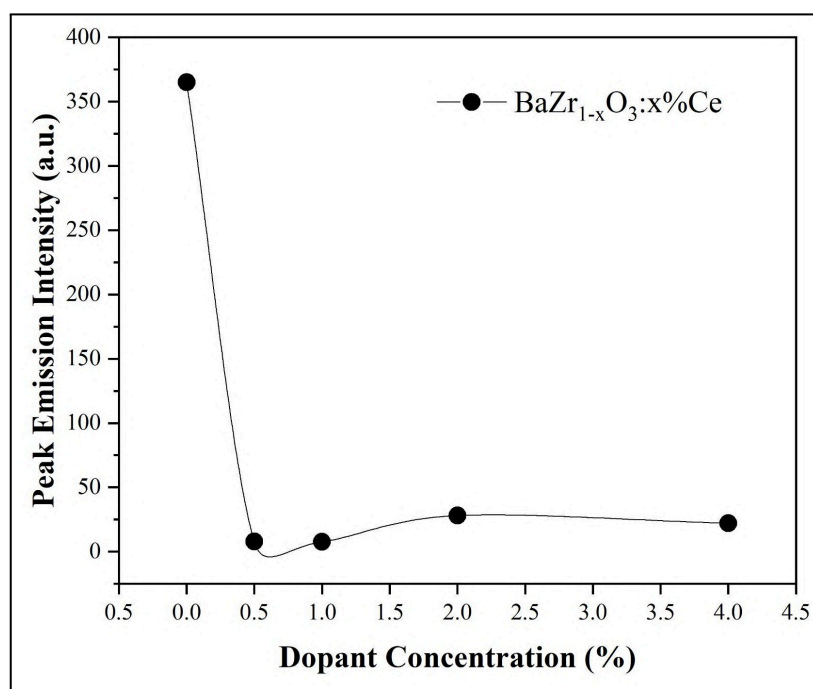


Figure 5: Radioluminescence intensity as a function of dopant concentration.

The incorporation of the rare earths suggests that this modification in the crystal structure passivate the intrinsic defects. The observed disappearance of the broad emission band upon cerium doping reflects a passivation mechanism that suppresses radioluminescence not by eliminating defects but by rendering them optically inactive. This interpretation aligns with similar findings in halide perovskites, where Eu^{3+} – Eu^{2+} interactions lead to passivation through charge transfer (Sun et al., 2025). Similar behavior was reported in YAG:Ce scintillators, where first-principles calculation by Linderälv and collaborators (2021) demonstrated that oxygen vacancies act as killer centers by facilitating non-radiative energy transfer, thereby reducing luminescence efficiency. Importantly, this luminescence passivation results from a charge transfer process: Ce^{3+} ions transfer charge to intrinsic defects such as oxygen vacancies during thermal relaxation, forming Ce^{4+} states that are optically inactive. This charge transfer, combined with strong electron-phonon coupling, leads to the dissipation of excitation energy as lattice vibrations rather than photon emission (Linderälv et al., 2021).

The suppression of radioluminescence observed upon cerium doping can also be linked to the formation of structural defects, particularly oxygen vacancies and F^+ centers, as revealed by recent EPR spectroscopy studies on Ce-doped BZO (Mesquita et al., 2025). These defects introduce mid-gap electronic states that promote non-radiative recombination processes, thereby suppressing the luminescent emission. Moreover, $\text{Ce}^{3+}/\text{Ce}^{4+}$ valence fluctuations and dopant clustering, also evidenced by EPR, likely

enhance charge carrier trapping and lattice disorder, further compromising the material's radiative efficiency, consistent with the charge transfer mechanism described by Linderälv et al. (2021).

3.4 Count Analysis under Various Radioisotope Excitations

The scintillation performance of the BZO-based samples was evaluated under excitation from radioactive sources placed in close proximity to the scintillating BZO films. The measurements were carried out by recording the count per second (cps) for each sample using ^{137}Cs , ^{22}Na and ^{90}Sr , with corresponding background signals recorded under identical conditions (Table 2).

Table 2: Count measured (cps) for BZO-based samples under excitation with different radioactive sources (^{137}Cs , ^{22}Na and ^{90}Sr). Measurements were performed under identical conditions, and cps values are used as relative indicators of light output.

Sample	Background Cps	^{137}Cs Cps	^{90}Sr Cps	^{22}Na Cps
BZO	0.2	14	160	909
BZO: 0.5%Ce	0.4	8	150	685
BZO: 2%Ce	0.6	10	135	527
BZO: 4%Ce	0.2	9	105	491

Upon exposure to the ^{137}Cs source, no significant changes were observed in the count rate as a function of dopant concentration in the BZO films. The overall response remained at background levels across all compositions, indicating the absence of detectable scintillation under this excitation. This suggests that the gamma radiation from ^{137}Cs did not effectively induce luminescence in the thin BZO layers.

The use of ^{137}Cs is mainly justified by the gamma emission attributed to its daughter isotope, $^{137\text{m}}\text{Ba}$, which emits gamma radiation at 662 keV. However, due to the low activity of the source and the limited thickness of the BZO films, the gamma radiation was not capable of efficiently stopping or interacting with this material. In addition, ^{137}Cs emits β^- particles (electrons) with a maximum energy of about 511 keV and an average energy of approximately 174 keV. Given the low average energy and the low source activity, it is likely that only a small number of electrons reach the BZO film with sufficient energy to produce detectable scintillation.

Oxide scintillators such as GGAG:Ce and LuAG:Ce have been successfully employed as films for the detection of charged particles, such as electrons, alphas and betas. Single-crystalline GAGG:Ce films have demonstrated high efficiency in electron detection for scanning electron microscopy (Bok et al., 2016), while highly luminescent LuAG:Ce thick films have been explored for radiation imaging and particle detection (Matsumoto & Ito, 2022). The density and band gap of BZO ($\sim 6.2 \text{ g/cm}^3$ and $\sim 5 \text{ eV}$) are comparable to those of these established scintillators, GGAG:Ce ($\sim 6.6 \text{ g/cm}^3$ with $\sim 6.5 \text{ eV}$) and LuAG:Ce ($\sim 6.7 \text{ g/cm}^3$ and $\sim 7.8 \text{ eV}$) (Moreira et al., 2011; Lecoq, 2016). This suggests that the limited interaction observed in this study is more likely related to the thin film geometry than to any intrinsic limitation of the material in terms of density or stopping power.

When the ^{90}Sr source was applied, which had the same activity as the ^{137}Cs source, the energy from β^- particles is slightly higher but still comparable to those emitted by ^{137}Cs . However, the high-energy beta particles (up to 2.28 MeV) are primarily due to its daughter isotope ^{90}Y , which is almost instantly formed after the decay of ^{90}Sr and governs the maximum energy emitted in the process. These high-energy particles readily penetrate the 50 μm BZO layer. As the energy is much higher, more light is emitted in the films, increasing the probability that a trigger is produced in the SiPM by efficiently depositing energy in the film and producing detectable scintillation.

In contrast, the significantly higher cps measured for ^{22}Na (~ 900 cps) can be attributed to its activity (10 μCi), which is a hundred times higher than that of the other sources used in this experiment (0.1 μCi). Although the positron energy emitted by ^{22}Na is in a similar range on the order of hundreds of keV as the beta particles from ^{137}Cs , the markedly higher activity of the ^{22}Na source is the dominant factor explaining

the increased response. The fact that the ^{22}Na source produced a count two orders of magnitude higher than ^{137}Cs under the same measurement conditions reinforces this interpretation.

^{22}Na emits gamma radiation at 511 keV (from positron-electron annihilation) and 1275 keV (from nuclear de-excitation of ^{22}Ne). Due to the low thickness of the BZO films, the probability of gamma interaction and energy deposition is greatly limited. Nevertheless, the contribution of these gamma photons cannot be entirely ruled out.

Therefore, under the given conditions, excitation of the BZO layer by ^{22}Na is predominantly driven by positrons particles. Due to the short source-to-film distance, positrons are able to reach the scintillator and deposit energy directly into the material before annihilation.

Beyond the differences among radiation sources, the effect of dopant concentration on the scintillation response showed distinct behaviors depending on the excitation type. Under X-ray excitation (up to 45 keV), the introduction of dopants strongly suppressed the radioluminescence, with the emission becoming nearly undetectable. In contrast, when excited by radioactive sources such as ^{22}Na and ^{90}Sr , the scintillation signal persisted across all samples, although a clear reduction in count was observed with increasing dopant concentration. This decline may indicate a decrease in light output, potentially associated with the increased non-radiative pathways. These contrasting trends highlight that the luminescence mechanisms and the role of dopants in BZO films are strongly dependent on interaction mechanisms and the nature of the radiation.

4 Conclusions

In this work, BaZrO_3 (BZO) thin films, both undoped and doped with rare-earth elements, were synthesized via a microwave-assisted hydrothermal method and applied as scintillating layers. The materials were confirmed to be stoichiometric, and their luminescent behavior was evaluated under different excitation conditions. Radioluminescence measurements using X-rays revealed that doping significantly suppressed the emission. In contrast, radioactive source excitation resulted in measurable scintillation for both undoped and doped samples, although higher counts were consistently observed for the undoped samples. The clear scintillation response under exposure to positrons from the ^{22}Na source provides a relevant proof for the viability of BZO in positron and electron detection. These results emphasize the critical role of experimental geometry, excitation energy, and dopant concentration in modulating scintillation efficiency. Therefore, count values obtained under these conditions should be interpreted as relative indicators of light output rather than absolute light yield. Importantly, the use of BZO aligns with growing efforts to replace lead-based scintillators with non-toxic and environmentally sustainable alternatives. Although further optimization is needed, particularly regarding material thickness, absolute light yield, and decay time, this study supports the feasibility of employing BZO thin films in radiation detection systems, including compact dosimetric devices and nuclear instrumentation technologies.

Acknowledgements

The authors acknowledge the federal public institutions on Brasil and in Spain for providing research facilities and access to characterization infrastructure as: Federal University of Pelotas (UFPel), Federal University of Sergipe (UFS), Laboratory of Interdisciplinary Electrochemistry and Ceramics (LIEC) at the Federal University of São Carlos (UFSCar), University Complutense Madrid (UCM) and Consejo Superior de Investigaciones Científicas (CSIC). The authors acknowledge the support of the Brazilian funding agency Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES)—Finance Code 001, the Spanish Council of Science and Innovation through the Grant PID2023-147569NB-C21 funded by MICIU/AEI/10.13039/501100011033 and FEDER, UE.

References

Broadcom Inc. (2024). *Silicon Photomultiplier Module: AFBR-S4E001* [Image]. Retrieved from <https://www.broadcom.com/products/optical-sensors/silicon-photomultiplier-sipm/afbr-s4e001>

- Bok, J., Lalinský, O., Hanuš, M., Onderišinová, Z., Kellar, J., & Kučera, M. (2016). GAGG:Ce single crystalline films: New perspective scintillators for electron detection in SEM. *Ultramicroscopy*, 163, 1–5.
- Cheng, S., Nikl, M., Beitlerová, A., Kucerková, R., Du, X., Niu, G., Jia, Y., Tang, J., Ren, G., & Wu, Y. (2021). Ultrabright and highly efficient all-inorganic zero-dimensional perovskite scintillators. *Advanced Optical Materials*, 9, 2100460.
- Fassbender, R. U., Teixeira, V. C., Galante, D., Ferrer, M., Jardim, P. L. G., Ratmann, C. R., Cava, S. S., & Moreira, M. L. (2021). Correlation between local structure and electronic properties of BaZrO₃:Tb,Yb optical ceramics. *Journal of Electron Spectroscopy and Related Phenomena*, 251, 147106.
- Fujishiro, F., Yamamoto, S., Yahata, T., & Oishi, M. (2024). Variable photoluminescence intensity ratio with the excitation wavelength in Eu³⁺-doped perovskite-type alkaline earth zirconates: Possibility of a unique visualization of ultraviolet light. *Inorganic Chemistry*, 63(13), 5865–5871.
- Hunyadi, M., Samu, G. F., Csige, L., Csík, A., Buga, C., Janáky, C. (2022). Scintillator of Polycrystalline Perovskites for High-Sensitivity Detection of Charged-Particle Radiations. *Advanced Functional Materials*, 32, 2206645.
- Lecoq, P. (2016). Development of new scintillators for medical applications. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, 809, 130–139.
- Linderälv, C., Åberg, D., & Erhart, P. (2021). Luminescence quenching via deep defect states: A recombination pathway via oxygen vacancies in Ce-doped YAG. *Chemistry of Materials*, 33, 73–80.
- Liu, R., Li, F., Zeng, F., Zhao, R., & Zheng, R. (2024). Halide perovskite X-ray detectors: Fundamentals, progress, and outlook. *Applied Physics Reviews*, 11, 021327.
- Mahadik, P., Sengupta, B., Vishwanadh, S., Mishra, S. K., Sudarsan, V., & Dey, G. K. (2017). Effect of γ -irradiation on structure and properties of Nd³⁺-doped perovskite. *Radiation Physics and Chemistry*, 139, 152–156.
- Matsumoto, S., & Ito, A. (2022). High-throughput production of LuAG-based highly luminescent thick film scintillators for radiation detection and imaging. *Scientific Reports*, 12, 19319.
- Mesquita, D., Moreira, M., Villegas, J., Gomes, M., Pimentel, J.R., Longo, E., Silva, P. (2025, July 21). *Defect-Induced Magnetism in Cerium-Doped Barium Zirconate Perovskites: A Systematic EPR Study of Oxygen Vacancies and Dopant Effects* [Preprint, Authorea].
- Moreira, M. L., Volanti, D. P., Andrés, J., Montes, P. J.R., Valerio, M. E.G., Varela, J. A., Longo, E. (2011). Microwave-assisted hydrothermal synthesis of rare-earth-doped BaZrO₃ powders. *Scripta Materialia*, 64, 118–121.
- Moreira, M. L., Andrés, J., Mastelaro, V. R., Varela, J. A., & Longo, E. (2011). On the reversed crystal growth of BaZrO₃ decaoctahedron: Shape evolution and mechanism. *CrystEngComm*, 13, 5818–5824.
- Moseley, O. D. I., Doherty, T. A. S., Parmee, R., Anaya, M., Stranks, S. D. (2021). Recent advances in halide perovskite scintillators for radiation detection. *Journal of Materials Chemistry C*, 9, 11588–11604.
- Pagano, F., Král, J., Děcká, K., Pizzichemi, M., Mihóková, E., Čuba, V., & Auffray, E. (2024). Nanocrystalline lead halide perovskites to boost time-of-flight performance of medical imaging detectors. *Advanced Materials Interfaces*, 11, 2300659.
- Sun, P.-P., Shi, Y., Wang, G., Deng, K., Wu, J., Zeng, C., & Chi, W. (2025). Minimizing vacancy defects with Eu passivation strategy to enable highly efficient perovskite photovoltaics. *Applied Surface Science*, 684, 161922.

Feasibility of Retrospective Dosimetry Using Ceramic Material from Spark Plugs

C. R. E. Silva*, L. C. Meira- Belo

Centro de Desenvolvimento da Tecnologia Nuclear, 30270-901, Belo Horizonte, Minas Gerais,
BRAZIL

claudeteroberta@gmail.com/ lcmb@cdtn.br

Abstract

Retrospective dosimetry aims to estimate the radiation doses absorbed by individuals or objects in past events by using physical, chemical, biological, or historical evidence. Among the available techniques, solid-state dosimetry methods such as Thermoluminescence (TL) and Optically Stimulated Luminescence (OSL) play a key role in reconstructing exposure scenarios when conventional dosimeters were not present at the time of the incident. In this study, we investigated the feasibility of using automotive spark plugs as retrospective dosimeters, taking advantage of the luminescent properties of their ceramic insulator, which is predominantly composed of alumina (Al_2O_3). The high operating temperatures to which spark plugs are routinely exposed act as a natural thermal reset, effectively removing pre-existing signals from environmental radiation and ensuring that any detectable luminescent response primarily reflects accidental or unusual radiation exposures following installation. The additive dose method (ADM) was applied using both TL and OSL techniques to characterize the dosimetric behavior of the ceramic material. The experimental procedures included controlled laboratory irradiation, stepwise additive dose increments, and subsequent analysis of glow curves and OSL decay profiles. The results revealed well-defined dose–response relationships, good reproducibility, and stable signals after irradiation, indicating that the ceramic insulator possesses trapping and recombination centers suitable for dosimetric purposes. Moreover, the absence of significant fading during the evaluation period corroborates its potential for long-term dose reconstruction. Overall, these findings demonstrate that ceramic materials from automotive spark plugs can serve as reliable retrospective dosimeters, capable of mapping dose distributions in scenarios of accidental or unexpected radiation exposure. This approach broadens the range of materials available for post-event dose assessment and provides a practical option for retrospective dosimetry in radiological emergencies or forensic investigations.

Keywords: Retrospective dosimetry; Automotive spark plugs; Optically stimulated luminescence; Thermally stimulated luminescence.

1 Introduction

Retrospective dosimetry is an essential methodology for reconstructing the distribution of absorbed dose in individuals or in the environment after accidental or intentional radiation events. Accurate dose determination is crucial in nuclear or radiological incidents, both to guide appropriate medical interventions and to mitigate the effects of exposure to ionizing radiation. Furthermore, such data are indispensable for epidemiological studies assessing the long-term effects of radiation on exposed populations (Wintle, 1997).

Traditional retrospective dosimetry approaches rely on the analysis of radiation-induced damage to biological tissues, such as blood, teeth, and nails (Trompier et al., 2009; Marciniak and Ciesielski, 2015). However, environmental materials such as brick, tile, marble, granite, quartz, and personal belongings can also serve as effective retrospective dosimeters (Bøtter-Jensen and Murray, 2002; Fattibene and Callens, 2010). Ideal dosimetric materials should be readily available, radiation-sensitive, homogeneous, and capable of retaining absorbed dose information until the time of analysis. A common challenge in this area is the presence of natural luminescent signals in the materials (spurious background signals), which can make it difficult to isolate and quantify the signal specifically induced by a hypothetical nuclear or radiological accident (Yukihara et al., 2022; Thomsen, 2004).

Thermoluminescence (TL) and optically stimulated luminescence (OSL) techniques are widely used to overcome these challenges by exploiting the physical properties of irradiated materials to quantify the absorbed dose. TL is a physical phenomenon in which irradiated materials emit light when heated in a controlled manner. This emission results from defects in the material's crystal lattice that act as traps for radiation-induced charge carriers (electrons and holes). When heated, these charges are released and recombine at luminescent centers, emitting photons in specific energy ranges. The resulting glow curve is proportional to the absorbed dose (McKeever, 1985; Bos, 2006). OSL operates similarly, except that the stimulation to release trapped charges is provided by optical radiation rather than heat. The stability of the charge traps in both methods depends on the material properties and environmental conditions (Jain and Ankjærgaard, 2010). Although traditionally applied to dedicated dosimeters, TL and OSL can also be employed with unconventional materials exposed to radiation in accidental events (Giussani et al., 2020).

This study focuses on investigating unconventional dosimetric materials, particularly the ceramic components of automotive spark plugs, and their potential for dose reconstruction based on their luminescent properties (Wintle, 1997; Bøtter-Jensen and Murray, 1999; Yukihara and McKeever, 2011; Giussani et al., 2020). Among automotive components, spark plugs have characteristics that make them promising retrospective dosimeters in accidental irradiation scenarios. The ceramic insulator, composed primarily of alumina (Al_2O_3), acts as the main dosimetric element. Alumina combines high thermal resistance and structural stability with the ability to retain information on prior exposure to ionizing radiation in its crystal structure (Yukihara and McKeever, 2011).

During normal engine operation, spark plugs are exposed to temperatures ranging from 500 °C to 950 °C (Knol et al., 2024). Under these conditions, charge carriers trapped in defects within the alumina crystal structure by ambient radiation tend to be thermally released, effectively resetting the dosimetric signal. However, in the event of a radiological or nuclear accident near a parked vehicle, the ceramic insulator remains at room temperature, allowing the trapped charges to be preserved. This phenomenon enables radiation-induced signals to be retained for long periods, thus making spark plugs a promising material for retrospective dosimetry. Moreover, spark plugs are shielded from ambient light by both the engine compartment and the spark plug cap, thereby minimizing the loss of dosimetric information from unintended optical stimulation. With proper collection and light-protected transport, subsequent TL or OSL analyses can accurately determine the absorbed dose.

2 Materials and Methods

Automotive spark plugs from different manufacturers and models were collected from local automotive workshops. The selection aimed to represent a variety of manufacturers and vehicle usage histories, ensuring diversity in operational conditions. Although the ceramic composition proved to be predominantly constant among the samples, variations in manufacturing processes and design were considered in order to encompass possible differences in dosimetric behavior. From this set, a single spark plug was chosen for testing, serving as the basis for all analyses and evaluations presented in this study.

Each spark plug was initially cleaned with a hydrocarbon-based solvent to remove oily residues. The components were then washed with water and soap, rinsed, and air-dried. After cleaning, the spark plugs were fractured in a bench vise, producing flat ceramic fragments with an average mass of (6.0 ± 0.1) mg.

TL and OSL measurements, as well as sample irradiations, were performed using a TL/OSL reader model DA-20 (RISØ National Laboratory, Denmark), located at the Luminescent Dosimetry Laboratory, Section of Radiation Dosimetry, Centro de Desenvolvimento da Tecnologia Nuclear (LDL/SECDOS/CDTN). The equipment allows precise control of irradiation time and enables the delivery of fractionated doses to the samples. The reader is equipped with a $^{90}\text{Sr}/^{90}\text{Y}$ beta source, with an initial activity of 1.48 GBq in 2010 and a maximum energy of 2.27 MeV.

The additive dose method (ADM) was applied by preparing multiple aliquots from the sample whose accumulated dose was to be estimated. In one aliquot, the natural luminescence intensity (corresponding to the natural dose or, in this context, the incidental dose) was measured. The remaining aliquots were irradiated with different known doses to obtain an additive dose line, providing the relationship between luminescence intensity and the sum of the natural dose plus the added laboratory doses. The method assumes that the sensitivity of the sample during laboratory irradiation is the same as when it was exposed to environmental radiation. The accumulated dose from an accidental exposure can then be estimated by interpolating the additive dose line.

2.1 Additive Dose Method by TL technique

To evaluate the accuracy and feasibility of using automotive spark plugs for retrospective dosimetry via the additive dose method with TL, 15 aliquots of approximately 6 mg each were prepared. First, the samples were annealed at 400 °C in the RISØ reader to erase any residual luminescent signals from natural background radiation. A preliminary reading was performed to ensure that any residual signals were removed. One aliquot was used to simulate the natural dose, while the remaining 14 aliquots were irradiated individually.

The natural dose was simulated with a known dose of 0.140 Gy, and the other aliquots were irradiated with additional doses ranging from 0.28 Gy to 4.0 Gy to obtain individual glow curves and subsequently construct the additive dose line. All irradiations were carried out in the RISØ reader. TL readings were performed over a temperature range of 20 °C to 380 °C with a heating rate of 5 °C/s, acquiring 400 data points. After TL measurements, the additive dose line was obtained from the integral of the area under each glow curve. A summary of the procedure sequence is presented in Table 1.

Table 1: Sequence of TL procedure using the ADM

Step	Procedure
1	Isothermal cleaning
2	TL reading for reference zero
3	Beta irradiation to simulate accidental dose (0.14 Gy)
4	TL reading of the simulated accidental dose
5	Irradiation Da + 0.28Gy + TL reading
6	Irradiation Da + 0.57Gy + TL reading
7	Irradiation Da + 0.86Gy + TL reading
8	Irradiation Da + 1.14Gy + TL reading
9	Irradiation Da + 1.43Gy + TL reading
10	Irradiation Da + 1.43Gy + TL reading
11	Irradiation Da + 1.73Gy + TL reading
12	Irradiation Da + 2.00Gy + TL reading
13	Irradiation Da + 2.28Gy + TL reading
14	Irradiation Da + 2.57Gy + TL reading
15	Irradiation Da + 3.15Gy + TL reading
16	Irradiation Da + 3.43Gy + TL reading
17	Irradiation Da + 3.72Gy + TL reading
18	Irradiation Da + 4.00Gy + TL reading

2.2 Additive Dose Method by the OSL technique

The accumulated dose using the ADM with the OSL technique was determined from 15 aliquots of approximately 6 mg each. All aliquots underwent an isothermal treatment in the RISØ reader at 450 °C with a heating rate of 10 °C, followed by blue LED stimulation at 90% power for 60 s to erase any latent signals from natural background radiation.

Subsequently, one aliquot was irradiated to simulate the natural dose (0.56 Gy), while the remaining aliquots were irradiated individually with additional doses ranging from 0.611 Gy to 7.80 Gy to obtain OSL emission curves and construct the additive dose line. All irradiations were performed using the $^{90}\text{Sr}/^{90}\text{Y}$ beta source integrated into the RISØ reader. OSL readings were carried out using blue LED stimulation at 90% power, with the temperature maintained at 25 °C, a readout time of 40 s, and 400 acquisition points.

This procedure was repeated three times to obtain the mean OSL response and its standard deviation for each radiation dose. Due to variations in aliquot mass and sensitivity, a sensitivity correction factor was applied. For this purpose, all aliquots were irradiated with a dose of 0.62 Gy and read using the same parameters applied for the additive dose line measurements. The individual sensitivity correction factor was determined by the ratio of the integral of the area under each glow curve to the integral used in the additive dose line. After applying sensitivity corrections, the final additive dose line was established. A summary of the procedure sequence is presented in Table 2.

Table 2: Sequence of TL procedure using the ADM

Step	Procedure
1	Isothermal cleaning
2	TL reading para o zero de referência
3	Irradiação beta para simular dose natural (0.56 Gy)
4	Leitura dose natural
5	Irradiation Da + 0,611Gy + OSL reading
6	Irradiation Da + 1,13 Gy + OSL reading
7	Irradiation Da + 1,68 Gy + OSL reading
8	Irradiation Da + 2,25 Gy + OSL reading
9	Irradiation Da + 2,81 Gy + OSL reading
10	Irradiation Da + 3,37 Gy + OSL reading
11	Irradiation Da + 3,93 Gy + OSL reading
12	Irradiation Da + 4,50 Gy + OSL reading
13	Irradiation Da + 5,06 Gy + OSL reading
14	Irradiation Da + 5,62 Gy + OSL reading
15	Irradiation Da + 6,19 Gy + OSL reading
16	Irradiation Da + 6,75 Gy + OSL reading
17	Irradiation Da + 7,31 Gy + OSL reading
18	Irradiation Da + 7,87 Gy + OSL reading

3 Results

The additive dose method (ADM) uses multiple aliquots of the same material, each with slightly different masses and intrinsic sensitivities. As a result, a variation in sensitivity was observed among the aliquots, requiring a correction by applying an individual sensitivity correction factor. For this purpose, all aliquots were irradiated with a dose of 0.23 Gy and read under the same parameters used for the additive dose line. The individual sensitivity correction factor for each aliquot was determined by the ratio between the integral of the area under its glow curve and the integral of the area under the reference glow curve used in the additive dose line. It should be noted that the non-integer dose values reported throughout the results are expected, as they arise from the characteristics of the fixed-distance $^{90}\text{Sr}/^{90}\text{Y}$ irradiator combined with time-controlled exposures. Because each irradiation includes a decay correction for the source, the calculated doses naturally exhibit fractional values rather than rounded integers.

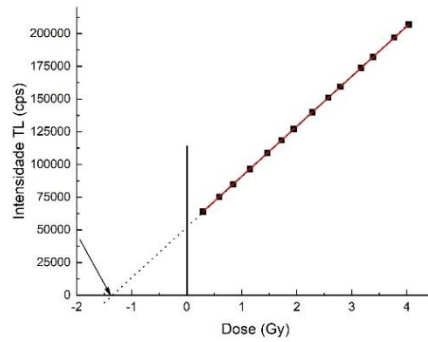


Figure 1: Additive dose line constructed from the integrated TL emission curves obtained through the additive dose method (ADM).

Once the additive dose line was obtained, the accumulated dose was estimated by extrapolating the linear fit to the x-axis. The resulting value was compared with the nominal dose of 0.14 Gy that had been applied to the aliquot to simulate the natural exposure. Figure 2 shows the additive dose line for TL emission, based on aliquots irradiated with additional doses ranging from 0.28 Gy to 4.0 Gy. The thermoluminescent response using the ADM exhibited a linear behavior, as observed in Figure 2. The linear model was applied to describe the relationship between the integrated area of the TL glow curves and the irradiation dose. The slope of the linear fit was determined as $a = (3828 \pm 3.75) \text{ cps} \cdot ^\circ\text{C} \cdot \text{Gy}^{-1}$, and the intercept as $b = (52\,641 \pm 6.76) \text{ cps} \cdot ^\circ\text{C}$. The estimated accidental dose was 0.14 Gy with an uncertainty of 2.1%, which shows excellent agreement with the nominal natural dose.

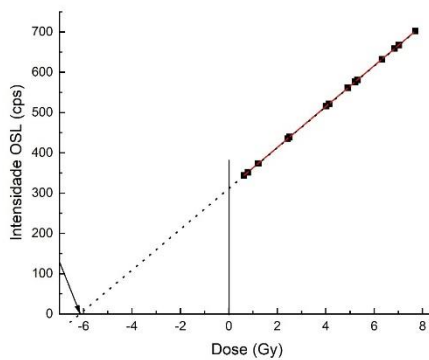


Figure 2: Additive dose line constructed from the integrated intensity of the OSL emission curves obtained through the additive dose method (ADM).

For the OSL measurements, the additive dose readings were performed using time windows corresponding to the first 0.8 s of stimulation for signal integration, while the background was calculated from the last 10 s of the OSL signal. The resulting additive dose line for the OSL technique is shown in Figure 3. From this line, the natural dose of the sample was determined. By extrapolating the additive dose line, the estimated dose was approximately 0.62 Gy, with an uncertainty of 8.7% and a coefficient of variation of 4.8%. When compared with the nominal dose of 0.56 Gy applied to simulate the natural exposure, the estimated value was slightly higher but still in good agreement with the expected dose.

4 Conclusions

The results obtained in this study demonstrate the potential use of ceramic components from automotive spark plugs as TL and OSL materials when applying the additive dose method. The spark-plug samples, composed of aluminum oxide as the electrical insulator, exhibited luminescent responses suitable for absorbed dose estimation under controlled laboratory conditions.

The additive dose method yielded percentage variations of approximately 2.1% for the thermoluminescence (TL) measurements and around 8.6% for the optically stimulated luminescence (OSL) measurements. These consistent responses highlight the feasibility and promise of using spark-plug ceramics as candidate materials for retrospective dosimetry, particularly in accidental irradiation scenarios.

Applying TL and OSL techniques to unconventional materials, such as spark plugs, represents an innovative approach to retrospective dosimetry and broadens the range of materials available for dose reconstruction in radiological emergencies. While further studies are needed to assess factors such as long-term stability, environmental effects, and reproducibility in real-field conditions, the findings presented here establish a solid basis for considering automotive spark plugs as potential low-cost alternatives for retrospective dosimetry.

Acknowledgements

The authors gratefully acknowledge the financial support provided by the Brazilian National Council for Scientific and Technological Development (CNPq), the National Institute of Science and Technology – INCT–INAIS (Instrumentation for Nuclear Applications with Quality in Industry and Health), and the Foundation for Research Support of the State of Minas Gerais (FAPEMIG).

References

- Bøtter-Jensen, L., Murray, A.S., (1999). Developments in optically stimulated luminescence techniques for dating and retrospective dosimetry. *Radiation Protection Dosimetry* 84, 307–315. <https://doi.org/10.1093/oxfordjournals.rpd.a032745>.
- Bos A.J.J. Theory of thermoluminescence. *Radiat. Meas.* 2007; 41(Suppl 6): S45-S56. <https://doi.org/10.1016/j.radmeas.2007.01.003>.
- Bøtter-Jensen L, Murray A.S., Optically stimulated luminescence in retrospective dosimetry. *Radiat. Prot. Dosim.* 2002; 101(1-4):309-314.
- Fattibene P., Callens F. EPR dosimetry with tooth enamel: A review. *Appl. Radiat. Isot.* 2010; 68:2033-2116. <https://doi.org/10.1016/j.apradiso.2010.05.017>.
- Furetta, C., 2003. *Handbook of Thermoluminescence*. World Scientific, Singapore. <https://doi.org/10.1142/5167>.
- Giussani A., Lopez M.A., et al. EURADOS review of retrospective dosimetry techniques for internal exposures to ionising radiation and their applications. *Radiat. Environ. Biophys.* 2020; 59(3):357-387. <https://doi.org/10.1007/s00411-020-00845-y>.
- Jain M, Ankjærgaard C. Towards a non-fading signal in feldspar: Insight into charge transport and tunnelling from time-resolved optically stimulated luminescence. *Radiat. Meas.* 2011; 46(3):292-309. <https://doi.org/10.1016/j.radmeas.2010.12.004>.

Unsupervised Learning in preparation for Pattern Recognition in k-Feldspar Using Cathodoluminescence Emission and EDX Composition Mapping

A. L. T. e Silva^{1*}, L.C. Meira-Belo¹, R. Kumar², M. Kook², M. Jain²

¹ Centro de Desenvolvimento da Tecnologia Nuclear, Radiation Dosimetry Section, 30161-970, Belo Horizonte, Brazil

² Technical University of Denmark, Department of Physics, 4000, Roskilde, Denmark

* aluiztasil@yahoo.com.br

Abstract

Unsupervised learning provides an effective strategy for identifying patterns in complex datasets, especially where labeled data is unavailable. This study applies unsupervised machine learning to pattern recognition in K-feldspar samples using integrated cathodoluminescence (CL) spectra and Energy Dispersive X-ray Spectroscopy (EDX) composition maps. A dataset of 92×128 pixels was analyzed, with each pixel containing a CL emission spectrum. The initial dataset formed a (11776, 1631) matrix and was highly redundant, with PCA showing over 85% of the variance explained by the first component—indicating low representational diversity for clustering. To overcome this, spectral data were enriched by extracting key features such as peak position, intensity, area, and full width at half maximum (FWHM). These were combined with normalized spatial coordinates and EDX data to create a richer, lower-dimensional feature space. PCA on the enriched dataset showed a more balanced variance distribution, with six components capturing over 80% of the variance, reflecting increased structural complexity. DBSCAN was used to identify clusters based on combined CL, spatial, and EDX features. The resulting segmentation revealed distinct emission-based domains and enabled the comparison of luminescence patterns with local composition. This workflow demonstrates a scalable approach to integrating multi-modal microscopy data and enhances interpretability for materials characterization. The results provide a strong benchmark for future mineralogical studies using unsupervised learning.

Keywords: *Unsupervised Learning; Cathodoluminescence; Feldspar.*

1 Introduction

Feldspars are abundant rock-forming minerals commonly used in luminescence dating techniques such as thermoluminescence (TL) and optically stimulated luminescence (OSL), which estimate the time since mineral grains were last exposed to heat or sunlight. These methods are widely applied in geological and archaeological studies (Pressad. et al., 2017). However, accurate age determination can be hindered by complex luminescence behaviors, including anomalous fading and compositional variability. To improve the understanding of these phenomena, advanced characterization techniques such as cathodoluminescence (CL) and energy dispersive X-ray spectroscopy (EDX) are increasingly employed.

CL microscopy (Götze, 2012) enables spatially resolved investigations of luminescent centers in feldspar minerals, offering insight into the interplay between crystal chemistry and metastable states. In particular, the study by Kumar et al. (2020) demonstrated that variations in infrared CL emission—especially at ~ 880 nm and ~ 955 nm—are closely correlated with local potassium and sodium content. Their results showed

that different emission centers are associated with distinct lattice sites, and that the shifts in emission barycenters reflect compositional gradients across the samples.

Building upon this foundation, the present study integrates spectrally enriched CL data with elemental maps obtained via energy-dispersive X-ray spectroscopy (EDX). By extracting pixel-level spectral features—such as peak positions, FWHM, and barycenters—and combining them with elemental composition, we apply unsupervised machine learning techniques to identify spatial emission domains. This approach advances the interpretability of CL data and offers a robust framework for mapping luminescent behavior in relation to feldspar microstructure (Kumar et al. 2020).

Unsupervised machine learning is particularly useful in large datasets like these, where patterns emerge without predefined labels (Wu Xi, et al 2022). Combined with data enrichment—merging spectral and chemical features—it enables deeper insights into the relationships between emission behavior and composition across thousands of measurements.

The objective of this paper is to present the data enrichment of a raw CL/EDX dataset to generate machine learning features that enhance the analysis of large luminescent datasets.

2 Materials and Methods

In this work, we focus on **Sample R47**, a well-characterized sedimentary feldspar of Tanzanian origin previously studied by Kumar et al. (2020). The sample exhibits a grain size distribution of 120–300 μm and was analyzed using CL microscopy at 80 $\times$ magnification with a spatial resolution of 12.6 μm . Elemental mapping via EDX reveals a potassium-rich composition, with a $\text{K}/(\text{K} + \text{Na})$ ratio exceeding 0.8 across most of the mapped region. R47 is particularly suited for this analysis due to the strong presence of both $\sim 880\text{ nm}$ and $\sim 955\text{ nm}$ near-infrared CL emissions, which are spatially associated with Na- and K-rich domains, respectively. The dataset used in this study is exclusively derived from CL and EDX measurements and chemical composition (Table 1) of this sample, allowing for a detailed investigation of its emission behavior and compositional heterogeneity at the pixel level.

Table 1 – R47 Chemical Composition

Element	Atomic %
C	18.35 ± 0.12
O	53.62 ± 0.26
Na	0.36 ± 0.01
Mg	0.06 ± 0.01
Al	6.10 ± 0.03
Si	17.00 ± 0.06
K	4.43 ± 0.02
Ca	0.04 ± 0.01
Fe	0.02 ± 0.01
Ba	0.03 ± 0.01
Total	100.00

2.1 The Raw Dataset

The original CL emission spectra from all pixels were combined into a high-dimensional matrix of shape (11776, 1600), where each row represents a pixel and each column a spectral channel. This raw dataset, though comprehensive, contains more information than necessary to characterize the relevant behaviors

for combined CL and EDX analysis (Extra 27 columns added to the original dataset resulting into a (11776, 1627) dataset). A preliminary Principal Component analysis (10 dimensions-PCA) revealed that the variance is not well-captured by a low number of components, suggesting redundancy and possible noise as reported on Figure 1 Scree Plot. These findings suggest that the raw dataset requires additional preprocessing and dimensionality reduction before it can be effectively utilized in machine learning applications.

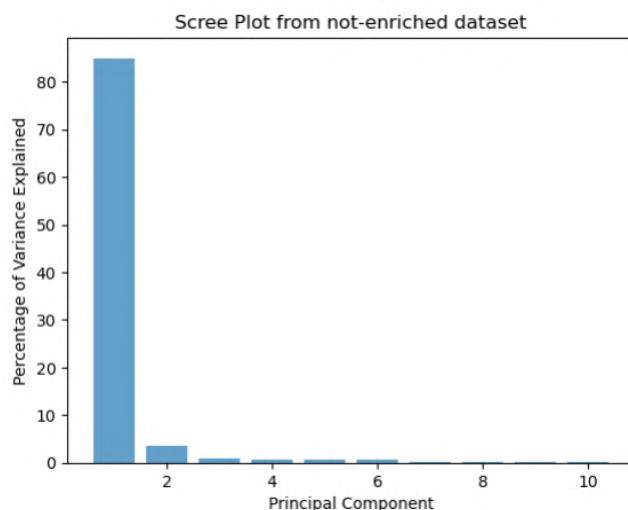


Figure 1 - Scree Plot from PCA of Raw dataset

2.2 Data Treatment and Feature Engineering

To process and analyze the large volume (Figure 2) of cathodoluminescence (CL) and energy-dispersive X-ray spectroscopy (EDX) data from feldspar samples, a structured data enrichment pipeline was developed. This workflow integrates glow curve spectra (11776×1600) with corresponding elemental composition data ($11,776 \times 27$) through a series of steps including peak extraction, full width at half maximum (FWHM) calculations, peak ratio computation, and additional spectral descriptors. The result is a unified dataset with 104 engineered features per pixel, capturing both optical and chemical characteristics.

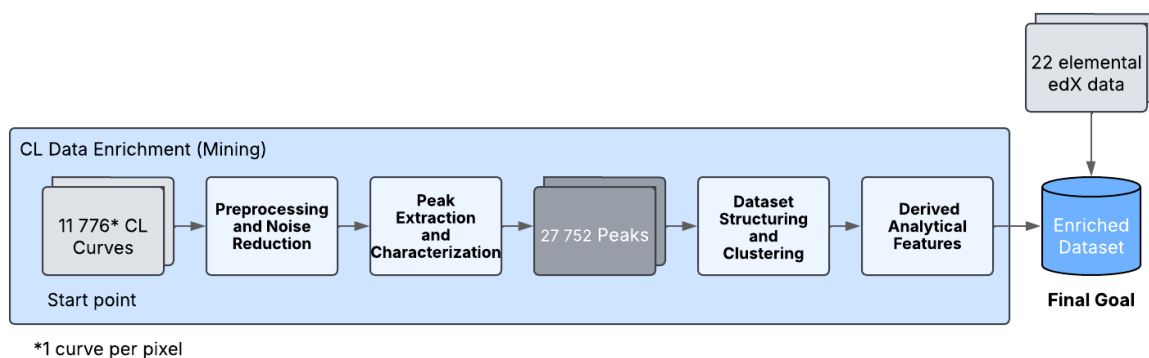


Figure 2 - Data Enrichment Pipeline

This enriched dataset serves as the foundation for downstream analysis using unsupervised and supervised machine learning models (Figure 3). The goal is to identify spatial and spectral emission patterns, support pattern recognition, and enable predictive modeling of luminescent behavior linked to microstructural and compositional variation in feldspar.

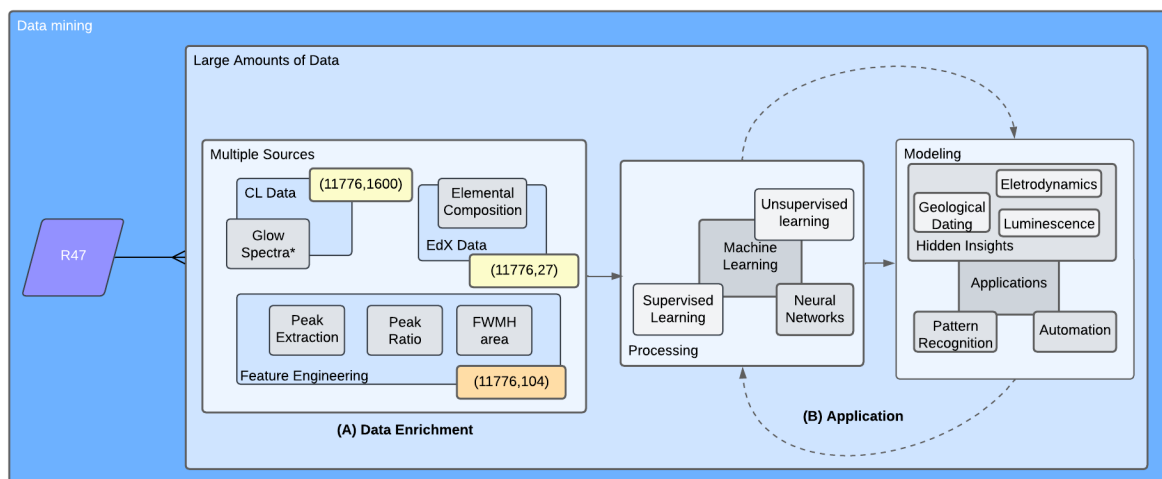


Figure 3 - Workflow for data enrichment and application of machine learning in CL/EDX analysis.

2.3 Preprocessing and Noise Reduction

Each raw CL spectrum (Figure 4) was first smoothed using a Savitzky–Golay filter (window length = 50, polynomial order = 2) to minimize high-frequency noise while preserving peak structure. This step enhanced signal clarity for downstream peak detection and analysis.

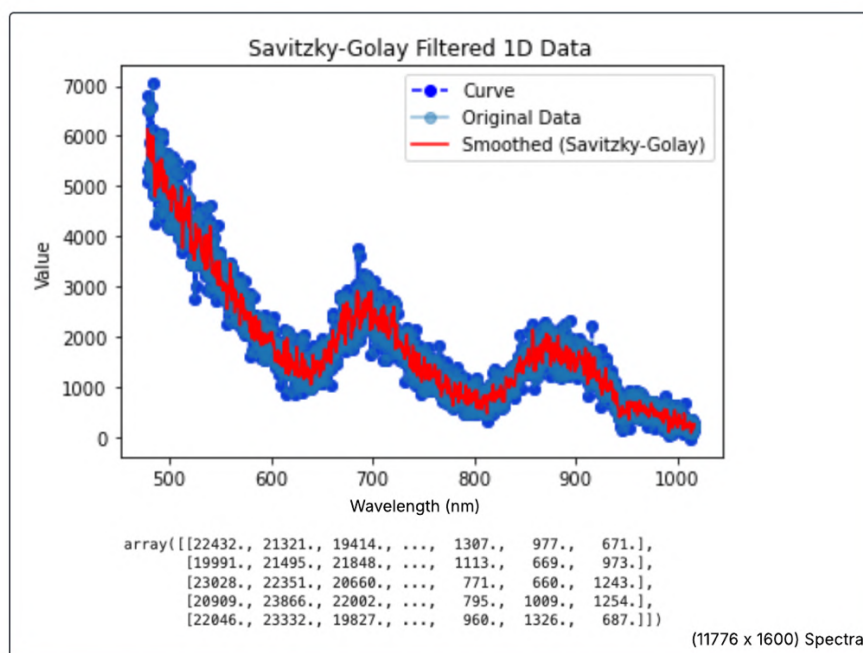


Figure 4 - Raw data versus Smoothed data using Slavitzky-Goulay Filter

2.4 Peak Extraction and Characterization

For each denoised spectrum, peak detection was performed using `scipy.signal.find_peaks` (SciPy, 2024), with parameters tuned to $\text{height} \geq 40$, $\text{distance} \geq 150$, and $\text{prominence} \geq 90$. Out of the 11776 CL Curves a total of 27,752 peaks were detected across all CL curves. Peak attributes were calculated for each detected peak:

- **Peak position (nm)**
- **Peak intensity**
- **Full Width at Half Maximum (FWHM)**
- **Left and right boundaries**
- **Baricenter**
- **Area under curve (integrated around FWHM)**

2.5 Dataset Structuring and Clustering

Before merging EDX elemental data, a separate clustering step was applied to the individual peaks extracted from the CL spectra. Each peak was defined by its wavelength (`peak_x`) and clustered using DBSCAN ($\text{eps} = 9$, $\text{min_samples} = 20$). DBSCAN was chosen because it does not require setting the number of clusters in advance and can identify groups of peaks with different shapes and densities while ignoring noise. This helped separate the main emission bands—such as those near 720 nm, 880 nm, and 955 nm—based purely on their spectral characteristics. The result provided a clearer structure for further analysis and made feature extraction more consistent before combining with the EDX data.

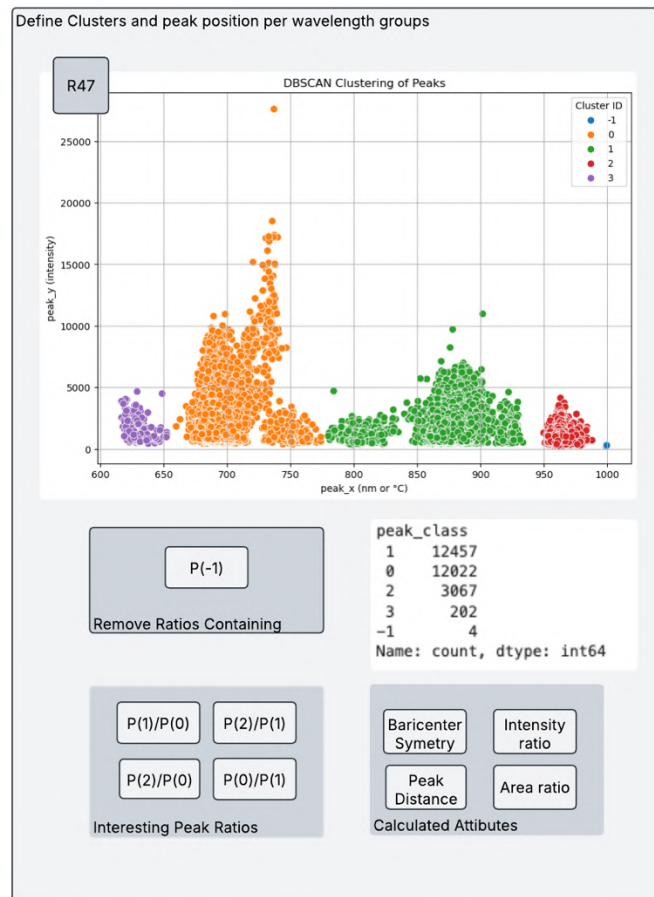


Figure 5 - Peak Clustering based on wavelength of incidence using DB Scan

This density-based method effectively identified distinct emission domains in the spectral space:

- Cluster 3 (Purple): 620–660 nm — likely related to Fe^{3+} transitions.
- Cluster 0 (Orange): Broad 670–780 nm band — related to Fe^{3+} and mixed emission states.
- Cluster 1 (Green): 800–930 nm — corresponds to IRCL emission centered around 880 nm.
- Cluster 2 (Red): 940–980 nm — associated with the 955 nm band typical of K-rich feldspar.
- Noise (ID -1, Blue): Outliers and low-signal peaks not assigned to any dense region.

This clustering served two purposes:

1. It **reduced peak variability** by assigning each peak to a consistent spectral group.
2. It enabled **group-wise feature pivoting**, making it possible to engineer domain-specific features like ratios (e.g., $P[2]/P[0]$) and cluster-based intensity distributions.

Only after this classification step was completed were peak-level descriptors aggregated per pixel, and **merged with EDX composition data** to form the final dataset used for broader spatial and compositional analysis.

To represent each pixel's spectrum as a fixed-size feature vector, the extracted peak features were pivoted into structured columns based on their clusters (e.g., P0, P1, P2). This accounted for the irregular number of peaks per spectrum. Missing values and sparsity were handled by removing rare peak positions (e.g. P-1), and maintaining consistent dimensionality across samples.

2.6 Derived Analytical Features

To enrich the representation of CL behavior, several derived features were calculated:

- **Peak ratios:** $P(1)/P(0)$, $P(2)/P(0)$, $P(2)/P(1)$, etc.
- **Symmetry descriptors:** Comparison of distances from peak to left and right boundaries.
- **Intensity and area ratios**
- **Peak distances and separations**

These features capture not only the location but also the relative shape and prominence of the peaks across the spectrum.

3 Enriched Dataset Performance

The enriched dataset used for unsupervised analysis in cathodoluminescence (CL) and EDX mapping is composed of 103 features distributed across eight categories (Table 2). The Peak Position Features (14) capture the spectral location of emission peaks, including primary and secondary peaks from the different clusters, which are crucial for identifying luminescent centers. Peak Shape Descriptors (29) quantify geometrical aspects such as full width at half maximum (FWHM), and baricenters providing detailed insights into emission characteristics. Area Metrics (7) represent the integral under the curve or within peak regions, serving as a proxy for emission intensity. Peak Count (2) tracks the number of detected peaks per spectrum, offering a quick summary of spectral complexity. Derived Analytical features (22) include calculated ratios, distances, and boolean flags (e.g., presence of IR peaks), enriching the dataset with comparative and interpretable variables. Coordinates (2) preserve the spatial context of each spectrum, enabling pixel-wise spatial analysis. Finally, EDX Elements (27) contribute the compositional data at each pixel, essential for correlating spectral features with elemental distribution. This structured

and multi-modal feature set can support robust pattern recognition and clustering in complex mineralogical samples.

Table 2 – Enriched Dataset Feature Types

Category	Number of Features
Peak Position Features	14
Peak Shape Descriptors	29
Area Metrics	7
Peak Count	2
Derived Analytical	22
Coodinates	2
EDX Elements	27
Total	103

To assess the impact of this feature engineering process, a principal component analysis (PCA) was applied to both the raw and enriched datasets. While the original high-resolution CL spectra were heavily dominated by the first principal component—accounting for over 85% of the total variance and indicating strong redundancy—the enriched dataset displayed a significantly more distributed variance profile. Approximately six components were required to explain 80% of the variance, reflecting a meaningful increase in dimensional complexity. This shift confirms that the addition of structured spectral and compositional features enhances the dataset’s suitability for machine learning tasks by improving separability and interpretability (Figure 6).

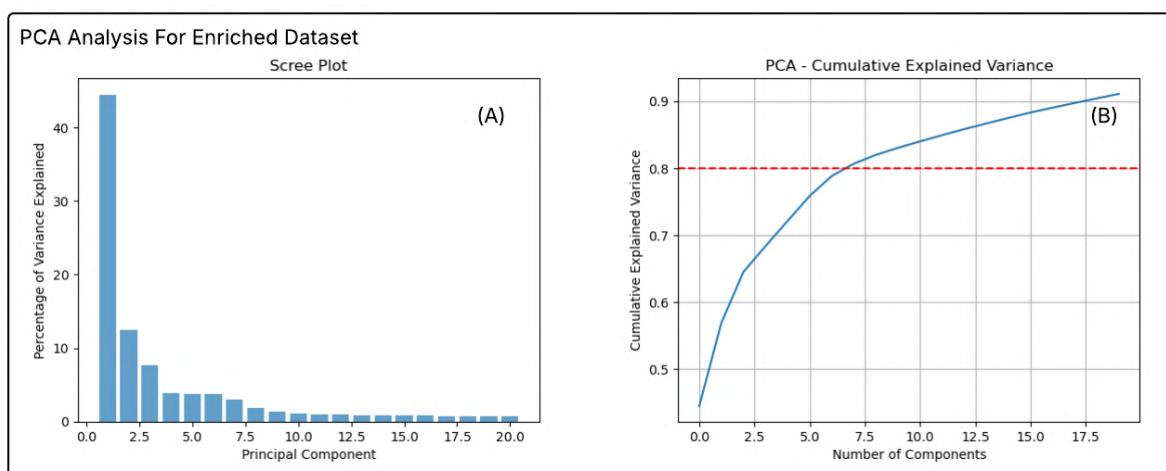


Figure 6 - Principal Component Analysis (PCA) of enriched dataset

4. Conclusion

This study addressed the challenge of enriching a high-dimensional raw dataset composed of cathodoluminescence (CL) spectra and EDX composition maps to improve its suitability for unsupervised machine learning. The initial dataset, while rich in spectral detail, exhibited high redundancy and low structural variance, limiting its utility for pattern recognition. Through systematic preprocessing—including spectral smoothing, peak extraction, and the integration of spatial and compositional attributes—the dataset was transformed into a lower-dimensional, feature-enriched representation. Principal Component Analysis (PCA) confirmed the effectiveness of this enrichment, revealing a more

balanced distribution of variance and greater potential for unsupervised methods. Clustering via DBSCAN successfully grouped CL emission pixels by emission peak and compositional similarity, enabling the identification of emission-based domains.

The proposed workflow supports scalable, interpretable processing of large luminescent datasets and provides a valuable framework for linking optical and chemical information at the microscale. These results establish a benchmark on a machine learning friendly dataset structuring for future studies in mineralogical classification and materials characterization, where automated pattern discovery from multi-modal microscopy data is increasingly vital. Optimization of peak extraction parameters and a definition of the clusters region of interest can reduce errors and standardize the approach for peak classification.

Future applications may extend this framework to other luminescent materials, explore kinetic or time-resolved CL behavior, or integrate deep learning models for domain generalization and segmentation tasks in Earth and materials sciences.

Acknowledgements

FAPEMIG, CNPQ and CAPES and CDTN/CNEN for sponsoring this work.

References

Arquero, O.; Berenguer-Antequera, J.; Benavente, J.F. (2024). *Radiat. Meas.* 178: 107293. <https://doi.org/10.1016/j.radmeas.2024.107293>.

Götze, J. Application of cathodoluminescence microscopy and spectroscopy in geosciences. (2012). *Microsc. Microanal.* 18, 1270–1284

Kumar, R.; Martin, L.I.D.J.; Poelman, D.; Vandenberghe, D.; De Grave, J.; Kook, M.; Jain, M. (2020). *Sci. Rep.* 10: 15650. <https://doi.org/10.1038/s41598-020-72334-7>.

Prasad, A. K., Poolton, N. R. J., Kook, M. & Jain, M. Optical dating in a new light: a direct, non-destructive probe of trapped electrons. *Sci. Rep.* 7, 1–15 (2017).

SciPy Community. (2024). *SciPy User Guide*. <https://docs.scipy.org/doc/scipy/tutorial/index.html#user-guide>.

Wu, X.; Liu, X.; Zhou, Y. (2022). *Lecture Notes in Electrical Engineering* 804: 576–590. doi:10.1007/978-981-16-6324-6_59

Respuesta Termoluminiscente ante rayos gamma de la HAp pura y dopada con tierras raras (Dy^{3+} y Eu^{3+}) a dos concentraciones 0.1 M y 0.5M

**Torres-Cortés, C.O.^{1*}, Pérez-Arrieta, M.L.¹, Montalvo-Rivera, T.²,
Vicencio-Hernández, S.Y.², Vega-Carrillo H.R.³**

¹Universidad Autónoma de Zacatecas, Unidad Académica de Física, 98000, Zacatecas, México

²CICATA-IPN, Unidad Legaria, 11500, Cd. de México, México

³Universidad Autónoma de Zacatecas, Unidad Académica de Estudios Nucleares, 98068, Zacatecas, México

* carina.torres@uaz.edu.mx

Abstract

La síntesis de materiales con propiedades termoluminiscentes es una actividad científica de relevancia para el desarrollo de nuevos o mejorados materiales, con el objetivo de ser utilizados en áreas específicas como es la dosimetría de radiación ionizante. La Hidroxiapatita (HAp) es un compuesto a base de fosfatos de calcio que se caracteriza por su biocompatibilidad con el tejido humano y que, al momento de incorporar impurezas en la estructura del compuesto, se espera que haya el aumento de la respuesta termoluminiscente debido a que de acuerdo con la literatura los lantánidos cuentan con una alta eficiencia luminiscente. En este trabajo se sintetizaron materiales termoluminiscentes de HAp pura y dopada con Dy^{3+} y Eu^{3+} a dos concentraciones (HAp:Dy 0.1, 0.5 M y HAp:Eu 0.1, 0.5 M) mediante el método hidrotérmico; los materiales fueron irradiados con una fuente de Co-60 con una actividad de 88 TBq a 80 cm de distancia durante 3.41 min. con una dosis de 199.65 cGy. La respuesta termoluminiscente se exhibe en las curvas de brillo, donde la HAp:Dy 0.1 M presenta la mayor respuesta TL mostrando un pico con mejor definición a la temperatura de 241 °C. Al aumentar la concentración de 0.1 a 0.5 M las curvas de brillo lucen similares donde los picos TL se desplazan a mayores temperaturas.

Palabras clave: Respuesta Termoluminiscente, Rayos Gamma, Hidroxiapatita, Tierras Raras.

1 Introducción

El uso controlado de las radiaciones ionizantes con objetivos benéficos y pacíficos es imprescindible en áreas médicas, industria alimentaria, nucleoelectricas y medio ambientales. Así, la detección y monitorización de este tipo de radiación es una práctica de gran importancia por el riesgo-beneficio que pueden tener y para ello, los materiales termoluminiscentes juegan un papel importante en la detección y cuantificación de dosis altas y bajas de diversos tipos de radiación ionizante [Chand *et al.*, 2022].

Durante varios años se han desarrollado diferentes materiales basados en la termoluminiscencia (TL), y el estudio de estos materiales (propiedades, comportamiento ante diversos tipos de radiación ionizante, aplicaciones dosimétricas, etc.) es de suma importancia para innovar en nuevos o en mejoras específicas [Chand *et al.*, 2022]. La termoluminiscencia es un mecanismo físico que implica la emisión de fotones de luz visible cuando un material semiconductor preferentemente o aislante, es irradiado con algún tipo de radiación ionizante, como partículas α , partículas β , rayos γ , neutrones, entre otras. La energía de estas radiaciones es absorbida por el material o bien llamado fósforo, creando defectos y portadores de carga, siendo atrapados en los centros de captura en estados metaestables. Para la expulsión o liberación de los portadores de carga, se requiere suministrar energía calorífica y calentarse, estos se recombinan en un estado de menor energía y así se emite la energía previamente absorbida en forma de luz visible (señal TL) [Bos, 2007; Furetta, 2010].

Este trabajo se centra en la detección de la radiación gamma; se considera la forma más energética de todas las ondas electromagnéticas. Debido a su alto nivel de energía, los rayos gamma poseen una gran capacidad de penetración y pueden ionizar al tejido humano al eliminar los electrones del material, motivos por los cuales es de interés científico cuantificarla. El origen natural de este tipo de radiación ionizante es en las galaxias y la radiación solar debido a la desintegración nuclear de materiales radiactivos, asimismo, existen emisores artificiales como fuentes de Co-60 con energía de 1.17 y 1.22 MeV o Cs-137 (0.662 MeV), que tienen aplicaciones industriales y médicas [Katiyar *et al.*, 2022]. Por otra parte, los materiales TL por excelencia o comúnmente utilizados se enlistan en la Tabla 1.

Tabla 1: Materiales TL como dosímetros termoluminiscentes (TLD) más empleados [Kry, *et al.*, 2020; Omanwar *et al.*, 2014]

Fórmula química	Nombre comercial	Z_{eff}	Pico principal en la curva de brillo (°C)
LiF:Mg,Ti	TLD-100	8.31	235
$^6\text{LiF:Mg,Ti}$	TLD-600	8.31	235
$^7\text{LiF:Mg,Ti}$	TLD-700	8.31	235
LiF:Mg,Cu,P	TLD-100H	8.31	200
$\text{CaF}_2\text{:Dy}$	TLD-200	16.90	160-290
$\text{Li}_2\text{B}_4\text{O}_7\text{:Mn}$	TLD-800	7.32	185
$\text{CaSO}_4\text{:Dy}$	TLD-900	15.62	220

Otro de los materiales con propiedades termoluminiscentes reportado en la literatura es la Hidroxiapatita (HAp), es un biomaterial que proviene del mineral natural apatita, fosfato más abundante en las rocas y huesos en el sistema biológico. La HAp es el principal componente del tejido calcificado humano, que constituye entre el 95 % y 97 % del esmalte, aproximadamente del 70 % al 75 % del hueso dental y entre el 60 % y el 70 % de los huesos corporales, debido a su gran biocompatibilidad con el tejido biológico es de interés para ser estudiado como un dosímetro termoluminiscente (TLD) ante distintos tipos de radiación ionizante [El-Gebaly *et al.*, 2022]. En este sentido, se han podido dar pasos científicos significativos en el desarrollo de materiales termoluminiscentes durante las últimas dos décadas [Rivera, 2012]. Alvarez *et al.*, (2014) sintetizaron HAp mediante el método sol-gel para el estudio de las

propiedades TL y concluyeron que la hidroxiapatita obtenida podría utilizarse en aplicaciones de dosimetría de radiación gamma.

Daneshvar *et al.*, (2019) sintetizaron la HAp dopada con varios lantánidos como el disprosio (Dy), europio (Eu), gadolinio (Gd) y lantano (La) al 1%mol mediante el método de hidrólisis, obtuvieron nanopulvos y mostraron buena respuesta para rayos gamma con una dosis de 25 Gy con una mayor intensidad TL el HAp:Dy. La HAp sintetizada por reacción en estado sólido fue por Taghipour *et al.*, (2022) y se realizó el estudio de la respuesta TL ante rayos gamma con una dosis de 800 Gy a diferentes temperaturas de tratamiento térmico, mostrando mayor intensidad el material con cocimiento a 900 °C.

Uno de los trabajos más recientes relacionados con la HAp fue hecho por Rivera *et al.*, (2025) donde se empleó el método de sol-gel, las curvas TL de los materiales se obtuvieron mediante la irradiación con fuentes de radiación gamma de Co-60, un haz de electrones de alta energía y una fuente emisora de neutrones. Los resultados proponen que la HAp puede ser un buen material en aplicaciones de dosimetría retrospectiva. El objetivo de este trabajo fue obtener la respuesta termoluminiscente ante rayos gamma de la HAp dopada con Dy^{3+} y Eu^{3+} con una concentración de 0.1 y 0.5M en ambos dopantes.

2 Materiales y métodos

La síntesis se realizó mediante el método hidrotérmico con los precursores: $Ca(NO_3)_2 \cdot 4H_2O$, NH_4HPO_4 y NH_4OH para el material puro y para los materiales dopados se empleó $Dy(NO_3)_3$ y $Eu(NO_3)_3$ para obtener el dopaje a dos concentraciones 0.1 y 0.5 M, la reacción se llevó a cabo en un reactor de presión controlada a una temperatura de 180 °C por 2 horas y a un pH=9, en seguida se realizaron varios lavados a cada material TL y se secaron a 200°C. El material en polvo se sometió a un tratamiento térmico a 700°C por 2 horas, en la Figura 1 se muestran algunas imágenes del proceso de síntesis.

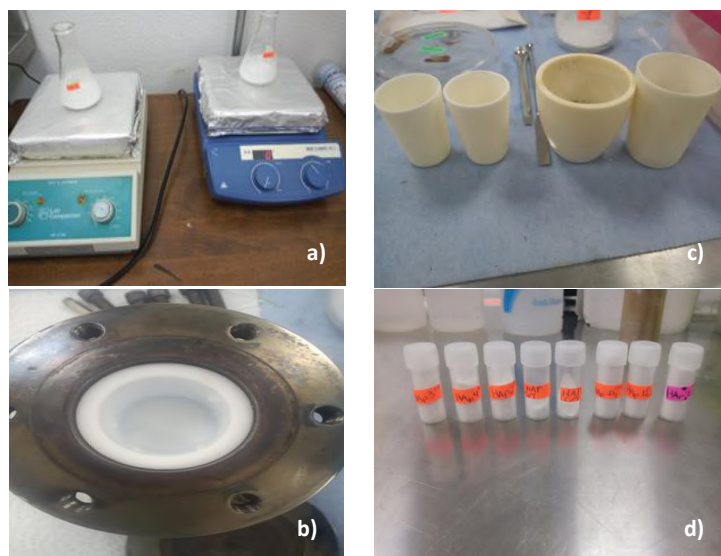


Figura 1: Imágenes del proceso de síntesis mediante el método hidrotérmico: a) Inicio de mezcla y reacción de los precursores; b) Reactor de teflón y autoclave para el inicio de reacción; c) Crisoles para el tratamiento térmico de los materiales TL; d) Materiales TL obtenidos en polvo.

Para determinar la respuesta termoluminiscente se irradiaron los 5 materiales (HAp pura, HAp:Dy 0.1M, HAp:Dy 0.5M, HAp:Eu 0.1M y HAp:Eu 0.5M) con una fuente de Co-60 con una actividad de 88 TBq, los materiales se expusieron a la radiación gamma por un tiempo de 3.41 min. a una distancia de 80 cm con una dosis de 200 cGy. Posteriormente a la irradiación se leyeron los materiales en un equipo TLD

Harshaw 5500 con una rampa de calentamiento de 10 °C/s hasta 350 °C para obtener las curvas de brillo (respuesta TL) correspondiente a cada material TL (Figura 2, equipos de irradiación a) y de lectura b)).

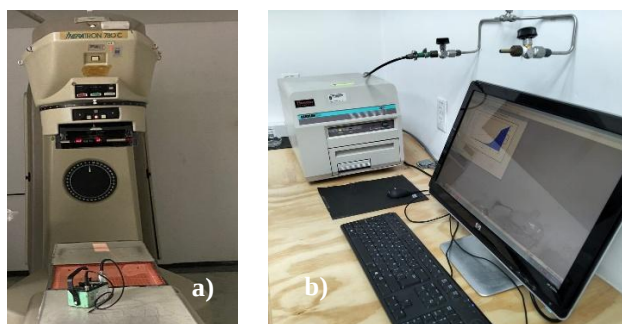


Figura 2: Imágenes a) Unidad de Co-60 THERATRON 780C y b) TLD Harshaw 5500

3 Resultados y discusión

En la Figura 3 se observa la respuesta TL del material de HAp pura donde no hay la presencia de picos y existe un corrimiento hacia la derecha de la intensidad TL, por lo que la absorción de la energía por el material puro no es adecuada. De acuerdo con trabajos reportados la HAp sin impurezas puede presentar respuesta TL para dosis más altas como es en la investigación reportada por Taghipour *et al.*, (2022), mostrando mejor respuesta la HAp con tratamiento térmico a 900°C y ante una fuente de Co-60 con una dosis de 1500 Gy.

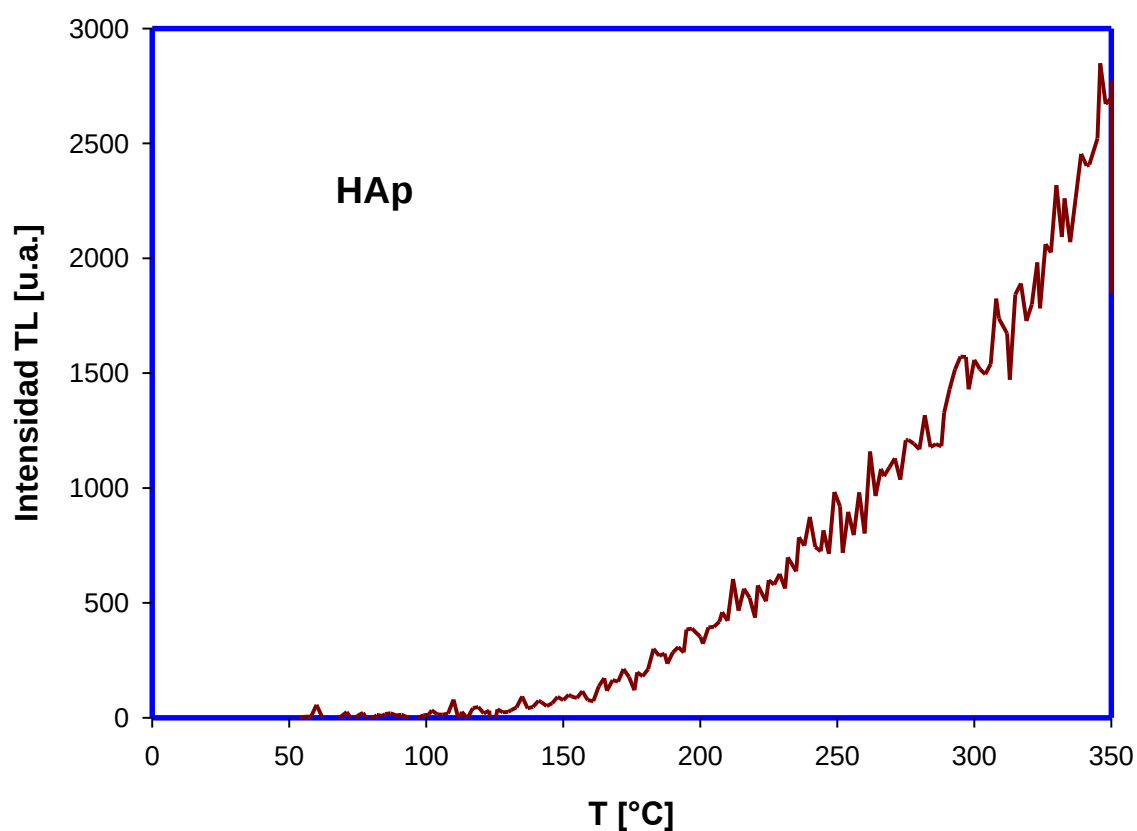


Figura 3. Respuesta Termoluminiscente de la HAp pura

La Figura 4 exhibe la curva de brillo para la HAp dopada con Dy^{3+} al 0.1M, se pronuncian dos picos evidentes y definidos al incorporar un dopante mostrando el de mayor intensidad a 238 °C y un segundo pico a una temperatura de 179 °C. La curva de brillo de la HAp:Dy 0.5M es muy similar a la forma de la curva de la HAp:Dy sin embargo, la intensidad es menor y los picos menos definidos y débiles. De acuerdo con Daneshvar *et al.*, (2019) al contrastar el material TL HAp:Dy 1 %mol obtenido a partir del método de síntesis por hidrólisis, también presenta dos picos, uno alrededor de 110 °C y el pico principal a 198°C, es comparable la forma de la curva de brillo conseguida para la HAp:Dy 0.1M. Es importante mencionar que las características TL son directamente dependientes del método de síntesis empleado, del tratamiento térmico para obtener la estructura cristalina deseada, cantidad de dopaje, así como también de la morfología y tamaño de partículas de las cuales están constituidas el material TL, en este sentido, el aumento de la cantidad de impureza dentro del cristal es directamente proporcional con el número de trampas de electrones generadas, lo que produce mayor cantidad de centros de recombinación [Daneshvar *et al.*, 2019; Bos 2017; Ziaie *et al.*, 2014].

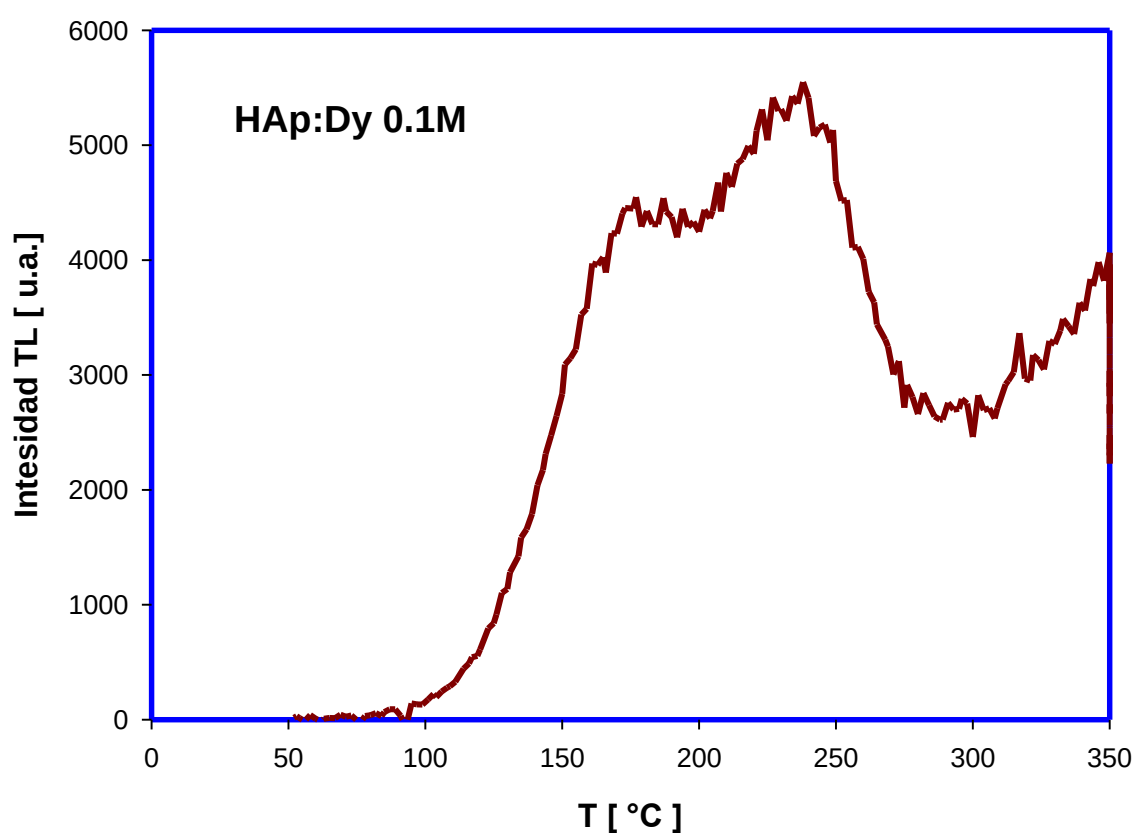


Figura 4. Curva de brillo de la HAp:Dy³⁺ con una concentración de 0.1M

En las Figuras 5 y 6 se muestran las curvas de brillo resultantes para la HAp dopada con Eu a dos concentraciones (0.1M y 0.5M); la HAp:Eu³⁺ 0.1M presenta un pico principal definido a aproximadamente 212 °C y otro menos intenso y más irregular a 286 °C. Mendoza-Anaya *et al.*, (2018) sinterizaron hidroxiapatita dopada con Eu a distintas concentraciones y expuestos a diversas dosis de radiación gamma. El material TL con mejor respuesta fue la HAp:Eu a 0.5 % con un pico TL principal en aproximadamente 130 °C.

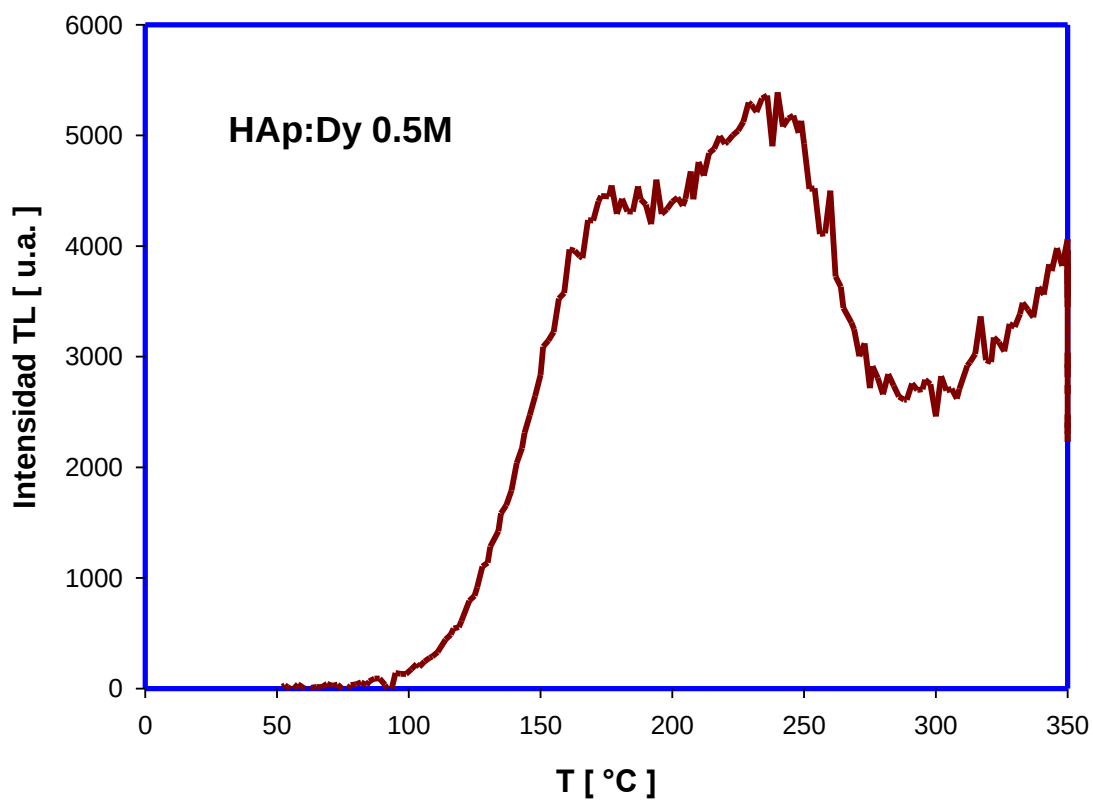


Figura 5. Curva de brillo del material TL de HAp:Dy³⁺ 0.5M

La Figura 6 muestra la respuesta TL de la HAp dopada con Eu³⁺ en una concentración de 0.1M.

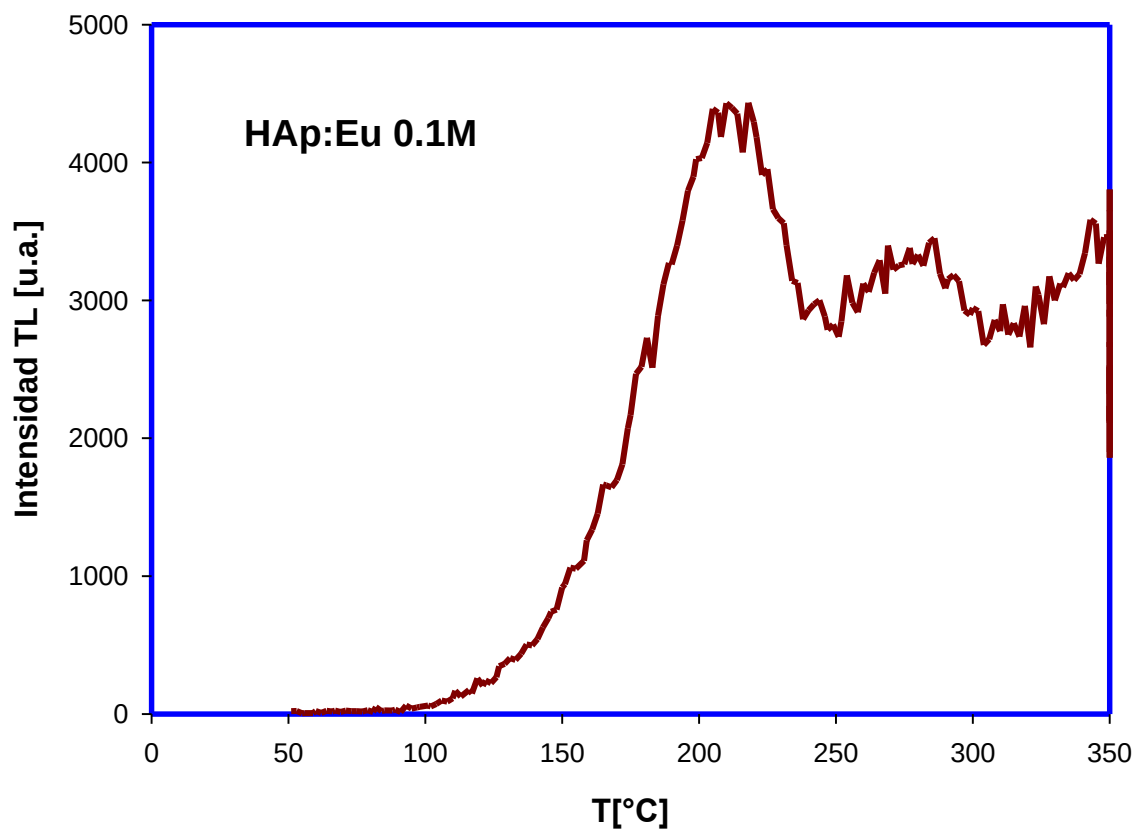


Figura 6. Curva de brillo de HAp:Eu³⁺ 0.1M

En la Figura 7 se puede observar la curva de brillo obtenida para la HAp:Eu³⁺ dopada con una concatenación de 0.5M. No hay la formación de um pico de brillo evidente, sin embargo, se observa una protuberancia ancha alrededor de 250 °C.

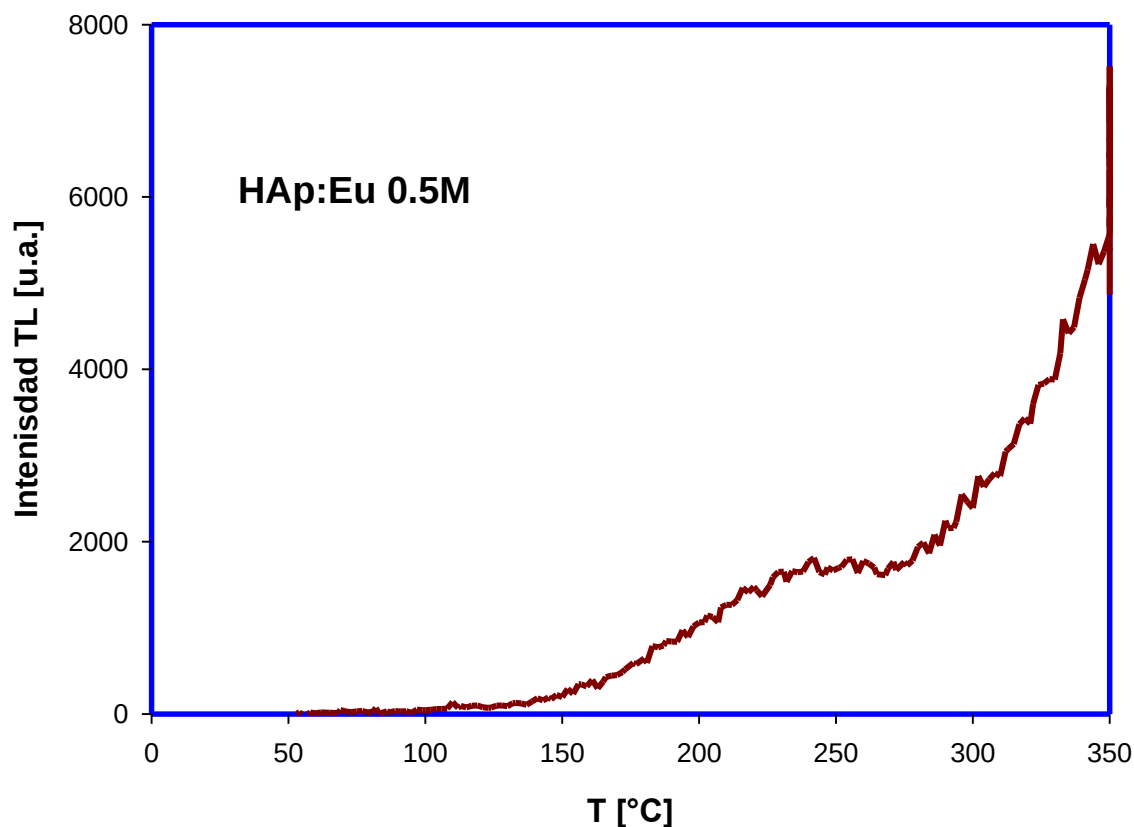


Figura 7. Curva de brillo del material TL HAp:Eu³⁺ 0.5M

Al comparar la respuesta dada con Dy³⁺ exhiben menor intensidad TL en los materiales dopados con Eu³⁺. Cabe mencionar que la literatura reporta mayor eficiencia TL como dopante el Dy mostrando picos TL más intensos y definidos a temperaturas ideales para aplicaciones dosimétricas. De acuerdo con Furetta and Kitis (2004), los materiales TL con fines de dosimetría usualmente deben presentar picos definidos en un rango de temperatura de 150 a 250 °C.

4 Conclusiones

Se obtuvieron 5 materiales TL con base a HAp tanto pura como dopada con dos tierras raras Dy³⁺ y Eu³⁺ a dos concentraciones 0.1M y 0.5 M. Se sometieron a irradiación con una fuente de rayos gamma (Co-60) y exhibieron curvas de brillo con presencia de picos TL. La HAp:Dy³⁺ 0.1M muestra la mejor curva de brillo con dos picos, el principal a una temperatura de 238 °C. A pesar de que en la literatura se reporta que a mayor concentración de dopaje habrá mayor producción de centros de recombinación, en este trabajo se mostró que la concentración óptima es 0.1M en ambos dopantes. De acuerdo con los resultados la HAp sintetizada mediante el método hidrotérmico y dopada con Dy³⁺ y Eu³⁺ es un material TL prometedor para dosimetría ante rayos gamma.

Agradecimientos

Carina Oliva Torres Cortés y sus colaboradores agradecen a la SECIHTI por el apoyo en convocatoria de posdoctorado, a la Unidad Académica de Física, UAZ como centro de trabajo principal, así como también al Centro de Investigación de Ciencia y Tecnología Aplicada IPN, Unidad Legaria por el acceso a equipos para la realización del proyecto.

Referencias

- Alvarez R., Rivera T., Guzman J., Piña-Barba M. C. and Azorin J. (2014). Thermoluminescent characteristics of synthetic hydroxyapatite (SHAp). *Applied Radiation and Isotopes*, 83:192-195.
- Bos A. J. J. (2017). Thermoluminescence as a Research Tool to Investigate Luminescence Mechanisms. *Materials*, 10:135.
- Chand S., Mehra R. and Chopra V. (2022). Recent developments in phosphate materials for their thermoluminescence dosimeter (TLD) applications. *LUMINESCENCE*, 252;119383.
- Daneshvar H., Shafaei M., Manouchehri F., Kakaei S. and Ziaie F. (2019). The role of La, Eu, Gd, and Dy lanthanides on thermoluminescence characteristics of nano-hydroxyapatite induced by gamma radiation. *SN Applied Sciences*, 1:1146.
- El-Gebaly Sh., El-Faramawy N. and Kahil H. (2022). Thermoluminescence response of bone hydroxyapatite. *Optical Materials*, 134:113089.
- Furetta C. (2010). Handbook of thermoluminescence. *World Scientific Publishing Singapore*.
- Furetta C. and Kitis G. (2004). Models in thermoluminescence. *Journal of Materials Science*, 39:2277-2294.
- Katiyar P., Pandey N. and Keshavkant S. (2022). Gamma radiation: A potential tool for abiotic stress mitigation and management of agroecosystem. *Plant Stress*, 5:100089.
- Kry S.F., Alvarez, P., Cygler J. E., DeWerd L. A., Howell R. M., Meeks S., O'Daniel J., Reft Ch., Sawacuchi G., Yukihara E. G. and Mihailidis D. (2020). Clinical use of luminescent dosimeters: TLDs and OSLDs. *Medical Physics*, 47:e19-e51.
- Mendoza-Anaya D., Flores-Díaz E., Mondragón-Galicia G., Fernández-García M. E., Salinas-Rodríguez E., Karthik T. V. K. and Rodríguez-Lugo V. (2018). The role of Eu on the thermoluminescence induced by gamma radiation in nano hydroxyapatite. *Journal of Materials Science: Materials in Electronics*, 29:15579–15586.
- Omanwar S. K., Koparkar K. A. and Virk H. S. (2014). Recent Advances and Opportunities in TLD Materials: A Review. *Defect and Diffusion Forum*, 347:75-110.
- Rivera T. (2012). Thermoluminescence in medical dosimetry. *Applied Radiation and Isotopes*, 71:30-34.
- Rivera-Montalvo T., Alvarez Romero R., Vicencio Hernandez S. J., Correcher Delgado V. (2025). Synthesis and thermoluminescent characterization of synthetic hydroxyapatite as a detector for retrospective TL dosimetry, *Applied Radiation and Isotopes*, 220: 111790.
- Taghipour P., Zolfagharpour F. and Hamideh Daneshvar H. (2022). hydroxyapatite is a solid-state reaction method and study of its thermoluminescence dosimetric properties against gamma rays. *Radiation Physics and Engineering*, 3: 7-10.
- Ziaie F., Farhadi Moein N. and Shafaei M. (2014). Thermoluminescent characteristics of nano-structure hydroxyapatite:Dy. *KERNTECHNIK, Ed. Carl Hanser Verlag GmbH & Co. KG*, 79: 500-503.